Polyfold Theory, proof by proof

The formal proof of Hofer–Wysocki–Zehnder's Polyfold and Fredholm Theory, Part I, in words

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Part I. The sc-calculus (HWZ Chapter 1, §§1.1–1.4)

Chapter 1 of Part I fixes the vocabulary of the whole theory: sc-Banach spaces, sc-operators, sc-subspaces and quotients, sc-Fredholm operators and their stability under sc^+-perturbations, partial quadrants, sc^0 and sc^1 maps, the tangent construction and the inductive definitions of sc^k and sc-smooth, the relation between sc-differentiability and classical differentiability on the levels (Propositions 1.16–1.19 and Corollary 1.20, quoted from the 2010 paper), the chain rule, the degeneracy index of a point of a partial quadrant and boundary recognition. The formal proof establishes all of it, Proposition 1.19 and Corollary 1.20 under the text's own hypothesis (C^{l+1} from U_{m+l} to F_m for $l \le k$), by the 2010 paper's induction (S_k) on elementary maps (§I.17). Example 1.2, the weighted Sobolev spaces, is not formalized.

Two decisions shape everything. First, an sc-Banach space is an *indexed family* of Banach spaces E_m with injective compact inclusions $E_{m+1} \to E_m$, not one space with a chain of subspaces (§I.1). "The point x of E_0 lies in E_m" then means that x is in the image of E_m, and every map has level maps $f_m : E_m \to F_m$ determined by injectivity. Second, the text's standing hypothesis "U relatively open in a partial quadrant" enters the proofs of §§1.2–1.3 only through two facts about each level set U_m: it is the intersection of an open and a convex set, and derivatives within it are unique. These two facts make up the notion of an *sc-domain* (§I.10), which the tangent construction and the level shift preserve by a two-line argument, so the inductions of §1.2 run on sc-domains, and the text's tuples (U, C, E) are a special case.

Four proofs take a different road from the text's. The chain rule replaces the two integrals of §1.4.2 by the one-dimensional mean value inequality along a segment, and does not need the first integral at all (§I.12). The compactness argument behind Propositions 1.16–1.18, the subsequence $h_n \to h$ in E_m for unit vectors $h_n \in E_{m+1}$, is isolated once as the *compact-open lemma*, and its C^k version replaces the multilinear maps $\Gamma(x)$ and the differentiation under the integral sign of the 2010 paper (§I.13). Theorem 1.25, which the text quotes from another paper, is proved by the argument of §2.3, for sc^1 germs (§I.19). And the induction (S_k) behind Proposition 1.19 runs on the order s over all elementary maps $d^j f(\pi_0 z)(\pi_1 z, \dots, \pi_j z)$ at once, rather than on a description of the 2^{k+1} components of $T^{k+1} f$ (§I.17). Smaller departures are flagged where they occur: the complement in Proposition 1.10 is built from a right inverse rather than a lifted basis; the regularizing step of §1.4.1 is a bootstrap; $k = k'$ in Lemma 1.24 is derived from the lemma, not proved first. What Mathlib lacked, the Riesz–Schauder theorem, the stability of Fredholm operators under compact perturbations and the invariance of the Fredholm index, is proved in §I.7; the index statements of the exposition's Lemma 2.D and Proposition 1.10 follow in §I.6 and §I.9. The lemmas that the exposition adds to make the text's remarks precise are proved where they belong: uniqueness of the sc-structure in finite dimensions (§I.1), sc-splittings as sc-projections (§I.3), the smooth points of a quotient (§I.5), the shift and tangent of a tuple (§I.10), compact-open continuity and $\mathrm{sc}^1 = C^1$ on quadrants in finite dimensions (§I.11).

Map.

- `Chapter1/ScBanach`: scales and sc-Banach spaces, the constant structure, sums and shifts; sc-operators, sc-isomorphisms, sc^+-operators; sc-subspaces, sc-splittings, Proposition 1.6; quotients,

Proposition 1.7.

- `Chapter1/FredholmPerturbation`: Fredholm operators between Banach spaces; Riesz–Schauder; compact perturbations.
- `Chapter1/FredholmIndex`: the index of a linear map, its additivity, its invariance under small and compact perturbations; the index on the levels of an sc-Fredholm operator; the index in Proposition 1.10.
- `Chapter1/ExpositionLemmas`: the exposition's Lemmas 1.A(2)–(3), 2.B, 2.C, 3.F, and $\mathrm{sc}^1 = C^1$ in finite dimensions on quadrants.
- `Chapter1/ScFredholm`: Definition 1.8, the index, the level-wise Fredholm property; Propositions 1.9 and 1.10.
- `Chapter1/ScSmooth`: level sets, the tangent TE and TU, level maps, sc^0; sc^1 with its derivative family, uniqueness of Df, sc^k and sc-smooth; partial quadrants, tuples, sc-domains.
- `Chapter1/ChainRule`: Theorem 1.21 and its iterates.
- `Chapter1/Properties`: the compact-open continuity of Df; the base cases of Propositions 1.16, 1.19 and Corollary 1.20.
- `Chapter1/Levels`: the compact-open lemma; the level operators $Df(x) : E_m \to F_m$; Propositions 1.16–1.18 on all levels; Proposition 1.19 and Corollary 1.20 under the stronger hypothesis "C^{k+1} on every level".
- `Chapter1/LevelsWeak`: Proposition 1.19 and Corollary 1.20 as stated: the derivative tower, elementary maps, their sc^1 step and tangent maps, the sum rule for sc^k maps, the induction (S_k).
- `Chapter1/Degeneracy`: Definition 1.23 and Lemma 1.24.
- `Chapter2/DegeneracyIndex`: Theorem 1.25, by the argument of §2.3.

I.1 Scales and sc-Banach spaces (Definition 1.1)

The scale. A **scale of Banach spaces** consists of real Banach spaces E_m, one for each $m \in \mathbb{N}$, together with continuous linear maps $\mathrm{incl}_m : E_{m+1} \to E_m$ that are injective and compact operators `Chapter1/ScBanach · ScScale`. Compactness is Mathlib's: the preimage of some neighbourhood of 0 is mapped into a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator`. The composite inclusion $\iota_m : E_m \to E_0$ is defined by recursion, $\iota_0 = \mathrm{id}$ and $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$ `Chapter1/ScBanach · ScScale.toBase`; it is injective `Chapter1/ScBanach · ScScale.toBase_injective`, and the composite $E_{m+k} \to E_m$ is defined likewise `Chapter1/ScBanach · ScScale.inclK` and is compatible with ι `Chapter1/ScBanach · ScScale.toBase_inclK`. The text's E_m is here the image $\iota_m(E_m) \subseteq E_0$; the text's $E_m \supset E_{m+1}$ is the inclusion of images, and "$x \in E_m$" for $x \in E_0$ is "$x \in \mathrm{range}\,\iota_m$". The **smooth points** E_∞ form the subspace of E_0 of points that lift to every level, $E_\infty = \bigcap_m \mathrm{range}\,\iota_m$ `Chapter1/ScBanach · ScScale.smooth` `Chapter1/ScBanach · ScScale.mem_smooth_iff`, and $E_\infty \cap E_m$ is its preimage in E_m `Chapter1/ScBanach · ScScale.smoothLvl`; a smooth point of E_m lifts to every E_{m+k} `Chapter1/ScBanach · ScScale.exists_inclK_of_mem_smoothLvl`.

The density axiom. An **sc-Banach space** is a scale in which $E_\infty \cap E_m$ is dense in E_m for every m `Chapter1/ScBanach · ScBanach`. This is condition (2) of Definition 1.1 exactly as the text states it. The weaker statement that E_{m+k} has dense image in E_m follows `Chapter1/ScBanach · ScBanach.denseRange_inclK` `Chapter1/ScBanach · ScBanach.denseRange_incl`; it does not obviously imply the axiom, which is why the strong form is the definition.

Why a family. Nothing in Chapter 1 uses more than the family: the text's identification of E_{m+1} with a subset of E_m is the injectivity of incl_m, a field of the structure, and its "the inclusion is continuous" is part of compactness. The gain is that each level is a Banach space in its own right, with its own norm, and Mathlib's analysis applies on each level without carrying a filtration along.

The constant structure. For a finite-dimensional normed space V the family $E_m = V$ with identity inclusions is a scale: the identity of a finite-dimensional space is a compact operator `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · isCompactOperator_id_iff_finiteDimensional` `Chapter1/ScBanach · ScBanach.constScale`. Here ι_m is the identity `Chapter1/ScBanach · ScBanach.constScale_toBase`, every point is smooth

Chapter1/ScBanach · ScBanach.constScale_smoothLvl, and density is trivial, so V is an sc-Banach space Chapter1/ScBanach · ScBanach.const Chapter1/ScBanach · ScBanach.const_toBase.

Only the constant structure (Lemma 1.A(2)–(3) of the exposition). The text remarks that a finite-dimensional space has precisely one sc-structure. For families this reads: if E is an sc-Banach space with E_0 finite-dimensional, the inclusions identify E with the constant structure on E_0. First, every point is smooth: E_∞ is a subspace of the finite-dimensional E_0, hence closed Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional, and dense by the axiom at level 0, hence everything Chapter1/ExpositionLemmas · ScBanach.smooth_eq_top_of_finiteDimensional. So every $x \in E_0$ lifts to every level, and $\iota_m : E_m \to E_0$, injective by construction, is bijective Chapter1/ExpositionLemmas · ScBanach.toBase_bijective_of_finiteDimensional. A continuous linear bijection between Banach spaces is an isomorphism (open mapping theorem) Mathlib/Analysis/Normed/Operator/Banach · ContinuousLinearEquiv.ofBijective; the isomorphisms ι_m commute with the inclusions by the very definition $\iota_{m+1} = \iota_m \circ \mathrm{incl}_m$, so they form an sc-isomorphism of E onto the constant structure on E_0 Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional_apply. In particular all the norms $|\cdot|_m$ are equivalent to $|\cdot|_0$. The second remark, that on an infinite-dimensional space the constant structure fails the compactness axiom, becomes: if an inclusion $\mathrm{incl}_m : E_{m+1} \to E_m$ of a scale is onto, then E_m is finite-dimensional. Indeed incl_m is then a continuous linear bijection, hence an isomorphism e, and $\mathrm{id}_{E_m} = \mathrm{incl}_m \circ e^{-1}$ is compact as a compact operator composed with a bounded one Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm; a Banach space with compact identity is finite-dimensional Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id Chapter1/ExpositionLemmas · ScScale.finiteDimensional_of_surjective_incl.

Direct sums. For scales E, F, the family $E_m \times F_m$ with the product inclusions is a scale: the product of two compact operators is compact, because the preimage of a product of neighbourhoods is mapped into the product of two compact sets Chapter1/ScBanach · ScBanach.prodScale. Its ι_m is the pair of the two Chapter1/ScBanach · ScBanach.prodScale_toBase, and its smooth points are the pairs of smooth points Chapter1/ScBanach · ScBanach.prodScale_smoothLvl; since a product of dense sets is dense, $E \oplus F$ is an sc-Banach space Chapter1/ScBanach · ScBanach.prod Chapter1/ScBanach · ScBanach.prod_smoothLvl.

The level shift. The family E_{m+1} with the inclusions incl_{m+1} is a scale E^1 Chapter1/ScBanach · ScBanach.shiftScale. Its composite inclusion to its base E_1, followed by ι_1, is ι_{m+1} Chapter1/ScBanach · ScBanach.shiftScale_toBase, and its smooth points at level m are the smooth points of E at level $m+1$ Chapter1/ScBanach · ScBanach.shiftScale_smoothLvl, so E^1 is an sc-Banach space Chapter1/ScBanach · ScBanach.shift Chapter1/ScBanach · ScBanach.shift_toBase Chapter1/ScBanach · ScBanach.shift_smoothLvl; E^k is the k-fold iterate Chapter1/ScBanach · ScBanach.shiftK.

Level sets. For $U \subseteq E_0$ the level set $U_m \subseteq E_m$ is the preimage $\iota_m^{-1}(U)$ Chapter1/ScSmooth · ScBanach.lvlSet; $U_0 = U$ Chapter1/ScSmooth · ScBanach.lvlSet_zero, $U_{m+1} = \mathrm{incl}_m^{-1}(U_m)$ Chapter1/ScSmooth · ScBanach.lvlSet_succ, and level sets preserve inclusions, intersections, openness, closedness and convexity Chapter1/ScSmooth · ScBanach.lvlSet_mono Chapter1/ScSmooth · ScBanach.lvlSet_inter Chapter1/ScSmooth · ScBanach.isOpen_lvlSet Chapter1/ScSmooth · ScBanach.isClosed_lvlSet Chapter1/ScSmooth · ScBanach.convex_lvlSet. This is the text's filtration $A_m = A \cap E_m$ of a subset. The level sets of $U_1 \subseteq E_1$, taken in the shifted space E^1, are the U_{m+1} Chapter1/Levels · ScBanach.shift_lvlSet, which is the text's $(A^1)_m = A_{1+m}$.

I.2 Sc-operators, sc-isomorphisms, sc⁺-operators (Definitions 1.3–1.4)

An **sc-operator** $T : E \to F$ is a family of continuous linear maps $T_m : E_m \to F_m$ commuting with the inclusions, $\mathrm{incl}_m^F \circ T_{m+1} = T_m \circ \mathrm{incl}_m^E$ Chapter1/ScBanach · ScOperator. Then $\iota_m T_m = T_0 \iota_m$ by induction Chapter1/ScBanach · ScOperator.toBase_comm, so the whole family is determined by the base map T_0 Chapter1/ScBanach · ScOperator.ext_base: this is the sense in which the text's single linear map T "with $T(E_m) \subset F_m$ and $T : E_m \to F_m$ continuous" is the same thing. The text's formulation is also available as a predicate on a continuous linear $T_0 : E_0 \to F_0$, "for every m there is a

continuous linear T_m with $\iota_m T_m = T_0 \iota_m$" Chapter1/ScBanach · IsScOperator, and the two notions pass into each other Chapter1/ScBanach · ScOperator.isScOperator Chapter1/ScBanach · IsScOperator.toScOperator Chapter1/ScBanach · IsScOperator.toScOperator_zero. The identity is an sc-operator, sc-operators compose and add Chapter1/ScBanach · ScOperator.id Chapter1/ScBanach · ScOperator.comp Chapter1/ScBanach · ScOperator.add, and they map smooth points to smooth points, since a lift of x to level j is carried by T_j to a lift of $T_0 x$ Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl.

An **sc-isomorphism** is a family of continuous linear equivalences $E_m \simeq F_m$ commuting with the inclusions Chapter1/ScBanach · ScIso; it has an underlying sc-operator Chapter1/ScBanach · ScIso.toScOperator, an inverse Chapter1/ScBanach · ScIso.symm, composites Chapter1/ScBanach · ScIso.trans and the identity Chapter1/ScBanach · ScIso.refl, and ι_m commutes with it and with its inverse Chapter1/ScBanach · ScIso.toBase_comm Chapter1/ScBanach · ScIso.symm_toBase_comm. The text defines an sc-isomorphism as a bijective sc-operator whose inverse is an sc-operator; the continuity of the inverse on each level, which there follows from the open mapping theorem, is here part of the data.

An sc$^+$**-operator** $E \to F$ is an sc-operator $E \to F^1$ into the shifted space, i.e. continuous linear maps $S_m : E_m \to F_{m+1}$ commuting with the inclusions Chapter1/ScBanach · ScPlus. The text's first condition, $S(E_m) \subset F_{m+1}$, is contained in the second and is dropped. The inclusion $F^1 \to F$, level by level $\mathrm{incl}_m : F_{m+1} \to F_m$, is itself an sc-operator Chapter1/ScBanach · ScBanach.shiftIncl, and composing gives the sc-operator $E \to F$ underlying S Chapter1/ScBanach · ScPlus.toScOperator Chapter1/ScBanach · ScPlus.toScOperator_apply. Its level-m map is the compact incl_m after the bounded S_m, hence compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.comp_clm Chapter1/ScBanach · ScPlus.isCompactOperator: the text's remark that sc$^+$-operators are sc-compact. In terms of images, S raises the level: if $x \in \operatorname{range} \iota_m$ then $S_0 x \in \operatorname{range} \iota_{m+1}$ Chapter1/ScFredholm · ScPlus.toScOperator_mem_range_toBase_succ.

I.3 Sc-subspaces and sc-splittings (Definition 1.5)

For a subspace $A \subseteq E_0$ its **level-m part** $A_m \subseteq E_m$ is the preimage of A under ι_m Chapter1/ScBanach · ScBanach.levelSub Chapter1/ScBanach · ScBanach.mem_levelSub; $A_0 = A$ Chapter1/ScBanach · ScBanach.levelSub_zero, the inclusions map A_{m+1} into A_m Chapter1/ScBanach · ScBanach.incl_mem_levelSub, and A_m is closed when A is, as the preimage of a closed set under a continuous map Chapter1/ScBanach · ScBanach.isClosed_levelSub.

A is an **sc-subspace** if A is closed and, for every m, A_m lies in the closure of $A_m \cap E_\infty$ Chapter1/ScBanach · ScBanach.IsScSubspace Chapter1/ScBanach · ScBanach.IsScSubspace.isClosed. The text asks that (A_m) be an sc-structure on A; of the conditions in that phrase, the closedness of A_m in E_m was just proved, and the compactness of the restricted inclusions $A_{m+1} \to A_m$ is automatic, since the restriction of a compact operator to a subspace, with values in a closed subspace, is compact. This is how the scale structure on a closed A is built: levels A_m, inclusions the restrictions of incl_m, injective because incl_m is, compact by the remark Chapter1/ScBanach · ScBanach.ofSubspaceScale. Its composite inclusion is that of E on the underlying points Chapter1/ScBanach · ScBanach.ofSubspaceScale_coe_toBase, and its smooth points are the points of A_m that are smooth in E Chapter1/ScBanach · ScBanach.mem_ofSubspaceScale_smoothLvl Chapter1/ScBanach · ScBanach.ofSubspaceScale_smoothLvl. The density condition of `IsScSubspace` is then exactly the density axiom for this scale, so an sc-subspace is an sc-Banach space with the induced structure Chapter1/ScBanach · ScBanach.ofSubspace.

$E = F \oplus G$ is an **sc-splitting** if F and G are sc-subspaces and, for every m, F_m and G_m are complementary subspaces of E_m Chapter1/ScBanach · ScBanach.IsScSplitting. The text asks for a *topological* direct sum $E_m = F_m \oplus G_m$. Since F_m and G_m are closed and complementary, the projection onto F_m along G_m is continuous by the closed graph theorem, which is Mathlib's "closed complemented from complementary closed subspaces" Chapter1/ScBanach · ScBanach.IsScSplitting.closedComplemented; so nothing is lost by asking only for complementarity. The splitting is symmetric in F and G Chapter1/ScBanach · ScBanach.IsScSplitting.symm.

Sc-splittings are sc-projections (Lemma 2.B of the exposition). The proof of Proposition 1.6 in the text passes from "$P \circ P = P$, continuous on every level" to "a sc-projection" and an sc-complement $(1-P)(E)$ without comment. Both directions are proved. (i) Given an sc-splitting $E = A \oplus B$, on each level A_m and B_m are complementary and closed, hence topological complements `Mathlib/Analysis/Normed/Module/Complemented · Submodule.IsCompl.isTopCompl_of_isClosed` `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.isTopCompl`, so the projection P_m of E_m onto A_m along B_m is continuous `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL`. These projections commute with the inclusions: write $x = P_{m+1}x + (x - P_{m+1}x)$ with the first term in A_{m+1} and the second in B_{m+1} `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_sub_projectionL`; incl_m carries them into A_m and B_m (§I.3), where P_m is the identity, resp. zero `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_eq_self_iff` `Mathlib/Topology/Algebra/Module/Complement · Submodule.projectionL_apply_eq_zero_iff`, so $P_m(\mathrm{incl}_m x) = \mathrm{incl}_m(P_{m+1}x)$. Hence the P_m form an sc-operator P `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj` `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_apply` with $P_m^2 = P_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.scProj_idem`, $P_m(E_m) = A_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.range_scProj` and $\ker P_m = B_m$ `Chapter1/ExpositionLemmas · ScBanach.IsScSplitting.ker_scProj`. (ii) Conversely let P be an sc-operator with $P_0^2 = P_0$. Idempotence passes to every level by injectivity of ι_m, since $\iota_m P_m P_m x = P_0 P_0 \iota_m x = P_0 \iota_m x = \iota_m P_m x$ `Chapter1/ExpositionLemmas · ScOperator.idem_lvl`. The level-m part of $P_0(E_0)$ is $P_m(E_m)$: if $\iota_m x = P_0 y$ then $P_0 \iota_m x = \iota_m x$, i.e. $\iota_m P_m x = \iota_m x$, so $x = P_m x$ `Chapter1/ExpositionLemmas · ScOperator.levelSub_range_eq`; the level-m part of $\ker P_0$ is $\ker P_m$ (§I.6, `Chapter1/ScFredholm · ScOperator.ker_lvl`). $P_0(E_0) = \{x : P_0 x = x\}$ is closed, and $\ker P_0$ is closed. Density of smooth points: $x \in P_m(E_m)$ satisfies $x = P_m x$ and is the limit of $P_m z_n$ for smooth $z_n \to x$, and the $P_m z_n$ are smooth points of $P_m(E_m)$ (§I.2); $x \in \ker P_m$ satisfies $x = x - P_m x$ and is the limit of $z_n - P_m z_n$, smooth points of $\ker P_m$. Finally $P_m(E_m)$ and $\ker P_m$ are complementary because P_m is idempotent `Mathlib/LinearAlgebra/Projection · LinearMap.IsIdempotentElem.isCompl`. So $E = P_0(E_0) \oplus \ker P_0$ is an sc-splitting `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem`, and together: $E = A \oplus B$ is an sc-splitting iff an idempotent sc-operator has range A and kernel B `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`. The text's $(1-P)(E)$ is $\ker P$. The formal Proposition 1.6 (§I.4) builds its complement directly and does not use this.

I.4 Proposition 1.6: finite-dimensional sc-subspaces

Statement. Let $F \subseteq E_0$ be finite-dimensional. Then F is an sc-subspace if and only if $F \le E_\infty$ `Chapter1/ScBanach · ScBanach.proposition_1_6_iff`; and if $F \le E_\infty$ there is a G with $E = F \oplus G$ an sc-splitting `Chapter1/ScBanach · ScBanach.proposition_1_6`.

The equivalence. If F is an sc-subspace, the density condition at level 0 says $F \subseteq \overline{F \cap E_\infty}$. But $F \cap E_\infty$ is a subspace of the finite-dimensional F, hence finite-dimensional, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.closed_of_finiteDimensional`; so $F \subseteq F \cap E_\infty$, i.e. $F \le E_\infty$. Conversely, if $F \le E_\infty$ then F is closed (finite-dimensional) and every point of F_m is already a smooth point of F_m, so the density condition holds trivially.

The complement. Mathlib provides a continuous linear projection $p : E_0 \to F$ with $p|_F = \mathrm{id}$, from the closed complementation of finite-dimensional subspaces (Hahn–Banach) `Mathlib/Analysis/LocallyConvex/HahnBanach · Submodule.ClosedComplemented.of_finiteDimensional`. This is the text's $P(x) = \sum \lambda_i(x) e_i$. Put $G := \ker p$. Since $F \le E_\infty$, the restriction of ι_m to F_m is a bijection onto F (injective by injectivity of ι_m, surjective because every point of F lifts to level m) `Chapter1/ScBanach · ScBanach.levelRestrict` `Chapter1/ScBanach · ScBanach.levelRestrict_bijective` `Chapter1/ScBanach · ScBanach.levelEquiv`; through this linear isomorphism $F_m \cong F$ the projection lifts to a continuous linear $P_m : E_m \to F_m$, $P_m = (\iota_m|_{F_m})^{-1} \circ p \circ \iota_m$ (continuous because F_m is finite-dimensional) `Chapter1/ScBanach · ScBanach.finProj`. It fixes F_m `Chapter1/ScBanach · ScBanach.finProj_apply_of_mem`, and $P_m x = 0$ iff $p(\iota_m x) = 0$ `Chapter1/ScBanach · ScBanach.finProj_eq_zero_iff`, so $G_m = \ker P_m$. Now F is an sc-subspace by the first part, and G is closed as a kernel. For the density of $G_m \cap E_\infty$ in G_m: given $g \in G_m$, let $Q := 1 - P_m$ on E_m; then $Qg = g$, Q maps E_m into G_m, and Q maps smooth points to smooth points, since $P_m x$ is smooth ($F \le E_\infty$) and smooth

points form a subspace Chapter1/ScBanach · ScScale.smoothLvl_submodule_closed_sub. Hence $g = Qg$ lies in the closure of $Q(E_\infty \cap E_m) \subseteq G_m \cap E_\infty$, because $E_\infty \cap E_m$ is dense and Q is continuous. This is the text's $g_n = (1-P)f_n \to g$. Finally F_m and $G_m = \ker P_m$ are complementary because P_m is a projection onto F_m Mathlib/LinearAlgebra/Projection · isCompl_of_proj.

I.5 Proposition 1.7: quotients

Statement. Let $A \subseteq E_0$ be closed. The quotients E_m/A_m form an sc-Banach space E/A Chapter1/ScBanach · ScBanach.quotient, the quotient maps form an sc-operator $E \to E/A$ Chapter1/ScBanach · ScBanach.quotientMap, and, as a statement, there are an sc-Banach space Q and an sc-operator $q : E \to Q$ with every q_m surjective and $\ker q_m = A_m$ Chapter1/ScBanach · ScBanach.proposition_1_7. The text assumes A is an sc-subspace; only closedness is used.

The inclusions. A_m is closed in E_m (§I.3), so E_m/A_m is a Banach space. The map $E_{m+1}/A_{m+1} \to E_m/A_m$ is induced by incl_m: it is well defined because incl_m maps A_{m+1} into A_m Chapter1/ScBanach · ScBanach.quotIncl Chapter1/ScBanach · ScBanach.quotIncl_mk. It is injective: if $\mathrm{incl}_m x \in A_m$ then $\iota_{m+1}x = \iota_m(\mathrm{incl}_m x) \in A$, so $x \in A_{m+1}$ Chapter1/ScBanach · ScBanach.quotIncl_injective. This injectivity is the text's identification of E_m/A_m with the subspace $\{x + A : x \in E_m\}$ of E/A, and it holds precisely because A_m is the *full* preimage of A. It is compact: the quotient map $E_m \to E_m/A_m$ after the compact incl_m is compact, and the induced map on E_{m+1}/A_{m+1} sends the image of a neighbourhood of 0 (which is a neighbourhood, since quotient maps are open) into the same compact set Chapter1/ScBanach · ScBanach.quotIncl_isCompactOperator. So the quotients form a scale Chapter1/ScBanach · ScBanach.quotientScale, whose composite inclusion sends the class of x to the class of $\iota_m x$ Chapter1/ScBanach · ScBanach.quotientScale_toBase_mk. The text's argument with a sequence $x_k + A_{m+1}$ of norm ≤ 1 and representatives of norm ≤ 2 is the sequential form of the same fact.

Density. The quotient map $E_m \to E_m/A_m$ is continuous and surjective, so the image of the dense set $E_\infty \cap E_m$ is dense. A class $[x]$ with x smooth is a smooth point of the quotient scale: a lift y of x to level j gives the lift $[y]$ of $[x]$. So the smooth points of E_m/A_m are dense Chapter1/ScBanach · ScBanach.quotient. This is the inclusion "$\{x + A : x \in E_\infty\} \subseteq (E/A)_\infty$" of the text's proof, which is all its density claim needs.

The smooth points of the quotient (Lemma 2.C of the exposition). The text also asserts that the two sets are equal. Let A be an sc-subspace and $x \in E_0$ with, for every j, a point $x_j \in E_j$ such that $\iota_j x_j - x \in A$. We find a smooth y with $y - x \in A$ Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset. *The differences.* $d_j := \mathrm{incl}_j x_{j+1} - x_j \in E_j$ has $\iota_j d_j = \iota_{j+1}x_{j+1} - \iota_j x_j = (\iota_{j+1}x_{j+1} - x) - (\iota_j x_j - x) \in A$, so $d_j \in A_j$. *A uniform bound for the inclusions.* Put $P_j := \prod_{i<j} \max(1, \|\mathrm{incl}_i\|) \ge 1$ Chapter1/ExpositionLemmas · ScBanach.inclBound Chapter1/ExpositionLemmas · ScBanach.one_le_inclBound. By induction, $\|\iota_j v\| \le P_j \|v\|$ for $v \in E_j$ Chapter1/ExpositionLemmas · ScBanach.norm_toBase_le and $\|\mathrm{incl}_J^{J+k} v\| \le P_{J+k}\|v\|$ for the composite inclusion $E_{J+k} \to E_J$ Chapter1/ExpositionLemmas · ScBanach.norm_inclK_le, each step multiplying by $\|\mathrm{incl}_i\| \le \max(1, \|\mathrm{incl}_i\|)$ Chapter1/ExpositionLemmas · ScBanach.norm_incl_le. This replaces the exposition's $M_j = \max_{i \le j} \|E_j \to E_i\|$, and avoids comparing levels $J+k$ with j. *Smooth corrections.* Since A is an sc-subspace, $A_j \cap E_\infty$ is dense in A_j, so there are $c_j \in A_j \cap E_\infty$ with $\|d_j - c_j\| \le 2^{-j}/P_j$ Mathlib/Topology/MetricSpace/Pseudo/Defs · Metric.mem_closure_iff. Then $b_j := d_j - c_j \in A_j$ has $P_j\|b_j\| \le 2^{-j}$. *The series.* $\sum_j \iota_j b_j$ converges absolutely in E_0, its terms being bounded by 2^{-j} Mathlib/Analysis/Normed/Group/InfiniteSum · Summable.of_norm_bounded; let S be its sum and $y := x_0 + S$. Its partial sums lie in A, which is closed, so $S \in A$ (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · HasProd.tendsto_prod_nat), and $y - x = (x_0 - x) + S \in A$. *$y$ is smooth.* Fix J. The tail $t := \sum_k \mathrm{incl}_J^{J+k} b_{J+k}$ converges in E_J (terms bounded by $2^{-J}2^{-k}$), and $\iota_J t = \sum_k \iota_{J+k} b_{J+k}$ because ι_J is continuous and linear Mathlib/Topology/Algebra/InfiniteSum/Module · ContinuousLinearMap.map_tsum and $\iota_J \circ \mathrm{incl}_J^{J+k} = \iota_{J+k}$ (§I.1). The head telescopes: $\sum_{j<J} \iota_j d_j = \iota_J x_J - x_0$ (the additive form of Mathlib/Algebra/BigOperators/Group/Finset/

Basic · Finset.prod_range_div), so $\sum_{j<J} \iota_j b_j = (\iota_J x_J - x_0) - \sum_{j<J} \iota_j c_j$, and the last sum is smooth, hence $\iota_J w$ for some $w \in E_J$. Splitting S into head and tail (the additive form of Mathlib/Topology/Algebra/InfiniteSum/NatInt · Multipliable.prod_mul_tprod_nat_add) gives $y = \iota_J(x_J - w + t)$. As J was arbitrary, $y \in E_\infty$. *As a statement about E/A.* The class of x is a smooth point of E/A iff for every j it is the image of a class $[z]$, $z \in E_j$, i.e. iff $\iota_j z - x \in A$ for some z (§I.5 and Mathlib/LinearAlgebra/Quotient/Defs · Submodule.Quotient.eq); by the above this happens iff $x + A$ contains a smooth point Chapter1/ExpositionLemmas · ScBanach.mk_mem_quotient_smooth_iff. This is the text's equality.

I.6 Definition 1.8: sc-Fredholm operators

Level kernels and ranges. For an sc-operator T, the kernel of T_m is the level-m part of the kernel of T_0 Chapter1/ScFredholm · ScOperator.ker_lvl, and the range of T_m lies in the level-m part of the range of T_0 Chapter1/ScFredholm · ScOperator.range_lvl_le; both from $\iota_m T_m = T_0 \iota_m$ and injectivity.

The definition. An **sc-Fredholm splitting** of $T : E \to F$ consists of subspaces $K, X \subseteq E_0$ and $C, Y \subseteq F_0$ such that $E = K \oplus X$ and $F = C \oplus Y$ are sc-splittings, $K = \ker T_0$, $Y = \operatorname{range} T_0$, K and C are finite-dimensional, and T_m restricts to a bijection $X_m \to Y_m$ for every m Chapter1/ScFredholm · ScFredholmSplitting; T is **sc-Fredholm** if such a splitting exists Chapter1/ScFredholm · IsScFredholm. The text's condition (3), "$T : X \to Y$ is an sc-isomorphism", is here the bijectivity on each level; the continuity of the inverses $Y_m \to X_m$ is automatic by the open mapping theorem between the closed subspaces X_m and Y_m, so nothing is lost.

The index. The index of a splitting is $\dim K - \dim C$ Chapter1/ScFredholm · ScFredholmSplitting.index, and the index of T is $\dim \ker T_0 - \dim(F_0/\operatorname{range} T_0)$ Chapter1/ScFredholm · scIndex. They agree: $K = \ker T_0$, and C is a complement of $Y = \operatorname{range} T_0$ in F_0, so $C \cong F_0/Y$ Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl Chapter1/ScFredholm · ScFredholmSplitting.index_eq. So the index does not depend on the splitting, which the text does not state.

The remarks after the definition. K is a finite-dimensional sc-subspace, hence $K \le E_\infty$ by Proposition 1.6 Chapter1/ScFredholm · ScFredholmSplitting.K_le_smooth. $\ker T_m = K_m$ Chapter1/ScFredholm · ScFredholmSplitting.ker_lvl and $\operatorname{range} T_m = Y_m$: one inclusion from above, the other because T_m maps X_m onto Y_m Chapter1/ScFredholm · ScFredholmSplitting.range_lvl. The text's "$F = C \oplus T(E_m)$" (read F_m) is the level-m part of the splitting $F = C \oplus Y$ together with $\operatorname{range} T_m = Y_m$. Finally, every T_m is a Fredholm operator in Mathlib's sense: its kernel K_m injects into the finite-dimensional K, its range Y_m is closed, and Y_m has the finite-dimensional complement C_m (which injects into C), so its range has finite codimension Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_isCompl Chapter1/ScFredholm · ScFredholmSplitting.isFredholm, using the criterion of §I.7. This is the fact the text uses in the first line of §1.4.1.

The index on every level (Lemma 2.D of the exposition). C is a finite-dimensional sc-subspace, so $C \le F_\infty$ by Proposition 1.6 Chapter1/FredholmIndex · ScFredholmSplitting.C_le_smooth. For any subspace $A \le E_\infty$, ι_m restricts to a bijection $A_m \to A$ (injective as ι_m is, surjective because every point of A lifts), so $\dim A_m = \dim A$ Chapter1/FredholmIndex · ScBanach.finrank_levelSub_of_le_smooth. Hence $\dim \ker T_m = \dim K_m = \dim K$, and $F_m/\operatorname{range} T_m = F_m/Y_m \cong C_m$ (complements, Mathlib/LinearAlgebra/Projection · quotientEquivOfIsCompl) has dimension $\dim C$. So T_m is Fredholm with index $\dim K - \dim C$ in the sense of §I.7 Chapter1/FredholmIndex · ScFredholmSplitting.fredholmIndex_lvl, which by the previous paragraph is the index of T for every splitting Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl; on level 0 this is the definition, $\operatorname{ind}(T)$ being the Fredholm index of T_0 Chapter1/FredholmIndex · scIndex_eq_fredholmIndex. The text states the splittings on each level but not the index.

I.7 Fredholm operators between Banach spaces: Riesz–Schauder and compact perturbations

Mathlib has a notion of Fredholm operator `Mathlib/Analysis/Normed/Operator/Fredholm/Basic · IsFredholm`: a continuous linear $u : E \to F$ that is a strict map, with closed range, finite-dimensional kernel, range of finite codimension `Mathlib/RingTheory/Finiteness/Cofinite · CoFG` and closed complemented kernel. It does not have the two theorems that §1.4.1 quotes as classical, that $1 +$ compact is Fredholm and that compact perturbations preserve the property. Both are proved here, for real Banach spaces.

The criterion. A continuous linear map with closed range between Banach spaces is a strict map `Mathlib/Topology/Maps/Strict/Basic · IsStrictMap`: by the open mapping theorem its corestriction to the range is an open quotient map `Mathlib/Topology/Maps/Strict/Module · isStrictMap_iff_isOpenQuotientMap_rangeRestrict` `Chapter1/FredholmPerturbation · isStrictMap_of_isClosed_range`. Hence a continuous linear map with finite-dimensional kernel, closed range and range of finite codimension is Fredholm, the kernel being closed complemented because it is finite-dimensional `Chapter1/FredholmPerturbation · isFredholm_of_banach`. A finite-rank operator is compact: the image of the unit ball is bounded in the finite-dimensional range, so its closure is compact `Chapter1/FredholmPerturbation · isCompactOperator_of_finiteDimensional_range`.

Riesz–Schauder: the kernel. Let $c : E \to E$ be compact and $N := \ker(1 + c)$. On N the operator c is $-\mathrm{id}$, and c restricts to a compact operator of the closed subspace N `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.restrict`; so id_N is compact, and N is finite-dimensional `Mathlib/Analysis/Normed/Operator/Compact/FiniteDimension · FiniteDimensional.of_isCompactOperator_id` `Chapter1/FredholmPerturbation · finiteDimensional_ker_one_add`.

The range is closed. Take a closed complement G of N (which exists since N is finite-dimensional). $1 + c$ is bounded below on G: otherwise there are unit vectors $g_n \in G$ with $(1 + c)g_n \to 0$; the cg_n lie in a compact set `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact`, so along a subsequence $cg_n \to y$ `Mathlib/Topology/Sequences · IsCompact.tendsto_subseq`, hence $g_n = (1+c)g_n - cg_n \to -y$; then $-y \in G$ (closed), $\|-y\| = 1$, and $(1+c)(-y) = 0$, so $-y \in G \cap N = 0$, a contradiction `Chapter1/FredholmPerturbation · exists_bound_below_on_compl`. A bounded-below operator from a Banach space has closed range (it is antilipschitz and uniformly continuous), and $(1+c)(G) = (1+c)(E)$ since $1 + c$ kills N `Chapter1/FredholmPerturbation · isClosed_range_one_add`.

The range has finite codimension. Suppose $R := \mathrm{range}(1+c)$ has infinite codimension. Pick, for every proper subspace, a vector outside it `Chapter1/FredholmPerturbation · pick` `Chapter1/FredholmPerturbation · pick_notMem`, and build the chain $M_0 = R$, $M_{j+1} = M_j + \mathbb{R}\,\mathrm{pick}(M_j)$ `Chapter1/FredholmPerturbation · chain`. Each M_j is R plus a finite-dimensional subspace `Chapter1/FredholmPerturbation · chain_eq_sup`, hence closed `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_sup_finiteDimensional` `Chapter1/FredholmPerturbation · chain_isClosed`, and proper, since R has infinite codimension `Mathlib/RingTheory/Finiteness/Cofinite · FG.cofg_of_codisjoint` `Chapter1/FredholmPerturbation · chain_ne_top`; the chain is increasing `Chapter1/FredholmPerturbation · chain_mono`. Riesz's lemma inside a subspace: for closed $M \subsetneq M'$ there is a unit vector of M' at distance $\geq 1/2$ from M (take $x \in M' \setminus M$, a point $y \in M$ with $\|x-y\| < 2\,\mathrm{dist}(x, M)$ `Mathlib/Topology/MetricSpace/HausdorffDistance · infDist_lt_iff`, and normalize $x-y$) `Chapter1/FredholmPerturbation · exists_unit_far`. This gives unit vectors $u_j \in M_{j+1}$ at distance $\geq 1/2$ from M_j. For $j < l$, $u_j + (1+c)u_l - (1+c)u_j \in M_l$, and $u_l - [u_j + (1+c)u_l - (1+c)u_j] = -(cu_l - cu_j)$, so $\|cu_l - cu_j\| \geq 1/2$. But the cu_j lie in a compact set and so have a convergent subsequence, two of whose terms are within $1/2$ of each other: a contradiction `Chapter1/FredholmPerturbation · coFG_range_one_add`. Hence $1 + c$ is Fredholm `Chapter1/FredholmPerturbation · isFredholm_one_add`.

Compact perturbations. Let $u : E \to F$ be Fredholm and k compact. Mathlib gives a quasi-inverse $v : F \to E$ of u, with $vu - 1$ and $uv - 1$ of finite rank `Mathlib/Analysis/Normed/Operator/Fredholm/Basic · isFredholm_iff_exists_isQuasiInverse`; a finite-rank difference from the identity is compact `Chapter1/FredholmPerturbation · isCompactOperator_sub_one_of_equiv`. Then $v(u + k) = 1 + c_1$ and $(u + k)v = 1 + c_2$ with $c_1 = (vu - 1) + vk$ and $c_2 = (uv - 1) + kv$ compact. The kernel of $u + k$ lies in the finite-

dimensional kernel of $1 + c_1$; the range of $u + k$ contains the range of $1 + c_2$, which is closed and of finite codimension, so range$(u + k)$ has finite codimension `Mathlib/RingTheory/Finiteness/Cofinite · CoFG.of_le` and is closed, as a subspace containing a closed subspace of finite codimension `Mathlib/Topology/Algebra/Module/FiniteDimension · Submodule.isClosed_mono_of_finiteDimensional_quotient`. By the criterion, $u + k$ is Fredholm `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`.

The index. Mathlib has no index; it is built here, first for linear maps over a field. For $u : E \to F$ put $\operatorname{ind}(u) := \dim \ker u - \dim(F/\operatorname{range} u) \in \mathbb{Z}$ `Chapter1/FredholmIndex · linIndex`, meaningful when u has *finite index*, i.e. finite-dimensional kernel and cokernel `Chapter1/FredholmIndex · HasFiniteIndex`. *Additivity.* If $u : E \to F$ and $v : F \to G$ have finite index, so has vu, and $\operatorname{ind}(vu) = \operatorname{ind}(u) + \operatorname{ind}(v)$ `Chapter1/FredholmIndex · HasFiniteIndex.comp_and_linIndex Chapter1/FredholmIndex · HasFiniteIndex.comp Chapter1/FredholmIndex · linIndex_comp`. The proof is the dimension count along the exact sequence $0 \to \ker u \to \ker vu \to \operatorname{range} u \cap \ker v \to F/\operatorname{range} u \to G/\operatorname{range} vu \to G/\operatorname{range} v \to 0$, done as four applications of rank–nullity `Mathlib/LinearAlgebra/FiniteDimensional/Lemmas · LinearMap.finrank_range_add_finrank_ker` and two of the third isomorphism theorem `Mathlib/LinearAlgebra/Isomorphisms · Submodule.quotientQuotientEquivQuotient`: to u restricted to $\ker vu$ (kernel $\ker u$, range $\operatorname{range} u \cap \ker v$), to the quotient map $F \to F/\operatorname{range} u$ restricted to $\ker v$, and to v followed by $G \to G/\operatorname{range} vu$ (kernel $\operatorname{range} u + \ker v$); finiteness of the new kernel and cokernel comes out of the same maps `Chapter1/FredholmIndex · finite_of_finite_ker_range Chapter1/FredholmIndex · finite_of_finite_quotient`. Conversely, if vu and uv have finite index then so has u, since $\ker u \subseteq \ker vu$ and $\operatorname{range} uv \subseteq \operatorname{range} u$ `Chapter1/FredholmIndex · finiteDimensional_ker_of_comp Chapter1/FredholmIndex · finiteDimensional_coker_of_comp Chapter1/FredholmIndex · HasFiniteIndex.of_comp_of_comp`. *Index zero.* A linear equivalence has index 0 `Chapter1/FredholmIndex · HasFiniteIndex.equiv Chapter1/FredholmIndex · linIndex_equiv`. If $w - 1$ has finite-dimensional range R, then $\ker w \subseteq R$, $\operatorname{range} w + R = E$ and $R \cap \operatorname{range} w = w(R)$, so $\dim \ker w$ and $\dim(E/\operatorname{range} w)$ both equal $\dim R - \dim w(R)$ and $\operatorname{ind}(w) = 0$ `Chapter1/FredholmIndex · HasFiniteIndex.of_finiteRange_sub_id_and_linIndex Chapter1/FredholmIndex · linIndex_eq_zero_of_finiteRange_sub_id`. If a is bijective and r has finite rank, $a + r = a(1 + a^{-1}r)$ has index $0 + 0$ by additivity `Chapter1/FredholmIndex · HasFiniteIndex.of_bijective_add_and_linIndex`.

Banach spaces. For $u : E \to F$ continuous linear, the **Fredholm index** is $\operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex`; a Fredholm operator has finite index `Chapter1/FredholmIndex · IsFredholm.hasFiniteIndex`. *Small perturbations.* Let v be a quasi-inverse of u and $\|v\|\,\|s\| < 1$. Then $\|vs\|, \|sv\| < 1$ `Mathlib/Analysis/Normed/Operator/Basic · ContinuousLinearMap.opNorm_comp_le`, so $1 + vs$ and $1 + sv$ are invertible by the Neumann series `Mathlib/Analysis/SpecificLimits/Normed · isUnit_one_sub_of_norm_lt_one Chapter1/FredholmIndex · bijective_one_add_of_norm_lt`. Since $vu - 1$ and $uv - 1$ have finite rank `Chapter1/FredholmIndex · finiteDimensional_range_sub_id_of_equiv`, $v(u + s) = (1 + vs) + (vu - 1)$ and $(u + s)v = (1 + sv) + (uv - 1)$ are bijections plus finite-rank maps, hence of index 0 `Chapter1/FredholmIndex · hasFiniteIndex_comp_add_of_norm_lt`; the same with $s = 0$ for vu and uv. By the converse above, $u + s$, u and v have finite index, and additivity gives $\operatorname{ind}(u + s) + \operatorname{ind}(v) = 0 = \operatorname{ind}(u) + \operatorname{ind}(v)$, so $\operatorname{ind}(u + s) = \operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex_add_of_norm_lt`. *Compact perturbations.* Let u be Fredholm and k compact, and $\varphi(t) := \operatorname{ind}(u + tk)$ for $t \in \mathbb{R}$. At each t_0, $u + t_0 k$ is Fredholm (the previous paragraph, $t_0 k$ being compact `Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.smul`), with a quasi-inverse v_0; for t so close to t_0 that $\|v_0\|\,\|(t - t_0)k\| < 1$, $\varphi(t) = \operatorname{ind}((u + t_0 k) + (t - t_0)k) = \varphi(t_0)$. So φ is locally constant `Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.iff_eventually_eq` on the connected $\mathbb{R}$, hence constant `Mathlib/Topology/LocallyConstant/Basic · IsLocallyConstant.apply_eq_of_preconnectedSpace`, and $\operatorname{ind}(u + k) = \varphi(1) = \varphi(0) = \operatorname{ind}(u)$ `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. This replaces the citation of Kato IV.5.26 in the exposition's (F2).

I.8 Proposition 1.9: sc-Fredholm operators are regularizing

Statement. Let T have an sc-Fredholm splitting, $e \in E_0$, and $T_0 e \in \operatorname{range} \iota_m$. Then $e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · proposition_1_9`.

Proof (the text's). Write $T_0 e = \iota_m y$. Since $F_m = C_m \oplus Y_m$, $y = c + y'$ with $c \in C_m$ and $y' \in Y_m$, and $y' = T_m x'$ for some $x' \in X_m$ by the bijectivity of $T_m : X_m \to Y_m$. Then $T_0(e - \iota_m x') = \iota_m c$, so $\iota_m c$ lies in C and in $Y = \operatorname{range} T_0$; these are complementary on level 0, so $\iota_m c = 0$. Hence $e - \iota_m x' \in \ker T_0 = K \leq E_\infty$ (§I.6), so it lifts to some $z \in E_m$, and $e = \iota_m(z + x')$. The step "$c = 0$" is where $C \cap Y = 0$ enters, as the exposition notes.

I.9 Proposition 1.10: compact perturbations of sc-Fredholm operators

Statement. If $T : E \to F$ is sc-Fredholm and $S : E \to F$ is an sc$^+$-operator, then $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_10`. Here S is read as the sc-operator $E \to F$ underlying it (§I.2), and $T + S$ is the sum of sc-operators.

The bootstrap. The text says: if $y = (T + S)e \in Y_m$ then, since T is regularizing, $e \in E_m$. But $Te = y - Se$ with Se only one level up. The formal proof iterates: if $(T + S)_0 e \in \operatorname{range} \iota_{m+1}$, then a fortiori $(T + S)_0 e \in \operatorname{range} \iota_m$ `Chapter1/ScFredholm · ScBanach.range_toBase_succ_le`, so $e \in \operatorname{range} \iota_m$ by induction, so $S_0 e \in \operatorname{range} \iota_{m+1}$ (§I.2), so $T_0 e = (T + S)_0 e - S_0 e \in \operatorname{range} \iota_{m+1}$, and Proposition 1.9 gives $e \in \operatorname{range} \iota_{m+1}$ `Chapter1/ScFredholm · ScPlus.regularizing_add`. So $T + S$ is regularizing.

The kernel. Each $(T + S)_m = T_m + S_m$ is Fredholm, as a compact perturbation (§I.7) of the Fredholm operator T_m (§I.6) `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`. Let $K' := \ker(T + S)_0$, finite-dimensional. Every $x \in K'$ is smooth: $(T + S)_0 x = 0 \in \operatorname{range} \iota_m$ for every m, so $x \in \operatorname{range} \iota_m$ by the bootstrap. By Proposition 1.6 there is an sc-splitting $E = K' \oplus X'$ `Chapter1/ScBanach · ScBanach.proposition_1_6`. (The text's $K_m = K_{m+1}$ is the same bootstrap.)

The range. Let $Y' := \operatorname{range}(T + S)_0$, closed since $(T + S)_0$ is Fredholm. The range of $(T + S)_m$ is exactly Y'_m: one inclusion from §I.6; conversely if $\iota_m y = (T + S)_0 e$ then e lifts to level m by the bootstrap, and $y = (T + S)_m e'$ by injectivity of ι_m. Y' is an sc-subspace: for $y = (T + S)_m e \in Y'_m$, approximate e by smooth points $e_n \in E_m$; then $(T+S)_m e_n \to y$ and the $(T+S)_m e_n$ are smooth points of Y'_m (sc-operators preserve smooth points, §I.2). This is the text's argument "$y_n = (T + S)e_n \in Y_\infty$, $y_n \to y$".

The complement. F_0/Y' is finite-dimensional. Let $p : F_0 \to F_0/Y'$ be the quotient map. The image $p(F_\infty)$ is a subspace of the finite-dimensional F_0/Y', dense because F_∞ is dense in F_0 and p is a continuous surjection, hence closed, hence everything. Instead of lifting a basis, take a linear right inverse g of $p|_{F_\infty}$ and let $\sigma := F_\infty \hookrightarrow F_0$ composed with g, so $p \circ \sigma = \mathrm{id}$; put $C' := \operatorname{range} \sigma$, a finite-dimensional subspace of F_∞. On every level Y'_m and C'_m are complementary: disjoint because an element of $C'_m \cap Y'_m$ is $\sigma(q)$ with $p\sigma(q) = q = 0$; spanning because for $z \in F_m$ the point $c := \sigma(p(\iota_m z))$ is smooth, so it lifts to level m, and $z - c \in Y'_m$. So $F = C' \oplus Y'$ is an sc-splitting (C' is an sc-subspace by Proposition 1.6 `Chapter1/ScBanach · ScBanach.proposition_1_6_iff`).

The bijections. $(T + S)_m$ maps X'_m into Y'_m; it is injective on X'_m because its kernel on level m is K'_m, which meets X'_m in 0; and it is onto Y'_m because every $e \in E_m$ splits as $k + x$ with $k \in K'_m$, $x \in X'_m$ and $(T + S)_m e = (T + S)_m x$. This is condition (3) of Definition 1.8, which the text does not verify; the exposition supplies it with the open mapping theorem, the formalization needs only the bijections (§I.6). Hence $T + S$ is sc-Fredholm `Chapter1/ScFredholm · proposition_1_10`.

The index. The text does not say that the index is preserved; the exposition does. The index of $T+S$ is the Fredholm index of $(T+S)_0 = T_0+S_0$ (§I.6); T_0 is Fredholm and S_0 compact, so by the invariance under compact perturbations (§I.7) $\operatorname{ind}(T+S) = \operatorname{ind}(T)$ `Chapter1/FredholmIndex · proposition_1_10_index`. On every level, $T + S$ being sc-Fredholm, its level map $(T + S)_m$ is Fredholm of index $\operatorname{ind}(T + S) = \operatorname{ind}(T)$ by the index on every level (§I.6) `Chapter1/FredholmIndex · proposition_1_10_index_lvl`.

I.10 Partial quadrants, tuples and sc-domains (Definition 1.11)

The quadrant. In $\mathbb{R}^n \oplus W$, with $\mathbb{R}^n$ carrying its constant structure, the quadrant on level m is the set of (a, w) with $a_i \geq 0$ for all i `Chapter1/ScSmooth · quadrant`; it is convex `Chapter1/ScSmooth · quadrant_convex`, closed `Chapter1/ScSmooth · quadrant_isClosed`, has non-empty interior (the point $(1, \dots, 1, 0)$) `Chapter1/ScSmooth · quadrant_interior_nonempty`, and the level sets of the level-0 quadrant are the level quadrants `Chapter1/ScSmooth · lvlSet_quadrant`.

Partial quadrants. $C \subseteq E_0$ is a **partial quadrant** if $C = T_0^{-1}(\text{quadrant})$ for some n, some sc-Banach space W and some sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ `Chapter1/ScSmooth · IsPartialQuadrant`. The text asks in addition that C be closed and convex; both follow. Indeed the level set C_m is T_m^{-1} of the level-m quadrant `Chapter1/ScSmooth · IsPartialQuadrant.lvlSet_eq`, hence convex `Chapter1/ScSmooth · IsPartialQuadrant.convex_lvlSet`, closed `Chapter1/ScSmooth · IsPartialQuadrant.isClosed_lvlSet` and with non-empty interior `Chapter1/ScSmooth · IsPartialQuadrant.interior_lvlSet_nonempty`; for $m = 0$ this is the text's "closed convex" `Chapter1/ScSmooth · IsPartialQuadrant.convex` `Chapter1/ScSmooth · IsPartialQuadrant.isClosed`. A convex set with non-empty interior has unique differentiability at every point of its closure `Mathlib/Analysis/Calculus/TangentCone/Real · uniqueDiffWithinAt_convex`, so C_m has unique differentiability at each of its points `Chapter1/ScSmooth · IsPartialQuadrant.uniqueDiffWithinAt_lvlSet`; this is the exposition's Lemma 3.D, that derivatives on a quadrant are unique, in Mathlib's language. The whole space is a partial quadrant with $n = 0$ `Chapter1/ScSmooth · isPartialQuadrant_univ`.

Tuples. (U, C, E) is a tuple, U a **relatively open subset of the partial quadrant** C, if C is a partial quadrant and $U = O \cap C$ for an open $O \subseteq E_0$ `Chapter1/ScSmooth · IsRelOpenInQuadrant`.

Sc-domains. $U \subseteq E_0$ is an **sc-domain** if for every m the level set U_m is the intersection of an open and a convex subset of E_m, and U_m has unique differentiability at each of its points `Chapter1/ScSmooth · IsScDomain`. A relatively open subset of a partial quadrant is an sc-domain: $U_m = O_m \cap C_m$ with O_m open and C_m convex, and unique differentiability of C_m at a point survives intersection with a neighbourhood `Chapter1/ScSmooth · IsRelOpenInQuadrant.isScDomain`. The property passes to the tangent: $(TU)_m = U_{m+1} \times E_m$ (§I.11) is $(O \times E_m) \cap (C \times E_m)$, and unique differentiability of a product is the product of the two `Chapter1/ScSmooth · IsScDomain.tangentSet`; and to the level shift, since the level sets of U_1 in E^1 are the U_{m+1} `Chapter1/Levels · IsScDomain.shift`. This is what the proofs of §§1.2–1.3 use in place of the text's claim that $T(U, C, E)$ is again a tuple (Definition 1.13). Every theorem of §§1.2–1.3 below is stated for sc-domains; the tuples are a special case.

The shift and the tangent of a tuple (Lemmas 3.A(2) and 3.B of the exposition). The text's claim is nevertheless true, and is proved with Chapter 2, where retracts need it. If $T : E \to \mathbb{R}^n \oplus W$ presents C, its levels T_{m+1} form an sc-isomorphism $T^1 : E^1 \to \mathbb{R}^n \oplus W^1$ `Chapter2/LevelPolyfold · ScIso.shift`, and $C_1 = T_1^{-1}(\text{quadrant})$ by the level description of a partial quadrant above, so C_1 is a partial quadrant in E^1 `Chapter2/LevelPolyfold · IsPartialQuadrant.shift`; with $U = O \cap C$, $U_1 = O_1 \cap C_1$ is relatively open in it `Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift`. For the tangent, $T_{m+1} \times T_m$ maps $(TE)_m = E_{m+1} \times E_m$ onto $(\mathbb{R}^n \times W_{m+1}) \times (\mathbb{R}^n \times W_m)$, and the reassociation $((a, w), q) \mapsto (a, (w, q))$ is an sc-isomorphism onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso`; the composite presents $TC = \{(x, h) : x \in C_1\}$, since the quadrant condition only looks at the $\mathbb{R}^n$-coordinates of $T_1 x$ `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. And $TU = (O_1 \times E_0) \cap TC$ is relatively open in TC `Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet`. The identity $T(U^1) = (TU)^1$ is §I.11's `Chapter1/Levels · ScBanach.shift_tangentSet`.

I.11 Sc⁰ maps, the tangent, sc¹ maps, sc^k (Definitions 1.12–1.15)

Level maps. For $f : E_0 \to F_0$ and $x \in E_m$, the level map $f_m(x)$ is the inverse image under ι_m of $f(\iota_m x)$, when $f(\iota_m x) \in \text{range } \iota_m$, and a junk value otherwise `Chapter1/ScSmooth · levelMap`; so $\iota_m f_m(x) = f(\iota_m x)$ whenever the right side lifts `Chapter1/ScSmooth · toBase_levelMap`, $f_0 = f$ `Chapter1/ScSmooth · levelMap_`

zero, and $f_m(x)$ is characterized by $\iota_m y = f(\iota_m x)$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq. Maps are total functions on E_0; only the values on the domain U will matter.

Sc⁰. f is sc^0 **on** U if for every m and $x \in U_m$ the point $f(\iota_m x)$ lifts to F_m, and f_m is continuous on U_m Chapter1/ScSmooth · IsSc0 Chapter1/ScSmooth · IsSc0.toBase_levelMap. Then f is continuous on U Chapter1/ScSmooth · IsSc0.continuousOn_base, f_m maps U_m into V_m when f maps U into V Chapter1/ScSmooth · IsSc0.levelMap_mapsTo, the property passes to subsets Chapter1/ScSmooth · IsSc0.mono and depends only on the values on U Chapter1/ScSmooth · IsSc0.congr. Level maps compose: $(g \circ f)_m = g_m \circ f_m$ on U_m Chapter1/ScSmooth · IsSc0.levelMap_comp, so compositions of sc^0 maps are sc^0 Chapter1/ScSmooth · IsSc0.comp. The identity is sc^0 Chapter1/ScSmooth · isSc0_id, and an sc-operator is sc^0 on all of E_0 with level maps T_m Chapter1/ScSmooth · ScOperator.isSc0 Chapter1/ScSmooth · ScOperator.levelMap_eq. The text's definition says "$f(U_m) \subset V_m$" with V undefined; V is U', and the target tuple enters only as $f(U) \subseteq U'$.

The tangent. $TE := E^1 \oplus E$, with levels $E_{m+1} \times E_m$ Chapter1/ScSmooth · ScBanach.tangent Chapter1/ScSmooth · ScBanach.tangent_toBase; $TU \subseteq (TE)_0 = E_1 \times E_0$ is the set of (x, h) with $x \in U_1$ Chapter1/ScSmooth · ScBanach.tangentSet, with level-m part $U_{m+1} \times E_m$ Chapter1/ScSmooth · ScBanach.mem_lvlSet_tangentSet Chapter1/ScSmooth · ScBanach.lvlSet_tangentSet, monotone in U Chapter1/ScSmooth · ScBanach.tangentSet_mono. The level-1 part of TU is $T(U_1)$ in $T(E^1) = (TE)^1$, the identity $T(U^1) = (TU)^1$ that the inductions of §1.2 use Chapter1/Levels · ScBanach.shift_tangentSet.

Sc¹. Given a family $D : E_1 \to \mathcal{L}(E_0, F_0)$, the tangent map is $T_D f(x, h) = (f_1(x), D(x)h)$ Chapter1/ScSmooth · tangentMapWith. f is sc^1 **on** U **with derivatives** D if (1) f is sc^0 on U; (2) for every $x \in U_1$ the map $f \circ \iota_1 : E_1 \to F_0$ has derivative $D(x) \circ \iota_1$ at x within U_1 Mathlib/Analysis/Calculus/FDeriv/Defs · HasFDerivWithinAt; (3) $T_D f$ is sc^0 on TU Chapter1/ScSmooth · IsSc1With. Condition (2) is the text's limit $|f(x + h) - f(x) - Df(x)h|_0/|h|_1 \to 0$ over $h \in E_1$ with $x + h \in U_1$, in Mathlib's words. f is sc^1 **on** U if some D works Chapter1/ScSmooth · IsSc1; `scDeriv` is a chosen such D Chapter1/ScSmooth · scDeriv Chapter1/ScSmooth · IsSc1.isSc1With_scDeriv and `tangentMap` the tangent map built from it Chapter1/ScSmooth · tangentMap Chapter1/ScSmooth · IsSc1.tangentMap_isSc0. If f maps U into V then $T_D f$ maps TU into TV Chapter1/ScSmooth · IsSc1With.tangentMapWith_mapsTo Chapter1/ScSmooth · IsSc1.tangentMap_mapsTo; level 0 of (3) says $(x, h) \mapsto D(x)h$ is continuous on $U_1 \times E_0$ Chapter1/ScSmooth · IsSc1With.continuousOn_deriv; and both notions depend only on the values of f on U Chapter1/ScSmooth · IsSc1With.congr Chapter1/ScSmooth · IsSc1.congr.

Uniqueness of $Df(x)$. The text writes "the" operator $Df(x)$ and "the" tangent map. If D and D' both satisfy (2) at $x \in U_1$ and U_1 has unique differentiability at x, then $D(x) \circ \iota_1 = D'(x) \circ \iota_1$ Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq; since ι_1 has dense range (§I.1) and both operators are continuous, $D(x) = D'(x)$ Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt. Hence on a set U whose U_1 has unique differentiability everywhere (an sc-domain), the chosen derivative agrees on U_1 with any admissible family Chapter1/ScSmooth · IsSc1.scDeriv_eqOn and the chosen tangent map agrees on TU with any $T_D f$ Chapter1/ScSmooth · IsSc1.tangentMap_eqOn. This is the exposition's Lemma 3.E, and the formal proofs invoke it wherever the text says "the tangent map". The uniform version of (3) at level 0: for $x \in U_1$, a compact $K \subseteq E_0$ and $c > 0$ there is $\delta > 0$ such that $\|D(x')w - D(x)w\| \le c$ for all $x' \in U_1$ within δ of x and all $w \in K$; this is the continuity of $(x', w) \mapsto D(x')w - D(x)w$ on $U_1 \times E_0$ turned uniform over the compact K Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually Chapter1/Properties · IsSc1With.exists_delta_deriv, and it is the text's remark that $x \mapsto Df(x)$ is continuous for the compact-open topology.

Compact-open continuity as an equivalence (Lemma 3.F of the exposition). For a metric space X, $s \subseteq X$ and $A : X \to \mathcal{L}(Y, Z)$: $(x, h) \mapsto A(x)h$ is continuous on $s \times Y$ if and only if for every $x \in s$, compact $K \subseteq Y$ and $\varepsilon > 0$, eventually for x' near x in s, $\|A(x')w - A(x)w\| < \varepsilon$ for all $w \in K$ Chapter1/ExpositionLemmas · continuousOn_apply_iff_uniformOnCompact. (□) As above, on the subtype s: for each $w \in K$, $(x', w') \mapsto A(x')w' - A(x)w'$ is continuous with value 0 at (x, w), so it is small near (x, w), and compactness of K makes the neighbourhood of x uniform in w Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually; neighbourhoods in the subtype are neighbourhoods within s Math-

lib/Topology/NhdsWithin · eventually_nhds_subtype_iff. (□) Continuity within $s \times Y$ at (x, w) is tested on sequences Mathlib/Order/Filter/AtTopBot/CountablyGenerated · tendsto_iff_seq_tendsto: let $(x_n, w_n) \to (x, w)$ with $x_n \in s$. The set $K = \{w\} \cup \{w_n\}$ is compact Mathlib/Topology/Compactness/Compact · Filter.Tendsto.isCompact_insert_range, so $\|A(x_n)w_n - A(x)w_n\| \to 0$ by the hypothesis, and $A(x)w_n \to A(x)w$ by continuity of $A(x)$; adding, $A(x_n)w_n \to A(x)w$. For an sc^1 map, level 0 of condition (3) is the left side for Df on U_1, so $x \mapsto Df(x)$ is compact-open continuous on U_1 Chapter1/ExpositionLemmas · IsSc1With.deriv_uniformOnCompact. The text's other half, that norm continuity can fail, needs the shift map of the exposition's §6 and is not formalized.

Sc^k and sc-smooth. By recursion on k: sc^0 is sc^0, and f is sc^{k+1} on U if it is sc^1 on U and the chosen tangent map is sc^k on TU Chapter1/ScSmooth · IsSck; f is **sc-smooth** on U if it is sc^k for every k Chapter1/ScSmooth · IsScSmooth. An sc^k map is sc^0 Chapter1/ScSmooth · IsSck.isSc0, sc^{k+1} implies sc^k Chapter1/ScSmooth · IsSck.of_succ Chapter1/ScSmooth · IsSck.of_le, and f is sc-smooth iff it is sc^1 with sc-smooth tangent map Chapter1/ScSmooth · isScSmooth_iff. On an sc-domain the notion depends only on the values on U, by induction with the uniqueness of the tangent map Chapter1/ChainRule · IsSc1.tangentMap_congr Chapter1/ChainRule · IsSck.congr. **Finite dimensions.** The text's remark that sc^1 maps between finite-dimensional spaces are the C^1 maps holds on every sc-domain, in particular on relatively open subsets of quadrants, with boundary. Let V, W carry their constant structures and $U \subseteq V$ be an sc-domain. All level sets of U are U itself Chapter1/ExpositionLemmas · ScBanach.lvlSet_const and all $f \circ \iota_m$ are f. If f is sc^1, then $f \circ \iota_1 = f$ is C^1 within $U_1 = U$ by Proposition 1.16 (§I.14, Chapter1/Properties · IsSc1With.contDiffOn_base); if f is C^1 within U, it is C^1 on every level, so sc^1 by Corollary 1.20 at $k = 0$ (§I.17, Chapter1/Properties · corollary_1_20_zero) Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_const. Mathlib's "C^1 within U" means, on a set with unique derivatives, differentiable within U at every point with $x \mapsto$ fderivWithin $f\, U\, x$ continuous on U Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_fderivWithin Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const: at a boundary point of a quadrant this is the one-sided derivative along U, unique because the quadrant has interior (§I.10). Likewise sc-smooth is C^∞ within U, by Proposition 1.18 at $m = 0$ (§I.16) and the sc-smooth case of Corollary 1.20 Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const. For the text's tuples: Chapter1/ExpositionLemmas · isSc1_iff_contDiffOn_of_isRelOpenInQuadrant Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant. The open case is also checked in the test file Test · isSc1_iff_contDiffOn_of_finiteDimensional Test · isScSmooth_iff_contDiffOn_of_finiteDimensional. By §I.1 every sc-structure on a finite-dimensional space is the constant one up to sc-isomorphism.

I.12 Theorem 1.21: the chain rule

Statement. Let $U \subseteq C \subseteq E$ and $V \subseteq D \subseteq F$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq V$, and g sc^1 on V. Then $g \circ f$ is sc^1 on U and $T(g \circ f) = Tg \circ Tf$ on TU Chapter1/ChainRule · theorem_1_21. The same holds for sc-domains Chapter1/ChainRule · theorem_1_21', and the existence part needs still less: if g is sc^1 on V with derivatives Dg, f is sc^1 on U with derivatives Df, $f(U) \subseteq V$, and V_1 is open-in-convex, then $g \circ f$ is sc^1 on U with derivatives $x \mapsto Dg(f_1(x)) \circ Df(x)$, with no hypothesis on U Chapter1/ChainRule · IsSc1With.comp. The text's third tuple $W \subset Q \subset G$ plays no role.

The derivative. Fix $x \in U_1$, let $y_0 := f_1(x) \in V_1$ and K a compact set containing $Df(x)(\iota_1 \bar{B}_1)$, which exists because ι_1 is compact Mathlib/Analysis/Normed/Operator/Compact/Basic · IsCompactOperator.image_closedBall_subset_compact. Write $V_1 = O \cap C$ with O open and C convex, and choose $\rho > 0$ with the ρ-ball around y_0 inside O. Given $c > 0$, two tolerances are fixed. First, by the uniform smallness of $[Dg(y) - Dg(y_0)](w + b)$ for $y \in V_1$ near y_0, $w \in K$ and b small, which is the continuity of $(y, b, w) \mapsto [Dg(y) - Dg(y_0)](w + b)$ on $V_1 \times F_0 \times K$ made uniform over K Mathlib/Topology/Compactness/Compact · IsCompact.eventually_forall_of_forall_eventually Chapter1/ChainRule · IsSc1With.exists_delta, there is δ with this expression of norm $\le c/2$. Second, $\varepsilon := \min(\delta/2, (c/2)/(\|Dg(y_0)\| + 1))$. For $z \in U_1$ close to x three things hold eventually: $\|f_1(z) - y_0\|_1 < \min(\rho, \delta)$ (continuity of f_1 on U_1), and $r := f(\iota z) - f(\iota x) - Df(x)\iota(z - x)$ has $\|r\| \le \varepsilon \|z - x\|_1$ (condition (2) for f Mathlib/Analysis/Calculus/FDeriv/Defs · hasFDerivWithinAt_iff_isLittleO). Let $k := f_1(z) - y_0 \in F_1$. The segment $y_0 + tk$, $t \in [0, 1]$, lies in V_1: inside the ρ-ball, hence in O, and in C by convexity, since both endpoints are in C. The text

states this inclusion without comment; it is the one place where the structure of V enters.

The mean value inequality. The text's second integral is replaced by: for $\varphi(t) := g(\iota(y_0 + tk)) - t\,Dg(y_0)(\iota k)$ on $[0,1]$, the derivative within $[0,1]$ is $[Dg(y_0 + tk) - Dg(y_0)](\iota k)$ (condition (2) for g along the segment, composed with the affine curve), and if its norm is $\le C$ on $[0,1]$ then $\|\varphi(1) - \varphi(0)\| \le C$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le Chapter1/ChainRule · IsSc1With.norm_sub_le_of_segment`. Now write $\iota k = f(\iota z) - f(\iota x) = s(w_1 + b)$ with $s := \|z - x\|_1$, $w_1 := Df(x)\iota(u)$ for the unit vector $u := s^{-1}(z - x)$, so $w_1 \in K$, and $b := s^{-1}r$, so $\|b\| \le \varepsilon \le \delta/2 < \delta$. For every t, $y_0 + tk$ is within δ of y_0, so $\|[Dg(y_0 + tk) - Dg(y_0)](\iota k)\| \le (c/2)\,s$, and the mean value inequality gives

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)(\iota k)\| \le \tfrac{c}{2}\|z - x\|_1.$$

(When $z = x$ both sides vanish.) The text's first integral is not needed: $\|Dg(y_0)r\| \le \|Dg(y_0)\|\,\varepsilon\|z - x\|_1 \le \frac{c}{2}\|z - x\|_1$ directly, because $Dg(y_0)$ is bounded on F_0. Adding the two estimates, since $\iota k = Df(x)\iota(z - x) + r$,

$$\|g(f(\iota z)) - g(f(\iota x)) - Dg(y_0)Df(x)\iota(z - x)\| \le c\|z - x\|_1$$

eventually, which is condition (2) for $g \circ f$ at x with the operator $Dg(y_0) \circ Df(x)$ `Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp`. The text's subsequence argument for the uniform convergence of the second integrand is the compactness step above; the text's appeal to Proposition 1.18 to make $g : V_1 \to G_0$ of class C^1 is not used.

The tangent map. $g \circ f$ is sc^0 as a composite. The tangent map of $g \circ f$ with the composed derivatives equals $T_{Dg} \circ T_D f$ on TU, since $(g \circ f)_1 = g_1 \circ f_1$ on U_1 `Chapter1/ChainRule · IsSc1With.tangentMapWith_comp`; the right side is a composite of sc^0 maps ($T_D f$ maps TU into TV), hence sc^0: condition (3) `Chapter1/ChainRule · IsSc1With.comp`. For the chosen derivatives, $T(g \circ f) = Tg \circ Tf$ on TU follows from the uniqueness of the derivative on U_1 (§I.11), which is where the quadrant structure of U enters `Chapter1/ChainRule · theorem_1_21' Chapter1/ChainRule · theorem_1_21`.

Iterating. If U, V are sc-domains, f is sc^k on U with $f(U) \subseteq V$ and g is sc^k on V, then $g \circ f$ is sc^k on U: induction on k, the step being the theorem for the sc^1 parts, the induction hypothesis for $Tg \circ Tf$ on the sc-domains TU, TV, and the identity $T(g \circ f) = Tg \circ Tf$ with the independence of sc^k from the values off TU `Chapter1/ChainRule · IsSck.comp Chapter1/ChainRule · IsSck.comp_of_isRelOpenInQuadrant`. Sc-smooth maps compose to sc-smooth maps `Chapter1/ChainRule · IsScSmooth.comp`. This is the exposition's Corollary 4.A.

I.13 The compact-open lemma and the level operators $Df(x) : E_m \to F_m$

The proofs of Propositions 1.16–1.18 in the 2010 paper all rest on one mechanism, which the formal proof isolates.

The lemma, $k = 0$. Let $A : X \to \mathcal{L}(Y, Z)$ with $(x, y) \mapsto A(x)y$ continuous on $s \times Y$, and $\iota : Y' \to Y$ a compact operator. Then $x \mapsto A(x) \circ \iota$ is continuous on s into $\mathcal{L}(Y', Z)$ with the operator norm `Chapter1/Levels · continuousOn_clm_comp_of_isCompactOperator`. Proof: the image of the unit ball of Y' under ι lies in a compact K; by uniform continuity on compact sets `Chapter1/Levels · exists_delta_of_continuousOn`, for $x' \in s$ near x and all $w \in K$, $\|A(x')w - A(x)w\| \le \varepsilon/2$, so $\|(A(x') - A(x)) \circ \iota\| \le \varepsilon/2$ on unit vectors. This is the 2010 paper's subsequence argument (unit vectors $h_n \in E_{m+1}$ have a subsequence converging in E_m) in uniform form.

The lemma, C^k. Let s be open-in-convex with unique differentiability at every point, and $(x, y) \mapsto A(x)y$ of class C^k on $s \times Y$. Then $x \mapsto A(x) \circ \iota$ is C^k on s `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. Induction on k over all Z and A. For $k + 1$: the partial derivative in x, $B(x)y := D_x(A(x)y)$, is C^k in (x, y) `Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn`, linear and continuous in y for $x \in s$ (differentiate the linearity identities and use uniqueness of derivatives), so

$B : s \to \mathcal{L}(Y, \mathcal{L}(X, Z))$; by the induction hypothesis $x \mapsto B(x) \circ \iota$ is C^k, and so is its flip $G'(x) \in \mathcal{L}(X, \mathcal{L}(Y', Z))$. G' is the derivative of $x \mapsto A(x) \circ \iota$: for $z \in s$ near x the segment from x to z lies in s (open-in-convex), and for a unit vector $v \in Y'$ the mean value inequality applied to $\tau \mapsto A(x + \tau(z - x))(\iota v) - \tau B(x)(\iota v)(z - x)$ `Mathlib/Analysis/Calculus/MeanValue · Convex.norm_image_sub_le_of_norm_hasDerivWithin_le`, with the uniform continuity of B on $\{x\} \times \iota(\bar{B}_1)$, bounds $\|A(z)\iota v - A(x)\iota v - B(x)(\iota v)(z - x)\| \le c\|z - x\|$ uniformly in v. So $x \mapsto A(x) \circ \iota$ is differentiable within s with a C^k derivative, hence C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. The 2010 paper's Proposition 2.3 carries the same information in the multilinear maps $\Gamma(x)(h_1, \dots, h_k, h)$ and a differentiation under the integral sign; the lemma replaces both.

The level operators. Let f be sc^1 on U with derivatives D, and $x \in E_{m+1}$. The **level-m operator** $Df(x) : E_m \to F_m$ is the continuous linear T with $\iota_m \circ T = D(\iota x) \circ \iota_m$, when such T exists (it is unique by injectivity), and 0 otherwise `Chapter1/Levels · levelDeriv` `Chapter1/Levels · toBase_levelDeriv`; on level 0 it is $D(x)$ itself `Chapter1/Levels · levelDeriv_zero`. For $x \in U_{m+1}$ it exists: the level-m tangent map sends $(x, v) \in U_{m+1} \times E_m$ to a pair whose second component $T_m(v)$ satisfies $\iota_m T_m(v) = D(\iota x)\iota_m v$; T_m is linear (by injectivity of ι_m) and continuous (condition (3) on level m) `Chapter1/Levels · IsSc1With.exists_levelDeriv` `Chapter1/Levels · IsSc1With.toBase_levelDeriv`. So the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.levelMap_tangentMapWith`, and condition (3) says exactly that $(x, h) \mapsto Df(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv`: this is condition (2) of Proposition 1.16. By the compact-open lemma with $\iota = \mathrm{incl}_m : E_{m+1} \to E_m$, the map $x \mapsto Df(x) \circ \mathrm{incl}_m$ is continuous $U_{m+1} \to \mathcal{L}(E_{m+1}, F_m)$ in the operator norm `Chapter1/Levels · IsSc1With.continuousOn_levelDeriv_clm`: the continuity of df in condition (1) of Proposition 1.16. The base case, $x \mapsto D(x) \circ \iota_1$ continuous $U_1 \to \mathcal{L}(E_1, F_0)$, is also proved directly `Chapter1/Properties · IsSc1With.continuousOn_deriv_clm`, and with it $f \circ \iota_1 : U_1 \to F_0$ is C^1 `Chapter1/Properties · IsSc1With.contDiffOn_base`.

Level-raising: differentiability of $f : U_{m+1} \to F_m$. Let U be an sc-domain. For $x \in U_{m+1}$, the map $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ has derivative $Df(x) \circ \mathrm{incl}_m$ at x within U_{m+1} `Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level`. This is the 2010 paper's integral representation $f(x + k) - f(x) = \int_0^1 Df(x + \tau k)k \, d\tau$. For $z \in U_{m+1}$ near x and $k := z - x$, the segment $x + \tau k$ lies in U_{m+1} (open-in-convex). The integrand $g(\tau) := Df(x + \tau k)(\mathrm{incl}_m k)$ is continuous into F_m by condition (3) on level m, so its Bochner integral over $[0, 1]$ exists in F_m. Its image under ι_m is the integral of $D(\iota(x + \tau k))\iota_{m+1}k$ in F_0, which by the fundamental theorem of calculus for the C^1 curve $\tau \mapsto f(\iota(x + \tau k))$ (condition (2) for f, composed with the affine curve) `Mathlib/MeasureTheory/Integral/IntervalIntegral/FundThmCalculus · integral_eq_sub_of_hasDerivAt_of_le` equals $f(\iota z) - f(\iota x) = \iota_m(f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x))$. By injectivity, $f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) = \int_0^1 g$ in F_m. Then

$$\left\| f_m(\mathrm{incl}_m z) - f_m(\mathrm{incl}_m x) - Df(x)\mathrm{incl}_m k \right\|_m \le \sup_\tau \|(Df(x + \tau k) - Df(x)) \circ \mathrm{incl}_m\| \, \|k\|_{m+1},$$

and the supremum tends to 0 by the norm continuity just proved. The 2010 paper forms the integral in F_0 and remarks that the integrand is continuous into F_m; here it is formed in F_m and pushed down. Consequently $f_m \circ \mathrm{incl}_m : U_{m+1} \to F_m$ is C^1 within U_{m+1} `Chapter1/Levels · IsSc1With.contDiffOn_level`, which is condition (1) of Proposition 1.16 in full.

The level operator is the derivative of f_m. If $f_m : U_m \to F_m$ happens to be differentiable at $\mathrm{incl}_m x$ within U_m with derivative L, then $L = Df(x)$: both $\iota_m \circ L \circ \mathrm{incl}_m$ and $D(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1}, hence equal by uniqueness; and since incl_m has dense range, $\iota_m \circ L = D(\iota x) \circ \iota_m$ on all of E_m `Mathlib/Topology/Algebra/Module/ContinuousLinearMap/Basic · ext_on` `Chapter1/Levels · IsSc1With.levelDeriv_eq`. This is the exposition's Lemma 3.H and the key to Proposition 1.19.

I.14 Proposition 1.16: sc¹ in terms of the levels

The level conditions. For a family $D_m(x) \in \mathcal{L}(E_m, F_m)$, $x \in E_{m+1}$, the **level conditions** are: (a) for every m and $x \in U_{m+1}$, $f_m \circ \mathrm{incl}_m$ has derivative $D_m(x) \circ \mathrm{incl}_m$ at x within U_{m+1}; (b) for every m, $(x,h) \mapsto D_m(x)h$ is continuous on $U_{m+1} \times E_m$ `Chapter1/Levels · IsSc1Levels`. These are the text's (1) and (2) with $m-1$ renamed m and the extensions $Df(x)$ made an explicit family. From them, $x \mapsto D_m(x) \circ \mathrm{incl}_m$ is norm continuous (compact-open lemma) `Chapter1/Levels · IsSc1Levels.continuousOn_clm`, so $f : U_{m+1} \to F_m$ is C^1 on an sc-domain `Chapter1/Levels · IsSc1Levels.contDiffOn`: the rest of the text's (1).

Statement. For an sc-domain U and f sc^0 on U: f is sc^1 on U iff some family D_m satisfies the level conditions `Chapter1/Levels · proposition_1_16`.

(□). The level operators of §I.13 satisfy (a) (level-raising) and (b) (condition (3)) `Chapter1/Levels · IsSc1With.isSc1Levels`.

(□), and what the text hides. The 2010 paper calls this direction clear. It is, once one knows that the operators $D_m(x)$ for different m are restrictions of one operator, which the text's notation $Df(x)$ assumes. Proof: for $x \in U_{m+1}$, both $\iota_m \circ D_m(x) \circ \mathrm{incl}_m$ and $D_0(\iota x) \circ \iota_{m+1}$ are derivatives of $f \circ \iota_{m+1}$ at x within U_{m+1} (compose (a) at level m with ι_m, and (a) at level 0 with the inclusion $E_{m+1} \to E_1$), so they agree by uniqueness, and since incl_m has dense range, $\iota_m \circ D_m(x) = D_0(\iota x) \circ \iota_m$ `Chapter1/Levels · IsSc1Levels.toBase_apply`. Then f is sc^1 with derivatives D_0: condition (2) is (a) at level 0; condition (3) at level m reads, after the compatibility, $(x,h) \mapsto (f_{m+1}(x), D_m(x)h)$ on $U_{m+1} \times E_m$, which is continuous by (b) and the continuity of f_{m+1} `Chapter1/Levels · IsSc1Levels.isSc1With`.

The "in particular". If f is sc^1 with derivatives D and $x \in U_1$ is a smooth point, then $D(x) : E_0 \to F_0$ is an sc-operator: lift x to $y \in U_{m+1}$; the level-m tangent map at (y,v) has second component $T_m(v)$ with $\iota_m T_m(v) = D(x)\iota_m v$, linear and continuous in v, which is the level-m map required by `IsScOperator` `Chapter1/Properties · IsSc1With.scOperator_deriv`.

I.15 Proposition 1.17: lifting the index

Statement. If U is an sc-domain in E and f is sc^k on U, then $f_1 : E_1 \to F_1$ is sc^k on U_1, as a map between the shifted spaces E^1 and F^1 `Chapter1/Levels · IsSck.shift`; sc-smooth maps lift to sc-smooth maps `Chapter1/Levels · IsScSmooth.shift`. The target V of the text plays no role.

Proof. The level maps of f_1, computed in the shifted spaces, are the level maps of f shifted by one `Chapter1/Levels · IsSc0.levelMap_shift`, so an sc^0 map lifts to an sc^0 map, with no hypothesis on U `Chapter1/Levels · IsSc0.shift`. For sc^1 with derivatives D: the derivatives of f_1 are the level-1 operators $Df(x) : E_1 \to F_1$, $x \in U_2$; condition (2) for f_1 is the level-raising statement of §I.13 at $m=1$, and condition (3) is the shift of condition (3) for f, since $T(U_1) = (TU)^1$ (§I.11) and the level tangent maps are $(x,h) \mapsto (f_{m+2}(x), Df(x)h)$ `Chapter1/Levels · IsSc1With.shift`. For sc^{k+1}: the tangent map of f_1 agrees on $T(U_1)$ with the level-1 map of Tf (uniqueness of derivatives on the sc-domain U_1), which is sc^k by the induction hypothesis applied to Tf on TU `Chapter1/Levels · IsSck.shift`. This is the 2010 paper's argument, with the compatibility $T(U^1) = (TU)^1$ and the restriction of Df proved rather than assumed.

I.16 Proposition 1.18: sc^k maps are C^k from U_{m+k} to F_m

Statement. Let U be an sc-domain and f sc^k on U. Then $f_m \circ (E_{m+k} \to E_m) : U_{m+k} \to F_m$ is C^k within U_{m+k} for every m `Chapter1/Levels · IsSck.contDiffOn_levels`; $f : U_{m+l} \to F_m$ is C^l for $l \le k$ `Chapter1/Levels · IsSck.contDiffOn_levels_of_le`; in particular $f \circ \iota_k : U_k \to F_0$ is C^k `Chapter1/Levels · IsSck.contDiffOn_base`; and an sc-smooth map is C^k from U_{m+k} to F_m for all m, k `Chapter1/Levels · IsScSmooth.contDiffOn_levels`.

Proof. The 2010 paper proves the case $m = 0$ and reduces to it with Proposition 1.17 applied m times. The formal proof handles all levels at once, by induction on k over all sc-domains, with a compatible level-lowering operator $\rho : E_n \to E_m$, $n = m + k$, $\iota_m \rho = \iota_n$, as a parameter `Chapter1/Levels · IsSck.contDiffOn_levels'`; this avoids identifying $(E^m)_k$ with E_{m+k}. For $k = 0$, ρ is the identity and the claim is the continuity of f_m. For $k + 1$: write $\rho = \mathrm{incl}_m \circ \rho'$ with $\rho' : E_{m+k+1} \to E_{m+1}$ `Chapter1/Levels · ScBanach.incl_shift_inclK` `Chapter1/Levels · ScBanach.shift_inclK_mem_lvlSet`. By the level-raising statement of §I.13, $f_m \circ \rho$ has derivative $x \mapsto Df(\rho' x) \circ \rho$ on U_{m+k+1}, where $Df(\cdot)$ is the level-m operator. It remains to see that this derivative is C^k into $\mathcal{L}(E_{m+k+1}, F_m)$. The induction hypothesis for the sc^k map Tf on the sc-domain TU, with the level-lowering operator of TE (which is the pair of those of E^1 and E `Chapter1/Levels · ScBanach.tangent_inclK`), says that the level-m tangent map composed with it, $(x, h) \mapsto (f_{m+1}(\rho' x), Df(\rho' x)(E_{m+k} \to E_m)h)$, is C^k on $U_{m+k+1} \times E_{m+k}$; its second component is $(x, h) \mapsto A(x)h$ with $A(x) := Df(\rho' x) \circ (E_{m+k} \to E_m)$. The compact-open lemma with $\iota = \mathrm{incl}_{m+k} : E_{m+k+1} \to E_{m+k}$ makes $x \mapsto A(x) \circ \mathrm{incl}_{m+k} = Df(\rho' x) \circ \rho$ of class C^k `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator`. A map that is differentiable within an sc-domain with a C^k derivative is C^{k+1} `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`.

I.17 Proposition 1.19 and Corollary 1.20: sc^{k+1} from classical smoothness on the levels

Statement. Let U be an sc-domain and f sc^0 on U. If for every $m \geq 0$ and $0 \leq l \leq k$ the map $f_m \circ (E_{m+l} \to E_m) : U_{m+l} \to F_m$ is C^{l+1} within U_{m+l}, then f is sc^{k+1} on U `Chapter1/LevelsWeak · proposition_1_19_text`. This is the text's statement; its "f induces a map $f : U_{m+l} \to F_m$" is the sc^0 hypothesis (values of level m on U_m; the continuity is then the case $l = 0$). The proof is the 2010 paper's (its Proposition 2.4) and lives in `Chapter1/LevelsWeak`; an earlier, shorter proof under a stronger hypothesis is kept and described at the end of this section.

Why the hypothesis does not induct. The obvious induction would show that f is sc^1 and that Tf satisfies the hypothesis with $k - 1$. It does not. The map $Tf : (TU)_{m+l} \to (TF)_m$ has the second component $(x, h) \mapsto Df(x)h$ on $U_{m+l+1} \times E_{m+l} \to F_m$, and $Df(x)$ on E_{m+l} is the derivative of $f : U_{m+l} \to F_m$ (§I.13), a C^{l+1} map; so this component is only C^l, one order short of the C^{l+1} that the hypothesis for Tf would ask. What survives the induction is a description of the components of the iterated tangent maps: they are built from the derivatives $d^j f(x)$ of f itself, evaluated at points and vectors of various levels, and the regularity of f on the coarser levels supplies the missing order. The 2010 paper's inductive statement (S_k) carries exactly this description. The formal proof organizes it in six steps: (1) the derivatives $d^j f(x)$ on all level pairs, with their compatibility, form a *derivative tower*; (2) an *elementary map* is $d^j f$ evaluated at sc-operators; (3) it is sc^0, and sc^1 with an explicit derivative, by the level criterion of Proposition 1.16; (4) its tangent map is the pair of an elementary map and a sum of elementary maps; (5) by induction on s, every elementary map with $j + s \leq k + 1$ is sc^s; (6) f is the elementary map with $j = 0$.

The derivative tower. A derivative tower of order $k + 1$ for f on U `Chapter1/LevelsWeak · IsDerivTower` consists of base-level maps $\Phi_i(x)(h_1, \dots, h_i) \in F_0$ ($x, h_a \in E_0$) and, for $x \in U_{m+l}$, continuous i-linear maps $\Psi_{i,m,l}(x)$ on E_{m+l} with values in F_m, given for $i \leq l + 1$ and $i \leq k + 1$, such that: $\Phi_0 = f$; $\iota_m \Psi_{i,m,l}(x)(h_1, \dots, h_i) = \Phi_i(\iota_{m+l} x)(\iota_{m+l} h_1, \dots, \iota_{m+l} h_i)$, i.e. all the Ψ are one map seen on different levels; $\Psi_{i,m,l}$ is continuous on U_{m+l}; and for $i \leq l$, $i \leq k$ it is differentiable within U_{m+l} at every point, with derivative $\xi \mapsto \Psi_{i+1,m,l}(x)(\xi, \cdot)$. The ranges of i are exactly what "$f : U_{m+l} \to F_m$ is C^{l+1}" provides; the compatibility is the text's use of one symbol $d^j f(x)$ on all levels. Two values with the same images in E_0 agree, by injectivity of ι_m `Chapter1/LevelsWeak · IsDerivTower.eq_of_toBase`.

The tower of a map satisfying the hypothesis. $\Psi_{i,m,l}$ is Mathlib's i-th derivative within U_{m+l} of $f_m \circ (E_{m+l} \to E_m)$ `Chapter1/LevelsWeak · levelDerivTower` `Mathlib/Analysis/Calculus/ContDiff/`

FTaylorSeries · iteratedFDerivWithin, and Φ_i is the i-th derivative of $f : U_{i-1} \to F_0$ at the lifts of its arguments to E_{i-1} (arbitrary off the image of E_{i-1}) Chapter1/LevelsWeak · baseDerivTower. For $l > k$ the hypothesis is used with order $k+1$: $f : U_{m+l} \to F_m$ is the C^{k+1} map $f : U_{m+k} \to F_m$ restricted along the inclusion $E_{m+l} \to E_{m+k}$, and U_{m+l} is the preimage of U_{m+k} Chapter1/LevelsWeak · ScBanach.inclLE Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet; so $f : U_{m+l} \to F_m$ is $C^{\min(l,k)+1}$ for all m, l Chapter1/LevelsWeak · contDiffOn_levels_min. Continuity of $\Psi_{i,m,l}$ for $i \le \min(l,k)+1$ is the continuity of the iterated derivatives of a $C^{\min(l,k)+1}$ map Mathlib/Analysis/Calculus/ContDiff/Defs · ContDiffOn.continuousOn_iteratedFDerivWithin; for $i \le \min(l,k)$ the i-th derivative is C^1 Mathlib/Analysis/Calculus/ContDiff/Comp · ContDiffWithinAt.iteratedFDerivWithin_right, and its derivative is the $(i+1)$-st derivative with the new vector in front Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_succ_apply_left. Both use the uniqueness of derivatives on the level sets of the sc-domain. The compatibility Chapter1/LevelsWeak · levelDerivTower_toBase is proved in three steps, for $i \le l+1$ (so that $i-1 \le m+l$): (i) composing on the left with the continuous linear ι_m commutes with iterated derivatives within a set Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_left; (ii) on U_{m+l}, $\iota_m \circ f_m \circ (E_{m+l} \to E_m)$ equals $(f : U_{i-1} \to F_0) \circ (E_{m+l} \to E_{i-1})$, since f is sc^0 Chapter1/ScSmooth · IsSc0.toBase_levelMap, and a derivative within a set depends only on the values on the set Mathlib/Analysis/Calculus/ContDiff/FTaylorSeries · iteratedFDerivWithin_congr; (iii) composing on the right with the continuous linear inclusion $E_{m+l} \to E_{i-1}$ commutes as well, the preimage of U_{i-1} being U_{m+l} Mathlib/Analysis/Calculus/ContDiff/Basic · ContinuousLinearMap.iteratedFDerivWithin_comp_right. With the lifts in the definition of Φ_i Chapter1/LevelsWeak · baseDerivTower_toBase this gives the tower Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels.

Shifted spaces. E^p is the sc-Banach space with $(E^p)_l = E_{p+l}$ by definition Chapter1/LevelsWeak · ScBanach.shiftAdd (the p-fold iterate of the shift of §I.1 has these levels only up to rewriting $p+l$). The identifications $E \cong E^0$ and $(E^p)^1 \cong E^{p+1}$ are sc-isomorphisms whose level maps are identities transported along $l = 0+l$, resp. $p+(l+1) = p+1+l$ Chapter1/LevelsWeak · ScBanach.castCLE Chapter1/LevelsWeak · ScBanach.shiftAddZeroIso Chapter1/LevelsWeak · ScBanach.shiftAddShiftIso; they are used only through their compatibility with the inclusions into the base level Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm. For $b \le a$ the inclusion $E_a \to E_b$ is compatible with the ι's Chapter1/LevelsWeak · ScBanach.toBase_inclLE.

Elementary maps. Elementary data on an sc-Banach space X Chapter1/LevelsWeak · ElemData are an sc-operator $\pi_0 : X \to E^p$ (the base point), i sc-operators $\pi_a : X \to E^{n_a}$ (the vectors) and a target shift m. The elementary map is

$$g(z) = d^i f(\pi_0 z)(\pi_1 z, \dots, \pi_i z) := \Phi_i(\pi_0 z)(\pi_1 z, \dots, \pi_i z),$$

lifted to $F_m = (F^m)_0$ Chapter1/LevelsWeak · ElemData.map, on the domain $W = \pi_0^{-1}(U_p)$ Chapter1/LevelsWeak · ElemData.dom, whose level-l set is $\pi_0^{-1}(U_{p+l})$ Chapter1/LevelsWeak · ElemData.mem_lvlSet_dom. The exposition's map of type $(p, l; n_1, \dots, n_j)$ on $U^p \oplus E^{n_1} \oplus \cdots \oplus E^{n_j}$ is the case $X = E^p \oplus E^{n_1} \oplus \cdots$ with the projections as slots. The conditions are Chapter1/LevelsWeak · ElemData.Good

$$m+i \le p, \qquad m+i \le n_a+1 \ (1 \le a \le i), \qquad i \le k.$$

The second is the text's $j+l-1 \le n_i$. The first is $j+l \le k+1$ for the base point $x \in U_{k+1}$, which the text derives from $j+l-1 \le n_i \le k$ (and states separately, as $l \le k+1$, only for $j=0$ in the exposition). The third bounds the order of derivative that will be used. The text's upper bound $n_i \le k$ is not needed.

On level l, the base point and the vectors pushed down to a common level q ($q \le p+l$, $q \le n_a+l$) are $\rho_0 = (E_{p+l} \to E_q) \circ \pi_0$ and $\rho_a = (E_{n_a+l} \to E_q) \circ \pi_a$ Chapter1/LevelsWeak · ElemData.ρ₀ Chapter1/LevelsWeak · ElemData.ρ. Seen in F_0, g at a point of level l is Φ_i at the base-level images of $\pi_0 z$ and $\pi_a z$ Chapter1/LevelsWeak · ElemData.map_toBase, and its level-l map is $z \mapsto \Psi_{i,m+l,\lambda}(\rho_0 z)(\rho_1 z, \dots, \rho_i z)$ for any λ with $i \le \lambda+1$, $m+l+\lambda \le p+l$ and $m+l+\lambda \le n_a+l$ Chapter1/LevelsWeak · ElemData.levelMap_eq. The first two conditions make $\lambda = i-1$ admissible, and $i \le k$ gives $i \le k+1$, so the tower value exists. Hence g is sc^0: its values at level-l points lie in F_{m+l}, and its level-l map is continuous on W_l as the evaluation of the continuous $\Psi_{i,m+l,i-1} \circ \rho_0$ at the continuous $(\rho_1, \dots, \rho_i)$ Chapter1/LevelsWeak · ElemData.isSc0.

The sc¹ step. For $z \in W_{l+1}$ the candidate derivative $Dg(z) : X_l \to F_{m+l}$ is the derivative (3.5) of the exposition `Chapter1/LevelsWeak · ElemData.deriv` `Chapter1/LevelsWeak · ElemData.deriv_apply`:

$$Dg(z)\zeta = \Psi_{i+1,m+l,i}(\rho_0 z)(\rho_0\zeta, \rho_1 z, \dots, \rho_i z) + \sum_{a=1}^{i} \Psi_{i,m+l,i-1}(\rho_0 z)(\rho_1 z, \dots, \rho_a\zeta, \dots, \rho_i z),$$

the first term on the uniform level $m+l+i$, the sum on $m+l+i-1$. The conditions put every argument where it belongs: $\rho_0 z$ and $\rho_a z$ (with z of level $l+1$) in E_{m+l+i}, because $m+i \le p+1$ and $m+i \le n_a+1$; $\rho_0\zeta$ (with ζ of level l) in E_{m+l+i}, because $m+i \le p$; $\rho_a\zeta$ in $E_{m+l+i-1}$, because $m+i-1 \le n_a$; and $\Psi_{i+1,m+l,i}$ exists because $i+1 \le k+1$, which is where $i \le k$ enters. On the base level, $Dg(z)\zeta$ is $\Phi_{i+1}(\pi_0 z)(\pi_0\zeta, \pi z) + \sum_a \Phi_i(\pi_0 z)(\pi z$ with $\pi_a z := \pi_a\zeta)$ `Chapter1/LevelsWeak · ElemData.deriv_toBase`. The level criterion of Proposition 1.16 (§I.14) asks two things.

Differentiability of $g : W_{l+1} \to F_{m+l}$ with derivative $Dg(z)$ restricted to X_{l+1}. On W_{l+1}, the level-l map of g is $z' \mapsto \Psi_{i,m+l,i}(\rho_0 z')(\rho_1 z', \dots, \rho_i z')$ with the slots taken at level $l+1$ (the formula above with $\lambda = i$) `Chapter1/LevelsWeak · ElemData.levelMap_incl_eq`. In the tower, $\Psi_{i,m+l,i}$ is differentiable with derivative $\Psi_{i+1,m+l,i}$ ($i \le i$ and $i \le k$); the slots are continuous linear; and the evaluation $(A, h) \mapsto A(h_1, \dots, h_i)$ of continuous multilinear maps is differentiable with derivative $(B, \eta) \mapsto B(h) + \sum_a A(h$ with $h_a := \eta_a)$ `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`, proved from Mathlib's derivative of a multilinear map in its arguments `Mathlib/Analysis/Calculus/FDeriv/Analytic · ContinuousMultilinearMap.hasFDerivAt` and the linearity in A (the evaluation written as a flipped multilinear map `Chapter1/LevelsWeak · cmm_eval_eq`). The chain rule gives a derivative whose first term is that of Dg (the base point restricted from level $l+1$ to level l is the same vector `Chapter1/LevelsWeak · ElemData.ρ₀_incl`) and whose sum terms are values of $\Psi_{i,m+l,i}$; these equal the $\Psi_{i,m+l,i-1}$ terms of Dg because they have the same images in E_0 (the tower's compatibility) `Chapter1/LevelsWeak · ElemData.hasFDerivWithinAt_level`.

Continuity of $(z, \zeta) \mapsto Dg(z)\zeta$ on $W_{l+1} \times X_l$: $\Psi_{i+1,m+l,i}$ and $\Psi_{i,m+l,i-1}$ are continuous on their level sets, and the slots, the insertion of $\rho_a\zeta$ and the evaluation are continuous `Chapter1/LevelsWeak · ElemData.continuousOn_deriv`.

So g satisfies the level conditions `Chapter1/LevelsWeak · ElemData.isSc1Levels` `Chapter1/Levels · IsSc1Levels`, and when W is an sc-domain it is sc^1 with derivative Dg at level 0 `Chapter1/Levels · IsSc1Levels.isSc1With` `Chapter1/LevelsWeak · ElemData.isSc1With`. This is the text's "Γ is sc^1", with the level criterion in place of the direct check of Definition 1.14 that the exposition carries out.

The tangent map of an elementary map. On $TX = X^1 \oplus X$ a slot $\pi : X \to E^n$ gives two slots: π on the first factor, $TX \to E^{n+1}$ through $(E^n)^1 \cong E^{n+1}$ `Chapter1/LevelsWeak · ScOperator.tangentX`, and π on the second factor, $TX \to E^n$ `Chapter1/LevelsWeak · ScOperator.tangentH`. Three kinds of elementary data on TX arise from g. The first, g', has all slots on the first factor, with levels $p+1$, n_a+1 and target shift $m+1$ `Chapter1/LevelsWeak · ElemData.shiftData`: it is g one level up. The second, g_0, has the base point π_0 on the first factor (level $p+1$), $i+1$ vectors, namely π_0 on the second factor (level p) followed by the π_a on the first (levels n_a+1), and target shift m `Chapter1/LevelsWeak · ElemData.derivData`: it is the term $d^{i+1}f(x)(\delta x, h)$. The third, g_a for $1 \le a \le i$, is like g' but with π_a moved to the second factor (level n_a) and target shift m `Chapter1/LevelsWeak · ElemData.slotData`: it is the term $d^i f(x)(h$ with $h_a := \delta h_a)$. All satisfy the conditions, which is the text's "one checks that the new indices satisfy the required inequalities": for g', $m+1+i \le p+1$ and $m+1+i \le n_a+2$ `Chapter1/LevelsWeak · ElemData.Good.shiftData`; for g_0, $m+i+1 \le p+1$ (for the base point and for the new first vector, of level p) and $m+i+1 \le n_a+2$, and $i+1 \le k$, which has to be assumed `Chapter1/LevelsWeak · ElemData.Good.derivData`; for g_a, $m+i \le p+1$, $m+i \le n_a+1$ for the moved slot and $m+i \le n_b+2$ for the others `Chapter1/LevelsWeak · ElemData.Good.slotData`. The domain of all three (they share the base point π_0 on the first factor) is TW `Chapter1/LevelsWeak · ElemData.tangentSet_dom`.

If W is an sc-domain and $i+1 \le k$, the tangent map of g agrees on TW with the pair (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$) `Chapter1/LevelsWeak · ElemData.tangentMap_eqOn`. Proof: by the uniqueness of the derivative on the sc-domain W, the tangent map is (g on level 1, Dg) `Chapter1/ScSmooth · IsSc1.tangentMap_`

eqOn; the first components agree in F_0, both being Φ_i at the base-level images of the level-1 slots Chapter1/LevelsWeak · ElemData.map_toBase Chapter1/LevelsWeak · ScBanach.toBase_shiftAddShiftIso_symm; the second components agree in F_0 term by term, by the base-level formula for Dg Chapter1/LevelsWeak · ElemData.deriv_toBase.

Sums. The constant map 0 is sc^k, being the zero sc-operator Chapter1/LevelsWeak · ScOperator.zeroOp Chapter1/LevelsWeak · IsSck.zero. If g_1, g_2 are sc^k on an sc-domain, $g_1 + g_2$ is the addition $Y \oplus Y \to Y$ Chapter1/LevelsWeak · ScOperator.addOp, an sc-operator and hence sc-smooth Chapter2/TangentBundle · ScOperator.isScSmooth, composed with the pairing (g_1, g_2), which is sc^k Chapter2/StrongBundle · IsSck.scPair; so it is sc^k by the chain rule (§I.12, Chapter1/ChainRule · IsSck.comp) Chapter1/LevelsWeak · IsSck.add. Finite sums follow by induction over the index set Chapter1/LevelsWeak · IsSck.sum. Scalar multiples are not needed: the text's "linear combinations" are sums.

The induction (S_k). *Claim:* for every s, every sc-Banach space X and all elementary data on X satisfying the conditions, with sc-domain W and $i + s \le k + 1$, the elementary map is sc^s on W Chapter1/LevelsWeak · ElemData.isSck. The induction is on s, for all X, i and data at once. For $s = 0$ this is the sc^0 statement. For $s + 1$: g is sc^1 by the sc^1 step, and it remains to show that its tangent map is sc^s. If $s = 0$, the tangent map of an sc^1 map is sc^0 Chapter1/ScSmooth · IsSc1.tangentMap_isSc0. If $s \ge 1$, then $i + 1 \le k$ (from $i + s + 1 \le k + 1$), so the tangent map agrees on TW with (g' carried along $(F^m)^1 \cong F^{m+1}$, $g_0 + \sum_a g_a$); TW is an sc-domain Chapter1/ScSmooth · IsScDomain.tangentSet, equal to the domain of the three new data Chapter1/LevelsWeak · IsScDomain.of_eq Chapter1/LevelsWeak · IsSck.of_eq_set, and sc^s passes to maps that agree on the domain Chapter1/ChainRule · IsSck.congr. The pair is sc^s if its two components are Chapter2/StrongBundle · IsSck.scPair. Now g' and the g_a have i vectors and $i + s \le k + 1$, and g_0 has $i + 1$ vectors and $(i + 1) + s \le k + 1$, so all are sc^s by the induction hypothesis (applied on $X' = TX$); g' composed with the sc-isomorphism $(F^m)^1 \cong F^{m+1}$, an sc-operator Chapter1/ScBanach · ScIso.toScOperator, is sc^s by the chain rule; and $g_0 + \sum_a g_a$ by the sum rule.

This is the text's induction, run on s over all elementary maps rather than on the description of $T^{k+1}f$. The 2010 paper proves that every term of $\pi \circ T^{k+1}f$ is sc^1, hence $T^{k+1}f$ is, and that the terms of $T^{k+2}f$ are again elementary with admissible indices; the formal proof shows that an elementary map is sc^{s+1} because its tangent map is made of elementary maps that are sc^s. The computations are the same — the derivative (3.5), the three kinds of new terms and their indices — and the formal version never names $T^{k+1}f$ or its 2^{k+1} components.

Proposition 1.19. f is the elementary map on $X = E$ with no vectors, $p = m = 0$ and $\pi_0 : E \cong E^0$ Chapter1/LevelsWeak · ElemData.base; its domain is U Chapter1/LevelsWeak · ElemData.base_dom and the conditions hold trivially Chapter1/LevelsWeak · ElemData.base_good. With the tower of f (step 1) and $s = k + 1$, $0 + (k + 1) \le k + 1$, this elementary map is sc^{k+1}; composed with the sc-isomorphism $F^0 \cong F$ it agrees with f on U, since $\Phi_0 = f$; so f is sc^{k+1} Chapter1/LevelsWeak · proposition_1_19_text. If the hypothesis holds for every l, then f is sc^{k+1} for every k, i.e. sc-smooth Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_text.

Corollary 1.20. For a finite-dimensional target V with its constant structure, the level maps of $f : E_0 \to V$ are $f \circ \iota_m$ Chapter1/ScSmooth · levelMap_eq_of_toBase_eq Chapter1/ScBanach · ScBanach.const_toBase. Suppose $f \circ \iota_l$ is C^{l+1} within U_l for $0 \le l \le k$. Then f is sc^0 with no further hypothesis: f is continuous on $U = U_0$ (the case $l = 0$), so each $f \circ \iota_m$ is continuous on U_m. And the hypothesis of Proposition 1.19 holds: $f_m \circ (E_{m+l} \to E_m) = (f \circ \iota_l) \circ (E_{m+l} \to E_l)$, a C^{l+1} map composed with a continuous linear one, on U_{m+l}, the preimage of U_l Chapter1/LevelsWeak · ScBanach.preimage_inclLE_lvlSet. So f is sc^{k+1} Chapter1/LevelsWeak · corollary_1_20_text. If $f \circ \iota_l$ is C^{l+1} within U_l for every l, then f is sc-smooth (for $k = 0$ the corollary gives sc^1, hence sc^0) Chapter1/LevelsWeak · IsScSmooth.of_contDiffOn_levels_finiteDimensional_text; and if f is C^∞ on U, it is sc-smooth Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional (proved with the stronger version below, since C^∞ on U restricts to C^{k+1} on every U_m). The two sc-smooth statements are the exposition's; the second is what the theory uses (bump functions, Chapter 5).

The earlier version, under a stronger hypothesis. Chapter1/Levels proves Proposition 1.19 under the hypothesis that every level map $f_m : U_m \to F_m$ is C^{k+1} within U_m Chapter1/Levels · proposition_1_19.

For $k \geq 1$ this asks more than the text (C^{k+1} instead of C^1 on $U_m \to F_m$ itself), and in exchange the induction needs no (S_k); for $k = 0$ the hypotheses coincide.

$k = 0$. Suppose every f_m is C^1 within U_m. Put $Df(x) :=$ the derivative of $f = f_0$ within U at $\iota_1 x$, for $x \in E_1$ `Mathlib/Analysis/Calculus/FDeriv/Defs · fderivWithin`. Condition (2) holds because $f \circ \iota_1$ is the restriction of the differentiable f along the continuous linear ι_1. For condition (3) at level m: the derivative $df_m(x)$ of f_m within U_m at $x \in U_m$ is compatible with Df, $\iota_m \circ df_m(x) = Df(\iota_m x) \circ \iota_m$, because $\iota_m \circ f_m = f \circ \iota_m$ on U_m and derivatives within U_m are unique `Mathlib/Analysis/Calculus/FDeriv/Basic · UniqueDiffWithinAt.eq`; so the level-m tangent map is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$, continuous on $U_{m+1} \times E_m$ because f_m is C^1 (its derivative is continuous into $\mathcal{L}(E_m, F_m)$, and evaluation is continuous) `Chapter1/Properties · proposition_1_19_zero`. This is the text's proof of (S_0).

$k + 1$ *from* k. Suppose every f_m is C^{k+2}. Then f is sc^1 by the case $k = 0$, with the derivative Df above, and we must show Tf is sc^{k+1} on TU. By the induction hypothesis (for the sc-domain TU and the sc^0 map Tf) it suffices that the level maps of Tf are C^{k+1}. The level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), Df(x)h)$ with the level operator (§I.13), and the level operator at $x \in U_{m+1}$ is the derivative of f_m at $\mathrm{incl}_m x$ (Lemma 3.H, `Chapter1/Levels · IsSc1With.levelDeriv_eq`). So the level-m map of Tf is $(x, h) \mapsto (f_{m+1}(x), df_m(\mathrm{incl}_m x)h)$ on $U_{m+1} \times E_m$, whose components are C^{k+1}: the first as f_{m+1} is, the second as the derivative of a C^{k+2} map is C^{k+1} and evaluation is bilinear `Chapter1/Levels · proposition_1_19`. This is where the stronger hypothesis is used, and where the text's hypothesis would fail (see "Why the hypothesis does not induct").

Sc-smooth, and the corollary. If f maps each U_m into F_m and every f_m is C^∞, then f is sc-smooth `Chapter1/Levels · IsScSmooth.of_contDiffOn_levels`. For a finite-dimensional target: C^1 on every U_m gives sc^1 `Chapter1/Properties · corollary_1_20_zero`, C^{k+1} on every U_m gives sc^{k+1} `Chapter1/Levels · corollary_1_20`, in particular C^{k+1} on $U = U_0$ does, since restricting along ι_m preserves C^{k+1} `Chapter1/Levels · corollary_1_20_base`.

I.18 Definition 1.23 and Lemma 1.24: the degeneracy index

Definitions. For $p = (a, w) \in \mathbb{R}^n \oplus W$ let $\#(p)$ be the number of indices i with $a_i = 0$ `Chapter1/Degeneracy · quadrantDeg`. For an sc-isomorphism $T : E \to \mathbb{R}^n \oplus W$ presenting a partial quadrant C and $x \in E_0$, let $d_T(x) := \#(T_0 x)$ `Chapter1/Degeneracy · degIndexWith`. The **degeneracy index** $d_C(x)$ is $d_T(x)$ for a presentation T of C chosen once and for all, and 0 if C is not a partial quadrant `Chapter1/Degeneracy · degIndex`; by Lemma 1.24 below it equals $d_T(x)$ for every presentation and every $x \in C$ `Chapter1/Degeneracy · degIndex_eq`, and on the quadrant itself it is $\#$ `Chapter1/Degeneracy · degIndex_quadrant`. The text's interior points, boundary points and corners are the points with $d_C = 0$, 1, ≥ 2; Chapter 2 makes the boundary a definition.

Lemma 1.24, core form. Let $S : \mathbb{R}^n \times W_0 \to \mathbb{R}^{n'} \times W_0'$ be a continuous linear equivalence with $q \in Q \iff Sq \in Q'$ for the two quadrants. Then $\#(p) = \#(Sp)$ for every $p \in Q$ `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`. Only level 0 and linear algebra are used; the sc-structure plays no role. Proof. For $p \in Q$ let $E_p := \{(b, v) : b_i = 0 \text{ whenever } p_i = 0\}$, the kernel of the surjection $(b, v) \mapsto (b_i)_{i \in I(p)}$ `Chapter1/Degeneracy · coordMap` `Chapter1/Degeneracy · coordMap_surjective` `Chapter1/Degeneracy · mem_ker_coordMap`, so that $\dim(V/E_p) = \#(p)$ `Chapter1/Degeneracy · finrank_quotient_ker_coordMap`: this makes the text's "codimension of E_x" precise. For $u \in E_p$, $p + tu \in Q$ for all t near 0: coordinate by coordinate, if $p_i = 0$ then $u_i = 0$ and the coordinate stays 0; if $p_i > 0$ it stays positive for small t by continuity `Chapter1/Degeneracy · eventually_add_smul_mem_quadrant`. Hence $Sp + tSu \in Q'$ for t near 0, so for each i with $(Sp)_i = 0$ we have $0 \leq t(Su)_i$ for all t near 0, of both signs, which forces $(Su)_i = 0$ `Chapter1/Degeneracy · eq_zero_of_eventually_nonneg_mul`; so $S(E_p) \subseteq E'_{Sp}$ `Chapter1/Degeneracy · map_ker_coordMap_le`. The same argument for S^{-1} gives the reverse inclusion (the text concludes equality from one inclusion "since S is an isomorphism", which for infinite-dimensional W needs this second step), so $S(E_p) = E'_{Sp}$ and S induces a linear isomorphism $V/E_p \cong V'/E'_{Sp}$; comparing dimensions, $\#(p) = \#(Sp)$.

Lemma 1.24. If T and T' both present C, then $S := T_0' \circ T_0^{-1}$ carries the quadrant onto the quadrant (both are the image of C), and the core form at $p = T_0 x$ gives $d_T(x) = d_{T'}(x)$ for $x \in C$ `Chapter1/Degeneracy · lemma_1_24`. The text proves first that $k = k'$; here this is a consequence: $T_0^{-1}(0) \in C$ and $\#(0) = n$ in $\mathbb{R}^n \oplus W$ `Chapter1/Degeneracy · quadrantDeg_zero`, so the lemma at this point gives $n = n'$ `Chapter1/Degeneracy · lemma_1_24_dim`.

I.19 Theorem 1.25: boundary recognition

Statement. Let $U \subseteq C \subseteq E$ and $U' \subseteq C' \subseteq E'$ be relatively open subsets of partial quadrants, f sc^1 on U with $f(U) \subseteq U'$, g sc^1 on U' with $g(U') \subseteq U$, $g \circ f = \mathrm{id}$ on U and $f \circ g = \mathrm{id}$ on U'. Then $d_C(x) = d_{C'}(f(x))$ for every $x \in U$ `Chapter2/DegeneracyIndex · theorem_1_25`. The text's "germ of a local sc-diffeomorphism" is undefined in Chapter 1 and is read this way, with U, U' as small as one likes; the text has sc-smooth where sc^1 suffices, so the formal statement is stronger. The text gives no proof, referring to "[20], Theorem 1.19"; the formal proof is the argument of §2.3 (Lemma 2.29, Theorem 2.31, Corollary 2.32) specialized to the identity retraction, and lives in the Chapter 2 file because it uses the retraction vocabulary. Smooth points are not needed; level-1 points suffice.

The subspaces E_x. For a presentation T of C and $x \in E_0$, let $E_x := \{v \in E_0 : (T_0 v)_i = 0 \text{ whenever } (T_0 x)_i = 0\}$, the kernel of a continuous linear map to a finite-dimensional space `Chapter2/DegeneracyIndex · minimalSubspaceWith`; its codimension is $d_T(x)$ `Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith`. Two facts about the index: it is upper semicontinuous, $d_T(y) \le d_T(x)$ for y near x (a nonzero coordinate stays nonzero) `Chapter2/DegeneracyIndex · eventually_degIndexWith_le`; and every $x \in C$ is the limit of points of C on level 1 with the same index, namely $T_0^{-1}(a, w_n)$ with $w_n \in W_1$ converging to w in W_0, which exist by density `Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith`. These are the exposition's Lemma 5.A, with level 1 in place of smooth.

The sign argument. Let f be sc^1 on $U \subseteq C$ with values in C', $x \in U_1$, and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for t near 0 (the quadrant coordinates as in §I.18, and the openness of U_1) `Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet`, so $f(\iota(x + tv)) \in C'$ and, for each j with $(T_0' f(x))_j = 0$, the function $t \mapsto (T_0' f(\iota(x + tv)))_j$ is ≥ 0 with a zero at $t = 0$: a local minimum, so its derivative at 0, which is $(T_0' Df(x) \iota_1 v)_j$ by condition (2), vanishes `Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1`. So $Df(x)$ maps $E_x \cap \iota_1(E_1)$ into $E'_{f(x)}$. Since f is differentiable only along E_1, the density of $E_x \cap \iota_1(E_1)$ in E_x is needed to conclude $Df(x)(E_x) \subseteq E'_{f(x)}$; it holds because E_x is the kernel of a continuous linear map to a finite-dimensional space and $\iota_1(E_1)$ is dense `Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`. The text asserts the corresponding density ("the closure of $(T_x^R O)_\infty$ is $T_x^R O$") without proof.

One inequality at level-1 points. With g the inverse, the chain rule gives $Dg(f_1(x)) \circ Df(x) = \mathrm{id}$ on E_0 at $x \in U_1$ `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`, so $Df(x)$ is injective and $E_x \cong Df(x)(E_x) \le E'_{f(x)}$ sits inside a subspace of codimension $d_{C'}(f(x))$, while $Dg(f_1(x))$ maps $E'_{f(x)}$ into E_x by the sign argument for g; comparing codimensions, $d_C(x) \le d_{C'}(f(x))$ `Chapter2/DegeneracyIndex · degIndexWith_le_of_germ`. Applying this to g at $f_1(x)$ gives the reverse inequality, so $d_C(x) = d_{C'}(f(x))$ for $x \in U_1$ `Chapter2/DegeneracyIndex · theorem_1_25_lvl1`.

General points. For $x \in U$, approximate x by level-1 points $x_n \in U$ of C with $d_C(x_n) = d_C(x)$; then $f(x_n) \to f(x)$ (continuity of f on U), so $d_{C'}(f(x_n)) \le d_{C'}(f(x))$ eventually by semicontinuity, and $d_{C'}(f(x_n)) = d_C(x_n) = d_C(x)$ by the level-1 case; hence $d_C(x) \le d_{C'}(f(x))$. The symmetric argument for g at $f(x)$ gives equality `Chapter2/DegeneracyIndex · theorem_1_25`.

I.20 What the formalization taught us

- **The chain rule needs one integral less than the text writes, and one hypothesis more than it states.** The term $Dg(f(x))[f(x+h)-f(x)-Df(x)h]$ is $o(|h|_1)$ because $Dg(f(x))$ is bounded, with no integral `Chapter1/ChainRule · IsSc1With.hasFDerivWithinAt_comp`; the segment from $f(x)$ to $f(x+h)$ lies in V_1 only because V_1 is open-in-convex and $f : U_1 \to F_1$ is continuous, which the text does not say. The whole proof uses, of the quadrant structure of U, only the uniqueness of derivatives, and that only for $T(g\circ f) = Tg\circ Tf$ `Chapter1/ChainRule · theorem_1_21'`.
- **"The" derivative $Df(x)$ is a theorem, not a notation.** It is unique because E_1 is dense in E_0 and U_1 has unique differentiability `Chapter1/ScSmooth · IsSc1With.eq_of_uniqueDiffWithinAt`; without it the tangent map, hence sc^2, is undefined. The same density argument shows that the extensions $Df(x) : E_{m-1} \to F_{m-1}$ of Proposition 1.16, written with one symbol in the text, are restrictions of one operator `Chapter1/Levels · IsSc1Levels.toBase_apply`, which is what makes the "clear" direction of that proposition work.
- **One compactness argument carries §1.2.** The compact-open lemma `Chapter1/Levels · contDiffOn_clm_comp_of_isCompactOperator` turns joint continuity (or C^k-ness) of $(x,h) \mapsto Df(x)h$ into norm continuity (or C^k-ness) of $x \mapsto Df(x)|_{E_{m+1}}$; Propositions 1.16, 1.17 and 1.18 are this lemma plus bookkeeping of levels.
- **Proposition 1.19 does not induct on its own hypothesis.** If f satisfies it with k, then Tf in general does not satisfy it with $k-1$: the second component $(x,h) \mapsto Df(x)h : U_{m+l+1} \times E_{m+l} \to F_m$ is one order short. The stronger hypothesis "C^{k+1} on every level" makes the induction two lines `Chapter1/Levels · proposition_1_19`; the text's needs the 2010 paper's (S_k), which remembers f through the elementary maps $d^j f$ on mixed levels `Chapter1/LevelsWeak · ElemData.isSck Chapter1/LevelsWeak · proposition_1_19_text`. Written out, (S_k) needs three things the text leaves implicit: one $d^j f(x)$ on all levels `Chapter1/LevelsWeak · isDerivTower_of_contDiffOn_levels`, the base-point condition $j+l \le k+1$ in place of the bound $n_i \le k$ (which is not needed) `Chapter1/LevelsWeak · ElemData.Good`, and the derivative of the multilinear evaluation `Chapter1/LevelsWeak · hasFDerivAt_cmm_eval`.
- **Regularizing is a bootstrap.** In §1.4.1 the step "$e \in E_m$ by Proposition 1.9" needs m applications, since Se is only one level above e `Chapter1/ScFredholm · ScPlus.regularizing_add`.
- **The index is an addition, and an easy one once additivity is there.** Neither Proposition 1.10 nor the remark after Definition 1.8 mentions the index of $T+S$ or of the level operators; with the additivity of the index `Chapter1/FredholmIndex · linIndex_comp` and a quasi-inverse, the index is locally constant along $u+tk$, hence invariant `Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`, and the sc-statements follow on every level `Chapter1/FredholmIndex · proposition_1_10_index_lvl`.
- **The text's quick remarks are all true, and some need real arguments.** "Precisely one sc-structure" in finite dimensions `Chapter1/ExpositionLemmas · ScBanach.isoConstOfFiniteDimensional`; "$(E/A)_\infty$ consists of the classes of E_∞", a Mittag-Leffler series `Chapter1/ExpositionLemmas · ScBanach.exists_smooth_mem_coset`; "it is a sc-projection" `Chapter1/ExpositionLemmas · ScBanach.isScSplitting_iff_exists_scProj`; "sc^1 is C^1 in finite dimensions", which at the boundary of a quadrant means derivatives within U `Chapter1/ExpositionLemmas · isSc1_iff_fderivWithin_const`.
- **Mathlib had Fredholm operators but not Riesz–Schauder.** The theorem that $1+$ compact is Fredholm, and the stability under compact perturbations, had to be proved `Chapter1/FredholmPerturbation · isFredholm_one_add Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator`; the finite codimension of the range comes from a chain of closed subspaces and Riesz's lemma `Chapter1/FredholmPerturbation · coFG_range_one_add`.
- **The degeneracy index knows nothing of the sc-structure.** Lemma 1.24 holds for any linear equivalence of the base levels carrying one quadrant onto the other `Chapter1/Degeneracy · quadrantDeg_eq_of_equiv`, and $k = k'$ is its value at the origin `Chapter1/Degeneracy · lemma_1_24_dim`. Boundary recognition needs only sc^1 and only level-1 points, not smooth ones `Chapter2/DegeneracyIndex · theorem_1_25`.

Part II. Differential geometry based on retracts (HWZ Chapter 2, §§2.1–2.6)

Chapter 2 builds a differential geometry whose local models are not open sets of Banach spaces but images of sc-smooth retractions, sets whose "dimension" may jump. The formal proof establishes the chapter in its order: retractions and retracts and the independence of their tangent from the retraction (§§II.2–II.4), M-polyfolds with their filtration, metrizability and tangent bundle (§§II.5–II.9), sub-M-polyfolds (§II.10), the degeneracy index with the boundary recognition argument (§§II.11–II.13), tame retracts and the faces of tame M-polyfolds (§§II.14–II.16), products and fibred products (§II.17), and strong bundles with their pull-backs (§§II.18–II.20). What is not formalized is the classical Proposition 2.2 (motivation only), Lemma 2.29 in its formulation by sc-smooth paths (its content, the sign arguments, is formalized and used instead), the second statement of Theorem 2.31 in general (for tame retracts its conclusion, Proposition 2.36's partial quadrant, is proved directly), the openness of $p : TX \to X_1$, and the categories $\mathcal{R}$, $\mathcal{BR}$, $\mathcal{SBR}$.

Three choices shape the formal route. First, a retraction is a total map $r : E_0 \to E_0$ with an explicit domain U, and the tangent TO of a retract is described as the *fixed-point set* of Tr in TU (§II.2); this description, which the text's proofs use without stating it, makes Propositions 2.4 and 2.6 one-line consequences of the chain rule. Second, an M-polyfold is a space with *one* atlas, and every notion is defined through the maximal atlas of all charts compatible with it (§II.5); this needs the fact, not in the text, that compatibility is a local property, proved from the chain rule and the locality of sc-smoothness (§II.1). Third, where the text computes in the model $C = [0,\infty)^k \oplus W$, the formal statements carry a presentation $T : E \cong \mathbb{R}^n \oplus W$ of the partial quadrant and its coordinate functionals $\lambda_i = (T\,\cdot)_i$; the subspaces E_x and cones C_x are defined through T, and are shown not to depend on it (§II.14).

Six proofs take a different road from the text's (or the exposition's). Theorem 2.16 (metrizability of the levels) is proved without Lemma 2.56 and Proposition 2.57, which need a regularity of X_m the text does not check: a partition of unity of X pulled back to X_m gives an explicit metric, and the same argument proves Smirnov's theorem, which Mathlib lacks (§II.7). Proposition 2.30 is proved from the sign arguments at points on level 1, not from Lemma 2.29 at smooth points (§II.13). Theorem 2.31 is proved for every chart of every M-polyfold by a direct isomorphism of quotients, without reducing to a chart that realizes the index (§II.13). Proposition 2.35 uses only the second condition of tameness (§II.14). Proposition 2.36's partial quadrant is proved for tame retracts without the equality case of Theorem 2.31, the smooth vectors of its proof being written down explicitly (§II.14). The second sentence of Proposition 2.44 is proved by removing all coordinates of σ at once rather than by the exposition's induction on $\#\sigma$ (§II.16).

Map.

- `Chapter2/Retracts`: restriction and locality of sc^k maps; retractions, retracts, Proposition 2.4, Definitions 2.5, 2.7, Propositions 2.6, 2.10, Theorem 2.8; local models.
- `Chapter2/TangentBundle`: the tangent of a retract as a retract, the tangent functor, locality of Tf; the tangent bundle TX, Proposition 2.19 (1)–(2), TX as an M-polyfold.

- `Chapter2/MPolyfold`: charts, atlases, M-polyfolds, Definition 2.14, the levels X_m, Lemma 2.15, sub-M-polyfolds (Definition 2.20), a retract as an M-polyfold.
- `Chapter2/LevelPolyfold`: X^1 as an M-polyfold; the locality of Definition 2.14.
- `Chapter2/Metrizable`: Smirnov's theorem, Theorem 2.16, Proposition 2.19 (3), Proposition 2.21 (3)–(4).
- `Chapter2/TangentMap`: the tangent map $Tf : TX \to TY$ and the chain rule.
- `Chapter2/SubMPolyfold`: Proposition 2.21 (1)–(2), Lemma 2.26.
- `Chapter2/DegeneracyIndex`: E_x, the two-sided sign argument, the core of Theorem 2.31, Theorem 1.25, Definition 2.22, Lemma 2.23, Proposition 2.24, Corollary 2.32.
- `Chapter2/Boundary`: the boundary, the one-sided sign argument, Proposition 2.30, Theorem 2.31.
- `Chapter2/Tame`: the definitions of §2.4 and of §2.5's local objects.
- `Chapter2/TameRetracts`: Propositions 2.35, 2.36, the local geometry of faces, Lemma 2.40, Corollary 2.41, Propositions 2.42, 2.43 (the retraction onto a face).
- `Chapter2/TameFaces`: independence of the presentation; Proposition 2.43 (faces are tame), Proposition 2.44 (first sentence), Theorem 2.45.
- `Chapter2/PartialCone`: Proposition 2.36, $T^R_x O$ an sc-subspace and $C_x O$ a partial quadrant.
- `Chapter2/CornerGerms`: Proposition 2.44 (second sentence), the corners f_σ.
- `Chapter2/FibredProduct`: products of M-polyfolds (the exposition's Lemma 4.D), Proposition 2.46.
- `Chapter2/StrongBundle`, `Chapter2/StrongBundlePullback`: §2.5.
- `Chapter2/StrongBundleOne`: the M-polyfold $Y[1]$, $P[1]$, the chart independence of $Y_{m,k}$, sc$^+$-sections into $Y[1]$.

II.1 Restriction and locality of sc^k maps

The text restricts sc-smooth maps to smaller sets ("$r : U \to U \cap U'$ is sc-smooth") and glues local sc-smoothness without comment. Both need an argument, because the tangent map of an sc^1 map is built from a *chosen* derivative family (Part I, §I.11), and a chosen derivative on U and one on $U' \subseteq U$ need not be the same objects.

Restriction. If f is sc^1 on U with derivative family D and $U' \subseteq U$, the same D works on U' `Chapter2/Retracts · IsSc1With.of_subset` `Chapter2/Retracts · IsSc1.of_subset`. When U' is an sc-domain (unique differentiability on every level, Part I, §I.10), derivatives at points of U'_1 are unique, so the tangent maps built on U' and on U agree on TU' `Chapter2/Retracts · IsSc1.tangentMap_eqOn_of_subset`. By induction on k, using that TU' is again an sc-domain and that the tangent maps agree there, an sc^k map restricts to sc^k on every smaller sc-domain `Chapter2/Retracts · IsSck.of_subset`, and so does an sc-smooth map `Chapter2/Retracts · IsScSmooth.of_subset`; sc-smoothness depends only on the values on U `Chapter2/Retracts · IsScSmooth.congr`. Relatively open subsets of quadrants, and their intersections with open sets, are sc-domains `Chapter2/Retracts · IsRelOpenInQuadrant.inter_isOpen` `Chapter2/Retracts · IsScDomain.inter_isOpen`, and $T(U \cap W) = TU \cap (W_1 \times E_0)$ `Chapter2/Retracts · ScBanach.tangentSet_inter_isOpen`.

Locality. sc^0 is local: continuity on each level is a local property `Chapter2/Retracts · IsSc0.of_locally`. For sc^1, suppose every $x \in U$ has an open $W_x \ni x$ with f sc^1 on $U \cap W_x$. Two such local structures give the same derivative at a common point, by uniqueness on the sc-domain `Chapter2/Retracts · IsSc1With.eq_of_eq_near`. Glue: at a point $x \in U_1$ take the derivative of the structure on $U \cap W_{\iota_1 x}$. This family D agrees near every point with a local derivative family; the three conditions of Definition 1.14 (the sc^0 property, differentiability on U_1, and the sc^0 property of the tangent map on TU) are local and hold `Chapter2/Retracts · IsSc1.of_locally`. The induction on k applies the same argument to the tangent maps on TU `Chapter2/Retracts · IsSck.of_locally`, and gives the sc-smooth case `Chapter2/Retracts · IsScSmooth.of_locally`. The identity is sc^1 with derivative the identity `Chapter2/Retracts · isSc1With_id`, its tangent map is the identity on TU `Chapter2/Retracts · tangentMap_id_eqOn`, and it is sc-smooth on every sc-domain `Chapter2/Retracts · isSck_id` `Chapter2/Retracts · isScSmooth_id`.

II.2 Retractions, retracts and their tangent (Definitions 2.1, 2.3, 2.5; Proposition 2.4)

Definition 2.1. An *sc-smooth retraction* on a partial quadrant $C \subseteq E_0$ is a set U relatively open in C and a map $r : E_0 \to E_0$ with $r(U) \subseteq U$, r sc-smooth on U and $r(r(x)) = r(x)$ for $x \in U$ `Chapter2/Retracts · ScRetraction`. Since U is relatively open, it is an sc-domain and r is sc^1 `Chapter2/Retracts · ScRetraction.isScDomain Chapter2/Retracts · ScRetraction.isSc1`. The *retract* is $O = r(U)$ `Chapter2/Retracts · ScRetraction.retract`. It is contained in U and is exactly the fixed-point set of r in U: if $x = r(y)$ then $r(x) = r(r(y)) = x$ `Chapter2/Retracts · ScRetraction.retract_subset Chapter2/Retracts · ScRetraction.apply_eq_self_of_mem_retract Chapter2/Retracts · ScRetraction.mem_retract_iff`. On level 1, r_1 fixes the points of O_1 `Chapter2/Retracts · ScRetraction.levelMap_one_eq_self`, and $O_m \subseteq U_m$ `Chapter2/Retracts · ScRetraction.lvlSet_retract_subset`.

Definition 2.3. (O, C, E) is an *sc-smooth retract* if O is the retract of some sc-smooth retraction on C `Chapter2/Retracts · IsScRetract Chapter2/Retracts · ScRetraction.isScRetract`. Every relatively open $U \subseteq C$ is one, with $r = \mathrm{id}$ `Chapter2/Retracts · ScRetraction.id Chapter2/Retracts · IsRelOpenInQuadrant.isScRetract`. A *local model* is a triple (E, C, O) with C a partial quadrant and O a retract in C `Chapter2/Retracts · LocalModel`; these are the models of M-polyfold charts.

The tangent of a retraction. Put $Tr(TU) := \{Tr(p) : p \in TU\} \subseteq E_1 \times E_0$ `Chapter2/Retracts · ScRetraction.tangent`. The chain rule applied to $r \circ r = r$ (Part I, Theorem 1.21) gives $Tr \circ Tr = Tr$ on TU `Chapter2/Retracts · ScRetraction.tangentMap_tangentMap Chapter1/ChainRule · theorem_1_21`, since Tr maps TU into TU `Chapter2/Retracts · ScRetraction.tangentMap_mapsTo`. Hence the **fixed-point description**: $p \in Tr(TU)$ iff $p \in TU$ and $Tr(p) = p$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff`. Base points of $Tr(TU)$ lie in O_1 `Chapter2/Retracts · ScRetraction.fst_mem_of_mem_tangent`. For $x \in O_1$ the fixed-point condition reads $Dr(x)h = h$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_of_mem`, and the second component of $Tr \circ Tr = Tr$ at (x, h) gives $Dr(x) \circ Dr(x) = Dr(x)$ `Chapter2/Retracts · ScRetraction.scDeriv_comp_self` — the remark after Definition 2.3 that $Dr(x)$ is a projection. So for $x \in O_1$, $(x, h) \in Tr(TU)$ iff h is in the range of $Dr(x)$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_mem_range`, and $Tr(TU) = \{(r(x), Dr(x)h) : x \in U_1, h \in E_0\}$ `Chapter2/Retracts · ScRetraction.mem_tangent_iff_exists`.

Proposition 2.4. *If $r(U) \subseteq r'(U')$ then $r' \circ r = r$ on U*: for $y \in U$, $r(y) \in r'(U')$ is a fixed point of r' `Chapter2/Retracts · ScRetraction.comp_eq_self_of_retract_subset`. *And then $Tr(TU) \subseteq Tr'(TU')$*: let $p \in Tr(TU)$, so $p \in TU$ and $Tr(p) = p$. Its base point lies in $O_1 \subseteq U_1'$, so $p \in TU'$. The map r sends U into U' (into $O \subseteq U'$), so the chain rule applies to $r' \circ r$ on U: $T(r' \circ r)(p) = Tr'(Tr(p)) = Tr'(p)$; and $T(r' \circ r) = Tr$ on TU because $r' \circ r = r$ on U (tangent maps depend only on the values on U). So $Tr'(p) = Tr(p) = p$, which is the fixed-point condition for r' `Chapter2/Retracts · ScRetraction.tangent_subset_of_retract_subset`. Applied in both directions, two retractions with the same retract have the same tangent image `Chapter2/Retracts · proposition_2_4`. This is the text's proof, with its step "$Tr'(x, h) = (x, h)$" made the definition of membership.

Definition 2.5. TO is $Tr(TU)$ for a retraction chosen (by choice) among those onto O, and the empty set if O is not a retract `Chapter2/Retracts · retractTangent`; by Proposition 2.4 it equals $Tr(TU)$ for *every* retraction onto O `Chapter2/Retracts · retractTangent_eq`. The tangent space at $x \in O_1$ is the fibre $T_xO = \{h : (x, h) \in TO\}$ `Chapter2/Retracts · tangentSpace`, the range of $Dr(x)$ `Chapter2/Retracts · tangentSpace_eq_range`, a linear subspace `Chapter2/Retracts · tangentSpace_eq_coe_submodule` (closed, as the range of a continuous projection).

II.3 Sc-smooth maps between retracts (Proposition 2.6, Definition 2.7, Theorem 2.8, Proposition 2.10)

Proposition 2.6. Let r on U and s on V be retractions onto the same O. Since s takes values in O, which r fixes, $(f \circ r) \circ s = f \circ s$ on V `Chapter2/Retracts · ScRetraction.comp_comp_eqOn`. (1) If $f \circ r$ is sc-

smooth on U, then $f \circ r \circ s$ is sc-smooth on V by the chain rule (s maps V into $O \subseteq U$), hence so is $f \circ s$ Chapter2/Retracts · proposition_2_6_1 Chapter1/ChainRule · IsScSmooth.comp. (2) For $p \in TO$: $p \in TV$ and $Ts(p) = p$ (fixed-point description for s, after Proposition 2.4); so $T(f \circ r)(p) = T(f \circ r)(Ts(p)) = T((f \circ r) \circ s)(p) = T(f \circ s)(p)$ Chapter2/Retracts · proposition_2_6_2. (3) If f maps O into $O' = r'(U')$, then $r' \circ f \circ r = f \circ r$ on U; the chain rule gives $Tr'(T(f \circ r)(p)) = T(f \circ r)(p)$ for $p \in TU$, and $T(f \circ r)(p) \in TU'$, so $T(f \circ r)(p) \in TO'$ by the fixed-point description Chapter2/Retracts · proposition_2_6_3.

Definition 2.7. f is *sc-smooth from* (O, C, E) *to* (O', C', E') if $f(O) \subseteq O'$ and $f \circ r$ is sc-smooth on U for every retraction r onto O Chapter2/Retracts · IsScSmoothRetractMap. For a retract O this is equivalent to "$f(O) \subseteq O'$ and $f \circ r$ sc-smooth for one r", by Proposition 2.6 (1) Chapter2/Retracts · isScSmoothRetractMap_iff_exists. Quantifying over all retractions makes the definition independent of choices without further argument; the target triple enters only through $f(O) \subseteq O'$. The *tangent map* Tf is $T(f \circ r)$ for the chosen retraction Chapter2/Retracts · retractTangentMap; on TO it equals $T(f \circ r)$ for every r onto O, by Proposition 2.6 (2) Chapter2/Retracts · retractTangentMap_eqOn, it maps TO into TO' by (3) Chapter2/Retracts · IsScSmoothRetractMap.tangentMap_mapsTo, and $Tf(x, h) = (f(x), D(f \circ r)(x)h)$ Chapter2/Retracts · retractTangentMap_apply. An sc-smooth map between retracts is continuous on O (level 0 of the sc^0 property of $f \circ r$, and $f = f \circ r$ on O) Chapter2/Retracts · IsScSmoothRetractMap.continuousOn and maps points of O on level m to points on level m Chapter2/Retracts · IsScSmoothRetractMap.mem_range_toBase.

Theorem 2.8. Let $O = r(U)$, $O' = r'(U')$ be retracts, f sc-smooth from O to O' and g from O' to O''. Since $f \circ r$ takes values in O', which r' fixes, $(g \circ r') \circ (f \circ r) = (g \circ f) \circ r$ on U Chapter2/Retracts · ScRetraction.comp_comp_comp_eqOn. The left side is sc-smooth by the chain rule ($f \circ r$ maps U into $O' \subseteq U'$, and $g \circ r'$ is sc-smooth on U'), so $g \circ f$ is sc-smooth from O to O'' Chapter2/Retracts · IsScSmoothRetractMap.comp. For the tangent maps, at $p \in TO$: $T(g \circ f)(p) = T((g \circ f) \circ r)(p) = T(g \circ r')(T(f \circ r)(p)) = Tg(Tf(p))$, the last step because $Tf(p) \in TO'$ Chapter2/Retracts · retractTangentMap_comp. Together Chapter2/Retracts · theorem_2_8. The text gives no proof ("follows from Theorem 1.21"); this is the one it has in mind, with $r' \circ f \circ r = f \circ r$ made explicit. Further: the identity is sc-smooth Chapter2/Retracts · isScSmoothRetractMap_id, a retraction is sc-smooth as a map $(U, C, E) \to (O, C, E)$ and the inclusion $O \to U$ as a map $(O, C, E) \to (U, C, E)$ Chapter2/Retracts · ScRetraction.isScSmoothRetractMap Chapter2/Retracts · ScRetraction.isScSmoothRetractMap_inclusion, and sc-smoothness depends only on $f|_O$ Chapter2/Retracts · IsScSmoothRetractMap.congr.

Proposition 2.10. (1) For $W \subseteq E_0$ open, $U \cap r^{-1}(W)$ is relatively open in C because r is continuous on U Chapter2/Retracts · ScRetraction.relOpen_inter_preimage; the restriction of r to it is a retraction Chapter2/Retracts · ScRetraction.restrict (sc-smooth by §II.1) with retract $O \cap W$ Chapter2/Retracts · ScRetraction.restrict_retract. So $(O \cap W, C, E)$ is a retract Chapter2/Retracts · proposition_2_10_1 Chapter2/Retracts · LocalModel.restrict; "an open subset O' of O" is $O \cap W$ in the subspace topology. (2) If $V = r(U)$ and s is sc-smooth from (V, C, E) to itself with $s \circ s = s$ on V, then $\rho = s \circ r$ is sc-smooth on U (that is what sc-smoothness of s on the retract V means), maps U into $s(V) \subseteq V \subseteq U$, and is idempotent: $\rho(\rho(x)) = s(r(s(r(x)))) = s(s(r(x))) = s(r(x))$ because r fixes $s(r(x)) \in V$. Its retract is $s(V)$ Chapter2/Retracts · proposition_2_10_2. As the text's proof.

Restriction and locality between retracts. An sc-smooth map between retracts restricts to $O \cap W$: a retraction onto $O \cap W$ composed with r is handled by the chain rule Chapter2/Retracts · IsScSmoothRetractMap.restrict; the target may be shrunk to any set containing the image Chapter2/Retracts · IsScSmoothRetractMap.mono_target. Conversely, if every point of O has an open W with f sc-smooth on $(O \cap W, C, E)$, then f is sc-smooth on (O, C, E): for a retraction r onto O and $y \in U$, the set W around $r(y)$ pulls back under the continuous r to an open $u \ni y$; on $U \cap u$ the map $f \circ r$ equals $(f \circ r') \circ r$ with r' the restriction of r onto $O \cap W$, which is sc-smooth, and the locality of §II.1 concludes Chapter2/Retracts · IsScSmoothRetractMap.of_locally. The text uses this silently every time it says "the definition does not depend on the choice of charts".

II.4 The tangent functor on retracts

The tangent of a partial quadrant. If $C = T_0^{-1}(\text{quadrant})$ for $T : E \cong \mathbb{R}^n \oplus W$, then $TC = C_1 \times E_0$ is a partial quadrant in $TE = E^1 \oplus E$: compose the tangent of T, an sc-isomorphism $TE \cong T(\mathbb{R}^n \oplus W)$, with the re-association $((a,w),(b,v)) \mapsto (a,(w,(b,v)))$ onto $\mathbb{R}^n \oplus (W^1 \oplus (\mathbb{R}^n \oplus W))$ `Chapter2/TangentBundle · tangentQuadrantIso` `Chapter2/TangentBundle · IsPartialQuadrant.tangentSet`. TU is relatively open in TC `Chapter2/TangentBundle · IsRelOpenInQuadrant.tangentSet` (this is Lemma 3.B of the exposition of Chapter 1; Part I, V1.36). Sc-operators are sc-smooth on every sc-domain `Chapter2/TangentBundle · ScOperator.isScSmooth`.

Tr is a retraction. Tr is sc-smooth on TU (Part I: the tangent map of an sc-smooth map is sc-smooth) `Chapter2/Retracts · LocalModel.tangent_isScSmooth`, maps TU into itself and is idempotent; so it is an sc-smooth retraction on TC `Chapter2/TangentBundle · ScRetraction.tangentRetraction` with retract TO `Chapter2/TangentBundle · ScRetraction.tangentRetraction_retract`. Hence (TO, TC, TE) is a retract `Chapter2/TangentBundle · IsScRetract.tangentSet` `Chapter2/TangentBundle · LocalModel.tangent`, base points of TO lie in O_1 and $(x,0) \in TO$ for $x \in O_1$ `Chapter2/TangentBundle · retractTangent_fst_mem` `Chapter2/TangentBundle · mk_zero_mem_retractTangent`.

Tf is sc-smooth. If f is sc-smooth from O to O' then Tf is sc-smooth from (TO, TC, TE) to (TO', TC', TE'): with the retraction Tr onto TO, $Tf \circ Tr = T(f \circ r) \circ Tr = T(f \circ r \circ r) = T(f \circ r)$ on TU, which is sc-smooth `Chapter2/TangentBundle · IsScSmoothRetractMap.tangent`. Tf depends only on $f|_O$ `Chapter2/TangentBundle · retractTangentMap_congr`, and a map equal to the identity on O has tangent map the identity on TO `Chapter2/TangentBundle · retractTangentMap_id`. With Theorem 2.8 this is the tangent functor $T : \mathcal{R} \to \mathcal{R}$ of the text; the bundle structure ($\mathcal{BR}$, Definition 2.9) is not formalized.

Locality of the tangent. For W open, $T(O \cap W) = TO \cap (W_1 \times E_0)$: with the restricted retraction, a fixed point of Tr over a base point in W_1 is a fixed point of the restricted Tr, the two tangent maps agreeing on the smaller domain (§II.1) `Chapter2/TangentBundle · retractTangent_inter_isOpen` `Chapter2/TangentBundle · retractTangent_mono_inter`. And Tf computed on $(O \cap W, C, E)$ agrees with Tf computed on (O, C, E) on $T(O \cap W)$ `Chapter2/TangentBundle · retractTangentMap_inter_isOpen`. These two facts are what Definitions 2.17–2.18 use silently when the chain rule is applied on a triple overlap.

II.5 Charts, atlases and M-polyfolds (Definitions 2.11–2.14)

Definition 2.11. A *chart* on X is a local model (O, C, E) and a partial equivalence φ between X and E_0 with open source V and target O, continuous on V, with inverse continuous on O `Chapter2/MPolyfold · MChart`. Images of open subsets of V are open subsets $O \cap W$ of O `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq`, hence retracts (Proposition 2.10 (1)) `Chapter2/MPolyfold · MChart.isScRetract_image`, and charts restrict to open subsets of their sources `Chapter2/MPolyfold · MChart.restrict`. The overlap $\varphi(V \cap V')$ `Chapter2/MPolyfold · MChart.overlap` is a retract `Chapter2/MPolyfold · MChart.isScRetract_overlap`, the transition map is $\psi \circ \varphi^{-1}$ `Chapter2/MPolyfold · MChart.transition`, mapping $\varphi(V \cap V')$ into $\psi(V \cap V')$ `Chapter2/MPolyfold · MChart.transition_mapsTo`, and two charts are *compatible* if both transition maps are sc-smooth between the overlap retracts `Chapter2/MPolyfold · MChart.Compatible`. Compatibility is symmetric, reflexive, and passes to restrictions `Chapter2/MPolyfold · MChart.Compatible.symm` `Chapter2/MPolyfold · MChart.compatible_self` `Chapter2/MPolyfold · MChart.Compatible.restrict`.

Compatibility is local. Let φ, ψ be charts and S a family of charts each compatible with φ and with ψ, whose sources cover $V \cap V'$. Then φ and ψ are compatible `Chapter2/MPolyfold · MChart.compatible_of_locally`. By the locality between retracts (§II.3) it suffices to show that $\psi \circ \varphi^{-1}$ is sc-smooth near every point $\varphi(x)$ of the overlap. Take $\chi \in S$ around x and the triple overlap $s = V \cap V' \cap V''$, with $\varphi(s) = \varphi(V \cap V') \cap W$ for an open W. On $\varphi(s)$, $\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1})$ `Chapter2/MPolyfold · MChart.transition_eq_comp_on`; the two factors are sc-smooth on the restricted retracts $\varphi(s)$ and $\chi(s)$, and Theorem 2.8 composes them. The reverse transition is the same argument with the roles exchanged. This is the step that makes "the equivalence class of atlases" a usable notion; the text does not discuss it.

Definition 2.12. An *atlas* is a set of pairwise compatible charts covering X `Chapter2/MPolyfold · MAtlas`; two atlases are *equivalent* if their union is an atlas `Chapter2/MPolyfold · MAtlas.Equiv`, i.e. if every chart of one is compatible with every chart of the other `Chapter2/MPolyfold · MAtlas.equiv_iff`. The relation is reflexive and symmetric `Chapter2/MPolyfold · MAtlas.Equiv.refl` `Chapter2/MPolyfold · MAtlas.Equiv.symm`.

Definition 2.13. An *M-polyfold* structure on a Hausdorff paracompact space is an atlas `Chapter2/MPolyfold · MPolyfold`. Its *charts* are all charts compatible with every chart of the atlas `Chapter2/MPolyfold · MPolyfold.IsChart` `Chapter2/MPolyfold · MPolyfold.isChart_of_mem`. By locality of compatibility (with S the atlas), any two of them are compatible `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`; restrictions of charts are charts `Chapter2/MPolyfold · MPolyfold.IsChart.restrict`; every point lies in one `Chapter2/MPolyfold · MPolyfold.exists_isChart_mem_source`; and equivalent atlases have the same charts, again by locality `Chapter2/MPolyfold · MPolyfold.isChart_congr`. So the maximal atlas represents the equivalence class, and every later definition (levels, index, tangent bundle) quantifies over it.

Definition 2.14. $f : X \to Y$ is *sc-smooth on the open set* U if it is continuous on U and for all charts φ of X, ψ of Y and every open $V \subseteq U \cap \operatorname{dom}\varphi$ with $f(V) \subseteq \operatorname{dom}\psi$, the representation $\psi \circ f \circ \varphi^{-1}$ is sc-smooth from $(\varphi(V), C, E)$ into (O', C', E') `Chapter2/MPolyfold · MPolyfold.IsScSmoothOn`; *sc-smooth* means sc-smooth on X `Chapter2/MPolyfold · MPolyfold.IsScSmoothMap`. Continuity is a field: the text's pairs of charts with $f(V) \subseteq V'$ exist around every point only for continuous f. The notion passes to open subsets and to maps agreeing on U `Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.mono` `Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.congr`. *One pair of charts suffices*: if every $x \in U$ has charts φ, ψ and an open $V \ni x$ with $\psi \circ f \circ \varphi^{-1}$ sc-smooth on $\varphi(V)$, then f is sc-smooth on U `Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts`. For arbitrary charts φ', ψ' and V', locality reduces to a neighbourhood $\varphi'(V \cap V')$ of each point, where $\psi' \circ f \circ \varphi'^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ \varphi'^{-1})$, three sc-smooth maps between retracts composed by Theorem 2.8. This is the text's "the definition does not depend on the choice of charts". The identity is sc-smooth `Chapter2/MPolyfold · MPolyfold.isScSmoothMap_id`, and compositions are sc-smooth by the same three-factor argument with a chart of the middle space `Chapter2/MPolyfold · MPolyfold.IsScSmoothOn.comp` `Chapter2/MPolyfold · MPolyfold.IsScSmoothMap.comp`.

II.6 The filtration X_m and the M-polyfold X^1 (Lemma 2.15)

Levels. x is on level m if $\varphi(x) \in \iota_m(E_m)$ for some chart φ around x `Chapter2/MPolyfold · MPolyfold.levelSet`. *Independence of the chart*: if ψ is another chart around x, the transition map $\varphi \circ \psi^{-1}$ is sc-smooth between the overlap retracts, hence maps points on level m to points on level m (§II.3), and $\varphi(x) = (\varphi \circ \psi^{-1})(\psi(x))$ `Chapter2/MPolyfold · MPolyfold.mem_levelSet_iff`. $X_0 = X$, $X_{m+1} \subseteq X_m$, and $X_\infty = \bigcap_m X_m$ `Chapter2/MPolyfold · MPolyfold.levelSet_zero` `Chapter2/MPolyfold · MPolyfold.levelSet_succ_subset` `Chapter2/MPolyfold · MPolyfold.smoothSet`. For $x \in V_m$ the lift of $\varphi(x)$ to E_m `Chapter2/MPolyfold · MPolyfold.levelLift` satisfies $\iota_m(\text{lift}) = \varphi(x)$ and is compatible with the inclusions `Chapter2/MPolyfold · MPolyfold.toBase_levelLift` `Chapter2/MPolyfold · MPolyfold.levelLift_eq_incl`. X_m carries the topology generated by the sets $\{x \in V : \text{lift}(x) \in W\}$, W open in E_m `Chapter2/MPolyfold · MPolyfold.levelBasis` `Chapter2/MPolyfold · MPolyfold.levelTopology`; that this family is a basis is not needed.

Lemma 2.15. The preimage under $X_{m+1} \to X_m$ `Chapter2/MPolyfold · MPolyfold.levelIncl` of a generating set $\{\text{lift}_m \in W\}$ is the generating set $\{\text{lift}_{m+1} \in \text{incl}_m^{-1}(W)\}$, by the compatibility of lifts with inclusions; so the inclusion is continuous `Chapter2/MPolyfold · lemma_2_15`. This is the text's proof.

The space X_m. As a type, X_m is the subtype of points on level m with this topology `Chapter2/LevelPolyfold · MPolyfold.Level`. The inclusion $X_m \to X$ is continuous: for V open in X and $x \in X_m$, a chart φ around x gives $\varphi(V \cap \operatorname{dom}\varphi) = O \cap W$, and the generating set $\{\text{lift}_m \in W_m\}$ is a neighbourhood of x inside the preimage of V `Chapter2/LevelPolyfold · MPolyfold.continuous_levelVal` `Chapter2/LevelPolyfold · MPolyfold.isOpen_level_basic`. Since this inclusion is injective and continuous into the Hausdorff X, X_m is Hausdorff (an instance in `LevelPolyfold`, the text's argument with disjoint neighbourhoods).

The level shift of the local data. An sc-isomorphism shifts to $E^1 \to F^1$ `Chapter2/LevelPolyfold · ScIso.shift`, so C_1 is a partial quadrant in E^1 and U_1 relatively open in it `Chapter2/LevelPolyfold · IsPartialQuadrant.shift` `Chapter2/LevelPolyfold · IsRelOpenInQuadrant.shift`. A retraction r induces the retraction r_1 on U_1 — sc-smooth by Proposition 1.17 (Part I, §I.15), idempotent by the injectivity of ι_1 `Chapter2/LevelPolyfold · ScRetraction.shift` — with retract O_1 `Chapter2/LevelPolyfold · ScRetraction.shift_retract` `Chapter2/LevelPolyfold · IsScRetract.shift` `Chapter2/LevelPolyfold · LocalModel.shift`. An sc-smooth map between retracts is sc^0 on O, maps O_m into O'_m and is continuous there `Chapter2/LevelPolyfold · IsScSmoothRetractMap.isSc0` `Chapter2/LevelPolyfold · IsScSmoothRetractMap.mapsTo_levelMap` `Chapter2/LevelPolyfold · IsScSmoothRetractMap.continuousOn_levelMap`, and its level-1 map is sc-smooth from O_1 to O'_1: with the retraction r_1, $f_1 \circ r_1 = (f \circ r)_1$, sc-smooth by Proposition 1.17 `Chapter2/LevelPolyfold · IsScSmoothRetractMap.shift`.

The M-polyfold X^1. For a chart φ of X, $x \mapsto \mathrm{lift}_1(x)$ on V_1 with model (O_1, C_1, E^1) is a chart of X_1 `Chapter2/LevelPolyfold · MPolyfold.levelChart` (with a junk value for the inverse off O_1, since X_1 might be empty `Chapter2/LevelPolyfold · MPolyfold.levelChartInv`); its overlap with another level chart is the level-1 part of the overlap `Chapter2/LevelPolyfold · MPolyfold.levelChart_overlap`, its transition map the level-1 map of the transition map `Chapter2/LevelPolyfold · MPolyfold.levelChart_transition`, and so any two level charts are compatible by the level shift of sc-smooth maps `Chapter2/LevelPolyfold · MPolyfold.levelChart_compatible`. They form an atlas `Chapter2/LevelPolyfold · MPolyfold.levelAtlas`, and X_1 with it is the M-polyfold X^1 `Chapter2/LevelPolyfold · MPolyfold.shift` `Chapter2/LevelPolyfold · MPolyfold.shift_isChart`. Its paracompactness, an instance argument, is supplied by Theorem 2.16 (§II.7). Only $m = 1$ is constructed.

II.7 Metrizability: Smirnov's theorem and Theorem 2.16 (Theorem 2.55)

The text proves Theorem 2.16 with Smirnov's theorem (2.55), Michael's characterization of paracompactness (Lemma 2.56) and Proposition 2.57 (a regular space with a locally finite closed cover by paracompact subspaces is paracompact). Proposition 2.57 needs X_m to be regular, which the text does not check. The formal proof needs neither 2.56 nor 2.57.

The metrization lemma. Let Y have open subspaces V_i, each metrizable with a metric d_i, and continuous $\rho_i : Y \to [0, \infty)$ with $\operatorname{supp} \rho_i \subseteq V_i$, the supports locally finite and $\rho_i(y) > 0$ for some i at every y `Chapter2/Metrizable · metrizableSpace_of_partition`. Put $\delta_i(x, y) := \min(d_i(x, y), 1)$ for $x, y \in V_i$ and 1 otherwise `Chapter2/Metrizable · truncDist` — nonnegative, at most 1, symmetric, zero only on the diagonal, and satisfying the triangle inequality inside V_i `Chapter2/Metrizable · truncDist_nonneg` `Chapter2/Metrizable · truncDist_le_one` `Chapter2/Metrizable · truncDist_triangle` `Chapter2/Metrizable · truncDist_eq_zero` — and the *cone metric*

$$e_i(x, y) := |\rho_i(x) - \rho_i(y)| + \min(\rho_i(x), \rho_i(y))\, \delta_i(x, y), \qquad D(x, y) := \sum_i e_i(x, y).$$

The sum is finite (only the i with x or y in $\operatorname{supp} \rho_i$ contribute). $D(x, x) = 0$ and D is symmetric. *Triangle inequality:* termwise, by an elementary inequality for $a, b, c \geq 0$ and δ's bounded by 1, which needs the triangle inequality for δ_i only when $\min(\rho_i(x), \rho_i(z)) > 0$ and $\rho_i(y) > 0$, i.e. when all three points lie in V_i `Chapter2/Metrizable · cone_triangle`. *Separation:* if $D(x, y) = 0$, take i with $\rho_i(x) > 0$; then $\rho_i(y) = \rho_i(x)$ and $\delta_i(x, y) = 0$, so $x = y$. *Topology:* $y \mapsto D(x, y)$ is continuous (locally a finite sum of continuous functions; the term $\min(\rho_i(x), \rho_i(y))\delta_i(x, y)$ is continuous where $y \in V_i$ and vanishes near points outside $\operatorname{supp} \rho_i$), so D-balls are open; conversely, if s is open and $x \in s$, take i with $\rho_i(x) > 0$ and a d_i-ball of radius $r \leq 1$ around x inside s; then $D(x, y) < \frac{1}{2}\rho_i(x) r$ forces $\rho_i(y) > \frac{1}{2}\rho_i(x)$ (so $y \in V_i$) and $\frac{1}{2}\rho_i(x)\delta_i(x, y) < \frac{1}{2}\rho_i(x) r$, so $d_i(x, y) < r$ and $y \in s$. Mathlib then builds a metric space structure with this distance and the given topology `Mathlib/Topology/MetricSpace/Defs · MetricSpace.ofDistTopology`.

Pulled back, and Smirnov. If $\pi : Y \to X$ is continuous into a paracompact Hausdorff X and every point of X has an open U with $\pi^{-1}(U)$ metrizable, then Y is metrizable: a partition of unity of X

subordinate to these U Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate, composed with π, satisfies the hypotheses of the lemma with $V_i = \pi^{-1}(U_i)$ Chapter2/Metrizable · metrizableSpace_of_pullback. With $\pi = \mathrm{id}$ this is **Smirnov's theorem** (Theorem 2.55) Chapter2/Metrizable · metrizableSpace_of_locally. Metrizable spaces are paracompact (Stone; in Mathlib through the instance for pseudo-emetric spaces) Chapter2/Metrizable · paracompactSpace_of_metrizableSpace.

Theorem 2.16. *X is metrizable*: chart domains are homeomorphic to subsets O of the Banach space E_0 Chapter2/Metrizable · MChart.sourceHomeomorph Chapter2/Metrizable · MChart.metrizableSpace_source, and Smirnov applies Chapter2/Metrizable · MPolyfold.metrizableSpace. $V_m \cong O_m$: the lift $x \mapsto \mathrm{lift}_m(x)$ is a topological embedding of $V \cap X_m$ into E_m Chapter2/Metrizable · MPolyfold.isEmbedding_levelLift. Its injectivity is that of φ; the induced topology is finer than that of X_m because generating sets of X_m inside V are preimages of open sets; it is coarser because a generating set $\{\mathrm{lift}_m^{\psi} \in W\}$ of another chart, intersected with V, is the preimage under $\mathrm{lift}_m^{\varphi}$ of the open set $\{z \in \varphi(V \cap V')_m : (\psi \circ \varphi^{-1})_m(z) \in W\}$, open since the level-$m$ map of the transition is continuous on the level-m part of the overlap retract. *X_m is metrizable*: apply the pulled-back Smirnov to the continuous inclusion $X_m \to X$ and the chart domains V, whose preimages $V \cap X_m$ embed in E_m Chapter2/Metrizable · MPolyfold.metrizableSpace_level; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_level. X_∞: the text does not define its topology; it is taken to be the initial topology of the inclusions $X_\infty \to X_m$ Chapter2/Metrizable · MPolyfold.SmoothLevel Chapter2/Metrizable · MPolyfold.smoothIncl Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq. It embeds in the countable product $\prod_m X_m$, which is metrizable Chapter2/Metrizable · metrizableSpace_pi_nat, so X_∞ is metrizable Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel. Together Chapter2/Metrizable · theorem_2_16. *No hypothesis is added.* As a by-product every subset of an M-polyfold is metrizable, hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype, which discharges the paracompactness of sub-M-polyfolds (§II.10).

II.8 The tangent bundle TX (Definitions 2.17–2.18, Proposition 2.19)

Tuples and transport. A tangent tuple is (x, φ, h) with $x \in X_1$, φ a chart of the structure around x and $(\mathrm{lift}_1 x, h) \in TO$ Chapter2/TangentBundle · MPolyfold.TangentPre; its vector is $(\mathrm{lift}_1 x, h)$ Chapter2/TangentBundle · MPolyfold.TangentPre.vec Chapter2/TangentBundle · MPolyfold.TangentPre.vec_mem. For a vector of TO over a point of an open $\varphi(s)$, membership in TO and in $T(\varphi(s))$ agree (locality of the tangent, §II.4) Chapter2/TangentBundle · MPolyfold.TangentPre.mem_retractTangent_image_iff. The transport of the tuple to a chart ψ around x is $T(\psi \circ \varphi^{-1})(\mathrm{vec})$, the tangent map computed on the overlap retract Chapter2/TangentBundle · MPolyfold.TangentPre.transport. Transport to φ itself does nothing, by $T(\mathrm{id}) = \mathrm{id}$ on the overlap $\varphi(V)$ Chapter2/TangentBundle · MPolyfold.TangentPre.transport_self Chapter2/TangentBundle · MChart.transition_self_eqOn; the transported vector lies in TO' Chapter2/TangentBundle · MPolyfold.TangentPre.transport_mem and has base point $\mathrm{lift}_1^{\psi}(x)$ Chapter2/TangentBundle · MPolyfold.TangentPre.transport_fst.

The chain rule for transports. Transporting to ψ and then by $T(\chi \circ \psi^{-1})$ is transporting to χ Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport. On the triple overlap $s = V \cap V' \cap V''$ the transition maps are sc-smooth from $\varphi(s)$ to $\psi(s)$ and from $\psi(s)$ to $\chi(s)$ Chapter2/TangentBundle · MChart.Compatible.transition_image Chapter2/TangentBundle · MChart.image_eq_overlap_inter, $\chi \circ \varphi^{-1} = (\chi \circ \psi^{-1}) \circ (\psi \circ \varphi^{-1})$ there, and Theorem 2.8 gives $T(\chi \circ \varphi^{-1}) = T(\chi \circ \psi^{-1}) \circ T(\psi \circ \varphi^{-1})$ on $T(\varphi(s))$. The three tangent maps here are computed on $\varphi(s)$ and $\psi(s)$, the transports on the full overlaps; the locality of Tf (§II.4) identifies them. The text never checks this.

Definition 2.17. Two tuples are *equivalent* if their points agree and the transport of the first to the chart of the second is the second Chapter2/TangentBundle · MPolyfold.TangentPre.Rel. Equivalent tuples transport to the same vector in every chart (chain rule for transports) Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.transport; hence the relation is reflexive, symmetric and transitive Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.refl Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.symm Chapter2/TangentBundle · MPolyfold.TangentPre.Rel.trans, an equivalence relation Chapter2/TangentBundle · MPolyfold.TangentPre.setoid.

Definition 2.18. TX is the quotient Chapter2/TangentBundle · MPolyfold.TangentSpace Chapter2/TangentBundle · MPolyfold.TangentSpace.mk, with $p : TX \to X^1$, $[(x,\varphi,h)] \mapsto x$ Chapter2/TangentBundle · MPolyfold.tangentProj, and $T_xX = p^{-1}(x)$ Chapter2/TangentBundle · MPolyfold.tangentFiber. For a chart φ: $TV = p^{-1}(V_1)$ Chapter2/TangentBundle · MPolyfold.tangentChartSet and $T\varphi : TV \to TO$ sends a class to the transport of a representative to φ, well defined by the previous paragraph Chapter2/TangentBundle · MPolyfold.tangentChartMap Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk; $T\varphi[(x,\varphi,h)] = (\varphi(x),h)$ Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_self, $T\varphi$ lands in TO over the lift of the base point Chapter2/TangentBundle · MPolyfold.tangentChartMap_mem Chapter2/TangentBundle · MPolyfold.tangentChartMap_fst, and it is injective on TV: equal images have equal base points (injectivity of φ and ι_1) and then the representatives are equivalent Chapter2/TangentBundle · MPolyfold.tangentChartMap_injOn. Its inverse on TO is $a \mapsto [(\varphi^{-1}(\iota_1 a_1), \varphi, a_2)]$ Chapter2/TangentBundle · MPolyfold.TangentPre.ofVec Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec Chapter2/TangentBundle · MPolyfold.mk_ofVec_tangentChartMap, and in another chart $T\psi \circ (T\varphi)^{-1} = T(\psi \circ \varphi^{-1})$ on $T(\varphi(V \cap V'))$ Chapter2/TangentBundle · MPolyfold.tangentChartMap_mk_ofVec_of_mem Chapter2/TangentBundle · MPolyfold.mk_ofVec_mem_tangentChartSet_iff. The topology is generated by the sets $\{\alpha \in TV : T\varphi(\alpha) \in W\}$, W open Chapter2/TangentBundle · MPolyfold.tangentBasis Chapter2/TangentBundle · MPolyfold.isOpen_tangent_basic; TV itself is open Chapter2/TangentBundle · MPolyfold.isOpen_tangentChartSet. The vector space structure of T_xX is not formalized; T_xX is identified with $T_{\varphi(x)}O$ by $T\varphi$.

Proposition 2.19. (2) *p is continuous*: the preimage of a generating set $\{\mathrm{lift}_1^{\psi} \in W\}$ of X_1 is the generating set of TX given by ψ and $W \times E_0$ Chapter2/TangentBundle · MPolyfold.continuous_tangentProj; so is $TX \to X$ Chapter2/TangentBundle · MPolyfold.continuous_levelVal_tangentProj. Openness of p is not proved. (1) *TX is Hausdorff* (the instance `T2Space` in `TangentBundle`): two classes over the same point lie in one TV and have distinct images under the injective $T\varphi$, separated by open sets of $E_1 \times E_0$ whose preimages are basic; classes over distinct points are separated by preimages of disjoint open sets of X under the continuous $TX \to X$. (3) *TX is metrizable*: apply the pulled-back Smirnov theorem to $TX \to X$ and the chart domains V, whose preimages TV are homeomorphic to subsets of $E_1 \times E_0$ through the charts of §II.8's next paragraph Chapter2/Metrizable · MPolyfold.metrizableSpace_tangentSpace Chapter2/Metrizable · MChart.metrizableSpace_source; hence paracompact Chapter2/Metrizable · MPolyfold.paracompactSpace_tangentSpace Chapter2/Metrizable · proposition_2_19_3.

TX as an M-polyfold. For a chart φ of X, $(TV, T\varphi, (TO, TC, TE))$ is a chart of TX Chapter2/TangentBundle · MPolyfold.tangentChart Chapter2/TangentBundle · MPolyfold.tangentChartInv; its overlap with another is $T(\varphi(V \cap V'))$ Chapter2/TangentBundle · MPolyfold.tangentChart_overlap and its transition map is $T(\psi \circ \varphi^{-1})$ — formula (5) Chapter2/TangentBundle · MPolyfold.tangentChart_transition — which is sc-smooth between the tangent retracts by the tangent functor (§II.4); so the charts are compatible Chapter2/TangentBundle · MPolyfold.tangentChart_compatible, form an atlas Chapter2/TangentBundle · MPolyfold.tangentAtlas and make TX an M-polyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold Chapter2/TangentBundle · MPolyfold.tangentPolyfold_isChart. *$p : TX \to X^1$ is sc-smooth*: in the charts $T\varphi$ and the level chart of φ it is the projection $TO \to O_1$, $(a,h) \mapsto a$, which is sc-smooth between the retracts (TO, TC, TE) and (O_1, C_1, E^1) because, with the retraction Tr onto TO, $\mathrm{pr}_1 \circ Tr = r_1 \circ \mathrm{pr}_1$ on TU, a composite of the sc-operator pr_1 Chapter2/TangentBundle · ScBanach.tangentFst and the sc-smooth r_1 Chapter2/TangentBundle · MPolyfold.isScSmoothRetractMap_tangent_fst; one pair of charts per point suffices (§II.5) Chapter2/TangentBundle · MPolyfold.tangentProj_isScSmooth.

II.9 The tangent map of an sc-smooth map between M-polyfolds

The text uses Tf for maps between M-polyfolds from §2.3 on without defining it. *The chain rule on chart images, with locality*: for F sc-smooth on $\varphi(s_1)$ into a retract O_1 mapping $\varphi(t)$ into O_1', G sc-smooth on O_1' and H sc-smooth on $\varphi(s_2)$ with $H = G \circ F$ on $\varphi(t)$ ($t \subseteq s_1 \cap s_2$ open), $TH = TG \circ TF$ on $T(\varphi(t))$, the three tangent maps computed on their own domains Chapter2/TangentMap · MChart.retractTangentMap_comp_image — Theorem 2.8 on $\varphi(t)$ plus the locality of Tf. *Levels*: an sc-smooth map maps X_m into

Y_m, because its coordinate representations do Chapter2/TangentMap · MPolyfold.IsScSmoothOn.mapsTo_levelSet Chapter2/TangentMap · MPolyfold.IsScSmoothMap.mapsTo_levelSet. *The image vector*: for a tuple (x, φ, h) and a chart ψ around $f(x)$, $T(\psi \circ f \circ \varphi^{-1})(\mathrm{vec})$, computed on $\varphi(V \cap f^{-1}(V'))$ Chapter2/TangentMap · MPolyfold.TangentPre.fdom Chapter2/TangentMap · MPolyfold.TangentPre.ftransport, lies in TO' over $\psi(f(x))$ Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_mem Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_fst; changing ψ changes it by T of the transition map Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_transport, changing φ by transport Chapter2/TangentMap · MPolyfold.TangentPre.ftransport_eq_of_transport, and equivalent tuples give the same image Chapter2/TangentMap · MPolyfold.TangentPre.Rel.ftransport — all by the chain rule with locality. The image tuple $(f(x), \psi, \cdot)$ Chapter2/TangentMap · MPolyfold.TangentPre.fimageWith therefore has a class independent of ψ and of the representative Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq Chapter2/TangentMap · MPolyfold.TangentPre.mk_fimageWith_eq_of_rel.

Tf. With a chosen ψ, $Tf[(x, \varphi, h)]$ is the class of the image tuple Chapter2/TangentMap · MPolyfold.tangentMap, computable in any chart around $f(x)$ Chapter2/TangentMap · MPolyfold.tangentMap_mk. It covers f Chapter2/TangentMap · MPolyfold.levelVal_tangentProj_tangentMap, maps T_xX into $T_{f(x)}Y$ Chapter2/TangentMap · MPolyfold.tangentMap_mapsTo_fiber, and in charts $T\psi \circ Tf = T(\psi \circ f \circ \varphi^{-1}) \circ T\varphi$ Chapter2/TangentMap · MPolyfold.tangentChartMap_tangentMap. $T(\mathrm{id}) = \mathrm{id}$ Chapter2/TangentMap · MPolyfold.tangentMap_id, and **the chain rule** $T(g \circ f) = Tg \circ Tf$ holds: with charts ψ around $f(x)$ and χ around $g(f(x))$ and the open set $t = \mathrm{fdom}(\varphi, f, \psi) \cap f^{-1}(\mathrm{fdom}(\psi, g, \chi))$, the chain rule with locality gives $T(\chi \circ g \circ f \circ \varphi^{-1}) = T(\chi \circ g \circ \psi^{-1}) \circ T(\psi \circ f \circ \varphi^{-1})$ on $T(\varphi(t))$ Chapter2/TangentMap · MPolyfold.tangentMap_comp. Tf is continuous: near a class in TV, the preimage of a generating set of TY contains the generating set of TX given by φ and the preimage under the continuous $T(\psi \circ f \circ \varphi^{-1})$ Chapter2/TangentMap · MPolyfold.continuous_tangentMap; and Tf is sc-smooth $TX \to TY$, its representation in the charts $T\varphi$, $T\psi$ being $T(\psi \circ f \circ \varphi^{-1})$, sc-smooth by the tangent functor Chapter2/TangentMap · MPolyfold.tangentMap_isScSmooth.

II.10 Sub-M-polyfolds (Definition 2.20, Proposition 2.21)

Definition 2.20. $A \subseteq X$ is a *sub-M-polyfold* if every $a \in A$ has an open $V \ni a$ and a map r with $r(V) \subseteq V$, sc-smooth on V, idempotent on V, with $r(V) = A \cap V$ Chapter2/MPolyfold · MPolyfold.IsSubMPolyfold; X is one Chapter2/MPolyfold · MPolyfold.isSubMPolyfold_univ. The data at a point are packaged Chapter2/SubMPolyfold · MPolyfold.SubData Chapter2/SubMPolyfold · MPolyfold.IsSubMPolyfold.data: r maps V into A, fixes $A \cap V$, and is continuous on V Chapter2/SubMPolyfold · MPolyfold.SubData.apply_mem Chapter2/SubMPolyfold · MPolyfold.SubData.apply_eq_self Chapter2/SubMPolyfold · MPolyfold.SubData.continuousOn.

Proposition 2.21, the structure. *Shrinking.* For a chart φ, put $W := (V \cap \mathrm{dom}\,\varphi) \cap r^{-1}(V \cap \mathrm{dom}\,\varphi)$ Chapter2/SubMPolyfold · MPolyfold.SubData.shrink: open (continuity of r) Chapter2/SubMPolyfold · MPolyfold.SubData.isOpen_shrink, containing every $a \in A \cap V \cap \mathrm{dom}\,\varphi$ Chapter2/SubMPolyfold · MPolyfold.SubData.mem_shrink, with $r(W) \subseteq W$ and $r(W) = W \cap A$ Chapter2/SubMPolyfold · MPolyfold.SubData.mapsTo_shrink Chapter2/SubMPolyfold · MPolyfold.SubData.image_shrink. *The retract.* $\rho := \varphi \circ r \circ \varphi^{-1}$ is sc-smooth from $\varphi(W)$ to itself — Definition 2.14 for r in the charts φ, φ — idempotent, and $\rho(\varphi(W)) = \varphi(W \cap A)$ Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image; so $\varphi(W \cap A)$ is a retract by Proposition 2.10 (2) Chapter2/SubMPolyfold · MPolyfold.SubData.isScRetract_image. *The chart.* With the subspace topology on A, φ restricted to $W \cap A$ is a chart of A with model $(\varphi(W \cap A), C, E)$ Chapter2/SubMPolyfold · MPolyfold.subChart Chapter2/SubMPolyfold · MPolyfold.subChartInv. *Compatibility.* The overlap of two such charts is $\varphi((W \cap W') \cap A)$, a subset of $\varphi(V \cap V')$ Chapter2/SubMPolyfold · MPolyfold.subChart_overlap. An sc-smooth map between retracts is sc-smooth on every sub-retract $O_1 \subseteq O$: for a retraction s onto O_1, $f \circ s = (f \circ r) \circ s$, sc-smooth by the chain rule Chapter2/SubMPolyfold · IsScSmoothRetractMap.mono_source — this is the text's "$(\psi' \circ \psi^{-1}) \circ s'' = (\varphi' \circ \varphi^{-1}) \circ s''$". The transition maps of the sub-charts are restrictions of those of X, hence sc-smooth Chapter2/SubMPolyfold · MPolyfold.subChart_compatible. They form an atlas Chapter2/SubMPolyfold · MPolyfold.subAtlas, and A is an M-polyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart. The structure requires A to be paracompact (an instance argument); it is, as a subset of the metrizable X (§II.7), and for closed A also elementarily Chapter2/

SubMPolyfold · MPolyfold.paracompactSpace_of_isClosed.

(1) and (2). The inclusion $A \to X$ is sc-smooth: in the sub-chart and φ it is the identity of $\varphi(W \cap A)$ into O; one pair of charts per point suffices Chapter2/SubMPolyfold · MPolyfold.subPolyfold_inclusion_isScSmooth; it is a homeomorphism onto its image by the choice of topology. For any data (V, r), the map $i^{-1} \circ r$ (a total map with a junk value off V) Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto Chapter2/SubMPolyfold · MPolyfold.SubData.retractInto_val is sc-smooth on V: near $x \in V$ take a chart φ' around x, a chart φ around $r(x)$ and $V_0 = V \cap \operatorname{dom} \varphi' \cap r^{-1}(\operatorname{dom} \varphi)$; on $\varphi'(V_0)$ the representation in φ' and the sub-chart on φ is $\varphi \circ r \circ \varphi'^{-1}$, sc-smooth by Definition 2.14 for r Chapter2/SubMPolyfold · MPolyfold.subPolyfold_retraction_isScSmooth.

Proposition 2.21 (3), (4), in a chart. Let r be a retraction onto O, s sc-smooth from (O, C, E) to itself with $s \circ s = s$, and $t := s \circ r$, a retraction onto $s(O)$ (Proposition 2.10 (2)) Chapter2/Metrizable · ScRetraction.compRetraction Chapter2/Metrizable · ScRetraction.compRetraction_retract. For $a \in s(O)_1$ put $P := Dr(a)$, $Q := Dt(a) = D(s \circ r)(a)$ — the text's "$Ds(a)$". Both are idempotent (§II.2). From $r \circ t = t$ and $t \circ r = t$ on U (s takes values in O, which r fixes; and $r \circ r = r$), the chain rule at the point a Chapter2/DegeneracyIndex · scDeriv_comp_apply gives $PQ = Q$ and $QP = Q$. If a is smooth, P and Q are sc-operators (Part I, Proposition 1.16), and so is $P - Q$; identities between base maps of sc-operators hold on every level Chapter2/Metrizable · ScOperator.lvl_comp_of_base Chapter2/Metrizable · ScOperator.lvl_comp_eq_zero_of_base Chapter2/Metrizable · ScOperator.lvl_eq_add_of_base. Then Q and $P - Q$ are commuting idempotent sc-operators with $Q(P - Q) = 0$, their ranges are sc-subspaces (Part I, §I.3: sc-splittings and sc-projections Chapter1/ExpositionLemmas · ScBanach.isScSplitting_of_idem), and on every level the level parts of QE and $(P - Q)E$ are disjoint with sum the level part of PE. Since $T_aO = PE$ and $T_a s(O) = QE$ (§II.2), this is the sc-splitting $T_aO = T_aA \oplus (Dr(a) - Dt(a))E$ Chapter2/Metrizable · proposition_2_21_3. For (4), s is sc-smooth from O onto $s(O)$ (shrinking the target), and $Ts(a, h) = (a, Qh)$ for $h \in T_aO$ because $Qh = QPh$; as Q maps PE onto $QE = Q(PE)$, $Ts(a)$ maps T_aO onto $T_a s(O)$ Chapter2/Metrizable · proposition_2_21_4. The transport to T_aX is not formalized.

II.11 The degeneracy index of an M-polyfold (Definitions 2.22, 2.25; Lemmas 2.23, 2.26; Proposition 2.24)

A retract is contained in its quadrant, and chart values lie in C Chapter2/DegeneracyIndex · LocalModel.O_subset_C Chapter2/DegeneracyIndex · MChart.apply_mem_C Chapter2/DegeneracyIndex · LocalModel.isPartialQuadrant. **Definition 2.22.** $d(x, \varphi) := d_C(\varphi(x))$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex, and $d_X(x)$ is the least $d(x, \varphi)$ over the charts of the structure around x Chapter2/DegeneracyIndex · MPolyfold.degIndex; $d_X(x) \le d(x, \varphi)$ for every such chart and the minimum is attained Chapter2/DegeneracyIndex · MPolyfold.degIndex_le Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq. **Lemma 2.23.** d_C is upper semicontinuous: coordinates positive at x stay positive nearby, so the vanishing set can only shrink Chapter2/DegeneracyIndex · eventually_degIndexWith_le; composed with the continuous chart, $d(\cdot, \varphi)$ is upper semicontinuous on V Chapter2/DegeneracyIndex · MPolyfold.eventually_chartDegIndex_le, and with a chart realizing $d_X(x)$, $d_X(y) \le d(y, \varphi) \le d(x, \varphi) = d_X(x)$ near x Chapter2/DegeneracyIndex · lemma_2_23. Hence $\{d_X < k\}$ is open and $\{d_X \ge k\}$ closed Chapter2/Boundary · MPolyfold.isOpen_setOf_degIndex_lt Chapter2/Boundary · MPolyfold.isClosed_setOf_le_degIndex.

Proposition 2.24. An sc-diffeomorphism of open sets is f sc-smooth on U, g sc-smooth on V, mapping into each other and mutually inverse Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn Chapter2/DegeneracyIndex · MPolyfold.IsScDiffeoOn.symm. For a chart ψ of Y, $\psi \circ f$ on $U \cap f^{-1}(\operatorname{dom} \psi)$, with inverse $g \circ \psi^{-1}$, is a chart of X Chapter2/DegeneracyIndex · MPolyfold.transportChart; it is compatible with every chart φ of the atlas of X, because its transition maps with φ are $\psi \circ f \circ \varphi^{-1}$ and $\varphi \circ g \circ \psi^{-1}$, sc-smooth by Definition 2.14 for f and for g Chapter2/DegeneracyIndex · MPolyfold.transportChart_isChart. Its index at x is $d(f(x), \psi)$ Chapter2/DegeneracyIndex · MPolyfold.chartDegIndex_transportChart. With ψ realizing $d_Y(f(x))$: $d_X(x) \le d_Y(f(x))$ Chapter2/DegeneracyIndex · MPolyfold.degIndex_le_of_isScDiffeoOn, and with g the reverse Chapter2/DegeneracyIndex · proposition_2_24. The text's "from the definitions one deduces immediately" is this construction.

Definition 2.25. $\partial X = \{d_X \geq 1\}$ Chapter2/DegeneracyIndex · MPolyfold.boundary; without boundary means $d_X \equiv 0$, equivalently $\partial X = \emptyset$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary Chapter2/DegeneracyIndex · MPolyfold.isWithoutBoundary_iff. ∂X and the corners $\{d_X \geq 2\}$ are closed, the interior $\{d_X = 0\}$ is open Chapter2/Boundary · MPolyfold.isClosed_boundary Chapter2/Boundary · MPolyfold.corners Chapter2/Boundary · MPolyfold.isClosed_corners Chapter2/Boundary · MPolyfold.isOpen_interior, and sc-diffeomorphisms preserve ∂X by Proposition 2.24 Chapter2/Boundary · MPolyfold.IsScDiffeoOn.mem_boundary_iff.

Lemma 2.26. For a sub-M-polyfold A and $a \in A$: take a chart φ of X realizing $d_X(a)$ and the data at a; the sub-chart built on φ has the same quadrant C and the same value $\varphi(a)$, so $d_A(a) \leq d_C(\varphi(a)) = d_X(a)$ Chapter2/SubMPolyfold · lemma_2_26.

A retract as an M-polyfold. A nonempty retract O, as a subtype, has the single chart id Chapter2/MPolyfold · LocalModel.selfChart, compatible with itself, and is an M-polyfold Chapter2/MPolyfold · LocalModel.toMPolyfold (Hausdorff and paracompact as a subspace of E_0); so is a nonempty relatively open $U \subseteq C$ Chapter2/MPolyfold · IsRelOpenInQuadrant.toMPolyfold Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.selfChart_isChart — the text's X_C.

II.12 The subspaces E_x, the cones C_x and the sign arguments

Coordinates. For a presentation T of C and $x \in E_0$, the coordinates vanishing at x form a continuous linear surjection $E_0 \to \mathbb{R}^{I(x)}$, $v \mapsto ((Tv)_i)_{i \in I(x)}$ Chapter2/DegeneracyIndex · coordProj Chapter2/DegeneracyIndex · coordMapL Chapter2/DegeneracyIndex · coordMapL_surjective. Its kernel is E_x Chapter2/DegeneracyIndex · minimalSubspaceWith Chapter2/DegeneracyIndex · mem_minimalSubspaceWith, closed Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith, and by the first isomorphism theorem $E_0/E_x \cong \mathbb{R}^{I(x)}$, so $\dim(E_0/E_x) = d_C(x)$ Chapter2/DegeneracyIndex · finrank_quotient_minimalSubspaceWith. The cone $C_x = \{v : (Tv)_i \geq 0 \text{ for } i \in I(x)\}$ Chapter2/Tame · partialQuadrantAtWith is convex and closed and contains E_x Chapter2/Tame · convex_partialQuadrantAtWith Chapter2/Boundary · isClosed_partialQuadrantAtWith Chapter2/Tame · minimalSubspaceWith_subset_partialQuadrantAtWith. *Approximation*: if $Tx = (a, w)$, the points $T^{-1}(a, w')$ with $w' \in W_1$ (resp. W_∞) converge to x as $w' \to w$ and have the same vanishing coordinates, so x is a limit of points on level 1 (resp. smooth points) of the same index, in C iff x is Chapter2/DegeneracyIndex · exists_tendsto_lvl1_degIndexWith Chapter2/TameRetracts · exists_tendsto_smooth_degIndexWith.

Density on level 1. For a continuous linear $L : E_0 \to V$, V finite-dimensional, the level-1 points of $\ker L$ are dense in $\ker L$ Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase. The image S of E_1 is dense in E_0; L maps S onto the finite-dimensional $L(S)$, which has a linear, hence continuous, right inverse g with values in S. For $v \in \ker L$ take $u_k \in S$ with $u_k \to v$; then $L(u_k) \to 0$, $g(L(u_k)) \to 0$, and $u_k - g(L(u_k)) \in S \cap \ker L$ converges to v. This is the text's "the closure of $(T^R_x O)_\infty$ is $T^R_x O$" in the form the proofs need; the level-1 points of C_x are dense in C_x likewise Chapter2/Boundary · partialQuadrantAtWith_subset_closure.

The two-sided sign argument. Let f be sc¹ on U relatively open in C, with values in C', $x \in U_1$ and $v \in E_1$ with $\iota_1 v \in E_x$. Then $x + tv \in U_1$ for $|t|$ small: the coordinates in $I(x)$ stay 0, the others stay positive, and U_1 is relatively open Chapter2/DegeneracyIndex · eventually_add_smul_mem_lvlSet. For j with $(T' f(x))_j = 0$ the real function $g(t) = (T' f(\iota_1(x + tv)))_j$ is ≥ 0 near 0 (values in C'), vanishes at 0, and has derivative $(T' Df(x)v)_j$ at 0 (Definition 1.14's derivative along E_1, composed with the path); a local minimum has zero derivative Mathlib/Analysis/Calculus/LocalExtr/Basic · IsLocalMin.hasDerivAt_eq_zero. So $Df(x)v \in E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.deriv_mem_minimalSubspace_of_lvl1, and by density and continuity of $Df(x)$ on E_0, $Df(x)$ maps E_x into $E'_{f(x)}$ Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace.

The one-sided sign argument. For $v \in E_1$ with $\iota_1 v \in C_x$, $x + tv \in U_1$ for small $t \geq 0$ Chapter2/Boundary · eventually_add_smul_mem_lvlSet_of_cone, the same g is ≥ 0 on $[0, \varepsilon)$ with $g(0) = 0$, so its one-sided derivative is ≥ 0 Chapter2/Boundary · nonneg_of_hasDerivWithinAt_Ici; hence $Df(x)v \in C'_{f(x)}$

Chapter2/Boundary · IsSc1With.deriv_mem_partialQuadrant_of_lvl1, and $Df(x)$ maps C_x into $C'_{f(x)}$ Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant. These are the derivative-level content of Lemma 2.29 (1) and (2); the lemma itself, about sc-smooth paths, is not formalized.

The chain rule at a point. For sc¹ maps f, g on relatively open sets with $f(U) \subseteq U'$, $Dg(f(x))Df(x) = D(g \circ f)(x)$ at $x \in U_1$ Chapter2/DegeneracyIndex · scDeriv_comp_apply; if $g \circ f = \mathrm{id}$ on U then $g_1(f_1(x)) = x$ and $Dg(f(x))Df(x) = \mathrm{id}$ Chapter2/DegeneracyIndex · levelMap_comp_eq_self Chapter2/DegeneracyIndex · scDeriv_comp_eq_self.

II.13 Proposition 2.30, Theorem 2.31, Corollary 2.32

Definition 2.27, Remark 2.28. For a retraction r, a presentation T and $x \in O_1$: $T^R_x O := Dr(x)E_0 \cap E_x$ and $C_x O := Dr(x)E_0 \cap C_x$ Chapter2/Tame · reducedTangentSpaceWith Chapter2/Tame · partialConeWith, defined at every point on level 1; $T^R_x O \subseteq C_x O$ Chapter2/Tame · reducedTangentSpaceWith_subset_partialConeWith. The number $\dim(T_x O / T^R_x O)$ is $\dim(A/(A \cap B))$ for $A = T_x O$, $B = E_x$ Chapter2/Boundary · quotDim; since $A \to E_0/E_x$ has kernel $A \cap E_x$, it is at most $d_C(x)$ and finite Chapter2/Boundary · quotDim_inf_minimalSubspace_le Chapter2/Boundary · quotDim_reducedTangentSpaceWith_le.

Germs between retracts. A germ of sc-diffeomorphisms between $O = r(U)$ and $O' = r'(U')$ is a pair F, G with $F \circ r$, $G \circ r'$ sc¹ (sc¹ suffices here; the text has sc-smooth), $F(O) \subseteq O'$, $G(O') \subseteq O$, inverse to each other on O, O' Chapter2/Boundary · ScRetraction.IsDiffeoGerm; an sc-smooth map between retracts with an sc-smooth inverse is one Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm, and two compatible charts give one between the overlap retracts, with the restricted retractions Chapter2/Boundary · MChart.Compatible.isDiffeoGerm. Since $(G \circ r') \circ (F \circ r) = r$ on U Chapter2/Boundary · ScRetraction.IsDiffeoGerm.comp_eq, the chain rule at $x \in O_1$, $y = (F \circ r)_1(x) \in O'_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_mem Chapter2/Boundary · ScRetraction.IsDiffeoGerm.image_image, gives

$$D(G \circ r')(y)\, D(F \circ r)(x) = Dr(x), \qquad D(F \circ r)(x)\, D(G \circ r')(y) = Dr'(y)$$

Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_deriv_inv. So $D := D(F \circ r)(x)$ maps $T_x O$ into $T_y O'$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range, and by the sign arguments for $F \circ r$ (with values in $O' \subseteq C'$) it maps E_x into E'_y and C_x into C'_y Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_minimalSubspace Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mapsTo_partialQuadrant.

Proposition 2.30. $D(T^R_x O) = T^R_y O'$ and $D(C_x O) = C_y O'$, with $Tf(x) = D$ Chapter2/Boundary · proposition_2_30_reduced Chapter2/Boundary · proposition_2_30_cone. "$\subseteq$": D maps $T_x O$ into $T_y O'$ and E_x (resp. C_x) into E'_y (resp. C'_y). "$\supseteq$": for $w \in T^R_y O'$, the vector $D'w$, $D' = D(G \circ r')(y)$, lies in $T_x O$ and in E_x by the same facts for the inverse germ, and $D(D'w) = Dr'(y)w = w$ because w is in the range of the projection $Dr'(y)$. This proof does not pass through Lemma 2.29.

The core of Theorem 2.31. Along a germ F, $\dim(T_x O / T^R_x O) \le d_{C'}(F(x))$ for $x \in O_1$ Chapter2/Boundary · ScRetraction.IsDiffeoGerm.quotDim_le. With $P = Dr(x)$, $P' = Dr'(y)$: D maps $T_x = PE$ into $T_y = P'E'$ and D' maps back, $D'D = P$ is the identity on T_x and $DD' = P'$ the identity on T_y. The map $\Phi : T_x \to T_y/(T_y \cap E'_y)$, $v \mapsto [Dv]$, is onto ($[w] = \Phi(D'w)$) with kernel $T_x \cap E_x$ ($Dv \in E'_y$ implies $v = D'Dv \in E_x$). Hence $T_x/T^R_x \cong T_y/T^R_y$, which embeds in E'_0/E'_y, of dimension $d_{C'}(y)$. The same argument with U_0 relatively open in C in place of the retract O is the form used for Corollary 2.32 and Theorem 1.25 Chapter2/DegeneracyIndex · degIndexWith_le_of_germ: there $D'D = \mathrm{id}$, so $E_0/E_x \cong T_y/T^R_y \hookrightarrow E'_0/E'_y$ and $d_C(x) \le d_{C'}(F(x))$.

Theorem 2.31. Let φ be a chart of an M-polyfold around x with $\varphi(x) = \iota_1 z$, r a retraction onto its model and T a presentation. For every other chart ψ around x, the overlaps are $O \cap W$ and $O' \cap W'$, the transition maps form a germ between the restricted retractions, and the restricted retraction has the same derivative at z Chapter2/Boundary · ScRetraction.scDeriv_restrict Chapter2/Boundary · reducedTangentSpaceWith_

restrict; so the core gives $\dim(T_zO/T_z^RO) \le d(x,\psi)$ Chapter2/Boundary · MPolyfold.quotDim_le_chartDegIndex. Taking ψ realizing $d_X(x)$: $\dim(T_zO/T_z^RO) \le d_X(x)$ Chapter2/Boundary · theorem_2_31. For a retract O as an M-polyfold with the chart id this is the text's statement $\dim(T_xO/T_x^RO) \le d_O(x)$, at points on level 1 Chapter2/Boundary · theorem_2_31_retract. The text instead reduces to a chart realizing $d_O(x)$ and counts linearly independent $\mathbb{R}^d$-components; the quotient isomorphism makes the reduction unnecessary. The second statement (equality makes C_xO a partial quadrant) is not formalized: it needs the sc-structure of T_xO and smooth lifts of a basis of T_xO/T_x^RO.

Corollary 2.32. Let U be nonempty and relatively open in C, and X_U its M-polyfold Chapter2/DegeneracyIndex · corollary_2_32. The chart id gives $d_{X_U} \le d_C$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le. At a point x on level 1, a chart ψ realizing $d_{X_U}(x)$ is compatible with id, so $F = \psi$ (on $U_0 = \operatorname{dom}\psi$, relatively open in C) and $G = \psi^{-1}$ form a germ from U_0 to the retract $\psi(V)$, and the core gives $d_C(x) \le d_{C'}(\psi(x)) = d_{X_U}(x)$ Chapter2/DegeneracyIndex · IsRelOpenInQuadrant.degIndex_le_degIndex_of_lvl1. For a general x, approximate by level-1 points $\sigma_i \in U$ with $d_C(\sigma_i) = d_C(x)$ (§II.12); by Lemma 2.23, $d_{X_U}(\sigma_i) \le d_{X_U}(x)$ eventually, so $d_C(x) = d_C(\sigma_i) \le d_{X_U}(\sigma_i) \le d_{X_U}(x)$. The text approximates by smooth points; level 1 suffices. **Theorem 1.25** (Part I, §I.19) is proved in the same file by the same argument for a germ between relatively open subsets of two quadrants Chapter2/DegeneracyIndex · theorem_1_25_lvl1 Chapter2/DegeneracyIndex · theorem_1_25.

II.14 Tame retractions; Propositions 2.35 and 2.36

Definition 2.33. C_e and E_e are the cone and subspace of §II.12 for the presentation T. They do not depend on it: for two presentations T, T' of C and $e \in C$, $v \in E_e$ (for T) gives $e + tv \in C$ for $|t|$ small Chapter2/TameFaces · eventually_add_smul_mem_of_minimalSubspace; for j with $(T'e)_j = 0$, $(T'(e+tv))_j = t(T'v)_j \ge 0$ for $t = \pm\delta/2$, so $(T'v)_j = 0$ Chapter2/TameFaces · minimalSubspaceWith_le; by symmetry the two subspaces agree Chapter2/TameFaces · minimalSubspaceWith_eq. This is the argument of Lemma 1.24.

Definition 2.34. r is *tame for* T if (1) $d_C(r(x)) = d_C(x)$ on U and (2) at every smooth $x \in O_1$ some $A \le E_x$ makes $E = Dr(x)E_0 \oplus A$ an sc-splitting Chapter2/Tame · ScRetraction.IsTameWith; *tame* if tame for every presentation Chapter2/Tame · ScRetraction.IsTame, and by the independence of E_x tameness for one presentation suffices Chapter2/TameFaces · ScRetraction.IsTameWith.isTame. A *tame retract* is the retract of a tame retraction Chapter2/Tame · IsTameRetract Chapter2/Tame · IsTameRetract.isScRetract. Identities of relatively open sets are tame ($T_xU = E$, $A = 0$) Chapter2/Tame · ScRetraction.id_isTame, and restrictions of tame retractions to $U \cap r^{-1}(V)$ are tame (same derivatives) Chapter2/TameFaces · ScRetraction.IsTameWith.restrict.

Proposition 2.35. *$Dr(x)E_x \subseteq E_x$ for every retraction at every $x \in O_1$*: the two-sided sign argument for r, with $r(x) = x$ Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace. At a smooth x, $Dr(x)$ is an sc-operator Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv, so $E = Dr(x)E \oplus (1 - Dr(x))E$ is an sc-splitting Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv. *The proposition* Chapter2/TameRetracts · proposition_2_35: with $A \le E_x$ from tameness (2), $E_0 = Dr(x)E_0 \oplus A$ algebraically (the level-0 part of the sc-splitting); write $v = t + a$; then $(1 - Dr(x))v = (1 - Dr(x))a = a - Dr(x)a \in E_x$, because $a \in E_x$ and $Dr(x)a \in E_x$. Only condition (2) is used; the text uses condition (1) for $Dr(x)E_x \subseteq E_x$. In coordinates: $\lambda_j(Dr(x)v) = \lambda_j(v)$ for $j \in I(x)$ Chapter2/TameRetracts · coord_scDeriv_eq.

Proposition 2.36. *(13) at smooth points.* If $V = A' \oplus B$ and $B \le K$, then $A'/(A' \cap K) \cong V/K$ Chapter2/TameRetracts · quotDim_eq_finrank_quotient_of_isCompl; with $A' = T_xO$, $B = A$ and $K = E_x$ this gives $\dim(T_xO/T_x^RO) = \dim(E_0/E_x) = d_C(x)$ Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq — the text's equations (15)–(16). *A tame chart realizes the index* Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith: let φ be a chart of any M-polyfold whose model has a tame retraction r. At y with $\varphi(y)$ smooth, $d(y,\varphi) = \dim(T/T^R) \le d_X(y)$ by Theorem 2.31, and $d_X(y) \le d(y,\varphi)$ by definition. At a general y, with $\varphi(y) = o$, approximate o by smooth points σ_i of C with the same vanishing coordinates; then $r(\sigma_i)$ are smooth points of O (an sc^0 map preserves smoothness

Chapter2/TameRetracts · IsSc0.mem_smooth Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth) converging to $r(o) = o$, with $d_C(r(\sigma_i)) = d_C(\sigma_i) = d_C(o)$ by tameness (1); so the points $\varphi^{-1}(r(\sigma_i))$ converge to y and have $d_X = d(\cdot,\varphi) = d_C(o)$, and Lemma 2.23 gives $d_C(o) \le d_X(y)$. For charts of the structure modelled on a tame retract Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract. Applied to the retract itself as an M-polyfold Chapter2/TameRetracts · ScRetraction.toLocalModel Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith: (14) $d_O = d_C$ on O Chapter2/TameRetracts · proposition_2_36_degIndex and (13) $\dim(T_xO/T^R_xO) = d_O(x) = d_C(x)$ at smooth points Chapter2/TameRetracts · proposition_2_36. This is the text's proof, generalized from the retract to any chart; Proposition 2.42 needs the general form.

Splitting off finitely many coordinates. Let $B \le A$ be sc-subspaces of E, $\ell_1,\dots,\ell_d$ continuous functionals on E_0 with $B = A \cap \bigcap_j \ker \ell_j$, and $e_1,\dots,e_d \in A$ smooth with $\ell_i(e_j) = \delta_{ij}$. Each e_j has a lift to every level m, compatible with the inclusions Chapter2/PartialCone · smoothLift Chapter2/PartialCone · incl_smoothLift. On level m the maps $v \mapsto ((\ell_j v)_j,\ v - \sum_j \ell_j(v)e_j)$ and $(a,u) \mapsto u + \sum_j a_j e_j$ are continuous linear $A_m \to \mathbb{R}^d \times B_m$ and back Chapter2/PartialCone · splitFwd Chapter2/PartialCone · splitBwd — the bracket lies in B because every ℓ_i kills it Chapter2/PartialCone · sub_sum_mem — and mutually inverse Chapter2/PartialCone · splitBwd_splitFwd Chapter2/PartialCone · splitFwd_splitBwd; they commute with the inclusions, so they form an sc-isomorphism $A \cong \mathbb{R}^d \oplus B$ Chapter2/PartialCone · splitIso, A and B with the sc-structures induced from E Chapter1/ScBanach · ScBanach.ofSubspace. It carries $\{v \in A : \ell_j(v) \ge 0\ \forall j\}$ onto $[0,\infty)^d \oplus B$, so that set is a partial quadrant in A Chapter2/PartialCone · isPartialQuadrant_ofSubspace. This is the map S of the proof of Theorem 2.31, with the e_j smooth as the exposition requires.

Proposition 2.36, the remaining statements Chapter2/PartialCone · proposition_2_36_partialCone. Let r be tame for T, $x \in O_1$ smooth, $P = Dr(x)$ and $I(x) = \{i_1,\dots,i_d\}$. $T_xO = PE$ is an sc-subspace, the range of the sc-projection P (Proposition 2.35 above) Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv. The vectors $\hat e_j := P(T^{-1}(e_{i_j},0))$ Chapter2/TameRetracts · coordVec lie in T_xO; they are smooth, because $T^{-1}(e_{i_j},0)$ is smooth Chapter2/PartialCone · coordVec_mem_smooth and P is an sc-operator at the smooth point x Chapter2/TameRetracts · ScRetraction.isScOperator_scDeriv Chapter1/ScBanach · ScOperator.mapsTo_smoothLvl; and $\lambda_{i_k}(\hat e_j) = \lambda_{i_k}(T^{-1}(e_{i_j},0)) = \delta_{jk}$ by Proposition 2.35 in coordinates Chapter2/TameRetracts · coord_scDeriv_eq. By definition $T^R_xO = T_xO \cap \bigcap_k \ker \lambda_{i_k}$ Chapter2/TameRetracts · coordL. *T^R_xO is an sc-subspace.* It is closed, as an intersection of closed subspaces Chapter2/DegeneracyIndex · isClosed_minimalSubspaceWith. For the density of smooth points, let $u \in T^R_xO \cap E_m$ with $Tu = (a,w)$ on level m, and put $\phi(w') := P(T^{-1}(a,w'))$, a continuous map $W_m \to E_m$. For smooth w' the point $\phi(w')$ is smooth, lies in T_xO, and lies in E_x, because $T^{-1}(a,w')$ has the same vanishing coordinates in $I(x)$ as u and $PE_x \subseteq E_x$ Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace. And $\phi(w) = Pu = u$, since $u \in T_xO$ and P is a projection Chapter2/Retracts · ScRetraction.scDeriv_comp_self. As W_∞ is dense in W_m, u lies in the closure of $\phi(W_\infty) \subseteq T^R_xO \cap E_\infty$ (this is the exposition's Lemma 3.A (1) in the case at hand). *The quadrant.* The splitting with $A = T_xO$, $B = T^R_xO$, $\ell_j = \lambda_{i_j}$ and $e_j = \hat e_j$ makes $\{v \in T_xO : \lambda_{i_j}(v) \ge 0\ \forall j\}$ a partial quadrant in T_xO, and this set is $C_xO = T_xO \cap C_x$. The text deduces the claim from the equality case of Theorem 2.31, which needs smooth lifts $\hat v^j$ of a basis of T_xO/T^R_xO (the exposition chooses them by a density argument); for a tame retraction they are the explicit $\hat e_j$, and the general equality case, still not formalized, is not needed.

II.15 Faces: local geometry, Lemma 2.40, Corollary 2.41, Proposition 2.42

Definitions 2.37, 2.39. X is *tame* if an atlas equivalent to its atlas has only tame models Chapter2/Tame · MPolyfold.IsTame; charts of an equivalent atlas are charts of the structure Chapter2/TameRetracts · MPolyfold.isChart_of_mem_equiv, so every point lies in a chart of the structure with a tame model Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart. A *face* is the closure of the connected

component in $\{d_X = 1\}$ of a point with $d_X = 1$ Chapter2/Tame · MPolyfold.IsFace, closed Chapter2/Tame · MPolyfold.IsFace.isClosed and inside ∂X Chapter2/Boundary · MPolyfold.IsFace.subset_boundary; X is *face-structured* if every x lies in exactly $d_X(x)$ faces Chapter2/Tame · MPolyfold.IsFaceStructured.

Coordinates of faces. $I(z) = \{i : \lambda_i(z) = 0\}$ Chapter2/TameRetracts · zeroSet with $d_C = \#I$ Chapter2/TameRetracts · degIndexWith_eq_card_zeroSet; the extended face f_i is the hyperplane $\{\lambda_i = 0\}$, closed Chapter2/TameRetracts · coordHyper Chapter2/TameRetracts · isClosed_coordHyper; $I(z) \subseteq I(e)$ for z near e Chapter2/TameRetracts · eventually_zeroSet_subset. *A connected piece of $\{d_C = 1\}$ lies in one face* (exposition Lemma 4.A): if $S \subseteq C$ is preconnected with $d_C = 1$ on S and $\lambda_i(p) = 0$ at some $p \in S$, then $S \subseteq f_i$ Chapter2/TameRetracts · subset_coordHyper_of_isPreconnected. Points of S have $I = \{i\}$ or $\lambda_i > 0$ Chapter2/TameRetracts · zeroSet_eq_singleton, so S is covered by the open sets $u = \{\lambda_l > 0 \text{ for } l \neq i\}$ and $v = \{\lambda_i > 0\}$, which are disjoint on S (a point in both would have $d_C = 0$); preconnectedness puts S in one of them Mathlib/Topology/Connected/Clopen · isPreconnected_iff_subset_of_disjoint, and $p \in u$.

Rays stay in their face (exposition Lemma 4.B). If f is sc¹, $x + tv \in U_1$ for small $t \geq 0$ and a functional L vanishes on $f(x + tv)$ for $t \in [0, \varepsilon)$, then $L(Df(x)v) = 0$ Chapter2/TameRetracts · IsSc1With.apply_deriv_eq_zero_of_ray. Now let r be tame, $x \in O_1$ smooth, $i \in I(x)$, $\lambda_i(y) = 0$ and $\lambda_l(y) > 0$ for the other $l \in I(x)$ Chapter2/TameRetracts · ray_zeroSet. For small $\tau > 0$, $x + \tau y \in U$ and $I(x + \tau y) = \{i\}$, so $d_C(r(x + \tau y)) = 1$ by tameness (1), and $I(r(x + \tau y)) \subseteq I(x)$ by continuity of r at $x = r(x)$. The image of $(0, \tau_0]$ under $\tau \mapsto r(x + \tau y)$ is preconnected with $d_C = 1$, so it lies in one f_j with $j \in I(x)$. Differentiating $\lambda_j(r(x + \tau y)) = 0$ at $\tau = 0$ gives $\lambda_j(Dr(x)y) = 0$, and $\lambda_j(Dr(x)y) = \lambda_j(y)$ by Proposition 2.35 Chapter2/TameRetracts · coordL; as $\lambda_l(y) > 0$ for $l \in I(x) \setminus \{i\}$, $j = i$. This is the step of Lemma 2.40 that the text omits, and it needs tameness (2).

The local structure of a face (exposition Lemma 4.C) Chapter2/TameRetracts · local_face_structure. Let $S = \{z \in O : d_C(z) = 1\}$ Chapter2/TameRetracts · ScRetraction.faceStratum and $e \in O$ with $\lambda_i(e) = 0$. (a) Choose ε so that $B = B_\varepsilon(e) \cap C \subseteq U$ and $I(z), I(r(z)) \subseteq I(e)$ on B. (b) $A_i \cap B = \{z \in B : I(z) = \{i\}\}$ is convex Chapter2/TameRetracts · convex_ball_inter_zeroSet_eq, hence connected, and r maps it into S; a smooth $u = r(T^{-1}(a, w')) \in O \cap B$ with $I(u) = I(e)$ and the ray of Lemma 4.B from u in the direction $T^{-1}(b, 0)$, $b = 1$ on $I(e) \setminus \{i\}$ and 0 elsewhere Chapter2/TameRetracts · coordVec Chapter2/TameRetracts · rayDir Chapter2/TameRetracts · exists_lvl1_coordVec, produces a point $p \in S$ with $\lambda_i(p) = 0$ in $r(A_i \cap B)$; so $r(A_i \cap B)$ lies in the component of S through p. (c) A point $e' \in f_i \cap B$ is the limit of points of $A_i \cap B$ (perturb the coordinates of $I(e') \setminus \{i\}$ upward), so $r(e')$ lies in the closure of that component.

Lemma 2.40. A *face of the retract* is $\overline{\hat{F}} \cap O$ for a component $\hat{F}$ of S Chapter2/TameRetracts · ScRetraction.IsRetractFace. A component of S through a point with $\lambda_i = 0$ lies in f_i Chapter2/TameRetracts · faceStratum_degIndexWith Chapter2/TameRetracts · connectedComponentIn_faceStratum_subset, and a component of S in f_i whose closure contains e is the component of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_eq_of_mem_closure. *The face through $x \in S$, $\lambda_i(x) = 0$, is the component of $f_i \cap O$ through x* Chapter2/TameRetracts · closure_connectedComponentIn_faceStratum: it is connected, lies in $f_i \cap O$, is closed in it, and is relatively open in it by Lemma 4.C(c); a preconnected subset meeting a relatively clopen set lies in it Chapter2/TameRetracts · subset_of_relClopen_of_isPreconnected. Conversely, the component of $f_i \cap O$ through any $e \in O$ with $\lambda_i(e) = 0$ is the face through the point p of Lemma 4.C at e Chapter2/TameRetracts · connectedComponentIn_coordHyper_eq. Together: the faces of the retract are the components of the sets $f_i \cap O$ Chapter2/TameRetracts · lemma_2_40. If the components of $f_i \cap O$ and $f_j \cap O$ through x coincide, a point z of the face in S has $I(z) = \{i\}$ and is a limit of points with $I = \{j\}$, which near z forces $j = i$ Chapter2/TameRetracts · lemma_2_40_injective. The faces of the M-polyfold O (Definition 2.39, with $d_O = d_C$ by Proposition 2.36) are exactly the faces of the retract, transferred along the subtype inclusion Chapter2/TameRetracts · image_val_connectedComponentIn Chapter2/TameRetracts · image_val_closure Chapter2/TameRetracts · LocalModel.isFace_toMPolyfold_iff.

Corollary 2.41. The map $i \mapsto$ (component of $f_i \cap O$ through x) sends $I(x)$ onto the faces through x (Lemma 2.40) and is injective (its second statement), so x lies in exactly $\#I(x) = d_C(x)$ faces of the retract Chapter2/TameRetracts · corollary_2_41_retract, and the M-polyfold O is face-structured Chapter2/

TameRetracts · LocalModel.isFaceStructured_toMPolyfold Chapter2/TameRetracts · corollary_2_41.

Proposition 2.42. Let X be tame. *The global bound* Chapter2/TameRetracts · MPolyfold.faces_encard_le: in a tame chart φ around x, $o = \varphi(x)$, Lemma 4.C gives for each $i \in I(o)$ a component $\hat{F}_i$ of S containing all points of O near o with $I = \{i\}$. A face $F = \overline{G}$ through x has points $y \in G$ near x; $d_C(\varphi(y)) = d_X(y) = 1$ (the tame chart realizes d_X) and $I(\varphi y) \subseteq I(o)$, so $I(\varphi y) = \{i\}$ for some $i \in I(o)$, and the preconnected set $\varphi^{-1}(\hat{F}_i) \subseteq \{d_X = 1\}$ meets G, hence lies in G. Distinct faces give distinct i; so at most $d_X(x)$ faces pass through x. *The local statement*: the tame chart realizes d_X on its domain, the M-polyfold O has $d_O = d_C$ (Proposition 2.36), and O is face-structured (Corollary 2.41) Chapter2/TameRetracts · proposition_2_42. The neighbourhood V of the text is represented by O.

II.16 Faces are tame sub-M-polyfolds: Propositions 2.43, 2.44 and Theorem 2.45

The projection onto a face. $P_i := T^{-1} \circ (a \mapsto a \text{ with } a_i = 0) \circ T$ is an sc-operator Chapter2/TameRetracts · killCoordL Chapter2/TameRetracts · killCoordModel Chapter2/TameRetracts · killCoord which kills the i-th coordinate and keeps the others Chapter2/TameRetracts · killCoord_coord Chapter2/TameRetracts · killCoord_coord_self, fixes f_i Chapter2/TameRetracts · killCoord_eq_self and maps C into C Chapter2/TameRetracts · killCoord_mem.

Proposition 2.43, the retraction. Let F be a face of the retract, through $x \in S$ with $I(x) = \{i\}$, so $F \subseteq f_i$ Chapter2/TameRetracts · proposition_2_43. For each $e \in F$, Lemma 4.C gives ε_e with $B_e = B_{\varepsilon_e}(e) \cap C \subseteq U$ and $r(z) \in F$ for $z \in B_e \cap f_i$ (the component of Lemma 4.C at e is the one of F). Put $U^* := \bigcup_{e \in F}(B_{\varepsilon_e}(e) \cap P_i^{-1}(B_{\varepsilon_e}(e))) \cap C$: relatively open in C, contains F (as $P_i e = e$), and for $z \in U^*$ the point $P_i z$ lies in some $B_e \cap f_i$, so $r(P_i z) \in F$. Then $s := r \circ P_i$ is sc-smooth on U^* (chain rule), maps U^* into $F \subseteq U^*$, and fixes F; so it is a retraction onto F, and $U^* \subseteq U$. The preimages under P_i are needed because P_i need not be norm-decreasing for the norm of E_0; the exposition's "$|P_1 z - e|_0 \leq |z - e|_0$" holds only for special norms. *F is a sub-M-polyfold of O*: in the chart id of O, s restricted to $O \cap U^*$ is an sc-smooth retraction onto F Chapter2/TameRetracts · LocalModel.selfChart_symm_apply Chapter2/TameRetracts · LocalModel.isSubMPolyfold_of_isFace.

The face as a model space. Insert and delete a coordinate: $\mathbb{R}^k \to \mathbb{R}^{k+1}$, $b \mapsto (b_1, \dots, 0_i, \dots, b_k)$, and back Chapter2/TameFaces · insertCoordL Chapter2/TameFaces · removeCoordL, as sc-operators on $\mathbb{R}^k \oplus W$ Chapter2/TameFaces · insertCoordModel Chapter2/TameFaces · removeCoordModel; composed with T they give $\iota : \mathbb{R}^k \oplus W \to E$ onto f_i and $\pi : E \to \mathbb{R}^k \oplus W$ Chapter2/TameFaces · faceIncl Chapter2/TameFaces · faceProj, with $\pi\iota = 1$, $\iota\pi = 1$ on f_i Chapter2/TameFaces · faceProj_faceIncl Chapter2/TameFaces · faceIncl_faceProj, $\iota(C') \subseteq C$ and $\pi(C) \subseteq C'$ for $C' = [0,\infty)^k \oplus W$ Chapter2/TameFaces · faceIncl_mem Chapter2/TameFaces · faceProj_mem, and $d_C(y) = d_{C'}(\pi y) + 1$ on f_i Chapter2/TameFaces · degIndexWith_eq_quadrantDeg_faceProj_add_one. The derivative of an sc-operator is itself Chapter2/TameFaces · ScOperator.scDeriv_eq, and the chain rule at a point holds between arbitrary sc-domains Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain.

Proposition 2.43, the tame model Chapter2/TameFaces · proposition_2_43_tameModel. With U^* and s as above, put $U' := \iota^{-1}(U^*)$ (relatively open in C') and $r' := \pi \circ r \circ \iota$ on U'. Since $r \circ \iota$ maps U' into $F \subseteq f_i$, where $\iota\pi = 1$, r' is an sc-smooth retraction with retract $\pi(F)$, and ι maps it back into F. *Tameness (1)*: for $z \in U'$, $d_{C'}(r'z) + 1 = d_C(r(\iota z)) = d_C(\iota z) = d_{C'}(z) + 1$. *Tameness (2)*: at a smooth $x \in \pi(F)$, ιx is a smooth point of O and $Dr'(x)v = \pi(Dr(\iota x)(\iota v))$ by the chain rule; for j with $x_j = 0$ the coordinate $j' = i.\text{succAbove}(j)$ vanishes at ιx, and Proposition 2.35 for r gives $\lambda_{j'}(Dr(\iota x)\iota v) = \lambda_{j'}(\iota v) = v_j$; so $\lambda'_j \circ Dr'(x) = \lambda'_j$, and $(1 - Dr'(x))E'$ is an sc-complement of $T_x\pi(F)$ inside E'_x. The exposition instead intersects the complement A of $T_x O$ with f_1; that is not needed. And $d_{C'}(\pi y) + 1 = d_C(y)$ on F.

Proposition 2.43, faces are tame Chapter2/TameFaces · LocalModel.proposition_2_43_isTame. For a face F of a tame retract O (paracompact, being closed), the chart $\pi : F \to \pi(F)$ Chapter2/TameFaces · MPolyfold.faceChart Chapter2/TameFaces · MPolyfold.faceChartInv is compatible with the sub-charts of F built on the

chart id of O, its transition maps being the sc-operators ι and π `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`; a chart compatible with all sub-charts built on one chart of X is a chart of the induced structure, by locality of compatibility `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`. Its model is the tame retract $(\pi(F), C', E')$, so F is tame (a space all of whose points lie in tame charts is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`), and the tame chart realizes the index: $d_F(x) = d_{C'}(\pi x) = d_C(x) - 1 = d_O(x) - 1$ (equation (14) twice). The text calls this last statement obvious.

Proposition 2.44, first sentence `Chapter2/TameFaces · MPolyfold.proposition_2_44`. In a tame chart φ around x: $\varphi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O (Proposition 2.42 with Corollary 2.41), and each face of O is a tame sub-M-polyfold with $d_F + 1 = d_O$, where $d_O(\varphi x) = d_X(x)$.

The corner as a model space. For $\sigma \subseteq \{1, \dots, n\}$ list the coordinates outside σ as $j \mapsto l_j$ `Chapter2/CornerGerms · cornerIdx` `Chapter2/CornerGerms · cornerIdx_injective` `Chapter2/CornerGerms · exists_cornerIdx`. Deleting the coordinates in σ and inserting zeros there `Chapter2/CornerGerms · cornerRemoveL` `Chapter2/CornerGerms · cornerInsertL` are mutually inverse on $\{a : a_l = 0,\ l \in \sigma\}$ `Chapter2/CornerGerms · cornerRemoveL_cornerInsertL` `Chapter2/CornerGerms · cornerInsertL_cornerRemoveL`; as sc-operators on $\mathbb{R}^n \oplus W$ and $\mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerInsertModel` `Chapter2/CornerGerms · cornerRemoveModel`, composed with T, they give $\iota_\sigma : \mathbb{R}^{n-\#\sigma} \oplus W \to E$ onto the corner $f_\sigma = \bigcap_{i\in\sigma} f_i$ and $\pi_\sigma : E \to \mathbb{R}^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl` `Chapter2/CornerGerms · cornerProj`, with $\pi_\sigma\iota_\sigma = 1$ and $\iota_\sigma\pi_\sigma = 1$ on f_σ `Chapter2/CornerGerms · cornerProj_cornerIncl` `Chapter2/CornerGerms · cornerIncl_cornerProj`, $\iota_\sigma(C') \subseteq C$ and $\pi_\sigma(C) \subseteq C'$ for $C' = [0,\infty)^{n-\#\sigma} \oplus W$ `Chapter2/CornerGerms · cornerIncl_mem` `Chapter2/CornerGerms · cornerProj_mem`, and $d_C(y) = d_{C'}(\pi_\sigma y) + \#\sigma$ on f_σ: the vanishing coordinates of y are those of σ together with the images l_j of the vanishing coordinates of $\pi_\sigma y$, disjointly `Chapter2/CornerGerms · degIndexWith_eq_quadrantDeg_cornerProj_add_card`. The sc-projection $P_\sigma = \iota_\sigma \circ \pi_\sigma$ `Chapter2/CornerGerms · cornerKill` sets the coordinates in σ to zero `Chapter2/CornerGerms · cornerKill_coord_mem`, fixes f_σ `Chapter2/CornerGerms · cornerKill_eq_self` and maps C into C `Chapter2/CornerGerms · cornerKill_mem`. For $\sigma = \{i\}$ these are the ι, π and P_i of Proposition 2.43.

The retraction onto a corner `Chapter2/CornerGerms · corner_retraction`. Let r be tame for T, $o \in O$ and $\sigma \subseteq I(o)$. For each $i \in \sigma$, Lemma 4.C at o `Chapter2/TameRetracts · local_face_structure` gives a face F_i of the retract with $o \in F_i \subseteq f_i$, containing a point with $d_C = 1$, and a ball around o in which r maps the points of $C \cap f_i$ into F_i. Distinct i give distinct faces: a point $p \in F_i$ with $d_C(p) = 1$ has $I(p) = \{i\}$, so $F_i = F_j$ forces $i = j$ `Chapter2/TameRetracts · zeroSet_eq_singleton`. Take a ball B around o inside all these balls with $B \cap C \subseteq U$, an open W_s with $W_s \cap U = r^{-1}(B) \cap U$ (continuity of r), and put $V := B \cap P_\sigma^{-1}(B \cap W_s)$, an open set containing o (since $P_\sigma o = o = r(o)$). On $V \cap C$ the map $s := r \circ P_\sigma$ is sc-smooth by the chain rule. For $z \in V \cap C$, $P_\sigma z \in C \cap f_\sigma \cap B$, so $r(P_\sigma z) \in F_i \subseteq f_i$ for every $i \in \sigma$: $s(z) \in O \cap f_\sigma$. Moreover $s(z) \in B$ (as $P_\sigma z \in W_s$), $P_\sigma s(z) = s(z)$, and $s(z) \in W_s$ (as $r(s(z)) = s(z) \in B$); so s maps $V \cap C$ into itself. On $f_\sigma \cap O$, P_σ and r are the identity, so $s \circ s = s$: s is an sc-smooth retraction with retract $O \cap V \cap f_\sigma$, and this set equals $O \cap V \cap \bigcap_{i\in\sigma} F_i$ (a point of $O \cap V \cap f_\sigma$ is fixed by r, which maps it into each F_i; conversely $F_i \subseteq f_i$). The exposition proceeds by induction on $\#\sigma$, identifying the germ of $F_1 \cap F_2$ with a face of F_1; the formal proof removes all coordinates of σ at once.

The tame model of a corner `Chapter2/CornerGerms · corner_tameModel`. Let s be an sc-smooth retraction on $U_s \subseteq U$, equal to $r \circ P_\sigma$ there, with retract $F \subseteq O \cap f_\sigma$; write $U_s = W_0 \cap C$ with W_0 open, and put $U' := \iota_\sigma^{-1}(W_0) \cap C'$ and $r' := \pi_\sigma \circ r \circ \iota_\sigma$ on U'. For $z \in U'$, $r(\iota_\sigma z) = s(\iota_\sigma z) \in F$ (as $P_\sigma\iota_\sigma = \iota_\sigma$), so r' maps U' into $\pi_\sigma(F) \subseteq U'$; it is sc-smooth by the chain rule and idempotent because $\iota_\sigma\pi_\sigma = 1$ on F and $r = 1$ on $F \subseteq O$; its retract is $\pi_\sigma(F)$. *Tameness (1):* $d_{C'}(r'z) + \#\sigma = d_C(r(\iota_\sigma z)) = d_C(\iota_\sigma z) = d_{C'}(z) + \#\sigma$ (the index formula on f_σ twice, and tameness (1) of r) `Chapter1/Degeneracy · degIndex_eq`. *Tameness (2):* at a smooth x of the retract, $\iota_\sigma x$ is a smooth point of O, and $Dr'(x)v = \pi_\sigma(Dr(\iota_\sigma x)\,\iota_\sigma v)$ by the chain rule between sc-domains `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain` `Chapter2/TameFaces · ScOperator.scDeriv_eq`; for j with $x_j = 0$ the coordinate l_j vanishes at $\iota_\sigma x$ `Chapter2/CornerGerms · coord_cornerIncl_idx`, and Proposition 2.35 gives $\lambda_{l_j}(Dr(\iota_\sigma x)\,\iota_\sigma v) = \lambda_{l_j}(\iota_\sigma v) = v_j$ `Chapter2/TameRetracts · coord_scDeriv_eq`; so $(1 - Dr'(x))E'$ lies in E'_x and is an sc-complement of $T_x\pi_\sigma(F)$ `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`. And $d_{C'}(\pi_\sigma y) + \#\sigma = d_C(y)$ for $y \in F$. This is the proof of the tame model

of Proposition 2.43 with σ in place of $\{i\}$.

Proposition 2.44, second sentence `Chapter2/CornerGerms · MPolyfold.proposition_2_44_corner`. Let X be tame and $x \in X$, and take a chart $\varphi : V \to O$ of the structure around x with a tame retraction r onto O, tame for a presentation T `Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart`. The face germs at x are the faces F_i, $i \in I(\varphi x)$, of the retract (Corollary 2.41: there are $d_X(x)$ of them), and σ is a set of such indices. The retraction onto the corner gives the pairwise distinct faces $F_i \ni \varphi(x)$, $i \in \sigma$, an open $B \ni \varphi(x)$ and s with retract $O \cap B \cap \bigcap_{i\in\sigma} F_i$. Put $A := \{y \in V : \varphi(y) \in B,\ \varphi(y) \in F_i\ \forall i \in \sigma\}$; its germ at x is the germ F^σ_x. Since $\varphi(A)$ is the retract of s, A is a sub-M-polyfold by the local criterion of Theorem 2.45 `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local` (paracompact as every subset `Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype`). The chart $\pi_\sigma \circ \varphi$ of A onto the retract $\pi_\sigma(\varphi(A))$ of r' `Chapter2/TameFaces · MPolyfold.faceChart` is compatible with the sub-charts of A built on φ, its transitions being ι_σ and π_σ `Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart`, hence a chart of the induced structure `Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible`; its model is tame `Chapter2/TameFaces · ScRetraction.IsTameWith.isTame`, so A is tame `Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart`. Both tame charts realize the index `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so for $a \in A$: $d_A(a) + \#\sigma = d_{C'}(\pi_\sigma\varphi(a)) + \#\sigma = d_C(\varphi(a)) = d_X(a)$; in particular $d_A(x) = d_X(x) - \#\sigma$, and the codimension, the number of quadrant coordinates removed in the tame model, is $\#\sigma$. The text states the proposition without proof.

Theorem 2.45. *The interior of a face.* $F \cap \{d_X = 1\} = G$ for $F = \overline{G}$ `Chapter2/TameRetracts · MPolyfold.theorem_2_45_interior`: at $y \in F$ with $d_X(y) = 1$, in a tame chart with $I(\varphi y) = \{i\}$, all points of $\{d_X = 1\}$ near y lie in $\varphi^{-1}(\hat F)$ for the single component $\hat F$ of Lemma 4.C at $\varphi(y)$, a connected subset of $\{d_X = 1\}$ containing y; it meets G (as $y \in \overline{G}$), hence lies in G. *A local criterion for sub-M-polyfolds* `Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local`: if near each $a \in A$, in a chart φ, $\varphi(A \cap V)$ agrees on an open $B \ni \varphi(a)$ with the retract of an sc-smooth retraction s contained in O, then A is a sub-M-polyfold, with the retraction $\varphi^{-1} \circ s \circ \varphi$ on $V' = \varphi^{-1}(B \cap W_0 \cap s^{-1}(B))$ — shrunk so that s maps it into itself, a step the exposition's "transport by φ^{-1}" omits. *G is a sub-M-polyfold* `Chapter2/TameFaces · MPolyfold.theorem_2_45_isSubMPolyfold_interior`: at $y \in G$, the local face $\overline{\hat F} \cap O$ agrees with $\varphi(G)$ on the ball of Lemma 4.C (both consist of the points of S there), and Proposition 2.43 gives its retraction. *The face-structured case.* Near $x \in F$, in a tame chart, $\varphi(F)$ agrees on a ball with a face of the retract `Chapter2/TameFaces · MPolyfold.exists_localFace_of_isFaceStructured`: the map $i \mapsto F_i$ (the closure of the component of $\varphi^{-1}(\hat F_i)$) from $I(\varphi x)$ to the faces through x is onto (as in Proposition 2.42), and since there are exactly $d_X(x) = \#I(\varphi x)$ faces through x, it is injective; so the points of G near x lie in $\varphi^{-1}(\hat F_j)$ for the one j with $F = F_j$. Hence F is a sub-M-polyfold `Chapter2/TameFaces · MPolyfold.theorem_2_45_isSubMPolyfold_face`, and with the charts $\pi \circ \varphi$ onto the restrictions of the tame retractions r' it is tame with $d_X(x) = d_F(x) + 1$ `Chapter2/TameFaces · MPolyfold.theorem_2_45_isTame_face`. The local and global face counts of the theorem are Proposition 2.42 and the definition of face-structured.

II.17 Products and fibred products: Lemma 4.D and Proposition 2.46

Products of partial quadrants. For presentations $T : E \cong \mathbb{R}^n \oplus W$ and $T' : E' \cong \mathbb{R}^{n'} \oplus W'$ of C and C', concatenating the coordinates ($\mathbb{R}^n \times \mathbb{R}^{n'} \cong \mathbb{R}^{n+n'}$ `Chapter2/FibredProduct · finAppendL`) gives a presentation $E \oplus E' \cong \mathbb{R}^{n+n'} \oplus (W \oplus W')$ `Chapter2/FibredProduct · prodPresentation` of $C \times C'$ `Chapter2/FibredProduct · prod_eq_prodPresentation`, so $C \times C'$ is a partial quadrant `Chapter2/FibredProduct · IsPartialQuadrant.prod` and $U \times U'$ is relatively open in it `Chapter2/FibredProduct · IsRelOpenInQuadrant.prod`. The vanishing coordinates of (x, x') are those of x followed by those of x', so $d_{C\times C'}(x, x') = d_C(x) + d_{C'}(x')$ `Chapter2/FibredProduct · degIndexWith_prodPresentation` `Chapter2/FibredProduct · degIndex_prod`, and the minimal subspace is $E_x \oplus E'_{x'}$, `Chapter2/FibredProduct · minimalSubspaceWith_prodPresentation`.

Products of retractions. $f \times f'$ is sc-smooth when f and f' are `Chapter2/FibredProduct · IsScSmooth.prodMap`,

with derivative $Df(x) \times Df'(x')$ Chapter2/FibredProduct · scDeriv_prodMap; so $r \times r'$ is an sc-smooth retraction Chapter2/FibredProduct · ScRetraction.prod with retract $O \times O'$ Chapter2/FibredProduct · ScRetraction.prod_retract Chapter2/FibredProduct · IsScRetract.prod, and products of sc-smooth maps between retracts are sc-smooth Chapter2/FibredProduct · IsScSmoothRetractMap.prodMap. Products of sc-subspaces and of sc-splittings are again such Chapter2/FibredProduct · IsScSubspace.prod Chapter2/FibredProduct · IsScSplitting.prod; hence if r, r' are tame for T, T', then $r \times r'$ is tame for the product presentation Chapter2/FibredProduct · ScRetraction.IsTameWith.prod: (1) is the additivity of the index, and for (2) the complement of $T_{(x,x')}(O \times O') = T_xO \oplus T_{x'}O'$ is $A \oplus A'$ inside $E_x \oplus E'_{x'}$.

Lemma 4.D (the product M-polyfold). $X \times Y$ is paracompact, being metrizable by Theorem 2.16 Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod. The product charts Chapter2/FibredProduct · MChart.prod have the product retracts as models Chapter2/FibredProduct · LocalModel.prod, their overlaps are products Chapter2/FibredProduct · MChart.overlap_prod, and products of compatible charts are compatible Chapter2/FibredProduct · MChart.Compatible.prod; so they form an atlas of an M-polyfold structure on $X \times Y$ Chapter2/FibredProduct · MPolyfold.prod, in which products of charts of the factors are charts Chapter2/FibredProduct · MPolyfold.IsChart.prod. In a product chart the index is additive Chapter2/FibredProduct · MPolyfold.chartDegIndex_prod; choosing charts that realize $d_X(x)$ and $d_Y(y)$ gives $d_{X\times Y}(x,y) \le d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.degIndex_prod_le. If X and Y are tame, products of tame charts are tame charts, so $X \times Y$ is tame, and since a tame chart realizes the index (§II.14), $d_{X\times Y}(x,y) = d_X(x) + d_Y(y)$ Chapter2/FibredProduct · MPolyfold.isTame_prod. The text's "$d_{X\times Y} = d_X + d_Y$" is thus a statement about tame spaces. The projections are sc-smooth Chapter2/FibredProduct · MPolyfold.isScSmoothMap_fst Chapter2/FibredProduct · MPolyfold.isScSmoothMap_snd and so is the pairing $z \mapsto (f(z), g(z))$ of sc-smooth maps Chapter2/FibredProduct · MPolyfold.IsScSmoothOn.prodMk.

Proposition 2.46. A *local sc-diffeomorphism* $f : X \to Z$ is read as: every point has an open neighbourhood U that f maps sc-diffeomorphically onto an open subset Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo; such an f is continuous and sc-smooth Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.continuous Chapter2/FibredProduct · MPolyfold.IsLocalScDiffeo.isScSmoothMap. The fibred product $X_f \times_g Y = \{(x,y) : f(x) = g(y)\}$ Chapter2/FibredProduct · MPolyfold.fibredProduct is closed in $X \times Y$ Chapter2/FibredProduct · MPolyfold.isClosed_fibredProduct, hence paracompact, which the text does not mention. *Sub-M-polyfold.* Near (x,y), with $f|_U : U \to W$ an sc-diffeomorphism and $V = g^{-1}(W)$, the map $R(x',y') = ((f|_U)^{-1}(g(y')), y')$ on $U \times V$ is sc-smooth, idempotent, with image the fibred product near (x,y); in a product chart it is an sc-smooth retraction Chapter2/FibredProduct · MPolyfold.fibredSubDataL, and so the fibred product is a sub-M-polyfold Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct (symmetrically when g is the local sc-diffeomorphism Chapter2/FibredProduct · MPolyfold.fibredSubDataR Chapter2/FibredProduct · MPolyfold.isSubMPolyfold_fibredProduct'). *π_2 is a local sc-diffeomorphism.* On the fibred product near (x,y), π_2 is sc-smooth with inverse $y' \mapsto ((f|_U)^{-1}(g(y')), y')$, a pairing of sc-smooth maps with values in the fibred product, hence sc-smooth into the sub-M-polyfold by Proposition 2.21 (2) Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct; symmetrically for π_1 when g is a local sc-diffeomorphism Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_fst_fibredProduct. The exposition factors this inverse through $y' \mapsto (x,y')$, which is not sc-smooth unless x is a smooth point (a constant map is sc^0 only when its value lies in X_∞); the formula above has the same values and avoids it. *The index.* Proposition 2.24 along π_2 gives $d_{X_f\times_g Y}(x,y) = d_Y(y)$ for arbitrary Y Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_snd, and along π_1, when g is a local sc-diffeomorphism, $= d_X(x)$ Chapter2/FibredProduct · MPolyfold.degIndex_fibredProduct_eq_fst. *Tameness.* If Y is tame, the charts $\psi \circ \pi_2$, ψ a tame chart of Y restricted to $g^{-1}(W)$, are charts of the fibred product with tame models Chapter2/FibredProduct · MPolyfold.isTame_fibredProduct. Together: Proposition 2.46 Chapter2/FibredProduct · MPolyfold.proposition_2_46, and in its last statement $d = d_X = d_Y$ for arbitrary X, Y and, for tame X, Y, $2d = d_{X\times Y}$ by Lemma 4.D Chapter2/FibredProduct · MPolyfold.proposition_2_46_of_isLocalScDiffeo.

II.18 Strong bundle retractions (Definitions 2.47–2.49)

$U \lhd F$. The double filtration is $U_m \times F_k$ `Chapter2/Tame · doubleLevel`. $C \times F_0$ is a partial quadrant in $E \oplus F$, through $(\mathbb{R}^n \oplus W) \oplus F \cong \mathbb{R}^n \oplus (W \oplus F)$ `Chapter2/StrongBundle · prodQuadrantIso` `Chapter2/StrongBundle · IsPartialQuadrant.prod_univ`, and $U \times F_0$ is relatively open in it `Chapter2/StrongBundle · IsRelOpenInQuadrant.prod_univ` — so "sc-smooth on $(U \lhd F)[i] = U \oplus F^i$" makes sense (exposition Lemma 1.B of Chapter 1). Pairings of sc^k maps are sc^k, with the tangent map rearranged by an sc-operator `Chapter2/StrongBundle · scPair` `Chapter2/StrongBundle · tangentProdSwap` `Chapter2/StrongBundle · IsSck.scPair` `Chapter2/StrongBundle · IsScSmooth.scPair`, and the projections and inclusions of products are sc-operators `Chapter2/StrongBundle · ScBanach.prodFstOp` `Chapter2/StrongBundle · ScBanach.prodSndOp` `Chapter2/StrongBundle · ScBanach.prodInlOp` `Chapter2/StrongBundle · ScBanach.prodShiftInclOp`.

Definition 2.47. A strong bundle map maps $U \times F_0$ into $U' \times F_0'$, has first component independent of h and second linear in h, maps $U_m \times F_k$ into $U_m' \times F_k'$ for $k \le m+1$, is sc-smooth on $U \times F_0$, and its fibre-lifted map on $U \times F_1$ `Chapter2/Tame · shiftFiber` is sc-smooth for $E \oplus F^1$ `Chapter2/Tame · IsStrongBundleMap`; a strong bundle retraction is an idempotent one `Chapter2/Tame · IsStrongBundleRetraction`. *Lemma 5.A.* The base $r(u) = R(u,0)_1$ `Chapter2/Tame · IsStrongBundleMap.base` `Chapter2/StrongBundle · IsStrongBundleRetraction.apply_fst` is an sc-smooth retraction on U: sc-smooth as R composed with $u \mapsto (u,0)$ and the first projection, idempotent by the first component of $R \circ R = R$ `Chapter2/StrongBundle · IsStrongBundleRetraction.baseRetraction`. For $u \in O$, $R(u,h) = (u, \Gamma(u,h))$, so the second component of $R \circ R = R$ reads $\Gamma(u,\Gamma(u,h)) = \Gamma(u,h)$: $\Gamma(u,\cdot)$ is a projection `Chapter2/StrongBundle · IsStrongBundleRetraction.fiber_idem`. R is an sc-smooth retraction on $U \times F_0$ in $C \times F_0$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetraction` and, through the lift of the fibre component to F_1 (possible since R preserves the double filtration at $(0,1)$ `Chapter2/StrongBundle · IsStrongBundleRetraction.snd_mem_range` `Chapter2/StrongBundle · IsStrongBundleRetraction.toBase_shiftFiber_snd`), on $U \times F_1$ for $E \oplus F^1$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne`.

Definition 2.48. K is a local strong bundle retract if $K = R(U \times F_0)$ `Chapter2/Tame · IsStrongBundleRetract`, with $K_{m,k}$, $K[i]_m$ and $K[1] \subseteq E_0 \times F_1$ `Chapter2/Tame · strongLevel` `Chapter2/Tame · strongLevelShift` `Chapter2/Tame · strongShiftOne` and the projection p `Chapter2/Tame · strongProj`. $K = \{(u,h) : u \in O, \Gamma(u,h) = h\}$ `Chapter2/StrongBundle · IsStrongBundleRetraction.mem_image_iff`. *Lemma 5.B.* $K[1]$ is the image of R on $U \times F_1$ `Chapter2/StrongBundle · IsStrongBundleRetraction.toScRetractionOne_retract`, so K and $K[1]$ are retracts `Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract` `Chapter2/StrongBundle · IsStrongBundleRetract.isScRetract_one`; the base $p(K)$ is O `Chapter2/StrongBundle · IsStrongBundleRetract.base` `Chapter2/StrongBundle · IsStrongBundleRetract.base_eq`; and $p : K[i] \to O$ is sc-smooth, because $p \circ R = r \circ \mathrm{pr}_1$ is `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst` `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one`, as is the inclusion $K[1] \to K[0]$ `Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl`.

Definition 2.49. An sc-smooth section is sc-smooth from (O, C, E) into $(K, C \times F_0, E \oplus F)$ with $f(x)_1 = x$ `Chapter2/Tame · IsScSmoothSection`; an sc⁺-section is sc-smooth into $K[1]$ `Chapter2/Tame · IsScPlusSection` `Chapter2/Tame · IsScPlusSection.mem`. Composed with the inclusion $K[1] \to K[0]$ (Theorem 2.8), an sc⁺-section is an sc-smooth section `Chapter2/StrongBundle · IsScPlusSection.isScSmoothSection`.

II.19 Strong bundles over M-polyfolds (Definitions 2.50–2.53)

Definition 2.50. For $P : Y \to X$ with linear structures on the fibres, a strong bundle chart is a chart φ of X (of the structure — the text says only "a homeomorphism onto O"), a strong bundle retraction R on $U \subseteq C$ with base retract O, and a homeomorphism $\Phi : P^{-1}(V) \to K$ with $\Phi(y)_1 = \varphi(P(y))$, linear on fibres `Chapter2/StrongBundle · StrongBundleChart` `Chapter2/StrongBundle · StrongBundleChart.K`. K is a local strong bundle retract and a retract on $[0]$ and $[1]$ `Chapter2/StrongBundle · StrongBundleChart.isStrongBundleRetract` `Chapter2/StrongBundle · StrongBundleChart.isScRetract` `Chapter2/StrongBundle · StrongBundleChart.isScRetract_one` `Chapter2/StrongBundle · StrongBundleChart.model`, the

source $P^{-1}(V)$ is open `Chapter2/StrongBundle · StrongBundleChart.isOpen_source`, $\Phi(P^{-1}(V)) = K$ and $p(K) = O$ `Chapter2/StrongBundle · StrongBundleChart.image_source` `Chapter2/StrongBundle · StrongBundleChart.base_K`. "Bounded on the fibres" needs no hypothesis: the restriction of the continuous Φ to a fibre is continuous `Chapter2/StrongBundle · StrongBundleChart.continuous_fiber`. Two charts are compatible if the transitions $\Phi' \circ \Phi^{-1}$ `Chapter2/StrongBundle · StrongBundleChart.overlap` `Chapter2/StrongBundle · StrongBundleChart.transition` and back are sc-smooth between the overlaps on $[0]$ and, fibre-lifted, on $[1]$, and carry fibre components in F_1 to F_1' `Chapter2/StrongBundle · StrongBundleChart.Compatible` `Chapter2/StrongBundle · StrongBundleChart.Compatible.symm`; the last condition, implicit in the text, is what makes $(\Phi' \circ \Phi^{-1})[1]$ the restriction of $\Phi' \circ \Phi^{-1}$. Atlases and their equivalence `Chapter2/StrongBundle · StrongBundleAtlas` `Chapter2/StrongBundle · StrongBundleAtlas.Equiv`; Banachable fibres `Chapter2/StrongBundle · IsBanachable`.

Definition 2.51. A strong bundle is a continuous surjection with Banachable fibres and a strong bundle atlas, Y Hausdorff paracompact `Chapter2/StrongBundle · StrongBundle`. Each Φ is a chart of Y with model $(K, C \times F_0, E \oplus F)$ `Chapter2/StrongBundle · StrongBundleChart.toMChart`, compatibility of strong bundle charts on $[0]$ is compatibility of these charts `Chapter2/StrongBundle · StrongBundleChart.Compatible.toMChart`, and they form the atlas of the M-polyfold $Y[0]$ `Chapter2/StrongBundle · StrongBundle.atlas0` `Chapter2/StrongBundle · StrongBundle.toMPolyfold0` `Chapter2/StrongBundle · StrongBundle.isChart_toMChart`. $P : Y[0] \to X$ is sc-smooth: in the charts Φ and φ it is $p : K \to O$ (Lemma 5.B), and one pair of charts per point suffices `Chapter2/StrongBundle · StrongBundle.isScSmoothMap_proj`. The double filtration $Y_{m,k}$ is defined through all the charts of the atlas around a point `Chapter2/StrongBundle · StrongBundle.doubleLevel`.

$Y_{m,k}$ does not depend on the chart `Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff`. Let y lie in the domain of a chart Φ of the atlas with $\Phi(y) \in E_m \times F_k$, $k \le m+1$, and let Φ' be another chart around y; then $\Phi'(y) = (\Phi' \circ \Phi^{-1})(\Phi(y))$ `Chapter2/StrongBundlePullback · StrongBundleChart.transition_apply`. *The base component:* $\Phi'(y)_1 = \varphi'(\varphi^{-1}(\Phi(y)_1))$, and the base transition, sc-smooth and so sc^0, maps points of level m to points of level m `Chapter2/MPolyfold · MChart.transition_apply`. *The fibre component, $k \le m$:* $\Phi(y)$ lies on level k of $E \oplus F$ (levels decrease `Chapter2/StrongBundleOne · StrongBundle.range_toBase_mono`), and the $[0]$-transition, sc^0 on the overlap retract `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap`, keeps it on level k. *The case $k = m+1$:* the fibre component of $\Phi(y)$ lies in F_1, so its lift $\Phi(y)[1]$ to $E_0 \times F_1$ `Chapter2/StrongBundleOne · StrongBundleChart.liftOne` `Chapter2/StrongBundleOne · StrongBundleChart.toBase_liftOne` is a point of level m of $E \oplus F^1$ (level m of F^1 is F_{m+1} `Chapter1/ScBanach · ScBanach.shiftScale_toBase`); the $[1]$-transition, sc^0 for $E \oplus F^1$ on the overlap `Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one`, keeps it on level m, and its value is the lift $\Phi'(y)[1]$, because the F_1-condition of compatibility puts the fibre component of $\Phi'(y)$ in F_1'. So $\Phi'(y) \in E_m' \times F_{m+1}'$. The case $k \le m$ uses the $[0]$-transition only; the case $k = m+1$ uses exactly the F_1-condition made explicit in Definition 2.50.

The set $Y[1]$ and its topology. $Y_{0,1}$ is the set of points whose fibre component lies in F_1 in every chart of the atlas around them `Chapter2/StrongBundleOne · StrongBundle.oneSet`; by the F_1-condition one chart suffices `Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff`. $Y[1]$ is $Y_{0,1}$ as a type `Chapter2/StrongBundleOne · StrongBundle.One` with the topology generated by the sets $\{y \in \operatorname{dom}\Phi : \Phi(y)[1] \in W\}$, W open in $E_0 \times F_1$, as for X_m in §2.2 `Chapter2/StrongBundleOne · StrongBundle.isOpen_one_basic`. The inclusion $Y[1] \to Y$ is continuous `Chapter2/StrongBundleOne · StrongBundle.continuous_oneVal`: inside the domain of a chart Φ, an open subset of Y is the preimage under Φ of an open $W \subseteq E_0 \times F_0$ (Φ is a homeomorphism onto K `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq`), and its points in $Y[1]$ form the basic set of $(1 \times \iota)^{-1}(W)$. A continuous injection into the Hausdorff space Y makes $Y[1]$ Hausdorff `Mathlib/Topology/Separation/Hausdorff · T2Space.of_injective_continuous`.

The charts $\Phi[1]$ `Chapter2/StrongBundleOne · StrongBundle.oneChart`. $y \mapsto \Phi(y)[1]$ maps $P^{-1}(V) \cap Y[1]$ into $K[1]$ `Chapter2/StrongBundleOne · StrongBundleChart.liftOne_mem`, with inverse $z \mapsto \Phi^{-1}(z_1, \iota z_2)$, a point of $Y_{0,1}$ `Chapter2/StrongBundleOne · StrongBundle.symm_mem_oneSet` `Chapter2/StrongBundleOne · StrongBundle.oneChartInv` `Chapter2/StrongBundleOne · StrongBundle.liftOne_symm`. It is continuous by the definition of the topology. Its inverse is continuous: a basic open set $\{\Phi'(y)[1] \in W\}$ pulls back to the set of $z \in K[1]$ in the $[1]$-part of the overlap with $(\Phi' \circ \Phi^{-1})[1](z) \in W$; the overlap is $K \cap W_0$ for an open

W_0 Chapter2/StrongBundlePullback · StrongBundleChart.overlap_eq_inter and the $[1]$-transition is continuous on it, so this set is open in $K[1]$. The overlap of $\Phi[1]$ and $\Phi'[1]$ is the $[1]$-part of the overlap of Φ and Φ' Chapter2/StrongBundleOne · StrongBundle.oneChart_overlap, and their transition map is the $[1]$-transition $(\Phi' \circ \Phi^{-1})[1]$ Chapter2/StrongBundleOne · StrongBundle.oneChart_transition, sc-smooth in both directions by Definition 2.50; so the charts $\Phi[1]$ are compatible Chapter2/StrongBundleOne · StrongBundle.oneChart_compatible and form an atlas of $Y[1]$ Chapter2/StrongBundleOne · StrongBundle.atlas1. $Y[1]$ is metrizable: P restricted to $Y[1]$ is continuous into the paracompact Hausdorff X, and over the domain V of a chart of X its preimage is the source of a chart $\Phi[1]$, homeomorphic to a retract and so metrizable Chapter2/Metrizable · MChart.metrizableSpace_source; the pull-back form of Smirnov's theorem applies Chapter2/Metrizable · metrizableSpace_of_pullback Chapter2/StrongBundleOne · StrongBundle.metrizableSpace_one. Hence $Y[1]$ is paracompact Chapter2/StrongBundleOne · StrongBundle.paracompactSpace_one, which an M-polyfold must be and the text does not mention, and $Y[1]$ **is an M-polyfold** Chapter2/StrongBundleOne · StrongBundle.toMPolyfold1 whose charts include the $\Phi[1]$ Chapter2/StrongBundleOne · StrongBundle.isChart_oneChart.

$P[1]$ and the inclusion $Y[1] \to Y[0]$. In the charts $\Phi[1]$ and φ, $P[1]$ is the projection $K[1] \to O$ of Lemma 5.B Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_fst_one, and one pair of charts per point suffices Chapter2/LevelPolyfold · MPolyfold.IsScSmoothOn.of_exists_charts; so $P[1] : Y[1] \to X$ **is sc-smooth** Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_proj1. In the charts $\Phi[1]$ and Φ the inclusion $Y[1] \to Y[0]$ is $(u, h) \mapsto (u, \iota h)$ Chapter2/StrongBundle · IsStrongBundleRetract.isScSmoothRetractMap_shiftIncl, so it is sc-smooth Chapter2/StrongBundleOne · StrongBundle.isScSmoothMap_oneVal (not stated in the text; it gives that an sc⁺-section is an sc-smooth section).

Definition 2.52. A section has $P \circ f = \mathrm{id}$ Chapter2/StrongBundle · IsSection; it is sc-smooth if sc-smooth into $Y[0]$ Chapter2/StrongBundle · StrongBundle.IsScSmoothSection, and an sc⁺-section if in every chart its fibre values lie in F_1 and its fibre-lifted representative Chapter2/StrongBundle · StrongBundle.localRepOne is sc-smooth into $K[1]$ Chapter2/StrongBundle · StrongBundle.IsScPlusSection. *This is the text's definition* Chapter2/StrongBundleOne · StrongBundle.isScPlusSection_iff: f is an sc⁺-section iff it is a section and $f = \iota \circ g$ for an sc-smooth $g : X \to Y[1]$, ι the inclusion. Forward: the values of f lie in $Y_{0,1}$ (one chart suffices), which defines g; on the domain of a chart φ of X under a strong bundle chart Φ, $g = \Phi[1]^{-1} \circ (\text{the local representative}) \circ \varphi$, which gives the continuity of g, and the representative of g in the charts φ, $\Phi[1]$ is the chart-wise representative, sc-smooth by hypothesis. Backward: the fibre values lie in F_1 because g takes values in $Y_{0,1}$, and the chart-wise representative is the representative of g in φ, $\Phi[1]$, sc-smooth by Definition 2.14 (a chart with empty domain is handled separately).

Definition 2.53. $f^*Y = \{(z, y) : P(y) = f(z)\}$ Chapter2/StrongBundle · pullbackSpace with the projections Chapter2/StrongBundle · pullbackProj Chapter2/StrongBundle · pullbackProj', $P \circ P' = f \circ P_f$ Chapter2/StrongBundle · pullback_comm; P_f is continuous and f^*Y closed in $Z \times Y$ Chapter2/StrongBundle · continuous_pullbackProj Chapter2/StrongBundle · isClosed_pullbackSpace.

II.20 Proposition 2.54: the pull-back strong bundle

The pulled-back retraction. For a retraction r' on U' and $g : O' \to O$ sc-smooth between retracts, $R_g(w, h) := (r'(w), \Gamma(g(r'(w)), h))$ Chapter2/StrongBundle · pullbackRetraction, where $g(r'(w)) \in O \subseteq U$ Chapter2/StrongBundle · pullback_mem_O Chapter2/StrongBundle · pullback_mem_U. It is a strong bundle retraction Chapter2/StrongBundle · isStrongBundleRetraction_pullback: it maps $U' \times F_0$ into itself; its first component does not depend on h and its second is linear in h; it preserves the double filtration (r' and $g \circ r'$ preserve levels, and R preserves the double filtration); on $[0]$ it is the pairing of $r' \circ \mathrm{pr}_1$ with the second component of $R \circ ((g \circ r') \circ \mathrm{pr}_1, \mathrm{pr}_2)$, sc-smooth by the chain rule and the pairing lemma, and likewise on $[1]$; and $R_g \circ R_g = R_g$ because $r' \circ r' = r'$ and $\Gamma(o, \cdot)$ is a projection for $o = g(r'(w)) \in O$ (Lemma 5.A). Its image is $K_g = \{(w, h) : w \in O', (g(w), h) \in K\}$ Chapter2/StrongBundle · pullback_image Chapter2/StrongBundle · isStrongBundleRetract_pullback.

The fibres. $P_f^{-1}(z) \to P^{-1}(f(z))$, $(z, y) \mapsto y$, is a bijection Chapter2/StrongBundlePullback · pullbackFiberEquiv;

transporting the linear structure along it Chapter2/StrongBundlePullback · pullbackFiber_addCommGroup Chapter2/StrongBundlePullback · pullbackFiber_module makes it a linear homeomorphism Chapter2/StrongBundlePullback · pullbackFiberCLE, so the fibres of f^*Y are Banachable.

The charts. The data: a strong bundle chart c of Y, a chart ψ of Z with $f(\operatorname{dom}\psi) \subseteq \operatorname{dom}\varphi$, a retraction r_Z onto the model of ψ, and the sc-smoothness of $g = \varphi \circ f \circ \psi^{-1}$ Chapter2/StrongBundlePullback · PullbackChartData Chapter2/StrongBundlePullback · PullbackChartData.g Chapter2/StrongBundlePullback · PullbackChartData.g_smooth. The chart $\Phi_f(z,y) = (\psi(z), \operatorname{pr}_F\Phi(y))$ maps $P_f^{-1}(\operatorname{dom}\psi)$ into K_g — because $\Phi(y) = (g(\psi(z)),\cdot) \in K$ Chapter2/StrongBundlePullback · PullbackChartData.g_apply Chapter2/StrongBundlePullback · PullbackChartData.mem_Kg — with inverse $(w,h) \mapsto (\psi^{-1}(w), \Phi^{-1}(g(w),h))$ Chapter2/StrongBundlePullback · PullbackChartData.symm_mem_pullback Chapter2/StrongBundlePullback · PullbackChartData.invFun; both are continuous, it covers ψ and is linear on fibres, so it is a strong bundle chart of f^*Y Chapter2/StrongBundlePullback · PullbackChartData.toChart Chapter2/StrongBundlePullback · PullbackChartData.image_eq_Kg.

Compatibility. Sc-operators and pairings are sc-smooth between retracts Chapter2/StrongBundlePullback · ScOperator.isScSmoothRetractMap Chapter2/StrongBundlePullback · IsScSmoothRetractMap.scPair Chapter2/StrongBundlePullback · isScRetract_univ, and overlaps of strong bundle charts are retracts on $[0]$ and $[1]$ Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap Chapter2/StrongBundlePullback · StrongBundleChart.isScRetract_overlap_one. On the overlap of two pulled-back charts the transition map is

$$(w,h) \mapsto (\psi_2\psi_1^{-1}(w),\ \operatorname{pr}_F(\Phi_2\Phi_1^{-1})(g_1(w),h))$$

Chapter2/StrongBundlePullback · PullbackChartData.transitionFormula Chapter2/StrongBundlePullback · PullbackChartData.transition_eqOn: its first component is the transition of Z composed with the projection, its second the fibre component of the transition of Y composed with $(w,h) \mapsto (g_1(w),h)$ Chapter2/StrongBundlePullback · PullbackChartData.gPair, which maps the overlap into the overlap of Y's charts Chapter2/StrongBundlePullback · PullbackChartData.gPair_mem. Each factor is sc-smooth between retracts, and Theorem 2.8 and the pairing lemma give the sc-smoothness on $[0]$ Chapter2/StrongBundlePullback · PullbackChartData.transition_isScSmooth; the same with F^1 on $[1]$ Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_eqOn Chapter2/StrongBundlePullback · PullbackChartData.transitionOne_isScSmooth; fibre components in F_1 go to F_1 because the transitions of Y have this property Chapter2/StrongBundlePullback · PullbackChartData.transition_snd_mem_range. So any two pulled-back charts are compatible Chapter2/StrongBundlePullback · PullbackChartData.compatible_toChart.

Proposition 2.54. f^*Y is paracompact: Z and $Y[0]$ are metrizable (Theorem 2.16), so $Z \times Y$ and its subspace f^*Y are metrizable, hence paracompact Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace — a requirement of Definition 2.51 that the text does not mention. The pulled-back charts, built around every point from a chart of Y's atlas and a chart of Z restricted to $f^{-1}(\operatorname{dom}\varphi)$, cover f^*Y and form an atlas Chapter2/StrongBundlePullback · pullbackAtlas. With the continuous surjection P_f and its Banachable fibres this is a strong bundle structure on $f^*Y \to Z$ Chapter2/StrongBundlePullback · proposition_2_54 ("the easy proof is left to the reader").

II.21 What the formalization taught us

- **Theorem 2.16 needs no added hypothesis, and not the text's route.** The text's proof applies Proposition 2.57 to X_m without checking that X_m is regular. A partition of unity of X, pulled back along the continuous inclusion, metrizes X_m directly Chapter2/Metrizable · metrizableSpace_of_pullback, and the same lemma proves Smirnov's theorem Chapter2/Metrizable · metrizableSpace_of_locally, Proposition 2.19 (3) Chapter2/Metrizable · proposition_2_19_3 and the paracompactness of f^*Y Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace. With it every paracompactness hypothesis carried by the constructions is discharged Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype.
- **Compatibility is local, and the text needs it.** That the charts compatible with an atlas are compatible with each other is not in the text; it is the chain rule plus the locality of sc-smoothness Chapter2/

MPolyfold · MChart.compatible_of_locally, and every chart-independence claim of §2.2 rests on it.

- TO **is a fixed-point set.** Proposition 2.4 and Proposition 2.6 are one-line consequences of $p \in Tr(TU) \iff Tr(p) = p$ Chapter2/Retracts · ScRetraction.mem_tangent_iff, which the text's proofs use without stating.
- **The tangent bundle needs the locality of** Tf**.** That Definition 2.17's relation is an equivalence relation needs the chain rule on the triple overlap and the agreement of Tf computed on an open subset with Tf computed on the whole retract Chapter2/TangentBundle · retractTangentMap_inter_isOpen Chapter2/TangentBundle · MPolyfold.TangentPre.transport_transport.
- **Level** 1 **suffices for §2.3.** Proposition 2.30, Theorem 2.31 and Corollary 2.32 hold at points on level 1 and for sc^1 germs Chapter2/Boundary · proposition_2_30_reduced Chapter2/Boundary · theorem_2_31 Chapter2/DegeneracyIndex · corollary_2_32; the text's "closure of $(T^R_x O)_\infty$" is the density of level-1 points in the kernel of a finite-rank map Chapter2/DegeneracyIndex · ScBanach.subset_closure_ker_inter_range_toBase.
- **Proposition 2.35 needs only tameness (2), and Lemma 2.40 needs (2) through it.** $Dr(x)E_x \subseteq E_x$ holds for every retraction Chapter2/TameRetracts · ScRetraction.scDeriv_mapsTo_minimalSubspace; that the retraction keeps points of a face in that face is the ray lemma Chapter2/TameRetracts · ray_zeroSet, which uses Proposition 2.35.
- **Three slips in the exposition.** "$|P_1 z - e|_0 \le |z - e|_0$" in the exposition's proof of Proposition 2.43 is false for a general norm on E_0; the formal neighbourhood intersects balls with their P_i-preimages Chapter2/TameRetracts · proposition_2_43. The "transport by φ^{-1}" in Theorem 2.45 needs a shrinking so that the retraction maps its domain into itself Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local. In Proposition 2.46 the inverse of π_2 cannot be factored through the constant-in-x map $y' \mapsto (x, y')$, which is not sc-smooth at a non-smooth x Chapter2/FibredProduct · MPolyfold.isScDiffeoOn_snd_fibredProduct.
- **Tameness is needed only for the additivity of the index under products.** $d_{X\times Y} \le d_X + d_Y$ in general Chapter2/FibredProduct · MPolyfold.degIndex_prod_le, with equality for tame factors Chapter2/FibredProduct · MPolyfold.isTame_prod; the identities $d = d_X = d_Y$ of Proposition 2.46 hold without tameness Chapter2/FibredProduct · MPolyfold.proposition_2_46_of_isLocalScDiffeo.
- **Definition 2.50 needs two explicit conditions.** The base homeomorphism of a strong bundle chart must be a chart of X, and the $[1]$-transition must preserve fibre components in F_1 Chapter2/StrongBundle · StrongBundleChart.Compatible. The second condition is exactly what makes $Y_{0,1}$, and so the M-polyfold $Y[1]$, independent of the chart Chapter2/StrongBundleOne · StrongBundle.mem_oneSet_iff Chapter2/StrongBundleOne · StrongBundle.mem_doubleLevel_iff.
- **Proposition 2.36 does not need the equality case of Theorem 2.31.** For a tame retraction the smooth vectors of the splitting are $Dr(x)T^{-1}(e_{i_j}, 0)$, dual to the coordinates by Proposition 2.35, and $T^R_x O$ is an sc-subspace because $Dr(x)$ preserves E_x Chapter2/PartialCone · proposition_2_36_partialCone.
- **Corners at once, not by induction.** The intersections F^σ_x of Proposition 2.44 are retracts of $r \circ P_\sigma$, by Lemma 4.C for each $i \in \sigma$; no identification of the germ of $F_1 \cap F_2$ with a face of F_1 is needed Chapter2/CornerGerms · corner_retraction.

Part III. Sc-Fredholm theory and the implicit function theorem (HWZ Chapter 3, §§3.1–3.8)

Chapter 3 is the reason for the first two chapters: it defines the sections of strong bundles to which an implicit function theorem applies, the *sc-Fredholm sections*, and proves that theorem. The engine is a fixed point problem with finitely many parameters, $w = B(a, w)$, $a \in \widetilde{C} \subseteq \mathbb{R}^n$, $w \in W$, where $B(a, \cdot)$ contracts with an arbitrarily small constant near 0 *on every level* W_m of the scale. Its solution $w = \delta(a)$ is the *solution germ*; the germ implicit function theorem (Theorem 3.33) says that δ is as regular as B. Around it the chapter builds: germs and their calculus (§III.1), sections of strong bundles in charts (§III.2), fillers (§§III.3–III.4), basic germs and their linearization (§III.5), sc-Fredholm germs and sections with the stability theorem 3.15 (§III.6); the germ implicit function theorem itself (§§III.7–III.11); local regularity and compactness, Theorems 3.40 and 3.12 (§§III.12–III.13); the finite-dimensional geometry of good position and the inverse and implicit function theorems in partial quadrants, which the formalization proved first and the rest uses (§§III.14–III.17); the basic class and local solvability (§§III.18–III.20); the linearization of an sc-Fredholm section at a zero, good position, and the graph description of the solution set of Theorem 3.54 (§§III.21–III.23); the tangent germ property, its invariance, and the manifold structure of solution sets (§III.24); and Lemmas 3.59–3.61 of the basic germ criterion (§III.25). Sections III.27–III.39 were added after the part was first assembled, for the Lean written since: tameness under sc-diffeomorphisms (§III.27), Corollary 3.39 (§III.28), good complements whose projection maps the quadrant into itself and local solvability with such a complement (§III.29), the index in Proposition 3.52 (§III.30), the linearization at a zero in charts and its chart independence (§III.31), Theorem 3.54 (1) at the nearby zeros (§§III.32–III.33), Theorem 3.54 (2) (§§III.34–III.36), the implicit function theorems 3.54, 3.23, 3.19, 3.24, 3.56, 3.57 assembled (§III.37), the basic germ criterion, Theorem 3.58 (§III.38), and what this addendum taught us (§III.39). Sections III.40–III.46 were added in a second addendum: Theorem 3.18 (§III.40); Theorem 3.54 (1) with the text's hypotheses, by an implicit function theorem at the nearby zero (§§III.41–III.43); the dimension of the solution set as the Fredholm index and its local constancy (§III.44); the kernel as tangent space, good position at interior zeros, and the filler condition of §III.33 as a theorem (§III.45); and what the second addendum taught us (§III.46). Section III.47 repairs a tameness hypothesis, and §III.48 records the switch of Definition 3.11 to the text's germ reading; the passages above that use the filling or the sc$^+$-section have been rewritten for it.

What is proved, and what is not. The germ implicit function theorem is proved *with the text's hypotheses*: no hypothesis had to be added to Theorems 3.29–3.35. What the formalization adds is precision about *where* the solution germ is regular. A germ is formalized as a fixed sequence of sets, and the proof shows that δ is sc^k on the germ shifted by $s_k = 2^k - 1$ levels; "if f is sc-smooth so is its solution germ" therefore holds for every k on a germ depending on k (§III.11). Elsewhere hypotheses had to be added: the good complement must be *closed* in Lemma 3.55 and in Definition 3.25; and Theorem 3.27 needs two regularity conditions on its data, each of which the text's proof asserts and each of which fails in an example (§III.24) — for solution sets both are theorems (§§III.29, III.35–III.36). These are the only ones. Definition 3.11 is

formalized as the text writes it, with the filling and the sc⁺-section sc-smooth *germs* whose levels may shrink, and every theorem built on it holds in that reading with the text's conclusions (§§III.6, III.48); the first formalization's whole-chart reading was narrower. Definition 6.1 (the index of the germ at a smooth point) and Proposition 6.2 (its constancy along sc-smooth paths), from Chapter 6 of the text, are proved with the text's hypotheses — tame germ data, no bump functions, the locally defined sc⁺-sections of Definition 6.1 — for paths on $\mathbb{R}$, by the text's device run on the restriction of the bundle to a chart domain; the first proof, with global sections, assumed bump functions (§III.49). The implicit function theorems of §§III.23–III.45 were first stated with a tameness hypothesis that no strong bundle over a nonempty base satisfies, and were vacuous; they now assume germ data and good position in tame charts, as the text does (§III.47). Theorem 3.54 (1) was first proved only under condition (3) of Definition 3.4 at the nearby zeros (§III.33); it is now proved with the text's hypotheses, by an argument that is not the text's (§§III.41–III.43), and the condition is a theorem (§III.45). The local solvability theorems 3.43 and 3.47 are proved in full, their clause (1) ("the solution map is an sc-smooth germ") in the level-wise sense with k-dependent neighbourhoods (§III.20). Of Theorem 3.54 the graph description of the solution set near a zero in good position is §III.23. Its conclusion (2), the manifold structure, and the global Theorems 3.56, 3.57 and Corollary 3.24 are proved with the text's hypotheses (§§III.34–III.37); its conclusion (1), good position at the nearby zeros, first under the added filler condition (§III.33), then with the text's hypotheses (§III.43), and with it the local Theorems 3.54, 3.23 and 3.19 (§§III.37, III.43), with the dimension of the solution set equal to the Fredholm index (§III.44). Theorem 3.18 is §III.40. The basic germ criterion (Theorem 3.58) is proved in the coordinates of its proof (§III.38), with Lemmas 3.59–3.61 (§III.25).

Seven structural choices. (1) Germs are fixed sequences of sets, and the calculus the text uses on them silently is proved (§III.1). (2) Contraction germs carry explicit contraction regions instead of the text's balls, because the regions of the iterated germ $f^{(1)}$ must be unbounded in the tangent directions (§III.7). (3) "A strong bundle chart" in the definitions of §3.1 is a chart *compatible* with the atlas (§III.2). (4) Tameness, which the text assumes throughout, is used only in Theorem 3.54, and only for the chart of the germ data at the zero: through Proposition 2.35, $(1 - Dr(0))E \subseteq E_0$, and through the tameness of the model of the chart $\psi \circ \varphi$ (§III.27). The theorems assume germ data and good position in tame charts, as Definitions 3.50, 3.51 and 3.53 do; the first formalization assumed every compatible chart tame, which cannot be met (§III.47). (5) The coordinate change of Theorem 3.38 is linear, which makes the transport of solution sets in §3.5 linear algebra. (6) Classical regularity in the appendix is Mathlib's C^k within a set, and the iterations of the text's Lemmas 3.76 and 3.82 are replaced by bootstrapping a derivative formula. (7) In Definition 3.11 the filling g and the sc⁺-section s are germs on germs of neighbourhoods of their own, and the basic germ is shrunk to the germ G'' of the points that ψ^{-1} sends into both, so that the nearby zeros produced by the implicit function theorem are points where g and s are sc¹ (§§III.6, III.48).

Map.

- `Chapter3/Germ`, `Chapter3/GermLevels`: Definitions 3.1, 3.2; the calculus of germs; Propositions 1.16–1.18 for germs.
- `Chapter3/StrongBundleSections`: the section algebra, compatible charts, Definitions 3.3, 3.6, 3.11, 3.14, Theorem 3.15.
- `Chapter3/Filler`, `Chapter3/FillerLinearAlgebra`, `Chapter3/FillerLinearization`: Definition 3.4, Lemmas 3.16, 3.17, Proposition 3.5, Remark 3.49.
- `Chapter3/BasicGermLinearization`: Lemma 3.9 at 0, Proposition 3.10.
- `Chapter3/ContractionGerm`, `Chapter3/GermIFT`: Definition 3.28, Theorems 3.29, 3.30, Lemma 3.31, the induction (Lemma 3.32, Theorem 3.33), Corollary 3.34, Theorem 3.35.
- `Chapter3/LocalCompactness`, `Chapter3/SectionCompactness`: the parametrized fixed point theorem, Definition 3.7, Theorem 3.40, Theorem 3.12, Corollary 3.13.
- `Chapter3/GoodPosition`, `Chapter3/GoodPositionGraph`: Definition 3.21, Proposition 3.22, Lemmas 3.45, 3.65, Proposition 3.69, Lemma 3.70.
- `Chapter3/QuadrantInverse`, `Chapter3/QuadrantIFT`: Theorems 3.72, 3.78 with their lemmas.
- `Chapter3/BasicClass`, `Chapter3/SolutionLinearization`, `Chapter3/LocalSolvability`: Theorem 3.38, Corollary 3.41, Lemmas 3.42, 3.44, 3.46, Theorem 3.43, Corollary 3.47.

- `Chapter3/FredholmGermLinearization`, `Chapter3/GoodPositionGerm`, `Chapter3/GoodPositionOpen`, `Chapter3/GlobalIFT`: Proposition 3.52, Definition 3.53, Lemma 3.55, Theorem 3.54 (the linear part and the graph description), Theorems 3.56, 3.57 from the tangent germ property.
- `Chapter3/TangentGerm`: Definition 3.25, Proposition 3.26, Theorem 3.27.
- `Chapter3/BasicGermCriterion`, `Chapter3/InverseFamily`: Lemmas 3.59, 3.60 (abstract form), Lemma 3.61.
- Added later (§§III.27–III.39): `Chapter2/TameInvariance` (tameness under sc-diffeomorphisms); `Chapter3/BasicClassComposition` (Corollary 3.39, the calculus of strong bundle isomorphisms); `Chapter3/GoodComplementProjection`, `Chapter3/LocalSolvabilityClosed` (good complements with a projection, Theorems 3.43, 3.47 with a closed projecting complement); `Chapter3/FredholmGermIndex` (Proposition 3.52 with its index); `Chapter3/LinearizationCharts` (the linearization at a zero in charts, chart independence, Definition 3.53 from a non-centred chart); `Chapter3/GlobalIFTFull`, `Chapter3/GlobalIFTNearbyGraph`, `Chapter3/GlobalIFTNearbyKernel`, `Chapter3/GlobalIFTNearbySurj`, `Chapter3/GlobalIFTNearbyPerturb` (Theorem 3.54 (1) at the nearby zeros); `Chapter3/GlobalIFTCenter`, `Chapter3/SolutionGermRegularity` (Theorem 3.54 (2) at the centre); `Chapter3/SolutionGermRecentre` (the re-centring, regular data, Proposition 3.73 / Lemma 3.77 with weak hypotheses); `Chapter3/GlobalIFTTheorems` (Theorem 3.27 with locally constant dimension, Theorems 3.54, 3.23, 3.19, Corollary 3.24, Theorems 3.56, 3.57); `Chapter3/BasicGermCriterionProof`, `Chapter3/BasicGermCriterionIndex` (Theorem 3.58).
- Added in the second addendum (§§III.40–III.46): `Chapter3/ParamPerturbation` (Theorem 3.18, the pull-back bundle, inert parameters, the linearization of the perturbed section); `Chapter3/GlobalIFTNearbyOntoLemmas`, `Chapter3/GlobalIFTNearbyOntoGerm`, `Chapter3/GlobalIFTNearbyOntoXi`, `Chapter3/GlobalIFTNearbyOntoCurve`, `Chapter3/GlobalIFTNearbyOnto` (Theorem 3.54 (1) with the text's hypotheses, Theorems 3.54, 3.23, 3.19 with them); `Chapter3/GlobalIFTIndex` (the index of the linearization in every chart, dimension = index, the index locally constant, the kernel as tangent space, good position at interior zeros, the filler condition as a theorem).
- Added with the repair of the tameness hypothesis (§III.47): `Chapter3/TameGerms` (tame charts, Definitions 3.50, 3.51, 3.53 in a tame chart, good position read in a tame chart, Theorem 3.15 for tame germ data); `Chapter3/TameVacuity` (a non-tame retraction onto a product with a point).
- Added with the germ reading of Definition 3.11 (§III.48): `Chapter3/GermVsChart` (a germ sc-smooth on no neighbourhood); `Chapter3/GermFilling`, `Chapter3/GermFillingChart` (Definitions 3.6 and 3.11 on germs, the bridges to the whole-chart notions); `Chapter3/GermCalculus`, `Chapter3/GermCalculusParam`, `Chapter3/GermPushforward`, `Chapter3/GermFillerLinearization` (sums, chain rules with whole-chart maps, the fill defect, push-forwards and products of germs, Proposition 3.5 for filling germs); `Chapter3/BasicGermShrink`, `Chapter3/BasicGermShrinkIFT` (the shrunk basic germ G'' and Corollary 3.41 on it); and the files `Chapter3/…G` (`FredholmSectionsG`, `TameGermsG`, `GoodPositionGermG`, `FredholmGermLinearizationG`, `GlobalIFTNearbyGraphG`, `GlobalIFTTheoremsG`, …), where the canonical names of Definitions 3.11, 3.14, 3.50, 3.51, 3.53 and of the theorems built on them now live. Their original files (`StrongBundleSections`, `GlobalIFT`, …) keep the whole-chart versions, named `…WholeChart` / `…_wholeChart`.
- Added in the fourth addendum (§§III.47, III.49): `Chapter3/TameGermData` (condition (3) of Definition 3.4 as a condition on the chart; a tame atlas whose chart carries no germ data); `Chapter3/IndexAlongPaths`, `Chapter3/IndexAlongPathsDevice` (Definition 6.1, bump functions on $\mathbb{R}^n \times X$, the section along a path, the slice of a section of the pull-back, Proposition 6.2).
- Added in the fifth addendum (§III.49): `Chapter3/RestrictOpen` (the restriction of M-polyfolds and strong bundles to open sets); `Chapter3/RestrictOpenGerms` (germ data and the index on the restriction); `Chapter3/IndexLocallyConstantLocal` (sc$^+$-sections through the fibre points from a chart over the whole base, the index along the zero set without bump functions); `Chapter3/IndexAlongPathsLocal` (Definition 6.1 with local sections, Proposition 6.2 with the text's hypotheses); `Chapter3/IndexAlongPathsLocalBump` (the two readings of Definition 6.1 agree under bump functions).

Throughout, P and Q are finite-dimensional normed spaces with the constant sc-structure (the text's $\mathbb{R}^n$ and $\mathbb{R}^N$), $\widetilde{C} \subseteq P$ a set containing 0, convex, with nonempty interior (the text's $[0,\infty)^k \oplus \mathbb{R}^{n-k}$), W an sc-Banach space, $E = P \oplus W$ with levels $P \times W_m$ `Chapter3/ContractionGerm · ScBanach.paramProd`. Because all levels of P coincide, the condition "$\widetilde{C} \times W$ is quadrant-like" is just "$0 \in \widetilde{C}$, convex, nonempty interior" `Chapter3/ContractionGerm · isQuadrantLike_const_iff` `Chapter3/ContractionGerm · IsQuadrantLike.paramProd`.

III.1 Germs and their calculus (Definitions 3.1, 3.2)

Definition 3.1. An *sc-germ of neighbourhoods* around $0 \in C$ is a family of sets $U_m \subseteq E_m$, each of the form $O \cap C_m$ with O open in E_m, containing 0, and nested along the inclusions: $\iota U_{m+1} \subseteq U_m$ `Chapter3/Germ · ScNbhdGerm`. Its *tangent germ* has levels $U_{m+1} \times E_m$ `Chapter3/Germ · ScNbhdGerm.tangent`. The text's example, a relatively open $U \ni 0$ with $U_m = U \cap E_m$, is `ofSet` `Chapter3/Germ · ScNbhdGerm.ofSet`. The germ calculus needs one property of C: every level C_m is convex with nonempty interior and $0 \in C$ ("quadrant-like") `Chapter3/Germ · IsQuadrantLike`. Partial quadrants have it `Chapter3/Germ · IsPartialQuadrant.isQuadrantLike`, and it passes to TC `Chapter3/Germ · IsQuadrantLike.tangentSet`. It makes every U_m a set of unique differentiability `Chapter3/Germ · ScNbhdGerm.uniqueDiffWithinAt`, the intersection of an open and a convex set `Chapter3/Germ · ScNbhdGerm.relOpenConvex`.

A germ is a *fixed* sequence of sets. The text says "the size of the sets U_m does not matter", i.e. it reads germs up to shrinking; the formalization cannot, and must say on which sequence each statement holds. One trivial remark becomes important in §III.9: the re-indexed sequence $(U_{m+1})_m$ is in general not a germ, since U_{m+1} is relatively open in C_{m+1} and not in C_m; it is one when all levels coincide, i.e. for germs in the finite-dimensional parameter space P.

Definition 3.2. f is an *sc⁰-germ* on $G = (U_m)$ if $f(U_m) \subseteq F_m$ and the level map $f_m : U_m \to F_m$ is continuous for every m `Chapter3/Germ · IsSc0Germ`. It is an *sc¹-germ with derivative family* D if it is sc⁰, differentiable at every $x \in U_1$ within U_1 in the sense of Definition 1.14 (increments in E_1, remainder measured in F_0) with derivative $D(x)$, and its tangent map $(x,h) \mapsto (f(x), D(x)h)$ is an sc⁰-germ on the tangent germ `Chapter3/Germ · IsSc1GermWith`; an sc¹-germ if some such D exists `Chapter3/Germ · IsSc1Germ`, in which case a derivative family is chosen `Chapter3/Germ · scDerivGerm` and gives the tangent germ `Chapter3/Germ · tangentMapGerm`. On quadrant-like C the derivative is unique on U_1 `Chapter3/Germ · IsSc1GermWith.eq_of_uniqueDiffWithinAt`, so the choice does not matter there `Chapter3/Germ · IsSc1Germ.tangentMapGerm_eqOn`. Then sc^k is defined by recursion — sc⁰ for $k = 0$, and sc¹ with sc^k tangent germ for $k+1$ `Chapter3/Germ · IsSckGerm` — and sc-smooth means sc^k for all k `Chapter3/Germ · IsScSmoothGerm`. These are the notions of Part I, §I.11, with $U \cap E_m$ replaced by U_m, and the bridge is exact: an sc^k map on a relatively open $U \ni 0$ is an sc^k-germ on `ofSet U` `Chapter3/Germ · IsSck.isSckGerm` `Chapter3/Germ · IsScSmooth.isScSmoothGerm`.

The calculus of germs. The text restricts, composes and pairs germs without comment. All of it is proved, with the proofs of Part I, §§I.11–I.12. (1) sc^k-germs depend only on the values on U_0 `Chapter3/Germ · IsSckGerm.congr` and restrict to smaller germs `Chapter3/Germ · IsSckGerm.mono` — by induction on k, using that the tangent germs of the smaller germ and of the larger one agree on the smaller tangent germ `Chapter3/Germ · IsSc1Germ.tangentMapGerm_mono`. (2) *The chain rule.* A composition $g \circ f$ makes sense level by level only if f maps U_m into the domain V_m of g for every m `Chapter3/Germ · ScNbhdGerm.MapsToGerm`; then sc⁰-germs compose `Chapter3/Germ · IsSc0Germ.comp`, the derivative of $g \circ f$ at $x \in U_1$ is $Dg(f(x)) \circ Df(x)$ — proved as in §I.12 with the mean value inequality along segments in V_1 and the uniform smallness of $Dg(y) - Dg(y_0)$ on compact sets `Chapter3/Germ · IsSc1GermWith.hasFDerivWithinAt_comp` — so $g \circ f$ is sc¹ with $T(g \circ f) = Tg \circ Tf$ on the tangent germ `Chapter3/Germ · IsSc1GermWith.comp` `Chapter3/Germ · IsSc1Germ.comp`, and by induction on k it is sc^k `Chapter3/Germ · IsSckGerm.comp` `Chapter3/Germ · IsScSmoothGerm.comp`. (3) An sc-operator T is an sc¹-germ with derivative T itself `Chapter3/Germ · ScOperator.isSc1GermWith`, its tangent map is the sc-operator (T,T) `Chapter3/Germ · ScOperator.tangentOp`, and so it is an sc-smooth germ on every germ `Chapter3/Germ · ScOperator.isSckGerm` `Chapter3/Germ · ScOperator.isScSmoothGerm`; composing with one on the outside preserves sc^k `Chapter3/Germ · IsSckGerm.operator_comp`. (4) The pairing (f,g) of two sc^k-germs is an sc^k-germ `Chapter3/Germ · pairMap` `Chapter3/Germ · IsSckGerm.pair`, through the reshuffle $TF \oplus TG \cong T(F \oplus G)$

Chapter3/Germ · tangentProdComm. (5) Where the text says "near 0" it shrinks germs: for an sc⁰-germ f with $f(0)=0$ and a germ G' the sets $\{x \in U_m : f_m(x) \in U'_m\}$ form the largest sub-germ on which f maps into G' Chapter3/GermFilling · ScNbhdGerm.preimage Chapter3/GermFilling · ScNbhdGerm.preimage_mapsToGerm.

Propositions 1.16–1.18 for germs. The text applies these to germs ("recalling Proposition 1.17" in Lemma 3.9; Lemma 3.31; Corollary 3.34) but states them for relatively open sets. For an sc¹-germ with derivative family D and $x \in U_{m+1}$, $D(x)$ restricts to a bounded level operator $E_m \to F_m$ Chapter3/GermLevels · IsSc1GermWith.exists_levelDeriv Chapter3/GermLevels · IsSc1GermWith.toBase_levelDeriv, the level-m tangent map is $(x,h) \mapsto (f_{m+1}(x), D_m(x)h)$ Chapter3/GermLevels · IsSc1GermWith.levelMap_tangentMapWith, jointly continuous Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv and continuous in x in the operator norm (the compact-open lemma of §I.13) Chapter3/GermLevels · IsSc1GermWith.continuousOn_levelDeriv_clm; and $f : U_{m+1} \to F_m$ is differentiable within U_{m+1} with derivative $D_m(x)$, by the integral representation $f(x+k) - f(x) = \int_0^1 D(x+\tau k)k\,d\tau$ in F_m (Proposition 1.16) Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level. The shifted sequence $(U_{m+1})_m$ is a germ in E^1 around $0 \in C^1$ Chapter3/GermLevels · ScNbhdGerm.shift, and an sc^k-germ on G is an sc^k-germ $G^1 \to F^1$ (Proposition 1.17) Chapter3/GermLevels · IsSc0Germ.shift Chapter3/GermLevels · IsSc1GermWith.shift Chapter3/GermLevels · IsSckGerm.shift. An sc^k-germ is C^k from U_{m+k} to F_m (Proposition 1.18), by the proof of §I.16 with U_m for $U \cap E_m$ Chapter3/GermLevels · IsSckGerm.contDiffOn_levels' Chapter3/GermLevels · IsSckGerm.contDiffOn_levels. sc^k depends on a germ only through its sets Chapter3/GermLevels · IsSckGerm.of_U_eq.

III.2 Sections of strong bundles in charts (Definition 3.3)

Let $P : Y \to X$ be a strong bundle over an M-polyfold (Part II, §II.19). A strong bundle chart $(\Phi, P^{-1}(V), K, U \lhd F)$ has a model $K = R(U \lhd F)$ for a strong bundle retraction R. Its base and fibre parts are read off: $r(u) = (R(u,0))_1$ Chapter3/Filler · StrongRetr.r, $\rho(u)h = (R(u,h))_2$ Chapter3/Filler · StrongRetr.ρ, and $R(u,h) = (r(u), \rho(u)h)$ because the base component of a strong bundle map does not depend on the fibre variable Chapter3/Filler · StrongRetr.apply_eq. Comparing components of $R \circ R = R$ gives $r \circ r = r$ Chapter3/Filler · StrongRetr.r_r and $\rho(r(u))\rho(u) = \rho(u)$ Chapter3/Filler · StrongRetr.ρ_r_ρ; for $x \in O = r(U)$, $r(x) = x$ Chapter3/Filler · StrongRetr.r_eq_self and $\rho(x)$ is a projection Chapter3/Filler · StrongRetr.ρ_ρ; R fixes the points of K Chapter3/Filler · StrongRetr.apply_eq_self_of_mem_K, so $\rho(x)h = h$ for $(x,h) \in K$ Chapter3/Filler · StrongRetr.ρ_eq_self_of_mem_K. $\rho(u)$ is linear by Definition 2.47 and continuous because R is sc⁰ on level 0 Chapter3/Filler · StrongRetr.ρ_continuous; it is a bounded operator Chapter3/Filler · StrongRetr.ρCLM — the text's "$\rho(u) : F \to F$ is a bounded linear operator", without comment.

Principal parts and the section algebra. For a section f (a map with $P \circ f = \mathrm{id}$), $\Phi \circ f \circ \varphi^{-1}(z) = (z, \mathbf{f}(z))$; $\mathbf{f}$ is its *principal part* Chapter3/StrongBundleSections · StrongBundleChart.principal Chapter3/StrongBundleSections · StrongBundleChart.apply_section. The fibrewise sum $f + s$ is defined with the vector space structure of the fibres $P^{-1}(x)$ Chapter3/StrongBundleSections · sectionAdd and is a section Chapter3/StrongBundleSections · IsSection.sectionAdd. Since Φ is linear on the fibres Chapter3/StrongBundleSections · StrongBundleChart.snd_add and maps the zero of $P^{-1}(x)$ to $(\varphi(x), 0)$ Chapter3/StrongBundleSections · StrongBundleChart.snd_fiberZero, the principal part of $f + s$ is $\mathbf{f} + \mathbf{s}$ Chapter3/StrongBundleSections · StrongBundleChart.principal_sectionAdd and $f(x) = 0$ iff $\mathbf{f}(\varphi(x)) = 0$ Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff Chapter3/StrongBundleSections · IsZeroAt. The fibres of K over O are linear: $(u,a),(u,b) \in K$ give $(u, a+b) \in K$, because membership is $\rho(u)a = a$ Chapter3/StrongBundleSections · StrongBundleChart.add_mem_K. *The sum rule*, used silently in "$f + s$ is an sc-smooth section", is pairing (Part I) followed by the addition sc-operator $F \oplus F \to F$ Chapter3/StrongBundleSections · ScBanach.addOp Chapter3/StrongBundleSections · IsScSmooth.add Chapter3/StrongBundleSections · IsScSmooth.sub Chapter3/StrongBundleSections · IsScSmooth.neg; a map sc-smooth into F^1 is sc-smooth into F (compose with the inclusion) Chapter3/StrongBundleSections · IsScSmooth.toBase_shift.

Which charts. "A strong bundle chart" in Definitions 3.3, 3.6 and 3.11 must mean a chart *compatible* with

the atlas of P Chapter3/StrongBundleSections · StrongBundle.IsChart Chapter3/StrongBundleSections · StrongBundle.isChart_of_mem: a chart of the atlas need not be centred at the point in question, and the notions must not depend on the atlas within its class. A compatible chart is a chart of the M-polyfold $Y[0]$ Chapter3/StrongBundleSections · StrongBundle.IsChart.toMChart. The price is that representatives of sections must be sc-smooth in every compatible chart. For an sc-smooth section, $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$ by Definition 2.14 as it stands Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothRetractMap. For an sc⁺-section, sc-smooth into $Y[1]$, the principal part takes values in F_1 in every compatible chart Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.principal_mem_range, and $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$: in a chart Φ' of the atlas this is the definition, and in the given chart it is the composition with the change of chart $(\Phi \circ \Phi'^{-1})[1]$ on $K[1]$, glued over the atlas by locality Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.isScSmoothRetractMap_one. This is precisely the compatibility on $[1]$ of Definition 2.50, used here for the first time.

Definition 3.3. (f, x_0) is an *sc-smooth section germ* if x_0 is smooth and in some compatible strong bundle chart with $\varphi(x_0) = 0$ the map $\mathbf{f} \circ r$ is an sc-smooth germ on some sc-germ of neighbourhoods G of 0 with $G_0 \subseteq U$, whose levels may shrink with m Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSectionGerm — the text's germ reading, as for Definition 3.11 (§III.48); until the fourth addendum the germ was the chart's $U \cap E_m$, which is stricter (§III.48). The text's "continuous section on a neighbourhood V" is the section on the chart domain; near x_0 continuity is a consequence. An sc-smooth section is an sc-smooth section germ at every smooth point with a centred chart: its representative composed with r is sc-smooth on the relatively open U, and §III.1's bridge applies on the germ $U \cap E_m$ Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.isScSmoothSectionGerm. No theorem assumes the definition.

III.3 Fillers (Definitions 3.4, 3.6; Lemmas 3.16, 3.17; Remark 3.49)

Definition 3.4. For the principal parts $f, g : E_0 \to F_0$ (with $O = r(U)$, $K = R(U \lhd F)$, $0 \in U$), g *fills* f if (1) $f = g$ on O near 0; (2) a point $y \in U$ near 0 with $g(y) = \rho(r(y))g(y)$ lies in O; (3) the map $y \mapsto [1 - \rho(r(y))]g(y)$ is sc¹ on U, and its derivative at 0 restricts to a topological linear isomorphism $\ker Dr(0) \to \ker \rho(0)$ Chapter3/Filler · IsFillerOf. The text's condition (3) presupposes the linearization; requiring the map to be sc¹ loses nothing, since it is sc-smooth when g and R are. The sc-smoothness of g itself is added in Definition 3.6. **Definition 3.6**: a *filled version* of $\mathbf{f}$ is a germ $(g, 0)$, and it is formalized as one, a *filling germ* Chapter3/GermFilling · IsFillingGerm: g is an sc-smooth germ (§III.1) on a germ G with $G_0 \subseteq U$, whose levels may shrink with m; (1) and (2) hold near 0; and in (3) the fill defect $k(y) = [1 - \rho(r(y))]g(y)$ Chapter3/GermFilling · fillDefect is an sc¹-germ on G with a derivative family Dk whose value $Dk(0)$ — unique (§III.1) — restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$. The first formalization took a filler sc-smooth on the whole U of the chart Chapter3/StrongBundleSections · IsFilledVersion, on the ground that a chart can always be restricted to $U \cap W$. That is the special case `$G = $ ofSet U` Chapter3/GermFillingChart · IsFilledVersion.isFillingGerm Chapter3/GermFillingChart · IsFillingGerm.isFilledVersion, and it is narrower: restricting a chart on level 0 never makes the levels shrink, and there are sc-smooth germs with shrinking levels that are sc-smooth on no neighbourhood of 0 Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth (§III.48). The lemmas below about fillers on U are the whole-chart ones; their germ forms, which the canonical notions use, are cited next to them.

Lemma 3.16. For a section s of $K \to O$ (that is, $(x, s(x)) \in K$ for $x \in O$), the extension $\tilde{s} = s \circ r$ satisfies $\tilde{s} = s$ on O (as $r(x) = x$ there), and $R(r(u), s(r(u))) = (r(u), s(r(u)))$, i.e. $\rho(r(u))\tilde{s}(u) = \tilde{s}(u)$, because $s(r(u))$ is a point of K and R fixes K Chapter3/Filler · lemma_3_16. The text's computation uses $s(x) \in K$ without saying so; it is the hypothesis that s is a section. **Lemma 3.17.** If g fills f and s is a section of $K \to O$, then $g + s \circ r$ fills $f + s$ Chapter3/Filler · lemma_3_17: (1) on O, $g + s \circ r = f + s$; (2) by Lemma 3.16, $(g+\tilde{s})(y) = \rho(r(y))(g+\tilde{s})(y)$ is equivalent to $g(y) = \rho(r(y))g(y)$; (3) by the same identity $[1-\rho(r(y))](g+\tilde{s})(y) = [1-\rho(r(y))]g(y)$ on U, so the two maps of (3) coincide on U and have the same derivative at 0 (derivatives on U_1 are unique). *For a filling germ* g on G and s sc-smooth on U the proof

is the same: the identity of (3) holds on U, hence on G_0, so the derivative family Dk of the fill defect of g is one of $g + s \circ r$, a filling germ on the same G `Chapter3/GermCalculus · IsFillingGerm.add_section`; for s itself a germ on G_s the filling lives on $G \cap r^{-1}(G_s)$ `Chapter3/GermCalculus · IsFillingGerm.add_section_germ`. The regularity statements of both lemmas — $\tilde{s}$ sc$^+$, $g + \tilde{s}$ sc-smooth — are proved where they are used, in Theorem 3.15 (§III.6). Condition (1) only involves f on O `Chapter3/StrongBundleSections · IsFillerOf.congr_left`.

Remark 3.49. On a smaller neighbourhood W of 0 condition (1) holds on all of $O \cap W$ and (2) for all $y \in U \cap W$: intersect the two neighbourhoods given by (1) and (2) `Chapter3/FillerLinearization · IsFillerOf.remark_3_49`.

III.4 The linearization of a filler (Proposition 3.5)

The text derives, "by linearization at $y = 0$", the matrix of $Dg(0)$ in the splittings $E = Dr(0)E \oplus \ker Dr(0)$, $F = \rho(0)F \oplus \ker \rho(0)$, and from it Proposition 3.5. The formal proof has two halves: the derivative identities, and the linear algebra.

The derivative identities. r is sc-smooth on U `Chapter3/FillerLinearization · StrongRetr.r_isScSmooth` with $r(0) = 0$ since $0 \in O$ `Chapter3/FillerLinearization · StrongRetr.r_zero`, and the chain rule applied to $r \circ r = r$ makes $Dr(0)$ idempotent `Chapter3/FillerLinearization · StrongRetr.scDeriv_r_idem`; $\rho(0)$ is idempotent `Chapter3/FillerLinearization · StrongRetr.ρCLM_idem`. The fibre part $(u, k) \mapsto \rho(u)k$ is sc^1 on $U \oplus F$ `Chapter3/FillerLinearization · StrongRetr.fiber_isSc1`, and its derivative at $(0, 0)$ is $(h, k) \mapsto \rho(0)k$: in the fibre direction because $k \mapsto \rho(0)k$ is linear, in the base direction because $\rho(u)0 = 0$; both directional derivatives are computed on the dense level 1 and extended by continuity `Chapter3/FillerLinearization · StrongRetr.scDeriv_fiber_zero`. By the chain rule for $(u, k) \mapsto \rho(u)k$ composed with $y \mapsto (r(y), g(y))$ at the point $(r(0), g(0)) = (0, 0)$ — this is where $g(0) = 0$ enters — the derivative of $y \mapsto [1 - \rho(r(y))]g(y)$ at 0 is $(1 - \rho(0))Dg(0)$ `Chapter3/FillerLinearization · IsFillerOf.scDeriv_sub_eq` `Chapter3/FillerLinearization · scDeriv_comp_apply'`. Next, $[1 - \rho(r(y))]g(r(y)) = 0$ near 0, from $g = f$ on O and $\rho(x)f(x) = f(x)$ on O (f a section of K) and the continuity of r; linearizing gives $(1 - \rho(0))Dg(0)Dr(0) = 0$ `Chapter3/FillerLinearization · IsFillerOf.deriv_comp_r_eq_zero`, i.e. $\rho(0)Dg(0)Dr(0) = Dg(0)Dr(0)$ `Chapter3/FillerLinearization · IsFillerOf.ρ_deriv_comp_r`. Condition (3) becomes: $(1 - \rho(0))Dg(0)$ restricts to a topological isomorphism $\ker Dr(0) \to \ker \rho(0)$ `Chapter3/FillerLinearization · IsFillerOf.exists_kerEquiv`. Finally $f \circ r = g \circ r$ near 0, so $D(f \circ r)(0) = Dg(0)Dr(0)$ — the text's "$Tg(0) = Tf(0)$ on T_0O" `Chapter3/FillerLinearization · IsFillerOf.scDeriv_section_comp_r`.

The linear algebra. Let p, q be bounded idempotents on Banach spaces E, F and $L : E \to F$ bounded with $(1 - q)Lp = 0$ and $(1 - q)L : \ker p \to \ker q$ a topological isomorphism C. Then $A := L|_{\operatorname{range} p} : \operatorname{range} p \to \operatorname{range} q$ is well defined `Chapter3/FillerLinearAlgebra · fillerRestrict`. Two facts carry everything: if $(1 - q)Lw = 0$ then $w \in \operatorname{range} p$ (injectivity of C, applied to the $\ker p$-component of w) `Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_of`; and $(1 - q)Lz = y$ is solvable in $\ker p$ for $y \in \ker q$ (surjectivity of C) `Chapter3/FillerLinearAlgebra · fillerRestrict_exists_of`. From them: A is onto iff L is `Chapter3/FillerLinearAlgebra · fillerRestrict_surjective_iff`; the inclusion $\ker A \to \ker L$ is a bijection — the text's $\ker Dg(0) = \ker Tf(0) \oplus \{0\}$ `Chapter3/FillerLinearAlgebra · fillerRestrict_kerEquiv`; $\operatorname{range} A = \operatorname{range} L \cap \operatorname{range} q$ `Chapter3/FillerLinearAlgebra · fillerRestrict_mem_range_iff` and $\operatorname{range} q / \operatorname{range} A \cong F / \operatorname{range} L$ `Chapter3/FillerLinearAlgebra · fillerRestrict_cokerEquiv`. So the indices are equal as $\dim \ker - \dim \operatorname{coker}$ with no finiteness assumption `Chapter3/FillerLinearAlgebra · fredholmIndex_fillerRestrict`. Closedness of the ranges transfers both ways: $\operatorname{range} A$ is closed in $\operatorname{range} q$ when $\operatorname{range} L$ is `Chapter3/FillerLinearAlgebra · fillerRestrict_isClosed_range_of`, and conversely $\operatorname{range} L$ is the preimage of $\operatorname{range} A$ under the continuous $M(y) = y - LC^{-1}(1 - q)y$ `Chapter3/FillerLinearAlgebra · isClosed_range_of_fillerRestrict` — this is where "topological" in (3) is used. Hence A is Fredholm iff L is `Chapter3/FillerLinearAlgebra · fillerRestrict_isFredholm_iff`, and the three statements together `Chapter3/FillerLinearAlgebra · proposition_3_5`. The text's proof factorizes $\begin{pmatrix} A & B \\ 0 & C \end{pmatrix} = \begin{pmatrix} 1 & BC^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & C \end{pmatrix}$ and

uses the additivity of the index; the isomorphisms of kernels and cokernels avoid both.

Proposition 3.5. For a filler g of the section f of $K \to O$, g sc-smooth with $g(0) = 0$: with $p = Dr(0)$, $q = \rho(0)$, $L = Dg(0)$ the hypotheses of the linear algebra hold by the identities above, and $Tf(0) := Dg(0)|_{Dr(0)E} : Dr(0)E \to \rho(0)F$ is onto iff $Dg(0)$ is, Fredholm iff $Dg(0)$ is, with equal index `Chapter3/FillerLinearization · IsFillerOf.proposition_3_5`. (The text writes the target of the lower right entry as "$(1 - D\rho(0))F$"; it is $(1 - \rho(0))F$.) *For a filling germ* (§III.3) the proof is the same with any derivative family Dg of g on any germ inside U: the fill defect is an sc-smooth germ on the germ of g, by the chain rule with the whole-chart outer map R (§III.48) `Chapter3/GermCalculus · fillDefect_isScSmoothGerm`, and the value at 0 of its derivative family is $(1 - \rho(0))Dg(0)$ `Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho`, which replaces `scDeriv_sub_eq`; so $Tf(0) = Dg(0)|_{Dr(0)E}$ is onto, Fredholm, of the index of $Dg(0)$ iff $Dg(0)$ is `Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_5`. $Dg(0)$ does not depend on the family or the germ `Chapter3/GoodPositionGermG · IsSc1GermWith.deriv_zero_eq`.

III.5 Basic germs, Lemma 3.9 and Proposition 3.10

Definition 3.7 (and 3.37). $f : P \times W_0 \to Q \times W_0$ is a *basic germ* on a germ G around 0 in $\widetilde{C} \oplus W$, with contraction map B, if f is an sc-smooth germ with $f(0) = 0$ whose W-component is $w - B(a, w)$ on U_0, and B is an sc^0-contraction germ in the sense of Definition 3.28 (§III.7) `Chapter3/LocalCompactness · IsBasicGerm`. ($B = w - f_W$ is then itself an sc-smooth germ `Chapter3/LocalSolvability · IsBasicGerm.B_isScSmoothGerm`.)

Lemma 3.9. The text: for every $\varepsilon > 0$ and m, $|D_2B(a, w)\zeta|_m \le \varepsilon|\zeta|_m$ when $(a, w) \in E_{m+1}$ is close to 0 in E_m; in particular $D_2B(0, 0) = 0$. Its proof: for $\zeta \in W_{m+1}$ the difference quotient of $t \mapsto B(a, w + t\zeta)$ is bounded by the contraction estimate, and the derivative is the limit (this needs $B : U_{m+1} \to W_m$ differentiable, Proposition 1.16 for germs — "recalling Proposition 1.17" in the text); then density of W_{m+1} in W_m. Only sc^1 is used. The formalization proves the three forms the chapter needs. (a) On level m in the working form of §III.7, with the constant $\frac{1}{2}$: at $(a, w) \in U_{m+1}$ with $a \in N(m+1) \cap \widetilde{C}$, $w \in O(m+1)$ and $\iota w \in O(m)$, $\|D_2B(a, w)\|_{L(W_m)} \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_levelDeriv_snd_le`; a linear map of norm $\le \frac{1}{2}$ then has no nonzero fixed point `Chapter3/ContractionGerm · ContractionGerm.eq_zero_of_norm_le_half`. (b) $D_2B(0, 0) = 0$, from the difference quotients along $t \mapsto (0, t\zeta)$, bounded by $\varepsilon|\zeta|$ for every ε, then density `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`. (c) On level 0 with any ε: $|D_2B(x)\zeta|_0 \le \varepsilon|\zeta|_0$ for $x \in U_1$ small *on level* 1 `Chapter3/SolutionLinearization · IsSc0ContractionGerm.exists_norm_deriv_snd_le` — the curve $t \mapsto x + t(0, \zeta)$, $\zeta \in W_1$, must stay in U_1, where B is differentiable.

Proposition 3.10. Let $\widetilde{C}$ be a partial quadrant, h a basic germ, s an sc^1-germ into $(Q \oplus W)^1$ (an sc$^+$-germ) with derivative family Ds, and $f = h + \iota s$ sc^1 with derivative family Df. Then $Df(0)$ is an sc-operator, sc-Fredholm of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · proposition_3_10`.

Proof. The derivative of an sc^1-germ at the smooth point 0 is an sc-operator, its level-m map being the fibre component of the level-m tangent map at $(0, \cdot)$ (the remark after Proposition 1.16) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`. The operator (32) of the text, $T_0(\delta a, \delta w) = (0, \delta w)$ `Chapter3/BasicGermLinearization · ScBanach.projWOp`, is sc-Fredholm with kernel $P \oplus 0$ and cokernel $Q \oplus 0$ `Chapter3/BasicGermLinearization · ScBanach.projWSplitting` `Chapter3/BasicGermLinearization · ScBanach.isScFredholm_projWOp`, of index $\dim P - \dim Q$ `Chapter3/BasicGermLinearization · ScBanach.scIndex_projWOp`. The text says "$Df(0)$ is an sc$^+$-perturbation of (32)"; the formal proof builds the sc$^+$-operator $S = Df(0) - T_0$ level by level `Chapter3/BasicGermLinearization · proposition_3_10_scPlus`: its Q-component is sc$^+$ because Q has all levels equal; $\mathrm{pr}_W Dh(0)(\delta a, 0) = -D_1B(0)\delta a = -DB(0)(\delta a, 0)$ has values in W_∞ ($(\delta a, 0)$ is smooth and $DB(0)$ an sc-operator) and its level maps $E_m \to W_{m+1}$ factor through $DB(0)_{m+1}$; $\iota Ds(0)$ is sc$^+$ because $Ds(0)$ is an sc-operator into $(Q \oplus W)^1$; and $\mathrm{pr}_W Dh(0)(0, \delta w) = \delta w$ because $D_2B(0) = 0$ (Lemma 3.9 (b)). The identities $Df(0) = Dh(0) + \iota Ds(0)$ and $\mathrm{pr}_W Dh(0) = \mathrm{pr}_W - DB(0)$ are derivative identities within U_1, extended to E_0 because two bounded operators agreeing on the dense E_1 are equal `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. Proposition 1.10 with its index statement concludes `Chapter1/ScFredholm · proposition_1_10` `Chapter1/FredholmIndex · proposition_`

1_10_index. For a basic germ alone ($s = 0$) Chapter3/FredholmGermLinearization · ScOperator.zero Chapter3/FredholmGermLinearization · proposition_3_10_basic, with its level-0 consequence: $Dh(0)$ is Fredholm of index $\dim P - \dim Q$ Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero.

The second statement of Proposition 3.10 ($Df(a,w)$ Fredholm on every level m for (a,w) small) is proved on level 0 for the germs of Theorem 3.38, at points of U_1 small on level 1 (§III.19); on levels $m \geq 1$ it is not formalized.

III.6 Sc-Fredholm germs and sections; Theorem 3.15 (Definitions 3.11, 3.14, 3.50, 3.51)

Definition 3.11. A *strong bundle isomorphism* $\Psi : U \lhd F \to U' \lhd F'$ is a pair of strong bundle maps (Definition 2.47) inverse to each other on $U \oplus F$ and $U' \oplus F'$ Chapter3/StrongBundleSections · IsStrongBundleIso; its base map is ψ. A principal part $k : U \to F$ is *conjugated to a basic germ* if there are a strong bundle isomorphism with $\psi(0) = 0$ onto U' relatively open in $\widetilde{C} \oplus W$ ($\widetilde{C}$ a partial quadrant of a finite-dimensional P), a basic germ h on a germ G with $G_0 \subseteq U'$, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0 Chapter3/StrongBundleSections · IsConjugateToBasicGerm; this depends only on k on U Chapter3/StrongBundleSections · IsConjugateToBasicGerm.congr. (f, x_0) is an *sc-Fredholm germ* if in a compatible chart centred at x_0 the principal part has a filling germ g (§III.3), and there is a local sc⁺-section s — an sc-smooth germ into F^1 on a germ G_s with $(G_s)_0 \subseteq U$ Chapter3/GermFilling · IsScPlusGerm — with $\iota s(0) = g(0)$ such that $g - \iota s$ is conjugated to a basic germ Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm. This is the text's reading: g and s are germs, on germs of neighbourhoods of their own whose levels may shrink with the level, while Ψ is an isomorphism of the whole local bundles and the conjugation identity is asked near 0 only, so it depends only on the germ of $g - \iota s$ Chapter3/GermFillingChart · IsConjugateToBasicGerm.congr_nhds. The first formalization asked g and s to be sc-smooth on $U \cap E_m$ for every m Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart. That notion implies the germ reading Chapter3/GermFillingChart · StrongBundle.IsScFredholmGermWholeChart.isScFredholmGerm and is exactly the germ reading with g and s sc-smooth on U Chapter3/GermFillingChart · StrongBundle.isScFredholmGerm_iff_germG_wholeChart; whether every germ datum can be traded for a whole-chart one is open (§III.48). Definitions 3.14, 3.50, 3.51, 3.53 and every theorem of this part built on them are stated for the germ reading; the whole-chart notions and theorems are kept under the names …WholeChart, …_wholeChart. The text's $C' = T\psi(0)C$, used in the proof of Theorem 3.54, is not part of the data; it is a theorem (§III.23).

Definition 3.14. f is an *sc-Fredholm section* if it is sc-smooth, regularizing ($x \in X_m$ and $f(x) \in Y_{m,m+1}$ imply $x \in X_{m+1}$) and an sc-Fredholm germ at every smooth point Chapter3/TameGermsG · StrongBundle.IsScFredholmSection. The text assumes a tame bundle; nothing in Definitions 3.3–3.14 or in Theorem 3.15 uses tameness, so it is not assumed. (Definitions 3.50, 3.51 restate 3.11, 3.14.) A zero of a regularizing section is a smooth point: the zero section lies in $Y_{m,m+1}$ over X_m Chapter3/SectionCompactness · StrongBundle.fiberZero_mem_doubleLevel, so a zero on level m is on level $m+1$, and by induction on every level Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt.

Theorem 3.15. If f is an sc-Fredholm section and s an sc⁺-section, then $f + s$ is an sc-Fredholm section Chapter3/FredholmSectionsG · theorem_3_15. *Proof.* The text says "$f + s$ is an sc-smooth section which is also regularizing" and checks the germ property; each clause is a proof.

Sc-smooth. In a chart the representative of $f+s$ is $z \mapsto (z, \mathbf{f}(z)+\mathbf{s}(z))$. The map $z \mapsto (z, \mathbf{f}(z))$ is sc-smooth $O \to K$, $z \mapsto (z, \mathbf{s}_1(z))$ is sc-smooth $O \to K[1]$ (§III.2), and composing the latter with $K[1] \subseteq K$ and adding fibre components (sum rule) gives an sc-smooth map with values in K, since its fibres are linear Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.add_isScSmoothRetractMap; continuity of $f + s$ on each chart domain is read off the same representative Chapter3/StrongBundleSections · StrongBundle.continuousOn_sectionAdd Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.continuousOn_source. So $f + s$ is sc-smooth Chapter3/StrongBundleSections · StrongBundle.IsScSmoothSection.sectionAdd.

Regularizing. For $x \in X_m$, the representative of s maps O_m into $K[1]_m$, so $\mathbf{s}(\varphi(x)) \in F_{m+1}$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.snd_mem_range`. If $(f+s)(x) \in Y_{m,m+1}$ then $\mathbf{f}(\varphi(x)) = (\mathbf{f}+\mathbf{s})(\varphi(x)) - \mathbf{s}(\varphi(x)) \in F_{m+1}$, i.e. $f(x) \in Y_{m,m+1}$ (Φ is linear on fibres `Chapter3/StrongBundleSections · StrongBundleChart.snd_sub`), and $x \in X_{m+1}$ `Chapter3/FredholmSectionsG · StrongBundle.regularizing_sectionAdd`.

The germ property at a smooth point x `Chapter3/FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd`. Take the chart, filling germ g on G_g, sc⁺-germ t_0 on G_{t_0} with $\iota t_0(0) = g(0)$ and conjugation Ψ for (f,x). Let $t = \mathbf{s}$ be the principal part of s in this chart; it is a section of $K \to O$ `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.mem_K` with lift t_1 to F_1, $\iota t_1 = t$ on O `Chapter3/StrongBundleSections · StrongBundle.IsScPlusSection.toBase_principalLift`. Then: $\tilde{t} = t \circ r$ is sc-smooth on the whole U, and $t_1 \circ r$ is sc-smooth into F^1 on U (sc-smoothness on the retract O, Definition 2.7) — s is a section of the bundle, not a germ; $g + \tilde{t}$ is a filling germ of $\mathbf{f}+\mathbf{s}$ on the same G_g (Lemma 3.17 for germs, §III.3 `Chapter3/GermCalculus · IsFillingGerm.add_section`, with the filler condition (1) transported along $\mathbf{f}+\mathbf{s}$ on O `Chapter3/FredholmSectionsG · IsFillingGerm.congr`); $t_1 \circ r + t_0$ is an sc⁺-germ on the same G_{t_0}, a germ plus a whole-chart map (§III.48), with value $(g+\tilde{t})(0)$ at 0 `Chapter3/FredholmSectionsG · StrongBundle.sectionAdd_chartData_germ`; and $(g+\tilde{t}) - \iota(t_1 \circ r + t_0) = g - \iota t_0$ *on* U (where $\iota t_1(r(u)) = t(r(u))$ because $r(u) \in O$), which is conjugated to a basic germ by the same Ψ. This last identity holds on U only, which is why the conjugation condition is stated for $u \in U$. The germs of the data do not change, so nothing of g or t_0 away from G_g, G_{t_0} is used; the whole-chart proof `Chapter3/StrongBundleSections · StrongBundle.IsScFredholmGermWholeChart.sectionAdd` is the same with the sum rule on U in place of the sum of a germ and a whole-chart map. □

III.7 Contraction germs: Definition 3.28 and its working form

Definition 3.28. For a germ G around 0 in $\widetilde{C} \oplus W$, $B : P \times W_0 \to W_0$ is an *sc⁰-contraction germ* (i.e. $f(a,w) = w - B(a,w)$ is one `Chapter3/ContractionGerm · contractionMap`) if B is an sc⁰-germ on G, $B(0,0) = 0$, and for every $\varepsilon > 0$ and m there is $r > 0$ such that every (a,w) with $a \in \widetilde{C}$, $|a| < r$, $|w|_m < r$ lies in U_m and $|B_m(a,w) - B_m(a,w')|_m \le \varepsilon |w - w'|_m$ for such points `Chapter3/ContractionGerm · IsSc0ContractionGerm`. (The text: "for all (a,w), (a,w') on level m sufficiently close to $(0,0)$"; that these points are in the domain U_m is presupposed.) B_m is the level-m map `Chapter3/ContractionGerm · levelB`, with $B_m(0) = 0$ `Chapter3/ContractionGerm · levelB_zero`.

The working form. The induction of §III.10 builds from f a contraction germ $f^{(1)}$ whose fibre is $TW = W^1 \oplus W$ and whose map is affine in the second factor; its natural contraction region is unbounded there, which balls cannot express. So the proofs work with explicit regions: a *contraction germ in working form* consists of B and G as above with open neighbourhoods $N(0) \supseteq N(1) \supseteq \cdots$ of 0 in P, open $O(m) \ni 0$ and closed $K(m) \subseteq O(m)$ in W_m with $\iota K(m+1) \subseteq K(m)$, such that for $a \in N(m) \cap \widetilde{C}$: $\{a\} \times O(m) \subseteq U_m$, $B_m(a,\cdot)$ maps $K(m)$ into $K(m)$ and is a $\frac{1}{2}$-contraction on $O(m)$ `Chapter3/ContractionGerm · ContractionGerm`.

Lemma: every sc⁰-contraction germ has a working form `Chapter3/GermIFT · IsSc0ContractionGerm.toContractionGerm`. Apply Definition 3.28 with $\varepsilon = \frac{1}{2}$ to get radii r_m `Chapter3/GermIFT · IsSc0ContractionGerm.rad` `Chapter3/GermIFT · IsSc0ContractionGerm.rad_spec`. Make them decrease fast enough for the inclusions: $r'_0 = r_0$, $r'_{m+1} = \min(r_{m+1}, r'_m/(\|\iota_m\| + 1))$ `Chapter3/GermIFT · IsSc0ContractionGerm.rad'` `Chapter3/GermIFT · IsSc0ContractionGerm.norm_incl_le_of_le`. Put $O(m) = \{|w|_m < r'_m\}$, $K(m) = \{|w|_m \le r'_m/2\}$; then $|\iota w|_m \le \|\iota_m\| r'_{m+1}/2 \le r'_m/2$ for $w \in K(m+1)$. Since B_m is continuous at 0 with $B_m(0) = 0$ `Chapter3/GermIFT · IsSc0ContractionGerm.levelB_zero_eq`, there is $\rho^0_m > 0$ with $|B_m(x)|_m < r'_m/4$ for $x \in U_m$, $|x|_m < \rho^0_m$ `Chapter3/GermIFT · IsSc0ContractionGerm.exists_cont_rad` `Chapter3/GermIFT · IsSc0ContractionGerm.contRad`. Let ρ_m decrease with $\rho_m \le \min(\rho^0_m, r'_m)$ `Chapter3/GermIFT · IsSc0ContractionGerm.prad` `Chapter3/GermIFT · IsSc0ContractionGerm.prad_anti` and $N(m) = \{|a| < \rho_m\}$. For $a \in N(m) \cap \widetilde{C}$ and $w \in K(m)$,

$|B_m(a,w)|_m \le |B_m(a,w)-B_m(a,0)|_m + |B_m(a,0)|_m \le \frac{1}{2}\cdot\frac{r'_m}{2} + \frac{r'_m}{4} = \frac{r'_m}{2}$.

III.8 Existence of the solution germ: the parametrized fixed point theorem and Theorem 3.29

The parametrized Banach fixed point theorem. The text invokes "the parametrized version of Banach's fixed point theorem" in Theorems 3.29 and 3.40 without stating it; Mathlib has the theorem without the parameter. Let X be a complete metric space, $K \subseteq X$ closed and nonempty, and $g_l : K \to K$ ($l \in \Lambda'$, a subset of a topological space) $\frac{1}{2}$-contractions such that $l \mapsto g_l(x)$ is continuous for every $x \in K$. Then the fixed point $\phi(l) \in K$ is unique in K and continuous in l: from $d(\phi(l),\phi(l_0)) \le \frac{1}{2}d(\phi(l),\phi(l_0)) + d(g_l\phi(l_0), g_{l_0}\phi(l_0))$ one gets $d(\phi(l),\phi(l_0)) \le 2d(g_l(\phi(l_0)), g_{l_0}(\phi(l_0)))$, which tends to 0 by continuity at the single point $\phi(l_0)$ `Chapter3/LocalCompactness · exists_fixedPoint_continuousOn`.

Theorem 3.29. In the working form, for $a \in N(m) \cap \widetilde{C}$ the map $B_m(a,\cdot) : K(m) \to K(m)$ is a $\frac{1}{2}$-contraction of a closed subset of W_m, with a fixed point $\delta_m(a) \in K(m)$ `Chapter3/ContractionGerm · ContractionGerm.exists_fixed` `Chapter3/ContractionGerm · ContractionGerm.fixLvl`, which is the only fixed point in the larger set $O(m)$ (two fixed points in $O(m)$ satisfy $|w-w'|_m \le \frac{1}{2}|w-w'|_m$) `Chapter3/ContractionGerm · ContractionGerm.fixed_unique` `Chapter3/ContractionGerm · ContractionGerm.eq_fixLvl`. For $a \in N(m+1) \cap \widetilde{C}$, $\iota\delta_{m+1}(a) \in K(m) \subseteq O(m)$ is a fixed point of $B_m(a,\cdot)$ because B commutes with the inclusions; hence $\iota\delta_{m+1}(a) = \delta_m(a)$ `Chapter3/ContractionGerm · ContractionGerm.incl_fixLvl`. So $\delta := \delta_0$ `Chapter3/ContractionGerm · ContractionGerm.sol` maps $V_m := N(m) \cap \widetilde{C}$ into W_m with level-m map δ_m `Chapter3/ContractionGerm · ContractionGerm.levelMap_sol`. Each δ_m is continuous on V_m by the parametrized fixed point theorem ($g_a = B_m(a,\cdot)$ on $K(m)$, continuous in a because B_m is) `Chapter3/ContractionGerm · ContractionGerm.continuousOn_fixLvl`. Write $\mathrm{germ}(s) = (N(m+s) \cap \widetilde{C})_m$ `Chapter3/ContractionGerm · ContractionGerm.germ` — legitimately a germ because P is finite-dimensional — and call δ a *solution germ* on a germ V if it is an sc^0-germ there with $\delta(0) = 0$, $(a,\delta_m(a)) \in U_m$ on V_m and $B(a,\delta(a)) = \delta(a)$ on V_0 `Chapter3/ContractionGerm · ContractionGerm.IsSolutionGerm`. Then: δ is a solution germ on $\mathrm{germ}(0)$, and every $w \in O(0)$ with $B(a,w) = w$, $a \in N(0) \cap \widetilde{C}$, equals $\delta(a)$ `Chapter3/ContractionGerm · theorem_3_29` `Chapter3/ContractionGerm · ContractionGerm.eq_sol`. The property passes to every $\mathrm{germ}(s)$ `Chapter3/ContractionGerm · ContractionGerm.isSolutionGerm_germ`. In the text's form, from Definition 3.28 with balls: the working form of §III.7 gives the solution germ, unique among solutions w with $|w|_0 < r'_0$ `Chapter3/GermIFT · theorem_3_29_of_isSc0ContractionGerm`. The text's "uniquely determined" leaves this region implicit (Remark 3.36 names it, $|w|_0 \le \tau_0$).

III.9 The solution germ is sc^1: Theorem 3.30

The text cites [21, Theorem 2.3], "the hard part of the regularity theory", and gives no proof. The formal proof shows that δ is sc^1 on the germ shifted by one level, $V_m = N(m+1) \cap \widetilde{C}$, and the shift is forced by the argument.

The operators. Let D be the derivative family of B. For $a \in N(m+1) \cap \widetilde{C}$ the point $(a,\delta_{m+1}(a))$ lies in U_{m+1}, and the level-m operator of §III.1 there is $L_m(a) := DB(a,\delta_{m+1}(a)) : P \times W_m \to W_m$ `Chapter3/ContractionGerm · ContractionGerm.Lop`, with fibre part $T_m(a) = D_2B(a,\delta_{m+1}(a))$ `Chapter3/ContractionGerm · ContractionGerm.Top` and $L_m(a)(h,\zeta) = L_m(a)(h,0) + T_m(a)\zeta$ `Chapter3/ContractionGerm · ContractionGerm.Lop_apply`. By Lemma 3.9 (a), $\|T_m(a)\| \le \frac{1}{2}$ `Chapter3/ContractionGerm · ContractionGerm.norm_Top_le`, so $1 - T_m(a)$ is invertible (Neumann series) `Chapter3/ContractionGerm · ContractionGerm.isUnit_one_sub_Top`, and the candidate derivative is $A_m(a) := (1-T_m(a))^{-1}L_m(a)(\cdot,0) \in L(P,W_m)$ `Chapter3/ContractionGerm · ContractionGerm.Aop`. It satisfies the *key identity* $L_m(a)(h,A_m(a)h) = A_m(a)h$ `Chapter3/ContractionGerm · ContractionGerm.Lop_Aop`, and is the unique solution ζ of $L_m(a)(h,\zeta) = \zeta$ `Chapter3/ContractionGerm · ContractionGerm.eq_Aop`. Because

the level operators commute with the inclusions Chapter3/ContractionGerm · ContractionGerm.toBase_Lop, so do the A_m: $\iota A_m(a)h = A_0(a)h$ Chapter3/ContractionGerm · ContractionGerm.toBase_Aop.

Differentiability at $a_0 \in N(2) \cap \widetilde{C}$ Chapter3/ContractionGerm · ContractionGerm.hasFDerivWithinAt_sol. Let $x_1 = (a_0, \delta_1(a_0)) \in U_1$, $A_1 = A_1(a_0)$, $A_0 = A_0(a_0)$. For a near a_0 put $h = a - a_0$ and $k = (h, A_1 h) \in E_1$, so $|k|_1 \le (1 + \|A_1\|)|h|$. *This is where $a_0 \in N(2)$ is needed*: A_1 exists only because $(a_0, \delta_2(a_0)) \in U_2$, and without it the increment $A_0 h$ need not lie in W_1, so the sc¹-differentiability of B at x_1, which controls only increments in E_1, could not be used. For h small, $x_1 + k \in U_1$ and $y := \delta(a_0) + A_0 h \in O(0)$. By the sc¹ property, $B(a, y) = B(a_0, \delta(a_0)) + D(x_1)\iota k + \text{rem} = \delta(a_0) + L_0(a_0)(h, A_0 h) + \text{rem} = y + \text{rem}$ with $|\text{rem}|_0 = o(|h|)$, using the key identity on level 0. So y is an approximate fixed point of $B(a, \cdot)$, and the $\frac{1}{2}$-contraction on $O(0)$ gives $|\delta(a) - y|_0 \le \frac{1}{2}|\delta(a) - y|_0 + |\text{rem}|_0$, i.e. $|\delta(a) - \delta(a_0) - A_0 h|_0 \le 2|\text{rem}|_0 = o(|h|)$.

The tangent map is sc⁰. On level m of the tangent of germ(1), i.e. on $(N(m+2) \cap \widetilde{C}) \times P$, the tangent map is $(a, h) \mapsto (\delta_{m+1}(a), A_m(a)h)$ Chapter3/ContractionGerm · ContractionGerm.tangent_toBase_sol. Its first component is continuous (§III.8); the second because $(x, v) \mapsto D_m(x)v$ is jointly continuous and $\zeta(a) = A_m(a)h$ satisfies $\zeta(a) = L_m(a)(h, 0) + T_m(a)\zeta(a)$, whence $|\zeta(a) - \zeta(a')|_m \le 2|L_m(a)(h,0) - L_m(a')(h,0)|_m + 2|(T_m(a) - T_m(a'))\zeta(a')|_m$ Chapter3/ContractionGerm · ContractionGerm.continuousOn_Aop. With the derivative family $a \mapsto A_0(a)$ on germ(1) Chapter3/ContractionGerm · ContractionGerm.solDeriv, δ is an sc¹-germ there Chapter3/ContractionGerm · ContractionGerm.isSc1GermWith_sol — **Theorem 3.30**, on the shifted germ Chapter3/ContractionGerm · theorem_3_30. At parameters $a_0 \in N(1) \setminus N(2)$ the argument does not apply, and whether δ is differentiable there is neither needed nor addressed.

III.10 The contraction germ $f^{(1)}$: Lemma 3.31

Lemma 3.31. The text: if f is of class sc^k, $k \ge 1$, and δ of class sc^j, then $f^{(1)}(v, b, u, w) = (u - B(v, u), w - DB(v, \delta(v))(b, w))$ is an sc⁰-contraction germ of class sc^{min(k−1, j)} whose solution germ is $T\delta$ — "as one verifies by comparing Tf with $f^{(1)}$", and "from the uniqueness". The formal version: if B is sc^{k+1} with derivative family D and δ is sc^j on germ(s), then $f^{(1)}$ is a contraction germ in working form of class sc^{min(k, j)} with solution germ $T\delta$ on $(N(2+s) \cap \widetilde{C}) \times P$ Chapter3/GermIFT · ContractionGerm.lemma_3_31. The parameters are $(v, b) \in \widetilde{C} \times P \subseteq P \times P$ (quadrant-like Chapter3/GermIFT · ContractionGerm.isQuadrantLike_next), the unknowns $(u, w) \in TW$, with levels $W_{m+1} \times W_m$.

Step 1: the germ and the regions. The germ of $f^{(1)}$ is $U_m^{(1)} = \{((v, b), (u, w)) : (v, u) \in U_{m+1},\ v \in N(m+1+s)\}$ Chapter3/GermIFT · ContractionGerm.nextG, and the regions are $N^{(1)}(m) = N(m+2+s) \times P$, $O^{(1)}(m) = O(m+1) \times W_m$, $K^{(1)}(m) = K(m+1) \times W_m$ Chapter3/GermIFT · ContractionGerm.next. They are unbounded in b and w, necessarily: the solution must be $(\delta(v), D\delta(v)b)$ for all $b \in P$.

Step 2: the map. With the reshuffling sc-operator $\sigma((v, b), (u, w)) = ((v, u), (b, w))$ Chapter3/GermIFT · tangentShuffle and $\Gamma((v, b), (u, w)) = ((v, \delta(v)), (b, w))$, where $v \mapsto \delta(v)$ is regarded as the *shifted* germ into W^1 Chapter3/GermIFT · ContractionGerm.nextΓ, formula (38) is $B^{(1)} = (\text{pr}_1 \circ TB \circ \sigma, \text{pr}_2 \circ TB \circ \Gamma)$ Chapter3/GermIFT · ContractionGerm.nextB Chapter3/GermIFT · ContractionGerm.nextB_apply. On level m, for $v \in N(m+1) \cap \widetilde{C}$, $B_m^{(1)}((v, b), (u, w)) = (B_{m+1}(v, u), L_m(v)(b, w))$ Chapter3/GermIFT · ContractionGerm.levelB_next: the point $(v, \delta_{m+1}(v))$ lies in U_{m+1}, which is where the parameter region $N(m+1+\cdot)$ comes from.

Step 3: contraction and invariance. For $(v, b) \in N^{(1)}(m)$ and $(u, w), (u', w') \in O^{(1)}(m)$, the two components contract by $\frac{1}{2}$: $B_{m+1}(v, \cdot)$ on $O(m+1)$, and $T_m(v)$ with $\|T_m(v)\| \le \frac{1}{2}$ (Lemma 3.9 (a)); $K(m+1) \times W_m$ is mapped into itself; $B^{(1)}(0) = 0$ Chapter3/GermIFT · ContractionGerm.nextB_zero.

Step 4: the class of $B^{(1)}$. TB is an sc^k-germ on the tangent germ; σ, the projections $(v, b, u, w) \mapsto v$ into P^1 and $\mapsto (b, w)$ Chapter3/GermIFT · projV Chapter3/GermIFT · projBW and the identification $P^1 \oplus W^1 = (P \oplus$

$W)^1$ Chapter3/GermIFT · shiftParamOp are sc-operators; Γ is the pairing of (b, w) with $(v, \delta(v))$, which is sc^j because δ is and so is its shift into W^1 (Proposition 1.17 for germs) Chapter3/GermIFT · ContractionGerm.nextΓ_isSckGerm; and Γ maps $U_m^{(1)}$ into $U_{m+1} \times E_m$ level by level Chapter3/GermIFT · ContractionGerm.levelMap_nextΓ Chapter3/GermIFT · ContractionGerm.nextΓ_mapsToGerm, as does σ Chapter3/GermIFT · ContractionGerm.tangentShuffle_mapsToGerm. The chain rule and the pairing rule of §III.1 give $B^{(1)}$ sc^j Chapter3/GermIFT · ContractionGerm.nextB_isSckGerm.

Step 5: the solution germ is $T\delta$. For $v \in N(2+s) \cap \widetilde{C}$ and $b \in P$, the point $(\delta_1(v), A_0(v)b) \in O(1) \times W_0$ is a fixed point of $B^{(1)}(v, b, \cdot)$ — the first component by $B_1(v, \delta_1(v)) = \delta_1(v)$, the second by the key identity of §III.9 — so by the uniqueness of Theorem 3.29 *in* $O^{(1)}(0)$ it is the solution germ of $f^{(1)}$ at (v, b) Chapter3/GermIFT · ContractionGerm.next_sol, and since $A_0(v) = D\delta(v)$ (Theorem 3.30) this is $T\delta(v, b)$ Chapter3/GermIFT · ContractionGerm.next_sol_eq_tangentMapWith. The text's "from the uniqueness" is true because the regions of Step 1 contain $T\delta(v, b)$ for every b.

III.11 The induction: Lemma 3.32, Theorem 3.33, Corollary 3.34, Theorem 3.35

The text proves Lemma 3.32 by iterating $f \mapsto f^{(1)}$ and Theorem 3.33 by contradiction (if δ were sc^j but not sc^{j+1}, $f^{(j)}$ would be sc¹, so $T^j\delta$ would be sc¹ by Theorem 3.30). The formal proof is the same argument as one induction on j, with the statement quantified over *all* contraction germs, since $f^{(1)}$ has parameter space $P \times P$ and fibre TW and the induction hypothesis must apply to it. It also keeps track of the germ.

Claim (S_j): for every contraction germ in working form, over any finite-dimensional parameter space, of class sc^k with $k \geq j$, the solution germ is an sc^j-germ on germ(s) *for some* s Chapter3/GermIFT · ContractionGerm.sol_isSckGerm. For $j = 0$ take $s = 0$ (Theorem 3.29). *Step* $j \to j+1$ Chapter3/GermIFT · ContractionGerm.isSckGerm_succ_of_next: let B be sc^k, $k = k' + 1 \geq j + 1$. (1) By (S_j) for f, δ is sc^j on germ(s). (2) By Lemma 3.31 with this s, $f^{(1)}$ is a contraction germ of class sc^{min(k', j)} = sc^j. (3) By (S_j) for $f^{(1)}$, its solution germ is sc^j on its germ(s'), whose level m is $(N(m + s' + 2 + s) \cap \widetilde{C}) \times P$. (4) By Theorem 3.30 δ is sc¹ on germ(1), hence on germ$(s + s' + 1)$ (restriction), whose tangent germ has level m equal to the set in (3). (5) On that set the solution germ of $f^{(1)}$ is $T\delta$ (Lemma 3.31, Step 5); the identification of $T\widetilde{C} = \widetilde{C} \times P$ inside TP with $\widetilde{C} \times P$ inside $P \times P$ is the identity on every level, an sc-isomorphism Chapter3/GermIFT · constTangentOp' through which the chain rule transports the germ. (6) So δ is sc¹ with sc^j tangent germ: sc^{j+1} on germ$(s + s' + 1)$. Since the proof does the same for every contraction germ, s and s' are the same function of j, so $s_0 = 0$, $s_{j+1} = 2s_j + 1$: $s_j = 2^j - 1$.

Theorem 3.33 follows from (S_k): there is s such that δ is a solution germ and an sc^k-germ on germ(s) Chapter3/GermIFT · theorem_3_33; if B is sc-smooth this holds for every k Chapter3/GermIFT · theorem_3_33_smooth; and from Definition 3.28 with balls through the working form Chapter3/GermIFT · theorem_3_33_of_isSc0ContractionGerm. Lemma 3.32 is contained in the induction: its step from j to $j+1$ is items (2)–(3), without the iterated germs $f^{(j)}$. **Corollary 3.34**: if B is sc-smooth, for m and k take s for k; by Proposition 1.18 for germs, δ maps $V_{m,k} := N(m+k+s) \cap \widetilde{C}$ into W_m by a C^k map Chapter3/GermIFT · corollary_3_34 Chapter3/GermLevels · IsSckGerm.contDiffOn_levels, the inclusions of the constant structure being the identity Chapter3/GermIFT · ScBanach.const_inclK. **Theorem 3.35**: for $\widetilde{C}$ a partial quadrant (which is quadrant-like), the same Chapter3/GermIFT · theorem_3_35 Chapter3/GermIFT · theorem_3_35_smooth.

What the shrinking means. Definition 3.2 calls δ sc-smooth on a germ if it is sc^k *on that germ* for every k. The proof gives, for each k, the germ germ$(2^k - 1)$; these shrink with k, their intersection is in general not a neighbourhood of 0, and no diagonal choice gives one germ, because being sc^k on a germ involves every level of it. So Theorem 3.33 and the first sentence of Theorem 3.35 hold for germs read up to shrinking, and the last sentences of Corollary 3.34 and Theorem 3.35 hold in the sense of

Corollary 3.34 — for every k on a germ depending on k — and are proved neither by the text nor here as `IsScSmoothGerm` on one germ. In the text's reading of germs the two senses coincide. Whether the shrinking is a property of contraction germs or only of the method is not decided: in the toy model of the exposition (Chapter3.md §3) the solution germ is sc-smooth on the original germ.

III.12 Local regularity and compactness: Theorem 3.40

Theorem 3.40 (text): for $f = h + s$ on a relatively open $U \ni 0$ in C, h a basic germ, s an sc$^+$-germ with $s(0) = 0$, and $S = \{f = 0\}$, there are relatively open neighbourhoods $U \supseteq O(0) \supseteq O(1) \supseteq \cdots$ of 0 on level 0 with $\mathrm{cl}(S \cap O(m)) \subseteq C_m$. Of f only the W-component matters: S lies in the set of solutions of $w = B(a, w) - \mathbf{s}(a, w)$, $\mathbf{s} = P_W \circ s$. The formal statement uses only that B is an sc^0-contraction germ (not sc-smooth), that $\mathbf{s}$ is an sc^0-germ into W^1 with $\mathbf{s}(0) = 0$ (the sc$^+$ property level by level), and that $\widetilde{C}$ is closed (true for partial quadrants; the theorem is proved for convex $\widetilde{C}$, and closedness makes the solution set closed in E_0, where the text takes closures in C_0). It is *stronger* than the text's in the way Theorem 3.12 needs: for any set S of solutions in U_0 there are nested open $O(i) \ni 0$ in E_0 and compact $K_i \subseteq E_i$ with $\mathrm{cl}_{E_0}(S \cap O(i)) = \iota_i(K_i)$ `Chapter3/LocalCompactness · theorem_3_40_compactness`; in particular $\mathrm{cl}(S \cap O(i)) \subseteq E_i$ `Chapter3/LocalCompactness · theorem_3_40_regularity`.

Step 0: the level-m contractions with a parameter. Let r_m be the radii of Definition 3.28 for $\varepsilon = \frac{1}{2}$ and $c_m = \min(r_m, r_0/(\|\iota_m\| + 1))$, $\iota_m : W_m \to W_0$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.crad`, so that $|\iota_m y|_0 < r_0$ whenever $|y|_m \le c_m/2$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.norm_toBase_lt_rad` (the text's $\tau_1' < \tau_0/c_1$). Points with $a \in \widetilde{C}$, $|a|, |w|_m < r_m$ lie in U_m and satisfy the contraction estimate `Chapter3/LocalCompactness · IsSc0ContractionGerm.mem_G_of_lt` `Chapter3/LocalCompactness · IsSc0ContractionGerm.contr_of_lt`. Choose $\rho_m \in (0, c_m]$ with $|B_m(a, 0)|_m \le c_m/8$ for $|a| < \rho_m$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_cprad` `Chapter3/LocalCompactness · IsSc0ContractionGerm.cprad`. On $\Sigma_m = \{(a, z) : a \in \widetilde{C}, |a| < \rho_m, |z|_m \le c_m/8\}$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.Sigma` the maps $w \mapsto B_m(a, w) - z$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap` send the closed ball $\bar{B}_m(c_m/2)$ `Chapter3/LocalCompactness · IsSc0ContractionGerm.cBall` into itself ($\frac{1}{2} \cdot \frac{c_m}{2} + \frac{c_m}{8} + \frac{c_m}{8} = \frac{c_m}{2}$) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_mapsTo`, are $\frac{1}{2}$-contractions `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_contr` and depend continuously on (a, z) `Chapter3/LocalCompactness · IsSc0ContractionGerm.gmap_continuousOn`. The parametrized fixed point theorem gives a continuous $\delta_m : \Sigma_m \to \bar{B}_m(c_m/2)$ with $\delta_m(a, z) = B_m(a, \delta_m(a, z)) - z$, unique in the ball `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta` `Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_fixed`.

Step 1: the lifts — the induction the text leaves to the reader. Let $\mathrm{Sol}(\tau)$ be the solutions $x = (a, w)$ with $a \in \widetilde{C}$, $|a| \le \tau$, $|w|_0 \le \tau$ `Chapter3/LocalCompactness · solSet`. *Claim:* for every m there are $\tau_m > 0$ and a map $L_m : E_0 \to E_m$ such that for every $x \in \mathrm{Sol}(\tau_m)$: $\iota_m L_m x = x$, $L_m x \in U_m$, $|(L_m x)_W|_m \le c_m/2$, and L_m is continuous on $\mathrm{Sol}(\tau_m)$ *for the level-*0 *topology of the source* `Chapter3/LocalCompactness · IsSc0ContractionGerm.exists_lift`. For $m = 0$, $L_0 = \mathrm{id}$. Given L_m: by continuity of $\mathbf{s}_m : U_m \to W_{m+1}$ at 0 (where $\mathbf{s}_m(0) = 0$ `Chapter3/LocalCompactness · levelMap_s_zero`) there is η with $|\mathbf{s}_m(y)|_{m+1} < c_{m+1}/8$ for $|y|_m < \eta$; by continuity of L_m at 0 there is θ with $|L_m x|_m < \eta$ for $x \in \mathrm{Sol}(\tau_m)$, $|x|_0 < \theta$. Put $\tau_{m+1} = \min(\tau_m, \rho_{m+1}/2, \theta/2)$ and, for $x = (a, w) \in \mathrm{Sol}(\tau_{m+1})$, $z = \mathbf{s}_m(L_m x)$, so $(a, z) \in \Sigma_{m+1}$, and $L_{m+1} x := (a, \delta_{m+1}(a, z))$. *It lifts x:* $w' = \iota_{m+1}\delta_{m+1}(a, z)$ solves $w' = B(a, w') - \mathbf{s}(x)$ (apply ι_{m+1}, which commutes with B, and use $\iota_m L_m x = x$); so does w; both are smaller than r_0 on level 0 by the choice of c_{m+1}, so the level-0 contraction gives $w' = w$ — the text's (43), on every level. *Continuity:* $x \mapsto (a, \mathbf{s}_m(L_m x))$ is continuous from the level-0 topology into Σ_{m+1}, and δ_{m+1} is continuous. The induction step needs the continuity of L_m: the bound $|\mathbf{s}_m(L_m x)|_{m+1} \le c_{m+1}/8$ requires $L_m x$ small on level m, whereas x is small only on level 0. In the text's level-1 step $L_0 = \mathrm{id}$ and this does not show; existence, bound and continuity of the lifts have to be proved together.

Step 2: compactness. $\mathrm{Sol}(\tau)$ is closed in E_0 ($\widetilde{C}$ closed, B and $\mathbf{s}$ continuous on level 0) `Chapter3/`

LocalCompactness · IsSc0ContractionGerm.isClosed_solSet. For $x \in \mathrm{Sol}(\tau_{m+1})$ the W-component is ι_{m+1} of a vector of norm $\le c_{m+1}/2$, and $\iota_{m+1} : W_{m+1} \to W_0$ is compact Chapter3/LocalCompactness · ScBanach.toBase_succ_isCompactOperator; with $|a| \le \tau_{m+1}$ in the finite-dimensional P, a sequence of lifts has a subsequence converging in E_{m+1} (parameters in a compact ball, W_0-parts in a compact image, and the continuity of the lift for the level-0 topology) Chapter3/LocalCompactness · IsSc0ContractionGerm.isCompact_lift_image.

Step 3: the neighbourhoods. $O(i)$ are the balls of radius ϱ_i, $\varrho_0 = \tau_1$, $\varrho_{i+1} = \min(\varrho_i, \tau_{i+2})$ — one level ahead of the text's $O(m)$, because compactness in E_i comes from boundedness in E_{i+1}. Then $\mathrm{cl}(S \cap O(i)) \subseteq \mathrm{Sol}(\tau_{i+1})$, every point of it is ι_i of the image under $E_{i+1} \to E_i$ of its lift $L_{i+1}x$, and K_i is the compact set of these images intersected with the closed preimage of $\mathrm{cl}(S \cap O(i))$.

For a basic germ plus an sc⁺-germ (S the zero set of $h+\iota s$; its Q-component only shrinks S) this is the **local Theorem 3.12** Chapter3/LocalCompactness · theorem_3_12_local, and **Corollary 3.13 in local form**: solutions $x_k \to 0$ in E_0 lie in $\mathrm{Sol}(\tau_m)$ for k large, and $L_m x_k \to L_m 0 = 0$ in E_m Chapter3/LocalCompactness · corollary_3_13_local.

III.13 Theorem 3.12 and Corollary 3.13 for sections

Theorem 3.12. Let f be a section of a strong bundle with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. Then there are nested open $O(i) \ni x_0$ in X such that $\mathrm{cl}_X\{x \in O(i) \mid f(x) = 0\}$ is the image of a compact subset of X_i (with the level topology of Part II, §II.6) under the inclusion $X_i \to X$ Chapter3/SectionCompactnessG · theorem_3_12. Of "f an sc-smooth section of a tame strong bundle" only $P \circ f = \mathrm{id}$ is used; the regularity enters through the germ data, and of the filling germ g only its continuity on level 0 near 0. The text calls the theorem "an immediate consequence of Theorem 3.40"; the proof makes explicit six steps.

(i) *Normalizations.* $f(x_0) = 0$ gives $\mathbf{f}(0) = 0$, so $g(0) = 0$ (filler (1)) and $s(0) = 0$ ($\iota s(0) = g(0)$). (ii) *The zeros correspond.* $f(x) = 0 \iff \mathbf{f}(\varphi(x)) = 0$ (§III.2) $\iff g(\varphi(x)) = 0$ (filler (1)) near x_0; zeros of g in U near 0 lie in O (filler (2)); and $g(u) = 0 \iff (h + t)(\psi(u)) = 0$, $t(\psi(u)) := (\Psi(u, s(u)))_2$, because $\Psi(u, k) = (\psi(u), \Gamma(u, k))$ Chapter3/SectionCompactness · IsStrongBundleMap.apply_eq with $\Gamma(u, \cdot)$ linear Chapter3/SectionCompactness · IsStrongBundleMap.snd_sub and injective Chapter3/SectionCompactness · IsStrongBundleIso.snd_eq_zero_iff. (iii) *t is an sc⁺-germ with $t(0) = 0$:* the base map of a strong bundle map is sc-smooth Chapter3/SectionCompactness · IsStrongBundleMap.base_isScSmooth, $\psi'(0) = 0$ Chapter3/SectionCompactness · IsStrongBundleIso.base_zero', and the fibre component of $\Psi[1]$ lifts that of Ψ to F_1' Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd. Theorem 3.40 needs t only as an sc⁰-germ into F'^1 on the germ of the basic germ. With s a germ on G_s, t is one on the germ of the points of the germ of h that ψ^{-1} maps into G_s level by level Chapter3/GermFilling · ScNbhdGerm.preimage (germ composition with the whole-chart ψ^{-1} and $\Psi[1]$, §III.48), and the basic germ restricts to that germ Chapter3/BasicGermShrink · IsBasicGerm.mono (a form of the shrinking of §III.48). (iv) *Local compactness for a subset.* The local theorem for an arbitrary subset of the solution set of $h + \iota t$ Chapter3/SectionCompactness · theorem_3_12_local_subset is applied on the restricted basic germ to $\psi\varphi$ of the zeros near x_0; its $O'(i)$ are intersected with a small ball, so that E_0-closures stay inside $\psi(U)$ and the region of (ii) (the closure of a subset of the solution set is still compact). (v) *Back to X.* The E_0-closure consists of zeros of $h + \iota t$, a limit of zeros in the domain of continuity being a zero Chapter3/SectionCompactness · eq_of_mem_closure_of_continuousOn, hence of zeros of g in U, which lie in O. Here g must be continuous at the limits $\psi^{-1}(u')$: the level-0 neighbourhood of the filler and conjugation identities is chosen inside the open set O_g with $(G_g)_0 = O_g \cap C$, so the limits lie in $(G_g)_0$, where the sc⁰-germ g is continuous. The whole-chart theorem Chapter3/SectionCompactness · theorem_3_12_wholeChart took t sc-smooth on all of U' and g continuous on U; nothing of either away from a neighbourhood of 0 was used. An X-closure of a subset of the chart domain could a priori leave it; compactness in E_0 shows it does not. (vi) *Levels.* A compact subset of E_i is carried to a compact subset of X_i because ψ^{-1} is sc⁰ and the inverse chart $O_i \to X_i$ Chapter3/SectionCompactness · MPolyfold.levelChartSymm is continuous for the level topology of X_i, the transition

maps being sc^0 Chapter3/SectionCompactness · MPolyfold.continuousOn_levelChartSymm — the paragraph after Definition 2.14, for every level.

Corollary 3.13. If $f(x_k) = 0$ and $x_k \to x_0$, fix m and take $O(m)$ and the compact $K_m \subseteq X_m$; for k large $x_k \in O(m)$, so $x_k = \iota y_k$ with $y_k \in K_m$, and $x_0 = \iota y_0$. X_m is not known to be first countable here (its metrizability, Theorem 2.16, is not used), so subsequences are avoided: a sequence in a compact set whose images under a continuous injection converge, converges Chapter3/SectionCompactness · tendsto_of_isCompact_of_tendsto_map. So $y_k \to y_0$ in X_m Chapter3/SectionCompactnessG · corollary_3_13. The text states the corollary for an arbitrary limit x while the hypothesis concerns the germ at x_0; for an sc-Fredholm section and any limit x of zeros, x is a zero (the principal part is continuous in a chart Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto), hence a smooth point (regularizing), hence the germ property holds at x and the first part applies Chapter3/Chapter5BridgeG · corollary_3_13_section.

III.14 Good position (Definition 3.21, Lemma 3.65, Proposition 3.22, Lemma 3.45, Proposition 3.69)

Definition 3.21. P is a *good complement* of the finite-dimensional N with respect to C if $E_0 = N \oplus P$ algebraically and for some $\varepsilon > 0$, $|p|_0 \le \varepsilon |n|_0$ implies $(n \in C \iff n + p \in C)$ Chapter3/GoodPosition · IsGoodComplement; N is *in good position* if $N \cap C$ has nonempty interior in N and N has a good complement Chapter3/GoodPosition · IsInGoodPosition. The text asks for an sc-complement (or, for non-smooth N, a topological one); none of Proposition 3.22, Lemmas 3.45, 3.65 or Proposition 3.69 uses more than the algebraic splitting, so only that is required. Where closedness is needed (Lemma 3.55, Definition 3.25) it is added there.

Set-up in coordinates. After a presentation $E \cong \mathbb{R}^n \oplus W$, $C = \{\ell_i \ge 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$ Chapter3/GoodPosition · coordFun Chapter3/GoodPosition · coordFun_surj. Inside N, $Q := N \cap C$ is the cone $\{\ell_i \ge 0\}$ Chapter3/GoodPosition · quadCone, closed Chapter3/GoodPosition · quadCone_isClosed. Let Σ be the indices i with $\ell_i(a) = 0$ for some nonzero $a \in Q$ Chapter3/GoodPosition · vanIdx; by definition, for $i \notin \Sigma$ and $x \in Q \setminus 0$, $\ell_i(x) > 0$ Chapter3/GoodPosition · pos_of_not_mem_vanIdx. Let $N \cap W = \{x \in N : \ell_i(x) = 0 \ \forall i\}$ Chapter3/GoodPosition · nullSpace and $Q_\Sigma = (\ell_i)_{i \in \Sigma} : N \to \mathbb{R}^\Sigma$ Chapter3/GoodPosition · sigmaCoord.

Lemma 3.65. A good complement satisfies $\ell_i(p) = 0$ for $p \in P$, $i \in \Sigma$: take $a \in Q \setminus 0$ with $\ell_i(a) = 0$, scale p so that $|p| \le \varepsilon |a|$; $a \in C$ gives $a + p \in C$, so $\ell_i(p) = \ell_i(a + p) \ge 0$; the same for $-p$ Chapter3/GoodPosition · lemma_3_65. *Consequences.* Q_Σ is onto: for $k \in \mathbb{R}^\Sigma$ pick v with $\ell_i(v) = k_i$ and split $v = n + p$; then $Q_\Sigma(n) = k$ (the text's (55), p. 125) Chapter3/GoodPosition · sigmaCoord_surjective. If an interior point y of Q has $\ell_i(y) = 0$, then $\ell_i = 0$ on N ($y \pm tu \in Q$) Chapter3/GoodPosition · restrFun_eq_zero_of_interior.

Proposition 3.22, the normal form. There are k, a subspace N_0 and a linear isomorphism $e : N \to \mathbb{R}^k \times N_0$ with $Q = e^{-1}([0, \infty)^k \times N_0)$, $k = \dim N - \dim(N \cap W)$, the $\mathbb{R}^k$-component of e vanishing exactly on $N \cap W$, and the vanishing coordinates of $e(x)$ as many as the vanishing $\ell_i(x)$ for $x \in Q \setminus N \cap W$ Chapter3/GoodPosition · QuadrantNormalForm Chapter3/GoodPosition · exists_quadrantNormalForm. Three cases. *Case A, $Q = N$:* every ℓ_i is ≥ 0 on a vector space, hence 0; $k = 0$ Chapter3/GoodPosition · quadrantNormalForm_of_eq_univ (a case the text's case distinction does not single out). *Case B, $Q \ne N$, $\Sigma = \emptyset$:* an interior point y is nonzero with all $\ell_i(y) > 0$; for $u \in N$ the set $\{t : u + ty \in Q\}$ is closed, contains all large t and is bounded below; at its minimum t^* some $\ell_i(u + t^* y) = 0$, so $u + t^* y = 0$: N is the line $\mathbb{R}y$ and Q a half-line Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx_empty (the text's Lemma 3.68, second case). *Case C, $\Sigma \ne \emptyset$:* first $Q = \{x \in N : \ell_i(x) \ge 0 \text{ for } i \in \Sigma\}$ — move from such an x towards an interior point y along $x + sy$; at the first parameter s^* with $x + s^* y \in Q$, if $s^* > 0$ the point $z = x + s^* y$ is nonzero (else $\ell_i(x) < 0$ for $i \in \Sigma$), so it has $\ell_i(z) > 0$ for all i ($\ell_i(z) \ge s^* \ell_i(y) > 0$ in Σ, by definition of Σ outside), and finitely many strict inequalities survive a small step back, contradicting minimality Chapter3/GoodPosition · mem_quadCone_of_vanIdx. This replaces the

text's extreme rays (Lemmas 3.62–3.64, cited from [21], and 3.66–3.68). Then a linear right inverse g of Q_Σ gives $e(x) = (Q_\Sigma x, x - gQ_\Sigma x) \in \mathbb{R}^\Sigma \times \ker Q_\Sigma$ `Chapter3/GoodPosition · splitEquiv`, and $\ker Q_\Sigma = N \cap W$ `Chapter3/GoodPosition · quadrantNormalForm_of_vanIdx`. For a partial quadrant in good position this applies in a presentation `Chapter3/GoodPosition · IsInGoodPosition.quadrantNormalForm`, and choosing a basis of N_0 makes $(N, N \cap C)$ isomorphic to $(\mathbb{R}^{\dim N}, [0,\infty)^k \oplus \mathbb{R}^{\dim N - k})$ through an sc-isomorphism of constant structures `Chapter3/GoodPosition · scIsoOfLinearEquiv` `Chapter3/GoodPosition · preimage_eq_of_normalForm`: **$N \cap C$ is a partial quadrant in N** `Chapter3/GoodPosition · proposition_3_22`.

Lemma 3.45 (linear form). A continuous linear isomorphism $S : E_0 \to E_0'$ with $x \in C \iff Sx \in C'$ carries a good complement P of N to one of $S(N)$, with $\varepsilon' = \varepsilon/(\|S\|\|S^{-1}\|+1)$: for $|Sp| \le \varepsilon'|Sn|$, $|p| \le \|S^{-1}\|\varepsilon'\|S\||n| \le \varepsilon|n|$ `Chapter3/GoodPosition · IsGoodComplement.map`; and the interior condition is transported by the homeomorphism $S|_N$ `Chapter3/GoodPosition · IsInGoodPosition.map`. The text states it for $T\varphi(0)$; since the coordinate change of Theorem 3.38 is linear, this is the form §§III.18–III.23 need.

Proposition 3.69. In the model $\mathbb{R}^n \oplus W$, for $x \in N \cap C$: $d_{N\cap C}(x) = d_C(x)$ if $x \notin N \cap W$, and $d_{N\cap C}(x) + \dim(N \cap W) = \dim N$ if $x \in N \cap W$ `Chapter3/GoodPosition · proposition_3_69` — read off the normal form ($d_{N\cap C}$ does not depend on the presentation, Lemma 1.24).

III.15 The degeneracy index along a graph: Lemma 3.70

Lemma 3.70. In the model, for N in good position with good complement Y and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v + \tau(v) \in C$, and $d_{N\cap C}(v) = d_C(v + \tau(v))$ if $v \notin N \cap W$ `Chapter3/GoodPositionGraph · lemma_3_70`. The text distinguishes $\dim N = 1$ (a contradiction argument with $\tau(v) = o(|v|)$) and $\dim N \ge 2$ (the sign of a directional derivative). The formal proof unifies them.

Ray decomposition. From the normal form, $u_i = e^{-1}(\mathbf{e}_i, 0)$ are finitely many nonzero points of Q, and every $v \in Q \setminus N \cap W$ has some i and $s > 0$ with $v - su_i \in Q$ and $s|u_i| \le c|v|$ (take $s = \lambda_i > 0$ for $e(v) = (\lambda, n_0)$) `Chapter3/GoodPositionGraph · exists_rayDecomp` — the text's basis e^j of $\widetilde{N}$. *Smallness.* $D\tau(0) = 0$ and continuity of $D\tau$ on the convex Q make τ η-Lipschitz on $Q \cap B(\rho)$ for every η `Chapter3/GoodPositionGraph · exists_lipschitz_small` `Chapter3/GoodPositionGraph · quadCone_convex`. *The margin.* Choose $\eta \le \varepsilon$ with $\eta\|\ell_j\||u_i| < \ell_j(u_i)$ for all i and $j \notin \Sigma$ — a positive lower bound of finitely many positive numbers `Chapter3/GoodPositionGraph · exists_pos_le_of_forall_pos`. *Conclusion* `Chapter3/GoodPositionGraph · lemma_3_70_core`. For $|v| < \rho/(1+c)$: $|\tau(v)| \le \varepsilon|v|$, so $v + \tau(v) \in C$ (good complement). For $j \in \Sigma$, $\ell_j(\tau(v)) = 0$ (Lemma 3.65). For $v \notin N \cap W$, $w = v - su_i \in Q$, the segment $[w, v]$ lies in $Q \cap B(\rho)$ — which is why the radius is $\rho/(1+c)$, the text's "close to 0" — and for $j \notin \Sigma$,

$$\ell_j(v + \tau(v)) = \ell_j(w + \tau(w)) + s\ell_j(u_i) + \ell_j(\tau(v) - \tau(w)) \ge 0 + s\ell_j(u_i) - \|\ell_j\|\eta s|u_i| > 0.$$

So v and $v + \tau(v)$ have the same vanishing coordinates, and Proposition 3.69 (1) gives the lemma in the model, with the coordinate functionals `Chapter3/GoodPositionGraph · coordCLM`. (The level Y_1 of the text's τ plays no role.)

III.16 The quadrant inverse function theorem (Theorem 3.72, Proposition 3.73, Lemmas 3.74, 3.77, Corollary 3.75)

Interior points. For the model quadrant, x is interior iff all its half-line coordinates are positive `Chapter3/QuadrantInverse · interior_quadrant`, so $d_C(x) = 0$ exactly at interior points `Chapter3/QuadrantInverse · degIndex_eq_zero_iff`. A linear isomorphism S with $S(C) = C$ preserves d_C (Lemma 1.24 for the presentation $T \circ S$) `Chapter3/QuadrantInverse · ScIso.ofCLE` `Chapter3/QuadrantInverse · degIndex_map_eq`. A linear map carrying a set with nonempty interior onto itself is onto `Chapter3/QuadrantInverse · surjective_of_image_eq` — the text's "$\mathbb{R}^n = C - C$".

Proposition 3.73. The hypotheses (`InvData`): U relatively open and convex in the partial quadrant C of a finite-dimensional V; $f : U \to C$ of class C^j, $j \geq 1$; $\|Df(x) - 1\| \leq \frac{1}{2}$; and x interior iff $f(x)$ interior — the only consequence of $d_C(x) = d_C(f(x))$ the proof uses `Chapter3/QuadrantInverse · InvData`. The text's $0 \in U$, $f(0) = 0$ are not needed. Conclusion: $f(U)$ is relatively open in C, f is injective on U, with an inverse of class C^j on $f(U)$ `Chapter3/QuadrantInverse · proposition_3_73`.

Lemma 3.74. $x \mapsto x - f(x)$ is a $\frac{1}{2}$-contraction on the convex U (mean value inequality) `Chapter3/QuadrantInverse · InvData.norm_sub_sub_le`, which gives (70): $\frac{1}{2}|x - x'| \leq |f(x) - f(x')| \leq \frac{3}{2}|x - x'|$ `Chapter3/QuadrantInverse · InvData.half_norm_le` `Chapter3/QuadrantInverse · InvData.norm_sub_le`, and injectivity `Chapter3/QuadrantInverse · InvData.injOn`. *Openness*: if $B(x', 3r) \cap C \subseteq U$ then $B(f(x'), r) \cap C \subseteq f(U)$ `Chapter3/QuadrantInverse · InvData.ball_inter_subset_image`. The set $A = f(\bar{B}(x', 2r) \cap C)$ is compact ($\bar{B}(x', 2r) \cap C \subseteq U$ is compact in finite dimensions). Interior points first: for an interior $y_1 \in B(y', r)$ choose an interior $x_0 \in U$ with $f(x_0) \in B(y', r)$ — the text takes any interior $x_0 \in B(x', 3r) \cap C$, which does not suffice — and follow the segment y_t from $f(x_0)$ to y_1, which stays interior. $\{t : y_t \in A\}$ is closed and contains 0; at its maximum $t^* < 1$, $y_{t^*} = f(x^*)$ with x^* interior and $|x^* - x'| < 2r$, and at the interior point x^*, $Df(x^*)$ is a unit `Chapter3/QuadrantInverse · InvData.isUnit_deriv` and U a neighbourhood `Chapter3/QuadrantInverse · InvData.U_mem_nhds`, so the classical inverse function theorem maps neighbourhoods onto neighbourhoods `Chapter3/QuadrantInverse · InvData.map_nhds_eq`, pushing t^* further — a contradiction. The compactness of A replaces the text's Cauchy sequence and also gives the remaining points by density. So $f(U)$ is relatively open `Chapter3/QuadrantInverse · InvData.image_relOpen`. **Corollary 3.75**: f and its inverse `Chapter3/QuadrantInverse · InvData.inv` (2-Lipschitz by (70) `Chapter3/QuadrantInverse · InvData.norm_inv_sub_le`) are continuous and inverse to each other `Chapter3/QuadrantInverse · corollary_3_75`.

Lemma 3.77. The inverse g is differentiable within $f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$ `Chapter3/QuadrantInverse · InvData.hasFDerivWithinAt_inv`: with $x_0 = g(y_0)$, $x_0 + \delta = g(y_0 + h)$, the quotient $|g(y_0 + h) - g(y_0) - Df(x_0)^{-1}h|/|h|$ is $|\delta|/|h|$ (bounded by (70)) times the remainder quotient of f at x_0. The lemma's extra hypotheses (relatively open image, homeomorphism, invertible derivative) are consequences of those of Proposition 3.73. *Regularity (the text: Lemma 3.76 on the iterated tangent maps $T^k f$).* $x \mapsto [Df(x)]^{-1}$ is C^{j-1} (inversion is smooth on units) `Chapter3/QuadrantInverse · InvData.contDiffOn_inverse_deriv`, so $Dg = [Df]^{-1} \circ g$ and $g \in C^i$, $i < j$, give $g \in C^{i+1}$: g is C^j `Chapter3/QuadrantInverse · InvData.contDiffOn_inv`. Lemma 3.76 is not needed.

Theorem 3.72. U a relatively open neighbourhood of 0 in C, f of class C^j with $f(U) \subseteq C$, $f(0) = 0$, $Df(0)C = C$, $d_C(x) = d_C(f(x))$ on U: then f is a C^j-diffeomorphism between relatively open neighbourhoods U', V' of 0 `Chapter3/QuadrantInverse · theorem_3_72`. *Proof.* $Df(0)$ is onto, hence invertible, with $Df(0)^{-1}C = C$; $\tilde{f} = Df(0)^{-1}f$ maps into C, $D\tilde{f}(0) = 1$, and preserves d_C ($d_C \circ Df(0)^{-1} = d_C$), hence interior points. On $U' = B(0, \rho) \cap C$ with $\|D\tilde{f} - 1\| \leq \frac{1}{2}$, Proposition 3.73 applies, and $V' = Df(0)\tilde{f}(U')$ is relatively open since $Df(0)$ is a linear homeomorphism preserving C. The text's hypothesis (2) is used only through interior points; some such hypothesis is needed ($f(x, y) = (x, y+x^2)$ on $[0, \infty)^2$ satisfies (1) but $f(C)$ is not a neighbourhood of 0 in C).

III.17 The implicit function theorem in partial quadrants (Theorem 3.78, Lemmas 3.79–3.82)

Theorem 3.78. U a relatively open neighbourhood of 0 in the partial quadrant C of a finite-dimensional V, $f : U \to F$ of class C^j, $j \geq 1$, with $f(0) = 0$, $Df(0)$ onto, $K = \ker Df(0)$ in good position with good complement Y. Then there are a relatively open $V' \ni 0$ in $K \cap C$, $\tau : K \to Y$ of class C^j on V' and $\varepsilon, \sigma > 0$ with (1) $\tau(0) = 0$, $D\tau(0) = 0$, and $k + \tau(k) \in U$, $f(k + \tau(k)) = 0$ for $k \in V'$; (2) $|\tau(k)| \leq \sigma$ for $|k| \leq \varepsilon$; (3) if $k \in V'$, $|k| \leq \varepsilon$, $y \in Y$, $|y| \leq \sigma$, $k + y \in U$ and $f(k+y) = 0$, then $y = \tau(k)$ `Chapter3/QuadrantIFT · theorem_3_78`. The two bold clauses are not in the text; the first is the content of the theorem (its proof has $\tau(k) = B(k, \tau(k))$), the second is needed for $f(k + y)$ to make sense.

Normalization (71)–(72). With $L = (Df(0)|_Y)^{-1}$, in coordinates $x = k + y \in K \oplus Y$, $Lf(k+y) = y - B(k,y)$ with $B(k,y) := y - Lf(k+y)$, $B(0,0) = 0$, $DB(0,0) = 0$. The data after normalization: on $K \times Y$ a convex set C_p with nonempty interior, a relatively open $U_p \ni 0$, B of class C^j on U_p with $B(0) = 0$, $DB(0) = 0$, the good-position estimate and the interior condition `Chapter3/QuadrantIFT · ContractionData`. Only convexity and interior are used; the quadrant enters through good position.

Lemmas 3.79, 3.80: the constants `Chapter3/QuadrantIFT · Radii` `Chapter3/QuadrantIFT · ContractionData.exists_radii`. With ε the good-position constant and $c_0 = \varepsilon/(2(1+\varepsilon)) \le \frac{1}{2}$ `Chapter3/QuadrantIFT · Radii.c₀`, continuity of DB at 0 gives a, b with $\varepsilon a \le b$, the region $R = \{|k| \le a, |y| \le b\} \cap C_p \subseteq U_p$ `Chapter3/QuadrantIFT · Radii.R` `Chapter3/QuadrantIFT · Radii.R_sub_U` and $\|DB\| \le c_0$ on R. R is convex `Chapter3/QuadrantIFT · Radii.R_convex`, so the mean value inequality gives $|B(q) - B(p)| \le c_0|q - p|$ on R `Chapter3/QuadrantIFT · Radii.norm_sub_le` `Chapter3/QuadrantIFT · Radii.norm_B_sub_le_snd` and $|B(p)| \le c_0|p|$ `Chapter3/QuadrantIFT · Radii.norm_B_le`. For $k \in V_0 = \{k \in K \cap C : |k| < a\}$ `Chapter3/QuadrantIFT · Radii.V₀` and $|y| \le \varepsilon|k|$: good position puts $(k,y) \in C_p$, $|y| \le \varepsilon a \le b$, so $(k,y) \in R$ and $|B(k,y)| \le c_0(1+\varepsilon)|k| = \frac{\varepsilon}{2}|k|$. So $B(k,\cdot)$ maps $X_k = \bar{B}_Y(\varepsilon|k|)$ into itself `Chapter3/QuadrantIFT · Radii.mapsTo_ball` and has a fixed point $\tau(k)$ there `Chapter3/QuadrantIFT · Radii.exists_fixed` `Chapter3/QuadrantIFT · Radii.τ`; any fixed point y with $(k,y) \in R$ equals $\tau(k)$ `Chapter3/QuadrantIFT · Radii.eq_τ`. (In Lemma 3.80 the maximum is over $B((1+\varepsilon)a) \cap C$, where DB is defined.)

Lemma 3.81: continuity. $|\tau(k)| \le \frac{\varepsilon}{2}|k|$ `Chapter3/QuadrantIFT · Radii.norm_τ_le`. If $|k_0| \le 2|k|$, then $(k, \tau(k_0)) \in R$ and $|\tau(k) - \tau(k_0)| \le \frac{1}{2}|\tau(k) - \tau(k_0)| + c_0|k - k_0|$, so $|\tau(k) - \tau(k_0)| \le \varepsilon|k - k_0|$ — the text's estimate, whose argument needs $(k, \tau(k_0))$ in the domain. If $|k_0| > 2|k|$, then $|k - k_0| \ge \frac{1}{2}|k_0|$ and $|\tau(k) - \tau(k_0)| \le \frac{\varepsilon}{2}(|k| + |k_0|) \le \frac{3\varepsilon}{2}|k - k_0|$. In all cases the constant is $1 + 2\varepsilon$ `Chapter3/QuadrantIFT · Radii.norm_τ_sub_le`, and τ is continuous `Chapter3/QuadrantIFT · Radii.continuousOn_τ`. Since X_k depends on k, the parametrized fixed point theorem of §III.8 does not apply, and continuity is proved by hand, as in the text.

Lemma 3.82: regularity. With $D_1 B = DB \circ \mathrm{inl}$, $D_2 B = DB \circ \mathrm{inr}$ `Chapter3/QuadrantIFT · Radii.D₁` `Chapter3/QuadrantIFT · Radii.D₂`, $\|D_2 B\| \le \frac{1}{2}$ on R `Chapter3/QuadrantIFT · Radii.norm_D₂_le`, $M = (1 - D_2 B)^{-1}$ (Neumann) `Chapter3/QuadrantIFT · Radii.M` and $\Phi = M D_1 B$ `Chapter3/QuadrantIFT · Radii.Φ`: from the fixed-point identities and the little-o composition of B with the Lipschitz $k' \mapsto (k', \tau(k'))$, τ is differentiable within V_0 with $D\tau(k) = \Phi(k, \tau(k))$ `Chapter3/QuadrantIFT · Radii.hasFDerivWithinAt_τ` `Chapter3/QuadrantIFT · Radii.fderivWithin_τ`. Φ is C^{j-1} on R `Chapter3/QuadrantIFT · Radii.contDiffOn_Φ`, so $\tau \in C^i$, $i < j$, gives $D\tau = \Phi \circ (\mathrm{id}, \tau) \in C^i$ and $\tau \in C^{i+1}$: τ is C^j on all of V_0 `Chapter3/QuadrantIFT · Radii.contDiffOn_τ`. The text iterates $f \mapsto f^{(1)}$ (the finite-dimensional model of Lemma 3.31) and concludes "on a possibly smaller neighbourhood"; in finite dimensions there are no levels to shift and no shrinking is needed. *Assembly* `Chapter3/QuadrantIFT · ContractionData.exists_solution`: $V' = V_0$, $\sigma = b$, ε below a; (1) $Lf(k + \tau(k)) = \tau(k) - B(k, \tau(k)) = 0$ and L is injective; (3) $f(k+y) = 0$ means $B(k,y) = y$ with $(k,y) \in R$.

III.18 Weak stability of basic germs: Theorem 3.38

Theorem 3.38 (text): for a basic germ f and the principal part s of an sc⁺-section with $s(0) = 0$ there is a strong bundle isomorphism Φ covering an sc-diffeomorphism φ such that $(\Phi_*(f+s))^1$, with the levels raised by one, is a basic germ, and $n - N = n' - N'$. The text's proof: with $A = P_W \circ D_2 s(0)$, $1 + A$ is sc-Fredholm of index 0, with sc-splittings $W = C \oplus X = R \oplus Z$, $C = \ker(1+A)$, $R = \mathrm{range}(1+A)$; $P_W \circ s(a,w) = Aw + S(a,w)$ with S a small contraction in w on every level $m \ge 1$; and with $L = (1+A)|_X : X \to R$ the equation becomes $x - L^{-1} P_2 \bar{B}(a, c + x) = 0$ in the new parameters (a,c) and the new fibre variable x. The formal proof follows it with two observations: the coordinate change is *linear*, and the index raise is a *change of sc-structure*.

The linear part of s. For an sc¹-germ s into W^1, $A = D_2 s(0)$ is an sc⁺-operator whose level-m map is the level operator of Proposition 1.16 at $0 \in U_{m+1}$ `Chapter3/BasicClass · scPlusOfDeriv`. The remainder

$S(a,w) = s(a,w) - Aw$ on level $m+1$ Chapter3/BasicClass · scPlusRemainder is, for a and w small depending on m and ε, an ε-contraction in w: $D_2 S(0) = 0$ on level $m+1$, the derivative of $s : U_{m+1} \to W_{m+1}$ is continuous in the operator norm (§III.1), and the mean value inequality on a ball for fixed a concludes Chapter3/BasicClass · scPlusRemainder_contraction. The partial derivative of S in a does not vanish and plays no role. The text gives this one sentence ("S is … an arbitrary small contraction on every level $m \geq 1$").

The data and the splitting. The data of the theorem are the germ G, the W-component s of the sc$^+$-germ with $s(0) = 0$, and an sc-Fredholm splitting $W = C \oplus X = R \oplus Z$ of $T = 1 + A$ Chapter3/BasicClass · StabilityData. Such data exist: the identity is sc-Fredholm of index 0 (the trivial splitting) Chapter3/BasicClass · isScFredholm_id Chapter3/BasicClass · scIndex_id, so $1 + A$ is sc-Fredholm by Proposition 1.10 Chapter3/BasicClass · StabilityData.exists_of_scPlus. With the projections $P_1 : W_m \to X_m$ along C_m, $P_2 : W_m \to R_m$ along Z_m Chapter3/BasicClass · StabilityData.projX Chapter3/BasicClass · StabilityData.projR, $L = T|_X : X_m \to R_m$ is bijective Chapter3/BasicClass · StabilityData.Lm_bijective, a topological isomorphism by the open mapping theorem Chapter3/BasicClass · StabilityData.Lequiv, and $L^{-1}P_2$ is an sc-operator $W \to X$ Chapter3/BasicClass · StabilityData.LinvPOp. The finite-dimensional C, $Z \subseteq W_\infty$ lift to every level Chapter3/BasicClass · ScBanach.finLift.

The coordinate changes. On the index-raised spaces, the new source is $E' = (P \oplus C) \oplus X^1$ Chapter3/BasicClass · StabilityData.E' and the new target $F' = (Q \oplus Z) \oplus X^1$ Chapter3/BasicClass · StabilityData.F' — the text's $\tau : Z \to C$, making the new finite-dimensional target $\mathbb{R}^N \oplus C$, is not applied. The base change $\psi(a,w) = ((a,(1-P_1)w), P_1 w)$ Chapter3/BasicClass · StabilityData.split and its inverse $\psi^{-1}((a,c),x) = (a, c + x)$ Chapter3/BasicClass · StabilityData.glue are sc-operators inverse to each other Chapter3/BasicClass · StabilityData.glue_split Chapter3/BasicClass · StabilityData.split_glue; so are the fibre change $\Psi(q,w) = ((q,(1-P_2)w), L^{-1}P_2 w)$ Chapter3/BasicClass · StabilityData.fiber and $\Psi^{-1}((q,z),x) = (q, z + (1+A)x)$ Chapter3/BasicClass · StabilityData.fiberInv Chapter3/BasicClass · StabilityData.fiber_fiberInv Chapter3/BasicClass · StabilityData.fiberInv_fiber, and Ψ vanishes only at 0 Chapter3/BasicClass · StabilityData.fiber_eq_zero_iff. No general strong bundle isomorphism is needed.

The push-forward. The perturbed germ is $f + \iota s$ with the inclusion $W^1 \to Q \oplus W$ Chapter3/BasicClass · ScBanach.inrOp Chapter3/BasicClass · StabilityData.fsum, sc-smooth by the pairing and operator rules Chapter3/BasicClass · StabilityData.fsum_isScSmoothGerm. Its germ of neighbourhoods on E' is $\psi(U_{m+1})$, the preimage of U_{m+1} under ψ^{-1} Chapter3/BasicClass · StabilityData.pushGerm Chapter3/BasicClass · StabilityData.glue_mapsToGerm, and the push-forward is $(\Phi_*(f+s))^1 = \Psi \circ (f+\iota s)^1 \circ \psi^{-1}$ with the level-shifted operators Chapter3/BasicClass · ScOperator.shiftOp Chapter3/BasicClass · StabilityData.pushMap. It is sc-smooth: $(f+\iota s)^1$ is sc-smooth on the shifted germ (Proposition 1.17 for germs) and Ψ, ψ^{-1} are sc-operators Chapter3/BasicClass · StabilityData.pushMap_isScSmoothGerm. With $\bar{B} = B - S$ Chapter3/BasicClass · StabilityData.Bbar, the W-component of $(f+\iota s)^1$ at a level-1 point is $(1+A)w - \bar{B}(a,w)$ Chapter3/BasicClass · StabilityData.snd_levelMap_fsum, so the X-component of the push-forward is $x - \hat{B}((a,c),x)$ with $\hat{B}((a,c),x) = L^{-1}P_2\bar{B}(a, c+x)$ Chapter3/BasicClass · StabilityData.pushB Chapter3/BasicClass · StabilityData.snd_pushMap — the text's "desired formula". $\hat{B}$ is an sc^0-contraction germ on the index-raised structure: on level m of E', i.e. level $m+1$ of W, B contracts by Definition 3.7 and S by the remainder estimate, and $L^{-1}P_2$ is bounded Chapter3/BasicClass · StabilityData.levelMap_pushB Chapter3/BasicClass · StabilityData.pushB_isSc0ContractionGerm. So **the push-forward is a basic germ with contraction map** $\hat{B}$ Chapter3/BasicClass · theorem_3_38. Its zeros correspond to those of $f+\iota s$ under ψ, since Ψ is a linear isomorphism Chapter3/BasicClass · StabilityData.pushMap_eq_zero_iff. *The index*: $\operatorname{ind}(1+A) = \operatorname{ind}(1) = 0$ (Proposition 1.10 with its index), i.e. $\dim C = \dim Z$, so $n' - N' = (n + \dim C) - (N + \dim Z) = n - N$ Chapter3/BasicClass · StabilityData.finrank_Cs_eq_finrank_Zs.

The same on level 0. For §§III.19–III.20 the coordinate changes are also needed before the index raise: ψ, ψ^{-1}, Ψ, Ψ^{-1} on every level Chapter3/BasicClass · StabilityData.split₀ Chapter3/BasicClass · StabilityData.glue₀ Chapter3/BasicClass · StabilityData.fiber₀, the germ $\psi(U_m)$ Chapter3/BasicClass · StabilityData.pushGerm₀, and $g = \Phi_*(f+s) = \Psi \circ (f+\iota s) \circ \psi^{-1}$ on level 0 Chapter3/BasicClass · StabilityData.pushMap₀, sc-smooth Chapter3/BasicClass · StabilityData.pushMap₀_isScSmoothGerm, whose level-1 map is the index-raised push-forward

Chapter3/BasicClass · StabilityData.pushMap_eq_levelMap_pushMap₀.

III.19 The linearization at nearby solutions: Corollary 3.41

Corollary 3.41. For the data of Theorem 3.38 with $f := h$ basic and $D(h+\iota s)(0)$ surjective, there is $r > 0$ such that every zero x of $h+\iota s$ in U_0 with $|x|_0 < r$ is $\iota(x_1)$ with $x_1 \in U_1$, and $D(h+\iota s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$ Chapter3/SolutionLinearization · StabilityData.corollary_3_41. The text's sc⁺-map s has an $\mathbb{R}^N$-component; it can be absorbed into the basic germ (as in §III.23), so the data carry only the W-component.

The derivative on U_1. For $x \in U_1$, $D(h + \iota s)(x) = Dh(x) + \iota Ds(x)$ — uniqueness of the derivative within U_1, then density of E_1 Chapter3/SolutionLinearization · StabilityData.Dfs'_eq — and $\mathrm{pr}_W Dh(x) = \mathrm{pr}_W - DB(x)$, from $\mathrm{pr}_W h = \mathrm{pr}_W - B$ on U_0 Chapter3/SolutionLinearization · StabilityData.snd_Df_eq; $D_2B(0) = 0$ Chapter3/SolutionLinearization · StabilityData.DB_zero_snd. *The decomposition.* $D(h + \iota s)(x) = D(h + \iota s)(0) + S(x) + K(x)$ with $S(x)(\alpha, \zeta) = (0, -D_2B(x)\zeta)$ Chapter3/SolutionLinearization · StabilityData.Sop Chapter3/SolutionLinearization · StabilityData.Dfs'_eq_add of norm $\le \varepsilon$ for $x \in U_1$ small on level 1, by Lemma 3.9 (c) Chapter3/SolutionLinearization · StabilityData.exists_norm_Sop_le, and $K(x)$ Chapter3/SolutionLinearization · StabilityData.Kop compact: the sum of finite-rank operators through Q and through P and of $\iota(Ds(x) - Ds(0))$ through the compact inclusion $W_1 \to W_0$ Chapter3/SolutionLinearization · StabilityData.Kop_isCompactOperator.

Fredholm with index $n - N$ at every small $x \in U_1$. $D(h + \iota s)(0)$ is Fredholm of index $n - N$: $Dh(0)$ is (Proposition 3.10 for the basic germ, §III.5) and $\iota Ds(0)$ is compact Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs. A small perturbation of a Fredholm operator is Fredholm: if v is a quasi-inverse of u and $\|v\|\|s\| < 1$, then $(1 + vs)^{-1}v$ is a quasi-inverse of $u + s$ (Neumann series) Chapter3/SolutionLinearization · isFredholm_add_of_norm_lt, with the same index Chapter3/SolutionLinearization · isFredholm_fredholmIndex_add_of_norm_lt; a compact perturbation does not change the index Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator. So $D(h + \iota s)(x)$ is Fredholm of index $n - N$ for *every* $x \in U_1$ small on level 1, not only at solutions Chapter3/SolutionLinearization · StabilityData.exists_isFredholm_Dfs'.

Surjectivity: the compactness argument. Let N be a closed complement of $\ker D(h + \iota s)(0)$. Then $\ker D(h + \iota s)(x) \cap N = 0$ for $x \in U_1$ small on level 1 Chapter3/SolutionLinearization · StabilityData.exists_ker_Dfs'_inf_eq_bot. Otherwise there are $x_k \to 0$ in E_1 and unit vectors $(\alpha_k, \zeta_k) \in N$ in the kernels; the W-component of the equation reads $\zeta_k = D_1B(x_k)\alpha_k + D_2B(x_k)\zeta_k - \iota Ds(x_k)(\alpha_k, \zeta_k)$. α_k has a convergent subsequence (P finite-dimensional); $Ds(x_k)(\alpha_k, \zeta_k)$ is bounded in W_1 (the derivatives near 0 are uniformly bounded Chapter3/SolutionLinearization · IsSc1GermWith.exists_norm_deriv_le), so its image in W_0 has a convergent subsequence; $D_1B(x_k)\alpha_k \to D_1B(0)\alpha$ by the joint continuity of the tangent map Chapter3/SolutionLinearization · IsSc1GermWith.tendsto_deriv; and $\|D_2B(x_k)\| \to 0$. So ζ_k converges, $(\alpha_k, \zeta_k) \to (\alpha, z_0) \in N$ of norm 1 in $\ker D(h + \iota s)(0)$ — a contradiction. (The text uses the Neumann series $\zeta_k = \sum (D_2B(x_k))^l z_k$; here $|\zeta_k - z_k| \le \varepsilon_k \to 0$ suffices.) Then $\dim\ker D(h + \iota s)(x) \le \dim\ker D(h + \iota s)(0) = n - N = \mathrm{ind}$, and an operator of finite index with $\dim\ker \le \mathrm{ind}$ is onto Chapter3/SolutionLinearization · HasFiniteIndex.surjective_of_finrank_ker_le — instead of the text's "$Df(x)|_N$ has index 0 and is injective". *Zeros are on level 1 and small there*: by the lifts of Theorem 3.40, continuous at 0 (§III.12); the text's detour through "bounded on level 2, hence convergent on level 1" is not needed.

III.20 Local solvability: Lemma 3.46, Theorem 3.43, Lemmas 3.42, 3.44, Corollary 3.47

Lemma 3.46 (Appendix 3.8.2). If $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$, the image of the graph map Chapter3/LocalSolvability · graphMap (with $(\alpha, L\alpha) \in \widetilde{C} \oplus W$ iff $\alpha \in \widetilde{C}$ Chapter3/LocalSolvability · graphMap_mem_prod_

iff), has a good complement Y' with respect to $\widetilde{C}\oplus W$, then $\widetilde{K}$ has a good complement Z with respect to $\widetilde{C}$ such that $Z\times W$ is a good complement of K' Chapter3/LocalSolvability · lemma_3_46. *Proof.* $Z' := \mathrm{pr}_P(Y')$ satisfies $P = \widetilde{K} + Z'$; choose $Z \subseteq Z'$ with $P = \widetilde{K} \oplus Z$ and a linear right inverse $\rho : Z' \to Y'$ of the projection (bounded, finite dimensions); for $|z| \le \varepsilon'|\alpha|$, $\varepsilon' = \varepsilon/(\|\rho\| + 1)$, $y = \rho(z) = (z, w) \in Y'$ has $|y| \le \varepsilon|(\alpha, L\alpha)|$, so $(\alpha, L\alpha) \in C'$ iff $(\alpha, L\alpha) + y \in C'$, i.e. $\alpha \in \widetilde{C}$ iff $\alpha + z \in \widetilde{C}$. The text's intermediate complement Y_0 is not needed. Good position of K' gives good position of $\widetilde{K}$ Chapter3/LocalSolvability · lemma_3_46_isInGoodPosition.

Projecting solutions into the quadrant. A gap in the text's proof of Theorem 3.43 (3): a solution $a = k+z$ ($k \in \widetilde{K}$, $z \in Z$) of $H(a) = 0$ is matched with the solution of Theorem 3.78 by its uniqueness clause, which needs $k \in \widetilde{C}$ — not verified in the text. It holds because z is small compared with k: from $H(a) = 0 = H(0)$ and $DH(0)k = 0$, $|DH(0)z| = o(|a|)$; $DH(0)$ is injective on Z, hence bounded below there Chapter3/LocalSolvability · exists_norm_le_of_injOn; so $|z| \le \varepsilon|k|$ near 0, and the good-complement property gives $k \in \widetilde{C}$ from $k + z = a \in \widetilde{C}$ Chapter3/LocalSolvability · exists_proj_mem_of_eq_zero.

The data of Theorem 3.43. The text states it for $g = \Phi_*(f + s)$; the formal statement is for any sc-smooth germ g on G with $g(0) = 0$ whose index-raised germ g^1 on the shifted germ is a basic germ with contraction map B Chapter3/LocalSolvability · ShiftedBasicData. Its solution germ δ (of the working form of B) takes values in W_1 Chapter3/LocalSolvability · ShiftedBasicData.δ, with $\mathrm{pr}_W g(a, \delta(a)) = 0$ on level 0 Chapter3/LocalSolvability · ShiftedBasicData.snd_g_sol and uniqueness in the contraction region Chapter3/LocalSolvability · ShiftedBasicData.eq_δ. Put $H(a) = \mathrm{pr}_Q\, g(a, \delta(a))$ Chapter3/LocalSolvability · ShiftedBasicData.H. *Regularity of H* (the text: "H restricted to U_i' is of class C^{i-1}"): for every $j \ge 1$ there is a relatively open $V \ni 0$ in $\widetilde{C}$ on which H is C^j — by Corollary 3.34 for g^1, δ lifts to a C^j map into W_{2j}, and $g : E_{2j} \to F_j$ is C^j by Proposition 1.18 for germs Chapter3/LocalSolvability · ShiftedBasicData.H_contDiffOn. A single C^1 neighbourhood V_1 carries the derivative data Chapter3/LocalSolvability · ShiftedBasicData.Deriv Chapter3/LocalSolvability · ShiftedBasicData.exists_deriv: $D\delta(0) : P \to W_1$, $L = \iota D\delta(0)$, and $DH(0) = \mathrm{pr}_Q Dg(0)(\alpha, L\alpha)$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_H, by the chain rule for $a \mapsto g(a, \delta(a))$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.hasFDerivWithinAt_comp.

Lemma 3.44. "Recalling from Lemma 3.9 that $D_2B(0) = 0$": Lemma 3.9 gives this for g^1, on level 1; the kernel of $Dg(0)$ is computed on level 0. So first the W-component of $Dg(0)(0, \zeta)$ is ζ for $\zeta \in W_1$, by Lemma 3.9 (b) for g^1, and then for all $\zeta \in W_0$ by density Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_inr. Differentiating $\mathrm{pr}_W g(a, \delta(a)) = 0$ at 0 gives $\mathrm{pr}_W Dg(0)(\alpha, L\alpha) = 0$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.snd_Dg_graph. Hence $Dg(0)(\alpha, \zeta) = 0$ iff $\zeta = L\alpha$ and $DH(0)\alpha = 0$: $\ker Dg(0) = \{(\alpha, D\delta(0)\alpha) : \alpha \in \ker DH(0)\}$ Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq, and $DH(0)$ is onto when $Dg(0)$ is Chapter3/LocalSolvability · ShiftedBasicData.Deriv.DH₀_surjective.

Theorem 3.43 Chapter3/LocalSolvability · theorem_3_43. Hypotheses: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position to $\widetilde{C} \oplus W$, and *the regularity of Lemma 3.42 as an explicit hypothesis*: for every $\varepsilon > 0$ there is $r > 0$ such that every zero x of g in U_0 with $|x|_0 < r$ has W-component $\iota(w_1)$, $|w_1|_1 < \varepsilon$. *The choices* Chapter3/LocalSolvability · ShiftedBasicData.Sol Chapter3/LocalSolvability · ShiftedBasicData.exists_sol: $\widetilde{K} = \ker DH(0)$ with K' its graph (Lemma 3.44); the good complement Z of $\widetilde{K}$ with $Z \oplus W$ good for K' (Lemma 3.46); and the solution $\sigma : V' \to Z$ of $H(k + \sigma(k)) = 0$ with its uniqueness constants, from Theorem 3.78 applied to H (§III.17). Then $\tau(y) = (\sigma(v), \delta(v + \sigma(v)) - Lv)$ for $y = (v, Lv)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ on $V = \pi^{-1}(V')$, relatively open in $K' \cap C'$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.V Chapter3/LocalSolvability · ShiftedBasicData.Sol.V_relOpen, with values in $Z \oplus W$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_mem_Y, so that $y + \tau(y) = (a, \delta(a))$ with $a = v + \sigma(v)$ Chapter3/LocalSolvability · ShiftedBasicData.Sol.coe_add_τ. (3), first half: $y + \tau(y)$ lies in U_0 and is a zero,

since $\mathrm{pr}_W g(a,\delta(a)) = 0$ and $\mathrm{pr}_Q g(a,\delta(a)) = H(a) = 0$ `Chapter3/LocalSolvability · ShiftedBasicData.Sol.sol_mem_and_eq_zero`. Regularity: τ is C^1 on V `Chapter3/LocalSolvability · ShiftedBasicData.Sol.τ_contDiffOn` with W-component on level 1 `Chapter3/LocalSolvability · ShiftedBasicData.Sol.snd_τ_mem_range`, and $D\tau(0)\eta = -D\delta(0)\pi(\eta) + D\delta(0)\pi(\eta) = 0$ `Chapter3/LocalSolvability · ShiftedBasicData.Sol.hasFDerivWithinAt_τ` `Chapter3/LocalSolvability · ShiftedBasicData.Sol.fderivWithin_τ` — (2). (3), second half `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ`: a zero $x = (a, w)$ near 0 has, by the regularity hypothesis, w on level 1 and small there, so $w = \delta(a)$ by the uniqueness of the solution germ on level 1, so $H(a) = 0$; write $a = k + z$ with the projection along Z `Chapter3/LocalSolvability · ShiftedBasicData.Sol.projK`; then $k \in \widetilde{K} \cap \widetilde{C}$ (the gap above) and $z = \sigma(k)$ by the uniqueness clause of Theorem 3.78; so $x = y + \tau(y)$ for $y = (k, Lk)$. *Clause (1), "τ is an sc-smooth germ"*, is proved level-wise: for every $k \geq 1$ there are a relatively open $V_k \subseteq V$ containing 0 and a C^k map $\tau_k : V_k \to E_k = P \oplus W_k$ with $\iota_k \tau_k = \tau$ on V_k `Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level` — the reading of Corollary 3.34 and Definition 3.25 (ii). The proof applies Theorem 3.78 to H at regularity k on the neighbourhood of its regularity lemma, and identifies the C^k solution σ_k with σ near 0 through the uniqueness clause (3) of Theorem 3.78 — which needs σ continuous at 0 with $\sigma(0) = 0$, so that $|\sigma(y)|$ is below the uniqueness constant for y small; the text says nothing about why the solutions produced for the different k coincide. The W-component $\iota\delta(a) - L\pi(y)$, $a = \pi y + \sigma(\pi y)$, lifts to W_k through the C^k lift of δ into W_{2k}, with L the derivative of that lift at 0 composed with the inclusion (uniqueness of the derivative of $\iota\delta$ at 0). The full statement is `Chapter3/LocalSolvability · theorem_3_43_sc_smooth`; the statement without (1) is kept for Theorem 3.54.

Lemma 3.42. For the push-forward $g = \Phi_*(f + s)$ of Theorem 3.38 — packaged as the data of Theorem 3.43, g on level 0 with index-raised basic germ $(\Phi_*(f + s))^1$ `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData` — the regularity hypothesis holds: a zero x' of g in $\psi(U_0)$ with $|x'|_0$ small has $\psi^{-1}(x')$ a zero of $f + \iota s$, which by the lifts of Theorem 3.40, continuous at 0, lies on level 1 and is small there; ψ is linear on level 1 `Chapter3/LocalSolvability · StabilityData.lemma_3_42`.

Corollary 3.47 `Chapter3/LocalSolvability · theorem_3_47_sc_smooth` `Chapter3/LocalSolvability · theorem_3_47`. For a basic germ f and the data of Theorem 3.38 with $D(f + \iota s)(0)$ surjective and kernel K in good position to $C = \widetilde{C} \oplus W$: a good complement Y, V relatively open in $K \cap C$, $\sigma : V \to Y$ of class C^1 with values on level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$, the graph consisting of zeros of $f + \iota s$, an open $U'' \ni 0$ in which every zero lies on the graph, and for every k a C^k lift of σ into E_k near 0. *Proof.* Theorem 3.43 for $g = \Phi_*(f + s)$, with Lemma 3.42 for its regularity hypothesis. Since the coordinate change is linear, $Dg(0) = \Psi \circ D(f + \iota s)(0) \circ \psi^{-1}$ (chain rule for sc-operators) `Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀`, so surjectivity and the kernels correspond, $\ker Dg(0) = \psi(K)$; good position of $\psi(K)$ follows by Lemma 3.45 (finite dimensionality of K from Lemma 3.44); and the conclusion is transported back by ψ^{-1}: $Y = \psi^{-1}(Y')$, $V = \psi^{-1}(V')$, $\sigma = \psi^{-1}\tau\psi$. The text transports the solution set back with the nonlinear φ and justifies the good position with the degeneracy index (Lemma 3.70, Theorem 3.72); with ψ linear neither is needed. The transport uses that $\widetilde{C} \oplus C$ is a partial quadrant for finite-dimensional C `Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ`.

III.21 The linearization at a zero (Proposition 3.52) and good position of a germ (Definition 3.53)

Proposition 3.52 (text): if f is an sc-Fredholm section of a tame strong bundle and $f(x) = 0$, then $Tf(x) : T_xX \to Y_x$ is an sc-Fredholm operator. Formalized in the chart of Definition 3.11 centred at a zero x_0, with the level-0 Fredholm property `Chapter3/FredholmGermLinearizationG · proposition_3_52` `Chapter3/FredholmGermLinearizationG · proposition_3_52_local`: $Dg(0) : E_0 \to F_0$ and $Tf(x_0) = Dg(0)|_{T_0O} : T_0O \to K_0$ are Fredholm with $\mathrm{ind}\, Tf(x_0) = \mathrm{ind}\, Dg(0)$. Here $Dg(0)$ is the value at 0 of a derivative family Dg of the filling germ on a germ G with $G_0 \subseteq U$, the same for every family and every germ (§III.4); the statement exposes G and Dg, where the whole-chart statement `Chapter3/FredholmGermLinearization · proposition_`

3_52_wholeChart has the derivative on U, a junk value for a germ. (The sc-Fredholm property of $Tf(x)$ would need $T_0O = Dr(0)E$ and $K_0 = \rho(0)F$ as sc-Banach spaces, not set up. The statements quantify the index as "$n - N$ for some natural numbers n, N", which every integer satisfies; the identification with $\dim P - \dim Q$ of the conjugating basic germ is made inside the proof, by `proposition_3_10_basic_level_zero`, and not exported.)

Step (a): the derivative of the conjugation Chapter3/FredholmGermLinearizationG · IsConjugateToBasicGerm.scDeriv_isFredholm. Let $k = g - \iota s$, an sc¹-germ on $G_g \cap G_s$ (§III.48) with $k(0) = 0$ ($s(0) = g(0)$ in Definition 3.11, $g(0) = \mathbf{f}(0) = 0$ by the filler) and $(\Psi(u, k(u)))_2 = h(\psi(u))$ near 0. Linearizing needs: the chain rule for the germ $h \circ \psi$ on a germ small enough that ψ maps it into the germ of h (§III.1, `ScNbhdGerm.preimage`); the derivative of $u \mapsto \Gamma(u)k(u)$ at 0, where $\Gamma(u) = (\Psi(u, \cdot))_2$ is a bounded operator Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM and $D(\mathrm{pr}_{F'} \circ \Psi)(0,0)(h,k) = \Gamma(0)k$ (as in §III.4) Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_zero, so the derivative is $\Gamma(0)Dk(0)$ because $k(0) = 0$ Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp — for the germ k by the chain rule with the whole-chart outer map $\mathrm{pr}_{F'} \circ \Psi$, the two derivative families agreeing at 0 on $G_k \cap \psi^{-1}(G)$ Chapter3/FredholmGermLinearizationG · IsStrongBundleIso.fiberCLM_comp_deriv_zero_eq_germ, so no shrinking of the basic germ is needed at 0; and the invertibility of $D\psi(0)$ and $\Gamma(0)$ (inverses $D\psi'(0)$, $\Gamma'(0)$, chain rule for $\psi' \circ \psi = \mathrm{id}$). Then $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, so $Dk(0) = \Gamma(0)^{-1}Dh(0)D\psi(0)$ is Fredholm with the index of $Dh(0)$: Fredholm operators and their index are stable under composition with isomorphisms on both sides Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv, and $Dh(0)$ is Fredholm by Proposition 3.10 for the basic germ Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero. *Step (b)*: $Dg(0) = Dk(0) + \iota Ds(0)$ — for germs this is algebra at 0: $g - \iota s$ has the derivative family $Dg - \iota Ds$ on $G_g \cap G_s$ Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus (on U it was the uniqueness of the derivative within U_1 and density) — a compact perturbation since $\iota Ds(0)$ is the level-0 map of an sc⁺-operator (s is an sc¹-germ into F^1 and 0 is smooth); the index is unchanged (Part I, §I.9). *Step (c)*: Proposition 3.5 (§III.4) transfers both to $Tf(x_0)$; it needs $g(0) = 0$.

Definition 3.53 (text): (f, x) with $f(x) = 0$ is in good position if (1) $f'(x) : T_xX \to Y_x$ is onto and (2) if $d_X(x) \geq 1$, $\ker f'(x)$ is in good position to the partial cone C_xX in T_xX. In the chart: Definition 3.21 *inside a subspace* $T \subseteq E_0$ — $N \subseteq T$, interior of $N \cap C$ in N nonempty, and a good complement $P \subseteq T$ of N in T Chapter3/GoodPositionGerm · IsInGoodPositionIn (for $T = E_0$ it is Definition 3.21 Chapter3/GoodPositionGerm · isInGoodPositionIn_top_iff); and the germ is in good position if $Tf(0) = Dg(0)|_{T_0O}$ is onto and $\ker Dg(0)$ is in good position to $T_0O \cap C$ inside $T_0O = Dr(0)E$ Chapter3/GoodPositionGerm · IsGoodPositionAt Chapter3/GoodPositionGerm · IsGoodPositionAt.surjective_fillerRestrict. Here $\ker Dg(0) \subseteq T_0O$, because $(1 - \rho(0))Dg(0)$ is injective on $\ker Dr(0)$ (filler (3)) and $(1-\rho(0))Dg(0)Dr(0) = 0$ Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range, and it is the kernel of $Tf(0)$ Chapter3/GoodPositionGerm · IsFillerOf.map_ker_fillerRestrict — the text's "$\ker Tf(0) \oplus \{0\} = \ker Dg(0)$ by the properties of the filler". The case distinction $d_X(x) \geq 1$ is dropped: at an interior point the partial cone is the whole tangent space and (2) holds for every subspace. For a filling germ the condition is one on the operator $Dg(0)$ (the text's $f'(x)$ is defined through the filler at the centre only), which the chart notion takes as a parameter Chapter3/GoodPositionGermG · IsGoodPositionAtGerm — the notion above is the case of the derivative on U Chapter3/GoodPositionGermG · isGoodPositionAt_iff_germ — and the two kernel facts hold with the same proofs Chapter3/GoodPositionGermG · IsFillingGerm.ker_deriv_le_range Chapter3/GoodPositionGermG · IsFillingGerm.map_ker_fillerRestrict. For a section, (f, x_0) is in good position if it has the data of Definition 3.11 at x_0 and a derivative family Dg of the filling germ on a germ inside U, with the germ in good position at $Dg(0)$ in that chart Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm; as $Dg(0)$ does not depend on the family, "for some family" and "for every family" agree Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.forall_isGoodPositionAtGerm; it is then an sc-Fredholm germ Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm.isScFredholmGerm. The definition is in a chart, not in T_xX; the transport of C_xX to T_xX (Definition 2.38) is not formalized.

III.22 Good position along a graph: Lemma 3.55

Lemma 3.55 (text, Appendix 3.8.3): for N in good position to C with good complement Y, $V \subseteq N \cap C$ relatively open and $\tau : V \to Y_1$ of class C^1 with $\tau(0) = 0$, $D\tau(0) = 0$, for v near 0: (1) $v+\tau(v) \in C_1$; (2) Y is a topological complement of $N_v = \{n+D\tau(v)n\}$; (3) for some $\gamma_v > 0$, $n \in C_x \iff n+y \in C_x$ for $n \in N_v$, $y \in Y$, $|y|_0 \le \gamma_v |n|_0$, $x = v+\tau(v)$. *Formalized* `Chapter3/GoodPositionOpen · lemma_3_55` in the model $\mathbb{R}^n \oplus W$, with the core in coordinates for any C cut out by functionals `Chapter3/GoodPositionOpen · lemma_3_55_core`: for N in good position with good complement Y, Y **closed**, and $\tau : N \to E$ with values in Y, C^1 on $N \cap C$, $\tau(0) = 0$, $D\tau(0) = 0$: there is $\rho > 0$ such that for $v \in N \cap C$, $|v| < \rho$, $v+\tau(v) \in C$ and Y is a good complement of N_v (the subspace $\{n + Ln\}$ for $L = D\tau(v)$ `Chapter3/GoodPositionOpen · graphSpace`) with respect to the partial quadrant $C_{v+\tau(v)}$.

The added hypothesis. The step "$D\tau(v)n \in Y$ for all $n \in N$" needs Y closed: the derivative within $N \cap C$ at a point of unique differentiability of a map with values in a closed subspace takes values there, because the tangent cone of a subset of a closed subspace lies in it `Chapter3/GoodPositionOpen · tangentConeAt_subset_of_subset_closed` `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo`; for a non-closed algebraic complement it fails. The text's $Y = Y_0$ is a topological complement, hence closed. With Y closed, (2) is linear algebra: $E_0 = N_v \oplus Y$ whenever $E_0 = N \oplus Y$ and L takes values in Y `Chapter3/GoodPositionOpen · graphSpace_isCompl`, and N_v is closed (finite-dimensional) `Chapter3/GoodPositionOpen · graphSpace_isClosed`.

The estimate. (1) is Lemma 3.70's smallness: $|\tau(v)| \le \varepsilon|v|$ near 0 and good position. For (3), take $\varepsilon_0 = \min(\varepsilon, 1)$ and the neighbourhood where $\|D\tau(v)\| \le \varepsilon_0/4$; then $\gamma = \varepsilon_0/4$ works *for every* v, since $\gamma(1 + \varepsilon_0/4) + \varepsilon_0/4 \le \varepsilon_0$ (the text: $\gamma_v(1+\varepsilon) + \varepsilon < \gamma$, with γ_v chosen after v). If $v \in N \cap W$, $C_x = C$ and the estimate transfers the good-complement property from N to N_v. If $v \notin N \cap W$, no normal form is needed: if $\ell_i(v) = 0$ then $i \in \Sigma$, so ℓ_i vanishes on Y (Lemma 3.65), and $\ell_i(v + \tau(v)) = 0$ iff $\ell_i(v) = 0$ (Lemma 3.70); so $\ell_i(h+y) = \ell_i(h)$ for every coordinate i that defines C_x, and (3) holds for every γ. The text's cases $\dim N = 1$, $\dim N > 1$ are not needed, and the sc-level of the values of τ plays no role. The formal τ is given on all of $N \cap C$ rather than on a neighbourhood V; the conclusion is local.

III.23 Theorem 3.54: the linear part and the graph description of the solution set

Theorem 3.54 (text): for an sc-Fredholm section f of a tame strong bundle with $f(x) = 0$ and (f, x) in good position there is an open $V \ni x$ such that $S = \{y \in V : f(y) = 0\}$ satisfies (1) (f, y) is in good position at every $y \in S$, and (2) S is a sub-M-polyfold with the structure of a manifold with boundary with corners. The proof: the linear part (surjectivity and good position transported from T_xX to the chart and the push-forward); (3), the graph description of S from Corollary 3.47 for $\Psi_*(g) = h + t$; then Corollary 3.41 and Lemma 3.55 for (1), and Theorem 3.27 for (2). Formalized: the linear part and (3).

The linear part `Chapter3/GoodPositionGermG · theorem_3_54_linear`. In the chart, with the germ in good position and $(1-Dr(0))E \subseteq E_0$ (the minimal subspace at the corner 0; Proposition 2.35 for the tame retraction r `Chapter3/GoodPositionGermG · theorem_3_54_linear_of_isTameWith` `Chapter2/TameRetracts · proposition_2_35`): $Dg(0)$ is onto (Proposition 3.5 for the filling germ, §III.4, from the surjectivity of $Tf(0)$), and $\ker Dg(0)$ is in good position to C in E; only the filler at 0 enters, and r is whole-chart, so $Dr(0)$ is a genuine derivative. The latter is the *transport lemma*: for a bounded idempotent π with $(1-\pi)E_0$ inside the minimal subspace at 0, good position inside πE_0 gives good position in E_0, with the good complement $P \oplus (1-\pi)E_0$ and the constant $\varepsilon/(\|\pi\| + 1)$ `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition`. Two points the text passes over: for $p = z + k$ with $|p| \le \varepsilon'|n|$ the T_0O-component is $z = Dr(0)p$, so $|z| \le \|Dr(0)\|\varepsilon'|n|$ — the splitting must be topological; and $\ker Dr(0) \subseteq E_0$ is used only as "the $\mathbb{R}^n$-coordinates of k vanish, so $n + z + k \in C \iff n + z \in C$". *This inclusion is the only place where tameness enters.*

$C' = T\psi(0)C$. Not part of the data of a conjugation (§III.6); it is a theorem: for an sc-diffeomorphism ψ between relatively open subsets of partial quadrants with $\psi(0) = 0$, $x \in C \iff D\psi(0)x \in$

C', by the one-sided sign argument for ψ and ψ^{-1} (Part II, §II.12) `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant`, since $C_0 = C$ at the corner `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero` `Chapter3/GoodPositionGerm · scDeriv_mem_iff_of_inverse`.

The push-forward. $\Psi_*(g)(u') = (\Psi(\psi^{-1}(u'), g(\psi^{-1}(u'))))_2$ `Chapter3/GermPushforward · pushforward` is an sc-smooth germ on the germ of the points that ψ^{-1} maps into the germ of g `Chapter3/GermPushforward · IsStrongBundleIso.pullGerm` `Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isScSmoothGerm` (chain rules with the whole-chart Ψ and ψ^{-1}, §III.48), with $\Psi_*(g)(0) = 0$ and $D(\Psi_*g)(0) = \Gamma(0)Dg(0)D\psi(0)^{-1}$ for its explicit derivative family `Chapter3/GermPushforward · IsStrongBundleIso.pushforwardDeriv_zero` ($Dk(0) = \Gamma(0)Dg(0)$ for $k(u) = (\Psi(u, g(u)))_2$) `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization`; so it is onto `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_surjective`, its kernel $D\psi(0)(\ker Dg(0))$ is in good position to C' (Lemma 3.45) `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_ker_isInGoodPosition`, and it is Fredholm (transported from $Dg(0)$, Proposition 3.52) — the hypotheses of Corollary 3.47, for *any* conjugating isomorphism `Chapter3/GoodPositionGermG · theorem_3_54_pushforward`, and in bundle form `Chapter3/GoodPositionGermG · theorem_3_54_pushforward_section`. The text instead obtains the Fredholm property of $D(h+t)(0)$ from Propositions 3.10 and 1.10.

The graph description `Chapter3/GlobalIFTG · theorem_3_54_graph`. Let the charts of the bundle have tame base retractions (`htame`, the hypothesis of the first formalization; it cannot be met, and it is replaced by good position in a tame chart, §III.47), $f(x_0) = 0$, (f, x_0) in good position. The proof, in the order of the Lean.

1. *The chart and the identities.* Take the chart of Definition 3.53 centred at x_0, filling germ g on G_g, sc⁺-germ t_0 on G_s and conjugation Ψ covering ψ, with inverse Ψ' covering ψ'. *Shrink the basic germ* to the germ G'' of the points of G that ψ' sends into $G_g \cap G_s$ level by level: same Ψ, h, B and conjugation identity `Chapter3/BasicGermShrink · ConjugationData.shrink` (§III.48); below, G denotes G''. There is an open $N_1 \ni 0$ on which $\mathbf{f} = g$ on O, zeros of g in U lie in O, and $\Psi(u, (g - \iota t_0)(u))_2 = h(\psi(u))$ (filler (1), (2) and the conjugation).
2. *The sc⁺-germ t.* $t(u') = \Psi[1](\psi'(u'), t_0(\psi'(u')))_2$ is an sc-smooth germ into $(Q \oplus W)^1$ on G: $t_0 \circ \psi'$ is a germ on the preimage of G_s, which contains $G = G''$, and the whole-chart $\Psi[1]$ is composed on the outside (§III.48). (With t_0 sc-smooth on U, t was sc-smooth on all of U'.) On $U \cap N_1$, $\Psi_*(g)(\psi(u)) = h(\psi(u)) + \iota t(\psi(u))$, by the linearity of the fibre maps.
3. *Absorbing the $\mathbb{R}^N$-component.* The data of Theorem 3.38 carry only the W-component of an sc⁺-germ. Let t_Q, t_W be the two components of t, through the sc-operators $(q, w) \mapsto (q, 0)$ and $(q, w) \mapsto w$ `Chapter3/GlobalIFT · ScBanach.qPartOp` `Chapter3/GlobalIFT · ScBanach.wPartOp`. Then $h' = h + (t_Q, 0)$ is again a basic germ with the same contraction B (Definition 3.7 constrains only the W-component), and with the data $\mathcal{S}$ of Theorem 3.38 for $s = t_W$ (they exist, §III.18) one has $\mathcal{S}.\mathrm{fsum}\, h' = h' + \iota t_W = h + \iota t$.
4. *The hypotheses of Corollary 3.47.* Tameness gives $(1 - Dr(0))E \subseteq E_0$ (Proposition 2.35), so the push-forward step applies: $D(\Psi_*g)(0)$ is surjective with kernel in good position. Since $\mathcal{S}.\mathrm{fsum}\, h' = \Psi_*(g)$ only on a neighbourhood W' of 0, the two derivatives at 0 are identified by the uniqueness of the derivative within G_1'' and density of E_1 `Chapter3/GlobalIFT · scDeriv_eq_of_eqOn_nhds`; $\Psi_*(g)$ is a germ on the preimage of G_g, which contains G''.
5. *Corollary 3.47 for $h + t$* gives Y, V_0, σ (good complement, C^1, level 1, $\sigma(0) = 0$, $D\sigma(0) = 0$), the graph consisting of zeros, and U'' in which every zero lies on the graph.
6. *The neighbourhoods* (the text does not say which V). Let $\Omega = U'' \cap W'$ and $V' = V_0 \cap (1+\sigma)^{-1}(\Omega)$, relatively open in $K \cap C'$ because $v \mapsto v + \sigma(v)$ is continuous. Let $V_x = \{y \in \mathrm{dom}\,\varphi : \varphi(y) \in N_1,\ \psi\varphi(y) \in \Omega \cap G_0\}$, open and containing x_0 by continuity of $\psi \circ \varphi$. $D\sigma(0) = 0$ persists on the smaller V' (unique differentiability at 0 of V').
7. *The conclusions.* $\psi \circ \varphi$ is injective on V_x (φ is, and $\psi' \circ \psi = \mathrm{id}$ on U). For $y \in V_x$: $f(y) = 0 \iff \mathbf{f}(\varphi y) = 0 \iff g(\varphi y) = 0 \iff (h + \iota t)(\psi\varphi y) = 0$ (items 1–2). If $f(y) = 0$, then $\psi\varphi(y)$ is a zero in $U'' \cap G_0$, hence $v + \sigma(v)$ with $v \in V_0$, and $v \in V'$ since $v + \sigma(v) \in \Omega$. Conversely, for $v \in V'$, $u = \psi'(v + \sigma(v))$ lies in $U \cap N_1$, $g(u) = 0$, so $u \in O$ and $u = \varphi(y)$ for some y, which lies in V_x and is a zero.

So $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$. When this section was written, (1) and (2) were not formalized; they are now: the kernel and good position at the nearby zeros in §§III.32–III.33 (the surjectivity of $f'(y)$ under an added hypothesis), the data of Definition 3.25 at every zero in §§III.34–III.36, and the theorem in §III.37. The graph description also records, since then, that Y is closed, that the kernel is a smooth subspace and the level-wise lifts of σ (Corollary 3.47 (1)), and it is a corollary of the version with a good complement whose projection maps C' into itself (§III.29).

Theorems 3.56 and 3.57, the step from the tangent germ property. With $S = \{f = 0\}$ `Chapter3/GlobalIFT · solutionSet`: if $S \neq \emptyset$ has at every point regular data of Definition 3.25 (§III.24), then S is a sub-M-polyfold whose induced structure is tame and an n-dimensional manifold with boundary with corners (Theorem 3.27) `Chapter3/GlobalIFT · theorem_3_57_of_tangentGerm`; if X has no boundary, the induced structure is without boundary, since $d_S \leq d_X = 0$ (Lemma 2.26) `Chapter2/SubMPolyfold · lemma_2_26` `Chapter3/GlobalIFT · theorem_3_56_of_tangentGerm`. The first step of the text's proof — good position at every zero gives the tangent germ property — was not formalized when this section was written; the theorems with the text's hypotheses are in §III.37.

III.24 The tangent germ property: Definition 3.25 and Theorem 3.27

Definition 3.25. The data at $x \in M$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart`: a chart $\varphi : (U, x) \to (O, 0)$ onto a tame retract (O, C, E); an n-dimensional smooth $N \subseteq E$ in good position to C with a good complement Y, Y **closed**; a relatively open $V \ni 0$ in $N \cap C$; $\delta : V \to E$ continuous with values in Y; (i) $\varphi(M \cap U) = \{v + \delta(v) : v \in V\}$; (ii) $\delta(0) = 0$, $D\delta(0) = 0$ and δ an sc^∞-germ, read as: for every m, k a relatively open $V_{m,k} \ni 0$ in $N \cap C$ with $\delta(V_{m,k}) \subseteq E_m$ and $\delta : V_{m,k} \to E_m$ of class C^k `Chapter3/TangentGerm · IsScInftyGermAt`. This k-dependent reading is what the germ implicit function theorem delivers (§III.11); the reading "sc-smooth on one fixed germ" implies it by Proposition 1.18 for germs `Chapter3/TangentGerm · IsScSmoothGerm.isScInftyGermAt`, as does sc-smoothness on V as a set `Chapter3/TangentGerm · IsScSmooth.isScInftyGermAt`, so the theorem covers all three. M has the property if $M \subseteq X_\infty$ and every point has such data `Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty`. With these data: $N \cap C$ is a partial quadrant in N (Proposition 3.22) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isPartialQuadrant`, and N, Y are topological complements — this is where Y closed is used — so the projection $\pi : E \to N$ along Y is continuous `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isTopCompl` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π`; with $\Gamma(v) = v + \delta(v)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ`, $\pi\Gamma = \mathrm{id}$ and $\varphi(y) = \Gamma(\pi\varphi(y))$ for $y \in M \cap U$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`.

Proposition 3.26 `Chapter3/TangentGerm · proposition_3_26`. For an sc-diffeomorphism $\psi : X \to X'$ (sc-smooth on all of X with sc-smooth inverse ψ' on all of X') and $\psi(M) = M'$, M' has the n-dimensional tangent germ property iff M has it. One direction suffices, applied to ψ and ψ' `Chapter3/TangentGerm · proposition_3_26_image`: $\psi(M) \subseteq X'_\infty$ because sc-smooth maps preserve the levels, and the data at $\psi(x)$ are the transported chart $\varphi \circ \psi^{-1}$ `Chapter2/DegeneracyIndex · transportChart`, whose model is that of φ `Chapter3/TangentGerm · MPolyfold.transportChart_model_0`, with the same N, Y, V, δ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.transport`: condition (i) is transported because $\varphi\psi^{-1}(\psi(M) \cap \psi(U)) = \varphi(M \cap U)$. As in the text; tameness of X plays no role.

Two gaps in the text's proof of Theorem 3.27, and the conditions that close them `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. (a) The text asserts that δ is an sc^∞-germ near every $v_0 \in V$, by linearizing at v_0 and claiming that a certain map $N' \to N$ is an isomorphism. That needs δ differentiable at v_0, known only near 0, and fails: in $X = \mathbb{R}^2$ take M the graph of g with $g(s) = s^2$ near 0 and $g(s) = 4 + (s-2)^{1/3}$ near 2, a smooth embedded curve with a vertical tangent at $(2, 4)$; with $\varphi = \mathrm{id}$, N the horizontal axis, Y the vertical one and $\delta = g$, the data at 0 satisfy Definition 3.25, M has the tangent germ property (at $(2, 4)$ use a rotated chart), but δ is not differentiable at 2 and the chart $\pi\varphi$ on all of $M \cap U$ is not smoothly compatible with the chart at $(2, 4)$. So condition (a): δ is sc-smooth on V as a map on a set. (b) The text's $\Sigma = \{v + y \in C : v \in V, y \in Y\}$ is claimed

relatively open in C, so that $t(v+w) = v + \delta(v)$ is defined near 0 in O. False: $E = \mathbb{R}^3$, $C = [0,\infty)^3$, $N = \mathbb{R}(1,1,1)$, $Y = \{x+y-z=0\}$ is a good complement, and $\pi(x,y,z) = (x+y-z)(1,1,1)$, so $\pi z \in C$ iff $x + y \geq z$; Σ is not open at $(1,0,1)$ and points $(\varepsilon, 0, \varepsilon + \varepsilon^2) \notin \Sigma$ approach 0. So condition (b): $\pi(O) \subseteq C$; then $\Sigma \cap O = \{z \in O : \pi z \in V\}$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig` is relatively open in O `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Sig_eq`. (A good complement with $\pi'(C) \subseteq N \cap C$ always exists, via the normal form, but then $M \cap U$ must be rewritten as a graph along it with the quadrant inverse function theorem, and the regularity problem of (a) returns; this is not done.)

Theorem 3.27 `Chapter3/TangentGerm · theorem_3_27`. If $M \neq \emptyset$ has at every point data satisfying (a), (b), then:

M is a sub-M-polyfold `Chapter3/TangentGerm · theorem_3_27_isSubMPolyfold`, with the text's retraction. $t(v + w) = v + \delta(v)$, i.e. $t = \Gamma \circ \pi$ on $\Sigma \cap O$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t`, is idempotent with image $\Gamma(V)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_t` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_eq_self_of_mem`. It is sc-smooth: π is a continuous linear map into a finite-dimensional space, hence sc-smooth (Corollary 1.20) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmooth`; $\Gamma = \iota + \delta$ is sc-smooth on V by (a) and because the inclusion of a smooth finite-dimensional subspace is sc-smooth, its level maps being the linear lifts $N \to E_m$ `Chapter3/TangentGerm · finSmoothLift` `Chapter3/TangentGerm · isScSmooth_subtype_val` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmooth`; and for a retraction R onto $\Sigma \cap O$, $t \circ R = \Gamma \circ \pi \circ R$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.t_isScSmoothRetractMap`. The text says nothing about either. On $U_0 = \varphi^{-1}(\Sigma \cap O)$, open `Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀`, the map $s = \varphi^{-1} \circ t \circ \varphi$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s` is an sc-smooth retraction `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn` onto $M \cap U_0$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s` — Definition 2.20 at x.

The charts $\pi \circ \varphi$. On $M \cap U_0$, $y \mapsto \pi\varphi(y)$ is a chart with model $(V, N \cap C, N)$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.model`, a tame retract (the identity retraction) `Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract`, with inverse $v \mapsto \varphi^{-1}(v + \delta(v))$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chartInv` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart`, a finite-dimensional corner chart of dimension n `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner` containing x `Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source`. It is compatible with every chart $\varphi'|_{W \cap M}$ of the induced structure of Proposition 2.21: the transition maps are $\varphi' \circ \varphi^{-1} \circ \Gamma$ and $\pi \circ \varphi \circ \varphi'^{-1}$, compositions of sc-smooth maps between retracts `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_isScSmoothRetractMap` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_isScSmoothRetractMap` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart`. So it is a chart of the induced structure — **Theorem 3.27 (2)** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isChart` `Chapter2/SubMPolyfold · subPolyfold`; these charts are pairwise compatible by locality `Chapter3/TangentGerm · MPolyfold.tangentCharts_compatible` and form an atlas `Chapter3/TangentGerm · MPolyfold.germAtlas` equivalent to the induced one `Chapter3/TangentGerm · MPolyfold.theorem_3_27_atlas_equiv`, with tame models: **the induced structure is tame** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isTame`, and **sc-smoothly equivalent to an n-dimensional manifold with boundary with corners** `Chapter3/TangentGerm · MPolyfold.theorem_3_27_isManifoldWithCorners`. "Manifold with boundary with corners" belongs to Chapter 4; here it is an equivalent atlas of charts on relatively open subsets of n-dimensional partial quadrants `Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners`, whose transition maps are classically smooth because sc-smooth maps between finite-dimensional constant structures are C^∞ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`. Tameness of X is not used: the text's "since O is a tame retract, there exists … a tame sc-smooth retraction r" is Proposition 2.10 (1) for any retract. *The degeneracy index*: since $\pi\varphi$ has a tame model, $d_M(y) = d_{N \cap C}(\pi\varphi(y))$ (Proposition 2.36) `Chapter3/TangentGerm · MPolyfold.theorem_3_27_degIndex`; Lemma 3.70 relates this to $d_C(v + \delta(v)) = d_X(y)$ in the model $\mathbb{R}^n \oplus W$ only.

III.25 Lemmas 3.59–3.61: towards the basic germ criterion

Lemmas 3.59 and 3.60, abstract form. In the proof of Theorem 3.58 the family $L(b) = P_Y \circ Df(b)|_X : X \to Y$ is shown to be injective for b near 0 (Lemma 3.59) and to satisfy uniform lower bounds $|L(b)h|_m \geq c_m|h|_m$ (Lemma 3.60) — the hypotheses of Lemma 3.61. Both are sequence arguments, and all they use of the compactness condition (2) of Theorem 3.58 is a property of the family itself: if $b_j \to b$, (h_j) is bounded in X_m and $L(b_j)h_j \to 0$ in Y_m, then (h_j) has a convergent subsequence in X_m `Chapter3/BasicGermCriterion · FamilyCompactness`. (For $L = P_Y \circ Df|_X$ it follows from (2), since $Df(b_j)h_j - L(b_j)h_j = (1 - P_Y)Df(b_j)h_j$ lies in the finite-dimensional smooth complement A and is bounded on every level.) The two lemmas are formalized for an abstract family of sc-operators with jointly continuous level maps. *Lemma 3.59*: if $L(0)$ is injective, so is $L(b)$ for b near 0 — otherwise $b_j \to 0$ and unit vectors h_j with $L(b_j)h_j = 0$ give, by the compactness property, a limit h with $|h| = 1$ and $L(0)h = 0$ `Chapter3/BasicGermCriterion · lemma_3_59`. *Lemma 3.60*: on a compact parameter set K on which every $L(b)_m$ is injective there is $c_m > 0$ with $c_m|h|_m \leq |L(b)h|_m$ — otherwise $b_j \in K$ and unit h_j with $|L(b_j)h_j|_m \to 0$ give, after subsequences, $b_j \to b \in K$, $h_j \to h$, $|h|_m = 1$, $L(b)h = 0$ `Chapter3/BasicGermCriterion · lemma_3_60`. The text's step "$P_Y \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism" is needed only to feed Lemma 3.61. The setting of Theorem 3.58 (the splitting as coordinates, the complement A, the index computation), the strong bundle map $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ and the contraction estimate from condition (3) were not formalized when this section was written; they are now, in the coordinates of the text's proof, and there the hypotheses of the two lemmas are derived from those of the theorem (§III.38).

Lemma 3.61 (quoted by the text from [25, Proposition 4.8] without proof). Let V be a germ of relatively open neighbourhoods of 0 in a partial quadrant $\widetilde{C}$ of a finite-dimensional G, $V \oplus E$ the product germ with levels $V_m \times E_m$ `Chapter3/InverseFamily · ScNbhdGerm.prodFull`, and $L(v) : E \to F$ sc-isomorphisms such that $\Phi(v,h) = L(v)h$ `Chapter3/InverseFamily · famMap` is an sc-smooth germ on $V \oplus E$ and $c_m|h|_m \leq |L(v)h|_m$ for $v \in V_0$. Then $\Lambda(v,k) = L(v)^{-1}k$ `Chapter3/InverseFamily · famInv` is an sc-smooth germ on $V \oplus F$ `Chapter3/InverseFamily · lemma_3_61`. Operator-norm continuity of $v \mapsto L(v)$ is not assumed.

sc⁰. On level m, with $h_0 = L(v_0)^{-1}k_0$, $|L(v)^{-1}k - L(v_0)^{-1}k_0|_m \leq c_m^{-1}(|k-k_0|_m + |(L(v_0)-L(v))h_0|_m)$, and $v \mapsto L(v)h_0$ is continuous `Chapter3/InverseFamily · famInv_isSc0Germ`.

Differentiability `Chapter3/InverseFamily · famInv_hasFDerivWithinAt`. The candidate is $D\Lambda(v,k)(\delta v,\delta k) = L(v)^{-1}(\delta k - D_1\Phi(v,L(v)^{-1}k)\delta v)$ `Chapter3/InverseFamily · famInvDeriv`. With $h = \Lambda(v,k)$ and $w = D\Lambda(v,k)(\delta v,\delta k)$, $\Lambda(v',k') - \Lambda(v,k) - w = L(v')^{-1}[(L(v) - L(v'))w] - L(v')^{-1}r$, where $r = \Phi(v',h) - \Phi(v,h) - D_1\Phi(v,h)\delta v = o(|\delta v|)$ is the remainder of Φ along $(\delta v, 0)$ with h fixed, so that only the finite-dimensional variable moves. The term $(L(v) - L(v'))w$ is $o(|(\delta v,\delta k)|_1)$ *because the increment is measured on level* 1: $w/|(\delta v,\delta k)|_1$ lies in the compact set $D\Lambda(v,k)(\iota(B_1))$ ($\iota : F_1 \to F_0$ compact), and $(L(v) - L(v'))u \to 0$ uniformly for u in a compact set — a tube lemma for jointly continuous maps vanishing on $\{x_0\} \times K$ `Chapter3/InverseFamily · exists_forall_norm_le_of_isCompact`. Strong continuity plus the compactness of $F_1 \to F_0$ replace the operator-norm continuity; the differentiability of Λ as a map $(V \oplus F)_0 \to E_0$ would need the latter and is not claimed.

The tangent map and the induction. $T\Lambda((v,k),(\delta v,\delta k)) = (\Lambda(v,k), \Lambda(v,\delta k - (T\Phi((v,\Lambda(v,k)),(\delta v,0)))_2))$, a composition of Λ, $T\Phi$ and sc-operators (the projections `Chapter3/InverseFamily · tangentBaseOp` `Chapter3/InverseFamily · tangentInlOp` `Chapter3/InverseFamily · tangentFiberOp` and subtraction `Chapter3/InverseFamily · ScBanach.subOp`); it is sc⁰ `Chapter3/InverseFamily · famInv_tangent_isSc0`, so Λ is sc¹ `Chapter3/InverseFamily · famInv_isSc1GermWith`; and if Φ is sc^{k+1}, then Λ (sc^k by induction) and $T\Phi$ (sc^k) make $T\Lambda$ sc^k, so Λ is sc^{k+1} `Chapter3/InverseFamily · famInv_isSckGerm`. The sc¹ step uses no quadrant hypothesis; the induction uses the composition rules for germs.

III.26 What the formalization taught us

1. **The germ implicit function theorem holds with the text's hypotheses**, but on shrinking germs: the solution germ is sc^k on the germ shifted by $2^k - 1$ levels. In the text's reading of germs (up to shrinking) this is the theorem; read with fixed germs, "if f is sc-smooth so is its solution germ" holds for every k on a germ depending on k. Every later "sc-smooth germ" (Definition 3.25 (ii), Theorem 3.43 (1), Corollary 3.47 (1)) must be read the same way. Whether the shrinking is intrinsic is open.
2. **Theorem 3.30 has a short proof**, which the text cites from [21]: an approximate fixed point and the $\frac{1}{2}$-contraction on level 0, at parameters where the level-1 operators exist.
3. **The contraction regions must be explicit** for "from the uniqueness we conclude $\delta^{(1)} = T\delta$" (Lemma 3.31) to be true.
4. **Several "immediate consequences" are proofs of their own**: the compactness of Theorem 3.12 (not in Theorem 3.40), the induction of Theorem 3.40 (lifts, bounds and continuity together), the sc-smoothness and regularizing property in Theorem 3.15, the sc$^+$ property in Proposition 3.10, the projection of solutions into the quadrant in Theorem 3.43.
5. **Three added hypotheses**: Y closed in Lemma 3.55 and Definition 3.25; and the two regularity conditions of Theorem 3.27, each with a counterexample to the step of the text's proof it replaces.
6. **Tameness is nearly unused**: not in §3.1 at all, and in §3.6 only through $(1-Dr(0))E \subseteq E_0$ and, for the tangent germ property of the solution set, the tameness of the model of the chart $\psi \circ \varphi$ (§III.27).
7. **Linear coordinate changes** in Theorem 3.38 make the text's degeneracy-index argument for Corollary 3.47 unnecessary.
8. **The extreme-ray lemmas and the iterated tangent maps are not needed**: Proposition 3.22 by a segment argument, the regularity in Theorems 3.72 and 3.78 by bootstrapping, without shrinking.
9. **What remained of Theorem 3.54** when this list was written — the re-centring of good position at the nearby zeros, and the level-wise regularity of the solution map in the data of Definition 3.25 at every zero — is done in §§III.31–III.37, with one added hypothesis for Theorem 3.54 (1); see §III.39.

III.27 Tameness is carried by sc-diffeomorphisms, and by open subsets

The text never states that tameness is invariant under sc-diffeomorphisms, or that an open piece of a tame retract is a tame retract, but it uses both. In the proof of Theorem 3.54 the chart $\psi \circ \varphi$ onto $\psi(O)$ must be a chart of the tame M-polyfold X for Definition 3.25 to apply; in the proof of Proposition 3.26 it writes "$X' = \psi(X)$ is tame"; and wherever a chart is restricted to a smaller domain (the shrinking of the data of Definition 3.25 in §§III.36–III.37) the model of the restricted chart must again be a tame retract. Both facts are proved here, with Definition 2.34 as the formalization states it (Part II).

The objects. An sc-diffeomorphism between relatively open subsets $U \subseteq C$, $U' \subseteq C'$ of partial quadrants is a pair $\psi : U \to U'$, $\psi' : U' \to U$ of sc-smooth maps, each mapping its domain into the other's, inverse to each other on U and on U' `Chapter2/TameInvariance · IsScDiffeoQuadrant` (its inverse is again one `Chapter2/TameInvariance · IsScDiffeoQuadrant.symm`). A retraction $r : U \to U$ is *tame* for a presentation T of C if (1) $d_C(r(x)) = d_C(x)$ for $x \in U$ and (2) at every smooth point x of $O = r(U)$ on level 1 the tangent space $T_xO = Dr(x)E$ has an sc-complement A contained in the minimal subspace E_x `Chapter2/Tame · ScRetraction.IsTameWith`; it is tame if it is tame for every presentation `Chapter2/Tame · ScRetraction.IsTame`, and (O, C, E) is a tame retract if O is the image of a tame retraction `Chapter2/Tame · IsTameRetract`.

The transported retraction. For an sc-smooth retraction r with domain U, put $r' = \psi \circ r \circ \psi' : U' \to U'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport`. It maps U' into U' because $\psi'(U') \subseteq U$, $r(U) \subseteq U$ and $\psi(U) \subseteq U'$; it is sc-smooth by the chain rule applied twice; and it is idempotent, $r'(r'(u')) = \psi(r(\psi'\psi(r(\psi'u')))) = \psi(r(r(\psi'u'))) = r'(u')$, since $\psi'\psi = \mathrm{id}$ on U and r is idempotent. Its image is $\psi(O)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport_retract`: $r'(u') = \psi(r(\psi'u'))$ with $r(\psi'u') \in O$; conversely, for $z \in O$, $r'(\psi z) = \psi(r(z)) = \psi(z)$.

Condition (1). The degeneracy index is preserved by ψ, $d_{C'}(\psi(u)) = d_C(u)$ for $u \in U$: this is Theorem 1.25 `Chapter2/TameInvariance · IsScDiffeoQuadrant.degIndex_eq Chapter2/DegeneracyIndex · theorem_1_25`. Then for $u' \in U'$: $d_{C'}(r'(u')) = d_C(r(\psi' u')) = d_C(\psi' u') = d_{C'}(\psi\psi' u') = d_{C'}(u')$, the middle step being condition (1) for r.

The derivative of r'. On level 0, $D\psi'(\psi(x)) \circ D\psi(x) = 1$ for $x \in U_1$ and $D\psi(\psi'(y)) \circ D\psi'(y) = 1$ for $y \in U_1'$, by the chain rule for $\psi' \circ \psi = \mathrm{id}$ and $\psi \circ \psi' = \mathrm{id}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_inv_deriv Chapter2/TameInvariance · IsScDiffeoQuadrant.deriv_deriv_inv Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`. If $y \in U_1'$ and $\psi'(y) \in O$, then with $x = \psi_1'(y)$,

$$Dr'(y) = D\psi(x) \circ Dr(x) \circ D\psi'(y)$$

`Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport`. The chain rule is applied to $\psi \circ (r \circ \psi')$ and then to $r \circ \psi'$ `Chapter2/TameFaces · scDeriv_comp_apply_of_isScDomain`; the outer derivative of ψ is taken at the level-1 point of $r \circ \psi'$ over y, and this point is x because r fixes $\psi'(y) \in O$. Without $\psi'(y) \in O$ the formula would involve Dr at $\psi'(y)$ and $D\psi$ at the different point $r(\psi'(y))$.

Sc-isomorphisms carry sc-splittings. Let $D : E \to E'$ be an sc-operator with an sc-operator inverse D'. Its level maps are topological linear isomorphisms $E_m \cong E'_m$ `Chapter2/TameInvariance · scIsoLevel` (inverse to each other by the injectivity of the inclusions into level 0); they carry $A_m = A \cap E_m$ onto $(DA)_m$ `Chapter2/TameInvariance · levelSub_map_eq` and smooth points to smooth points `Chapter2/TameInvariance · mem_smoothLvl_of_isScOperator`. Hence an sc-subspace A (closed, with the smooth points of A_m dense in A_m for every m) is carried to the sc-subspace DA: $DA = (D')^{-1}(A)$ is closed, and the image of the closure of the smooth points of A_m lies in the closure of the image `Chapter2/TameInvariance · IsScSubspace.map`. An sc-splitting $E = F \oplus A$ (sc-subspaces, complementary on every level) is carried to the sc-splitting $E' = DF \oplus DA$, since an isomorphism of each level carries complements to complements `Chapter2/TameInvariance · IsScSplitting.map`.

Condition (2) `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameWith_transport`. Let T, T' be presentations of C, C', r tame for T, and y a smooth point of $\psi(O)$ on level 1. Then $x = \psi_1'(y)$ lies on level 1 of O (if $y = \psi(z)$ with $z \in O$, then $\psi'(y) = z$) and is smooth, because sc-smooth maps carry smooth points to smooth points. Tameness of r at x gives an sc-splitting $E = T_xO \oplus A$ with $A \subseteq E_x$. Because x and y are smooth points, $D\psi(x)$ and $D\psi'(y)$ are sc-operators (the remark after Proposition 1.16) `Chapter1/Properties · IsSc1With.scOperator_deriv` — at a point that is only on level 1 they need not be — and they are inverse to each other. From the formula for $Dr'(y)$ and $D\psi'(y)D\psi(x) = 1$: $T_y\psi(O) = \operatorname{range} Dr'(y) = D\psi(x)(\operatorname{range} Dr(x)) = D\psi(x)(T_xO)$. So $E' = T_y\psi(O) \oplus D\psi(x)A$ is an sc-splitting. Finally $D\psi(x)A \subseteq E'_y$: the derivative of an sc¹-map between relatively open subsets of partial quadrants maps the minimal subspace E_x into $E'_{\psi(x)}$ (the one-sided sign argument, and density) `Chapter2/DegeneracyIndex · IsSc1With.mapsTo_minimalSubspace`, which needs $\psi(U) \subseteq C'$, and $\psi(x) = y$. Since T' was arbitrary and the presentation T of C is available, the transported retraction of a tame retraction is tame `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport`, and **the image $(\psi(O), C', E')$ of a tame retract under an sc-diffeomorphism is a tame retract**, with retraction $\psi \circ r \circ \psi^{-1}$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image`.

Open subsets. For an open $W \subseteq E_0$, the restriction of r to $U \cap r^{-1}(W)$ is an sc-smooth retraction onto $O \cap W$ (Proposition 2.10 (1)) `Chapter2/Retracts · ScRetraction.restrict`. It is tame if r is `Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`: condition (1) is inherited pointwise; for (2), at a smooth point x of $O \cap W$ on level 1 the derivative of the restricted retraction is $Dr(x)$, because the two maps agree on the smaller relatively open domain, which contains x since $r(x) = x \in W$, and the derivative at a level-1 point depends only on the map near the point `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`; so the complement A of r at x serves. Hence $O \cap W$ is a tame retract whenever O is `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`. For a chart φ and an open s inside its source, $\varphi(s) = O \cap W$ for an open W `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq`, so the model of the restricted chart is a tame retract when that of φ is `Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`.

These are used in §III.34 (the chart $\psi\circ\varphi$ of Theorem 3.54 has the tame model $\psi(O)$) and in §§III.36–III.37 (the shrunk data of Definition 3.25 keep tame models). The text's sentence "$X' = \psi(X)$ is tame" in Proposition 3.26 is not needed there (§III.24): the tameness Definition 3.25 asks for is that of the chart models, which the transport of charts leaves unchanged.

III.28 Corollary 3.39: composition with a given conjugation

Corollary 3.39 (text): if the sc-germ g is equivalent, by a strong bundle isomorphism Φ, to a basic germ and s is an sc⁺-germ, there is a strong bundle map Ψ such that $((\Psi\circ\Phi)_*(g+s))^1$ is a basic germ. The text's proof: $\Phi_*(g) = h$ is basic, $t = \Phi_*(s)$ is an sc⁺-germ with $t(0) = 0$, $\Phi_*(g+s) = h + t$, and Theorem 3.38 gives Ψ with $(\Psi_*(h+t))^1$ basic; take $\Psi\circ\Phi$. Four things are silent: that Theorem 3.38's coordinate change is a strong bundle *isomorphism*; that the index raise of a strong bundle isomorphism is one between the shifted structures; that strong bundle isomorphisms and push-forwards compose; and that $\Phi_*(g+s) = h+t$ holds only where the conjugation identity holds. The formalization proves all four and states the corollary with the composite explicit.

Linear strong bundle isomorphisms. Two sc-operators $S : E \to E'$, $T : F \to F'$ give the sc-operator $S \oplus T$ `Chapter3/BasicClassComposition · ScOperator.prodMap`, and if $S(U) \subseteq U'$ the map $(u,k) \mapsto (Su, Tk)$ is a strong bundle map $U \lhd F \to U' \lhd F'$ (Definition 2.47 as formalized `Chapter2/Tame · IsStrongBundleMap`): its base component does not depend on k, it is linear in k, it preserves the double filtration because sc-operators preserve the levels, it is the sc-operator $S \oplus T$ on $(U \lhd F)[0]$, and on $(U \lhd F)[1]$ it is $S \oplus T^1$ with T^1 the shifted operator `Chapter3/BasicClassComposition · ScOperator.isStrongBundleMap_prodMap`. With inverse sc-operators S' (inverse to S on U, U') and T' it is a strong bundle isomorphism `Chapter3/BasicClassComposition · ScOperator.isStrongBundleIso_prodMap`.

Theorem 3.38's coordinate change as an isomorphism. In §III.18 the change of coordinates of Theorem 3.38 is the pair of linear sc-isomorphisms `$□ = $ split`, `$□ = $ fiber` (inverses `glue`, `fiberInv`) on the index-raised spaces `Chapter3/BasicClass · StabilityData.split` `Chapter3/BasicClass · StabilityData.fiber`. Put $\Lambda((a,w),(q,w')) = (\psi(a,w), \Psi(q,w'))$ `Chapter3/BasicClassComposition · StabilityData.linIso`. For every U relatively open in the shifted quadrant $(\widetilde{C}\oplus W)^1$, $\psi(U) = \mathrm{glue}^{-1}(U)$ is relatively open in $(\widetilde{C}\oplus C)\oplus X^1$ (`glue` is continuous and preserves the quadrant coordinates) `Chapter3/BasicClassComposition · StabilityData.isRelOpen_glue_preimage`, and $\Lambda : U \lhd (\mathbb{R}^N \oplus W)^1 \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z)\oplus X^1)$ is a strong bundle isomorphism by the preceding paragraph `Chapter3/BasicClassComposition · StabilityData.linIso_isStrongBundleIso`. Before the index raise, the same construction with the level-0 operators `split₀`, `fiber₀` gives Λ_0 `Chapter3/BasicClassComposition · StabilityData.linIso₀`, a strong bundle isomorphism $U \lhd (\mathbb{R}^N \oplus W) \to \psi(U) \lhd ((\mathbb{R}^N \oplus Z)\oplus X)$ — the text's "strong bundle map covering ψ" `Chapter3/BasicClassComposition · StabilityData.linIso₀_isStrongBundleIso` — and its index raise is Λ, because `split` and `fiber` are `split₀` and `fiber₀` on the levels $m+1$ `Chapter3/BasicClassComposition · StabilityData.strongShift_linIso₀`.

The index raise. For $\Psi : E_0 \times F_0 \to E_0' \times F_0'$ let Ψ^1 be its level-1 map on $E_1 \times F_1$ `Chapter3/BasicClassComposition · strongShift`. For a strong bundle map it exists on $U_1 \oplus F_1$ and satisfies $\iota\Psi^1(p) = \Psi(\iota p)$, because Ψ maps $U_1 \oplus F_1$ into $U_1' \oplus F_1'$ (the filtration with $k = 1 \le m+1$) `Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift`; its base map is the level-1 map of the base map of Ψ `Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_base_strongShift`. **Ψ^1 is a strong bundle map** $U^1 \lhd F^1 \to U'^1 \lhd F'^1$ of the shifted structures `Chapter3/BasicClassComposition · IsStrongBundleMap.shift`: independence of the base component and fibre linearity are read through the injective inclusions; the double filtration $(U^1 \lhd F^1)_{m,k} = U_{m+1} \oplus F_{k+1}$ is preserved because $k+1 \le (m+1)+1$ whenever $k \le m+1$, so the filtration property of Ψ applies; $(U^1 \lhd F^1)[0]$ is the shift of $(U \lhd F)[0]$, so Ψ^1 is sc-smooth there by Proposition 1.17 `Chapter1/Levels · IsScSmooth.shift`; and $(U^1 \lhd F^1)[1]$, with levels $E_{m+1} \oplus F_{m+2}$, is the shift of $(U \lhd F)[1]$, on which $(\Psi^1)[1]$ agrees with $(\Psi[1])^1$ (the lift of the fibre component to F_2' is the same element, by the injectivity of $F_2' \to F_1' \to F_0'$), so Proposition 1.17 applies again. For a strong bundle isomorphism with inverse Ψ', Ψ'^1 is the inverse of Ψ^1, again checked after the inclusion into level 0 `Chapter3/BasicClassComposition · IsStrongBundleIso.shift`. The text uses this whenever it writes $((\Psi\circ\Phi)_*(\cdot))^1$.

Composition. The base map of $\Lambda \circ \Phi$ is the composite of the base maps on U `Chapter3/BasicClassComposition · IsStrongBundleMap.base_comp`, and $(\Lambda \circ \Phi)[1] = \Lambda[1] \circ \Phi[1]$ on $U \oplus F_1$ `Chapter3/BasicClassComposition · IsStrongBundleMap.shiftFiber_comp`; so the composite is a strong bundle map — fibre linearity because over u it is the fibre map of Λ over $\varphi(u)$ after that of Φ over u, sc-smoothness on $[0]$ and $[1]$ by the chain rule `Chapter3/BasicClassComposition · IsStrongBundleMap.comp` — and the composite of strong bundle isomorphisms is one, with the inverses composed in the opposite order `Chapter3/BasicClassComposition · IsStrongBundleIso.comp`. Push-forwards compose: $(\Lambda \circ \Phi)_* g = \Lambda_*(\Phi_* g)$ on the target of Λ, since the base of $(\Lambda \circ \Phi)^{-1}$ is $\varphi^{-1} \circ \lambda^{-1}$ and $\Phi(\varphi^{-1}(w), \cdot)$ lies over w `Chapter3/BasicClassComposition · IsStrongBundleIso.pushforward_comp` `Chapter3/GermPushforward · pushforward`. The index raise commutes with composition, $(\Lambda \circ \Phi)^1 = \Lambda^1 \circ \Phi^1$ on $U_1 \oplus F_1$ `Chapter3/BasicClassComposition · IsStrongBundleMap.strongShift_comp`.

Corollary 3.39 as formalized `Chapter3/BasicClassComposition · corollary_3_39`. Let U be relatively open in a partial quadrant C of E with $0 \in U$, $g : U \to F$ sc-smooth (the text's "sc-germ"; sc-smoothness is what makes $(g+s)^1$ take values on level 1), d conjugation data of g (§III.30: a strong bundle isomorphism $\Phi : U \lhd F \to U' \lhd (\mathbb{R}^N \oplus W)$ covering φ with $\varphi(0) = 0$ and a basic germ h on a germ G inside U', with $(\Phi(u, g(u)))_2 = h(\varphi(u))$ for $u \in U$ near 0; the text's "equivalent by a strong bundle isomorphism to the basic germ $\Phi_* g$"), and $s : U \to F^1$ sc-smooth into F^1 with $s(0) = 0$ (the text's sc⁺-germ $s : O(C, 0) \to (F, 0)$). The proof, in the order of the Lean:

1. *The sc⁺-germ $t = \Phi[1]_* s$.* Put $t(u') = (\Phi[1](\varphi^{-1}u', s(\varphi^{-1}u')))_2$ `Chapter3/BasicClassComposition · ConjugationData.pushPlus`. It is sc-smooth from U' into $(\mathbb{R}^N \oplus W)^1$: the pairing of φ^{-1} with $s \circ \varphi^{-1}$ maps U' into $U \oplus F^1$, and $\Phi[1]$ is sc-smooth there by Definition 2.47; restricted to the germ $G \subseteq U'$ it is an sc-smooth germ. And $t(0) = 0$, from $\varphi^{-1}(0) = 0$, $s(0) = 0$ and $\Phi(0,0)_2 = 0$ (fibre linearity).
2. *Absorbing the $\mathbb{R}^N$-component.* As in Theorem 3.54 (§III.23, step 3), $h' = h + (t_{\mathbb{R}^N}, 0)$ `Chapter3/BasicClassComposition · ConjugationData.absorb` `Chapter3/GlobalIFT · ScBanach.qPartOp` is a basic germ with the same contraction map: it is an sc-smooth germ (pairing and the sum operator), $h'(0) = h(0) + 0 = 0$, and its W-component is that of h. The W-component t_W is an sc-smooth germ into W^1 vanishing at 0, so the data $\mathcal{S}$ of Theorem 3.38 exist for $s = t_W$ on the germ G of h `Chapter3/BasicClass · StabilityData.exists_of_scPlus`, and $\mathcal{S}.\mathrm{fsum}\, h' = h + \iota t$ `Chapter3/BasicClass · StabilityData.fsum`.
3. *$\Phi_*(g+s) = h + t$ near 0.* For $u \in U$ at which the conjugation identity holds, $\Phi_*(g + \iota s)(\varphi(u)) = (\Phi(u, g(u) + \iota s(u)))_2$ because $\varphi^{-1}\varphi(u) = u$; by fibre linearity this is $(\Phi(u, g(u)))_2 + (\Phi(u, \iota s(u)))_2 = h(\varphi(u)) + \iota t(\varphi(u))$, where $\iota(\Phi[1](u, s(u)))_2 = (\Phi(u, \iota s(u)))_2$ `Chapter3/SectionCompactness · IsStrongBundleMap.toBase_shiftFiber_snd`. So $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ at the points $\varphi(u)$, $u \in U$ near 0 — not everywhere, as the text's "clearly" suggests.
4. *The composite.* $\Gamma = \Lambda \circ \Phi^1$ `Chapter3/BasicClassComposition · ConjugationData.compIso`, with inverse $\Phi'^1 \circ \Lambda^{-1}$ `Chapter3/BasicClassComposition · ConjugationData.compIsoInv`, is a strong bundle isomorphism $U^1 \lhd F^1 \to \psi(U'^1) \lhd ((\mathbb{R}^N \oplus Z) \oplus X^1)$ `Chapter3/BasicClassComposition · ConjugationData.compDomain`: the index raise of Φ (relatively open U_1' in the shifted quadrant) followed by Λ, composed by the composition lemma. Its base map is $u \mapsto \psi(\varphi_1(u))$, which vanishes at 0; $\Gamma_* = \Lambda_* \circ \Phi^1_*$ (composition of push-forwards); and the germ of $\mathcal{S}.\mathrm{pushMap}\, h'$ lies in the target, because `glue` maps the level-1 part of G into U_1'.
5. *The basic germ.* $\mathcal{S}.\mathrm{pushMap}\, h' = \mathrm{fiber} \circ (\mathcal{S}.\mathrm{fsum}\, h')^1 \circ \mathrm{glue}$ `Chapter3/BasicClass · StabilityData.pushMap` is a basic germ by Theorem 3.38 (§III.18) `Chapter3/BasicClass · theorem_3_38`.
6. *The conjugation identity on level 1.* For $u \in U^1$ near 0: $(\Gamma(u, (g+s)^1(u)))_2 = \mathcal{S}.\mathrm{pushMap}\, h'(\gamma(u))$. Since $\gamma(u) = \mathrm{split}(\varphi_1(u))$ and $\mathrm{glue} \circ \mathrm{split} = \mathrm{id}$, both sides are `fiber` applied to elements of $(\mathbb{R}^N \oplus W)_1$, which are compared after the injective inclusion into level 0: $\iota(\Phi^1(u, (g+s)^1(u)))_2 = (\Phi(\iota u, (g+s)(\iota u)))_2$ `Chapter3/BasicClassComposition · IsStrongBundleMap.toBase_strongShift_snd`, which is $\mathcal{S}.\mathrm{fsum}\, h'(\varphi(\iota u))$ by step 3, and this is ι of $(\mathcal{S}.\mathrm{fsum}\, h')^1(\varphi_1(u))$. "Near 0" means: ιu lies in the neighbourhood where the conjugation identity holds (continuity of $\iota : E_1 \to E_0$), and $\varphi_1(u)$ lies in the level-1 part of the germ G, a relatively open condition at 0 by the continuity of φ on level 1.

The conclusion of `corollary_3_39` collects these: $\mathcal{S}$ with $\mathcal{S}.G = G$ and $\mathcal{S}.s = t_W$, $t(0) = 0$, h' basic, $\Phi_*(g+s) = \mathcal{S}.\mathrm{fsum}\, h'$ near 0, Γ a strong bundle isomorphism onto a relatively open subset of $(\widetilde{C} \oplus C) \oplus X^1$ with $\gamma(0) = 0$ and $\Gamma_* = \Lambda_* \circ \Phi_*^1$, the basic germ $\mathcal{S}.\mathrm{pushMap}\, h'$ inside the target, and the identity of step 6. In the form of Definition 3.11: $(g+s)^1 : U^1 \to F^1$ **is conjugated to a basic germ** on the shifted structures E^1, F^1 `Chapter3/BasicClassComposition · corollary_3_39_isConjugate`, with parameter space $\mathbb{R}^n \oplus C$, target $\mathbb{R}^N \oplus Z$ and fibre X^1 (the text's $\tau : Z \to C$ is not applied, as in §III.18). The text's form "$\Gamma = \Psi \circ \Phi$, then raise the index" is the same map: $\Lambda_0 \circ \Phi$ is a strong bundle isomorphism on level 0 and $(\Lambda_0 \circ \Phi)^1 = \Lambda_0^1 \circ \Phi^1 = \Lambda \circ \Phi^1$ on $U_1 \oplus F_1$ `Chapter3/BasicClassComposition · ConjugationData.compIso_eq_strongShift`. No hypothesis is added beyond the two readings ("sc-germ" as sc-smooth on U; "equivalent by a strong bundle isomorphism" as conjugation data). The first reading is the whole-chart one of §III.6, narrower than the text's germs; no theorem of the germ reading uses Corollary 3.39 (the facts at the nearby zeros come from the chart at x_0 through the linear coordinate change of Theorem 3.38, §III.43).

III.29 Good complements with a projection into the quadrant; Theorems 3.43, 3.47 and 3.54 (3) with a closed projecting complement

Two steps of the text depend on the choice of a good complement without saying so. In the proof of Theorem 3.27 the retraction $t(v+w) = v + \delta(v)$ is defined on $\Sigma = \{v + y \in C : v \in V,\ y \in Y\}$, which the text calls relatively open in C; §III.24 shows that this fails for the good complement $Y = \{x + y - z = 0\}$ of $N = \mathbb{R}(1,1,1)$ in $C = [0,\infty)^3$, and makes the remedy a hypothesis: condition (b) of the regularity conditions, $\pi(O) \subseteq C$ for the projection π onto N along Y `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular`. In the proof of Theorem 3.43 (Lemma 3.46) a good complement Z of $\widetilde{K} = \ker DH(0)$ is produced by projecting a good complement of K'. This section shows that for solution sets the complement can be chosen once and for all, at the bottom of the chain Theorem 3.78 □ Theorem 3.43 □ Corollary 3.47 □ Theorem 3.54 (3), so that its projection maps the partial quadrant into itself; condition (b) then holds for the chart of §III.34 without any hypothesis (Findings, Chapter 3, items 130–133). Along the way the closedness of the complement, which Lemma 3.55 and Definition 3.25 need (items 49, 81), is recorded in the statements (item 93).

The projection property. For subspaces N, Y of a vector space and a set C, *N projects C into itself along Y* if $z \in C$, $k \in N$ and $z - k \in Y$ imply $k \in C$ `Chapter3/GoodComplementProjection · ProjectsInto`. The definition avoids naming the projection; when $E = N \oplus Y$ it says exactly that the projection π onto N along Y maps C into C `Chapter3/GoodComplementProjection · ProjectsInto.projectionOnto_mem` `Chapter3/GoodComplementProjection · projectsInto_of_projectionOnto_mem` ($z - \pi z \in Y$, and conversely $z - k \in Y$, $k \in N$ force $\pi z = k$). It is carried by a linear isomorphism S with $x \in C \iff Sx \in C'$: given $z' \in C'$, $k \in N$ and $z' - Sk = Sy$ with $y \in Y$, the point $S^{-1}z'$ lies in C and $S^{-1}z' - k = y$, so $k \in C$ and $Sk \in C'$ `Chapter3/GoodComplementProjection · ProjectsInto.map`, in particular for continuous linear isomorphisms `Chapter3/GoodComplementProjection · ProjectsInto.mapL` — the analogue of Lemma 3.45 (§III.14).

The finite-dimensional lemma. *If the finite-dimensional subspace N is in good position to the partial quadrant C of the sc-Banach space E, then N has a closed good complement Y such that the projection onto N along Y maps C into $N \cap C$* `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto`. The text has no such statement; Definition 3.21 and Proposition 3.22 do not need it. The proof works in the coordinates of §III.14: after a presentation, $C = \{\ell_i \geq 0\}$ for continuous functionals $\ell_1, \dots, \ell_n$ jointly onto $\mathbb{R}^n$, the cone $Q = N \cap C$ is $\{x \in N : \ell_i(x) \geq 0\}$, Σ is the set of indices vanishing at some nonzero point of Q, and for a set S of indices $Q_S = (\ell_i)_{i \in S} : N \to \mathbb{R}^S$ `Chapter3/GoodComplementProjection · coordOn` (for $S = \Sigma$ it is the map Q_Σ of Proposition 3.22 `Chapter3/GoodComplementProjection · sigmaCoord_eq_coordOn`).

Step 1: the choice of coordinates `Chapter3/GoodComplementProjection · exists_coordSet`. Wanted is a set S of indices with (1) Q_S onto, (2) $Q = \{x \in N : \ell_i(x) \geq 0 \text{ for } i \in S\}$, (3) every ℓ_i, $i \notin S$, positive on $Q \setminus 0$. Property (3) holds for $S = \Sigma$ by the definition of Σ `Chapter3/GoodPosition · pos_of_not_mem_vanIdx`, and (1) for $S = \Sigma$ is

the consequence of Lemma 3.65 proved in §III.14 Chapter3/GoodPosition · sigmaCoord_surjective. Three cases, those of the normal form. If $Q = N$, take $S = \Sigma$; (2) is trivial. If $\Sigma \neq \emptyset$, take $S = \Sigma$; (2) is the segment argument of Proposition 3.22 Chapter3/GoodPosition · mem_quadCone_of_vanIdx. In the remaining case, $\Sigma = \emptyset$ and $Q \neq N$, the choice $S = \Sigma$ fails, and here the sketch of Finding 83, which takes $S = \Sigma$ throughout, misses a case (item 131). With $S = \emptyset$, (2) would say $Q = N$; and the formula of Step 4 below would reduce to an arbitrary projection onto N, which need not map C into C: for $C = [0,\infty)^2$ and $N = \mathbb{R}(1,1)$ every nonzero point of Q has both coordinates positive, so $\Sigma = \emptyset$; the line $\mathbb{R}(1,2)$ is a good complement of N (for $|s| \leq \varepsilon|t|$ the coordinates of $t(1,1) + s(1,2)$ have the sign of t), and the projection along it maps $(0,1) = -(1,1) + (1,2)$ to $-(1,1) \notin C$. The formal proof keeps one coordinate. Some $x_1 \in N$ lies outside C, so there is at least one coordinate, j_0 say, and $S = \{j_0\}$. An interior point y of Q is nonzero — if 0 were interior, Q would contain a ball of N around 0 and, being closed under positive scaling, all of N Chapter3/GoodComplementProjection · mem_quadCone_of_zero_mem_interior Chapter3/GoodPosition · smul_mem_quadCone — so $\ell_{j_0}(y) > 0$ by (3), and $Q_{\{j_0\}}$ is onto: $k \mapsto (k/\ell_{j_0}(y))\, y$ is a preimage. Property (2) for $S = \{j_0\}$ is the next step.

Step 2: the segment argument for any S Chapter3/GoodComplementProjection · mem_quadCone_of_coordSet. Suppose (3) holds for S, some $i_0 \in S$ is positive at a point $y \in Q$, and $x \in N$ has $\ell_i(x) \geq 0$ for $i \in S$. The set $T = \{s \geq 0 : x + sy \in Q\}$ is nonempty (for $i \in S$, $\ell_i(x) + s\ell_i(y) \geq 0$ for all $s \geq 0$; for $i \notin S$, $\ell_i(y) > 0$, so large s work), closed and bounded below; let $s^* = \min T$. If $s^* > 0$, the point $z = x + s^*y$ is nonzero, since $\ell_{i_0}(z) \geq s^*\ell_{i_0}(y) > 0$; so by (3) all ℓ_i, $i \notin S$, are positive at z and, by continuity, at $x + sy$ for s near s^*; the coordinates in S stay ≥ 0 for all $s \geq 0$; so a point $s_1 = \max(0, s^* - \delta/2) < s^*$ lies in T, contradicting minimality. Hence $s^* = 0$ and $x \in Q$. This is the argument of Proposition 3.22, Case C (§III.14), with the positivity outside S as the only input; with $S = \{j_0\}$ and y interior it gives (2) in the third case of Step 1.

Step 3: a lower bound Chapter3/GoodComplementProjection · exists_coord_ge. A coordinate positive on $Q \setminus 0$ is bounded below by $c|x|$ on Q, $c > 0$: its minimum on the compact set $Q \cap \{|x| = 1\}$ is positive, and Q is a cone.

Step 4: the projection and the estimate Chapter3/GoodComplementProjection · exists_proj_quadCone. Take S from Step 1, a continuous projection $p_N : E_0 \to N$ (Hahn–Banach, N finite-dimensional), a linear right inverse g of Q_S, and

$$\pi(z) = p_N(z) + g((\ell_i(z - p_N z))_{i \in S}).$$

Then π is continuous, $\pi = \mathrm{id}$ on N (the bracket vanishes), and $\ell_i(\pi z) = \ell_i(p_N z) + \ell_i(z - p_N z) = \ell_i(z)$ for $i \in S$, so for $z \in C$ the coordinates of πz in S are ≥ 0 and $\pi z \in Q$ by (2): $\pi(C) \subseteq N \cap C$. The kernel of π is a complement of N, and it is a good complement. For $p \in \ker\pi$ the coordinates in S vanish ($\ell_i(p) = \ell_i(\pi p) = 0$). Let c_i be the constants of Step 3 for $i \notin S$ and $\varepsilon' = \min_i c_i/(\|\ell_i\| + 1)$ Chapter3/GoodComplementProjection · exists_pos_le_of_finite. Let $x \in N$, $p \in \ker\pi$, $|p| \leq \varepsilon'|x|$. If $x \in C$: for $i \in S$, $\ell_i(x + p) = \ell_i(x) \geq 0$; for $i \notin S$, $\ell_i(x + p) \geq c_i|x| - \|\ell_i\|\,|p| \geq c_i|x| - \|\ell_i\|\varepsilon'|x| \geq 0$. If $x + p \in C$: the coordinates of x in S are those of $x + p$, hence ≥ 0, and $x \in Q$ by (2). The reverse implication needs no estimate.

Step 5: partial quadrants. For a partial quadrant presented by an sc-isomorphism T the functionals are the coordinate functions of T, continuous Chapter3/GoodComplementProjection · continuous_coordFun and jointly onto Chapter3/GoodPosition · coordFun_surj; good position supplies a good complement P with its constant and the interior point. Step 4 then gives a continuous projection $\pi : E_0 \to N$ with $\pi(C) \subseteq C$ whose kernel is a good complement Chapter3/GoodComplementProjection · exists_proj_goodComplement ($\ker\pi$ is a topological complement of N, as the kernel of a continuous projection). With $Y = \ker\pi$, closed as the kernel of a continuous map, the projection property is immediate: $z \in C$, $k \in N$, $z - k \in Y$ give $k = \pi k = \pi z \in C$ Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto.

Theorem 3.43 with a prescribed good complement. In the text's proof the complement Z of $\widetilde{K} = \ker DH(0)$ in $\mathbb{R}^{n'}$ comes from a good complement of $K' = \{(\alpha, L\alpha) : \alpha \in \widetilde{K}\}$ through Lemma 3.46 (§III.20). Nothing later uses that origin. *For every good complement Z of $\widetilde{K}$ (constant*

ε), $Z \oplus W$ is a good complement of K' with respect to $\widetilde{C} \oplus W$, with constant $\varepsilon/(\|L\| + 1)$ Chapter3/GoodComplementProjection · IsGoodComplement.prod_top_graph. The splitting: $K' \cap (Z \oplus W) = 0$ because $\widetilde{K} \cap Z = 0$, and $(a, w) = (\alpha, L\alpha) + (z, w - L\alpha)$ for $a = \alpha + z$. The estimate: with the maximum norm, $|(\alpha, L\alpha)| \le (\|L\| + 1)|\alpha|$, so $|(z, w)| \le \varepsilon/(\|L\| + 1) \cdot |(\alpha, L\alpha)|$ gives $|z| \le \varepsilon|\alpha|$, and $(\alpha, L\alpha)+(z, w) \in \widetilde{C}\oplus W$ iff $\alpha+z \in \widetilde{C}$ iff $\alpha \in \widetilde{C}$ iff $(\alpha, L\alpha) \in \widetilde{C}\oplus W$ Chapter3/LocalSolvability · graphMap_mem_prod_iff. This is the second half of the proof of Lemma 3.46 for an arbitrary Z; Lemma 3.46 itself is needed only for the good position of $\widetilde{K}$. The projection property passes the same way: if $z \in \widetilde{C} \oplus W$ and $z - (\alpha, L\alpha) \in Z \oplus W$, then $z_1 - \alpha \in Z$, so $\alpha \in \widetilde{C}$ when $(\widetilde{K}, Z)$ has the property, and $(\alpha, L\alpha) \in \widetilde{C}\oplus W$ Chapter3/GoodComplementProjection · ProjectsInto.prod_top_graph; the projection along $Z\oplus W$ is $(a, w) \mapsto (\pi_Z a, L\pi_Z a)$, and $\widetilde{C} \oplus W$ constrains only the first component.

So the data of the proof of Theorem 3.43 (§III.20: the derivative data, the graph description $K' = \mathrm{graph}(L|_{\widetilde{K}})$ of Lemma 3.44, a complement Z, and the solution σ of $H(k + \sigma(k)) = 0$ from Theorem 3.78 with its uniqueness constants Chapter3/LocalSolvability · ShiftedBasicData.Sol) exist with any prescribed good complement Z of $\widetilde{K}$ Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_of_goodComplement: K' is the graph Chapter3/LocalSolvability · ShiftedBasicData.Deriv.ker_Dg_eq, so $\widetilde{K}$ is in good position (Lemma 3.46) Chapter3/LocalSolvability · lemma_3_46_isInGoodPosition, $Z \oplus W$ is a good complement of K' by the lemma above, and Theorem 3.78 is applied to H with Z Chapter3/QuadrantIFT · theorem_3_78. Choosing Z by the finite-dimensional lemma in $\mathbb{R}^{n'}$ (constant sc-structure, partial quadrant $\widetilde{C}$) gives data whose Z projects $\widetilde{C}$ into $\widetilde{K} \cap \widetilde{C}$ Chapter3/GoodComplementProjection · ShiftedBasicData.exists_sol_proj.

Theorem 3.43, closed and projecting. With the hypotheses of Theorem 3.43 (§III.20: $\widetilde{C}$ a partial quadrant, $Dg(0)$ onto, $K' = \ker Dg(0)$ in good position, the regularity of Lemma 3.42 as a hypothesis), the conclusions of the level-wise form Chapter3/LocalSolvability · theorem_3_43_sc_smooth hold with a good complement $Y = Z \oplus W$ that is **closed** Chapter3/LocalSolvabilityClosed · theorem_3_43_closed — Z is finite-dimensional, hence closed, and $Z \times W_0$ is closed — and, with the data of the last paragraph, that has the **projection property** for $\widetilde{C} \oplus W$ Chapter3/GoodComplementProjection · theorem_3_43_proj (rewrite K' as the graph and apply the passage of the property to $Z \oplus W$). The other conclusions (V, τ with values in Y and W-components on level 1, C^1, $\tau(0) = 0$, $D\tau(0) = 0$, the graph of zeros, the C^k lifts τ_k, the neighbourhood U'' in which every zero lies on the graph) are those of §III.20, read off the same data Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_contDiffOn_level Chapter3/LocalSolvability · ShiftedBasicData.Sol.exists_eq_add_τ.

Corollary 3.47, closed and projecting Chapter3/LocalSolvabilityClosed · theorem_3_47_proj. For a basic germ f, the data $\mathcal{S}$ of Theorem 3.38 with $D(f + \iota s)(0)$ onto and kernel K in good position to $\widetilde{C} \oplus W$: the conclusions of the level-wise Corollary 3.47 Chapter3/LocalSolvability · theorem_3_47_sc_smooth hold with Y closed and K projecting $\widetilde{C}\oplus W$ into itself along Y. The proof is that of §III.20 with the two new properties carried along. Let S be the linear coordinate change $\psi(a, w) = ((a, (1-P_1)w), P_1 w)$ of Theorem 3.38 on level 0, a continuous linear isomorphism with inverse the gluing map. The push-forward $g = \Phi_*(f+s)$, packaged as data of Theorem 3.43 Chapter3/LocalSolvability · StabilityData.toShiftedBasicData, has $Dg(0) = \Psi \circ D(f+\iota s)(0) \circ S^{-1}$ Chapter3/LocalSolvability · StabilityData.scDerivGerm_pushMap₀, so $\ker Dg(0) = S(K)$, $Dg(0)$ is onto, and K is finite-dimensional (Lemma 3.44 for g and the injectivity of S^{-1}). S carries $\widetilde{C} \oplus W$ onto $(\widetilde{C} \oplus C_s) \oplus X$, where C_s is the finite-dimensional space of the splitting of Theorem 3.38 (§III.18) and $\widetilde{C}\oplus C_s$ is a partial quadrant of $\mathbb{R}^n \oplus C_s$ Chapter3/LocalSolvability · IsPartialQuadrant.const_prod_univ; so $S(K)$ is in good position (Lemma 3.45, §III.14). Theorem 3.43, closed and projecting, for g, with Lemma 3.42 for its regularity hypothesis Chapter3/LocalSolvability · StabilityData.lemma_3_42, gives Y', V', τ. Transported back: $Y = S^{-1}(Y')$ is a good complement of $K = S^{-1}(S(K))$ (Lemma 3.45) Chapter3/GoodPosition · IsGoodComplement.map, closed as the image of a closed set under a linear homeomorphism, and projecting, by the transport of the projection property Chapter3/GoodComplementProjection · ProjectsInto.mapL; $V = (S|_K)^{-1}(V')$ and $\sigma = S^{-1} \circ \tau \circ S|_K$. The W-component of $\sigma(y)$ is $c + w$ with $c \in C_s \subseteq W_\infty$

`Chapter3/BasicClass · StabilityData.Cs_le_smooth` and w the level-1 X-component of τ, so it lies on level 1; $D\sigma(0) = S^{-1} \circ D\tau(0) \circ S|_K = 0$, by the chain rule within V, which has unique differentiability at 0 as a relatively open subset of the convex cone $K \cap C$ with interior; the graph points are zeros because the linear fibre map of Theorem 3.38 is injective; the lifts are ψ^{-1} on level k applied to τ_k; and $U'' := S^{-1}(U'')$. The form with only the closedness is a corollary `Chapter3/LocalSolvabilityClosed · theorem_3_47_closed`.

Theorem 3.54 (3), closed and projecting `Chapter3/GlobalIFTG · theorem_3_54_graph_proj`. The graph description of §III.23 — for a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the chart φ centred at x_0, the conjugating sc-diffeomorphism ψ, the germ $h + t$ with surjective linearization and kernel K in good position, Y, V', σ, and the open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ — is proved with Corollary 3.47 closed and projecting in place of the plain Corollary 3.47, and the statement now records five more facts. (i) The inverse ψ' of ψ, sc-smooth on U', with $\psi'\psi = \mathrm{id}$ on U and $\psi\psi' = \mathrm{id}$ on U' (the base maps of the conjugating strong bundle isomorphism and its inverse); §III.34 needs it to make $\psi \circ \varphi$ a chart. (ii) Y is closed, and (iii) K projects $C' = \widetilde{C} \oplus W$ into itself along Y, both from Corollary 3.47 — the shrinking of V' in step 6 of §III.23 ($V' = V_0 \cap O_1$ with O_1 open, chosen so that $v + \sigma(v)$ stays in $\Omega = U'' \cap W'$) does not touch Y. (iv) K is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (proved in §III.34). (v) Every $v + \sigma(v)$, $v \in V'$, is a zero of $h + t$ in the germ, and for every $k \geq 1$ the C^k lift σ_k of Corollary 3.47 (1) survives the shrinking: on $V_k \cap O_1$, relatively open in $K \cap C'$, containing 0 and contained in V', it is still C^k and still lifts σ. The theorem of §III.23, with the same statement except the projection property, is a corollary `Chapter3/GlobalIFTG · theorem_3_54_graph`.

Where the formal route departs from the text. The text never chooses the good complement; Definition 3.25 lets it be any good complement, and for an arbitrary subset with the tangent germ property the repair of the claim "Σ is relatively open" would need a re-graphing over a better complement, which is not done ($\pi(O) \subseteq C$ remains a hypothesis of Theorem 3.27, §III.24). For solution sets the complement is not given but produced, at Theorem 3.78, and every link of the chain accepts a prescribed complement; so the repair is a choice made at the bottom of the chain, not a change in the proof of Theorem 3.27 (item 130). The closedness of Y is part of the text's sc-complement and is recorded because Lemma 3.55 and Definition 3.25 use it at the nearby zeros (items 49, 81, 93).

III.30 Proposition 3.52 with its index

Proposition 3.52 (text): the linearization $Tf(x) : T_xX \to Y_x$ of an sc-Fredholm section at a zero is an sc-Fredholm operator; in the proof (and in Theorem 3.54) its index is the index $n - N$ of the basic germ $h : O(([0,\infty)^k \oplus \mathbb{R}^{n-k}) \oplus W, 0) \to (\mathbb{R}^N \oplus W, 0)$ to which $g - s$ is conjugated. The statements of §III.21 state the index as "$n - N$ for some natural numbers n, N", which every integer is. The reason is that the conjugation of Definition 3.11 is an existential proposition `Chapter3/StrongBundleSections · IsConjugateToBasicGerm`, so the dimensions of the basic germ cannot be named outside it. Here the witnesses become data.

Conjugation data `Chapter3/FredholmGermIndex · ConjugationData`: finite-dimensional normed spaces P_n (the text's $\mathbb{R}^n$) and Q (the text's $\mathbb{R}^N$), an sc-Banach space W, a partial quadrant $\widetilde{C} \subseteq P_n$, a relatively open $U' \subseteq \widetilde{C} \oplus W$, a strong bundle isomorphism $\Psi : U \lhd F \to U' \lhd (Q \oplus W)$ with inverse Ψ' and $\psi(0) = 0$ for its base map, a germ G with $G_0 \subseteq U'$, a basic germ h on G with contraction map B, and $(\Psi(u, k(u)))_2 = h(\psi(u))$ for $u \in U$ near 0. A map is conjugated to a basic germ iff such data exist `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`.

The derivative of a conjugated map `Chapter3/FredholmGermIndexG · ConjugationData.scDeriv_isFredholm_index`. Let U be relatively open in the partial quadrant C, $0 \in U$, k an sc¹-germ with derivative family Dk on a germ G_k with $(G_k)_0 \subseteq U$, $k(0) = 0$, and d conjugation data of k. Then $Dk(0) : E_0 \to F_0$ is Fredholm of index $\dim P_n - \dim Q$; for k sc-smooth on U this is the whole-chart form `Chapter3/FredholmGermIndex · ConjugationData.scDeriv_isFredholm_index_wholeChart`. The proof is that of §III.21, step (a),

with the dimensions kept:

(a) $Dh(0)$ is Fredholm of index $\dim P_n - \dim Q$ (Proposition 3.10 for the basic germ on level 0) `Chapter3/FredholmGermLinearization · proposition_3_10_basic_level_zero`.

(b) $\Gamma(0)Dk(0) = Dh(0)D\psi(0)$, with $\Gamma(0) = (\Psi(0,\cdot))_2$ the fibre operator of Ψ at 0 `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM`. The left side is the derivative at 0 of $u \mapsto (\Psi(u,k(u)))_2$, because $k(0) = 0$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.scDeriv_fiber_comp`. For the right side, let N_1 be an open neighbourhood of 0 on which the conjugation identity holds; the germ $\{U \cap N_1\}$ has a subgerm G_1 mapped by ψ into G level by level `Chapter3/GermFilling · ScNbhdGerm.preimage`; on G_1 the chain rule for germs gives the derivative family $Dh(\psi_1(x))D\psi(x)$ of $h \circ \psi$ `Chapter3/Germ · IsSc1GermWith.comp`; and $u \mapsto (\Psi(u,k(u)))_2$, an sc¹-germ on G_k by the chain rule with the whole-chart outer map (§III.48), agrees with $h \circ \psi$ on $G_1 \cap G_k$, so the two derivative families agree at 0 (uniqueness of the derivative of a germ at points of U_1) `Chapter3/Germ · IsSc1Germ.scDerivGerm_eqOn`, with $\psi_1(0) = 0$.

(c) $D\psi(0)$ and $D\psi'(0)$ are inverse to each other (chain rule for $\psi' \circ \psi = \mathrm{id}$ on U and $\psi \circ \psi' = \mathrm{id}$ on U') `Chapter2/DegeneracyIndex · scDeriv_comp_eq_self`, and so are $\Gamma(0)$ and the fibre operator $\Gamma'(0)$ of Ψ' at 0: $\Psi'(\Psi(0,y)) = (0,y)$ with $\Psi(0,y) = (0,\Gamma(0)y)$ because $\psi(0) = 0$, and symmetrically.

(d) Hence $Dk(0) = \Gamma(0)^{-1} \circ Dh(0) \circ D\psi(0)$, a Fredholm operator composed with isomorphisms on both sides, of the same index `Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`.

Consequently **the index $n - N$ does not depend on the conjugation**: two conjugation data d, d' of the same k give $\dim d.P_n - \dim d.Q = \operatorname{ind} Dk(0) = \dim d'.P_n - \dim d'.Q$ `Chapter3/FredholmGermIndexG · ConjugationData.index_eq`. This is the text's remark after Theorem 3.38 that "the Fredholm index is invariant under strong bundle isomorphisms", for the conjugations of one germ.

Proposition 3.52 in the chart, with the index `Chapter3/FredholmGermIndexG · proposition_3_52_local_index`. Hypotheses: U relatively open in C, R a strong bundle retraction on $U \lhd F$ with base retraction r, $0 \in U$ and $0 \in O = r(U)$; g a filling germ of the principal part $\hat{f}$ (§III.3) and Dg a derivative family of g on a germ G with $G_0 \subseteq U$; $\hat{f}(0) = 0$ and $(x, \hat{f}(x)) \in K$ for $x \in O$; s an sc⁺-germ with derivative family Ds and $\iota s(0) = g(0)$; and conjugation data d of $k = g - \iota s$. Conclusion: $Dg(0) : E_0 \to F_0$ and $Tf(0) = Dg(0)|_{T_0O} : T_0O \to K_0$ `Chapter3/FillerLinearAlgebra · fillerRestrict` are Fredholm with index $\dim d.P_n - \dim d.Q$. Proof: $g(0) = \hat{f}(0) = 0$ (filler condition (1) at 0), so k is an sc¹-germ on $G \cap G_s$ with the family $Dg - \iota Ds$ and $k(0) = 0$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.sub_toBase_shift`, and $Dk(0)$ is Fredholm of index $\dim d.P_n - \dim d.Q$ by the preceding theorem. Since 0 is a smooth point, $Ds(0) : E \to F^1$ is an sc-operator (the remark after Proposition 1.16, for germs) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, so $\iota Ds(0)$ is the level-0 map of an sc⁺-operator and compact. $Dg(0) = Dk(0) + \iota Ds(0)$ is then the value at 0 of the identity $Dg = (Dg - \iota Ds) + \iota Ds$ `Chapter3/FredholmGermLinearizationG · IsSc1GermWith.deriv_zero_eq_add_scPlus`; on U it was the uniqueness of derivatives within U_1 and the density of E_1 `Chapter3/BasicGermLinearization · ContinuousLinearMap.ext_of_comp_toBase_one`. A compact perturbation of a Fredholm operator is Fredholm of the same index `Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator`. Proposition 3.5 (§III.4) transfers the Fredholm property and the index to $Tf(0)$ `Chapter3/GermFillerLinearization · IsFillingGerm.proposition_3_5`.

Proposition 3.52, with the index `Chapter3/FredholmGermIndexG · proposition_3_52_index`. Let f be a section of the strong bundle $P : Y \to X$ with (f, x_0) an sc-Fredholm germ and $f(x_0) = 0$. In the strong bundle chart of Definition 3.11 centred at x_0, with the filling germ g of $\hat{f}$, its germ G and a derivative family Dg, and the local sc⁺-germ s, conjugation data of $g - s$ exist, $Dg(0)$ and $Tf(x_0)$ are Fredholm, and **for every** conjugation data d of $g - s$ their index is $\dim d.P_n - \dim d.Q$. Proof: unpack Definition 3.11; $0 \in O$ because the chart is centred at x_0; $\hat{f}(0) = 0$ from $f(x_0) = 0$; $(x, \hat{f}(x)) \in K$ because the principal part of a section lies in the retract K of the chart; data d_0 exist by the equivalence above and give the Fredholm property; and the local theorem applied to an arbitrary d gives the index. No hypothesis is added. As in §III.21, only the level-0 Fredholm property is proved (T_0O and K_0 are not set up as sc-Banach spaces).

The statements of §III.21 stand next to these, with the same proofs and without the dimensions; the whole-chart forms are kept as well `Chapter3/FredholmGermIndex · proposition_3_52_index_wholeChart`.

III.31 The linearization at a zero in charts, its chart independence, and Definition 3.53 from a non-centred chart

Definition 3.53 is stated in T_xX, with the linearization $f'(x) : T_xX \to Y_x$ and the partial cone C_xX of Definition 2.38. The text defines $f'(x)$ only at the *centre* of a chart, before Definition 3.4, as the restriction of $D(\mathbf{f} \circ r)(0)$ to T_0O, identifies it with $Dg(0)|_{T_0O}$ for a filler g, and appeals to Proposition 2.30 for the independence of T^R_xX and C_xX of the chart. That $f'(x)$ itself, and the two conditions of Definition 3.53, do not depend on the chart is used — the proof of Theorem 3.54 repeats "as above" the argument at x_0 in a chart centred at a nearby zero y — but not proved (Finding 110). The formalization (§III.21) states Definition 3.53 in the chart of Definition 3.11 centred at the zero; this section proves the local form of the missing independence at a zero, in charts, and the assembly step that Theorem 3.54 (1) needs: Definition 3.53 at a zero y from data in a chart which is *not* centred at y (the chart at x_0). Throughout, z is a point on level 1 (the text has smooth points), and tameness is not used.

Linear algebra of conjugate operators. Let $L : T \to K$ and $L' : T' \to K'$ be bounded operators between normed spaces and $A : T \cong T'$, $\Gamma : K \cong K'$ topological linear isomorphisms with $L'A = \Gamma L$. Then $L' = \Gamma L A^{-1}$ `Chapter3/LinearizationCharts · eq_comp_of_conj`, so L is onto iff L' is `Chapter3/LinearizationCharts · surjective_iff_of_conj`; $\ker L' = A(\ker L)$ `Chapter3/LinearizationCharts · ker_eq_map_of_conj`; for complete spaces L is Fredholm iff L' is, with the same index, because Fredholm operators and their index are stable under composition with isomorphisms on both sides `Chapter3/LinearizationCharts · isFredholm_iff_of_conj Chapter3/FredholmGermLinearization · isFredholm_equiv_comp_equiv`; and if $D \subseteq K$ is dense and contained in the range of L, then $\Gamma(D)$ is dense in K' and contained in the range of L' `Chapter3/LinearizationCharts · dense_image_of_conj`. The isomorphisms that occur are restrictions of bounded operators $a : E_0 \to E_0'$, $a' : E_0' \to E_0$ which map a subspace T into T' and T' into T and are inverse to each other there `Chapter3/LinearizationCharts · restrictEquiv`. A subspace $N \subseteq E_0$ is transported by $A : T \cong T'$ to $A(N) = \{Ax : x \in T \cap N\} \subseteq T'$ `Chapter3/LinearizationCharts · transportSub`, with $y \in A(N) \iff A^{-1}y \in N$ for $y \in T'$ `Chapter3/LinearizationCharts · mem_transportSub_iff` and $A^{-1}(A(N)) = N$ for $N \subseteq T$ `Chapter3/LinearizationCharts · transportSub_symm`.

Good position inside a subspace is transported `Chapter3/LinearizationCharts · IsInGoodPositionIn.transport`. Good position inside T (§III.21, `Chapter3/GoodPositionGerm · IsInGoodPositionIn`) asks: $N \subseteq T$, the interior of $N \cap C$ in N nonempty, and a complement $P \subseteq T$ of N *in* T with the ε-condition of Definition 3.21. If $A : T \cong T'$ respects the cones ($x \in C \iff Ax \in C'$ for $x \in T$) and N is in good position to C inside T, then $A(N)$ is in good position to C' inside T'. *The interior*: A restricts to a homeomorphism $N \cong A(N)$ which carries $N \cap C$ onto $A(N) \cap C'$, so it carries interior points to interior points. *The complement*: $A(P)$; $A(N) \cap A(P) = 0$ and $A(N) + A(P) = T'$ by pulling back with A^{-1}. *The constant*: with $a = \|A\|$, $b = \|A^{-1}\|$, take $\varepsilon' = \varepsilon/(ab+1)$; for $y = Ax$, $q = Ap$ with $|q| \le \varepsilon'|y|$, $|p| \le b|q| \le b\varepsilon' a|x| \le \varepsilon|x|$, so $x \in C \iff x + p \in C$, i.e. $y \in C' \iff y + q \in C'$ — the constant of Lemma 3.45, no finite dimensionality needed (Finding 115). Applied to A and A^{-1} this gives an equivalence `Chapter3/LinearizationCharts · isInGoodPositionIn_transport_iff`.

Two derivative tools at a level-1 point. (i) *Uniqueness*: if $f_1 = f_2$ on a relatively open $V \subseteq U_1 \cap U_2$ containing z, then $Df_1(z) = Df_2(z)$: both are derivatives within V_1 (level 1), V is an sc-domain with unique differentiability, and two operators $E_0 \to F_0$ that agree after composition with the dense inclusion $E_1 \to E_0$ are equal `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`. (ii) *The fibre derivative at a point of the zero section* `Chapter3/LinearizationCharts · scDeriv_fiber_of_vanishing`: let Φ be sc¹ on a set $D \subseteq E \oplus F$ containing $\{(u,0) : u \in S\}$ (S relatively open, $z \in S_1$) and the whole fibre $\{z\} \times F$, with $\Phi(u,0) = 0$ on S and $\Phi(z,k) = \Gamma k$ for a bounded Γ. Then $D\Phi(z,0)(h,k) = \Gamma k$. Proof: along $k_1 \mapsto (z,k_1)$, $k_1 \in F_1$, the composite is $\Gamma \circ \iota$, so $D\Phi(z,0)$ restricted to the fibre direction is Γ after composition with the inclusion, hence equal to Γ by density; along $u_1 \mapsto (u_1,0)$, $u_1 \in S_1$, the composite

vanishes, and unique differentiability of S_1 at z gives zero in the base direction. This is the generalization from 0 to z of the identities of §III.4 and §III.21; it is the reason why only the fibre part survives at a zero (Finding 112).

The linearization at a point of the retract, without a filler. For the principal part $\mathbf{f}$ of a section in a strong bundle chart with retraction $R(u,h) = (r(u), \rho(u)h)$ and a level-1 point z of $O = r(U)$, put

$$\mathrm{Lin}(z) = D(\mathbf{f}\circ r)(z)|_{T_zO}, \qquad T_zO = Dr(z)E_0$$

`Chapter3/LinearizationCharts · linAt`; $Dr(z)$ is idempotent `Chapter3/LinearizationCharts · scDeriv_r_idem_at`. Three facts make it the right object.

Its values lie in K_z = $\rho(z)F$ at a zero `Chapter3/LinearizationCharts · rho_scDeriv_comp_r` (the text's "$\rho(0)Tf(0) = Tf(0)$", now at any zero). Since $(r(u), \mathbf{f}(r(u))) \in K$, $\rho(r(u))\mathbf{f}(r(u)) = \mathbf{f}(r(u))$ on U `Chapter3/Filler · ρ_eq_self_of_mem_K`; so $\mathbf{f}\circ r$ equals $\Psi_2(u) = (R(r(u), \mathbf{f}(r(u))))_2$, and their derivatives at z agree by (i). The chain rule for $\Psi_2 = (\text{fibre part of } R)\circ(r, \mathbf{f}\circ r)$ at z, where $(r, \mathbf{f}\circ r)$ passes through $(z,0)$ (because $r(z) = z$ and $\mathbf{f}(z) = 0$), and (ii) for the fibre map $(u,k) \mapsto \rho(u)k$, which vanishes on the zero section and is $\rho(z)$ on the fibre over z, give $D(\mathbf{f}\circ r)(z)v = \rho(z)D(\mathbf{f}\circ r)(z)v$. The fibre K_z is the fibre of K over z `Chapter3/LinearizationCharts · mem_rangeS_ρCLM_iff`.

The bridge to fillers `Chapter3/LinearizationCharts · scDeriv_comp_r_eq_of_eventuallyEq` (the text's "$Tg(0) = Tf(0)$ on T_0O", now at any point): if g is sc¹ and agrees with $\mathbf{f}$ on O near z, say on $O \cap W$ with W open, then $\mathbf{f}\circ r = g\circ r$ on the relatively open $U \cap r^{-1}(W)$, so by (i) and the chain rule (r maps z to itself on level 1) $D(\mathbf{f}\circ r)(z) = Dg(z)Dr(z)$. Hence $\mathrm{Lin}(z) = Dg(z)$ on T_zO `Chapter3/LinearizationCharts · linAt_eq_scDeriv`, by idempotence of $Dr(z)$. At a centre, for a filled version, $\mathrm{Lin}(0)$ is the $Tf(0) = Dg(0)|_{T_0O}$ of Proposition 3.5 `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt` `Chapter3/FillerLinearAlgebra · fillerRestrict`. At $z \neq 0$ the full $Dg(z) : E \to F$ is not related to $\mathrm{Lin}(z)$ (condition (3) of the filler holds at 0 only), so the relevant kernel is $\ker Dg(z) \cap T_zO = \ker \mathrm{Lin}(z)$ (Finding 111). *For a filling germ g* on G the bridge needs two things the whole-chart one did not: $Dg(z)$ exists only for z on level 1 of G, and the chain rule for $g\circ r$ is the inner one on the germ $r^{-1}(G)$, which is defined when $r(0) = 0$, i.e. $0 \in O$ `Chapter3/LinearizationChartsG · scDeriv_comp_r_eq_germ`. Under these two conditions $\mathrm{Lin}(z) = Dg(z)$ on T_zO `Chapter3/LinearizationChartsG · linAt_eq_germDeriv` `Chapter3/LinearizationChartsG · StrongBundle.linAt_eq_germDeriv_chart`, and at the centre, which lies in every germ, $\mathrm{Lin}(0) = Tf(0)$ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`. Both hold wherever the bridge is used: the chart is centred at a zero, and the points $z \neq 0$ are the level-1 lifts of nearby zeros, which lie in the germ of g by the shrinking of the basic germ (§§III.32, III.48).

Fibres of strong bundle charts and of transitions. For a strong bundle chart $\Phi : P^{-1}(V) \to K$ covering φ: points of K lie over O `Chapter3/LinearizationCharts · StrongBundleChart.fst_mem_0`; $\Phi^{-1}(x,\cdot)$ is additive and homogeneous on the fibre of K over x, since Φ is a fibrewise linear bijection of $P^{-1}(\varphi^{-1}(x))$ onto that fibre `Chapter3/LinearizationCharts · StrongBundleChart.symm_add` `Chapter3/LinearizationCharts · StrongBundleChart.symm_smul`; the fibre of K is closed under scalars and contains $(u, \rho(u)k)$ `Chapter3/LinearizationCharts · StrongBundleChart.mem_K_ρ`. For a second chart Φ' the transition $\Theta = \Phi'\circ\Phi^{-1}$ covers the base transition $\tau = \varphi'\circ\varphi^{-1}$ `Chapter3/LinearizationCharts · StrongBundleChart.transition_fst`, is linear on the fibres `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_add` `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_smul`, maps the zero section to the zero section `Chapter3/LinearizationCharts · StrongBundleChart.transition_snd_zero` and K into K' `Chapter3/LinearizationCharts · StrongBundleChart.transition_mem_K`, and is inverted by the reverse transition `Chapter3/LinearizationCharts · StrongBundleChart.transition_transition`. Over a point x of the base overlap, $\Gamma(x)k = (\Theta(x, \rho(x)k))_2$ is a bounded operator $F \to F'$ — linear by the fibre linearity, continuous as a composition of $\rho(x)$, Φ^{-1} and Φ' `Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM` — which on K_x is the transition itself `Chapter3/LinearizationCharts · StrongBundleChart.transitionFiberCLM_apply_of_mem`. The text's "bounded linear on the fibres" (Definition 2.50) is about Φ only; the linearity of Φ^{-1} on the fibres of K is this lemma (Finding 113). Finally, if $\varphi(V \cap V') = O \cap W$ with W open, the overlap of K is $K \cap (W \times F)$ `Chapter3/LinearizationCharts · StrongBundleChart.overlap_eq_inter_prod`.

The chart change at a zero `Chapter3/LinearizationCharts · linAt_chart_change`. *Statement.* Let c, c' be strong bundle charts whose transition Θ is sc-smooth on $[0]$ (as a map between the overlap retracts of K and K'), f a section whose representatives $z \mapsto (z, \mathbf{f}(z))$ are sc-smooth $O \to K$ and $O' \to K'$, y a zero of f in both chart domains, and z, z' level-1 points over $\varphi(y)$, $\varphi'(y)$. Then there are topological isomorphisms $A : T_zO \cong T_{z'}O'$ and $\Gamma : K_z \cong K'_{z'}$, with $\Gamma \circ \mathrm{Lin}(z) = \mathrm{Lin}'(z') \circ A$, and $x \in C_z \iff Ax \in C'_{z'}$ for $x \in T_zO$, where C_z is the partial quadrant at $\varphi(y)$ of a presentation of C `Chapter2/Tame · partialQuadrantAtWith`. The proof, in the order of the Lean:

1. *The base transition as a germ of sc-diffeomorphisms.* The base charts are compatible (both are charts of X, `Chapter2/MPolyfold · IsChart.compatible`). Choose open W, W' with $\varphi(V \cap V') = O \cap W$ and $\varphi'(V \cap V') = O' \cap W'$ `Chapter2/MPolyfold · exists_isOpen_image_eq` and restrict the base retractions to $U \cap r^{-1}(W)$, $U' \cap r'^{-1}(W')$ (Proposition 2.10 (1)); their retracts are the overlaps. Then τ, $\tau' = \varphi \circ \varphi'^{-1}$ form a germ of sc-diffeomorphisms between them `Chapter2/Boundary · ScRetraction.IsDiffeoGerm`, and $\tau \circ r_W$ maps z to z' on level 1. The derivatives of the restricted retraction and of $\mathbf{f} \circ r$ at z are those on the full domains, by (i) (Finding 114).
2. *The transition near the zero section.* Restrict the strong bundle retraction to $W \times F$; its retract is the overlap of K, so $\Theta \circ R_W$ is sc-smooth and its fibre part Φ is sc¹. $\Phi(z,k) = \Gamma(z)k$ because $R(z,k) = (z, \rho(z)k)$; $\Phi(u,0) = 0$ because $R(u,0) = (r(u),0)$ and Θ preserves the zero section.
3. *The section in the two charts.* For u in the restricted domain, $r(u) = \varphi(y_1)$ for some $y_1 \in V \cap V'$, and both $\mathbf{f}' \circ r' \circ \tau \circ r_W$ and $\Phi \circ (r, \mathbf{f} \circ r)$ take at u the value $(\Phi'(f(y_1)))_2$ (the principal parts are the fibre parts of $\Phi \circ f \circ \varphi^{-1}$ and $\Phi' \circ f \circ \varphi'^{-1}$, `Chapter3/StrongBundleSections · StrongBundleChart.apply_section'`). By (i) their derivatives at z agree.
4. *The key identity.* The chain rule on both sides, with $(r, \mathbf{f} \circ r)$ passing through $(z,0)$ since $\mathbf{f}(z) = 0$, and (ii) for Φ give
$$D(\mathbf{f}' \circ r')(z')\, D(\tau \circ r_W)(z)v = \Gamma(z)\, D(\mathbf{f} \circ r)(z)v.$$
The derivative of $x \mapsto \Gamma(x)\mathbf{f}(z)$ in the base direction has dropped out *because* $\mathbf{f}(z) = 0$; away from zeros the linearization is not chart independent (Finding 112).
5. *The isomorphism A.* $A_0 = D(\tau \circ r_W)(z)$ maps T_zO into $T_{z'}O'$ and $A'_0 = D(\tau' \circ r'_{W'})(z')$ maps back `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_mem_range`; on these subspaces they are inverse `Chapter2/Boundary · ScRetraction.IsDiffeoGerm.deriv_inv_deriv` (with the idempotence of $Dr(z)$); A is the induced isomorphism (`restrictEquiv`).
6. *The isomorphism Γ.* $\Gamma(z)$ maps K_z into $K'_{z'}$ (the transition maps K into K' and covers τ), the fibre operator $\Gamma'(z')$ of the reverse transition maps back, and they are inverse on these fibres (the reverse transition inverts the transition).
7. *Conclusions.* For $x \in T_zO$, $\mathrm{Lin}(z)x = D(\mathbf{f} \circ r)(z)x$ and $\mathrm{Lin}'(z')(Ax) = D(\mathbf{f}' \circ r')(z')A_0x$, so step 4 is $\Gamma\,\mathrm{Lin}(z) = \mathrm{Lin}'(z')A$. The cones: Proposition 2.30 for the partial cone, applied to the germ of step 1 `Chapter2/Boundary · proposition_2_30_cone`, gives $A_0(T_zO \cap C_z) = T_{z'}O' \cap C'_{z'}$ for the formalized partial cone `Chapter2/Tame · partialConeWith`, which is the cone condition on T_zO.

Hypotheses. Of the compatibility of Definition 2.50 only the sc-smoothness of Θ on $[0]$ is used — not on $[1]$, not the reverse transition (Finding 116). For two strong bundle charts of P it holds because both are charts of the M-polyfold $Y[0]$, hence compatible `Chapter3/LinearizationCharts · StrongBundle.IsChart.transition_isScSmoothRetractMap`; and for an sc-smooth section the representatives are sc-smooth in every strong bundle chart of P, so no hypothesis is added.

Chart independence `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. For two strong bundle charts of P around a zero y of an sc-smooth section, with level-1 points z, z' over $\varphi(y)$, $\varphi'(y)$: the linearizations $L = \mathrm{Lin}(z) : T_zO \to K_z$ and $L' = \mathrm{Lin}'(z') : T_{z'}O' \to K'_{z'}$ (corestricted to the fibres by the first fact above) are simultaneously surjective, simultaneously Fredholm with the same index, and $\ker L$ is in good position to C_z inside T_zO iff $\ker L'$ is in good position to $C'_{z'}$ inside $T_{z'}O'$. The four spaces are complete, being ranges of the idempotents $Dr(z)$, $\rho(z)$ and their primed versions `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`; the rest is the conjugation algebra, with $\ker L' = A(\ker L)$ and the transport of good position.

Definition 3.53 at a zero, from a non-centred chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`. *Statement.* Let f be an sc-smooth section, y a zero with (f, y) an sc-Fredholm germ (Definition 3.11), c a strong bundle chart of P around y (not necessarily centred at y), z on level 1 over $\varphi(y)$, and g sc¹ on U agreeing with $\mathbf{f}$ on O near $\varphi(y)$ — the case in view was the chart at x_0 with a whole-chart filled version. Assume (a) the range of $Dg(z)|_{T_zO}$ contains $\rho(z)\hat{k}$ for every smooth $\hat{k} \in F_\infty$, and (b) $\ker Dg(z) \cap T_zO$ is in good position to $C_{\varphi(y)}$ inside T_zO. Then (f, y) is in good position in the whole-chart sense `Chapter3/GoodPositionGerm · StrongBundle.IsGoodPositionGermWholeChart`, in the chart c' centred at y given by Definition 3.11, with its filled version g'. *Proof.* The chart change between c at z and c' at 0 gives A, Γ with $\Gamma \,\mathrm{Lin}(z) = \mathrm{Lin}'(0)A$; the bridges give $\mathrm{Lin}(z) = Dg(z)$ on T_zO and $\mathrm{Lin}'(0) = Dg'(0)$ on T_0O' (filler condition (1)). *Surjectivity of $Tf(y) = Dg'(0)|_{T_0O'}$ onto K_0'.* By Proposition 3.52 in the centred chart c' `Chapter3/FredholmGermLinearizationG · proposition_3_52_local` it is Fredholm, so its range S is closed in the closed subspace K_0', hence closed in F'. The map $k \mapsto \Gamma(\rho(z)k)$ is continuous $F \to F'$; by (a) and the conjugation it maps F_∞ into S; it maps F onto K_0' (preimages by Γ^{-1}, which lands in K_z where $\rho(z)$ is the identity); since F_∞ is dense in F, every point of K_0' is a limit of points of S, hence in S. So only a dense subset of K_z in the range is needed (Finding 117 (b)). *The kernel.* (b) is transported by A to good position inside T_0O' with respect to $C_0' = C'$ (the partial quadrant at the corner is C' itself `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`); and the transported subspace is $\ker Dg'(0)$: $\ker Dg'(0) \subseteq T_0O'$ by condition (3) of the filler g' *at its centre* `Chapter3/GoodPositionGerm · IsFillerOf.ker_deriv_le_range`, and for $w \in T_0O'$, $Dg'(0)w = 0 \iff \Gamma(Dg(z)A^{-1}w) = 0 \iff A^{-1}w \in \ker Dg(z)$. This is where the hypothesis is on $\ker Dg(z) \cap T_zO$ and not on $\ker Dg(z)$: at $z \neq 0$ condition (3) is not available and $\ker Dg(z)$ need not lie in T_zO (Finding 117 (a)). Variants: with all of K_z in the range `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_chart`, and with the hypotheses on $\mathrm{Lin}(z)$ itself, taking $g = \mathbf{f} \circ r$, which agrees with $\mathbf{f}$ on O because r is the identity there `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_linAt` `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj`.

In the germ reading the assembly is the same, with g a germ with derivative family Dg, z on level 1 of its germ, and the bridge for germs at z and at 0; the conclusion is good position at $Dg'(0)$ for the filling germ g' at y. The dense form needs the closed range of $Dg'(0)|_{T_0O'}$, which is now a hypothesis at chart level `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart_dense`, supplied at section level by Proposition 3.52 for germs in a chart centred at y `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. The forms with all of K_z in the range need no Proposition 3.52 at all: $\mathrm{Lin}'(0) = \Gamma \,\mathrm{Lin}(z)A^{-1}$ is then onto K_0' directly `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart` (the whole-chart variant had gone through the dense one).

What remains is to supply (a) and (b) in the chart at x_0, at the zeros $z_0 = \varphi(y)$ near 0: (b) with the text's hypotheses (§III.32), (a) first under an added hypothesis (§III.33) and then with the text's hypotheses (§III.43).

III.32 Theorem 3.54 (1) at the nearby zeros: Corollary 3.41 and Lemma 3.55 on the graph, and the kernel

The text proves property (4) of Theorem 3.54 — at every zero $y \in V$, with $\psi\varphi(y) = v + \sigma(v)$, "$\ker f'(y)$ is in good position to C_yX and $f'(y)$ is surjective" — in one sentence: it follows from Corollary 3.41 and Lemma 3.55. Unpacked, three steps are needed beyond the graph description of §III.23. First, Corollary 3.41 and Lemma 3.55 have to be applied at the graph points, Lemma 3.55 on the relatively open V' of Corollary 3.47 and for a general presentation of C' (Finding 92). Second, Lemma 3.55 is about the tangent space $N_v = \{n + D\sigma(v)n\}$ of the graph, not about a kernel; the identification $N_v = \ker D(h + t)(v + \sigma(v))$ is not commented on (Finding 91 (a), 101). Third, the statements live in the push-forward chart $\psi \circ \varphi$, about $D(h + t)$ on E', while Definition 3.53 at y (via §III.31) asks for statements about $Dg(z_0)$ on $T_{z_0}O$ in the chart at x_0 (Findings 102–104). This section proves all of it for the kernel; the surjectivity of

$f'(y)$ is §III.33. No hypothesis beyond those of the graph description is used.

Lemma 3.55 for a relatively open domain and a general presentation `Chapter3/GlobalIFTFull · lemma_3_55_relOpen`. *Statement.* Let $C = \{x : \ell_i(x) \geq 0\}$ be a partial quadrant presented by an sc-isomorphism T, with the coordinate functionals $\ell_i(x) = (Tx)_i$ `Chapter3/GlobalIFTFull · coordCLMWith`, N finite-dimensional in good position to C with a good complement Y, Y **closed**, V relatively open in $N \cap C$ with $0 \in V$, and $\sigma : N \to E_0$ with values in Y, C^1 on V, $\sigma(0) = 0$, $D\sigma(0) = 0$ (within V). Then there is $\rho > 0$ such that for $v \in V$ with $|v| < \rho$ and $z = v + \sigma(v)$: $z \in C$; Y is a good complement of $N_v = \{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace` with respect to the partial quadrant C_z; and N_v is in good position to C_z. *Proof.* $D = N \cap C$ is convex with nonempty interior in N (good position), so D and $V = D \cap O$ have unique differentiability. *The extension.* The text's Lemma 3.55 (§III.22, `Chapter3/GoodPositionOpen · lemma_3_55_core`) wants a map on the whole cone. Take a ball $B(0, r) \subseteq O$ and a smooth bump b equal to 1 on $\bar{B}(0, r/2)$ and 0 outside $B(0, r)$, and $\Phi(v) = b(v)v$. Φ maps D into V: D is closed under nonnegative scaling, and $|b(v)v| < r$. So $\tau = \sigma \circ \Phi$ is C^1 on the cone, agrees with σ on $B(0, r/2)$, and there its derivatives within the cone are those of σ within V (a neighbourhood of the point inside O does not change derivatives within a set); in particular $\tau(0) = 0$, $D\tau(0) = 0$. *Lemma 3.55 in coordinates* for τ, with the good-complement estimate of (N, Y) rewritten in the ℓ_i and the joint surjectivity of the coordinates ($T^{-1}(k, 0)$ has coordinates k), gives ρ_1, the membership $z \in C$, and the good complement Y of N_v with respect to C_z. *The interior*, which Lemma 3.55 does not state but good position requires: by the continuity of $D\sigma$ at 0 (a C^1 map on a set of unique differentiability) there is ρ_2 with $\|D\sigma(v)\| < \varepsilon$, the good-complement constant of (N, Y). $L = D\sigma(v)$ takes values in Y — this is where Y closed is used `Chapter3/GoodPositionOpen · fderivWithin_mem_of_mapsTo` — so $n \mapsto n + Ln$ is injective ($N \cap Y = 0$), a linear homeomorphism $N \cong N_v$ in finite dimensions. It carries the interior of D to an open subset of N_v contained in C_z: for w in the interior of D, $w \in C$ and $|Lw| \leq \varepsilon|w|$, so $w + Lw \in C$ by the good-complement property, and $C \subseteq C_z$. Take $\rho = \min(\rho_1, r/2, \rho_2)$.

Corollary 3.41 at the graph points `Chapter3/GlobalIFTFull · StabilityData.exists_surjective_Dfs'_graph`. For the data of Theorem 3.38 with $D(h + s)(0)$ surjective and a continuous σ on $V \ni 0$, $\sigma(0) = 0$, whose graph points $v + \sigma(v)$ are zeros in the germ: there is $\rho > 0$ such that for $v \in V$, $|v| < \rho$, the zero $v + \sigma(v)$ lies on level 1, at a point x_1 of the level-1 germ, and $D(h + s)(x_1) : E_0 \to F_0$ is a surjective Fredholm operator of index $\dim P - \dim Q$. Corollary 3.41 `Chapter3/SolutionLinearization · StabilityData.corollary_3_41` gives a radius r for the zeros, and continuity of $v \mapsto v + \sigma(v)$ at 0 gives ρ.

Property (4) in the push-forward chart `Chapter3/GlobalIFTFullG · theorem_3_54_nearby`. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description (§III.23, `Chapter3/GlobalIFTG · theorem_3_54_graph`: chart, ψ, $h + t$, K, closed Y, V', σ, V_x), a presentation T of $C' = \widetilde{C} \oplus W$, and an open $V_{x,1} \subseteq V_x$ containing x_0 such that for every zero $y \in V_{x,1}$, with $\psi\varphi(y) = v + \sigma(v)$, $v \in V'$: the point lies on level 1 and $D(h + t)$ there is surjective Fredholm of index $n - N$; $v + \sigma(v) \in C'$; Y is a good complement of N_v with respect to $C'_{v+\sigma(v)}$, and N_v is in good position to it. Only f a section is assumed, not an sc-Fredholm section. *Proof.* Take the radii ρ_1 (Corollary 3.41 at the graph points) and ρ_2 (Lemma 3.55, relatively open form), $\rho = \min(\rho_1, \rho_2)$, and the projection π onto K along Y, continuous because Y is closed and K finite-dimensional. Put $V_{x,1} = V_x \cap (\psi\varphi)^{-1}B(0, \rho/(\|\pi\| + 1))$, open since $\psi \circ \varphi$ is continuous on V_x. For a zero $y \in V_{x,1}$, $\psi\varphi(y) = v + \sigma(v)$ by the graph description, $v = \pi(\psi\varphi(y))$ because $\sigma(v) \in Y$, so $|v| \leq \|\pi\|\,|\psi\varphi(y)| < \rho$, and both lemmas apply at v.

Derivative identities at a nearby zero. Four lemmas move these statements from E' and N_v to E and $\ker Dg(z_0)$.

Uniqueness at a point of C_1 `Chapter3/GlobalIFTNearbyGraph · scDeriv_eq_of_eqOn_nhds_at`: two maps with sc¹-derivatives D_1, D_2 at $x \in C_1$ that agree on $W_1 \cap C_1$ for an open $W_1 \ni x$ of E_1 have $D_1 = D_2$ — uniqueness of the derivative within the quadrant on level 1, then density of E_1. The text writes $\Psi_*(g) = h + t$ as an identity; formally they agree only on $U' \cap W'$, and this lemma identifies their derivatives at every nearby point (Finding 104).

The fibre derivative of a strong bundle map at $(u_0, 0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_at`: for a strong bundle map Ψ with fibre operators $\Gamma(u) = (\Psi(u,\cdot))_2$ `Chapter3/FredholmGermLinearization · IsStrongBundleMap.fiberCLM` and $u_0 \in U_1$, $D(\mathrm{pr}_{F'} \circ \Psi)(u_0,0)(h,k) = \Gamma(u_0)k$; hence the derivative of $u \mapsto \Gamma(u)k(u)$ at a zero u_0 of k is $\Gamma(u_0)Dk(u_0)$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleMap.scDeriv_fiber_comp_at` — §III.21's identity at 0, moved to u_0.

The chain rule for the push-forward at a zero `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_pushforward_at`: for a strong bundle isomorphism Ψ covering ψ with inverse Ψ' covering ψ', and a zero $u_0 \in U_1$ of the sc-smooth g, $\Psi_*(g) = \kappa \circ \psi'$ with $\kappa(u) = (\Psi(u, g(u)))_2$, so

$$D(\Psi_* g)(\psi(u_0)) = \Gamma(u_0) \circ Dg(u_0) \circ D\psi'(\psi(u_0)).$$

$D\psi'(\psi(u_0))$ and $D\psi(u_0)$ are inverse on level 0 `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.scDeriv_base_inv_base`, and $\Gamma(u_0)$ is injective because Ψ' inverts Ψ on the fibres `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.fiberCLM_injective`; hence $\ker Dg(u_0) = D\psi'(\psi(u_0))(\ker D(\Psi_* g)(\psi(u_0)))$ `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.ker_scDeriv_eq_map`. *For a germ* g with derivative family Dg the formula is an identity of the explicit derivative family of $\Psi_*(g)$ (§III.23), valid at every level-1 zero u_0 and for every family `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.pushforwardDeriv_at`, and the kernel formula follows as before `Chapter3/GlobalIFTNearbyGraphG · IsStrongBundleIso.ker_deriv_eq_map_germ`; that g is sc¹ at u_0 then means that u_0 lies on level 1 of the germ of g.

Corresponding partial quadrants `Chapter3/GlobalIFTNearbyGraph · scDeriv_mem_partialQuadrantAtWith_iff`: for an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse ψ' and $z \in U_1'$, $x \in C_z' \iff D\psi'(z)x \in C_{\psi'(z)}$: the one-sided sign argument `Chapter2/Boundary · IsSc1With.mapsTo_partialQuadrant` for ψ' at z gives one direction, for ψ at $\psi'(z)$ the other, and $D\psi(\psi'(z))D\psi'(z) = 1$. At the centre this was "$C' = T\psi(0)C$" (§III.23).

Tangent vectors of a family of points of a retract `Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed`: let ρ be sc¹ on U and $\Theta_1 : N \to E_1$ differentiable at v within V (unique differentiability at v), with values in U_1 and $\rho(\iota\Theta_1(w)) = \iota\Theta_1(w)$ for $w \in V$. Then $D\rho(\Theta_1(v))$ fixes $\iota(D\Theta_1(v)n)$ for every n: differentiate $\rho \circ \iota \circ \Theta_1 = \iota \circ \Theta_1$ at v. For ρ a retraction onto O: the tangent vectors of a C^1 family of points of O lie in $T_{z_0}O$.

$N_v = \ker D(h+s)(v+\sigma(v))$ (Finding 101). *The inclusion* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_le_ker_Dfs'`. For the solution graph σ of Theorem 3.47 on V, a C^1-lift $\sigma_1 : V_1 \to E_1$ of σ on a relatively open $V_1 \subseteq V$ (Theorem 3.47 (1), §III.20), $v \in V_1$ and the level-1 point x_1 over $v+\sigma(v)$ in the germ: $N_v \subseteq \ker D(h+s)(x_1)$. *Proof.* $h+s$ is differentiable on level 1 only, so the graph is lifted: the kernel K is a smooth subspace `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`, hence lifts linearly to E_1 `Chapter3/BasicClass · finLift`, and $\Gamma_1 = \mathrm{lift} + \sigma_1$ is a C^1 map $V_1 \to E_1$ with $\iota\Gamma_1(w) = w + \sigma(w)$ and $\Gamma_1(v) = x_1$. $V_1 = V \cap O_1$ `Chapter3/GlobalIFTNearbyGraph · relOpen_cone_inter` has unique differentiability at its points `Chapter3/GlobalIFTNearbyGraph · uniqueDiffWithinAt_relOpen_cone`, and $D\sigma(v) = \iota \circ D\sigma_1(v)$ `Chapter3/GlobalIFTNearbyGraph · fderivWithin_eq_toBase_comp`. The composite $(h+s) \circ \iota \circ \Gamma_1$ vanishes on V_1, and near v (on $V_1 \cap \Gamma_1^{-1}$ of the open set defining the level-1 germ) Γ_1 maps into the level-1 germ, where $h+s$ has the derivative $D(h+s)(x_1)$ composed with ι `Chapter3/SolutionLinearization · StabilityData.isSc1GermWith_Dfs'`. The chain rule and uniqueness give $D(h+s)(x_1)(n + D\sigma(v)n) = 0$ for $n \in K$. *Equality* `Chapter3/GlobalIFTNearbyGraph · StabilityData.graphSpace_eq_ker_Dfs'` under the conclusions of Corollary 3.41 at x_1 (Fredholm, index $\dim P - \dim Q$, surjective) and with the closed good complement Y: $D\sigma(v)$ takes values in Y, so $\dim N_v = \dim K$ `Chapter3/GlobalIFTNearbyGraph · finrank_graphSpace_of_compl`; for a surjective Fredholm operator $\dim\ker = \mathrm{ind}$ `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`, so $\dim K = \mathrm{ind}\, D(h+s)(0) = \dim P - \dim Q = \mathrm{ind}\, D(h+s)(x_1) = \dim\ker D(h+s)(x_1)$, and an inclusion of finite-dimensional spaces of the same dimension is an equality.

Good position restricts to a subspace containing the kernel `Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le`: if N is in good position to C in E_0 and $N \subseteq T$, a good complement P restricts to the good complement $P \cap T$ of N in T, with the same constant ($n + p \in T$ and $n \in T$ give $p \in T$).

The graph description with the filler data exposed `Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph`. This is the body of the graph description (§III.23) stated in the chart, with the filling germ g of $\mathbf{f}$ on G_g and a derivative family Dg, the sc$^+$-germ t_0 on G_s, the conjugation Ψ (base ψ, inverse ψ'), the basic germ on a germ G that ψ' maps into $G_g \cap G_s$ level by level ($G \subseteq G''$, §III.48), and the tameness consequence $(1 - Dr(0))E \subseteq E_0$ as hypotheses, so that the later steps can use g at the points z_0. Its conclusions: the data $\mathcal{S}$ of Theorem 3.38 on G itself and the basic germ $h' = h + (t_{\mathbb{R}^N}, 0)$ with $\mathcal{S}.\mathrm{fsum}\, h' = \Psi_*(g)$ on $U' \cap W'$; $D(h+t)(0)$ surjective with kernel K in good position; Y (closed), V', σ with its C^k-lifts; an open $N_1 \ni 0$ on which the filler identities (1), (2) and the conjugation identity hold, with $\psi'(U' \cap W') \subseteq N_1$; and an open $N_2 \subseteq N_1$ with $\psi(U \cap N_2)$ inside the germ and $U' \cap W'$, every zero of g in $U \cap N_2$ in O, $g(u) = 0 \iff (h+t)(\psi(u)) = 0$ on $O \cap N_2$, and $\psi(\{u \in O \cap N_2 : g(u) = 0\}) = \{v + \sigma(v) : v \in V'\}$. The proof is that of §III.23 (items 1–7), in the chart; $D(h+t)(0) = D(\Psi_* g)(0)$ for the explicit family is part of the conclusion.

The kernel at the nearby zeros, in the chart at x_0 `Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel`. *Statement.* In the data of the preceding theorem there is an open $N_3 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_3$ of g there are $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point $z_{0,1}$ over z_0, **on level** 1 **of the germ** G_g **of the filling**, and $x_1 = \psi_1(z_{0,1})$ in the level-1 germ over $v + \sigma(v)$, with ($Dg(z_{0,1})$ the value of the derivative family): (i) $D(h+t)(x_1)$ surjective Fredholm of index $n - N$; (a) $N_v = \ker D(h+t)(x_1)$; (b) $D(\Psi_* g)(x_1) = D(h+t)(x_1)$ and $\ker Dg(z_{0,1}) = D\psi'(x_1)(\ker D(h+t)(x_1))$, and $Dg(z_{0,1})$ is surjective; (c) $\ker Dg(z_{0,1}) \subseteq T_{z_0} O = Dr(z_{0,1})E$; (d) $\ker Dg(z_{0,1})$ is in good position to C_{z_0} inside $T_{z_0} O$. *Proof.*

1. *The neighbourhood.* Take the C^1-lift σ_1 on $V_1 = V' \cap O_1$ (Theorem 3.47 (1) with $k = 1$), the radii ρ_1 (Corollary 3.41 at the graph points), ρ_2 (Lemma 3.55 relatively open, for the presentation $C' = \widetilde{C} \oplus W$), ρ_3 with $B(0, \rho_3) \subseteq O_1$, and $\rho = \min(\rho_1, \rho_2, \rho_3)$. With the projection π onto K along Y, let $N_3 = N_2 \cap O_3$ where $O_3 \cap U = \psi^{-1}(B(0, \rho/(\|\pi\| + 1))) \cap U$ (continuity of ψ on U). For a zero $z_0 \in O \cap N_3$ of g, $\psi(z_0) = v + \sigma(v)$ with $v \in V'$, and as in `theorem_3_54_nearby` $|v| < \rho$, so $v \in V_1$ and both lemmas apply.
2. *(i), and the level-*1 *points.* Corollary 3.41 at v gives x_1 on level 1 of $\mathcal{S}.G = G \subseteq G''$ (§III.48, `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_41_shrink`); x_1 lies in U_1', so $z_{0,1} = \psi_1'(x_1)$ is a level-1 point over $\psi'(\psi(z_0)) = z_0$, and $\psi_1(z_{0,1}) = x_1$ `Chapter2/DegeneracyIndex · levelMap_comp_eq_self`. Since ψ_1' maps G_1'' into $(G_g)_1$, $z_{0,1}$ lies in $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`: this is the step that needs the shrinking of the basic germ, since g is differentiable only there. (With g sc-smooth on U, any level-1 point would do.)
3. *(a)* is $N_v = \ker D(h+t)(x_1)$ above, fed with the lift and (i).
4. $D(\Psi_* g)(x_1) = D(h+t)(x_1)$: the two maps agree on $U' \cap W'$, so the uniqueness lemma at x_1 applies on the open set of E_1' cut out by $W' \cap U_o$ (with $U' = U_o \cap C'$) and the open set defining the level-1 germ G_1; $\Psi_*(g)$ is an sc^1-germ on the preimage of G_g, which contains G `Chapter3/GermPushforward · IsStrongBundleIso.pushforward_isSc1GermWith` (whole-chart: sc-smooth on U' `Chapter3/GoodPositionGermG · IsStrongBundleIso.pushforward_linearization`).
5. *(b)*: the kernel formula is the chain rule for the push-forward at the zero $z_{0,1}$ of g, in the germ form above. *Surjectivity of* $Dg(z_{0,1})$: given k, solve $D(h+t)(x_1)w = \Gamma(z_0)k$ by (i); then $\Gamma(z_0)Dg(z_{0,1})D\psi'(x_1)w = \Gamma(z_0)k$, and $\Gamma(z_0)$ is injective. This is the surjectivity of the *filled* $Dg(z_0) : E \to F$, not of $Tf(z_0) = Dg(z_0)|_{T_{z_0}O} : T_{z_0}O \to K_{z_0}$ (§III.33).
6. *(c), the tangency to* O (Finding 102). Condition (3) of Definition 3.4, which gave $\ker Dg(0) \subseteq T_0 O$ at the centre, is not available at z_0. Instead: the transported retraction $r' = \psi \circ r \circ \psi'$ on U' `Chapter2/TameInvariance · IsScDiffeoQuadrant.transport` fixes every graph point, because each $w + \sigma(w)$, $w \in V'$, is $\psi(u)$ for a zero $u \in O$ of g (the graph description), and $r(u) = u$. The lifted graph map $\Gamma_1 = \mathrm{lift}{+}\sigma_1$ maps V_1 into U_1', so the tangent-vector lemma for r' gives $Dr'(x_1)$ fixing $n + D\sigma(v)n$ for every $n \in K$, i.e. $Dr'(x_1)$ is the identity on N_v. The derivative of the transported retraction is $Dr'(x_1) = D\psi(z_{0,1})Dr(z_{0,1})D\psi'(x_1)$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.scDeriv_transport`

(§III.27); applying $D\psi'(x_1)$ and using that it inverts $D\psi(z_{0,1})$, $Dr(z_{0,1})$ fixes $D\psi'(x_1)m$ for every $m \in N_v$. By (a) and (b), these are exactly the elements of $\ker Dg(z_{0,1})$, so $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$.

7. *(d)* (Finding 103). Lemma 3.55 gives N_v in good position to $C'_{v+\sigma(v)}$ in E'. The isomorphism $S = D\psi'(x_1) : E'_0 \cong E_0$ (inverse $D\psi(z_{0,1})$, `Chapter3/GlobalIFTNearbyGraph · IsStrongBundleIso.derivInvEquiv`) carries $C'_{v+\sigma(v)}$ onto C_{z_0} by the corresponding-quadrants lemma, so Lemma 3.45 `Chapter3/GoodPosition · IsInGoodPosition.map` gives $S(N_v) = \ker Dg(z_{0,1})$ (by (a), (b)) in good position to C_{z_0} in E; by (c) and the restriction lemma it is in good position inside $T_{z_0}O$.

Theorem 3.54 (1) at the nearby zeros, the kernel, for the section `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_nearby_kernel`. *Statement.* For a section f of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position: the data of the graph description, with the chart c centred at x_0, its filling germ g with a derivative family and the sc⁺-germ t_0 exposed (and K smooth, Y closed, the lifts), and an open $V_{x,2} \subseteq V_x$ containing x_0, such that for every zero $y \in V_{x,2}$: $\mathbf{f} = g$ on O near $\varphi(y)$, $g(\varphi(y)) = 0$, and with $z_0 = \varphi(y)$ the conclusions (i), (a)–(d) of the chart statement hold, with $z_{0,1}$ on level 1 of the germ of g. *Proof.* Tameness gives $(1-Dr(0))E \subseteq E_0$ (Proposition 2.35, `Chapter2/TameRetracts · proposition_2_35`, at the smooth corner point 0 of the tame base retraction; §III.23). The conjugation data are first shrunk to G'' (§III.48), so that the chart statements apply; they give N_1, N_2, N_3, with $\psi(U \cap N_2) \subseteq G''_0$; put $V_x = \varphi^{-1}(N_2)$ and $V_{x,2} = \varphi^{-1}(N_3)$ within the chart domain, both open by continuity of φ. On V_x, $\mathbf{f}(\varphi(y)) = g(\varphi(y))$ (filler (1) on N_1), so $f(y) = 0 \iff g(\varphi(y)) = 0 \iff (h+t)(\psi\varphi(y)) = 0$ `Chapter3/StrongBundleSections · StrongBundleChart.isZeroAt_iff`; $\psi \circ \varphi$ is injective on V_x; and the image of the zero set is the graph. For a zero $y \in V_{x,2}$, $\varphi(y) \in O \cap N_3$ is a zero of g, and the kernel statement applies. The hypotheses are those of the graph description: **the kernel part of Theorem 3.54 (1) needs no added hypothesis**, and of f only that it is a section (the regularity enters through the filling germ g, at 0 and at the lifts $z_{0,1}$ of the nearby zeros, which lie in its germ).

III.33 Theorem 3.54 (1) under the filler condition at the nearby zeros, and the λ-extension

At a zero y near x_0, with $z_0 = \varphi(y)$ in the chart at x_0, §III.32 gives the kernel condition (b) of the assembly step of §III.31 and the surjectivity of the *filled* $Dg(z_0) : E \to F$. Condition (a) asks for more: that $Tf(z_0) = Dg(z_0)|_{T_{z_0}O}$ maps onto $K_{z_0} = \rho(z_0)F$ (or onto a dense part of it). The text obtains it from Corollary 3.41. This section shows exactly what is missing, proves Theorem 3.54 (1) under the **added hypothesis** that closes the gap, and records the formalized part of an attempt to close it without that hypothesis (Findings 100, 105–107).

The block form at a zero. With respect to $E = T_{z_0}O \oplus \ker Dr(z_0)$ and $F = K_{z_0} \oplus \ker \rho(z_0)$ ($Dr(z_0)$ and $\rho(z_0)$ are idempotent),

$$Dg(z_0) = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}, \qquad A = Tf(z_0), \quad C = (1-\rho(z_0))Dg(z_0)|_{\ker Dr(z_0)} : \ker Dr(z_0) \to \ker \rho(z_0).$$

The lower left block vanishes at a zero: $Dg(z_0)Dr(z_0) = D(\mathbf{f} \circ r)(z_0)$ (the bridge, §III.31) takes values in K_{z_0} (the values of the linearization at a zero, §III.31). If $Dg(z_0)$ is onto, C is onto, and (Finding 100) $b \mapsto [Bb]$ induces an isomorphism $\ker C \cong \operatorname{coker} A$: so $Tf(z_0)$ **is onto if and only if** C **is injective** — condition (3) of Definition 3.4 at z_0. Definition 3.4 imposes it at 0 only, and it does not propagate: $u \mapsto Dg(u)$ is continuous for the strong operator topology, not in norm, and a strongly continuous family of surjections may pass from an isomorphism to a non-injective operator. In index terms: Corollary 3.41 gives $\operatorname{ind} Dg(z_0) = \dim K$ at all nearby zeros, $\operatorname{ind} Dg(z_0) = \operatorname{ind} A + \operatorname{ind} C = \dim \ker A$ when C is onto, and $\operatorname{ind}(f, y) = \operatorname{ind} A = \dim K - \dim \operatorname{coker} A$; so $\operatorname{coker} Tf(z_0) = 0$ is the local constancy $\operatorname{ind}(f, y) = \operatorname{ind}(f, x_0)$ along the solution set — the index stability of Proposition 6.2 of the text, whose proof uses the same step ("F' surjective" for the intrinsic linearization, from the implicit function theorem for the filled

one). The text's "(4) follows from Corollary 3.41 and Lemma 3.55" therefore has a gap for the surjectivity of $f'(y)$; the same sentence stands in the book's arXiv version (Theorem 3.13 there, with Definition 3.13, the filling, still imposing (3) at 0 only; Finding 157). Formalized of this paragraph is the direction used:

Proposition 3.5 (1) at a zero of a filler `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. *Statement.* Let U be relatively open in a partial quadrant, R a strong bundle retraction on $U \lhd F$, $\mathbf{f}$ with $\mathbf{f}\circ r$ sc^1 and $(x, \mathbf{f}(x)) \in K$ on O, g sc^1, z on level 1 with $\iota z \in O$, $\mathbf{f}(\iota z) = 0$, $\mathbf{f} = g$ on O near ιz. If $Dg(z)$ is surjective and $(1 - \rho(z))Dg(z)$ is injective on $\ker Dr(z)$ (in the form: $Dr(z)b = 0$ and $\rho(z)Dg(z)b = Dg(z)b$ imply $b = 0$), then for every $k \in F$, $\rho(z)k = Dg(z)x$ for some $x \in T_zO$. For a germ g, with $Dg(z)$ the value of a derivative family at a level-1 point of its germ, the proof is the same with the bridge for germs (§III.31) `Chapter3/GlobalIFTNearbySurjG · exists_range_deriv_eq_ρ_of_normal_injective_germ`. *Proof.* Write $Dg(z)e = \rho(z)k$, $a = Dr(z)e$, $b = e - a \in \ker Dr(z)$ (idempotence). $Dg(z)a = D(\mathbf{f}\circ r)(z)e$ lies in K_z (bridge and values at a zero), so $\rho(z)Dg(z)a = Dg(z)a$; then $\rho(z)Dg(z)b = \rho(z)(\rho(z)k - Dg(z)a) = \rho(z)k - Dg(z)a = Dg(z)b$ ($\rho(z)$ is linear and idempotent on the fibre). So $b = 0$, $e = a \in T_zO$. The converse direction and the index identity above are not formal statements; they explain why the hypothesis is the right one.

The added hypothesis `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. (f, x_0) is in good position *with a filler satisfying condition (3) at the nearby zeros* if x_0 is smooth and there are the data of Definition 3.53 in the chart form (a strong bundle chart centred at x_0, tame since §III.47, a filling germ g of $\mathbf{f}$ with a derivative family Dg on a germ G inside U, a local sc$^+$-germ s with $s(0) = g(0)$ and $g - s$ conjugated to a basic germ, the germ in good position at $Dg(0)$) together with an open $N \ni 0$ such that at every level-1 point z_1 *of* G with $\iota z_1 \in O \cap N$ and $g(\iota z_1) = 0$, $(1 - \rho(\iota z_1))Dg(z_1)$ is injective on $\ker Dr(z_1)$. The condition is imposed only where Dg is defined, which is where the lifts of the nearby zeros lie (§III.32); the whole-chart form imposed it at every level-1 point over $O \cap N$ and implies it `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilledWholeChart.isGoodPositionGermFilled`. It implies Definition 3.53 `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm` (forget N). According to the docstring, the fillers of the applications satisfy (3) at every point; Definition 3.4 as stated does not ask for it. It is recorded as an added hypothesis in the docstrings and in the Findings (items 100, 153).

Theorem 3.54 (1) `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with a filler satisfying condition (3) at the nearby zeros. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position (Definition 3.53, `IsGoodPositionGerm`) at every zero $y \in V_x$. *Proof.*

1. In the chart c at x_0 with the filling germ g: $(1 - Dr(0))E \subseteq E_0$ from tameness (Proposition 2.35); the germ G of the condition is intersected with the germ of g, and the basic germ is shrunk to the points that ψ' maps into this intersection and into the germ of s (§III.48); then the graph description in the chart and the kernel statement of §III.32 give N_1, N_2, N_3. Put $V_x = \varphi^{-1}(N_3 \cap N)$ in the chart domain, with N the neighbourhood of the added hypothesis.
2. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O \cap N_3$, $\mathbf{f}(z_0) = 0$, and $g(z_0) = \mathbf{f}(z_0) = 0$ by filler (1) on N_1; $\mathbf{f} = g$ on O near z_0. The kernel statement gives the level-1 point $z_{0,1}$, on level 1 of G, $Dg(z_{0,1})$ surjective, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$, and $\ker Dg(z_{0,1})$ in good position to C_{z_0} inside $T_{z_0}O$.
3. y is a smooth point, because f is regularizing `Chapter3/TameGermsG · IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ (Definition 3.14); and $\mathbf{f}\circ r$ is sc^1 for an sc-smooth section `Chapter3/GlobalIFTNearbySurj · StrongBundle.IsScSmoothSection.isSc1_principal_comp_r`.
4. Apply the assembly step of §III.31 in the chart c at $z_{0,1}$ with g, in its germ form, the closed range in the tame chart at y coming from Proposition 3.52 for germs `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense`. *Condition (a)*: the added hypothesis at $z_{0,1}$ (with $\iota z_{0,1} = z_0 \in O \cap N$, $g(z_0) = 0$) is the injectivity of C, and Proposition 3.5 (1) at z_0 turns the surjectivity of $Dg(z_{0,1})$ into $\rho(z_0)k \in Dg(z_{0,1})(T_{z_0}O)$ for every k, smooth or not. *Condition (b)*: $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$ by the tangency (c), and the good position is (d).

So Theorem 3.54 (1) holds under condition (3) of the filler at the nearby zeros; the kernel half holds with the text's hypotheses (§III.32). When this section was written, **the surjectivity of $f'(y)$ at the nearby zeros with Definition 3.4 as stated was open in the formalization**; it is proved in §§III.41–III.43 by another argument, and the condition (3) at the nearby zeros turns out to be a consequence of good position (§III.45). The global theorems are not affected: they assume good position at every zero, which supplies at each x_0 the nearby good position that the regularity argument needs (§III.36–§III.37, Finding 156).

The λ-extension: what the perturbation argument of Proposition 6.2 gives. The text's device for index stability (Proposition 6.2) is to add finitely many parameters λ multiplying sc⁺-sections, so that the extended problem is surjective, and to apply the implicit function theorem to it. In the chart at x_0 it can be set up without new bundle isomorphisms: for a finite-dimensional $\Lambda_0 \subseteq F_\infty$, the sc⁺-germ $\tilde{e}(\lambda, u) = \rho(r(u))(\lambda_1(\pi_K \psi(u)) + \hat{\jmath}(\lambda_2))$ (Finding 105). Two points are not addressed by the text, and the first is formalized.

The obstruction at the corner `Chapter3/GlobalIFTNearbyPerturb · mem_lineality_of_goodComplement_prod`. Theorem 3.47 for the extended problem needs its kernel in good position to the extended partial quadrant at the corner. Let $C \subseteq V$ contain 0 and be closed under nonnegative scaling, $N \subseteq V$ a subspace, and $P \subseteq \mathbb{R} \times V$ a subspace with the ε-condition of Definition 3.21 for $\mathbb{R} \times N$ `Chapter3/GlobalIFTNearbyPerturb · prodSub` with respect to $\mathbb{R} \times C$ `Chapter3/GlobalIFTNearbyPerturb · prodCone`. Then every $p = (\mu, y) \in P$ has $y \in C$ and $-y \in C$. *Proof.* $n = (1, 0)$ lies in $\mathbb{R} \times N$ and in $\mathbb{R} \times C$, $|n| = 1$; for $t = \pm\varepsilon/|p|$, $|tp| \leq \varepsilon|n|$, so $n + tp \in \mathbb{R} \times C$, i.e. $ty \in C$, and scaling by $1/|t|$ gives $\pm y \in C$. Hence a good complement projects into the lineality space $C \cap (-C)$, and $\mathbb{R} \times N$ is in good position only if every $v \in V$ is $n + y$ with $n \in N$, $y \in C \cap (-C)$ `Chapter3/GlobalIFTNearbyPerturb · sup_lineality_eq_top_of_goodComplement_prod`. For the naive extension (parameters multiplying smooth vectors with a factor vanishing at 0) the kernel is $\mathbb{R} \times K$, so good position fails unless K maps onto all boundary coordinates — the text's "has a complement contained in T^R" is exactly the statement that the complement lies in the lineality space.

The fix: a tilted kernel `Chapter3/GlobalIFTNearbyPerturb · isInGoodPosition_extendedKernel`. In $(\Lambda \times P_n) \oplus W$ with the partial quadrant $(\Lambda \times \widetilde{C}) \oplus W$ (the parameters λ unconstrained), let K, Y' be subspaces of $P_n \oplus W$ and $j : \Lambda \to P_n \oplus W$ linear with $K \cap j(\Lambda) = 0$, $(K + j(\Lambda)) \oplus Y' = P_n \oplus W$, Y' in the lineality space of $\widetilde{C} \oplus W$ ($v + y \in \widetilde{C} \oplus W \iff v \in \widetilde{C} \oplus W$ for $y \in Y'$), and $K + j(\Lambda)$ containing an interior point of $\widetilde{C} \oplus W$. Then the kernel $\widetilde{N} = \{((\lambda, a), w) : (a, w) + j(\lambda) \in K\}$ of $((\lambda, a), w) \mapsto D(h + t)(0)((a, w) + j(\lambda))$ `Chapter3/GlobalIFTNearbyPerturb · extendedKernel Chapter3/GlobalIFTNearbyPerturb · extProj Chapter3/GlobalIFTNearbyPerturb · extParam` is in good position. *Proof.* Choose a complement Λ_c of $\ker j$; every element of $j(\Lambda)$ is $j(\mu)$ with $\mu \in \Lambda_c$. *Interior*: if $k_0 + j(\mu_0)$ is interior, $n_0 = ((-\mu_0, k_0 + j(\mu_0)))$ lies in $\widetilde{N}$ and its $(P_n \oplus W)$-part is interior, an open condition. *The complement* $\Lambda_c \times Y'$ `Chapter3/GlobalIFTNearbyPerturb · extendedCompl`: an element of both has $(a, w) \in (K + j(\Lambda)) \cap Y' = 0$, then $j(\lambda) \in K \cap j(\Lambda) = 0$, $\lambda \in \ker j \cap \Lambda_c = 0$; and every $x = ((\lambda, a), w)$ splits by decomposing $(a, w) + j(\lambda) = k + j(\mu) + y$. *The constant*: since Y' lies in the lineality space, adding an element of $\Lambda_c \times Y'$ never changes membership in the quadrant, so the ε-condition holds for every ε (the proof takes $\varepsilon = 1$). The directions $j(\Lambda) \subseteq T_0(\psi(O)) \cap E'_\infty$ with $K + j(\Lambda) + E'_0 = E'$ exist by tameness ($T_0 O + E_0 = E$) and would be realized by the sc⁺-perturbations $\rho(r(u)) Dg(0) D\psi^{-1}(0) j(\lambda)$.

Why the argument does not close the gap (Findings 105–106). Carried through (not formalized: the extended basic germ and its sc⁺-perturbation, its derivative at the zeros, Theorem 3.47 for it, and the tangency argument), the argument gives for each finite-dimensional $\Lambda_0 \subseteq F_\infty$ a neighbourhood $V(\Lambda_0)$ of x_0 with $\rho(z_0)\Lambda_0 \subseteq \operatorname{range} Tf(z_0)$ at every zero in $V(\Lambda_0)$. The assembly step at a fixed zero y needs $\rho(z_0)\hat{k}$ in the range for *all* smooth $\hat{k}$, and $V(\Lambda_0)$ comes from Theorem 3.47 for the Λ_0-extended problem with no control in Λ_0: the radii of Theorems 3.40, 3.43, 3.47 would have to be uniform in the finite-dimensional parameter space, which the statements do not provide; $\operatorname{coker} Tf(z_0)$ is finite-dimensional and spanned by smooth vectors, but the spanning set depends on y. The formalization of this route was stopped there; §§III.41–III.43 take another one.

III.34 Theorem 3.54 (2) at the centre: the data of Definition 3.25 for the solution set at x_0

The text ends the proof of Theorem 3.54 with $\psi \circ \varphi(\{y \in V : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$ and concludes that the solution set has the d-dimensional tangent germ property, $d = \dim\ker f'(x)$, so that Theorem 3.27 applies. Definition 3.25 asks for much more than the graph identity (§III.24): a chart of X centred at the point onto a *tame* retract, a *smooth* finite-dimensional subspace N in good position with a *closed* good complement Y, a relatively open $V \ni 0$ in $N \cap C$, and $\delta : V \to Y$ continuous with the graph identity, $\delta(0) = 0$, $D\delta(0) = 0$ within $N \cap C$, and the sc$^\infty$-germ property in the k-dependent reading `Chapter3/TangentGerm · MPolyfold.TangentGermChart` `Chapter3/TangentGerm · IsScInftyGermAt`. This section assembles all of it at the centre x_0 from the graph description of §III.29, adding the three facts the text leaves silent: the chart is $\psi \circ \varphi$ and it is a chart of X; its model $\psi(O)$ is a tame retract; and $K = \ker D(h + t)(0)$ consists of smooth points. With the projecting complement of §III.29 the data also satisfy condition (b) of the regularity conditions. The data at the other zeros and condition (a) come in §§III.36–III.37.

The sc$^\infty$-germ property from the level-wise lifts. Corollary 3.47 (1) gives, for every $k \geq 1$, a relatively open $V_k \ni 0$ in the cone and a C^k map $\sigma_k : V_k \to E_k$ with $\iota_k \sigma_k = \sigma$; Definition 3.25 (ii) asks, for every level m and every k, for a relatively open $V_{m,k} \ni 0$ on which σ takes values in E_m and is C^k into E_m. Given m and k, take $j = \max(m, k) + 1$, $V_{m,k} = V_j$, and compose σ_j with the inclusion $E_j \to E_m$, a continuous linear map compatible with the inclusions into E_0 `Chapter3/SolutionGermRegularity · exists_clm_toBase_eq` (an iterated composition of the level inclusions): it is a lift of σ into E_m, of class C^j, hence C^k, and it coincides with the level map of σ on V_j by the injectivity of the inclusions `Chapter3/SolutionGermRegularity · isScInftyGermAt_of_lifts`. So the level-wise reading of Theorem 3.47 (1) (§III.20) is exactly what Definition 3.25 (ii) needs (item 94 (a)).

The kernel is smooth `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth`. For a basic germ f with contraction map B and the data $\mathcal{S}$ of Theorem 3.38 (sc$^+$-germ s into W^1), every $x = (a, w) \in \ker D(f + \iota s)(0)$ is a smooth point of $\mathbb{R}^n \oplus W$. The text takes the kernel of an sc-Fredholm operator to be smooth without comment. *The kernel equation.* $D(f + \iota s)(0) = Df(0) + \iota Ds(0)$ `Chapter3/SolutionLinearization · StabilityData.Dfs'_eq`, and the W-component of $Df(0)x$ is $w - DB(0)x$ `Chapter3/SolutionLinearization · StabilityData.snd_Df_eq`. Lemma 3.9 at $(0, 0)$ gives $DB(0)(0, w) = 0$ `Chapter3/BasicGermLinearization · IsSc0ContractionGerm.deriv_zero_snd`, so $DB(0)x = DB(0)(a, 0)$, and the W-component of the kernel equation reads

$$w = DB(0)(a, 0) - \iota Ds(0)x.$$

The bootstrap. $DB(0)$ is an sc-operator (the derivative of an sc^1-germ at the smooth point 0) `Chapter3/BasicGermLinearization · IsSc1GermWith.isScOperator_deriv_zero`, and so is $Ds(0) : E \to W^1$, which raises the level by one. By induction on m, $w \in W_m$: if $w = \iota y$ with $y \in W_m$, then $(a, 0)$ lies on level $m + 1$ (the parameter space has all levels equal), so $DB(0)(a, 0) \in W_{m+1}$, and $(a, y) \in E_m$ gives $Ds(0)x \in W_{m+1}$; hence $w \in W_{m+1}$. So $x \in E_\infty$.

The chart $\psi \circ \varphi$. An sc-smooth map g on a relatively open U is an sc-smooth map between retracts $(O, C, E) \to (O', C', E')$ for every retract $O \subseteq U$ mapped into O': for every sc-smooth retraction R onto O, $g \circ R$ is sc-smooth by the chain rule `Chapter3/GlobalIFTCenter · IsScSmooth.isScSmoothRetractMap_of_subset`. For a chart $\varphi : V \to O$ of X and an sc-diffeomorphism $\psi : U \to U'$ between relatively open subsets of partial quadrants with inverse ψ' (§III.27) `Chapter2/TameInvariance · IsScDiffeoQuadrant` and $O \subseteq U$, given that $\psi(O)$ is a retract of C', the map $\psi \circ \varphi$ is a chart onto the model $(\psi(O), C', E')$, with inverse $\varphi^{-1} \circ \psi'$ and the source of φ `Chapter3/GlobalIFTCenter · MChart.compDiffeo` (continuity of both maps: ψ, ψ' are sc^0, hence continuous on level 0). *It is a chart of the M-polyfold structure* `Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart`: for a chart φ' of the atlas, the overlap of $\psi\varphi$ with φ' is ψ of the overlap of φ with φ', and the two transition maps are $(\varphi' \circ \varphi^{-1}) \circ \psi'$ and $\psi \circ (\varphi \circ \varphi'^{-1})$; both are compositions of sc-smooth maps between retracts — the transitions of φ and φ', compatible because both are charts of X, and ψ' resp. ψ restricted to the overlap retracts by the lemma above.

The tame model. For the base retraction r of the centred strong bundle chart, tame by the hypothesis on the charts, and the sc-diffeomorphism ψ of the conjugation, $(\psi(O), C', E')$ is a tame retract with the retraction $\psi \circ r \circ \psi'$ `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the text takes this for granted when it calls $\psi \circ \varphi$ a chart of the tame M-polyfold (item 95).

Condition (b) from the choice of complement. For data of Definition 3.25 whose complement Y has the projection property for C, $\pi(O) \subseteq C$: the retract lies in C, and π maps C into C `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.π_mem_of_projectsInto`. So the regularity conditions reduce to condition (a), the sc-smoothness of δ on V in the set sense `Chapter3/GlobalIFTCenter · MPolyfold.TangentGermChart.isRegular_of_projectsInto`.

Theorem 3.54 (2) at the centre `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj`. *Let f be a section of a strong bundle whose charts compatible with the atlas have tame base retractions, $f(x_0) = 0$ and (f, x_0) in good position. Then there are an open $V_x \ni x_0$ and a dimension d such that $S_{V_x} = \{y \in V_x : f(y) = 0\}$ has at x_0 the data of Definition 3.25 of dimension d, with $\pi(O) \subseteq C$ for their projection and retract.* The proof fills the fields of the data one by one from the graph description, closed and projecting (§III.29) `Chapter3/GlobalIFTG · theorem_3_54_graph_proj`:

1. *The chart.* ψ and ψ' form an sc-diffeomorphism $U \to U'$ (item (i) of §III.29); $O = r(U) \subseteq U$ since the retraction maps U into U `Chapter3/Filler · StrongRetr.r_mem`; $\psi(O)$ is a tame retract (above, with the retract of the base retraction identified with $O = r(U)$ `Chapter3/StrongBundleSections · StrongBundleChart.r_image`); the chart is $\chi = \psi \circ \varphi$, a chart of X, containing x_0, with $\chi(x_0) = \psi(0) = 0$.
2. *The subspace.* $N = K = \ker D(h + t)(0)$, finite-dimensional because $D(h + t)(0)$ is Fredholm `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`, smooth (above), in good position to C' (the hypothesis of Corollary 3.47 established in §III.23); $d := \dim K$.
3. *The complement and the domain.* Y closed, a good complement; $V = V'$, relatively open in $K \cap C'$, containing 0; $\delta = \sigma$, continuous (it is C^1), with values in Y.
4. *The graph identity.* $\chi(S_{V_x} \cap \text{source}) = \psi\varphi(S_{V_x})$, since V_x lies in the source of φ; this is the graph identity of Theorem 3.54 (3).
5. *Regularity at* 0. The sc$^\infty$-germ property from the lifts (above); $\sigma(0) = 0$; and $D\sigma(0) = 0$ within the whole cone $K \cap C'$, from $D\sigma(0) = 0$ within $V' = O \cap (K \cap C')$ with O open, since derivatives within a set do not change when the set is intersected with a neighbourhood of the point.
6. *Condition (b).* From the projection property of Y, by the lemma above.

Without the last item this is the plain form `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart`. *What the statement gives about d.* The dimension is existentially quantified in the statement; the proof takes $d = \dim\ker D(h + t)(0)$, the dimension of the kernel of the linearization of the push-forward, and the statement does not identify d with $\dim\ker f'(x_0)$ (the docstring's "$n = \dim\ker f'(x_0)$" is not part of the statement). The local constancy of §III.37 makes d intrinsic in another way: it is the dimension of every datum of Definition 3.25 for the solution set at points of the chart domain.

What is missing at this stage, and where it is supplied. Condition (a), the sc-smoothness of σ on V' in the set sense, is not provided by Theorem 3.47 (1), which gives the sc$^\infty$-germ property at 0 only (item 82); it is obtained in §III.36 by re-centring the data at the nearby zeros, which needs the data of this section at every zero near x_0, hence good position there (Theorem 3.54 (1), §§III.32–III.33). The solution set here is the one in V_x; §III.37 restricts the chart so that it describes the global solution set.

III.35 The re-centring lemma

The text's proof of Theorem 3.27 begins (p. 84) by showing that the map δ of Definition 3.25 is an sc$^\infty$-germ near *every* point $v_0 \in V$, not only near 0: take the point $x_0 = \varphi^{-1}(v_0 + \delta(v_0))$ of M, the data (ψ, N', Y', V', τ) of Definition 3.25 at x_0, and consider $w \mapsto \pi \circ \varphi \circ \psi^{-1}(w + \tau(w))$; its linearization at 0 is claimed to be an isomorphism $N' \to N$, so it is a local diffeomorphism near 0, and $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) =$

$(1-\pi)\varphi\psi^{-1}(1+\tau)$ shows that δ is regular near v_0. For an arbitrary set with the tangent germ property the claim is false: δ need not even be differentiable at v_0 (the curve with a vertical tangent of §III.24), and that is why condition (a) of `IsRegular` had to be added to Theorem 3.27. What the argument really needs is that δ be C^1 at v_0. The formal *re-centring lemma* is the text's argument under that hypothesis, for two graph descriptions of the same set: one in a chart χ_1 by a map σ_1 of class C^1 on all of its domain, one in a chart χ_2 centred at the point over v_0 by a map σ_2 with the level-wise C^k-lifts of Theorem 3.47 (1) at 0. Its conclusion is that σ_1 is C^k into every level $E_{1,m}$ near v_0, and that the two subspaces have the same dimension. Section III.36 applies it to the shrunk data of Definition 3.25; for the solution set of an sc-Fredholm section, §III.37 supplies the second description at every nearby zero. Four ingredients are proved first: the inverse function machinery of Proposition 3.73 with the hypotheses its proofs use, a local inverse theorem for C^k homeomorphisms between relatively open subsets of convex sets, Sard's lemma in fixed dimension, and two facts about relatively open subsets of convex sets.

The inverse function machinery with the hypotheses its proofs use `Chapter3/SolutionGermRecentre · InvDataW`. On a finite-dimensional space V, let $U \subseteq V$ be convex with unique differentiability at each of its points, $f : V \to V$ of class C^j on U with $j \geq 1$, $\|Df(x) - 1\| \leq \frac{1}{2}$ for $x \in U$ (derivatives within U), and let the image $f(U)$ have unique differentiability at each of its points. These are the data of Proposition 3.73 as formalized in §III.16 (`InvData`, `Chapter3/QuadrantInverse · InvData`) with the partial quadrant removed: `InvData` asks for a partial quadrant C, U relatively open in C and convex, $f(U) \subseteq C$, and that x be an interior point of C iff $f(x)$ is; from the last hypothesis it derives that $f(U)$ is relatively open in C `Chapter3/QuadrantInverse · InvData.image_relOpen`, and the only use of that openness in the regularity proof is the unique differentiability of $f(U)$. The weak data keep convexity and the derivative bound and replace the quadrant hypotheses by unique differentiability of U and of $f(U)$. The proofs are those of §III.16, retold here because they now stand on their own.

The bi-Lipschitz estimate (Lemma 3.74, (70)). $f - \mathrm{id}$ has derivative $Df - 1$ of norm $\leq \frac{1}{2}$ on the convex U, so by the mean value inequality it is a $\frac{1}{2}$-contraction there: $|(f(x') - x') - (f(x) - x)| \leq \frac{1}{2}|x' - x|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_sub_sub_le`. Writing $x' - x = (f(x') - f(x)) - ((f(x') - x') - (f(x) - x))$ gives $\frac{1}{2}|x' - x| \leq |f(x') - f(x)|$ `Chapter3/SolutionGermRecentre · InvDataW.half_norm_le`, hence f is injective on U `Chapter3/SolutionGermRecentre · InvDataW.injOn`. *The derivative is invertible*: $Df(x) = 1 - (1 - Df(x))$ with $\|1 - Df(x)\| \leq \frac{1}{2} < 1$ is a unit by the Neumann series `Chapter3/SolutionGermRecentre · InvDataW.derivUnit Chapter3/SolutionGermRecentre · InvDataW.isUnit_deriv`. *The inverse* g on $f(U)$ is the inverse of f on U (a choice of preimage, unique by injectivity) `Chapter3/SolutionGermRecentre · InvDataW.inv Chapter3/SolutionGermRecentre · InvDataW.inv_apply`; the estimate gives $|g(y') - g(y)| \leq 2|y' - y|$ `Chapter3/SolutionGermRecentre · InvDataW.norm_inv_sub_le`, so g is Lipschitz, hence continuous on $f(U)$ `Chapter3/SolutionGermRecentre · InvDataW.continuousOn_inv`.

Lemma 3.77 `Chapter3/SolutionGermRecentre · InvDataW.hasFDerivWithinAt_inv`: g is differentiable within $f(U)$ at every $y \in f(U)$ with $Dg(y) = [Df(g(y))]^{-1}$. With $x = g(y)$: as $y' \to y$ in $f(U)$, $g(y') \to x$ in U (continuity), so $f(g(y')) - f(x) - Df(x)(g(y') - x) = o(|g(y') - x|)$, and $|g(y') - x| \leq 2|y' - y|$ turns this into $o(|y' - y|)$; applying the bounded operator $Df(x)^{-1}$ and using $f(g(y')) = y'$ gives $g(y') - g(y) - Df(x)^{-1}(y' - y) = o(|y' - y|)$. No hypothesis on $f(U)$ enters here.

Proposition 3.73, the regularity of the inverse `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv`: g is C^j on $f(U)$. First, $x \mapsto [Df(x)]^{-1}$ is C^{j-1} on U: the derivative within U of a C^j map is C^{j-1} there because U has unique differentiability, and inversion is smooth near the units `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inverse_deriv`. Then by induction on $i \leq j$: g is C^0 (continuous); if g is C^i with $i < j$, then g has the derivative $y \mapsto [Df(g(y))]^{-1}$ at every point of $f(U)$, which is C^i as the composite of the C^{j-1} map $x \mapsto [Df(x)]^{-1}$ with g, and a map with a C^i derivative on a set of unique differentiability is C^{i+1} there `Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_succ_iff_hasFDerivWithinAt_of_uniqueDiffOn`. This last step is the one place where the unique differentiability of $f(U)$ is used. It is the bootstrapping of §III.16 in place of the text's iterated tangent maps (Lemma 3.76).

Relatively open subsets of convex sets. Let $D \subseteq V_1$ be convex with nonempty interior and $A = O \cap D$

with O open. Then A has unique differentiability at each of its points (a convex set with interior has it, and intersecting with an open set keeps it) `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`; and if A is nonempty, it has nonempty interior: for $a \in A$ and an interior point w of D, the points $a + t(w - a)$ with $0 < t$ small lie in O by continuity and in the interior of D, because the open segment from a point of the closure of a convex set to an interior point lies in the interior `Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex`.

Sard's lemma in fixed dimension `Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty`. If $f : V_1 \to V_2$ between finite-dimensional spaces is differentiable on s and $f(s)$ has nonempty interior, then $\dim V_2 \le \dim V_1$. Suppose $\dim V_1 < \dim V_2$. There is a linear surjection $p : V_2 \to V_1$ (send the first $\dim V_1$ vectors of a basis of V_2 to a basis of V_1 and the rest to 0) `Chapter3/SolutionGermRecentre · exists_linearMap_surjective`. The map $f \circ p$ on $p^{-1}(s) \subseteq V_2$ has the derivative $Df(p(x)) \circ p$, an endomorphism of V_2 whose range lies in the range of $Df(p(x))$, of dimension $\le \dim V_1 < \dim V_2$; so it is not injective and its determinant vanishes. By Mathlib's change-of-variables estimate, a set on which a map is differentiable with vanishing Jacobian determinant has an image of Haar measure zero `Mathlib/MeasureTheory/Function/Jacobian · MeasureTheory.addHaar_image_eq_zero_of_det_fderivWithin_eq_zero`. But $f(s) \subseteq (f \circ p)(p^{-1}(s))$ since p is onto, and the nonempty open interior of $f(s)$ has positive Haar measure: a contradiction. The lemma uses only differentiability (not continuity of the derivative) and only the finite dimension of both spaces.

A C^k homeomorphism with invertible derivative has a C^k inverse `Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse`. Let $D_1 \subseteq V_1$, $D_2 \subseteq V_2$ be convex with nonempty interior, A relatively open in D_1, β of class C^k on A with $k \ge 1$, $\beta(A)$ relatively open in D_2, α continuous on $\beta(A)$ with $\alpha(\beta(w)) = w$ for $w \in A$, and $D\beta(w_0) = L$ (derivative within A) a linear isomorphism $V_1 \to V_2$ at some $w_0 \in A$. Then there is a relatively open $B' \subseteq \beta(A)$ in D_2 containing $\beta(w_0)$ on which α is C^k. *Proof.* Normalize: $\hat\beta = L^{-1} \circ \beta$ is C^k on A with $D\hat\beta(w_0) = 1$. Its derivative is continuous within A (C^k, $k \ge 1$, on a set of unique differentiability), so $\|D\hat\beta(w) - 1\| \le \frac{1}{2}$ for $w \in A$ in a ball $B(w_0, r_1)$; take $r \le r_1$ so small that $B(w_0, r) \subseteq O$, where $A = O \cap D_1$. The set $U = D_1 \cap B(w_0, r)$ is convex, lies in A, is relatively open in D_1 (hence has unique differentiability), and the derivatives of $\hat\beta$ within U and within A agree there (locality of the derivative within a set), so $\|D\hat\beta - 1\| \le \frac{1}{2}$ on U. Its image $\beta(U) = \beta(A) \cap \alpha^{-1}(B(w_0, r))$ — because α inverts β on A — is relatively open in D_2 by the continuity of α, so $\beta(U)$, and with it $\hat\beta(U) = L^{-1}(\beta(U))$, has unique differentiability. So $(U, \hat\beta, k)$ are weak inverse data, and $\hat\beta^{-1}$ is C^k on $\hat\beta(U)$ (Proposition 3.73 above). On $B' = \beta(U)$, $\alpha = \hat\beta^{-1} \circ L^{-1}$, since $\alpha(\beta(w)) = w = \hat\beta^{-1}(\hat\beta(w))$; hence α is C^k on B'. The relative openness of $\beta(U)$ comes from the continuity of the given inverse α, not from a degeneracy index; that is the point of the weak data.

The re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. Let χ_1, χ_2 be charts of the M-polyfold X `Chapter2/MPolyfold · MPolyfold.IsChart.compatible`, with models (O_i, C_i, E_i), and $S \subseteq X$. Suppose:

- (in χ_1) $K_1 \subseteq E_1$ is a finite-dimensional smooth subspace in good position to C_1, Y_1 a closed algebraic complement of K_1, V_1 relatively open in $K_1 \cap C_1$, $\sigma_1 : K_1 \to E_{1,0}$ with values in Y_1 and of class C^1 on V_1; $V_{x,1}$ is open in X inside the domain of χ_1 and $\chi_1(S \cap V_{x,1}) = \{v + \sigma_1(v) \mid v \in V_1\}$;
- (in χ_2) the same kind of data K_2, Y_2, $V_2 \ni 0$, σ_2 with values in Y_2, continuous on V_2, $\sigma_2(0) = 0$, $D\sigma_2(0) = 0$ within V_2, and for every $k \ge 1$ a relatively open $V_{2,k} \subseteq V_2$ containing 0 and a map $\sigma_{2,k} : K_2 \to E_{2,k}$ of class C^k on $V_{2,k}$ with $\iota_k \sigma_{2,k} = \sigma_2$ there (the lifts of Theorem 3.47 (1)); $V_{x,2}$ open inside the domain of χ_2 with $\chi_2(S \cap V_{x,2}) = \{w + \sigma_2(w) \mid w \in V_2\}$;
- $y \in V_{x,1} \cap V_{x,2}$ with $\chi_2(y) = 0$, and $v_0 \in V_1$ with $\chi_1(y) = v_0 + \sigma_1(v_0)$.

Then $\dim K_1 = \dim K_2$, and for every level m and every k there is a relatively open $B' \subseteq V_1$ in $K_1 \cap C_1$ containing v_0 on which σ_1 takes values in $E_{1,m}$ and its level map $\sigma_1 : B' \to E_{1,m}$ is of class C^k.

The hypotheses are exactly what each step uses: the closedness of Y_i makes the projection π_i onto K_i along Y_i continuous; good position gives $K_i \cap C_i$ a nonempty interior in K_i (for unique differentiability and for Sard); smoothness of K_i gives the linear lifts of K_i to every level `Chapter3/BasicClass · ScBanach.finLift`. Note that V_1 need not contain 0 and σ_1 need not vanish anywhere: χ_1 is an arbitrary chart in which S is a C^1 graph.

Step 1: notation and the points of S. Write $\Gamma_1(v) = v + \sigma_1(v)$, $\Gamma_2(w) = w + \sigma_2(w)$, and π_1, π_2 for the continuous projections along Y_1, Y_2, so that $\pi_i\Gamma_i = \mathrm{id}$. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` For $v \in V_1$ the point $\chi_1^{-1}(\Gamma_1(v))$ lies in $S \cap V_{x,1}$ and χ_1 maps it back to $\Gamma_1(v)$, which therefore lies in O_1; likewise for $w \in V_2$. Since $\sigma_2(0) = 0$, $\Gamma_2(0) = 0$, and both $\chi_1^{-1}(\Gamma_1(v_0))$ and $\chi_2^{-1}(\Gamma_2(0))$ are y.

Step 2: the transition and the retractions onto the overlaps. The charts are compatible, so the transition $T = \chi_1 \circ \chi_2^{-1}$ is an sc-smooth map from the overlap retract $\chi_2(V_{\chi_2} \cap V_{\chi_1})$ to $\chi_1(V_{\chi_1} \cap V_{\chi_2})$, and $T' = \chi_2 \circ \chi_1^{-1}$ in the other direction `Chapter2/MPolyfold · MChart.transition` `Chapter2/MPolyfold · MChart.overlap`. The overlaps are sc-retracts `Chapter2/MPolyfold · MChart.isScRetract_overlap`: choose sc-smooth retractions r_2, r_1 onto them, with domains U_{r_2}, U_{r_1}. Then $T \circ r_2$ is sc-smooth on U_{r_2}, $T' \circ r_1$ on U_{r_1}, $T(\chi_2(q)) = \chi_1(q)$ on the domain of χ_2, and $T' \circ T = \mathrm{id}$ on the overlap. Here the formal proof departs from the text, which treats $\varphi \circ \psi^{-1}$ as a map one can differentiate along the graph; a transition is only an sc-smooth map between retracts, and every derivative below is taken of $T \circ r_2$ or $T' \circ r_1$, on relatively open subsets of partial quadrants.

Step 3: the parameter sets and the maps β, α. Let $A = \{w \in V_2 \mid \chi_2^{-1}(\Gamma_2(w)) \in V_{x,1}\}$ and $B_s = \{v \in V_1 \mid \chi_1^{-1}(\Gamma_1(v)) \in V_{x,2}\}$: the parameters whose points of S lie in both chart neighbourhoods. By continuity of $w \mapsto \chi_2^{-1}(\Gamma_2(w))$ on V_2 (and of its counterpart), A is relatively open in $K_2 \cap C_2$, B_s in $K_1 \cap C_1$; $0 \in A$ and $v_0 \in B_s$. Define

$$\beta(w) = \pi_1(T(\Gamma_2(w))), \qquad \alpha(v) = \pi_2(T'(\Gamma_1(v))).$$

For $w \in A$, with $q = \chi_2^{-1}(\Gamma_2(w)) \in S \cap V_{x,1} \cap V_{x,2}$: $T(\Gamma_2(w)) = \chi_1(q)$ lies on the graph of σ_1, say $= \Gamma_1(v)$ with $v \in V_1$; applying π_1 gives $\beta(w) = v$, so $T(\Gamma_2(w)) = \Gamma_1(\beta(w))$, $\beta(w) \in B_s$, $\alpha(\beta(w)) = w$, and $\Gamma_2(w)$ lies in the overlap. Symmetrically, for $v \in B_s$: $T'(\Gamma_1(v)) = \Gamma_2(\alpha(v))$, $\alpha(v) \in A$, $\beta(\alpha(v)) = v$. Moreover $\beta(0) = v_0$, and α is continuous on B_s ($\pi_2 \circ \chi_2 \circ \chi_1^{-1} \circ \Gamma_1$). Consequently, for every $A' \subseteq A$, $\beta(A') = \{v \in B_s \mid \alpha(v) \in A'\}$, which is relatively open in $K_1 \cap C_1$ whenever A' is relatively open in $K_2 \cap C_2$. This is the text's "local diffeomorphism" in its topological half, obtained without any derivative. `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`

Step 4: the lifted graph maps. For $j \geq 1$ let $A_j = A \cap V_{2,j}$ and $\Gamma_{2,j}(w) = \lambda_j(w) + \sigma_{2,j}(w) \in E_{2,j}$, where $\lambda_j : K_2 \to E_{2,j}$ is the linear lift `Chapter3/BasicClass · ScBanach.finLift`. Then A_j is relatively open and contains 0, $\Gamma_{2,j}$ is C^j on A_j with $\iota_j\Gamma_{2,j} = \Gamma_2$ there, $\Gamma_{2,j}$ maps A_j into the level-j part of the domain of r_2, and $r_2(\Gamma_2(w)) = \Gamma_2(w)$ for $w \in A_j$ (the graph points lie in the overlap, where r_2 is the identity). Its derivative within A_j at 0 is λ_j: indeed $\iota_j \circ D\sigma_{2,j}(0) = D\sigma_2(0)$ within A_j, which equals the derivative within V_2 (the two sets agree near 0) and so vanishes; ι_j is injective, so $D\sigma_{2,j}(0) = 0$. Here the unique differentiability of A_j at 0 is that of a relatively open subset of the convex cone $K_2 \cap C_2$ with interior `Chapter3/SolutionGermRecentre · uniqueDiffOn_relOpen_convex`.

Step 5: β is C^j on A_j. On A_j, $\beta = \pi_1 \circ ((T \circ r_2) \circ \iota_j) \circ \Gamma_{2,j}$, because r_2 fixes $\Gamma_2(w) = \iota_j\Gamma_{2,j}(w)$. The map $(T \circ r_2) \circ \iota_j$ is C^j on the level-j part of U_{r_2} — Proposition 1.18: an sc^j map is C^j from level j to level 0 `Chapter1/Levels · IsSck.contDiffOn_base` — and π_1 is continuous linear; so β is C^j on A_j. This is where the lifts are indispensable: σ_2 itself is only continuous into level 0, and $T \circ r_2$ is not differentiable on level 0.

Step 6: the chain rule at 0. Take $j = 2$; then β is C^2, in particular C^1, on A_2, with derivative $D\beta = D\beta(0)$ within A_2 at 0. Read the lift on level 1: $\Gamma_{2,1} = \iota_{1,2} \circ \Gamma_{2,2}$, with $\iota_1\Gamma_{2,1} = \Gamma_2$ on A_2, values in the level-1 part of U_{r_2}, derivative $\iota_{1,2}\lambda_2$ at 0, and $\iota_1\iota_{1,2}\lambda_2(n) = n$ for $n \in K_2$. Let $\mathcal{D} = D(T \circ r_2)(\Gamma_{2,1}(0))$, the sc-derivative at the level-1 point over 0. The map $w \mapsto T(r_2(\iota_1\Gamma_{2,1}(w)))$ has, by the chain rule for the

sc^1 map $T \circ r_2$ (differentiable from level 1 to level 0) and the differentiable $\Gamma_{2,1}$, the derivative $n \mapsto \mathcal{D}n$ at 0 within A_2. On A_2 this map equals $\Gamma_1 \circ \beta$ (Step 3 and $r_2 \circ \Gamma_2 = \Gamma_2$), and $\Gamma_1 \circ \beta$ has, at 0, the derivative $n \mapsto D\beta\, n + D\sigma_1(v_0) D\beta\, n$ by the chain rule — here the C^1-property of σ_1 *at* v_0, within V_1, is used, together with $\beta(A_2) \subseteq V_1$ and $\beta(0) = v_0$. By uniqueness of the derivative within A_2 at 0,

$$\mathcal{D}n = D\beta\, n + D\sigma_1(v_0)\, D\beta\, n \qquad (n \in K_2).$$

Step 7: $D\beta(0)$ is injective. Let $D\beta\, n = 0$; then $\mathcal{D}n = 0$. The composite $(T' \circ r_1) \circ (T \circ r_2)$ is sc-smooth on U_{r_2} ($T \circ r_2$ maps U_{r_2} into the overlap retract of χ_1, hence into U_{r_1}), and it equals r_2 there: $r_2(u)$ lies in the overlap, $T(r_2(u))$ in the other overlap where r_1 is the identity, and $T' \circ T = \mathrm{id}$ on the overlap. By the chain rule for sc^1 maps between relatively open subsets of partial quadrants `Chapter2/DegeneracyIndex · scDeriv_comp_apply`, $D((T' \circ r_1) \circ (T \circ r_2))(\Gamma_{2,1}(0))n = D(T' \circ r_1)(\cdot)\, \mathcal{D}n = 0$. Since the composite agrees with r_2 on the relatively open U_{r_2}, its derivative at the level-1 point is $Dr_2(\Gamma_{2,1}(0))$ `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen`. And $Dr_2(\Gamma_{2,1}(0))n = n$: the family $w \mapsto \iota_1 \Gamma_{2,1}(w)$ consists of fixed points of r_2, so the derivative of r_2 fixes its tangent vectors at 0, which are the vectors $\iota_1 \iota_{1,2} \lambda_2(n) = n$ `Chapter3/GlobalIFTNearbyGraph · scDeriv_apply_toBase_deriv_eq_of_fixed` — the tangent space K_2 of the graph lies in the tangent space of the retract. So $n = 0$. Hence $D\beta(0)$ is injective and $\dim K_2 \le \dim K_1$.

Step 8: surjectivity by counting dimensions. β is differentiable on A_2, and $\beta(A_2)$ is a nonempty relatively open subset of the convex cone $K_1 \cap C_1$ (Step 3; it contains $\beta(0) = v_0$), which has nonempty interior in K_1 by good position; so $\beta(A_2)$ has nonempty interior `Chapter3/SolutionGermRecentre · interior_nonempty_of_relOpen_convex`, and Sard's lemma in fixed dimension gives $\dim K_1 \le \dim K_2$ `Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty`. So $\dim K_1 = \dim K_2$, the injective $D\beta(0)$ is bijective, and $L = D\beta(0) : K_2 \to K_1$ is a linear isomorphism. This is the first conclusion of the lemma, and also the reason why the dimension of a set with two such descriptions cannot jump.

Step 9: the inverse, and the regularity of σ_1. Fix m and k and take $j = m + k + 2$. By Step 5, β is C^j, hence $C^{\max(k,1)}$, on A_j, $\beta(A_j) \subseteq B_s$, and its derivative within A_j at 0 is again L (the derivatives within A_j and within A_2 agree at 0, both sets being relatively open near 0 in the same cone). Now apply the local inverse theorem above with $D_1 = K_2 \cap C_2$, $D_2 = K_1 \cap C_1$ (convex with interior), $A = A_j$, the continuous inverse α on $\beta(A_j)$ and $w_0 = 0$ `Chapter3/SolutionGermRecentre · exists_contDiffOn_inverse`: there is a relatively open $B' \subseteq \beta(A_j)$ in $K_1 \cap C_1$ containing v_0 on which α is $C^{\max(k,1)}$, with $\alpha(B') \subseteq A_j$. Next the level-m map

$$\Psi(w) = (T \circ r_2)_m(\iota_{m,k+2}\, \Gamma_{2,j}(w)) - \lambda_m(\beta(w)) \in E_{1,m}, \qquad w \in A_j,$$

where $\iota_{m,k+2} : E_{2,m+k+2} \to E_{2,m}$ and λ_m is the lift of K_1 to level m. It is C^k on A_j: the sc^{k+2} map $T \circ r_2$ is C^{k+2} from level $m+k+2$ to level m (Proposition 1.18 `Chapter1/Levels · IsSck.contDiffOn_levels`), $\Gamma_{2,j}$ is C^j into level $j = m + k + 2$, and $\lambda_m \circ \beta$ is C^j. Including into level 0, $\iota_m \Psi(w) = T(r_2(\Gamma_2(w))) - \beta(w) = \Gamma_1(\beta(w)) - \beta(w) = \sigma_1(\beta(w))$. Hence for $v \in B'$, $v = \beta(\alpha(v))$ and $\sigma_1(v) = \iota_m \Psi(\alpha(v))$: σ_1 takes values in $E_{1,m}$ on B', its level-m map is $\Psi \circ \alpha$ `Chapter1/ScSmooth · levelMap_eq_of_toBase_eq`, which is C^k on B' as a composite of C^k maps, and $B' \subseteq B_s \subseteq V_1$. This is the text's identity $\delta \circ (\pi\varphi\psi^{-1}(1+\tau)) = (1-\pi)\varphi\psi^{-1}(1+\tau)$, written on level m through the lifts.

Comparison with the text. Four things in the formal argument are not in the sketch of p. 84. (a) The isomorphism claim needs σ_1 differentiable at v_0, which for an arbitrary set with the tangent germ property fails; here σ_1 is assumed C^1 on all of V_1, and §III.36 arranges this by shrinking V_1 to the neighbourhood of 0 on which the germ property makes δ of class C^1. (b) Only the *injectivity* of the linearization comes from the chain rule, and it uses that the tangent space K_2 of the second graph lies in the tangent space of the overlap retract (Step 7). The symmetric argument for surjectivity would differentiate $T' \circ \Gamma_1 = \Gamma_2 \circ \alpha$ at v_0; but $T' \circ r_1$ is differentiable only from level 1 to level 0, so this would need σ_1 differentiable *into level* 1 at v_0, which is known near the centre of χ_1 (Theorem 3.47 (1)) and not at v_0. Surjectivity therefore comes

from the dimension count by Sard's lemma (Step 8), which at the same time shows that the dimension is locally constant. (c) The regularity of $\beta^{-1} = \alpha$ is not the quadrant inverse function theorem 3.72 (§III.16): that theorem asks for $d_C(x) = d_C(\beta(x))$ (formally: interior points to interior points), which here would be a statement about the intrinsic degeneracy index of S — not available. It is Lemma 3.77 and the bootstrapping of Proposition 3.73 under the weak data, with the relative openness of $\beta(U)$ supplied by the continuity of the inverse α. (d) The neighbourhood B' depends on (m, k) through the lift $\Gamma_{2,m+k+2}$; this is harmless, because sc-smoothness in the set sense is a local property of the level maps (§III.36).

III.36 Regular data by re-centring, and Theorem 3.27 with locally constant dimension

This section turns the re-centring lemma of §III.35 into condition (a) of `IsRegular` (§III.24) and adapts Theorem 3.27 to data whose dimension varies from point to point. The tool is the shrinking of the data of Definition 3.25 (2): the chart is restricted to an open subset of its domain and the parameter set V to a smaller relatively open set, with the same N, Y and δ. The data at a point x are shrunk into the neighbourhood of 0 where δ is C^1; then, at every point of the shrunk graph, the data at that point (which a set with the tangent germ property has everywhere) provide the second graph description that the re-centring lemma needs.

Open subsets of tame retracts (told in §III.27). The restriction of a tame retraction $r : U \to U$ to $U \cap r^{-1}(W)$, W open (Proposition 2.10 (1)) `Chapter2/Retracts · ScRetraction.restrict`, is tame: the degeneracy-index condition of Definition 2.34 is inherited pointwise, and at a smooth point x of $O \cap W$ the sc-complement of T_xO of the original retraction serves, because the derivative of the restricted retraction at x is $Dr(x)$ — the two maps agree on a relatively open neighbourhood `Chapter3/LinearizationCharts · scDeriv_eq_of_eqOn_relOpen Chapter3/SolutionGermRecentre · ScRetraction.restrict_isTame`. So $O \cap W$ is a tame retract `Chapter3/SolutionGermRecentre · IsTameRetract.inter_isOpen`, and the model $\varphi(s)$ of a chart restricted to an open s of its domain is tame when the model of φ is, since $\varphi(s) = O \cap W$ for some open W `Chapter2/MPolyfold · MChart.exists_isOpen_image_eq Chapter3/SolutionGermRecentre · MChart.isTameRetract_image`. The text uses this silently whenever it shrinks a chart of a tame M-polyfold.

Shrinking the data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink`. Let $D = (\varphi, N, Y, V, \delta)$ be data of Definition 3.25 (2) for M at x of dimension n `Chapter3/TangentGerm · MPolyfold.TangentGermChart`, s an open subset of the domain of φ containing x, $V' \subseteq V$ relatively open in $N \cap C$ containing 0, and M' a set with $\varphi(M' \cap s) = \{v + \delta(v) \mid v \in V'\}$. Then the chart φ restricted to s `Chapter2/MPolyfold · MChart.restrict` — a chart of the structure `Chapter2/MPolyfold · MPolyfold.IsChart.restrict` with a tame model by the paragraph above — together with the same N, Y, δ and the parameter set V', are data of Definition 3.25 (2) for M' at x of dimension n. Every field is inherited: good position, the closed good complement, $\delta(0) = 0$, $D\delta(0) = 0$ and the germ property of δ concern N, Y, δ and the cone $N \cap C$ only, which are unchanged, and continuity on V' follows from continuity on V. The set M' may differ from M; this is how §III.37 passes between the solution set in a neighbourhood and the global solution set. The shrunk data have the source s, the model $\varphi(s)$, the same chart values, the same N, δ and projection π, and the parameter set V' `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_source Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_model_0 Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_φ_apply Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_N Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_V Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_δ Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π`. They are regular `Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular` as soon as $\delta$ is sc-smooth on $V'$ in the set sense and the original data satisfy condition (b), $\pi(O) \subseteq C$: condition (b) passes to the smaller retract $\varphi(s) \subseteq O$ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`.

The shrinking data `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_shrinkData`. For an open

$W \ni x$ in X and an open $O'' \ni 0$ in N, let

$$s = \{y \in \operatorname{dom}\varphi \cap W \mid \pi\varphi(y) \in O''\}, \qquad V' = \{v \in V \cap O'' \mid \varphi^{-1}(v + \delta(v)) \in W\}.$$

Then s is open (the preimage of O'' under the continuous $\pi \circ \varphi$ on the open $\operatorname{dom}\varphi \cap W$) and contains x ($\pi\varphi(x) = \pi(0) = 0$); V' is relatively open in $N \cap C$ because $v \mapsto \varphi^{-1}(v + \delta(v))$ is continuous on V `Chapter3/TangentGerm · MPolyfold.TangentGermChart.Γ_continuousOn`, and contains 0 ($\varphi^{-1}(0) = x \in W$); $s \subseteq W$, $V' \subseteq V \cap O''$; and $\varphi(M \cap s) = \{v + \delta(v) \mid v \in V'\}$. For the last identity: a point $y \in M \cap s$ has $\varphi(y) = \Gamma(v)$ with $v = \pi\varphi(y) \in V$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.exists_eq_Γ`, $v \in O''$ because $y \in s$, and $\varphi^{-1}(\Gamma(v)) = y \in W$; conversely for $v \in V'$ the point $\varphi^{-1}(\Gamma(v))$ lies in M and in the domain `Chapter3/TangentGerm · MPolyfold.TangentGermChart.symm_Γ_mem`, in W by definition of V', and $\pi\varphi$ of it is $v \in O''$ since $\pi\Gamma = \mathrm{id}$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.π_Γ`.

The lifts from the germ property. Definition 3.25 (ii), in the reading of §III.24 `Chapter3/TangentGerm · IsScInftyGermAt`, gives for every m, k a relatively open $V_{m,k} \ni 0$ on which δ takes values in E_m and is C^k into E_m. With $m = k$: for every $k \geq 1$ the level map of δ on $V_{k,k} \cap V$ is a C^k-lift δ_k of δ into E_k — the hypothesis of the re-centring lemma on the centred side `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_lift`. With $(m,k) = (0,1)$: δ is C^1 on a relatively open neighbourhood of 0 in $N \cap C$ `Chapter3/TangentGerm · IsScInftyGermAt.contDiffOn_one`. And $D\delta(0) = 0$ within V, since the data give it within $N \cap C$ and V agrees with $N \cap C$ near 0 `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.fderivWithin_V_zero`.

Condition (a) of `IsRegular` by re-centring `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`. Let X be an M-polyfold (Hausdorff, paracompact), $M \subseteq X$, D data of Definition 3.25 (2) for M at x of dimension n satisfying condition (b), $\pi(O) \subseteq C$, and $U \ni x$ open such that every point of $M \cap U$ has data of Definition 3.25 (2) for M of *some* dimension. Then there are data D' for M at x of dimension n which are regular (`IsRegular`: (a) and (b)), whose chart domain lies in U, and such that every datum of Definition 3.25 (2) for M at a point of M in the chart domain of D' has dimension n. (The proof constructs D' as the shrinking of D, with the same N, Y, δ; the statement records only the regularity, the domain and the dimension.)

Step 1: shrinking into the C^1 neighbourhood. By the germ property, δ is C^1 on $V_1 = O_1 \cap (N \cap C)$ with O_1 open and $0 \in O_1$. The shrinking data for $W = U$ and $O'' = O_1$ give $s \subseteq U$ and $V' \subseteq V \cap O_1$, so $V' \subseteq V_1$ `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_subset`, with $\varphi(M \cap s) = \Gamma(V')$; let D_1 be the shrunk data for M itself. Then δ is C^1 on all of D_1's parameter set V': this is hypothesis (a) of the re-centring lemma on the side of D_1, and the reason for the shrinking.

Step 2: the re-centring at every point. Let $y \in M \cap s$ and let D'' be any data of Definition 3.25 (2) for M at y, of dimension n'. Put $v_0 = \pi\varphi(y)$; then $v_0 \in V'$ and $\varphi(y) = \Gamma(v_0)$. The re-centring lemma applies with \$□ _1 = \$ the chart of D_1 (graph of δ over V' on its whole domain s, δ of class C^1 on V') and \$□ _2 = \$ the chart of D'', centred at y (graph of δ'' over V'' on its whole domain, $\delta''(0) = 0$, $D\delta''(0) = 0$ within V'', the lifts from the germ property of D''); all the remaining hypotheses — smooth finite-dimensional N, closed good complement, good position, relatively open parameter sets — are fields of the two data `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels`. It gives $n = \dim N = \dim N'' = n'$, and for every m, k a relatively open $B' \subseteq V'$ in $N \cap C$ containing v_0 on which δ takes values in E_m and is C^k into E_m.

Step 3: δ is sc-smooth on V' in the set sense. V' is a relatively open subset of the partial quadrant $N \cap C$ of N with the constant sc-structure `Chapter3/TangentGerm · MPolyfold.TangentGermChart.V_isRelOpenInQuadrant`, all of whose levels are V'. By Proposition 1.19 in the form "maps every level into the corresponding level, and the level maps are C^∞" `Chapter1/Levels · IsScSmooth.of_contDiffOn_levels`, it suffices that for every m: (i) $\delta(V') \subseteq E_m$, and (ii) the level map $\delta : V' \to E_m$ is C^k for every k. For $v \in V'$ the point $y = \varphi^{-1}(\Gamma(v))$ lies in $M \cap s \subseteq M \cap U$, so by hypothesis there are data at y of some dimension, and Step 2 applies with $v_0 = \pi\varphi(y) = v$. For (i), the set B' for $(m,0)$ contains v, and δ maps it into E_m. For (ii), fix k; around every $v \in V'$ the set $B' = O_B \cap (N \cap C)$ for (m,k) contains v, and the level map is C^k on

$O_B \cap V' \subseteq B'$; a map that is C^k near every point of a set is C^k on the set Mathlib/Analysis/Calculus/ContDiff/Defs · contDiffOn_of_locally_contDiffOn. Here the dependence of B' on (m,k) (§III.35, Step 9) does no harm: the property is checked one (m,k) at a time, and locally.

Step 4: conclusion. With (a) from Step 3 and (b) inherited from D, the shrunk data D_1 are regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular; their chart domain is $s \subseteq U$; and by Step 2 every datum at a point of $M \cap s$ has dimension n.

What the statement gives for an arbitrary subset. The hypotheses of the theorem are about M and D only; no solution set, section or Fredholm property enters. If M has the n-dimensional tangent germ property Chapter3/TangentGerm · MPolyfold.HasTangentGermProperty, every point of M has data of dimension n, so the hypothesis "every point of $M \cap U$ has data of some dimension" holds with $U = X$. Hence, read off the statement (this is not a separate Lean theorem): *for any M with the tangent germ property, any data at x that satisfy condition (b) can be shrunk to regular data at x, of the same dimension.* Condition (a) of `IsRegular` is therefore not an independent hypothesis of Theorem 3.27: if at every point of M some n-dimensional data satisfy (b), the regular data that Theorem 3.27 asks for exist at every point Chapter3/TangentGerm · theorem_3_27. (The inclusion $M \subseteq X_\infty$ of Definition 3.25 (1) is not used.) The counterexample to the text's step "δ is an sc$^\infty$-germ near every v_0" (§III.24) is not contradicted: there the data are regular only after shrinking V away from the point with a vertical tangent, which is what Step 1 does. What remains a genuine restriction is condition (b), which fails for some good complements (§III.24) and is obtained for solution sets by the choice of the complement (§III.29, §III.34).

The dimension is locally constant. The last clause of the theorem says more than the text: around every point where regular data exist, all data of Definition 3.25 (2) at nearby points of M have the same dimension. The dimension of a set with the tangent germ property at varying dimensions is therefore locally constant along M — but not necessarily globally constant, and Theorem 3.27 as formalized fixes n (§III.24). For the solution set of an sc-Fredholm section, whose dimension $\dim\ker f'(x)$ can change between components, the fixed-dimension theorem would need S split first; instead Theorem 3.27 is restated for locally constant dimension.

Manifolds with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc. An M-polyfold structure is that of a manifold with boundary with corners of locally constant dimension if it has an equivalent atlas all of whose charts have models (O, C, E) with O relatively open in the partial quadrant C and E_0 finite-dimensional — the definition of §III.24 Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners without the requirement that every chart have dimension n. A manifold with corners of dimension n is one of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCorners.isManifoldWithCornersLoc.

Theorem 3.27 with locally constant dimension Chapter3/GlobalIFTTheorems · theorem_3_27_loc. Let X be an M-polyfold, $M \subseteq X$ nonempty, and suppose every $x \in M$ has regular data of Definition 3.25 (2) for M of some dimension $n(x)$. Then M is a sub-M-polyfold of X; its induced structure (Proposition 2.21) is tame and that of a manifold with boundary with corners of locally constant dimension; and for every $x \in M$, all regular data d at x of any dimension and every base point $a_0 \in M$ (the value of the chart inverse off its domain), the chart $y \mapsto \pi\varphi(y)$ of d is a chart of the induced structure whose domain contains x. The proof is that of Theorem 3.27 (§III.24) with the dimension quantified pointwise; every step there is either pointwise or does not use n.

Sub-M-polyfold Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isSubMPolyfold. Definition 2.20 is pointwise: at $a \in M$ take regular data d (of whatever dimension); the open set $U_0 = \varphi^{-1}(\Sigma \cap O)$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.isOpen_U₀ Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_U₀ and the sc-smooth retraction $s = \varphi^{-1} \circ t \circ \varphi$ onto $M \cap U_0$ Chapter3/TangentGerm · MPolyfold.TangentGermChart.s Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_isScSmoothOn Chapter3/TangentGerm · MPolyfold.TangentGermChart.s_s Chapter3/TangentGerm · MPolyfold.TangentGermChart.image_s are those of §III.24, exactly as in the fixed-dimension proof Chapter3/TangentGerm · theorem_3_27_isSubMPolyfold.

The charts and the atlas. Let the charts be all maps $\pi\varphi$ of all regular data at all points of M, of all dimensions, with all base points `Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc` `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart`. Each is a chart of the induced structure `Chapter2/SubMPolyfold · subPolyfold`: it is compatible with every chart $\varphi'|_{W\cap M}$ of the induced atlas, by the lemma of §III.24, whose proof never compares dimensions `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_compatible_subChart` `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isChart`. Any two of them are compatible, by the locality of compatibility: both are compatible with every chart of the induced atlas, whose domains cover M `Chapter2/SubMPolyfold · MPolyfold.subAtlas` `Chapter2/MPolyfold · MChart.compatible_of_locally` `Chapter3/GlobalIFTTheorems · MPolyfold.tangentChartsLoc_compatible`. The argument never mentions the dimension of either chart. They form an atlas: every $y \in M$ has regular data, and the chart of these data, with the base point a fixed element of M (here $M \neq \emptyset$ is used), contains y `Chapter3/TangentGerm · MPolyfold.TangentGermChart.mem_chart_source` `Chapter3/GlobalIFTTheorems · MPolyfold.germAtlasLoc`. This atlas is equivalent to the induced atlas of Proposition 2.21, since each of its charts is a chart of the induced structure `Chapter2/MPolyfold · MAtlas.equiv_iff` `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_atlas_equiv`.

Tame, and a manifold with corners. The model of each chart $\pi\varphi$ is $(V, N\cap C, N)$ with the identity retraction, a tame retract `Chapter3/TangentGerm · MPolyfold.TangentGermChart.model_isTameRetract`; so the induced structure is tame, witnessed by the equivalent atlas just built `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isTame`. Each model is relatively open in the partial quadrant $N \cap C$ of the finite-dimensional N `Chapter3/TangentGerm · MPolyfold.TangentGermChart.chart_isCorner`; so the induced structure is that of a manifold with boundary with corners of locally constant dimension `Chapter3/GlobalIFTTheorems · MPolyfold.theorem_3_27_loc_isManifoldWithCornersLoc`. The assembled statement collects these, with the chart property of every $\pi\varphi$ `Chapter3/GlobalIFTTheorems · theorem_3_27_loc`.

As for Theorem 3.27, the hypotheses are the regularity conditions at every point; by the observation above, condition (a) there can be replaced by condition (b) plus shrinking. Theorem 3.57 (§III.37) applies this version to the solution set, where regular data exist at every zero by §§III.34–III.36 and the dimension $\dim\ker f'(x)$ is locally constant by Step 2 of the theorem above; the fixed-dimension statement for each part S_n of the solution set is obtained from Theorem 3.27 itself.

III.37 The implicit function theorems assembled: Theorems 3.54, 3.23, 3.19, Corollary 3.24, Theorems 3.56, 3.57

The text derives all its implicit function theorems from Theorem 3.54: (1) good position at every zero near x, (2) the sub-M-polyfold structure of the solution set near x, through the tangent germ property and Theorem 3.27. In the formalization the pieces are: the data of Definition 3.25 at the centre, with condition (b) (§III.34); good position at the nearby zeros, under an added hypothesis (§III.33); the recentring, which turns data at all nearby points into regular data and shows that the dimension is locally constant (§§III.35–III.36); and Theorem 3.27, with fixed dimension (§III.24) or with locally constant dimension `Chapter3/GlobalIFTTheorems · theorem_3_27_loc` (§III.36). This section assembles them. Throughout, f is a section of a strong bundle $P : Y \to X$ over an M-polyfold, "the charts have tame base retractions" means that the base retraction of every strong bundle chart compatible with the atlas is tame (the formal reading of "tame strong bundle", §III.23), $S = \{x \in X : f(x) = 0\}$ is the global solution set `Chapter3/GlobalIFT · solutionSet`, and good position is Definition 3.53 in the chart form `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm` (§III.21).

The data for the global solution set `Chapter3/GlobalIFTTheoremsG · theorem_3_54_tangentGermChart_solutionSet`. *If the charts have tame base retractions, $f(x_0) = 0$ and (f, x_0) is in good position, then S has at x_0 data of Definition 3.25 of some dimension d with $\pi(O) \subseteq C$.* §III.34 gives data D for $S_{V_x} = \{y \in V_x : f(y) = 0\}$. Restrict its chart to the open set $s = \text{source} \cap V_x \ni x_0$, keeping N, Y, V, δ `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink` `Chapter2/MPolyfold · MChart.restrict`: the graph identity holds for the global S because $S \cap s = S_{V_x} \cap \text{source}$. The restricted model $\varphi(s)$ is a tame retract (an open subset of a tame retract, §III.36), and condition (b) passes to it, since $\varphi(s) \subseteq O$ `Chapter2/MPolyfold · MChart.image_subset_model` and the

projection is unchanged Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_π.

Theorem 3.54 (2), from the nearby good position Chapter3/GlobalIFTTheoremsG · theorem_3_54_two. *Let the charts have tame base retractions, $f(x_0) = 0$, (f, x_0) in good position, and let every zero of f in an open $V_{x_0} \ni x_0$ be in good position. Then there is a dimension d and regular data of Definition 3.25 of dimension d for S at x_0 (condition (a): δ sc-smooth on V in the set sense; condition (b): $\pi(O) \subseteq C$) Chapter3/TangentGerm · MPolyfold.TangentGermChart.IsRegular, on a chart whose source lies in V_{x_0}; and every datum of Definition 3.25 for S at a point of S in that source has dimension d.* Only that f is a section is assumed; the regularity of f enters through the good position at x_0 and at the nearby zeros, which carry the sc-Fredholm germ data. *Proof.* Take the data D at x_0 above. Every $y \in S \cap V_{x_0}$ is a zero in good position, so the same theorem at y gives data of Definition 3.25 for S at y, of some dimension. These are the hypotheses of the re-centring theorem of §III.36 Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally for D (which satisfies (b)) and the open set V_{x_0}: it shrinks D to regular data D' with source in V_{x_0}, of the same dimension d, along which every datum has dimension d. This is the text's "proceeding as above we can construct a map σ" at every zero, made into a proof: the regularity condition (a), an added hypothesis of Theorem 3.27 for general sets (§III.24), is a theorem for solution sets (items 150–151).

Theorem 3.54 (1) and (2) Chapter3/GlobalIFTTheoremsG · theorem_3_54. *Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and let (f, x_0) be in good position* ***with a filler that satisfies condition (3) of Definition 3.4 at the zeros of the filler near*** 0 *Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled. Then there is an open $V_x \ni x_0$ such that (1) (f, y) is in good position at every zero $y \in V_x$, and (2) the conclusions of Theorem 3.54 (2) above hold with $V_{x_0} = V_x$.* (1) is §III.33 Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one; (2) is the previous theorem, fed with (1), with the strengthened good position weakened to good position Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled.isGoodPositionGerm and the section property of the sc-Fredholm section. The filler condition was an **added hypothesis** when this form was proved: Definition 3.4 imposes condition (3) at 0 only, and the surjectivity of $f'(y)$ at the nearby zeros is equivalent to it there (§III.33; items 100, 105–106); the kernel half of (1) needs no added hypothesis (§III.32). Since the second addendum the condition is a consequence of good position (§III.45), and Theorem 3.54 is proved with the text's hypotheses (§III.43). Part (3) of the text's proof, the graph description, is §III.29 Chapter3/GlobalIFTG · theorem_3_54_graph_proj, and part (4), the kernel at the nearby zeros in the chart at x_0, is §III.32 Chapter3/GlobalIFTNearbySurjG · theorem_3_54_nearby_kernel. The text's neighbourhood V is the V_x of (3), shrunk in (2) to the chart source of the re-centred data.

Theorem 3.23, boundary case, from the nearby good position Chapter3/GlobalIFTTheoremsG · theorem_3_23_of_nearby. *Let f be an sc-Fredholm section, the charts with tame base retractions, $f(x) = 0$, (f, x) in good position ($f'(x)$ onto with kernel in good position, Definition 3.53), and every zero in an open $V_{x_0} \ni x$ in good position. Then there are an open U with $x \in U \subseteq V_{x_0}$ and a dimension d such that every zero in U is a smooth point, $S(f, U) = \{y \in U : f(y) = 0\}$ is a sub-M-polyfold of X, and its induced M-polyfold structure (Proposition 2.21) Chapter2/SubMPolyfold · subPolyfold is tame and equivalent to that of a d-dimensional manifold with boundary with corners* Chapter3/TangentGerm · MPolyfold.IsManifoldWithCorners. *Proof.* Theorem 3.54 (2) at x gives regular data D of dimension d with source $U \subseteq V_{x_0}$ and the local constancy of the dimension; let $S' = S(f, U)$. At every $y \in S'$, Theorem 3.54 (2) at y, with the open neighbourhood $U \cap V_{x_0}$ in place of V_{x_0}, gives regular data D_y for S with source in $U \cap V_{x_0}$, of a dimension d_y; since $y \in S \cap U$, the local constancy along D gives $d_y = d$. Because the source of D_y lies in U, $S' \cap \text{source} = S \cap \text{source}$, so D_y restricted to its own source is a datum for S' Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink, still regular Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular. So every point of S' has regular data of dimension d, $S' \ni x$ is nonempty, and Theorem 3.27 with fixed dimension (§III.24) Chapter3/TangentGerm · theorem_3_27 gives the sub-M-polyfold with its tame induced structure equivalent to a d-dimensional manifold with boundary with corners. The zeros are smooth points because an sc-Fredholm section is regularizing Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt. The text's "a uniquely determined structure … the same as the one defined in Proposition 2.21" is the induced structure itself, with the equivalent atlas of the charts $\pi \circ \varphi$; uniqueness is not stated

separately.

Theorem 3.23 Chapter3/GlobalIFTTheoremsG · theorem_3_23. *The same under the added filler hypothesis at x instead of the nearby good position, with in addition: every zero in U is in good position.* Theorem 3.54 (1) (§III.33) supplies the open V_x of nearby good position, and the previous theorem applies with $V_{x_0} = V_x$.

Theorem 3.19, interior case. The first clause, "$f'(x_0)$ is sc-Fredholm", is Proposition 3.52 in the chart at x_0 Chapter3/GlobalIFTTheoremsG · theorem_3_19_linearization Chapter3/FredholmGermLinearizationG · proposition_3_52, applied to the sc-Fredholm germ contained in the good position: $Dg(0)$, for a derivative family Dg of the filling germ (§III.21), and $Tf(x_0) = Dg(0)|_{T_0O}$ are Fredholm on level 0 with the same index. The statement inherits the index clause of that proposition, "$= n - N$ for some natural numbers n, N", which carries no information; the index as the $n - N$ of the conjugating basic germ is §III.30 Chapter3/FredholmGermIndexG · proposition_3_52_index. *The theorem* Chapter3/GlobalIFTTheoremsG · theorem_3_19: *let $d_X \equiv 0$ Chapter2/DegeneracyIndex · MPolyfold.IsWithoutBoundary, f an sc-Fredholm section, the charts with tame base retractions, $f(x_0) = 0$, and (f, x_0) in good position with the filler condition at the nearby zeros. Then there are an open $U \ni x_0$ and d such that every zero in U is in good position (in particular $f'(y)$ is onto) and a smooth point, and $S(f, U)$ is a sub-M-polyfold whose induced structure is equivalent to a d-dimensional manifold with boundary with corners and has degeneracy index 0 everywhere. Proof.* Theorem 3.23 gives everything but the last clause; for that, Lemma 2.26, $d_S(y) \le d_X(y)$ Chapter2/SubMPolyfold · lemma_2_26, and $d_X = 0$. So the interior case is the boundary case plus Lemma 2.26; $d_X \equiv 0$ is used for nothing else. *Two readings and two omissions* (item 154), all resolved in the second addendum. (a) The text's hypothesis is "$f'(x_0)$ surjective"; good position adds condition (2) of Definition 3.53, which the text calls an empty condition when $d_X(x) = 0$. Formally condition (2) is stated in the chart for the chart's partial quadrant, so this form takes good position as in the boundary case; that the condition is void at $d_X = 0$ in a tame chart, and Theorem 3.19 with "$f'(x_0)$ surjective", are §III.45. The filler condition at the nearby zeros was the added hypothesis of Theorem 3.54 (1); the form with the text's hypotheses is §III.43. (b) "The dimension agrees with the Fredholm index": this statement quantifies d existentially; the proof takes $d = \dim\ker D(h+t)(0)$ (§III.34), and $\operatorname{ind} D(h+t)(0) = \dim P - \dim Q = \operatorname{ind} f'(x_0)$ holds for the conjugation data of the proof (§§III.18, III.30); the graph description with the conjugation exposed, and $\dim\ker = \operatorname{ind}$ for the surjective $D(h+t)(0)$, give the dimension as the index in §III.44. (c) "$\ker f'(y) = T_yS(f,U)$": in the chart at x_0, $\ker D(h+t)(z) = N_v$, the tangent space of the graph (§III.32); the chart form of the statement is §III.45, and the identification with T_yS as a subspace of T_yX needs the linear structure of T_yX (gap G.13 of Chapter 2) and is not formalized.

Theorem 3.57, with the text's hypotheses Chapter3/GlobalIFTTheoremsG · theorem_3_57. *Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with (f, x) in good position at every zero x, and let $S \neq \emptyset$. Then S is a sub-M-polyfold of X, its induced structure is tame and equivalent to that of a manifold with boundary with corners of locally constant dimension Chapter3/GlobalIFTTheorems · MPolyfold.IsManifoldWithCornersLoc (an equivalent atlas of charts on relatively open subsets of partial quadrants of finite-dimensional spaces), and at every $x \in S$ there are regular data of Definition 3.25 of a dimension $d(x)$ such that every datum at a point of S in their chart source has dimension $d(x)$. Proof.* At every $x \in S$, Theorem 3.54 (2) with $V_{x_0} = X$ — the hypothesis "every zero near x in good position" is part of the global hypothesis — gives the regular data and the local constancy. Theorem 3.27 with locally constant dimension (§III.36) Chapter3/GlobalIFTTheorems · theorem_3_27_loc then gives the sub-M-polyfold, the tameness and the corner atlas. **No hypothesis is added**: good position at every zero is the text's hypothesis, and it supplies at every point the nearby good position that the re-centring needs, so the gap of Theorem 3.54 (1) does not enter (item 156). $S \neq \emptyset$ is needed for the formal charts, whose inverses need a point of S as a value off the chart (for $S = \emptyset$ the statement says nothing). "Tame strong bundle" is read as above. The text states no dimension; the theorem says that the dimension is locally constant, which the text's tangent germ property of a fixed dimension d near each point implies but does not state (item 152).

Corollary 3.24 Chapter3/GlobalIFTTheoremsG · corollary_3_24 is Theorem 3.57 without the last clause: if at every zero $f'(x)$ is onto with kernel in good position, the solution set is a sub-M-polyfold with tame induced structure equivalent to a manifold with boundary with corners of locally constant dimension.

Theorem 3.56, $\partial X = \emptyset$, with the text's hypotheses `Chapter3/GlobalIFTTheoremsG · theorem_3_56`. *Under the hypotheses of Theorem 3.57 and $d_X \equiv 0$, S is a sub-M-polyfold whose induced structure is equivalent to a manifold with corners of locally constant dimension and has degeneracy index 0 everywhere*: Theorem 3.57 and Lemma 2.26, as for Theorem 3.19. (The statement keeps only these two conclusions; the tameness of the induced structure and the regular data are those of Theorem 3.57.) The text's hypothesis "$f'(x)$ surjective at every zero" is read as good position, which is the text's Definition 3.53 for $d_X(x) = 0$; that condition (2) is void in a tame chart at $d_X = 0$, and Theorem 3.56 with "$f'(x)$ surjective" and the tameness of the induced structure, are §III.45.

The parts of fixed dimension `Chapter3/GlobalIFTTheoremsG · theorem_3_57_dim`. Under the hypotheses of Theorem 3.57 (without $S \neq \emptyset$), let S_n be the set of points of S at which S has data of Definition 3.25 of dimension n `Chapter3/GlobalIFTTheorems · solutionSetDim`, and U_n the union of the sources of the regular data of dimension n along which the dimension is constantly n `Chapter3/GlobalIFTTheorems · solutionSetDimOpen`, an open set `Chapter3/GlobalIFTTheorems · isOpen_solutionSetDimOpen`. *Then S is the union of the S_n, the S_n are pairwise disjoint, $S_n = S \cap U_n$ and $S_n = S \cap F_n$ for a closed F_n (so S_n is open and closed in S), and each nonempty S_n is a sub-M-polyfold whose induced structure is tame and equivalent to an n-dimensional manifold with boundary with corners. Proof.* At every $x \in S$, Theorem 3.54 (2) with $V_{x_0} = X$ gives regular data D_x of a dimension $d(x)$ with the local constancy. *The dimension at a point is well defined*: two data at x of dimensions n, m both have the dimension $d(x)$ by the local constancy along D_x, since x lies in its source; so the S_n cover S (by D_x) and are pairwise disjoint. $S_n = S \cap U_n$: for $x \in S_n$, $d(x) = n$ and D_x is one of the data whose sources make up U_n; conversely, if $x \in S$ lies in the source of regular data of dimension n along which the dimension is n, then $d(x) = n$ and D_x witnesses $x \in S_n$. *S_n is closed in S*: $S_n = S \setminus \bigcup_{m \neq n} U_m$, since a point of S lies in exactly one S_m; take F_n the complement of that open union. *Each nonempty S_n*: at $x \in S_n$, restrict D_x (of dimension n) to a datum for S_n — $S_n \cap$ source $= S \cap$ source because every point of S in the source has dimension n — still regular `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.shrink_isRegular`, and apply Theorem 3.27 with fixed dimension n `Chapter3/TangentGerm · theorem_3_27`.

Where the formal route departs from the text. Two steps of the text's proof are replaced (items 150, 157): the regularity of the solution germ away from the centre, which the proof of Theorem 3.27 asserts for any set and which fails for general sets (§III.24), is proved for solution sets by re-centring with the C^1-property of σ on V' and the level-wise lifts at the nearby zero (§§III.35–III.36); and the relative openness of Σ, false for an arbitrary good complement, is obtained by choosing the complement (§III.29). With these, the global theorems 3.56, 3.57 and Corollary 3.24 are proved with the text's hypotheses, and the local Theorems 3.54, 3.23 and 3.19 under the hypothesis that the zeros near the point are in good position, which is Theorem 3.54 (1); this section proves that one under the filler condition at the nearby zeros, which Definition 3.4 does not impose, and §III.43 with the text's hypotheses. Not stated here: the uniqueness of the manifold structure separately from the equivalence of atlases; the dimension d of these statements is existentially quantified (only as the common dimension of all data of Definition 3.25 for S near the point) — its identification with $\dim \ker f'(x) = \operatorname{ind} f'(x)$ is §III.44, and $\ker f'(y) = T_y S$ in the chart is §III.45.

III.38 The basic germ criterion: Theorem 3.58

Theorem 3.58 (text): let U be a relatively open neighbourhood of 0 in a partial quadrant $C \subseteq E$ and $f : U \to F$ sc-smooth, with (1) $Df(x)$ sc-Fredholm at the smooth points, of constant index; (2) an sc-splitting $E = B \oplus X$, B finite-dimensional containing $\ker Df(0)$, with a compactness property: for $b_j \to b$ in $B \cap U$ small and (η_j) bounded in X_m with $Df(b_j)\eta_j = y_j + z_j$, $y_j \to 0$ in F_m, (z_j) bounded in F_{m+1}, (η_j) has a convergent subsequence in X_m; (3) for every m and $\varepsilon > 0$, $|[D_2 f(b,0) - D_2 f(b,x)]h|_m \leq \varepsilon |h|_m$ for b small and $x \in X_{m+1}$ small on level m. Then $f - f(0)$ is conjugated near 0 to a basic germ. The proof: $Y = Df(0)X$, $F = A \oplus Y$, P the projection onto Y; Lemmas 3.59, 3.60 and an index argument make $L(b) = P \circ Df(b)|_X$ sc-isomorphisms with uniform lower bounds; Lemma 3.61 makes $\Phi((b,x),(c,y)) = ((b,x),(c,L(b)^{-1}y))$ a strong bundle isomorphism; and

$H(b,x) = x - L(b)^{-1}Ph(b,x)$, $h = f - f(0)$, is a contraction by (3).

The coordinates. The formalization works in the coordinates of the text's proof: $E = \mathbb{R}^n \oplus X$ with $\mathbb{R}^n$ a finite-dimensional normed space P (constant sc-structure), $C = \widetilde{C} \oplus X$ with $\widetilde{C}$ a partial quadrant of P, $F = \mathbb{R}^N \oplus Y$ with $\mathbb{R}^N = Q$ finite-dimensional, and "$P\circ$" the Y-component. Condition (3) becomes `Chapter3/BasicGermCriterionProof · DerivCondition`: for every m and $\varepsilon > 0$ there is $r > 0$ with $|Df(b,0)(0,h) - Df(b,x)(0,h)|_m \le \varepsilon|h|_m$ for $h \in X_m$, $b \in \widetilde{C}$, $|b| < r$, $x \in X_{m+1}$, $|x|_m < r$, $(b,0)$ and (b,x) in U_{m+1} (the level operators of Proposition 1.16). Condition (2) becomes `Chapter3/BasicGermCriterionIndex · CompactnessCondition`: there is $r > 0$ such that for every m, every $b_j \to b_0$ in $\widetilde{C}$ with $(b_j,0) \in U$ and $|b_0| < r$, every bounded $(\eta_j) \subseteq X_m$ with $Df(b_j,0)(0,\eta_j) = y_j + \iota z_j$, $y_j \to 0$ in F_m and (z_j) bounded in F_{m+1}, (η_j) has a subsequence converging in X_m. (The smallness of b is read as uniform in m, as the order of the text's quantifiers suggests; Lemma 3.60 needs the property on every level on one neighbourhood.)

Theorem 3.58 as formalized `Chapter3/BasicGermCriterionIndex · theorem_3_58`. Let U be relatively open in $\widetilde{C} \oplus X$ with $0 \in U$ and $f : U \to Q \oplus Y$ sc-smooth such that (1) for every $b \in \widetilde{C}$ with $(b,0) \in U$ the linearization $Df(b,0)$, an sc-operator `Chapter3/BasicGermCriterionIndex · derivParamOp`, is sc-Fredholm with the sc-index of $Df(0)$; (2) $h \mapsto (Df(0)(0,h))_2$ is bijective $X_0 \to Y_0$, and the compactness condition holds; (3) the derivative condition holds. Then there are an open $V_o \ni 0$ in P and sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ for $b \in V_o \cap \widetilde{C}$, such that the push-forward $k(b,x) = ((f - f(0))_1(b,x), L(b)^{-1}(f - f(0))_2(b,x))$ is a basic germ on $U \cap (V_o \oplus X)$ with contraction map $H(b,x) = x - L(b)^{-1}(f(b,x) - f(0))_2$, Φ is a strong bundle isomorphism covering the identity, and $f - f(0)$ is conjugated to a basic germ (Definition 3.11). The hypotheses differ from the text's in three places: (1) is asked only at the points $(b,0)$, $b \in \widetilde{C}$, which is all the proof uses; "constant index" is equality with the index at 0; and the text's "$B \supseteq \ker Df(0)$, $Y = Df(0)X$" is replaced by its consequence, the bijectivity of $P \circ Df(0)|_X$ on level 0. The reduction of a general sc-splitting $E = B \oplus X$, $F = A \oplus Y$ to these coordinates is not formalized.

The proof has two halves, as in the text.

First half: Theorem 3.58 from the family of sc-isomorphisms.

Theorem 3.58, given the family `Chapter3/BasicGermCriterionProof · theorem_3_58_of_family`. In addition to the coordinates, f sc-smooth on U and (3), suppose: V_o open with $0 \in V_o$ and $(b,0) \in U$ for $b \in V_o \cap \widetilde{C}$; sc-isomorphisms $L(b) : X \to Y$ with $L(b)h = (Df(b,0)(0,h))_2$ on level 0 for $b \in V_o \cap \widetilde{C}$; and for every m a constant $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for $b \in V_o \cap \widetilde{C}$, $h \in X_m$. Then, with $U' = U \cap (V_o \oplus X)$, the conclusions above hold.

Step 1: $(b,h) \mapsto L(b)h$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus X$ `Chapter3/BasicGermCriterionProof · famMap_isScSmooth_of_deriv` (the text: "since B is a finite-dimensional sc-smooth space, the map $(b,x) \mapsto P \circ Df(b)x$ is sc-smooth"). It is $\mathrm{pr}_Y \circ Tf \circ J$ with the sc-operator $J(b,h) = ((b,0),(0,h))$ into the tangent space `Chapter3/BasicGermCriterionProof · tangentInjOp`; J maps into the tangent set because $(b,0) \in U$ lies on every level; and the tangent map of the sc-smooth f is sc-smooth.

Step 2: $(b,y) \mapsto L(b)^{-1}y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ `Chapter3/BasicGermCriterionProof · famInv_isScSmooth_of`. This is Lemma 3.61 (§III.25) `Chapter3/InverseFamily · lemma_3_61` for the germ $\{V_o \cap \widetilde{C}\}$ with the uniform lower bounds, translated between maps on sets and germs on the special germ of a set: an sc^k-germ on $\{U\}$ is an sc^k-map on U `Chapter3/BasicGermCriterionProof · IsSckGerm.isSck_ofSet` `Chapter3/BasicGermCriterionProof · IsScSmoothGerm.isScSmooth_ofSet`, by induction on k: the sc^0 case is a matter of definitions `Chapter3/BasicGermCriterionProof · IsSc0Germ.isSc0_ofSet`; for the step, the tangent germ of $\{U\}$ is the germ of the tangent set, and the derivative family of the germ agrees with the derivative of the map on the tangent set; the converse is `IsSck.isSckGerm` (§III.1).

Step 3: the strong bundle isomorphism Φ. For a family $M(b) : Y \to X$ of sc-isomorphisms put

$\Phi_M((b,z),(c,y)) = ((b,z),(c,M(b)y))$ Chapter3/BasicGermCriterionProof · fiberIsoMap. If $(b,y) \mapsto M(b)y$ is sc-smooth on $(V_o \cap \widetilde{C}) \oplus Y$ and the parameters of U' lie in $V_o \cap \widetilde{C}$, Φ_M is a strong bundle map $U' \lhd (Q \oplus Y) \to U' \lhd (Q \oplus X)$ covering the identity Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleMap: it is linear on the fibres; it preserves the filtration because $M(b)$ maps Y_k to X_k; on $[0]$ it is sc-smooth as the pairing of the projection with $(c, M(b)y)$, the latter the composite of the sc-operator $((b,z),(c,y)) \mapsto (b,y)$ Chapter3/BasicGermCriterionProof · paramFiberOp with $(b,y) \mapsto M(b)y$ Chapter3/BasicGermCriterionProof · fiberIsoMap_isScSmooth; on $[1]$, $\Phi_M[1]$ is Φ_{M^1} for the shifted family $M^1(b) = M(b) : Y^1 \to X^1$ Chapter3/BasicGermCriterionProof · shiftFam Chapter3/BasicGermCriterionProof · shiftFiber_fiberIsoMap, and $(b,y) \mapsto M^1(b)y$ is sc-smooth by Proposition 1.17 Chapter3/BasicGermCriterionProof · famMap_shiftFam_isScSmooth — the text's "if we raise the index by one, the map is also sc-smooth by Proposition 1.17". With $M = L^{-1}$ (Step 2) and $M = L$ (Step 1) the two maps are inverse to each other, so $\Phi = \Phi_{L^{-1}}$ is a strong bundle isomorphism Chapter3/BasicGermCriterionProof · fiberIsoMap_isStrongBundleIso.

Step 4: the push-forward and the contraction map. With $h = f - f(0)$ Chapter3/BasicGermCriterionProof · shiftedGerm, $\Phi_* h = k$, $k(b,x) = (h_1(b,x), L(b)^{-1}h_2(b,x))$ Chapter3/BasicGermCriterionProof · pushGerm358, and the X-component of k is $x - H(b,x)$ with $H(b,x) = x - L(b)^{-1}h_2(b,x)$ Chapter3/BasicGermCriterionProof · contrMap358 Chapter3/BasicGermCriterionProof · pushGerm358_snd. Both are sc-smooth on U': sums, pairings and the composite of $(b,y) \mapsto L(b)^{-1}y$ with $(b,x) \mapsto (b, h_2(b,x))$. On level m, $H_m(b,w) = w - L(b)^{-1}_m((f_m(b,w))_2 - (f_m(0))_2)$ Chapter3/BasicGermCriterionProof · levelMap_contrMap358, and $|L(b)^{-1}_m| \le c_m^{-1}$ Chapter3/BasicGermCriterionProof · norm_symm_le.

Step 5: the contraction estimate Chapter3/BasicGermCriterionProof · contrMap358_contraction. For every m and $\varepsilon > 0$ there is $r > 0$ such that $(a,w) \in U'_m$ and $|H(a,w) - H(a,w')|_m \le \varepsilon |w - w'|_m$ for $a \in \widetilde{C}$, $|a| < r$, $w, w' \in X_m$ small on level m. Choose r below the radius r_3 of (3) for εc_m, below the radius of a ball around 0 in V_o, and so small that $(a,z) \in U_m$ whenever $|a| < r$ and $|z|_m < r$ ($U = V \cap C$ with V open; the level-m norm of z controls its level-0 norm through the norm of the inclusion) Chapter3/BasicGermCriterionProof · mem_lvlSet_of_small. First let $w_1, w_1' \in X_{m+1}$ with $|w_1|_m, |w_1'|_m < r$. On the convex set $S = \{z \in X_{m+1} : |z|_m < r\}$ the map $g(z) = f_m(a, \iota z)$ has the derivative $Df(a,z)(0,\iota\,\cdot)$ (Proposition 1.16 on level m) Chapter1/Levels · IsSc1With.hasFDerivWithinAt_level, and by (3) $|Df(a,z)(0,v) - Df(a,0)(0,v)|_m \le \varepsilon c_m |v|_m$. **The mean value inequality with the derivative bound in the weaker norm** Chapter3/BasicGermCriterionProof · norm_sub_sub_le_of_hasFDerivWithinAt_segment — the one-dimensional mean value inequality for $t \mapsto g(w_1' + t(w_1 - w_1')) - T\iota(w_1' + t(w_1 - w_1'))$ on $[0,1]$, which measures the increment in the norm of X_m rather than of X_{m+1}, as condition (3) requires — gives

$$|g(w_1) - g(w_1') - T\iota(w_1 - w_1')|_m \le \varepsilon c_m |\iota(w_1 - w_1')|_m, \qquad T = Df(a,0)(0,\cdot).$$

The Y-component of T is $L(a)_m$ Chapter3/BasicGermCriterionProof · snd_levelDeriv_eq_L (the level-m operator of Proposition 1.16 restricts the level-0 one). Then $H(a,w_1) - H(a,w_1') = L(a)^{-1}[L(a)(w_1 - w_1') - (f(a,w_1) - f(a,w_1'))_2]$, whose norm is at most $c_m^{-1}\varepsilon c_m |w_1 - w_1'|_m$. This is the text's integral $\int_0^1 D_2 f(b, tx + (1-t)x')(x - x')\,dt$, made precise. For general $w, w' \in X_m$: approximate them by sequences from X_{m+1} (dense in X_m), small on level m for large indices, and pass to the limit with the continuity of H_m on U'_m — the text's "by the density of X_{m+1} in X_m".

Step 6: the basic germ and the conjugation. k is a basic germ on the germ $\{U'\}$ with contraction map H (Definition 3.7 as formalized Chapter3/LocalCompactness · IsBasicGerm): it is an sc-smooth germ (Step 4), $k(0) = 0$, its X-component is $x - H(b,x)$, and H is an sc^0-contraction germ — sc^0, $H(0) = 0$ since $L(0)^{-1}0 = 0$, and the contraction estimate on every level with every ε (Step 5). The conjugation identity $(\Phi(u, h(u)))_2 = k(u)$ holds on all of U', because Φ covers the identity; with Φ from Step 3 these are the data of Definition 3.11 for $h = f - f(0)$.

Second half: the family from conditions (1) and (2).

The family. At the smooth point $(b,0)$ the linearization $Df(b,0)$ is an sc-operator Chapter3/

BasicGermCriterionIndex · derivParamOp Chapter1/Properties · IsSc1With.scOperator_deriv whose level maps are the level operators of Proposition 1.16 Chapter3/BasicGermCriterionIndex · derivParamOp_lvl; $L(b) = \mathrm{pr}_Y \circ Df(b,0) \circ \mathrm{inr}_X$ is an sc-operator $X \to Y$ (set to 0 where $(b,0) \notin U$) Chapter3/BasicGermCriterionIndex · derivFamOp; and $(b,h) \mapsto L(b)h$ is jointly continuous on $O \times X_m$ for parameters with $(b,0) \in U$, by the continuity of the level operators (Proposition 1.16 (2)) Chapter3/BasicGermCriterionIndex · continuousOn_levelDeriv_param Chapter3/BasicGermCriterionIndex · continuousOn_derivFamOp.

The compactness property of the family Chapter3/BasicGermCriterionIndex · familyCompactness_derivFamOp. Let O be a set of parameters in $\widetilde{C}$ with $(b,0) \in U$ and $|b| < r$ (r from (2)), $b_j \to b_0$ in O, (h_j) bounded in X_m, $L(b_j)h_j \to 0$. Write $Df(b_j,0)(0,h_j) = (0, L(b_j)h_j) + ((Df(b_j,0)(0,h_j))_Q, 0)$. The first summand tends to 0 in F_m. The second lies in $Q \oplus 0$, where all levels coincide, and is bounded: a jointly continuous family of operators is uniformly bounded near a parameter Chapter3/BasicGermCriterionIndex · exists_bound_of_continuousOn_clm (continuity at $(b_0,0)$, then scaling), so $|Df(b_j,0)(0,\cdot)|$ is bounded for j large. Condition (2) applied to the tails of the sequences gives a convergent subsequence: this is the compactness property of the abstract Lemmas 3.59, 3.60 (§III.25) Chapter3/BasicGermCriterion · FamilyCompactness. The text's "$z_j = Df(b_j)h_j \in A$ is bounded on every level since A is a smooth subspace" omits the uniform operator bound.

Lemma 3.59. Let $V' = \{b : (b,0) \in V\}$ ($U = V \cap C$) and $O = V' \cap B(0,r) \cap \widetilde{C}$, which contains 0. $L(0)$ is injective on X_0 by (2), so Lemma 3.59 Chapter3/BasicGermCriterion · lemma_3_59 gives $r_1 > 0$ with $L(b)$ injective on X_0 for $b \in O$, $|b| < r_1$, hence on every level (the inclusions are injective and commute with $L(b)$) Chapter3/BasicGermCriterionIndex · ScOperator.injective_lvl.

Lemma 3.60. Take a closed ball $\bar{B}(0,r_1')$ inside $V' \cap B(0,r) \cap B(0,r_1)$ and $K = \bar{B}(0,r_1') \cap \widetilde{C}$, compact (since $\widetilde{C}$ is closed) and inside O. On K every $L(b)_m$ is injective, the family is jointly continuous and has the compactness property (restricted from O Chapter3/BasicGermCriterionIndex · FamilyCompactness.mono), so Lemma 3.60 Chapter3/BasicGermCriterion · lemma_3_60 gives $c_m > 0$ with $c_m|h|_m \le |L(b)h|_m$ for $b \in K$.

The index argument. The text: "$P \circ Df(b)|_X$ is an injective sc-Fredholm operator of index 0, hence an sc-isomorphism". The formal proof uses a block computation instead. **The index of a block operator** Chapter3/BasicGermCriterionIndex · fredholmIndex_block: for a Fredholm $T : P \oplus X \to Q \oplus Y$ with P, Q finite-dimensional and the block $L = \mathrm{pr}_Y \circ T \circ \mathrm{inr}_X$ of finite index, $\mathrm{ind}\, T = \dim P + \mathrm{ind}\, L - \dim Q$. Indeed $T - \mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is $(p,x) \mapsto T(p,0) + ((T(0,x))_Q, 0)$, of finite rank, hence compact, so $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ is Fredholm with the index of T Chapter1/FredholmPerturbation · isFredholm_add_isCompactOperator Chapter1/FredholmIndex · fredholmIndex_add_isCompactOperator; and the indices of pr_X and inr_Y are $\dim P$ and $-\dim Q$ Chapter3/BasicGermCriterionIndex · hasFiniteIndex_snd_and_linIndex Chapter3/BasicGermCriterionIndex · hasFiniteIndex_inr_and_linIndex, which add under composition Chapter1/FredholmIndex · linIndex_comp. At $b = 0$, $L(0)$ is bijective on level 0, of index 0, so $\mathrm{ind}\, Df(0) = \dim P - \dim Q$ on level 0, which is the sc-index (an sc-Fredholm operator has the same index on every level) Chapter1/FredholmIndex · IsScFredholm.fredholmIndex_lvl. **Surjectivity from the index** Chapter3/BasicGermCriterionIndex · surjective_of_fredholmIndex_block: if $\mathrm{ind}\, T = \dim P - \dim Q$ and the block L is injective, then L has finite index — its cokernel is a quotient of the cokernel of the Fredholm operator $\mathrm{inr}_Y \circ L \circ \mathrm{pr}_X$ — and index 0 by the block formula, so it is surjective. For $b \in K$ and every m, $Df(b,0)_m$ is Fredholm of index $\mathrm{ind}\, Df(0) = \dim P - \dim Q$ by (1) and the level-wise index, and $L(b)_m$ is injective; so $L(b)_m$ is bijective, a topological isomorphism by the open mapping theorem, and the $L(b)$, $b \in K$, are sc-isomorphisms. The sc-Fredholm property of the projection P and the composition rule for sc-Fredholm operators are not needed.

Conclusion. Extend $b \mapsto L(b)$ outside K by $L(0)$, put $V_o = B(0,r_1')$ (so $V_o \cap \widetilde{C} \subseteq K$), and apply the first half with the lower bounds of Lemma 3.60. The neighbourhood replacement $U' = U \cap (V_o \oplus X)$ is the text's "replacing, if necessary, U by a smaller neighbourhood".

III.39 What the addendum taught us

1. **The global implicit function theorems hold with the text's hypotheses.** Theorem 3.57, Corollary 3.24 and Theorem 3.56 are proved for an sc-Fredholm section of a strong bundle whose charts have tame base retractions, with good position at every zero `Chapter3/GlobalIFTTheoremsG · theorem_3_57` `Chapter3/GlobalIFTTheoremsG · corollary_3_24` `Chapter3/GlobalIFTTheoremsG · theorem_3_56` (§III.37). Theorem 3.56 reads "$f'(x)$ surjective" as good position, which the text's Definition 3.53 makes the same thing at points with $d_X(x) = 0$; that the kernel condition is void in the chart form is not formalized.
2. **Two steps of the text's proof had to be replaced, and both repairs are specific to solution sets.** The regularity of the solution germ away from the centre, which the proof of Theorem 3.27 asserts for any set and which fails in general (§III.24), follows for two graph descriptions of one set from the C^1-property of the first and the level-wise lifts of the second, by the re-centring lemma `Chapter3/SolutionGermRecentre · recentre_contDiffOn_levels` (§III.35). The relative openness of Σ, false for an arbitrary good complement, holds for a complement whose projection maps the partial quadrant into itself, and the proof of Theorem 3.54 can produce such a complement from the start `Chapter3/GoodComplementProjection · exists_goodComplement_projectsInto` (§III.29). Read off the statement of the regularity theorem `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`, the first repair works for every set with the tangent germ property whose data satisfy the second regularity condition (§III.36).
3. **The proof of the local Theorem 3.54 (1) has a gap, and it is the same gap as in the book's arXiv version.** At a nearby zero, Corollary 3.41 gives the surjectivity of the filled linearization; the surjectivity of $f'(y)$ needs condition (3) of Definition 3.4 at that zero, which the definition imposes at the centre only `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`. The local theorems were therefore first proved under that condition at the nearby zeros `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled` `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one` (§III.33); the kernel half needs nothing (§III.32). The global theorems do not need it, because their hypothesis already gives good position at every zero. The second addendum proves the surjectivity with Definition 3.4 as stated (§§III.41–III.43).
4. **Definition 3.53 presupposes a chart independence the text does not prove.** It holds at zeros, because only the fibre part of a chart transition survives there `Chapter3/LinearizationCharts · linAt_chart_change` (§III.31), and it is what lets the good position found in the chart at x_0 be transported to the chart centred at a nearby zero.
5. **Three silent steps are theorems of their own**: tameness is carried by sc-diffeomorphisms and by open subsets `Chapter2/TameInvariance · IsScDiffeoQuadrant.isTameRetract_image` (§III.27); the kernel of the linearization of a basic germ plus an sc$^+$-germ consists of smooth points `Chapter3/SolutionGermRegularity · StabilityData.ker_Dfs_le_smooth` (§III.34); and the index raise of a strong bundle isomorphism is one `Chapter3/BasicClassComposition · IsStrongBundleIso.shift` (§III.28).
6. **The dimension of the solution set is locally constant, not constant**, and the dimension count that proves the re-centring is also what shows it (Sard's lemma in fixed dimension `Chapter3/SolutionGermRecentre · finrank_le_of_interior_image_nonempty`, §III.35). Theorem 3.27 is therefore also proved with a dimension that may vary `Chapter3/GlobalIFTTheorems · theorem_3_27_loc` (§III.36), and the solution set of Theorem 3.57 splits into open and closed parts of fixed dimension `Chapter3/GlobalIFTTheoremsG · theorem_3_57_dim`.
7. **Proposition 3.52's index needs the conjugation as data.** With the witnesses of the conjugation named, the index of $Tf(x_0)$ is the $n - N$ of every conjugating basic germ, and that number does not depend on the conjugation `Chapter3/FredholmGermIndexG · proposition_3_52_index` `Chapter3/FredholmGermIndexG · ConjugationData.index_eq` (§III.30).
8. **The quadrant inverse function theorem is not the right tool away from the centre**: it needs the degeneracy index to be preserved, which for the solution set is unknown; Lemma 3.77's bootstrapping with the hypotheses its proof uses suffices `Chapter3/SolutionGermRecentre · InvDataW.contDiffOn_inv` (§III.35).
9. **The basic germ criterion is proved in the coordinates of its proof**, with condition (1) needed only along the finite-dimensional directions and the text's index argument replaced by a block index

formula Chapter3/BasicGermCriterionIndex · fredholmIndex_block Chapter3/BasicGermCriterionIndex · theorem_3_58 (§III.38).

10. **What was left of Chapter 3** after this addendum: the surjectivity in Theorem 3.54 (1) with Definition 3.4 as stated; in Theorem 3.19, the dimension as the Fredholm index and ker $f'(y) = T_y S$; Proposition 3.52 as an sc-Fredholm operator on $T_x X$; Theorem 3.58 for a general sc-splitting; Theorem 3.18; Lemma 3.70 for a general presentation of the partial quadrant; and Theorem 3.27 for sets whose data violate the second regularity condition. The second addendum (§§III.40–III.46) settles the first three items, the third in charts, and Theorem 3.18.

III.40 Theorem 3.18: parametrized perturbations

The text states Theorem 3.18 after a five-line argument: the section $\tilde{f}(r,x) = ((r,x), f(x))$ of the pull-back bundle $\pi^*P \to \mathbb{R}^n \times X$, $\pi(r,x) = x$, "is a sc-Fredholm section as is readily verified", $\tilde{s}(r,x) = ((r,x), \sum_i r_i s_i(x))$ is an sc$^+$-section, and Theorem 3.15 adds them. The formalization follows this route. What it has to build first is everything the five lines take for granted: $\mathbb{R}^n$ as an M-polyfold with enough charts, the pull-back bundle as a strong bundle, the sc-calculus for a scalar factor, the product chart with its filler, and the extension of a basic germ by n inert parameters (Findings 160–167). Tameness is not used anywhere, as for Theorem 3.15.

$\mathbb{R}^n$ and $\mathbb{R}^n \times X$. $\mathbb{R}^n$ carries the constant sc-structure Chapter3/ParamPerturbation · rnSc and the local model $(\mathbb{R}^n, \mathbb{R}^n, \mathbb{R}^n)$ Chapter3/ParamPerturbation · rnModel. For every a the translation $r \mapsto r - a$ is a chart Chapter3/ParamPerturbation · rnChart; two translation charts are compatible, because their transition maps are translations and a C^∞ map between finite-dimensional spaces is sc-smooth for the constant structures Chapter3/ParamPerturbation · isScSmoothRetractMap_const_of_contDiff Chapter3/ParamPerturbation · rnChart_compatible. The M-polyfold $\mathbb{R}^n$ has the identity chart as its atlas Chapter3/ParamPerturbation · rnPolyfold, and every translation chart is a chart of it Chapter3/ParamPerturbation · isChart_rnChart. These charts are needed: Definition 3.11 asks for a chart *centred* at the smooth point (r_0, x), and the product of the identity of $\mathbb{R}^n$ with a chart of X is centred only at $r_0 = 0$. $\mathbb{R}^n$ is tame Chapter3/ParamPerturbation · rnPolyfold_isTame and every point lies on every level Chapter3/ParamPerturbation · rnPolyfold_levelSet; so (r,x) is on level m of the product iff x is Chapter3/ParamPerturbation · prod_rn_mem_levelSet_iff, smooth iff x is Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff, and $\mathbb{R}^n \times X$ is tame when X is Chapter3/ParamPerturbation · prod_rn_isTame — which the text needs, since Definition 3.14 is stated for tame bundles.

Two rules of the sc-calculus. Scalar multiplication $\mathbb{R} \oplus F \to F$ is continuous and bilinear on every level, hence sc-smooth by Proposition 1.19 Chapter3/ParamPerturbation · isScSmooth_smul; composed with a pairing, $x \mapsto a(x)v(x)$ is sc-smooth for sc-smooth $a : U \to \mathbb{R}$ and $v : U \to F$ Chapter3/ParamPerturbation · IsScSmooth.smul; finite sums of sc-smooth maps are sc-smooth Chapter3/ParamPerturbation · IsScSmooth.finsetSum. The coordinate $r \mapsto r_i$ is an sc-operator Chapter3/ParamPerturbation · rnCoordOp. Chapter 1 states neither rule.

The pull-back bundle. Proposition 2.54 is formalized as an existence statement Chapter2/StrongBundlePullback · proposition_2_54; here it is made a structure, the strong bundle π^*P with the atlas of all pulled-back charts Chapter3/ParamPerturbation · StrongBundle.pullback, whose total space is paracompact by Theorem 2.16 Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace. A pulled-back chart is made from a chart ψ of the base Z around z and a strong bundle chart c of P around $\pi(z)$, with π mapping the domain of ψ into that of c Chapter3/ParamPerturbation · exists_pullbackChartData; such a chart is a chart of π^*P as soon as c is merely compatible with the atlas of P Chapter3/ParamPerturbation · PullbackChartData.isChart_toChart. A map $\tau : Z \to Y$ over π is a section $z \mapsto (z, \tau(z))$ of π^*Y Chapter3/ParamPerturbation · pullbackLift Chapter3/ParamPerturbation · isSection_pullbackLift, whose value in a pulled-back chart is $(\psi(z), \mathrm{pr}_F \Phi(\tau(z)))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift. For $\tau = \sigma \circ \pi$ the representative is $w \mapsto (w, \hat{\sigma}(g(w)))$ with $g = \varphi \circ \pi \circ \psi^{-1}$, which maps the retract of ψ into that of φ Chapter3/ParamPerturbation · PullbackChartData.g_mem_0 Chapter3/ParamPerturbation · PullbackChartData.toChart_pullbackLift_comp; so the pull-back of an sc-smooth section

along an sc-smooth map is sc-smooth Chapter3/ParamPerturbation · StrongBundle.IsScSmoothSection.pullback (Finding 167).

The three sections. $F(r,x) = f(x) + \sum_i r_i s_i(x)$ is computed in the fibre $P^{-1}(x)$, from the fibre combination $\sum_i r_i s_i(x)$ Chapter3/ParamPerturbation · fiberComb, and read in the fibre of π^*Y over (r,x) Chapter3/ParamPerturbation · paramSection; $\tilde{f}$ Chapter3/ParamPerturbation · paramLift and $\tilde{s}$ Chapter3/ParamPerturbation · paramComb are the text's sections, and $F = \tilde{f} + \tilde{s}$ in the fibres of π^*Y Chapter3/ParamPerturbation · paramSection_eq_sectionAdd.

$\tilde{s}$ is an sc⁺-section Chapter3/ParamPerturbation · StrongBundle.isScPlusSection_paramComb. The atlas of π^*P contains the charts pulled back through *every* chart ψ of $\mathbb{R}^n \times X$, not only product charts, so the check must work for an arbitrary ψ. In a pulled-back chart, by the linearity of Φ on the fibre over $x = \pi(z)$, $\tilde{s}(z)$ has the value $(\psi(z), \sum_i r_i \hat{s}_i(x))$ Chapter3/ParamPerturbation · PullbackChartData.toChart_paramComb, so the principal part is $w \mapsto \sum_i \lambda_i(w)\hat{s}_i(g(w))$ with $\lambda(w) = (\psi^{-1}(w))_1$. The map λ is sc-smooth on the retract of ψ: it is the projection $\mathbb{R}^n \times X \to \mathbb{R}^n$ (an sc-smooth map of M-polyfolds) read in ψ and in the identity chart of $\mathbb{R}^n$ Chapter3/ParamPerturbation · isScSmooth_chart_coord. Each $\hat{s}_i \circ g$ is sc-smooth into F^1, s_i being sc⁺. The product rule and the sum rule conclude.

The product chart and its filler. For a strong bundle chart c of P and $r_0 \in \mathbb{R}^n$, the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd is $\psi = (r \mapsto r - r_0) \times \varphi$ on $\mathbb{R}^n \times V$, with the retraction $1 \times r$ of $\mathbb{R}^n \times U$ onto $\mathbb{R}^n \times O$; there $g = \varphi \circ \pi \circ \psi^{-1}$ is the projection $(a,u) \mapsto u$. *Filler conditions.* If g fills $\hat{f}$ in c, then $(a,u) \mapsto g(u)$ fills $(a,u) \mapsto \hat{f}(u)$ in the product chart Chapter3/ParamPerturbation · PullbackChartData.ofProd_isFillerOf. Conditions (1) and (2) are those of g, because the fibre projection of the product retraction at (a,u) is $\rho(r(u))$ — which uses $\varphi(\varphi^{-1}(r(u))) = r(u)$ on O. Condition (3): $D(1 \times r)(0) = 1 \times Dr(0)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_scDeriv_r, so its kernel is $0 \times \ker Dr(0)$; the fibre projection at 0 is $\rho(0)$ (from $r(0) = 0$); and $[1 - \rho(r(u))]g(u)$ does not depend on a, so its derivative at 0 on $0 \times \ker Dr(0)$ is that of g. *For a filling germ g on G* the same holds on the product germ $\mathbb{R}^n \times G$, with levels $\mathbb{R}^n \times G_m$, unbounded in the parameter direction, which Definition 3.1 allows Chapter3/GermCalculusParam · ScNbhdGerm.univProd: $(a,u) \mapsto g(u)$ is a germ on it with the derivative family $Dg(u) \circ \mathrm{pr}_2$ Chapter3/GermCalculusParam · IsSc1GermWith.univProd, and the fill defect in the product chart is the pull-back along pr_2 of that of g, so condition (3) is the kernel identification just made, without a uniqueness argument Chapter3/GermCalculusParam · PullbackChartData.ofProd_isFillingGerm. *Principal parts* ($a = r - r_0$ the chart coordinate): $\tilde{s}$ has $(a,u) \mapsto \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramComb and F has $(a,u) \mapsto \hat{f}(u) + \sum_i (a + r_0)_i \hat{s}_i(u)$ Chapter3/ParamPerturbation · PullbackChartData.ofProd_principal_paramSection, the text's $(r,u) \mapsto \hat{f}(u) + \sum_i r_i \hat{s}_i(u)$.

Inert parameters. Let Λ be finite-dimensional. The product $1_\Lambda \times \Phi$, $((a,u),k) \mapsto ((a,\varphi(u)),\Gamma(u,k))$, of a map Φ of $U \lhd F$ with the identity of Λ Chapter3/ParamPerturbation · paramMap is a strong bundle map when Φ is: on $[0]$ and $[1]$ it is $1 \times \Phi$ up to the reassociations of the factors, which are sc-operators Chapter3/ParamPerturbation · ScBanach.assocOp Chapter3/ParamPerturbation · ScBanach.assocInvOp Chapter3/ParamPerturbation · IsStrongBundleMap.paramMap; and a strong bundle isomorphism, with inverse $1_\Lambda \times \Phi'$, when Φ is Chapter3/ParamPerturbation · IsStrongBundleIso.paramMap. The Λ-extension of a germ of neighbourhoods is $U''_m = \{((a,\lambda),w) \mid (a,w) \in U_m\}$ Chapter3/ParamPerturbation · ScNbhdGerm.paramExtend, unbounded in λ, which Definition 3.1 allows. *The Λ-extension of a basic germ is basic* Chapter3/ParamPerturbation · IsBasicGerm.paramExtend: if $h(a,w) = (h_1(a,w), w - B(a,w))$ is basic on $\widetilde{C} \oplus W$, then $h'((a,\lambda),w) = h(a,w)$ is basic on $(\widetilde{C} \times \Lambda) \oplus W$, with $B'((a,\lambda),w) = B(a,w)$: it is the composite of h with the projection forgetting λ Chapter3/ParamPerturbation · paramForgetOp, and the contraction estimate of Definition 3.28 holds with the same radius since $|a| \le |(a,\lambda)|$. *The Λ-extension of a conjugation* Chapter3/ParamPerturbation · IsConjugateToBasicGerm.paramExtend: if k is conjugated to h by Ψ, then $(\lambda,u) \mapsto k(u)$ is conjugated to h' by $\Psi''((\lambda,u),k) = ((((\psi(u))_1,\lambda),(\psi(u))_2),\Gamma(u,k))$, the composite Chapter3/BasicClassComposition · IsStrongBundleIso.comp of $1_\Lambda \times \Psi$ with the linear reassociation

$(\lambda,(a,w)) \mapsto ((a,\lambda),w)$ `Chapter3/ParamPerturbation · paramSwapOp` `Chapter3/ParamPerturbation · paramSwapInvOp`, a strong bundle isomorphism as a pair of sc-isomorphisms `Chapter3/BasicClassComposition · ScOperator. isStrongBundleIso_prodMap`. The parameter quadrant is $\widetilde{C} \times \Lambda$ in $P \times \Lambda$, put second so as to reuse the product of a partial quadrant with a whole space; the order is immaterial (Finding 163).

$\tilde{f}$ is an sc-Fredholm section `Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSection.param`. *Sc-smooth*: it is the pull-back of f along the projection. *Regularizing* `Chapter3/ParamPerturbationG · StrongBundle. regularizing_paramLift`: if (r,x) is on level m and $\tilde{f}(r,x)$ on level $(m,m+1)$ of π^*Y, then $x \in X_m$ and $f(x) \in Y_{m,m+1}$ — tested in the pulled-back charts, whose fibre parts are those of P — so $x \in X_{m+1}$ because f is regularizing, and (r,x) is on level $m+1$. *The germ at a smooth point* (r_0,x) `Chapter3/ ParamPerturbationG · StrongBundle.IsScFredholmGerm.param`: x is smooth, so Definition 3.11 at x gives a centred chart c, a filling germ g on G_g, an sc⁺-germ t_0 on G_{t_0} with $t_0(0) = g(0)$ and a conjugation of $g - t_0$ to a basic germ. In the product chart at (r_0,x), which is centred there, the principal part of $\tilde{f}$ is $(a,u) \mapsto \hat{f}(u)$, its filling germ $(a,u) \mapsto g(u)$ on $\mathbb{R}^n \times G_g$ (the filler conditions above), the sc⁺-germ to subtract $(a,u) \mapsto t_0(u)$ on $\mathbb{R}^n \times G_{t_0}$ `Chapter3/GermCalculusParam · IsScSmoothGerm.univProd`, and $(a,u) \mapsto (g-t_0)(u)$ is conjugated to the $\mathbb{R}^n$-extension of the basic germ; the conjugation concerns $g - t_0$ near 0 in U, and $1 \times \Psi$ and the extended basic germ are whole-chart objects, so no germ calculus enters it. This is the "readily verified".

Theorem 3.18 `Chapter3/ParamPerturbationG · theorem_3_18`. *Statement.* For a strong bundle $P : Y \to X$, an sc-Fredholm section f and sc⁺-sections $s_1,\dots,s_n$, the section F is an sc-Fredholm section of π^*P. *Proof.* $F = \tilde{f} + \tilde{s}$; $\tilde{f}$ is sc-Fredholm and $\tilde{s}$ is sc⁺; Theorem 3.15 `Chapter3/FredholmSectionsG · theorem_3_15`. The paracompactness of $\mathbb{R}^n \times X$ and of π^*Y (Definitions 2.13, 2.51) enters as instance arguments; the version without them supplies both by Theorem 2.16 `Chapter3/ParamPerturbationG · theorem_3_18' Chapter2/ FibredProduct · MPolyfold.paracompactSpace_prod`. The text's sketch of the germ of F — filled version $(r,u) \mapsto g(u) + \sum_i r_i \hat{s}_i(r(u))$, sc⁺-part $t_0(u) + \sum_i r_i \hat{s}_i(r(u))$, conjugation $\Psi'(r,u,k) = (r - r_0, \Psi(u,k))$ — is exactly what Theorem 3.15 produces from the germ data of $\tilde{f}$ and the principal part of $\tilde{s}$ `Chapter3/ FredholmSectionsG · StrongBundle.IsScFredholmGerm.sectionAdd`; the translation $r - r_0$ sits in the chart rather than in the conjugation (Finding 164).

The linearization (after Theorem 3.19 in the text, used for the finite-dimensional reduction and for transversality). In the product chart at (r_0,x), for a chart c with $0 \in O$, the derivative of $\hat{F} \circ (1 \times r)$ at 0 is

$$(\delta a, \delta u) \mapsto D(\hat{G} \circ r)(0)\delta u + \sum_i \delta a_i \hat{s}_i(0), \qquad \hat{G} = \hat{f} + \sum_i r_{0,i} \hat{s}_i$$

`Chapter3/ParamPerturbationG · scDeriv_paramSection`: the product rule at $a = 0$, where the terms $\sum_i a_i D(\hat{s}_i \circ r)$ vanish. The tangent space of the product retract at 0 lies in $\mathbb{R}^n \times T_0O$ `Chapter3/ ParamPerturbation · PullbackChartData.ofProd_mem_tangent`, and in the chart form of the linearization (§III.31) $\mathrm{Lin}_F(0)(\delta a, \delta u) = \mathrm{Lin}_{f+\sum r_{0,i}s_i}(0)\delta u + \sum_i \delta a_i \hat{s}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`. At $r_0 = 0$ this is the text's formula $\tilde{f}'(0,x)(h,u) = f'(x)u + \sum_i h_i e_i$ `Chapter3/ParamPerturbationG · linAt_ paramSection_zero`. The formula is the derivative at the centre of a chart and does not use $F(r_0,x) = 0$; at a general (r_0,x) its first term is the linearization of $f + \sum r_{0,i}s_i$, not $f'(x)$ (Finding 165). The text's "sc¹-sections s_i" (p. 75) must be sc⁺ for Theorem 3.18 to apply (Finding 166).

III.41 Theorem 3.54 (1) with the text's hypotheses: the plan and the elementary lemmas

§III.33 showed what the text's argument for property (4) of Theorem 3.54 leaves open: at a zero $z_0 = \varphi(y)$ near 0, in the chart at x_0, the filled operator $Dg(z_0)$ is onto (Corollary 3.41), but $f'(y) = Dg(z_0)|_{T_{z_0}O}$ is onto only if condition (3) of the filler holds at z_0, which Definition 3.4 does not ask; and the perturbation

argument of the text's Proposition 6.2 gives $\rho(z_0)\Lambda_0 \subseteq$ range $f'(y)$ only on a neighbourhood depending on the finite-dimensional Λ_0. This section and the next two prove the surjectivity anyway, with the text's hypotheses, by a new argument found while formalizing (Findings 180–189). The text's statement is true; its proof, and that of the book's arXiv version, is incomplete.

The plan. By the assembly step of §III.31 it suffices to show, in the chart at x_0 and for every zero z_0 of g in O near 0, that $\rho(z_0)k \in Dg(z_0)(T_{z_0}O)$ for every smooth $k \in F_\infty$; the closed range of $f'(y)$ (Proposition 3.52 at y) does the rest. Fix z_0 and k. Suppose $u(\mu)$, $0 \le \mu < \mu_0$, is a curve through $u(0) = z_0$ of zeros of the *perturbed filler* $g + \mu\rho(r(\cdot))k$, right differentiable at 0 into E_1 with velocity ξ. The curve lies in O (the first lemma below), so $r(u(\mu)) = u(\mu)$ and $Dr(z_0)\xi = \xi$, i.e. $\xi \in T_{z_0}O$; differentiating $g(u(\mu)) = -\mu\rho(r(u(\mu)))k$ at $\mu = 0$ gives $Dg(z_0)\xi = -\rho(z_0)k$. So one curve for each pair (z_0, k) suffices. It is produced by an implicit function theorem *at the nearby zero*, in the normal form of Theorem 3.38: its infinite-dimensional part is Theorem 3.30 for a contraction germ extended by the parameter μ, applied at the centre, whose regions in the parameter a do not depend on k (§III.42); its finite-dimensional part is a one-sided implicit function theorem at the image of z_0, in the direction of an interior vector of the partial quadrant that good position provides (this section and §III.43). The set of zeros to which the argument applies is cut out by conditions on the data at x_0 alone. This section proves the elementary lemmas, with no sc-structure involved.

Zeros of the perturbed filler lie in O `Chapter3/GlobalIFTNearbyOntoLemmas · StrongRetr.mem_O_of_perturbed_zero`. *Statement.* Let R be a strong bundle retraction on $U \lhd F$, $u \in U$, $\mu \in \mathbb{R}$, $k \in F$, and let condition (2) of Definition 3.4 hold at u ($g(u) = \rho(r(u))g(u)$ implies $u \in O$). If $g(u) + \mu\rho(r(u))k = 0$, then $u \in O$. *Proof.* $r(u) \in U$ `Chapter3/Filler · StrongRetr.r_mem` lies in O, $\rho(r(u))$ is linear `Chapter3/Filler · StrongRetr.ρ_isLinear` and idempotent on the fibre over a point of O `Chapter3/Filler · StrongRetr.ρ_ρ`; so $\rho(r(u))g(u) = -\mu\rho(r(u))\rho(r(u))k = -\mu\rho(r(u))k = g(u)$, and condition (2) applies. □

An inward direction from good position `Chapter3/GlobalIFTNearbyOntoLemmas · exists_inward_of_isInGoodPosition`. *Statement.* Let $C = \{x : \ell_i(x) \ge 0\}$ be a partial quadrant presented by an sc-isomorphism T, with coordinate functionals $\ell_i = (T\cdot)_i$, $z \in C$, and $N \ne 0$ a subspace in good position to the partial quadrant $C_z = \{v : \ell_i(v) \ge 0$ for the i with $\ell_i(z) = 0\}$. Then there are $v \in N$ and $\eta, t_0 > 0$ with $z + t(v + \zeta) \in C$ for all $t \in [0, t_0]$ and $|\zeta| < \eta$. *Proof.* Let v be an interior point of $N \cap C_z$ in N (good position). *For an active index i, $\ell_i(v) > 0$.* Otherwise $\ell_i(v) = 0$; for $u \in N$ and small t, $v \pm tu \in C_z$, so $\pm t\ell_i(u) \ge 0$ and ℓ_i vanishes on N. Let P be a good complement with constant ε and $p \in P$; choose $n_0 \ne 0$ in $N \cap C_z$ (it has interior and $N \ne 0$) and scale it so that $|p| \le \varepsilon|n_0|$. Good position gives $n_0 \pm p \in C_z$, so $\ell_i(\pm p) = \ell_i(n_0 \pm p) \ge 0$ and $\ell_i(p) = 0$. Then ℓ_i vanishes on $N \oplus P = E$, which contradicts the surjectivity of the coordinates of a presentation. *The estimates.* A finite family of positive numbers has a positive lower bound `Chapter3/GlobalIFTNearbyOntoLemmas · exists_pos_le_of_forall_pos_fin`: $\ell_i(v) \ge m$ for the active i and $\ell_i(z) \ge m$ for the others. For the active i, $\ell_i(z + t(v + \zeta)) = t(\ell_i(v) + \ell_i(\zeta)) \ge t(m - \|\ell_i\|\,|\zeta|) \ge 0$ when $|\zeta|$ is small; for the others, $\ell_i(z + t(v + \zeta)) \ge m - t\|\ell_i\|(|v| + |\zeta|) \ge 0$ when t is small. □ The lemma needs $N \ne 0$: for the zero subspace good position is an empty condition, and the case $\dim K = 0$ of Theorem 3.54 (1) is treated separately (§III.43). The note behind the formalization phrased it as "N meets the interior of C_z"; the cone condition is what transports along the linear coordinate change (Finding 185).

Inward directions absorb any vector after rescaling `Chapter3/GlobalIFTNearbyOntoLemmas · inward_add_smul`. If $a_0 + t(\alpha + \beta) \in \widetilde{C}$ for $t \in [0, t_0]$ and $|\beta| < \eta$, then for every a_1 and $s \ge (|a_1| + 1)/\eta$ the vector $\alpha_0 = a_1 + s\alpha$ satisfies $a_0 + t(\alpha_0 + \beta) \in \widetilde{C}$ for $t \in [0, t_0/s]$ and $|\beta| < 1$: indeed $a_0 + t(\alpha_0 + \beta) = a_0 + ts(\alpha + (a_1 + \beta)/s)$ with $ts \le t_0$ and $|(a_1 + \beta)/s| < (|a_1| + 1)/s \le \eta$.

A one-sided curve of zeros `Chapter3/GlobalIFTNearbyOntoLemmas · exists_curve_of_inward` (the finite-dimensional implicit function theorem at a boundary point, in an inward direction). *Statement.* Let $\widetilde{C} \subseteq P$ be convex, $\Omega \subseteq \mathbb{R} \times P$ open, $D = \Omega \cap (\mathbb{R} \times \widetilde{C})$, $a_0 \in \widetilde{C}$ with $(0, a_0) \in \Omega$, $H : \mathbb{R} \times P \to Q$ of class C^1 on D, $H(0, a_0) = 0$, $D_a H(0, a_0)$ onto (the derivative within D, restricted to $0 \times P$), and α_0 with $DH(0, a_0)(1, \alpha_0) = 0$ and $a_0 + t(\alpha_0 + \beta) \in \widetilde{C}$ for $t \in [0, t_0]$, $|\beta| < \eta$. Then there are $\mu_0 > 0$

and $a : \mathbb{R} \to P$ with $a(0) = a_0$, right derivative α_0 at 0, and $(\mu, a(\mu)) \in D$, $H(\mu, a(\mu)) = 0$ for $0 \le \mu < \mu_0$. *Proof. Unique derivatives.* The inward property puts a ball around $a_0 + t_0\alpha_0$ into $\widetilde{C}$, so $\mathbb{R} \times \widetilde{C}$ is convex with nonempty interior and D has unique derivatives; derivatives within D are well defined. *The contraction.* Let S be a right inverse of $M_0 = D_a H(0, a_0)$ and Z its range; SM_0 is the identity on Z. For $\mu \ge 0$ and $\zeta \in Z$ with $|\zeta| \le c\mu$, $c = \eta/2$, the point $(\mu, a_0 + \mu\alpha_0 + \zeta)$ lies in D and close to $(0, a_0)$ `Chapter3/GlobalIFTNearbyOntoLemmas · dist_curve_le`, and $a_0 + \mu\alpha_0 + \zeta = a_0 + \mu(\alpha_0 + \zeta/\mu)$ is in $\widetilde{C}$ by the inward property ($|\zeta/\mu| \le c < \eta$). Put $T_\mu(\zeta) = \zeta - SH(\mu, a_0 + \mu\alpha_0 + \zeta)$. Its derivative in ζ is $1 - SD_a H(\mu, a_0 + \mu\alpha_0 + \zeta)|_Z$, which tends to $1 - SM_0|_Z = 0$ as $(\mu, \zeta) \to 0$ by the continuity of DH at $(0, a_0)$; the mean value inequality on the convex ball makes T_μ a $\frac{1}{2}$-contraction for small μ. The differentiability of H at $(0, a_0)$ in the direction $(1, \alpha_0)$, with $DH(0, a_0)(1, \alpha_0) = 0$, gives $|T_\mu(0)| = |SH(\mu, a_0 + \mu\alpha_0)| = o(\mu)$; so $|T_\mu(0)| \le c\mu/2$ for small μ and T_μ maps the ball of radius $c\mu$ into itself. *The curve.* The fixed point $\zeta(\mu)$ has $SH(\mu, a_0 + \mu\alpha_0 + \zeta(\mu)) = 0$, hence $H = 0$ there (S is injective), and $|\zeta(\mu)| \le 2|T_\mu(0)| = o(\mu)$; so $a(\mu) = a_0 + \mu\alpha_0 + \zeta(\mu)$ has right derivative α_0 at 0. □

Two facts on right derivatives. If k is continuous at 0 from the right, $\mu \mapsto \mu k(\mu)$ has the right derivative $k(0)$ at 0 `Chapter3/GlobalIFTNearbyOntoLemmas · hasDerivWithinAt_smul_right_of_continuousWithinAt`; and right derivatives at 0 on $[0, \infty)$ are unique `Chapter3/GlobalIFTNearbyOntoLemmas · eq_of_hasDerivWithinAt_Ici`.

III.42 The λ-extended contraction germ and the perturbation in the normal form

Throughout, the normal form is the data of Theorem 3.38 from §III.20 `Chapter3/LocalSolvability · ShiftedBasicData`: a germ $g : O(\widetilde{C} \oplus W, 0) \to Q \oplus W$, sc-smooth, whose index raise $g^1 = (G_1, w - B(a, w))$ is a basic germ on the shifted structures; at the chart at x_0 it is obtained from the stability data of the graph description by Theorem 3.38 `Chapter3/LocalSolvability · StabilityData.toShiftedBasicData`. (In this section g and e denote the normal form and its perturbation; the filler of the chart at x_0 returns in §III.43.) The norms of w below are those of W_{m+1}, the shifted level m.

The perturbation on the shifted structure. For an sc⁺-perturbation $e : O(\widetilde{C} \oplus W, 0) \to (Q \oplus W)^1$, its W-component on level 1 `Chapter3/GlobalIFTNearbyOntoGerm · perturbEW` is sc-smooth on the index-raised germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_isScSmoothGerm`, by Proposition 1.17 for germs applied to the level-raised map `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelMap_e_isScSmoothGerm` `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.eW_eq_on`. Its level maps are those of e read on the shifted levels `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_eW`.

Contraction regions that do not depend on the perturbation `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii`. A system of regions consists of radii ν_m in P and ρ_m in W_{m+1} such that for $a \in \widetilde{C}$, $|a| < \nu_m$ and $|w|, |w'| < \rho_m$: (a, w) lies in the domain of g^1 on level m, $|B(a, w) - B(a, w')| \le \frac{1}{4}|w - w'|$, and $|B(a, 0)| \le \rho_m/8$; ν decreasing `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.Radii.ν_anti_le` and the ρ_m compatible with the inclusions. *They exist below any prescribed radii δ_m* `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii`: on each level, Definition 3.28 with $\varepsilon = \frac{1}{4}$ gives a radius on which $B(a, \cdot)$ is a $\frac{1}{4}$-contraction, and since $B(0, 0) = 0$ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_B_zero` and B is continuous at 0 on each level, $|B(a, 0)| \le \rho_m/8$ for $|a|$ small; the radii are then shrunk level by level to be monotone and below δ_m. Nothing here involves a perturbation.

The λ-extended germ. On $(\mathbb{R} \times \widetilde{C}) \oplus W^1$ take the germ of neighbourhoods $\{((\mu, a), w) \mid (a, w) \in U^1_m\}$ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbGerm`, with the projections forgetting μ and keeping μ as sc-operators `Chapter3/GlobalIFTNearbyOntoGerm · perturbProj` `Chapter3/GlobalIFTNearbyOntoGerm · perturbMuProj`

and $\mathbb{R} \times \widetilde{C}$ quadrant-like Chapter3/GlobalIFTNearbyOntoGerm · isQuadrantLike_perturb, and put

$$B_\mu(a, w) = B(a, w) - \mu\, Pe(a, w)$$

Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB, P the projection onto W, with $B_\mu(0, 0) = 0$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_zero. It is sc-smooth on the germ: B composed with the projection, minus the scalar product of the μ-coordinate with Pe composed with the projection, by the product rule with a scalar factor of §III.40 Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbB_isScSmoothGerm Chapter3/ParamPerturbation · IsScSmooth.smul; its level maps are $B_m - \mu(Pe)_m$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.levelB_perturbB. Given constants $c_m \geq 0$ with $|Pe(a, w)| \leq c_m$ and $|Pe(a, w) - Pe(a, w')| \leq c_m|w - w'|$ on the regions, define the μ-radii $\mu_m = \min_{j \leq m} \min(\frac{1}{4}, \rho_j/8)/(c_j + 1)$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu, positive and decreasing Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_pos Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_anti, with $\mu_m c_m \leq \frac{1}{4}$ and $\mu_m c_m \leq \rho_m/8$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbMu_mul_le.

Lemma: B_μ is a contraction germ in the working form Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction, with the parameter regions $(-\mu_m, \mu_m) \times \{|a| < \nu_m\}$ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.mem_perturbContraction_N_iff, the open balls $\{|w| < \rho_m\}$ and the closed balls $\{|w| \leq \rho_m/2\}$ (Definition 3.A, §III.7). *The contraction*: $|B_\mu(a, w) - B_\mu(a, w')| \leq (\frac{1}{4} + |\mu|c_m)|w - w'| \leq \frac{1}{2}|w - w'|$. *The closed balls are invariant*: for $|w| \leq \rho_m/2$, $|B_\mu(a, w)| \leq |B(a, w) - B(a, 0)| + |B(a, 0)| + |\mu||Pe(a, w)| \leq \frac{1}{4} \cdot \frac{\rho_m}{2} + \frac{\rho_m}{8} + |\mu|c_m \leq \frac{\rho_m}{2}$. The regions in a and w are those of B and do not depend on e; only the μ-intervals do. The general construction of a working form from Definition 3.28 Chapter3/GermIFT · toContractionGerm is *not* used: its radii in a would depend on e, and the finite-dimensional step needs the nearby zero inside the level-3 region for every perturbation (Finding 182).

Theorem 3.30 for B_μ Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_isSc1GermWith. The solution germ $\delta_\mu(\mu, a)$ is sc¹ on the germ re-indexed by one level, $((-\mu_{m+1}, \mu_{m+1}) \times \{|a| < \nu_{m+1}\}) \cap (\mathbb{R} \times \widetilde{C})$, with the derivative family of the proof of Theorem 3.30 Chapter3/ContractionGerm · theorem_3_30. *At $\mu = 0$* Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction_sol_zero: a fixed point $w \in W_1$ of $B(a, \cdot)$ with $|w| < \rho_0$, $|a| < \nu_0$, $a \in \widetilde{C}$, is $\delta_\mu(0, a)$, by the uniqueness of the fixed point in the open ball of level 0.

The perturbation in the normal form. Now the chart at x_0 returns: the strong bundle retraction R on $U \lhd F$, the conjugation Ψ covering ψ with inverse Ψ', and the linear maps glue, split, fiber of the coordinate change of Theorem 3.38 (§III.18). The perturbation $u \mapsto \rho(r(u))k$ of the filler, pushed to the normal form, is $\widehat{e}_k(y) = \Xi(y, k)$ with the *joint* map

$$\Xi(y, \kappa) = \operatorname{fiber}\Big(\Psi[1](\psi^{-1}(\operatorname{glue} y),\ (R[1](r(\psi^{-1}(\operatorname{glue} y)), \kappa))_F)\Big)_{F'}$$

Chapter3/GlobalIFTNearbyOntoXi · xiMap on $\psi(U) \oplus F^1$ Chapter3/GlobalIFTNearbyOntoXi · xiDom Chapter3/GlobalIFTNearbyOntoXi · mem_xiDom, a relatively open set ($\operatorname{glue}^{-1}(U')$ is relatively open in the quadrant of the normal form) Chapter3/GlobalIFTNearbyOntoXi · glue₀_preimage_relOpen Chapter3/GlobalIFTNearbyOntoXi · xiDom_relOpen. *Ξ is sc-smooth into the index-raised target* Chapter3/GlobalIFTNearbyOntoXi · xiMap_isScSmooth: it is the composite of the linear glue, $\psi^{-1} \times 1$, the map $(u, \kappa) \mapsto (u, (R[1](r(u), \kappa))_F)$, the strong bundle map $\Psi[1]$ on the index-raised bundle (Definition 2.47), a projection and the linear fiber. *Ξ is linear in κ* Chapter3/GlobalIFTNearbyOntoXi · xiMap_smul Chapter3/GlobalIFTNearbyOntoXi · xiMap_zero, also on every level Chapter3/GlobalIFTNearbyOntoXi · levelMap_xiMap_smul, since R and Ψ are linear on the fibres. For a smooth k with level-1 representative k_1, $\widehat{e}_k = \Xi(\cdot, k_1)$ Chapter3/GlobalIFTNearbyOntoXi · ehat is sc-smooth into the index-raised target Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmooth (a constant map with smooth value is sc-smooth Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth), an sc⁺-perturbation on the germ

of Theorem 3.38 Chapter3/GlobalIFTNearbyOntoXi · pushGerm₀_U_subset Chapter3/GlobalIFTNearbyOntoXi · ehat_isScSmoothGerm, with level maps $(\hat{e}_k)_m(y) = \Xi_m(y, k_{m+1})$ Chapter3/GlobalIFTNearbyOntoXi · levelMap_ehat.

Lemma: bounds uniform in k Chapter3/GlobalIFTNearbyOntoXi · exists_xi_bounds. *For every level m there are $\delta > 0$ and $M \geq 0$ such that for $|a| < \delta$, $|\kappa|_{m+2} < \delta$ and a convex set S of $w \in W_{m+1}$ with $|w| < \delta$ on S (inside the domain of Ξ on level $m+1$): $|\Xi_m((a,w),\kappa)| \leq 1$ and $|\Xi_m((a,w),\kappa) - \Xi_m((a,w'),\kappa)| \leq M|w-w'|$ on S. Proof.* By Proposition 1.18 Chapter1/Levels · IsScSmooth.contDiffOn_levels, Ξ is C^1 from level $m+1$ to level m of its index-raised target, which is the shifted level m of the normal form: on the shifted structure the perturbation is C^1 on every level, which is what contraction estimates on the shifted levels need. *This is where the index raise of Theorem 3.38 is used* (Finding 183). Ξ_m vanishes at $\kappa = 0$, so by continuity of Ξ_m and of its derivative at 0 both are small on a small ball, and the mean value inequality on the convex S gives the Lipschitz bound. □ *The constants for $\hat{e}_k$* Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds: there are radii $\delta_m > 0$, depending on Ξ only, such that for every system of regions with $\nu_m, \rho_m \leq \delta_m$ and every smooth k, the constants $c_m = (2|k|_{m+2}/\delta_m)(M_m + 1)$ bound $(P\hat{e}_k)_m$ and its Lipschitz constant in w on the regions. The proof is the *scaling* $\Xi_m(y,\kappa) = (2|\kappa|/\delta_m)\,\Xi_m(y, (\delta_m/2|\kappa|)\kappa)$, which needs only the linearity of Ξ in κ; the note behind the formalization used the linearity of $D_y\Xi$ in κ instead.

III.43 The curve of perturbed zeros, the surjectivity in the chart at x_0, and Theorem 3.54 (1)

The reduction at a nearby zero (in the normal form, with a perturbation e, regions R and constants c as in §III.42). Let $z = (a_0, w_0)$ be a level-1 zero of g in the domain of the germ with $a_0 \in \widetilde{C}$, $|a_0| < \nu_3$ and $|w_0|_1 < \rho_0$. Then w_0 is a fixed point of $B(a_0, \cdot)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.B_eq_of_zero (because $g^1 = (G_1, w - B)$), so $w_0 = \delta_\mu(0, a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.sol_zero_eq. On the domain $D_3 = (-\mu_3, \mu_3) \times (\{|a| < \nu_3\} \cap \widetilde{C})$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDom, which has unique derivatives Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.uniqueDiffOn_curveDom, put $(\mu, a) \mapsto (a, \delta_\mu(\mu, a)) \in E_1$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveLift_mem and

$$\widetilde{G}(\mu, a) = (g + \mu\, \iota e)(a, \delta_\mu(\mu, a))$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap. *Its W-component vanishes* Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveMap_snd: $\delta_\mu(\mu, a)$ is the fixed point of $B_\mu(a, \cdot)$, and the W-component of $g + \mu\iota e$ is $w - B_\mu(a, w)$ (read on level 1, Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.snd_g_toBase). So the whole equation is carried by $H = (\widetilde{G})_Q$. *$\widetilde{G}$ is C^1* on D_3 Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.contDiffOn_curveMap: δ_μ is C^1 into W_2 by Theorem 3.30 on the re-indexed germ (Proposition 1.16 on level 1), and g, e are C^1 from level 1 to level 0. *Its derivative at* $(0, a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_curveMap is

$$D\widetilde{G}(0, a_0)(\dot{\mu}, \alpha) = Dg(z)(\alpha,\ \iota D\delta_\mu(0, a_0)(\dot{\mu}, \alpha)) + \dot{\mu}\, \iota e(z)$$

Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solD Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv: the chain rule, with the derivative of the solution germ on level 1 Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level, the derivative of g at z Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.hasFDerivWithinAt_Dg', and the product rule for $\mu\, \iota e$, whose derivative at $\mu = 0$ in the base directions vanishes. Its W-component vanishes too Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_snd.

Lemma: $D_a H(0, a_0)$ is onto, by a dimension count Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq. *Statement.* Let moreover $\ker Dg(z)$ be finite-dimensional of dimension $\dim P - \dim Q$. Then $M_0 = D_a H(0, a_0) : P \to Q$ Chapter3/

GlobalIFTNearbyOntoCurve · ShiftedBasicData.DH is onto, and $\ker Dg(z)$ is the graph of $L = \iota D_a\delta_\mu(0,a_0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.solL over $\ker M_0$. *Proof.* At $\dot\mu = 0$ the derivative formula reads $Dg(z)(\alpha, L\alpha) = (M_0\alpha, 0)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.curveDeriv_inr Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.Dg'_graph, so the graph of L over $\ker M_0$ lies in $\ker Dg(z)$ Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.map_graph_le_ker. The graph map is injective, so $\dim\ker M_0 \le \dim\ker Dg(z) = \dim P - \dim Q$; rank–nullity gives $\operatorname{rank} M_0 \ge \dim Q$, so M_0 is onto, and then $\dim\ker M_0 = \dim\ker Dg(z)$ and the inclusion is an equality. □ The obvious route, eliminating the W-block with $(1 - D_2B(z))^{-1}$, needs $\|D_2B(z)\| \le \frac{1}{2}$ on the level where $Dg(z)$ is known to be onto, i.e. the surjectivity on the shifted structure; the dimension count uses only the kernel dimension on level 0 (Finding 184).

Proposition: the curve of perturbed zeros Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.exists_perturbed_curve. *Statement.* Under the hypotheses of the lemma, let $v \in \ker Dg(z)$ with $a_0 + t(v_1 + \beta) \in \widetilde{C}$ for $t \in [0, t_0]$, $|\beta| < \eta$ (v_1 the P-component). Then there are $\mu_0 > 0$ and a curve $u'(\mu) = (a(\mu), \delta_\mu(\mu, a(\mu)))$, $0 \le \mu < \mu_0$, on level 2, in the domain of the germ with $a(\mu) \in \widetilde{C}$, with $u'(0) = z$ (on level 1), right differentiable at 0 as a map into E_2, and $(g + \mu\iota e)(u'(\mu)) = 0$. *Proof.* (i) By the lemma, M_0 is onto and $v_1 \in \ker M_0$ (as $v = (v_1, Lv_1)$). (ii) Choose a_1 with $M_0a_1 = -D_\mu H(0,a_0)$, and $\alpha_0 = a_1 + sv_1$ with s from the rescaling lemma of §III.41: α_0 is inward at a_0 (with $|\beta| < 1$) and $DH(0,a_0)(1,\alpha_0) = D_\mu H(0,a_0) + M_0a_1 + sM_0v_1 = 0$. (iii) H is C^1 on D_3 with $H(0,a_0) = 0$, and $\widetilde{C}$ is convex; the one-sided curve lemma (§III.41) on $\$\square = \$$ the open region of level 3 gives $a(\mu)$ with right derivative α_0 and $H(\mu, a(\mu)) = 0$. (iv) The level-1 map $(\mu, a) \mapsto \delta_\mu(\mu, a) \in W_2$ is differentiable at $(0, a_0)$ within D_3 (Theorem 3.30, Chapter3/GermLevels · IsSc1GermWith.hasFDerivWithinAt_level), so by the chain rule for $\mu \mapsto (\mu, a(\mu))$ the curve u' is right differentiable at 0 into E_2. (v) At $\mu = 0$, $\delta_\mu(0,a_0) = w_0$; and $(g + \mu\iota e)(u'(\mu)) = \widetilde{G}(\mu, a(\mu)) = (H(\mu, a(\mu)), 0) = 0$. □

The normal form at a nearby point. For the chart at x_0: at every level-1 point y of $\psi(U_1)$ in the germ, $Dg(y) = \text{fiber} \circ D(h + t)(\text{glue}\, y) \circ \text{glue}$ for the normal form g of $h + t$ Chapter3/GlobalIFTNearbyOnto · StabilityData.scDerivGerm_pushMap₀_at (the coordinate change is linear), and the kernel of $\text{fiber} \circ T \circ \text{glue}$ is $\text{split}(\ker T)$ Chapter3/GlobalIFTNearbyOnto · StabilityData.ker_fiber₀_comp_glue₀, split being injective Chapter3/GlobalIFTNearbyOnto · StabilityData.split₀_injective. Corollary 3.39 is not used: the facts at the nearby zero come from the chart at x_0 through these linear maps (Finding 187).

Theorem A: the surjectivity in the chart at x_0 Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_chart_onto. *Statement.* Take the data of the graph description in the chart at x_0 (§III.29, Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph): the strong bundle retraction $R = (r, \rho)$ on U, a filling germ g of $\mathbf{f}$ on G_g with a derivative family Dg, $\mathbf{f}(0) = 0$, the conjugation Ψ covering ψ, $\psi(0) = 0$, the stability data on a germ that ψ' maps into G_g level by level (inside G'', §III.48) with the basic germ h', $D(h' + \iota t)(0)$ onto with kernel K in good position, a closed good complement Y of K, the C^1 graph $\sigma : V' \to Y$ with $\sigma(0) = 0$, $D\sigma(0) = 0$ and its C^k-lifts, an open W_o on which $h' + \iota t$ is the push-forward of g, an open $N_2 \ni 0$ on which ψ maps into the germ and into $U' \cap W_o$, and on which ψ maps the zeros of g in O onto the graph. Assume condition (2) of Definition 3.4 on $U \cap N_2$ and $K \neq 0$. Then there is an open $N_4 \subseteq N_2$ containing 0 such that for every zero $z_0 \in O \cap N_4$ of g, every level-1 point $z_{0,1}$ over z_0 and every smooth k, there is $x \in T_{z_0}O = Dr(z_{0,1})E_0$ with $Dg(z_{0,1})x = \rho(z_0)k$. *Proof*, in the order of the Lean:

1. *Data at x_0, independent of k.* Lemma 3.55 on the graph with its radius ρ_2 Chapter3/GlobalIFTFull · lemma_3_55_relOpen (§III.32); the kernel at the nearby zeros on an open $N_3 \subseteq N_2$ Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel (§III.32); the normal form $\mathcal{D}$ of Theorem 3.38 Chapter3/LocalSolvability · StabilityData.toShiftedBasicData; the radii δ_m of the bounds of Ξ Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds and contraction regions below them Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.exists_radii; and the radius r_{42} of Lemma 3.42 for the bound ρ_0 Chapter3/LocalSolvability · StabilityData.lemma_3_42: zeros of the normal form on level 0

smaller than r_{42} have their W-component on level 1, smaller than ρ_0 there.

2. *The neighbourhood.* Let π be the projection onto K along Y (a topological complement: Y closed, K finite-dimensional), $c_s = \|\mathrm{split}\| + 1$, and $r' = \min(\rho_2/(\|\pi\|+1), r_{42}/c_s, \nu_3/c_s)$. By the continuity of ψ on U, $\psi^{-1}(B(0,r')) \cap U = O_4 \cap U$ for an open $O_4 \ni 0$; put $N_4 = N_3 \cap O_4$. Nothing here depends on k.
3. *At a zero $z_0 \in O \cap N_4$.* The kernel theorem at z_0 gives $v \in V'$ with $\psi(z_0) = v + \sigma(v)$, the level-1 point x_1 over it, $D(h' + \iota t)(x_1)$ onto and Fredholm of index $\dim P - \dim Q$, its kernel N_v, and $\ker Dg(z_{0,1}) = D\psi^{-1}(x_1)N_v \subseteq T_{z_0}O$. Since $|v| = |\pi(\psi(z_0))| \le \|\pi\| r' < \rho_2$, Lemma 3.55 at v gives $x_1 \in C'$ and N_v in good position to C'_{x_1}.
4. *The zero of the normal form.* $z = \mathrm{split}(x_1)$ lies in the germ on level 1, with $a_0 = (x_1)_P \in \widetilde{C}$; $|\mathrm{split}\,\psi(z_0)| < c_s r'$, so $|a_0| < \nu_3$, and Lemma 3.42 gives $|w_0|_1 < \rho_0$; g vanishes at z because $h' + \iota t$ vanishes at the graph point.
5. *Its kernel.* $Dg(z) = \mathrm{fiber} \circ D(h' + \iota t)(x_1) \circ \mathrm{glue}$, so $\ker Dg(z) = \mathrm{split}(N_v)$, finite-dimensional of dimension $\dim\ker D(h' + \iota t)(x_1) = \dim P - \dim Q$ (for a surjective Fredholm operator the kernel dimension is the index, `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`).
6. *The inward direction.* $N_v \neq 0$: its dimension is $\dim P - \dim Q$, which is also $\dim K$ ($D(h' + \iota t)(0)$ is onto and Fredholm of that index, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`), and $K \neq 0$. The inward lemma at x_1 (§III.41) gives $v_v \in N_v$ with $x_1 + t(v_v + \zeta) \in C'$; then $\mathrm{split}\, v_v \in \ker Dg(z)$, and its P-component is inward at a_0 in the quadrant of the normal form for $|\beta| < \eta/(\|\mathrm{glue}\| + 1)$, since $\mathrm{glue}(\beta, 0)$ has norm at most $\|\mathrm{glue}\|\,|\beta|$.
7. *The perturbation and the curve.* Take the level-1 representative k_1 of k; $\hat{e}_k$ is an sc$^+$-perturbation (§III.42) with constants c_m on the regions; the proposition above gives μ_0 and the curve u' of zeros of $g + \mu\iota\hat{e}_k$ through z.
8. *Back in the chart.* $u(\mu) = \psi^{-1}(\mathrm{glue}\, u'(\mu))$ on level 1: ψ^{-1} is C^1 from level 2 to level 1 (Proposition 1.18, `Chapter1/Levels · IsScSmooth.contDiffOn_levels`), so u is right differentiable at 0 with some velocity ξ_1, $u(0) = z_{0,1}$, and its base points $u_\mu = \psi^{-1}(\mathrm{glue}\, \iota u'(\mu))$ lie in U.
9. *The perturbed equation.* By continuity at $\mu = 0$, $\mathrm{glue}\, \iota u'(\mu) \in W_o$ and $u_\mu \in N_2$ for $0 \le \mu < \mu_2$ `Chapter3/GlobalIFTNearbyOnto · exists_Ico_of_eventually_nhdsWithin_Ici`. There the push-forward identity of g, the description of $\hat{e}_k$ as the Ψ-fibre image of $\rho(r(u_\mu))k$ transported by the linear fiber, and the injectivity of the fibre maps of Ψ turn $(g + \mu\iota\hat{e}_k)(u'(\mu)) = 0$ into $g(u_\mu) + \mu\rho(r(u_\mu))k = 0$.
10. *Tangency.* By the first lemma of §III.41 and condition (2) on N_2, $u_\mu \in O$.
11. *Differentiation.* r is sc^1 on U, and g is an sc^1-germ on G_g; the curve $u(\mu)$ stays on level 1 of G_g, because $\mathrm{glue}\, \iota u'(\mu)$ lies on level 1 of the germ of the stability data and ψ_1' maps that into $(G_g)_1$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_inclK_mem_of_subset_shrinkGerm`. So both are differentiable from level 1 to level 0 at $z_{0,1}$ along the curve, g by the chain rule within $(G_g)_1$ with derivative $Dg(z_{0,1})$ (with g sc-smooth on U no such condition was needed). Along the curve, $r(u_\mu) = u_\mu$, so the right derivatives at 0 agree, $Dr(z_{0,1})\xi_0 = \xi_0$ ($\xi_0 = \iota\xi_1$; uniqueness of right derivatives, §III.41): $\xi_0 \in T_{z_0}O$. The map $\mu \mapsto \rho(r(u_\mu))k$ is continuous at 0 (R is continuous on level 0), so $\mu \mapsto \mu\rho(r(u_\mu))k$ has the right derivative $\rho(r(z_0))k = \rho(z_0)k$; differentiating $g(u_\mu) = -\mu\rho(r(u_\mu))k$ gives $Dg(z_{0,1})\xi_0 = -\rho(z_0)k$. Take $x = -\xi_0$. □

Where the hypotheses enter. Tameness and good position at x_0 through the graph description, Corollary 3.41 and Lemma 3.55 (steps 1, 3) and the inward direction at boundary zeros (step 6); condition (2) of the filler through the tangency (step 10); $K \neq 0$ for the inward direction. The sc-Fredholm property at y is not used here; it enters only through the closed range in the assembly step. Condition (3) of the filler at the nearby zeros is not used. *Why there is no uniformity problem.* The only k-dependent quantities are the constants c_m and the μ-intervals; N_4 is fixed in step 2 before k is chosen. The perturbation argument of §III.33 applied the implicit function theorem at x_0 to the Λ_0-extended problem, whose neighbourhood depends on Λ_0; here it is applied at the nearby zero (Finding 181).

Theorem 3.54 (1) with the text's hypotheses `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'`. *Statement.* Let f be an sc-Fredholm section of a strong bundle whose charts have tame base retractions, $f(x_0) = 0$,

and (f, x_0) in good position. Then there is an open $V_x \ni x_0$ such that (f, y) is in good position at every zero $y \in V_x$. *Proof.* Destructure the good position at x_0: the centred chart c, the filling germ g with its derivative family, the sc⁺-germ, the conjugation, shrunk to G'' (§III.48), and the good position in the chart. Tameness gives $(1 - Dr(0))E \subseteq E_0$ by Proposition 2.35 `Chapter2/TameRetracts · proposition_2_35`; the graph description in the chart `Chapter3/GlobalIFTNearbyGraphG · theorem_3_54_chart_graph` and the kernel at the nearby zeros `Chapter3/GlobalIFTNearbyKernelG · theorem_3_54_chart_kernel` give $N_1 \supseteq N_2 \supseteq N_3$. A zero y of f in the chart domain with $\varphi(y) \in N_2$ is a zero of g ($\mathbf{f} = g$ on $O \cap N_1$). *Case* $K = 0$. Put $V_x = \varphi^{-1}(N_2)$ in the chart domain. A zero $y \in V_x$ maps by $\psi\varphi$ onto a graph point $v + \sigma(v)$ with $v \in K = 0$, so $\psi(\varphi(y)) = \sigma(0) = 0 = \psi(0)$, $\varphi(y) = 0$ (as $\psi^{-1}\psi = 1$ on U), and $y = x_0$ by the injectivity of the chart: the only zero is x_0, in good position by hypothesis. *Case* $K \neq 0$. Theorem A gives N_4 (condition (2) on $U \cap N_2$ is the filler's, §III.29); put $V_x = \varphi^{-1}(N_3 \cap N_4)$ in the chart domain. For a zero $y \in V_x$: $z_0 = \varphi(y) \in O$ is a zero of g; the kernel theorem gives the level-1 point $z_{0,1}$, $\ker Dg(z_{0,1}) \subseteq T_{z_0}O$ and its good position inside $T_{z_0}O$; y is smooth `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection.mem_smoothSet_of_isZeroAt`, so (f, y) is an sc-Fredholm germ; $\mathbf{f} = g$ on O near z_0. The assembly step in its germ form `Chapter3/GlobalIFTNearbySurjG · StrongBundle.isGoodPositionGermTame_of_chart_dense` (§III.31; whole-chart `Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense`) applies at $z_{0,1} \in (G_g)_1$ with (a) from Theorem A — $\rho(z_0)\hat{k}$ in the range of $Dg(z_{0,1})|_{T_{z_0}O}$ for every smooth $\hat{k}$ — and (b) from the kernel theorem, $\ker Dg(z_{0,1}) \cap T_{z_0}O = \ker Dg(z_{0,1})$. □ **No hypothesis is added**: the statement has exactly the hypotheses of the kernel theorem of §III.32 plus the sc-Fredholm property (Finding 189).

Theorems 3.54, 3.23 and 3.19 with the text's hypotheses. Theorem 3.54 (1) and (2) `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54'`: (1) as above on V_x, and (2), the regular data of Definition 3.25 at x_0 of a dimension d that every nearby datum shares, by §III.37 `Chapter3/GlobalIFTTheoremsG · theorem_3_54_two` fed with (1). Theorem 3.23 `Chapter3/GlobalIFTNearbyOntoG · theorem_3_23'`: (1) supplies the open set of nearby good position that the boundary case from the nearby good position needs `Chapter3/GlobalIFTTheoremsG · theorem_3_23_of_nearby`, and every zero of the resulting U is in good position. Theorem 3.19 `Chapter3/GlobalIFTNearbyOntoG · theorem_3_19'`: Theorem 3.23 and Lemma 2.26, $d_S \leq d_X = 0$ `Chapter2/SubMPolyfold · lemma_2_26`. The first forms under the filler condition at the nearby zeros (§§III.33, III.37) are kept; that condition is now a theorem (§III.45).

III.44 The dimension of the solution set is the Fredholm index, and the index is locally constant

Theorem 3.19 says that the solution set is a manifold "whose dimension agrees with the Fredholm index", and the proof of Theorem 3.54 takes $d = \dim\ker f'(x)$. In the statements of §§III.34–III.37 the dimension was existential. This section identifies it, at every zero and in every chart, and deduces the local constancy of the index along the solution set (Findings 170–171).

"$f'(y)$ is onto, Fredholm, of index d", chart-intrinsically `Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex`: in every strong bundle chart around y compatible with the atlas, at the level-1 point z over $\varphi(y)$, every corestriction $L : T_zO \to K_z$ of the linearization $\mathrm{Lin}(z)$ (§III.31) is onto and Fredholm with $\mathrm{ind}\, L = \dim\ker L = d$. "$f'(y)$ is onto" alone `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt`: in *some* chart, $\mathrm{Lin}(z)$ maps onto the fibre $\{k \mid (\varphi(y), k) \in K\}$; by the chart independence it is then onto in every chart.

Proposition 3.52 in every chart `Chapter3/GlobalIFTIndexG · StrongBundle.linAt_index_of_conjugationData`. *Statement.* Let f be an sc-smooth section, x_0 a zero, c a strong bundle chart centred at x_0 with a filling germ g of $\mathbf{f}$ and a derivative family Dg on a germ inside U, an sc⁺-germ s with $s(0) = g(0)$, and conjugation data d_c of $g - s$ to a basic germ `Chapter3/FredholmGermIndex · ConjugationData` with finite-dimensional spaces P_n, Q. Then in every strong bundle chart c' around x_0, at the level-1 point z' over $\varphi'(x_0)$, every corestriction L' of $\mathrm{Lin}'(z')$ is Fredholm of index $\dim P_n - \dim Q$, and onto if the germ is in good position at $Dg(0)$ in c. *Proof.* Proposition 3.52 in c (§III.30, `Chapter3/FredholmGermIndexG · proposition_3_52_local_index`): $Tf(0)$ is

Fredholm of index $\dim P_n - \dim Q$. $Tf(0) = Dg(0)|_{T_0O}$ is Lin(0) by the bridge for germs at the centre, which lies in every germ `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt` (whole-chart: `Chapter3/LinearizationCharts · IsFilledVersion.fillerRestrict_eq_linAt`), up to the identification of the fibres $\rho(0)F$ and $\rho(\iota 0)F$, which are equal subspaces but different types in Lean `Chapter3/GlobalIFTIndex · exists_rangeS_ρCLM_equiv`. The chart independence (§III.31, `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`) carries the Fredholm property, the index and the surjectivity from c at 0 to c' at z'; the four spaces are complete as ranges of idempotents `Chapter3/FillerLinearAlgebra · isClosed_range_of_idem`. □ At a zero in good position this gives IsSurjLinOfIndex$(f, x_0, \dim P_n - \dim Q)$ `Chapter3/GlobalIFTIndexG · StrongBundle.isSurjLinOfIndex_of_conjugationData`, because for a surjective Fredholm operator the kernel dimension is the index `Chapter3/GlobalIFT · finrank_ker_eq_fredholmIndex_of_surjective`; and the d with IsSurjLinOfIndex(f, x_0, d) equals $\dim P_n - \dim Q$ for every conjugation in the chart of Definition 3.53 `Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.exists_isSurjLinOfIndex` (a conjugation exists, `Chapter3/FredholmGermIndex · isConjugateToBasicGerm_iff_nonempty`). The first clause of Theorem 3.19 with the index `Chapter3/GlobalIFTIndexG · theorem_3_19_linearization_index`: for an sc-Fredholm section and a zero in good position, in the chart of Definition 3.53, $Dg(0)$ and $Tf(x_0)$ are Fredholm, $Tf(x_0)$ is onto, and $\operatorname{ind} Dg(0) = \operatorname{ind} Tf(x_0) = \dim\ker Tf(x_0) = \dim d.P_n - \dim d.Q$ for every conjugation d — the old formulation's "$n - N$ for some natural numbers" made precise.

The conjugation exposed. The graph description and the tangent germ data at the centre are built from the chart of Definition 3.53 by destructuring the existential good position, so their statements did not record which conjugation they used. Their proofs now prove versions that expose it `Chapter3/GlobalIFTG · theorem_3_54_graph_proj_conj` `Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj_conj`: the filling germ with its germ and derivative family Dg, the sc$^+$-germ, the good position at $Dg(0)$ in the chart and conjugation data d_c, with $\dim K = \dim d_c.P_n - \dim d_c.Q$ for $K = \ker D(h+t)(0)$, because $D(h+t)(0)$ is onto and Fredholm of index $n - N$ (§III.18, `Chapter3/SolutionLinearization · StabilityData.isFredholm_Dfs`). The earlier statements are corollaries.

Dimension = index. *At the centre* `Chapter3/GlobalIFTIndexG · theorem_3_54_tangentGermChart_index`: for an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, the data of Definition 3.25 at x_0 for $\{y \in V_x : f(y) = 0\}$ (with $\pi(O) \subseteq C$) have a dimension d with IsSurjLinOfIndex(f, x_0, d): the exposed $d = \dim K = \dim d_c.P_n - \dim d_c.Q$ and the previous paragraph. The same for the global solution set `Chapter3/GlobalIFTIndexG · theorem_3_54_tangentGermChart_solutionSet_index` (restriction of the chart, §III.37) and for the regular data of Theorem 3.54 (2) `Chapter3/GlobalIFTIndexG · theorem_3_54_two_index` (the re-centring keeps the dimension, `Chapter3/SolutionGermRecentre · MPolyfold.TangentGermChart.exists_isRegular_of_locally`). *At a nearby zero* `Chapter3/GlobalIFTIndexG · isSurjLinOfIndex_of_forall_dim`: if y is a zero in good position at which every datum of Definition 3.25 for the solution set has dimension d, then IsSurjLinOfIndex(f, y, d) — the data at y from the centre theorem at y have dimension $\operatorname{ind} f'(y)$, and that dimension is d.

The index is locally constant along the solution set `Chapter3/GlobalIFTIndexG · theorem_3_54_index_locally_constant`. *Statement.* For an sc-Fredholm section of a strong bundle whose charts have tame base retractions, a zero x_0 in good position: there are an open $V_x \ni x_0$ and d with IsSurjLinOfIndex(f, y, d) at every zero $y \in V_x$; in particular $\operatorname{ind} f'(y) = \operatorname{ind} f'(x_0)$. *Proof.* Theorem 3.54 (1) (§III.43) gives an open $V_{x,1}$ of nearby good position; Theorem 3.54 (2) with the index on it gives regular data D of dimension d with source in $V_{x,1}$ along which every datum has dimension d; on the source of D, every zero is in good position with all its data of dimension d, and the previous paragraph applies. □ The proof does not repeat the kernel computation of the matrix argument (Finding 188): "the dimension is locally constant" (from the re-centring and Sard's lemma in fixed dimension, §III.35) and "the dimension is the index" give "the index is locally constant". Along the solution set this is the index stability of the text's Proposition 6.2, whose proof needs exactly the surjectivity of §III.43. *Theorem 3.54 with the index* `Chapter3/GlobalIFTIndexG · theorem_3_54_index`: on an open $V_x \ni x_0$ every zero is in good position with IsSurjLinOfIndex(f, y, d), and the solution set has at x_0 regular data of dimension d whose chart domain *is* V_x, every datum at a point of the solution set in V_x having dimension d.

Theorems 3.23, 3.19 and 3.57 with the index. Theorem 3.23 `Chapter3/GlobalIFTIndexG · theorem_3_23_index`:

for an sc-Fredholm section with tame base retractions and a zero x in good position, an open $U \ni x$ and d such that every zero $y \in U$ is in good position with IsSurjLinOfIndex(f, y, d), the zeros in U are smooth, and $S(f, U)$ is a sub-M-polyfold with tame induced structure equivalent to a d-dimensional manifold with boundary with corners — the proof of the boundary case from the nearby good position (§III.37) with Theorem 3.54 (2) with the index. Theorem 3.19 `Chapter3/GlobalIFTIndexG · theorem_3_19_index`: for $d_X \equiv 0$ the same, with degeneracy index 0 everywhere (Lemma 2.26): "a smooth manifold whose dimension agrees with the Fredholm index", now with the tameness of the induced structure. Theorem 3.57 `Chapter3/GlobalIFTIndexG · theorem_3_57_index`: under the hypotheses of §III.37 (good position at every zero, $S \neq \emptyset$), at every $x \in S$ there are regular data of a dimension d, shared by every datum at the points of S in their chart domain, with IsSurjLinOfIndex(f, y, d) at every zero y of that domain.

III.45 The kernel as tangent space, good position at interior zeros, and the filler condition as a theorem

"$\ker f'(y) = T_y S(f, U)$", in a chart `Chapter3/GlobalIFTIndexG · theorem_3_54_ker_eq_tangent`. *Statement.* For an sc-smooth section of a strong bundle whose charts have tame base retractions and a zero x_0 in good position, there are the strong bundle chart c centred at x_0 of Definition 3.53, the conjugating sc-diffeomorphism $\psi : U \to U' \subseteq C'$ with inverse ψ' and $\psi(0) = 0$, the finite-dimensional $K = \ker D(h + t)(0)$, the solution graph $\sigma : V' \to E'$ (C^1, $\sigma(0) = 0$, $D\sigma(0) = 0$, V' relatively open in $K \cap C'$), an open $V_x \ni x_0$ with $\psi\varphi(\{y \in V_x : f(y) = 0\}) = \{v + \sigma(v) : v \in V'\}$, and an open $V_{x,2} \subseteq V_x$ containing x_0, such that at every zero $y \in V_{x,2}$, with $\psi(\varphi(y)) = v + \sigma(v)$, and every level-1 point z over $\varphi(y)$:

$$D\psi(z)(\ker \mathrm{Lin}(z)) = N_v = \{n + D\sigma(v)n \mid n \in K\},$$

the tangent space at v of the graph, which is the solution set in the chart $\psi \circ \varphi$ of its tangent germ data. *Proof.* From the kernel statement at the nearby zeros (§III.32, `Chapter3/GlobalIFTNearbySurjG · theorem_3_54_nearby_kernel`): z lies on level 1 of the germ of g, $\ker Dg(z) = D\psi'(\psi z)N_v$, $\ker Dg(z) \subseteq T_z O$, and $\mathrm{Lin}(z) = Dg(z)$ on $T_z O$ (the bridge for germs, which needs exactly z in the germ and $0 \in O$, `Chapter3/LinearizationChartsG · linAt_eq_germDeriv`; whole-chart `Chapter3/LinearizationCharts · linAt_eq_scDeriv`), so $\ker \mathrm{Lin}(z) = \ker Dg(z)$; and $D\psi(z)D\psi'(\psi z) = 1$ on level 0 by the chain rule for $\psi \circ \psi' = 1$ on U'. □ The intrinsic statement, with $T_y S$ as a subspace of $T_y X$ (Definition 2.38), is not formalized: $T_y X$ has no linear structure in the formalization (Chapter 2 gap G.13); the chart form is the statement (Finding 172).

Good position at a zero with $d_X = 0$ is surjectivity. *Linear algebra* `Chapter3/GlobalIFTIndex · isInGoodPositionIn_of_subset`: a subspace N of $T \subseteq E_0$ is in good position inside T to every cone C containing T — $N \cap C = N$ has interior in N, and any algebraic complement of N in T is a good complement with any constant, both sides of the condition being true. *The statement* `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_linAt_surj_of_degIndex_eq_zero`: let f be an sc-smooth section, y a zero with (f, y) an sc-Fredholm germ and $d_X(y) = 0$, c a strong bundle chart around y with tame base retraction, and $\mathrm{Lin}(z)$ onto K_z at the level-1 point z over $\varphi(y)$; then (f, y) is in good position. *Proof.* The model of c is a tame retract, so $d_C(\varphi(y)) = d_X(y) = 0$ by the strong form of Proposition 2.36 `Chapter2/TameRetracts · degIndex_eq_chartDegIndex_of_isTameRetract`; no coordinate of the presentation vanishes at $\varphi(y)$, so the partial quadrant at $\varphi(y)$ is the whole space, and the kernel condition of the assembly step holds by the linear algebra. The assembly step with all of K_z in the range `Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj` concludes. □ For an sc-Fredholm section with tame base retractions, "$f'(y)$ onto in some chart" therefore gives good position at a zero with $d_X(y) = 0$ `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt` (y is smooth, so the germ is sc-Fredholm); conversely good position gives "$f'(y)$ onto" in the chart of Definition 3.53 `Chapter3/GlobalIFTIndexG · StrongBundle.IsGoodPositionGerm.isLinSurjAt`; so at such a zero the two are equivalent `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt` — the text's remark after Theorem 3.54. In a non-tame chart $d_C(\varphi(y))$ can exceed $d_X(y)$, and the cone would have to be transported from a chart realizing $d_X(y)$; the standing tameness hypothesis makes that unnecessary

(Finding 173).

Theorems 3.19 and 3.56 with "$f'(x)$ surjective". Theorem 3.19 `Chapter3/GlobalIFTIndexG · theorem_3_19_of_surjective`: for $d_X \equiv 0$, an sc-Fredholm section with tame base retractions, $f(x_0) = 0$ and $f'(x_0)$ onto in some chart: the equivalence gives good position at x_0, and Theorem 3.19 with the index (§III.44) applies. Theorem 3.56 with the tameness and the regular data `Chapter3/GlobalIFTIndexG · theorem_3_56_tame`: for $d_X \equiv 0$ and good position at every zero, $S \neq \emptyset$: Theorem 3.57 with the index (§III.44) and Lemma 2.26 — the solution set is a sub-M-polyfold with tame induced structure, a manifold of locally constant dimension without boundary, with regular data at every zero of a dimension equal to the index at the zeros of their chart domain. With "$f'(x)$ surjective at every zero" `Chapter3/GlobalIFTIndexG · theorem_3_56_of_surjective`: the equivalence at every zero, then the previous statement.

The converse of the matrix argument, and the filler condition as a theorem. *The converse* `Chapter3/GlobalIFTIndex · normal_injective_of_exists_range_scDeriv_eq_ρ`. Let z be a level-1 point over a point z_0 of the retract with $\ker Dg(z) \subseteq T_{z_0}O$, and suppose $\rho(z_0)k \in Dg(z)(T_{z_0}O)$ for every $k \in F$. Then $Dr(z)b = 0$ and $\rho(z_0)Dg(z)b = Dg(z)b$ imply $b = 0$: choose $a \in T_{z_0}O$ with $Dg(z)a = \rho(z_0)Dg(z)b = Dg(z)b$; then $a - b \in \ker Dg(z) \subseteq T_{z_0}O$, so $b \in T_{z_0}O$; writing $b = Dr(z)w$, idempotence gives $b = Dr(z)b = 0$ `Chapter3/LinearizationCharts · scDeriv_r_idem_at`. The surjectivity of $Dg(z)$ is not needed for this direction. With the direction of §III.33 `Chapter3/GlobalIFTNearbySurj · exists_range_scDeriv_eq_ρ_of_normal_injective`, under $Dg(z)$ onto and $\ker Dg(z) \subseteq T_{z_0}O$ (and g agreeing with $\mathbf{f}$ on O near a zero z_0 of $\mathbf{f}$): $\rho(z_0)F \subseteq Dg(z)(T_{z_0}O)$ iff condition (3) of Definition 3.4 holds at z_0 `Chapter3/GlobalIFTIndex · exists_range_scDeriv_eq_ρ_iff_normal_injective` — the "iff" of Finding 100, both directions formalized. Both are linear algebra in the operator $Dg(z)$ and hold for a germ at the points of its germ `Chapter3/GlobalIFTIndexG · normal_injective_of_exists_range_deriv_eq_ρ_germ` `Chapter3/GlobalIFTIndexG · exists_range_deriv_eq_ρ_iff_normal_injective_germ`. *The filler condition* `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`: for an sc-Fredholm section with tame base retractions and a zero x_0 in good position, (f, x_0) is in good position with the filler condition at the nearby zeros `Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled`. *Proof.* Take the chart and filler of the kernel statement (§III.32) and the open $V_{x,1}$ of nearby good position (§III.43); let N be an open neighbourhood of 0 on which $\mathbf{f} = g$ on O and $O \cap N = \varphi(V_{x,2} \cap V_{x,1})$ (`Chapter2/MPolyfold · exists_isOpen_image_eq`). At a level-1 zero z_1 of g over a point of $O \cap N$, on level 1 of the germ of g (the condition asks no more, §III.33): $z_0 = \varphi(y)$ for a zero y of f in $V_{x,2} \cap V_{x,1}$; y is in good position, so $\mathrm{Lin}(z_1)$ is onto K_{z_0} in the chart c (§III.44); the kernel statement gives $\ker Dg(z_1) \subseteq T_{z_0}O$; and the converse gives condition (3) at z_0. □ So the added hypothesis of the first forms of Theorems 3.54, 3.23, 3.19 (§§III.33, III.37) holds whenever their other hypotheses do: it was never a restriction, but proving that needed the surjectivity of §III.43 (Finding 174).

III.46 What the second addendum taught us

1. **The text's Theorem 3.54 (1) is true, and the gap of its proof can be closed.** The surjectivity of $f'(y)$ at the zeros near a zero in good position does not follow from Corollary 3.41 and Lemma 3.55, as the text (and the book's arXiv version) says; it follows from an implicit function theorem *at the nearby zero* for the perturbed filler $g + \mu\rho(r(\cdot))k$, one smooth direction k at a time, whose curve of zeros is tangent to the retract by condition (2) of the filler `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_chart_onto` `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'` (§§III.41–III.43). With it, Theorems 3.54, 3.23 and 3.19 hold with the text's hypotheses.
2. **The uniformity problem disappears when the contraction regions are written down.** Theorem 3.30 is applied at the centre to the k-extended contraction germ, whose regions in a are those of the unperturbed germ `Chapter3/GlobalIFTNearbyOntoGerm · ShiftedBasicData.perturbContraction`; only the μ-interval depends on k, and the bounds of the perturbation scale linearly in k `Chapter3/GlobalIFTNearbyOntoXi · exists_xi_radii_bounds`. The black-box use of Corollary 3.47 for an extended problem gives neighbourhoods depending on the extension.
3. **The index raise of Theorem 3.38 is what makes the perturbation C^1 on the shifted levels** (§III.42),

and the surjectivity of the reduced finite-dimensional problem comes from a dimension count on level 0 `Chapter3/GlobalIFTNearbyOntoCurve · ShiftedBasicData.surjective_DH_and_ker_eq`, not from inverting $1 - D_2B$.

4. **The index is locally constant along the solution set** `Chapter3/GlobalIFTIndexG · theorem_3_54_index_locally_constant`, the special case of the text's Proposition 6.2, and the dimension of the solution set is the Fredholm index at every zero, in every chart `Chapter3/GlobalIFTIndexG · theorem_3_19_index` (§III.44); $\ker f'(y)$ is the tangent space of the solution set in the chart of the proof `Chapter3/GlobalIFTIndexG · theorem_3_54_ker_eq_tangent` (§III.45).
5. **The filler condition at the nearby zeros was never a restriction** `Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm`, and good position at a zero with $d_X = 0$ is surjectivity for a tame strong bundle `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_iff_isLinSurjAt` (§III.45), so Theorems 3.19 and 3.56 hold with "$f'(x)$ surjective".
6. **Theorem 3.18's "readily verified" is a construction**: translation charts of $\mathbb{R}^n$, the pull-back bundle as a structure, the product rule for a scalar factor, and the extension of a basic germ by inert parameters `Chapter3/ParamPerturbation · IsBasicGerm.paramExtend` `Chapter3/ParamPerturbationG · theorem_3_18` (§III.40).
7. **What is left of Chapter 3**: Proposition 3.52 as an sc-Fredholm operator on T_xX; $\ker f'(y) = T_yS$ with $T_yS \subseteq T_yX$; Theorem 3.58 for a general sc-splitting; Lemma 3.70 for a general presentation of the partial quadrant; Theorem 3.27 for sets whose data violate the second regularity condition; and Proposition 6.2 beyond solution sets (since proved, with the text's hypotheses for paths on $\mathbb{R}$, §III.49). The only added hypotheses are the closedness of good complements and the regularity conditions of Theorem 3.27 for general subsets.

III.47 The tameness hypothesis repaired

The implicit function theorems of §§III.23 and III.32–III.45 were first formalized with the hypothesis, `htame` in the Lean, that the base retraction of **every** strong bundle chart compatible with the atlas is tame; the phrase "a strong bundle whose charts have tame base retractions" in those sections describes it. No strong bundle over a nonempty base satisfies it, so the statements that assumed it were vacuously true (Finding 190). This section records why, and what replaced it. The proofs told above are unchanged; the hypotheses and the charts of the conclusions changed, and the names of the theorems did not.

Why the hypothesis fails. Compatibility of strong bundle charts (Definition 2.50) constrains the transition maps between the retracts K, not the retraction whose image K is. From a compatible chart with base retraction r on $U \subseteq C$, the chart $(1, \varphi)$ onto $\{1\} \times O$, with model $\mathbb{R} \oplus E$, quadrant $[0,\infty) \times C$ and base retraction $r'(a,u) = (1, r(u))$ on $[0,\infty) \times U$, has the transition maps of the first chart composed with affine maps, so it is compatible with every chart the first one is; but r' lowers the degeneracy index at $(0,u)$. The last point is machine-checked. The constant retraction $(a,w) \mapsto (1,0)$ of $[0,\infty) \oplus W$ `Chapter3/TameVacuity · ScRetraction.halfLineConst` has retract $\{(1,0)\}$ `Chapter3/TameVacuity · ScRetraction.halfLineConst_retract`, and its product with any sc-smooth retraction r of a nonempty U is not tame `Chapter3/TameVacuity · ScRetraction.halfLineConst_prod_not_isTame`: the degeneracy index of a product is the sum of the indices `Chapter2/FibredProduct · degIndex_prod`, so it is $1 + d_C(u)$ at $((0,0),u)$, $u \in r(U)$, and $d_C(u)$ at the image $((1,0),u)$, against condition (1) of Definition 2.34. That the chart $(1,\varphi)$ is compatible is not formalized.

The new hypotheses. The text asks for the tameness of particular charts: the chart of an sc-Fredholm germ has "a tame strong local bundle" $K \to O$ (Definitions 3.6 and 3.50), and the proof of Proposition 3.52 starts from "a tame strong bundle chart". A *tame chart* is a compatible strong bundle chart whose base retraction is tame `Chapter3/TameGerms · StrongBundle.IsTameChart`. *Tame germ data* at x are the data of Definition 3.11 (§III.6) in a tame chart `Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame`; *good position in a tame chart* adds Definition 3.53 in that chart (§III.21) `Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame`; and an sc-Fredholm section *with tame germ data* has them at every smooth point `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionTame`. Each implies the notion without tame-

ness Chapter3/TameGermsG · StrongBundle.IsScFredholmGermTame.isScFredholmGerm Chapter3/TameGermsG · StrongBundle.IsGoodPositionGermTame.isGoodPositionGerm. They can be met: tameness depends only on the domain of a retraction and its values there Chapter3/TameGerms · ScRetraction.IsTame.congr, and the identity retraction is tame Chapter2/Tame · ScRetraction.id_isTame, so a compatible chart whose base retraction is the identity of U is tame Chapter3/TameGerms · StrongBundle.isTameChart_of_eqOn_id.

The restated theorems. The proofs used the tameness for three charts only (Finding 191): the chart of the germ data at x_0, for $(1 - Dr(0))E \subseteq E_0$ in the linear part (§III.23) Chapter3/GlobalIFTG · theorem_3_54_graph_proj_conj; the chart $\psi \circ \varphi$ built from it, whose model is tame because r is (§§III.27, III.34) Chapter3/GlobalIFTCenterG · theorem_3_54_tangentGermChart_proj_conj; and the chart in which $f'(y)$ is onto at a zero with $d_X(y) = 0$ (§III.45). So each theorem now takes (f, x_0) in good position in a tame chart in place of good position and `htame`; where it concludes something at the other zeros it takes an sc-Fredholm section with tame germ data; its conclusions are good position in tame charts, and the charts it produces are tame. This applies to Theorem 3.54 (1) with the text's hypotheses Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one', to its first form, whose filler condition is now stated in a tame chart Chapter3/GlobalIFTNearbySurjG · StrongBundle.IsGoodPositionGermFilled Chapter3/GlobalIFTNearbySurjG · theorem_3_54_one, to Theorems 3.23 and 3.19 Chapter3/GlobalIFTNearbyOntoG · theorem_3_23' Chapter3/GlobalIFTNearbyOntoG · theorem_3_19', to the global theorems, with good position in a tame chart at every zero Chapter3/GlobalIFTTheoremsG · theorem_3_57 Chapter3/GlobalIFTTheoremsG · theorem_3_56 Chapter3/GlobalIFTTheoremsG · corollary_3_24, to the forms with the index Chapter3/GlobalIFTIndexG · theorem_3_54_index, and to the filler condition as a theorem Chapter3/GlobalIFTIndexG · isGoodPositionGermFilled_of_isGoodPositionGerm.

Reading the conclusion in the tame chart. At a nearby zero y the good position is proved in the chart at x_0 and carried to a chart centred at y (§III.31). That chart was the one Definition 3.11 supplies at y, of which nothing says that it is tame. The chart change now gives the good position in **every** chart carrying germ data at y Chapter3/LinearizationCharts · StrongBundle.isGoodPositionAt_of_chart_dense, the statement of §III.31 being its corollary Chapter3/LinearizationCharts · StrongBundle.isGoodPositionGerm_of_chart_dense. A good position given in any chart therefore holds in the tame chart of the germ data Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame: in the chart of the given good position $Tf(0)$ maps onto $K_0 \supseteq \rho(0)F_\infty$, $\ker Dg(0) \subseteq T_0O$ and $C_0 = C$, which are the hypotheses of the chart change at $z = 0$; in the germ reading the point $z = 0$ lies in every germ, $Tf(0)$ is onto all of K_0, and the form of the chart change with all of K_0 in the range needs no Proposition 3.52 (§III.31) Chapter3/LinearizationChartsG · StrongBundle.isGoodPositionAtGerm_of_chart. The same step gives good position from "$f'(y)$ surjective" at a zero with $d_X(y) = 0$: the surjectivity, given in one chart, holds in every chart Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective, in particular in the tame chart of the germ data at y, where $d_C = d_X = 0$ Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt.

Theorems 3.15 and 3.18 keep tame germ data. The proof of Theorem 3.15 (§III.6) does not change the chart, so $f + s$ has germ data in the tame charts of f Chapter3/TameGermsAddG · StrongBundle.IsScFredholmGermTame.sectionAdd Chapter3/TameGermsAddG · theorem_3_15_tame. For Theorem 3.18 (§III.40) the germ of $\tilde{f}$ at (r_0, x) is taken in the product chart $(r \mapsto r - r_0) \times \varphi$, whose base retraction $1 \times r$ on $\mathbb{R}^n \times U$ is tame when r is: the identity retraction of $\mathbb{R}^n$ is tame, so $d(a, u) = d_C(u)$, and the minimal subspaces and tangent spaces of the product are products Chapter2/FibredProduct · ScRetraction.IsTameWith.prod Chapter3/ParamPerturbation · PullbackChartData.ofProd_isTame. Hence $\tilde{f}$ has tame germ data Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmGermTame.param Chapter3/ParamPerturbationG · StrongBundle.IsScFredholmSectionTame.param, and so has $F(r, x) = f(x) + \sum_i r_i s_i(x)$ Chapter3/ParamPerturbationG · theorem_3_18_tame, which is what the implicit function theorems need when applied to F.

Not done (Finding 199): the compatibility of the chart $(1, \varphi)$, that is, the vacuity of `htame` at the level of charts; and the passage from an atlas of tame charts to tame germ data (Definition 3.11 □ Definition 3.50). The text builds the tame chart into Definition 3.50, and so do the theorems.

Germ data and the charts of a tame atlas (Findings 350–353, fourth addendum). The passage was to be made by re-centring a tame chart of the atlas at the point and transporting the filler and the conju-

gation to it. That strategy fails, because a filling is bound to its chart. *Condition (3) of Definition 3.4 is a condition on the chart*: a filling germ in a chart with strong bundle retraction $R = (r, \rho)$ gives a topological isomorphism $\ker Dr(0) \cong \ker \rho(0)$, the linearization of the fill defect $y \mapsto [1 - \rho(r(y))]g(y)$ on $\ker Dr(0)$ `Chapter3/TameGermData · IsFillingGerm.nonempty_kerEquiv`; so in a chart with $\ker Dr(0) \neq 0$ and $\rho(0)$ injective no map fills anything `Chapter3/TameGermData · not_isFillingGerm_of_ker`. *A tame chart with no filling.* For the trivial bundle $\mathbb{R}^n \times \mathbb{R}^k \to \mathbb{R}^n$ take the model $\mathbb{R} \oplus \mathbb{R}^n$ without boundary, the base retraction $r(a, u) = (0, u)$, the product of the constant retraction of $\mathbb{R}$ onto 0 with the identity `Chapter3/TameGermData · TrivialBundle.Reven`, the chart $x \mapsto (0, x)$, and $R((a, u), h) = ((0, u), h)$ `Chapter3/TameGermData · TrivialBundle.evenBundleChart`. It is compatible with the translation charts `Chapter3/TameGermData · TrivialBundle.isChart_evenBundleChart` and tame — the degeneracy index of $\mathbb{R} \oplus \mathbb{R}^n$ is 0 everywhere and $\mathbb{R} \oplus 0$ is an sc-complement of $T_x O = 0 \oplus \mathbb{R}^n$ `Chapter3/TameGermData · TrivialBundle.Reven_isTame` `Chapter3/TameGermData · TrivialBundle.isTameChart_evenBundleChart` — but $\ker Dr(0) = \mathbb{R} \oplus 0$ while $\rho(0) = 1$, so no map fills the principal part of any section in it `Chapter3/TameGermData · TrivialBundle.not_isFillingGerm_evenBundleChart`. *A tame atlas without germ data in its chart.* With the one-chart atlas $\{\text{this chart}\}$ `Chapter3/TameGermData · TrivialBundle.bundleEven` every chart of the atlas is tame `Chapter3/TameGermData · TrivialBundle.bundleEven_atlas_isTameChart`, the graph section of a C^∞ map has tame germ data — in the compatible translation charts `Chapter3/TameGermData · TrivialBundle.isScFredholmSectionTame_graphSection_even` — and in the chart of the atlas no section has a filling `Chapter3/TameGermData · TrivialBundle.tame_atlas_no_germData_in_atlas_chart`. So Definition 3.50 must be the existential over all compatible tame charts, as formalized; read as "in a chart of the tame atlas" it would make every section of this bundle non-Fredholm. Whether germ data in some compatible chart give germ data in some compatible tame chart, for a tame atlas, stays open: the transition between two charts is defined on the retracts only, while the filling lives off them, so it cannot simply be pushed forward.

III.48 Definition 3.11 on germs

The text states Definitions 3.4, 3.6 and 3.11 for germs: the filled version is "a germ $(g, 0)$", the sc$^+$-section a local one, and a germ of neighbourhoods (Definition 3.1) is a sequence U_m that may shrink with m. The first formalization asked g and s to be sc-smooth on the whole chart domain U, on the ground that a chart can be restricted. This section records why that was narrower, and the two devices on which the proofs in the text's germ reading rest: the calculus of germs with whole-chart maps, and the shrinking of the basic germ (Findings 260–334).

Germs are not restrictions. On $\mathbb{R}$ with the constant structure, $f(x) = \sum_n c_n (x - a_n)^{n+1} |x - a_n|$, $a_n = 1/(n+2)$, $c_n = (4(n+2))^{-(n+2)}$, is C^k on $U_k = (-1/(k+2), 1/(k+2))$ — the terms $n \leq k$ have their singular points outside, the tail is C^{k+2} — and C^{n+2} on no open set containing a_n. So f is an sc-smooth germ on (U_m) and sc-smooth on no open $U \ni 0$ `Chapter3/GermVsChart · exists_isScSmoothGerm_not_isScSmooth`. Restricting a chart on level 0 never makes the levels shrink, so the whole-chart reading is a restriction of the class of sections; for a section sc-smooth on X the two readings of Definition 3.11 differ exactly in the regularity of g and s away from their germs (§III.6). Whether every germ datum can be traded for a whole-chart one is open: replacing the sc$^+$-germ by a whole-chart sc$^+$-section costs one level (the index raise of Theorem 3.38), and on level 0 no argument was found in either direction (Finding 264). No theorem depends on the answer, since the theorems are proved in the germ reading.

The calculus with whole-chart maps. The text forms $g + s$, $g - s$, $[1 - \rho(r(y))]g(y)$ and $\Psi_*(g)$ for germs without comment. Sums, differences, scalar multiples and compositions with sc-operators of sc^1-germs are sc^1-germs with the explicit derivative families, on $G_1 \cap G_2$ for germs on two germs `Chapter3/GermCalculus · IsSc1GermWith.add` `Chapter3/GermCalculus · IsScSmoothGerm.add_inter`. The chain rule of §III.1 composes two germs *at* 0, but R and Ψ are applied at $(r(y), g(y))$, which is $(0, 0)$ only when $g(0) = 0$ and $0 \in O$; so the chain rule is proved again with the outer map sc^1 on a set V and the inner one a germ mapping G_0 into V, with no condition at a base point `Chapter3/GermCalculus · IsSc1With.comp_germ` `Chapter3/GermCalculus · IsScSmooth.comp_germ`, and with a whole-chart inner map ψ, $\psi(0) = 0$, on the preimage germ `Chapter3/GermCalculus · IsScSmoothGerm.comp_preimage`. Hence the fill defect of a germ is a germ on the same

germ (§III.4); the push-forward $\Psi_*(g)$ is a germ on the germ of the points that ψ^{-1} maps into the germ of g, with $D(\Psi_* g)(0) = \Gamma(0) Dg(0) D\psi(0)^{-1}$ (§III.23); a germ plus a whole-chart map is a germ on the same germ (Theorem 3.15, §III.6); and $(a, u) \mapsto g(u)$ is a germ on $\mathbb{R}^n \times G$ (Theorem 3.18, §III.40). The strong bundle isomorphism Ψ and the basic germ h remain what they were.

A basic germ restricts. Definitions 3.7 and 3.28 pass to every smaller germ with the same h and B: the contraction field after shrinking its radius, since every level of a germ is a relatively open neighbourhood of 0, and the sc-smoothness by the uniqueness of derivatives in the quadrant `Chapter3/BasicGermShrink · IsBasicGerm.mono`.

The shrunk basic germ G''. The text does not relate the germ G of the basic germ to the germs G_g, G_s of g and s, and nothing forces $\psi^{-1}(G_m) \subseteq (G_g)_m$. For conjugation data with base map ψ, inverse ψ' and basic germ on G, let $G''_m = \{x \in G_m \mid \psi'_m(x) \in (G_g)_m \cap (G_s)_m\}$, the preimage germ along the sc⁰-germ $\psi'|_G$ (ψ' is sc-smooth on all of U' and $\psi'(0) = 0$) `Chapter3/BasicGermShrink · ConjugationData.shrinkGerm`. The conjugation data shrunk to G'' have the same Ψ, h, B and conjugation identity `Chapter3/BasicGermShrink · ConjugationData.shrink` `Chapter3/BasicGermShrink · ConjugationData.basic_shrinkGerm`; a point of G''_m is sent by ψ'_m into $(G_g)_m \cap (G_s)_m$ `Chapter3/BasicGermShrink · ConjugationData.levelMap_mem_inter_of_mem_shrinkGerm`, and read on the E-side, $u \in U_m$ with $\psi_m(u) \in G''_m$ lies in both germs `Chapter3/BasicGermShrink · ConjugationData.mem_of_levelMap_base_mem`. *Corollary 3.41 on G''*: every zero x of the normal form with $|x|_0$ small is ιx_1 with x_1 on level 1 of G'' — it is small on level 1 (Theorem 3.40), and the contraction field puts a level-1 ball around 0 inside G''_1 — so $\psi'_1(x_1)$ lies on level 1 of G_g and G_s `Chapter3/BasicGermShrinkIFT · ConjugationData.corollary_3_41_shrink`. This is the device of every argument at the nearby zeros: the level-1 lift $z_{0,1}$ of a zero near x_0 lies in the germ of g, where $Dg(z_{0,1})$ is defined (§§III.32, III.33, III.43, III.45). It relies on the lifts being small on level 1; a point small on level 0 only need not lie in G''_1. At the centre no shrinking is needed: 0 lies in every germ, and $Dg(0)$ is the same for all germs and families (§III.4).

What the proofs use of g and s. At 0: $Dg(0)$ and $Ds(0)$ (Propositions 3.5 and 3.52, Definition 3.53, the linear part of Theorem 3.54; §§III.4, III.21, III.23, III.30). On level 0 near 0: continuity of g and of the sc⁺-part (Theorem 3.12, §III.13; the local step of Theorem 5.5, §V.6). At the level-1 lifts of the nearby zeros: Dg, through G'' (Theorem 3.54 (1), §§III.32–III.33, III.43, III.45). Nowhere g or s on $U \cap E_m$ away from their germs. So every theorem of Chapter 3 holds in the germ reading with the text's hypotheses and its former conclusions, and no hypothesis is added; the statements that named the derivative of g on U at 0 (Proposition 3.52, the linearization statements of Theorems 3.19 and 3.54) now name the germ, a derivative family Dg and $Dg(0)$, and Corollary 5.8 in a chart has an sc⁺-germ on the shifted germ for its sc⁺-part (§V.7).

The switch. The canonical names now denote the germ reading: Definitions 3.11, 3.14, 3.50, 3.51 and 3.53 with the tame, filled and sc variants of good position `Chapter3/GermFillingChart · StrongBundle.IsScFredholmGerm` `Chapter3/TameGermsG · StrongBundle.IsScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGerm`, and the theorems built on them. The whole-chart notions and theorems keep their statements under the names `…WholeChart` and `…_wholeChart`, and each whole-chart notion implies its germ twin `Chapter3/TameGermsG · StrongBundle.IsScFredholmSectionWholeChart.isScFredholmSection` `Chapter3/GoodPositionGermG · StrongBundle.IsGoodPositionGermWholeChart.isGoodPositionGerm`. The non-vacuity witnesses are whole-chart data, hence germ data. Chapter 5 is stated with the germ notions: the local step of Theorem 5.5 and the spanning sections changed locally, and Corollary 5.8 in a chart changed its statement (§§V.6, V.12, V.7).

III.49 Definition 6.1 and Proposition 6.2: the index along paths

The text's Chapter 6 opens with the index of an sc-Fredholm germ at a smooth point which need not be a zero (Definition 6.1, pp. 200–201) and proves that it is constant along sc-smooth paths of smooth points (Proposition 6.2, p. 201). Its proof is the device that §III.44 used along the zero set: stabilize by

finitely many sc⁺-sections and apply the implicit function theorem. This section retells both: first with sc⁺-sections on all of X, which needs bump functions (Findings 340–348), then as the text does it, with locally defined sections on the restriction of the bundle to a chart domain and without bump functions (Findings 349, 360–367).

Definition 6.1 `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt`. *f has index i at x* if there is an sc⁺-section s of P with $(f+s)(x)=0$ and $(f+s)'(x)$ Fredholm of index i in every strong bundle chart around x (the chart-intrinsic statement of §III.44). The text's $f-s$ with $s(x)=f(x)$ is $f+(-s)$; the text's section is local, this one global; the reading with local sections, and the agreement of the two readings under bump functions, close this section. *Well defined.* At a zero of an sc-Fredholm section the index in `IsLinOfIndex` is unique, because in a chart $\mathrm{Lin}(z)$ is a bounded operator into K_z (§III.44), so the quantification over corestrictions is not vacuous `Chapter3/IndexAlongPaths · StrongBundle.IsLinOfIndex.unique`. Adding an sc⁺-section s with $s(x)=0$ at a zero x of f does not change the index: the germ datum of f at x, with conjugation data dc, gives the index $\dim P_n-\dim Q$ of $f'(x)$ (Proposition 3.52), and Theorem 3.15 gives germ data of $f+s$ in the same chart whose conjugation data are dc up to their values on U, since $(g+\hat{s}\circ r)-(\hat{s}_1\circ r+t_0)=g-t_0$ `Chapter3/IndexAlongPaths · StrongBundle.isLinOfIndex_sectionAdd_of_isZeroAt`. For two sections s,s' of the definition, $s'-s$ is an sc⁺-section vanishing at x `Chapter3/IndexAlongPaths · isZeroAt_sectionDiff_of_isZeroAt_sectionAdd` and $(f+s)+(s'-s)=f+s'$ `Chapter3/IndexAlongPaths · sectionAdd_sectionAdd_sectionDiff`, so the indices agree `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.unique`. The text argues instead that $(t-s)'(x)$ is an sc⁺-operator, hence compact on every level, and invokes Proposition 1.10; the project has no lemma that the linearization of an sc⁺-section is compact, and the germ data make it unnecessary. At a zero the index of the germ is that of $f'(x)$ `Chapter3/IndexAlongPaths · StrongBundle.isIndexAt_of_isZeroAt` `Chapter3/IndexAlongPaths · StrongBundle.IsIndexAt.isLinOfIndex_of_isZeroAt`. *Existence.* If X admits bump functions, every smooth point has an index: $f(x)$ is a smooth point of Y, Lemma 5.11 gives an sc⁺-section s_0 with $s_0(x)=f(x)$, and for $s=-s_0$ the germ data of the sc-Fredholm section $f+s$ at its zero x (Theorem 3.15) give the index `Chapter3/IndexAlongPaths · StrongBundle.exists_isIndexAt`.

Along the zero set, without good position (Findings 240–242). Let X admit bump functions, f be an sc-Fredholm section with tame germ data and $f(x_0)=0$; neither good position nor surjectivity at x_0 is assumed. In a chart centred at x_0, finitely many smooth vectors span a complement of the range of $\mathrm{Lin}(0)$, closed of finite codimension by Proposition 3.52, and Lemma 5.11 realizes them as allowable sc⁺-sections $s_1,\dots,s_k$ `Chapter3/IndexLocallyConstantG · exists_spanning_sections_germ`; with further sections that tilt the kernel into good position at a corner (the conditions (i)–(ii) of §V.19), the parametrized section $F(\lambda,x)=f(x)+\sum_i\lambda_i s_i(x)$ of Theorem 3.18 is in good position at $(0,x_0)$ in a tame chart `Chapter3/IndexLocallyConstantG · exists_sections_isParamGP_germ` `Chapter3/IndexLocallyConstantG · isGoodPositionGermTame_paramSection_of_isParamGP`. So the index of F' is one integer d at the zeros of F near $(0,x_0)$ (§III.44), among them $(0,y)$ for every zero y of f near x_0. At $(0,y)$, in the product chart, $\mathrm{Lin}_F(0)(\delta\lambda,\delta u)=\mathrm{Lin}_f(0)\delta u+\sum_i\delta\lambda_i\hat{s}_i(0)$, a finite-rank extension, so $d=k+\mathrm{ind}\,f'(y)$ `Chapter3/IndexLocallyConstant · fredholmIndex_eq_of_param` `Chapter3/IndexLocallyConstantG · StrongBundle.isLinOfIndex_of_paramSection_isLinOfIndex`. Hence $f'(y)$ has index $d-k$ in every chart at every zero y near x_0 `Chapter3/IndexLocallyConstantG · index_locally_constant_of_admitsBumpFunctions`, also as an sc-Fredholm operator on every level `Chapter3/IndexLocallyConstantG · theorem_3_54_index_locally_constant_general`. This is the text's device for Proposition 6.2 in the case of a zero.

The local form. For X with bump functions, f with tame germ data and an sc⁺-section s with $(f+s)(x_0)=0$, the index of the germ is one integer at the zeros of $f+s$ near x_0: $f+s$ has tame germ data (§III.47) and the previous paragraph applies to it `Chapter3/IndexAlongPaths · isIndexAt_locally_constant_of_isZeroAt_sectionAdd`. Hence along a path continuous at t_0 on which $f+s$ vanishes near t_0 the index is constant near t_0 `Chapter3/IndexAlongPaths · isIndexAt_eventually_const_of_path`, and on a preconnected parameter space a locally constant index is constant: by uniqueness the index is a locally constant function of t `Chapter3/IndexAlongPaths · isIndexAt_const_of_forall_eventually`. What remains is a section s, depending on t_0, that vanishes along the path — the text's $s(t,x)$ on $[0,1]\times X$ — and the passage from the product back to X.

Bump functions on $\mathbb{R}^n\times X$. The device works on the product of Theorem 3.18 (§III.40), which must

admit bump functions when X does. A C^∞ function on $\mathbb{R}^n$ is sc-smooth on the M-polyfold $\mathbb{R}^n$ Chapter3/IndexAlongPaths · isScSmoothMap_rn_real_of_contDiff, products of sc-smooth functions are sc-smooth by the product rule in every chart Chapter3/IndexAlongPaths · MPolyfold.IsScSmoothMap.mul, and $(r,x)\mapsto\beta_1(r)\beta_2(x)$ has support in $\operatorname{supp}\beta_1\times\operatorname{supp}\beta_2$ Chapter3/IndexAlongPaths · tsupport_prod_mul_subset; so $\mathbb{R}^n\times X$ admits bump functions Chapter3/IndexAlongPaths · MPolyfold.admitsBumpFunctions_prod_rn. Then for the pull-back $\tilde f = f\circ\pi$ and an sc$^+$-section S of π^*P with $\tilde f+S$ vanishing along the graph $t\mapsto(t,\varphi(t))$ near t_0, the index of $(\tilde f+S)'(t,\varphi(t))$ in every chart is one integer for t near t_0: $\tilde f+S$ has tame germ data (§§III.40, III.47), and the local constancy along its zero set applies Chapter3/IndexAlongPaths · isLinOfIndex_paramLift_sectionAdd_eventually_const. The proposition thereby reduces to two steps, the section along the path (E) and the index count on a slice (S) Chapter3/IndexAlongPaths · proposition_6_2_of_sections_of_slice.

(E): the text's section along the path Chapter3/IndexAlongPathsDevice · exists_isScPlusSection_pullback_of_path. For an sc-smooth section f (sc-Fredholm is not needed), an sc-smooth path $\varphi:\mathbb{R}^n\to X$ and t_0, there is an sc$^+$-section S of π^*P with $\tilde f+S=0$ along the graph near t_0. Take a strong bundle chart c around $\varphi(t_0)$. First the path is reparametrized by a C^∞ map σ of $\mathbb{R}^n$ equal to the identity near t_0 with values in $\varphi^{-1}(\operatorname{dom}\varphi_c)$, $\sigma(t)=t_0+b(t-t_0)(t-t_0)$ for a smooth bump b Chapter3/IndexAlongPathsDevice · exists_contDiff_eq_id_near_mapsTo; C^∞ self-maps of $\mathbb{R}^n$ are sc-smooth Chapter3/IndexAlongPathsDevice · isScSmoothMap_rn_of_contDiff, so $\psi=\varphi\circ\sigma$ is an sc-smooth path that stays in the chart. The representative $t\mapsto\Phi(f(\psi(t)))$ is sc-smooth from the constant structure $\mathbb{R}^n$ into $E\oplus F$; Proposition 1.17 makes it sc-smooth from $(\mathbb{R}^n)^1=\mathbb{R}^n$ into $(E\oplus F)^1=E^1\oplus F^1$, and its F^1-component $\hat e_1$ is the curve of smooth points that the text's $s(t,x)=R(x,f(\varphi(t)))$ needs in F^1 — no new analysis is required. In the product chart $(r\mapsto r-t_0)\times\varphi_c$ of π^*P the local section is $\tau(a,u)=-R[1](u,\hat e_1(a+t_0))_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is cut off by a bump of X equal to 1 near $\varphi(t_0)$ with support in the chart domain (Proposition 5.30) and extended by 0 (§V.9). Along the graph $\Phi(S(t,\varphi(t)))_2=-R(\varphi_c(\varphi(t)),\Phi(f(\varphi(t)))_2)_2=-\Phi(f(\varphi(t)))_2$, since $\Phi(f(\varphi(t)))\in K$ is fixed by R, so $\tilde f+S$ vanishes there for t near t_0, where $\sigma(t)=t$ and the bump is 1. The bump function and the reparametrization make the text's local section global; the text needs neither.

(S): the slice. The slice $S(t,\cdot)$ of a section of π^*P, read in Y through the fibres $(\pi^*Y)_{(t,x)}=P^{-1}(x)$, is a section of P Chapter3/IndexAlongPathsDevice · pullbackSlice Chapter3/IndexAlongPathsDevice · IsSection.pullbackSlice; $G(t,x)=0$ iff the slice vanishes at x Chapter3/IndexAlongPathsDevice · isZeroAt_pullback_iff_pullbackSlice, and the slice of $\tilde f+S$ is $f+S(t,\cdot)$ Chapter3/IndexAlongPathsDevice · pullbackSlice_sectionAdd_paramLift. *The slice of an sc$^+$-section is sc$^+$* Chapter3/IndexAlongPathsDevice · StrongBundle.IsScPlusSection.pullbackSlice: for a chart c of the atlas of P, the product chart $(r\mapsto r-t)\times\varphi_c$ is a chart of the atlas of π^*P, and the local representative of $S(t,\cdot)$ is that of S composed with the sc-operator $u\mapsto(0,u)$ Chapter3/IndexAlongPathsDevice · ScBanach.prodSndLeftOp and projected. *The linearization on the slice direction*: in the product chart at (t,x), whose base retraction is $(a,u)\mapsto(a,r(u))$ Chapter3/IndexAlongPathsDevice · PullbackChartData.r_ofProd, $\operatorname{Lin}_G(0)(0,\delta u)=\operatorname{Lin}_{G(t,\cdot)}(0)\delta u$, by the chain rule for $u\mapsto\hat G(0,r(u))$ Chapter3/IndexAlongPathsDevice · StrongBundle.linAt_ofProd_slice. *The index count*: if G is an sc-Fredholm section of π^*P whose slice $g=G(t,\cdot)$ is an sc-Fredholm section with $g(x)=0$, and $G'(t,x)$ is Fredholm of index i in every chart, then $g'(x)$ is Fredholm of index $i-n$ in every chart: $\operatorname{Lin}_G(0)(\delta a,\delta u)=\operatorname{Lin}_g(0)\delta u+\operatorname{Lin}_G(0)(\delta a,0)$ by linearity and the slice identity, the last term has rank at most n, and the parameter count of the zero-set case applies, the chart independence coming from the germ datum of g at x Chapter3/IndexAlongPathsDevice · StrongBundle.isLinOfIndex_pullbackSlice_of_isLinOfIndex. For $G=\tilde f+S$ ($n=1$) this is $\operatorname{ind}(f,x)=i-1$, with $S(t,\cdot)$ as the section of Definition 6.1 Chapter3/IndexAlongPathsDevice · StrongBundle.isIndexAt_of_pullback_isLinOfIndex.

Proposition 6.2 Chapter3/IndexAlongPathsDevice · proposition_6_2. *Let X admit sc-smooth bump functions, f be an sc-Fredholm section with tame germ data and $\varphi:\mathbb{R}\to X$ an sc-smooth path. Then* $\operatorname{ind}(f,\varphi(t))$ *is one integer for all t;* in particular $\operatorname{ind}(f,\varphi(0))=\operatorname{ind}(f,\varphi(1))$. For each t_0, (E) gives S; the index of $\tilde f+S$ is constant near t_0 along the graph; (S) turns it into $\operatorname{ind}(f,\varphi(t))+1$; and $\mathbb{R}$ is connected. The paracompactness of $\mathbb{R}\times X$ and of π^*Y is supplied by Theorem 2.16 Chapter3/IndexAlongPathsDevice · proposition_6_2'. *Against the*

text. (i) The path is defined on $\mathbb{R}$, not on $[0,1]$: $\mathbb{R}\times X$ is the product of Theorem 3.18 and has no boundary, and the text's "tame M-polyfold $[0,1]\times X$" is not used; a path on $[0,1]$ that extends sc-smoothly to $\mathbb{R}$ is covered, and by the reparametrization in (E) only a neighbourhood of each t_0 matters. (ii) The hypotheses are tame germ data, the reading of "tame strong bundle" of §III.47, and bump functions, which the text does not assume ("we do not need a sc-smooth partition of unity"): they are used for the stabilizing sections (Lemma 5.11) and for the cut-off of the section along the path, both globalizations of the text's locally defined sections. Without them the device runs on the restriction of P to a chart domain, as the text intends; that is the rest of this section. (iii) Definition 6.1 is proved well defined without Proposition 1.10. (iv) The two points the text's proof does not address — the tilting sections for the kernel condition at a corner and the uniformity in the zero — are those of the zero-set case above.

Without bump functions: the restriction to an open set (Findings 349, 360). The text restricts the bundle to a chart domain without comment ("in a chart, $s(y) = R(y, f(x))$"; "we do not need a sc-smooth partition of unity", p. 202). For an open $V \subseteq X$, a chart of X whose domain lies in V is transported to a chart of the subtype V with the same model `Chapter3/RestrictOpen · MChart.restrictTo`, compatibility is transported `Chapter3/RestrictOpen · MChart.Compatible.restrictTo`, and the charts of X inside V form the atlas of the M-polyfold V `Chapter3/RestrictOpen · MPolyfold.restrictOpen`, whose levels are the traces of those of X `Chapter3/RestrictOpen · MPolyfold.mem_levelSet_restrictOpen_iff`; an sc-smooth map with values in V is sc-smooth into V `Chapter3/RestrictOpen · MPolyfold.IsScSmoothMap.codRestrict_restrictOpen`. The restriction $P|V$ lives on the pull-back $\iota^*Y = \{(v,y) \mid P(y) = v\}$ of Definition 2.53, as the restricted sections of Definition 5.20 do (§V.23), and its fibres are those of P. A strong bundle chart of P with base domain inside V is transported with the same F, U, R, K `Chapter3/RestrictOpen · StrongBundleChart.restrictTo` `Chapter3/RestrictOpen · StrongBundleChart.Compatible.restrictTo`, and these charts form the atlas of $P|V$ `Chapter3/RestrictOpen · StrongBundle.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.restrictTo_mem_restrictOpen_atlas`; every chart of P restricts to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpen · StrongBundleChart.restrictInter`. The restriction $f|_V$ of an sc-smooth, sc$^+$ or regularizing section is again one `Chapter3/RestrictOpen · StrongBundle.IsScSmoothSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.IsScPlusSection.restrictOpen` `Chapter3/RestrictOpen · StrongBundle.regularizing_restrictOpen`. Two choices of the formalization: the paracompactness of V and of ι^*Y is supplied by Theorem 2.16, and V is nonempty, for the junk values of the transported inverse charts.

Germ data and the index on $P|V$ (Findings 361, 362). The data of Definition 3.11 on a relatively open U restrict to every relatively open $U_0 \subseteq U$ containing 0: the germs of the filling and of the sc$^+$-part are intersected with U_0, and $Dr(0)$ and $\rho(0)$ do not change, so the kernel isomorphism of condition (3) is the same `Chapter3/RestrictOpenGerms · IsFillingGerm.of_subset` `Chapter3/RestrictOpenGerms · IsScPlusGerm.of_subset`; the strong bundle isomorphism Ψ of the conjugation is restricted to $U_0 \lhd F \to \psi(U_0) \lhd F'$, where $\psi(U_0) = U' \cap \psi'^{-1}(U_0)$ is relatively open by the continuity of ψ', and the basic germ is shrunk to $G \cap \psi'^{-1}(U_0)$ as for G'' (§III.48), with the same P_n, Q, h, B `Chapter3/RestrictOpenGerms · ConjugationData.ofSubset` `Chapter3/RestrictOpenGerms · IsConjugateToBasicGerm.of_subset`. So a germ datum at $x \in V$ survives in its chart restricted to $\operatorname{dom}\varphi \cap V$ `Chapter3/RestrictOpenGerms · StrongBundleChart.germData_restrictInter`, which is tame when the chart is `Chapter3/RestrictOpenGerms · StrongBundle.IsTameChart.restrictInter`, and an sc-Fredholm section with tame germ data restricts to one on $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsScFredholmSectionTame.restrictOpen`. *The index.* At a zero $x \in V$, $f'(x)$ has index i in every chart of P iff $(f|_V)'(x)$ has index i in every chart of $P|V$ `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.restrictOpen` `Chapter3/RestrictOpenGerms · StrongBundle.IsLinOfIndex.of_restrictOpen`. The charts are not compared — a chart of $P|V$ need not come from one of P —: in either bundle the index is $\dim P_n - \dim Q$ for the germ datum (§III.44), whose restriction keeps P_n and Q `Chapter3/RestrictOpenGerms · StrongBundle.isLinOfIndex_restrictOpen_of_conjugationData`, and uniqueness identifies it with i. The same argument passes from $P|V$ to $P|W$ for an open $W \subseteq V$, where the chart restricted to $\operatorname{dom}\varphi \cap W$ is common to both and the two linearizations agree in it `Chapter3/IndexAlongPathsLocal · sectionRestrict` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsScPlusSection.restrictOpen_of_subset` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsLinOfIndex.restrictOpen_of_subset`; Definition 6.1 restricts accordingly `Chapter3/RestrictOpenGerms · StrongBundle.IsIndexAt.restrictOpen` `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.restrictOpen_of_subset`.

The stabilizing sections of a chart, without cut-off (Finding 363). In the zero-set argument above, bump functions enter only through Lemma 5.11, as sc$^+$-sections through given smooth points of the fibres. That property — through every smooth point e of the fibre over a smooth point x there is an sc$^+$-section s with $s(x) = e$ — is isolated `Chapter3/IndexLocallyConstantLocal · StrongBundle.HasScPlusSectionsThrough`. Bump functions give it by Lemma 5.11 `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`, and so does a chart of the atlas over the whole base, with no cut-off `Chapter3/IndexLocallyConstantLocal · StrongBundle.hasScPlusSectionsThrough_of_chart`: $\hat{e} = \Phi(e)_2$ is a smooth point of F, with a lift $\hat{e}_1 \in F^1$, and the section $y \mapsto \Phi^{-1}(\varphi(y), R(\varphi(y), \hat{e}))$ — the text's $R(\cdot, e)$ — has the local representative $z \mapsto R[1](z, \hat{e}_1)_2$, sc-smooth into $K[1]$ because the bundle retraction is sc-smooth on $[1]$; it is the cut-off section of Lemma 5.11 with $\beta = 1$ (§V.9), and its value at x is e since $\Phi(e) \in K$ is fixed by R. The restriction $P|V$ to a chart domain V has such a chart: the chart itself, transported to V. The three steps of the zero-set argument — the spanning sections, the tilting sections that put the parametrized section in good position, the local constancy — are repeated with this property in place of bump functions `Chapter3/IndexLocallyConstantLocal · exists_spanning_sections_of_sections` `Chapter3/IndexLocallyConstantLocal · exists_sections_isParamGP_of_sections` `Chapter3/IndexLocallyConstantLocal · index_locally_constant_of_sections`; the support and norm bounds of Lemma 5.11, which the index argument does not use, are dropped. *The index along the zero set with the text's hypotheses* `Chapter3/IndexLocallyConstantLocal · index_locally_constant_tame`: *let f be an sc-Fredholm section with tame germ data of any strong bundle and x_0 a zero; then there are an open $V \ni x_0$ and $i \in \mathbb{Z}$ with $f'(y)$ Fredholm of index i in every chart at every zero $y \in V$.* Take a chart around x_0 with domain V: $f|_V$ has tame germ data on $P|V$, which has sc$^+$-sections through every smooth fibre point, so the index of $f|_V$ is one integer at its zeros near x_0, and at each of them it is the index on P. In sc form, with the sc-Fredholm property of $\mathrm{Lin}(z)$ at every zero in every chart `Chapter3/LinearizationChartsScG · StrongBundle.linAt_isScFredholm`, $\mathrm{Lin}(z)$ is sc-Fredholm of index i on every level `Chapter3/IndexLocallyConstantLocal · theorem_3_54_index_locally_constant_general_tame`. This is the localization that Finding 343 described; the gap ind $C(z) = 0$ of Finding 243 remains only without tame germ data.

Definition 6.1 with local sections (Finding 364) `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal`. *f has index i at x* if there are an open $V \ni x$ and an sc$^+$-section s of $P|V$ with $(f|_V + s)(x) = 0$ and $(f|_V + s)'(x)$ Fredholm of index i in every chart of $P|V$: the global definition at the beginning of this section, on $P|V$. *Well defined.* Two local sections, on V and on V', both restrict to $V \cap V'$ (the restriction of Definition 6.1 to a smaller open set, above), where the uniqueness on $P|V \cap V'$ applies `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAtLocal.unique`. A global section gives a local one, with $V = X$ `Chapter3/IndexAlongPathsLocal · StrongBundle.IsIndexAt.isIndexAtLocal`, and at a zero the index is that of $f'(x)$ `Chapter3/IndexAlongPathsLocal · StrongBundle.isIndexAtLocal_of_isZeroAt`. *Existence, without bump functions.* On the domain V of a chart around x, the section through $f(x)$ of the previous paragraph, negated, is an sc$^+$-section s of $P|V$ with $(f|_V + s)(x) = 0$, and Theorem 3.15 gives the germ datum of $f|_V + s$ at x, hence an index `Chapter3/IndexAlongPathsLocal · StrongBundle.exists_isIndexAtLocal`. As for the global reading, a locally constant local index along a map from a preconnected space is constant `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_const_of_forall_eventually`.

Proposition 6.2 with the text's hypotheses (Finding 365) `Chapter3/IndexAlongPathsLocal · proposition_6_2_local`. *Let f be an sc-Fredholm section with tame germ data of a strong bundle $P : Y \to X$ — no bump functions — and $\varphi : \mathbb{R} \to X$ an sc-smooth path. Then* ind$(f, \varphi(t))$*, read with local sections, is one integer for all t*; any two values of the index at $\varphi(0)$ and at $\varphi(1)$ agree `Chapter3/IndexAlongPathsLocal · proposition_6_2_local'`. The proof is that of (E), (S) and Proposition 6.2 above, run near each t_0 on $P|V$, V the domain of a chart c around $\varphi(t_0)$ `Chapter3/IndexAlongPathsLocal · isIndexAtLocal_eventually_const_of_path`. The path is reparametrized into V as in (E), and is then an sc-smooth path of the M-polyfold V. The section along the path is the one of (E) with the cut-off as a datum `Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_bump`, taken equal to 1, which is possible because the chart $c|_V$ of $P|V$ covers its base `Chapter3/IndexAlongPathsLocal · exists_isScPlusSection_pullback_of_path_of_chart`. The product chart of $c|_V$ covers $\mathbb{R} \times V$, so the pull-back has sc$^+$-sections through every smooth fibre point without cut-off `Chapter3/IndexAlongPathsLocal · hasScPlusSectionsThrough_pullback_rn_of_chart`, and the index of $\tilde{f} + S$ is

constant along the graph near t_0 Chapter3/IndexAlongPathsLocal · isLinOfIndex_paramLift_sectionAdd_eventually_const_of_sections. The slice relation (S), applied to $P|V$ unchanged, turns it into $\mathrm{ind}(f|_V, \varphi(t)) = i - 1$ near t_0, a local index of f, and the connectedness of $\mathbb{R}$ ends the proof. *Against the text*, what remains is the domain of the path: it is $\mathbb{R}$, and a path sc-smooth on $[0,1]$ only is not covered, the project having no M-polyfold $[0,1]$. The hypotheses on f are the text's, tame germ data being the reading of "tame strong bundle" of §III.47.

The two readings agree under bump functions (Finding 367) Chapter3/IndexAlongPathsLocalBump · StrongBundle.isIndexAtLocal_iff_isIndexAt. If X admits bump functions, a local section of Definition 6.1 gives a global one with the same index Chapter3/IndexAlongPathsLocalBump · StrongBundle.IsIndexAtLocal.isIndexAt: the local section s on V is restricted to $W = \mathrm{dom}\,\varphi \cap V$ for a chart φ around x, read in that chart as $\tau : O \to F^1$, and cut off by a bump β supported in W and equal to 1 near x; the cut-off of Lemma 5.11 is made here in a chart of P that need not belong to the atlas Chapter3/IndexAlongPathsLocalBump · StrongBundle.cutoffSection_mem_oneSet_of_isChart Chapter3/IndexAlongPathsLocalBump · StrongBundle.isScPlusSection_cutoffSection_of_isChart. Then $f + \beta\tau$ vanishes at x, and its linearization there has the index of $(f|_W + s)'(x)$: the two sections agree where $\beta = 1$, so the two linearizations coincide in the chart restricted to that set, and the germ datum of $f + \beta\tau$ at x carries the index to every chart. So the global definition of the beginning of this section and the text's local one are the same notion whenever the hypotheses of Chapter 5 hold, and only the text's is available without them.

Part IV. Manifolds with boundary with corners (HWZ Chapter 4, §§4.1–4.4)

Chapter 4 asks when an M-polyfold — a space glued from images of sc-smooth retractions — is secretly an ordinary manifold with boundary with corners, and why the fibres of a family of such manifolds are good for almost every parameter. The pivot is the *sc⁺ condition*: a retraction that gains a level. At a smooth point its derivative is compact, so its retract has finite-dimensional tangent spaces (Lemma 4.7), and near a point where the tangent space is in good position to the partial quadrant the retract is the graph of a map over that tangent space (Proposition 4.9). Theorem 4.2 characterizes the manifolds with corners among M-polyfolds by these two properties; Theorem 4.18 applies the characterization to submanifolds in good position; Theorem 4.22 describes a submanifold near a point of general position; Theorem 4.23 combines that description with Sard's theorem on the strata of the corners. This Part retells: finite-dimensional sc-spaces and corner charts (§IV.1); sc⁺-maps (§IV.2); sc⁺-retractions, Lemmas 4.5 and 4.7 (§IV.3); Proposition 4.8 with its chart form and Remark 4.3 (§IV.4); charts centred at a smooth point, which the text uses and never constructs (§IV.5); Proposition 4.9 with Lemmas 4.11 and 4.12, by a direct proof (§§IV.6–IV.9); Theorem 4.2 (§IV.10); smooth finite-dimensional submanifolds and Remark 4.14 (§IV.11); Proposition 4.15 (§IV.12); good and general position in charts and their chart independence (§IV.13); Lemma 4.19 and Theorem 4.18 (§IV.14); Remark 4.20 (§IV.15); the linear algebra at the corner of a centred chart (§IV.16); Theorem 4.22 (§§IV.17–IV.18) and Remark 4.24 (§IV.19); Sard's theorem (§IV.20); the finite-dimensional core of Theorem 4.23 (§§IV.21–IV.22); the fibre over a regular value (§IV.23); Theorem 4.23 (§IV.24); what is not formalized (§IV.25); and what the formalization taught us (§IV.26).

What is proved, and what is not. Theorems 4.2, 4.18, 4.22 and 4.23 are proved with the text's hypotheses; Theorem 4.2 needs no tameness at all. Proposition 4.9 is proved with two hypotheses the text leaves implicit — the good complement Y is closed, and p is the projection onto T_0O *along* Y rather than $Dt(0)$ — and by a direct argument (§§IV.6–IV.9), because the text's route through Lemma 4.10 and Corollary 3.47 yields the graph map only as a germ over a complement of its own choosing; where Proposition 4.9 is applied (Theorems 4.2, 4.18, 4.22, Remark 4.20), the closed complement is chosen, so nothing is added there. One step of Lemma 4.19 is false and is repaired (§IV.14). Theorem 4.22 (4), which the text leaves to "the previous arguments", is proved (§IV.18). In Theorem 4.23 "regular value" has to be read as the stratified notion of its proof, and the finite cover of its proof is replaced by a countable one (§IV.24). Proved although the text does not construct it: charts and strong bundle charts centred at a smooth point (§IV.5), an sc⁺-retraction onto the fibre A_λ (§IV.23), Theorem 3.78 in C^∞ form (§IV.21), and Sard's theorem in all dimensions (§IV.20). Remark 4.3's "$X^1 \neq X$ is possible", for which the text gives no example, is proved by one, an M-polyfold whose retraction loses a level in its first derivative, with a new criterion for sc-smoothness (§IV.27). Not formalized: Lemma 4.10 (no longer needed), the whole of §4.4 on sc-differential forms, the open question after Proposition 4.15, and the chart independence of stratified regularity (§IV.25).

Five structural choices. (1) *sc⁺ is a property of maps.* The text defines it for retractions and operators and uses it for maps; here a map is sc⁺ when its lift into F^1 is sc-smooth, and the calculus of such maps is proved first (§IV.2). (2) *Tangent spaces of X are read in charts.* T_xX carries no linear structure in the

formalization (gap G.7 of Chapter 2), so T_aA, good position, general position and stratified regularity are stated in a chart, and — except for stratified regularity — proved independent of the chart (§IV.13). (3) *Centred charts are re-centred charts.* A chart with $\psi(x) = 0$ is the chart $\varphi - \varphi(x)$ on the set where the partial quadrant C agrees with $C_{\varphi(x)}$, which exists only at smooth points (§IV.5). (4) *Proposition 4.9 on fixed sets.* The sc-diffeomorphism $p : O \cap W \to V$ is proved with $\Gamma = p^{-1}$ sc-smooth on a *fixed* relatively open V, by contraction arguments on level 1, a frame formula for $D\Gamma$ and a C^k-bootstrap along the finite-dimensional tangent space. (5) *Stratified Sard.* Sard's theorem is applied stratum by stratum in the chart of Theorem 4.22, and the null sets of critical values of countably many charts are collected.

Map.

- `Chapter4/ScPlus`: sc$^+$-maps, Definitions 4.4, 4.6, Lemmas 4.5, 4.7, Proposition 4.8 and its chart form, Remark 4.3, finite-dimensional sc-spaces.
- `Chapter4/Recentre`: the partial quadrant C_e and its neighbourhood $G(e)$, translations, re-centred retracts, charts and strong bundle charts.
- `Chapter4/ScPlusGraph`: the mean value estimate for sc$^+$-retractions, Proposition 4.9 with Lemmas 4.11 and 4.12, the frame formula and the bootstrap.
- `Chapter4/Characterization`: Definition 4.1, corner charts, Theorem 4.2.
- `Chapter4/Submanifolds`: Definitions 4.13, 4.16, 4.17, 4.21, Remark 4.14, Proposition 4.15, Theorem 4.18, Lemma 4.19, Remark 4.20, the chart independence of good and general position.
- `Chapter4/LocalStructure`: Theorem 4.22 (1)–(4), Remark 4.24, the linear algebra of a centred chart, Lemmas 7.A and 7.B of the exposition.
- `Vendor/MilnorSard/…`: Sard's theorem, copied from our Milnor project (`math_faithful`) with the namespace renamed.
- `Chapter4/FamiliesLocal`, `Chapter4/Families`: the strata and Sard on them, Theorem 3.78 in C^∞ form, the fibre retraction, the sc$^+$-retraction onto A_λ and its general position, Theorem 4.23.
- Added in the fourth addendum (§IV.27): `Chapter4/RemarkFourThreeExample` (parameter families of level operators with one level of loss, their sc-smoothness, the reduction of Remark 4.3's example), `Chapter4/RemarkFourThreeFamily` (the family of moving projections, the M-polyfold of a retract, the example).

Throughout, $C = T^{-1}([0,\infty)^n \oplus W)$ is a partial quadrant presented by an sc-isomorphism $T : E \cong \mathbb{R}^n \oplus W$, with coordinates $\lambda_1, \dots, \lambda_n$; $I(e) = \{i : \lambda_i(e) = 0\}$, $d_C(e) = \#I(e)$, $C_e = \{\lambda_i \geq 0, i \in I(e)\}$ and $E_e = \{\lambda_i = 0, i \in I(e)\}$ (Definition 2.33, §II.12). For a retract $O = r(U)$ and a level-1 point z, $T_zO = Dr(z)E_0$ is the tangent space of Definition 2.17 and $T^R_zO = T_zO \cap E_z$ the reduced tangent space.

IV.1 Finite-dimensional sc-spaces and corner charts (Definition 4.1)

Finite-dimensional sc-spaces. Definition 4.1 models its charts on finite-dimensional spaces "with the constant sc-structure"; the text remarks after Definition 1.1 that a finite-dimensional space has only that one. This is proved. If E_0 is finite-dimensional, every level E_m is, since it injects into E_0 `Chapter4/ScPlus · ScBanach.finiteDimensional_lvl`. The range of $E_m \to E_0$ is then a finite-dimensional, hence closed, subspace; it contains the smooth points, which are dense (Definition 1.1 (2)); so it is all of E_0 `Chapter4/ScPlus · ScBanach.toBase_surjective`. The inclusions are therefore bijective, and they form an sc-isomorphism of E onto E_0 with the constant structure `Chapter4/ScPlus · ScBanach.scIsoConst`; in particular every point of E_0 is smooth `Chapter4/Characterization · ScBanach.mem_smooth_of_finiteDimensional`. The density axiom is what makes this work: $E_0 = \mathbb{R}$, $E_m = 0$ for $m \geq 1$ has compact injective inclusions and levels that differ (Findings, Chapter 4, item 6).

Definition 4.1. A chart of X is a *corner chart* if its local model is (U, C, V) with V a finite-dimensional real normed space carrying the constant sc-structure, C a partial quadrant in V, U relatively open in C, and the identity retraction `Chapter4/Characterization · MChart.IsCornerChart`. Its model is then relatively open in a partial quadrant `Chapter4/Characterization · MChart.IsCornerChart.isRelOpenInQuadrant`, finite-

dimensional Chapter4/Characterization · MChart.IsCornerChart.finiteDimensional, a tame retract — the identity retraction is tame Chapter2/Tame · ScRetraction.id_isTame Chapter4/Characterization · MChart.IsCornerChart.isTameRetract — and an sc⁺-retract by Remark 4.3 (§IV.4) Chapter4/Characterization · MChart.IsCornerChart.isScPlusRetract; every point of its model space is smooth Chapter4/Characterization · MChart.IsCornerChart.mem_smooth. *X has a corner structure* if it has an atlas of corner charts equivalent to its own atlas Chapter4/Characterization · MPolyfold.HasCornerStructure. The dimension of V may vary from chart to chart, as in the text. The text's "sc-smoothly compatible charts", "φ an sc-diffeomorphism", is read as compatibility with the atlas of X, as for tameness (Definition 2.37).

The definition is local Chapter4/Characterization · MPolyfold.hasCornerStructure_iff: X has a corner structure iff every point lies in a corner chart that is a chart of the M-polyfold structure. One direction takes a chart of the equivalent atlas through the point. For the other, the corner charts that are charts of X form an atlas: they cover X by hypothesis, any two charts of X are compatible Chapter2/MPolyfold · MPolyfold.IsChart.compatible, and the atlas is equivalent to that of X because each of its members is a chart of X. *A corner structure makes X tame* Chapter4/Characterization · MPolyfold.HasCornerStructure.isTame: the same atlas is an atlas of tame charts.

The transition maps are C^∞. For a chart whose model is relatively open in a partial quadrant, every overlap $\chi(V \cap V')$ is relatively open Chapter4/Characterization · MChart.isRelOpenInQuadrant_overlap. For compatible corner charts, the transition map is sc-smooth between the overlap retracts; composed with the identity retraction of the overlap it is sc-smooth on a relatively open subset of a partial quadrant in a finite-dimensional space with the constant structure, and such maps are exactly the C^∞ maps with derivatives within the set Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const Chapter4/Characterization · MChart.IsCornerChart.contDiffOn_transition. That is the text's "classically smooth"; "the atlas defines the structure of a smooth manifold with boundary with corners" is not formalized beyond it, because Mathlib's manifolds with corners have a single model space and here the partial quadrants vary from chart to chart.

Arbitrary finite-dimensional models. The constant structure is no restriction. A partial quadrant of a finite-dimensional E is one of E_0 with the constant structure Chapter4/Characterization · IsPartialQuadrant.toConst Chapter4/Characterization · IsRelOpenInQuadrant.toConst, and the identity of E_0 is an sc-diffeomorphism between the two models Chapter4/Characterization · IsRelOpenInQuadrant.isScDiffeoQuadrant_toConst. Composing a chart with it (the construction `compDiffeo` of Chapter 3) Chapter3/GlobalIFTCenter · MChart.compDiffeo gives, with the same map and domain, a corner chart of the same dimension Chapter4/Characterization · MChart.exists_isCornerChart_of_finiteDimensional. Hence X has a corner structure iff it has an equivalent atlas of charts onto relatively open subsets of partial quadrants in sc-Banach spaces with finite-dimensional E_0 Chapter4/Characterization · MPolyfold.hasCornerStructure_iff_finiteDimensional — verbatim the definition of the manifolds with corners of locally constant dimension of Theorem 3.27 (§III.36) — and the manifolds with corners of dimension n of Theorem 3.27 are the corner structures all of whose charts have dimension n Chapter4/Characterization · MPolyfold.isManifoldWithCorners_iff Chapter4/Characterization · MPolyfold.IsManifoldWithCorners.hasCornerStructure.

IV.2 Sc⁺-maps

The text uses "sc⁺-map" in the proof of Lemma 4.7 and composes sc⁺-retractions with sc-smooth maps in Lemma 4.5, but defines sc⁺ only for retractions (Definition 4.4) and operators (Definition 1.4). The formalization defines it for maps on any set, as the neighbouring notions of the project are (sc⁺-operators are sc-operators into F^1, sc⁺-sections sc-smooth maps into $K[1]$).

The lift. For $x \in E_1 \subseteq E_0$, $1_0^1 x$ is the point of E_1 over x, with a junk value off E_1 Chapter4/ScPlus · ScBanach.liftOne Chapter4/ScPlus · ScBanach.toBase_liftOne Chapter4/ScPlus · ScBanach.liftOne_toBase.

Definition. f is *sc⁺ on U* if $f(U) \subseteq F_1$ and the lift $x \mapsto 1_0^1 f(x)$, a map $U \to F^1$, is sc-smooth Chapter4/ScPlus · IsScPlus. Then $f(U_m) \subseteq F_{m+1}$: the lift is sc⁰, so it maps U_m into $(F^1)_m = F_{m+1}$ Chap-

ter4/ScPlus · IsScPlus.mem_range_succ. On an sc-domain the canonical lift can be replaced by any sc-smooth $g : U \to F^1$ with $\iota \circ g = f$, which is how the text phrases it Chapter4/ScPlus · IsScPlus.iff_exists. The calculus: an sc⁺-map is sc-smooth, being $\iota\circ$ (lift) with the sc-operator $\iota : F^1 \to F$ Chapter4/ScPlus · IsScPlus.isScSmooth; the property depends only on the values on U Chapter4/ScPlus · IsScPlus.congr and passes to smaller sc-domains Chapter4/ScPlus · IsScPlus.mono; *sc⁺ □ sc-smooth is sc⁺*, the lift of $g \circ f$ being (lift of g) $\circ f$, sc-smooth by the chain rule Chapter1/ChainRule · IsScSmooth.comp Chapter4/ScPlus · IsScPlus.comp_isScSmooth; *sc-smooth □ sc⁺ is sc⁺*, the lift of $g \circ f$ being $g^1\circ$ (lift of f) with $g^1 : V^1 \to G^1$ the map g induces on level 1, sc-smooth by Proposition 1.17 Chapter1/Levels · IsScSmooth.shift Chapter4/ScPlus · IsScSmooth.comp_isScPlus; and an sc⁺-operator is an sc⁺-map Chapter4/ScPlus · ScPlus.isScPlus.

The derivative at a smooth point Chapter4/ScPlus · IsScPlus.exists_scPlus_deriv. Let f be sc⁺ on an sc-domain U and $x \in U_1$ smooth. The lift g is sc-smooth into F^1, so its derivative at the smooth point x is an sc-operator $E \to F^1$ (the remark after Proposition 1.16), that is an sc⁺-operator $S : E \to F$. Since $f = \iota\circ g$ on U, the chain rule and the uniqueness of the derivative on an sc-domain give $Df(x) = \iota \circ Dg(x)$, the level-0 operator of S. This is the text's step "$Dr(o)$ is a sc⁺-operator, since r is a sc⁺-map". Consequently $Df(x)$ is compact, and so is every level operator of S Chapter4/ScPlus · IsScPlus.isCompactOperator_deriv.

A compact projection has finite-dimensional range Chapter4/ScPlus · finiteDimensional_range_of_isCompactOperator_of_idem: if $P = P \circ P$ is compact, its range is the closed set $\{v : Pv = v\}$, P restricts to a compact operator of the range which is the identity, and a normed space with compact identity is finite-dimensional (Riesz). The text takes this for granted.

IV.3 Sc⁺-retractions and sc⁺-retracts (Definitions 4.4, 4.6, Lemmas 4.5, 4.7)

Definition 4.4. An sc-smooth retraction $r : U \to U$ is an *sc⁺-retraction* if it is an sc⁺-map on U Chapter4/ScPlus · ScRetraction.IsScPlus. The text's condition $r(U_m) \subseteq U_{m+1}$ is then a consequence, not a hypothesis Chapter4/ScPlus · ScRetraction.IsScPlus.mem_lvlSet_succ. On an open subset V of an M-polyfold, an sc⁺-retraction is a map s, sc-smooth on V, with $s(V) \subseteq V$, $s \circ s = s$ on V, $s(V) \subseteq X_1$, and $1^1_0 \circ s : V \to X^1$ sc-smooth Chapter4/ScPlus · MPolyfold.IsScPlusRetractionOn; the junk value of 1^1_0 off X_1 is quantified over and plays no role.

Lemma 4.5 Chapter4/ScPlus · lemma_4_5. *If t is an sc⁺-retraction onto O and s an sc-smooth retraction (of a possibly different relatively open set) onto O, then s is sc⁺.* The text's proof: s maps its domain into O, which lies in the domain of t, and t fixes O, so $s = t \circ s$, which is sc⁺ □ sc-smooth. (The text writes "$s(V) = 0$" for $s(V) = O$.)

Definition 4.6. (O, C, E) is an *sc⁺-retract* if it is the retract of an sc⁺-retraction Chapter4/ScPlus · IsScPlusRetract; it is then a retract Chapter4/ScPlus · IsScPlusRetract.isScRetract, every sc-smooth retraction onto it is sc⁺ by Lemma 4.5 Chapter4/ScPlus · IsScPlusRetract.isScPlus, and so the property is intrinsic Chapter4/ScPlus · isScPlusRetract_iff_forall.

Lemma 4.7 (i) Chapter4/ScPlus · lemma_4_7_i. *The retract of an sc⁺-retraction consists of smooth points.* By induction on m: a point $y \in O$ lies in E_m; then $y \in U_m$, and $y = r(y) \in E_{m+1}$ by the level gain. (ii) Chapter4/ScPlus · lemma_4_7_ii. *At a level-1 point o over a point of O, $Dr(o)E_0$ is finite-dimensional.* o is smooth by (i), so $Dr(o)$ is compact (§IV.2); it is a projection because $r \circ r = r$ gives $Dr(o) \circ Dr(o) = Dr(o)$ at points of O Chapter2/Retracts · ScRetraction.scDeriv_comp_self; and a compact projection has finite-dimensional range. In the form with the tangent space of Definition 2.17 Chapter4/ScPlus · lemma_4_7_ii_tangentSpace: every $o \in O$ has a smooth lift $o_1 \in E_1$ Chapter2/TameRetracts · exists_smoothLvl_one_of_mem_smooth, and $T_{o_1}O$ is the range of $Dr(o_1)$ Chapter2/Retracts · tangentSpace_eq_coe_submodule, finite-dimensional. Tangent spaces are indexed by level-1 points, which is why the statement mentions the lift.

IV.4 Proposition 4.8, its chart form, and Remark 4.3

Proposition 4.8 `Chapter4/ScPlus · proposition_4_8`. *A retract (O, C, E) is an sc⁺-retract iff $O \subseteq E_\infty$ and the lift 1^1_0 is sc-smooth from (O, C, E) to (O_1, C_1, E^1)* in the sense of Definition 2.7. (□) $O \subseteq E_\infty$ is Lemma 4.7 (i); and by the criterion of Definition 2.7 through one retraction `Chapter2/Retracts · isScSmoothRetractMap_iff_exists`, the lift is sc-smooth between the retracts because $1^1_0 \circ r$ is the lift of the sc⁺-map r. (□) For a retraction r onto O, the composite $U \to O \to O_1 \to E^1$ is sc-smooth (Definition 2.7, the field `comp_isScSmooth` of `Chapter2/Retracts · IsScSmoothRetractMap`) and is the lift of r, whose values lie in E_1 because $O \subseteq E_\infty$. The text's proof is complete; it ends "the lemma is proved".

The chart form. Theorem 4.2 needs the statement for the charts of an M-polyfold, which the text does not state. The map $1^1_0 : X \to X^1$ is defined on X_1, with a junk value x_0 off it `Chapter4/ScPlus · MPolyfold.toLevelOne` `Chapter4/ScPlus · MPolyfold.levelVal_toLevelOne` `Chapter4/ScPlus · MPolyfold.toLevelOne_levelVal`. In the chart φ of X and the level chart $(V_1, \varphi, (O_1, C_1, E^1))$ of X^1 `Chapter2/LevelPolyfold · MPolyfold.levelChart` it is the lift $O \to O_1$ `Chapter4/ScPlus · MPolyfold.levelChart_toLevelOne`. A point of a chart domain is smooth iff its image is `Chapter4/ScPlus · MPolyfold.mem_smoothSet_iff`. So *the model of a chart $\varphi : V \to O$ is an sc⁺-retract iff $V \subseteq X_\infty$ and 1^1_0 is sc-smooth on V* `Chapter4/ScPlus · MPolyfold.proposition_4_8_chart`: (□) continuity on V comes from the lift read in the charts, and sc-smoothness is checked in the pair of charts φ, `levelChart` φ `Chapter2/LevelPolyfold · IsScSmoothOn.of_exists_charts`, where it is Proposition 4.8; (□) the same pair of charts turns the sc-smoothness of 1^1_0 on V into that of the lift on O. Hence "$X_0 = X_\infty$ and 1^1_0 sc-smooth" holds iff every chart of X is modelled on an sc⁺-retract `Chapter4/ScPlus · MPolyfold.isScSmoothMap_toLevelOne_iff`, and already if some cover of X by such charts exists `Chapter4/Characterization · MPolyfold.isScSmoothMap_toLevelOne_of_cover` — sc-smoothness being local.

Remark 4.3. The inclusion $\iota : (O_1, C_1, E^1) \to (O, C, E)$ is sc-smooth between retracts `Chapter4/ScPlus · isScSmoothRetractMap_toBase_one`, so $1^0_1 : X^1 \to X$ is always sc-smooth, being ι in the charts `levelChart` φ and φ `Chapter4/ScPlus · MPolyfold.remark_4_3_levelVal`. If every point lies in X_1 and 1^1_0 is sc-smooth, it is an sc-diffeomorphism with inverse 1^0_1 `Chapter4/ScPlus · MPolyfold.remark_4_3_isScDiffeo`. For a finite-dimensional V with the constant structure, $V^1 = V$ definitionally, the lift is the identity, and every retract in V is an sc⁺-retract, its retraction being its own lift `Chapter4/ScPlus · remark_4_3_const`. The remark's "it is possible that $X^1 \neq X$ as M-polyfolds although the sets agree" is given without an example; one is constructed in §IV.27.

IV.5 Charts centred at a smooth point

Theorem 4.2 (ii) asks for charts with $\psi(x) = 0$, the proofs of Theorems 4.18 and 4.22 and of Lemma 5.11 use them, and the text never says where they come from. Translating a chart $\varphi : V \to O \subseteq C$ by $e = \varphi(x)$ fails twice: translation by e preserves every level only if e is smooth, and $C - e$ is not a partial quadrant unless $e \in C \cap (-C)$. The construction (Findings, items 1, 2) uses the partial quadrant C_e and the neighbourhood on which it agrees with C.

The partial quadrant C_e and the neighbourhood $G(e)$. Let $G(e) = \{v : \lambda_i(v) > 0 \text{ whenever } \lambda_i(e) \neq 0\}$ `Chapter4/Recentre · recentreNbhd`, open `Chapter4/Recentre · isOpen_recentreNbhd` and containing e when $e \in C$ `Chapter4/Recentre · mem_recentreNbhd`. On it C and C_e agree: $G(e) \cap C = G(e) \cap C_e$, because on $G(e)$ the coordinates not vanishing at e are positive anyway `Chapter4/Recentre · recentreNbhd_inter_eq`. C_e is invariant under translation by e `Chapter4/Recentre · add_mem_partialQuadrantAtWith_iff`, so $C_e - e = C_e$ `Chapter4/Recentre · image_sub_partialQuadrantAtWith`. *C_e is a partial quadrant* — Definition 2.33 asserts it without proof: the vanishing indices $I(e)$ `Chapter4/Recentre · zeroIdx`, $d = \#I(e)$ of them `Chapter4/Recentre · recentreDim`, are put first, $\mathbb{R}^n \cong \mathbb{R}^{I(e)} \times \mathbb{R}^{\text{rest}}$ `Chapter4/Recentre · splitCoordsL`, which gives an sc-isomorphism $\mathbb{R}^n \oplus W \cong \mathbb{R}^{I(e)} \oplus (\mathbb{R}^{\text{rest}} \oplus W)$ `Chapter4/Recentre · splitCoordsIso` and the presentation T_e of C_e `Chapter4/Recentre · recentreIso` `Chapter4/Recentre · partialQuadrantAtWith_eq_preimage` `Chapter4/Recentre · isPartialQuadrant_partialQuadrantAtWith`: the coordinates not vanishing at e join the linear factor. *Re-centring keeps the tame data near e.* For $v \in G(e)$ the vanishing coordinates of $v - e$ for

T_e are those of v for T Chapter4/Recentre · recentreIso_fst_sub Chapter4/Recentre · zero_of_mem_recentreNbhd, so $d_{C_e}(v-e) = d_C(v)$ Chapter4/Recentre · degIndexWith_recentreIso Chapter4/Recentre · degIndex_sub_recentre, and the minimal subspaces and partial quadrants agree: $E^{T_e}_{v-e} = E^T_v$ Chapter4/Recentre · minimalSubspaceWith_recentreIso, $(C_e)_{v-e} = C_v$ Chapter4/Recentre · partialQuadrantAtWith_recentreIso.

Translations. Translation by a smooth vector a is sc-smooth on every sc-domain: its level maps are $x \mapsto x+a_m$, a_m the lift of a to E_m, and the criterion of Proposition 1.19 applies Chapter1/Levels · IsScSmooth.of_contDiffOn_levels Chapter4/Recentre · isScSmooth_add_const Chapter4/Recentre · isScSmooth_sub_const. For U relatively open in C with $U \subseteq G(e)$, the translate $U - e$ is relatively open in C_e Chapter4/Recentre · relOpen_translate and, for smooth e, $v \mapsto v - e$ is an sc-diffeomorphism $U \to U - e$ Chapter4/Recentre · isScDiffeoQuadrant_translate.

Re-centred retracts. A retraction r restricted to $U \cap r^{-1}(W) \cap W$, W open, is a retraction whose domain lies in W Chapter4/Recentre · ScRetraction.restrictIn, onto $r(U) \cap W$ Chapter4/Recentre · ScRetraction.restrictIn_retract, with the same derivative Chapter4/Recentre · ScRetraction.scDeriv_restrictIn, and tame if r is Chapter4/Recentre · ScRetraction.IsTameWith.restrictIn Chapter4/Recentre · ScRetraction.restrictIn_isTame. Transporting the restriction to $G(e)$ along the translation Chapter2/TameInvariance · IsScDiffeoQuadrant.transport shows that $((O \cap G(e)) - e, C_e, E)$ is a retract, with retraction $u \mapsto r(u+e) - e$ Chapter4/Recentre · isScRetract_recentre, and tame if (O, C, E) is Chapter2/TameInvariance · IsScDiffeoQuadrant.isTame_transport Chapter4/Recentre · isTameRetract_recentre — the transport of tameness uses exactly the invariance of d and of the minimal subspaces above.

Re-centred charts. For a chart φ, a presentation of its partial quadrant and a smooth e, the domain $V \cap \varphi^{-1}(G(e))$ Chapter4/Recentre · MChart.recentreSource is open, and the chart $\varphi_e = \varphi - e$ on it is φ restricted Chapter2/MPolyfold · MChart.restrict and composed with the translation Chapter4/Recentre · MChart.recentre, modelled on $((O \cap G(e)) - e, C_e, E)$. It is a chart of the M-polyfold, its transition maps being those of φ composed with translations Chapter3/GlobalIFTCenter · MChart.compDiffeo_isChart Chapter4/Recentre · MChart.recentre_isChart; $\varphi_e(x) = 0$ when $\varphi(x) = e$ Chapter4/Recentre · MChart.recentre_apply_self; its model is tame when that of φ is Chapter4/Recentre · MChart.isTameRetract_recentre; and it computes the same degeneracy index Chapter4/Recentre · MChart.chartDegIndex_recentre. So *every smooth point x of a chart domain is the centre of a chart of X* inside that domain, tame if the original is Chapter4/Recentre · MPolyfold.exists_centredChart, and *in a tame M-polyfold every smooth point is the centre of a tame chart* Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/Recentre · MPolyfold.IsTame.exists_centredChart. At the centre every coordinate of C_e vanishes, so the minimal subspace there is $\{\lambda = 0\}$, the text's "$\{0\} \oplus W$", which is what §§IV.14, IV.16 and IV.17 use.

Re-centred strong bundle charts (for Lemma 5.11). Only the base is translated: $(u, h) \mapsto (u \mp e, h)$ Chapter4/Recentre · subFst Chapter4/Recentre · addFst is sc-smooth on $E \oplus F$ and on $E \oplus F^1$ because $(e, 0)$ is smooth in both Chapter4/Recentre · mem_smooth_prod_zero Chapter4/Recentre · isScSmooth_addFst Chapter4/Recentre · isScSmooth_subFst. A strong bundle retraction restricts to $(U \cap r^{-1}(G) \cap G) \lhd F$ Chapter4/Recentre · IsStrongBundleRetraction.restrictIn and translates to $R_e(u, h) = (r(u+e) - e, \Gamma(u+e, h))$ Chapter4/Recentre · IsStrongBundleRetraction.translate; with $\Phi_e(y) = (\Phi(y)_1 - e, \Phi(y)_2)$ this is the re-centred chart Chapter4/Recentre · StrongBundleChart.recentre. Its overlaps are translates of open parts of the old ones Chapter4/Recentre · StrongBundleChart.recentre_overlap Chapter4/Recentre · StrongBundleChart.overlap_recentre_eq, its transition maps on $[0]$ and $[1]$ are the old ones composed with translations by $(e, 0)$, so compatibility passes to it Chapter4/Recentre · StrongBundleChart.Compatible.recentre, and it is a strong bundle chart Chapter4/Recentre · StrongBundle.IsChart.recentre. Every smooth point of the base is the centre of one Chapter4/Recentre · StrongBundle.IsChart.exists_centred. No hypothesis beyond the smoothness of the point is needed.

IV.6 Proposition 4.9: the data, and the mean value estimate

The text. Let $t : U \to U$ be an sc$^+$-retraction onto $O \ni 0$ with T_0O in good position to C, Y a good complement and $p = Dt(0)$ "the sc-projection". Then p maps a neighbourhood of 0 in O sc-diffeomorphically

onto one in $T_0O \cap C$. The proof: the section $\mathbf{f}(u) = (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm (Lemma 4.10), Corollary 3.47 describes its zeros near 0 as a graph $\{a + \delta(a)\}$, Lemma 4.11 says that every small zero is on the graph, Lemma 4.12 that the graph lies in O.

What changes, and why. (a) $Dt(0)$ is a projection onto T_0O along $\ker Dt(0)$, which nothing relates to Y; Lemma 4.12 writes $t(a+\delta(a)) = p\,t(a+\delta(a)) + \delta(a)$ with $\delta(a) \in Y$, which needs the projection *along* Y. (b) Y must be closed, or that projection is not continuous. (c) Corollary 3.47 gives δ over a complement of its own choosing and sc-smooth only as a germ — C^k into E_m on neighbourhoods shrinking with m and k (§III.20) — while the claim is an sc-diffeomorphism between fixed open sets. The formalization therefore takes p to be the projection along a closed Y and proves the proposition directly, with the sc$^+$ property doing the work of the Fredholm theory (Findings, items 20–28).

The data `Chapter4/ScPlusGraph · GraphData`: the sc$^+$-retraction t with $0 \in O$; N with $N = T_0O$, which equals $Dt(0)E_0$ `Chapter4/ScPlusGraph · GraphData.N_eq_range` and is finite-dimensional by Lemma 4.7 `Chapter4/ScPlusGraph · GraphData.finiteDimensional`; N in good position; a good complement Y, closed; and $p : E_0 \to N$ continuous linear with $p = 1$ on N, $p = 0$ on Y. The tangent space consists of smooth points: $Dt(0)$ is an sc$^+$-operator and N its fixed vectors `Chapter4/ScPlusGraph · GraphData.smooth`. $Y = \ker p$ `Chapter4/ScPlusGraph · GraphData.mem_Y_iff`. Because $N \subseteq E_\infty$ is finite-dimensional, it lifts continuously to every level `Chapter3/TangentGerm · finSmoothLift` `Chapter4/ScPlusGraph · GraphData.liftN`, and p, $q = 1 - p$ act on every level E_m `Chapter4/ScPlusGraph · GraphData.pL` `Chapter4/ScPlusGraph · GraphData.qL`. The identities the text uses hold for this p on every level: $(1-p)Dt(0) = 0$, because the range of $Dt(0)$ is N `Chapter4/ScPlusGraph · GraphData.qL_lvlDeriv_zero`, and $Dt(0) = 1$ on N `Chapter4/ScPlusGraph · GraphData.lvlDeriv_zero_of_mem_N`. The good complement has a constant ε: for $n \in N$, $y \in Y$, $|y|_0 \le \varepsilon|n|_0$, $n \in C \iff n + y \in C$ `Chapter4/ScPlusGraph · GraphData.eps` `Chapter4/ScPlusGraph · GraphData.eps_spec`. Its level-1 version is $\varepsilon' = \varepsilon/(\|\iota_1\|\|\mathrm{lift}_N\| + 1)$ `Chapter4/ScPlusGraph · GraphData.eps'`: $|y|_1 \le \varepsilon'|a|_1$ implies $|y|_0 \le \varepsilon|a|_0$ for $a \in N$ `Chapter4/ScPlusGraph · GraphData.norm_toBase_le_eps_mul`.

The level data of an sc$^+$-retraction. The lift $t^1 : U \to E^1$ `Chapter4/ScPlusGraph · ScRetraction.liftOne` is sc-smooth `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isScSmooth_liftOne`, sc^1 with derivative family Dt^1 `Chapter4/ScPlusGraph · ScRetraction.liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.isSc1With_liftOne`. By Proposition 1.16 for t^1, the map $t : U_{k+1} \to E_{k+1}$ `Chapter4/ScPlusGraph · ScRetraction.lvlMap` is differentiable within U_{k+1} with a derivative $Dt(x) \in \mathcal{L}(E_{k+1})$ `Chapter4/ScPlusGraph · ScRetraction.lvlDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.hasFDerivWithinAt_lvlMap` that depends continuously on x in the operator norm `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlDeriv`; t is continuous on U_{k+1} `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.continuousOn_lvlMap`, fixes O_{k+1} `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlMap_eq_self`, is t itself through the inclusions `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlMap`, and its level derivatives are restrictions of Dt `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_liftDeriv` `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.toBase_lvlDeriv`, projections at points of O `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.lvlDeriv_lvlDeriv`.

The mean value estimate (the text's (76)–(77)). The abstract form `Chapter4/ScPlusGraph · exists_norm_sub_sub_le_of_hasFDerivWithinAt`: if s is differentiable within a convex set S with derivative D continuous at $u_0 \in S$, then for every $\theta > 0$ there is $r > 0$ with $|s(z') - s(z) - D(u_0)(z' - z)| \le \theta|z' - z|$ for $z, z' \in S \cap B(u_0, r)$ — the mean value inequality for $s - D(u_0)$. For an sc$^+$-retraction `Chapter4/ScPlusGraph · ScRetraction.IsScPlus.exists_mvt`: at $u_0 \in U_{k+1}$ a ball meets C_{k+1} inside U_{k+1} `Chapter4/ScPlusGraph · ScRetraction.exists_ball_inter_subset` `Chapter4/ScPlusGraph · ScRetraction.lvlSet_U_subset_lvlSet_C`, the set $C_{k+1} \cap B(u_0, r_0)$ is convex, and the abstract form and the continuity of Dt give r with $C_{k+1} \cap B(u_0, r) \subseteq U_{k+1}$, $\|Dt(z) - Dt(u_0)\| \le \theta$ and the estimate with constant θ there. The text integrates along segments in U_{k+1}, which need not be convex; the convex set $C_{k+1} \cap B$ is the repair.

A shortcut does not work: inverting $p \circ t$ on the finite-dimensional quadrant $N \cap C$ by Proposition 3.73. In $\mathbb{R}^3$ with $C = [0,\infty)^2 \times \mathbb{R}$, $\Gamma(a) = (a_1, a_2, a_2^2)$ and $t(x) = \Gamma(x_1 + (x_3 - x_2^2)^2, x_2)$, t is a retraction with $N = \mathbb{R}^2 \times 0$, but $p\,t(x_1, x_2, 0) = (x_1 + x_2^4, x_2)$ misses $(0, y_2)$ for $y_2 > 0$.

IV.7 Lemmas 4.11 and 4.12: uniqueness and existence on level 1

Lemma 4.11, the contraction estimate `Chapter4/ScPlusGraph · GraphData.exists_contraction_radius`. For $\theta > 0$ the mean value estimate at $u_0 = 0$ gives $r > 0$ with the estimates of §IV.6 on $C_{k+1} \cap B(0,r)$ and, for $u, u' \in O_{k+1}$ with norms below r, $|(1-p)(u'-u)|_{k+1} \le \|1-p\|\theta|u'-u|_{k+1}$: since $u = t(u)$, $u' = t(u')$ and $(1-p)Dt(0) = 0$, $(1-p)(u'-u) = (1-p)[t(u') - t(u) - Dt(0)(u'-u)]$. **Uniqueness** `Chapter4/ScPlusGraph · GraphData.eq_of_p_eq`: if $p(u) = p(u')$ and $|(1-p)(u'-u)| \le c|u'-u|$ with $c < 1$, then $u'-u \in Y$, so $(1-p)(u'-u) = u'-u$ and $u' = u$. This is what the text's Lemma 4.11 is used for; its statement, "$y = \delta(a)$", presupposes the δ of Corollary 3.47.

Lemma 4.12, existence by a contraction `Chapter4/ScPlusGraph · GraphData.exists_graph_point`. Suppose the estimates hold on $C_1 \cap B(0,r)$ with $\theta < 1$, $\|1-p\|\theta < 1$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$. As $t(0) = 0$ and $t : U_1 \to E_1$ is continuous at 0, there is δ with $|t(u)|_1 < r$ for $|u|_1 < \delta$; put $r' = \min(r, \delta)$ and $\rho = r'/(1+\varepsilon')$. For $a \in N \cap C$ with $|a|_1 < \rho$:

1. The set $B_a = \{y \in Y_1 : |y|_1 \le \varepsilon'|a|_1\}$ is closed, hence complete, because Y is closed — the first use of that hypothesis.
2. For $y \in B_a$, $a+y \in C_1$ by the good complement property, $|y|_0 \le \varepsilon|a|_0$ coming from $|y|_1 \le \varepsilon'|a|_1$; and $|a+y|_1 \le (1+\varepsilon')|a|_1 < r'$. So $a+y$ lies in U_1, where t is defined — a point the text does not check.
3. $\Phi_a(y) = (1-p)t(a+y)$ maps B_a into itself: $\Phi_a(y) = (1-p)[t(a+y) - t(0) - Dt(0)(a+y)]$, of norm at most $\|1-p\|\theta(1+\varepsilon')|a|_1 \le \varepsilon'|a|_1$, and it lies in Y.
4. Φ_a is a contraction with constant $\|1-p\|\theta < 1$, by the estimate between $a+y$ and $a+y'$.
5. Its fixed point y (Banach's theorem on the complete B_a) gives $u = a+y$ with $(1-p)(t(u)-u) = 0$, i.e. $w - u \in N$ for $w = t(u)$. Now $w \in O$, $|w|_1 < r$ by the choice of δ, $t(w) = w$, and $Dt(0)(w-u) = w-u$. The estimate between u and w reads $|t(w) - t(u) - Dt(0)(w-u)|_1 = |w-u|_1 \le \theta|w-u|_1$, so $w = u$: $u = t(u) \in O$ — the text's argument "$a_1 = a$".

So $u \in O_1$, $p(u) = a$, $u - a \in Y$, $|u-a|_1 \le \varepsilon'|a|_1$ and $|u|_1 < r$. The text's "$|a(\tau)|_1 \le \varepsilon_1$ if $|a|_0 < \varepsilon$" is the level-1 continuity of t at 0 used in step 5.

IV.8 The step $p(O \cap W) \subseteq C$, and the neighbourhoods

p maps points of O near 0 into C `Chapter4/ScPlusGraph · GraphData.p_mem_C`, which the conclusion "$p : U_0 \to V \subseteq T_0O \cap C$" needs and the text never checks. If $u \in O_1$ has $|(1-p)u|_1 \le \kappa|u|_1$ with $\kappa \le \frac12$, then $|u|_1 \le |pu|_1 + \frac12|u|_1$, so $|u|_1 \le 2|pu|_1 \le 2\|\mathrm{lift}_N\||pu|_0$, and $|(1-p)u|_0 \le \|\iota_1\|\kappa|u|_1 \le 2\|\iota_1\|\kappa\|\mathrm{lift}_N\||pu|_0$; if that constant is at most ε, the good complement property gives $p(u) \in C \iff p(u) + (1-p)u = u \in C$, and $u \in O \subseteq C$. The hypothesis on u is the contraction estimate between 0 and u.

The neighbourhoods `Chapter4/ScPlusGraph · GraphData.Nbhd` `Chapter4/ScPlusGraph · GraphData.exists_nbhd`. For every $\theta_0 > 0$ the proof chooses, in this order:

1. $\theta = \min(\theta_0, \theta_1, \theta_2)$ with $\theta_1 = \varepsilon'/(2(\|1-p\|+1)(1+\varepsilon'))$ (so that $\|1-p\|\theta \le \frac12$ and $\|1-p\|\theta(1+\varepsilon') \le \varepsilon'$, for Lemmas 4.11 and 4.12) and $\theta_2 = \varepsilon/(2\|\iota_1\|\|\mathrm{lift}_N\|(\|1-p\|+1)+1)$ (for $p(O \cap W) \subseteq C$);
2. r from the contraction estimate on level 1, and ρ and the graph points $u(a)$ from Lemma 4.12;
3. δ_0 from the continuity of the lift $t^1 : U \to E_1$ at 0 on level 0: $|t(u)|_1 < r$ for $u \in U$, $|u|_0 < \delta_0$;
4. $\rho' = \min(\rho, r/(1+\varepsilon'), \delta_0/(\|\iota_1\|(1+\varepsilon')+1))$, the open $V' = \{a \in N : |a|_1 < \rho'\}$ of N, $V = V' \cap C$, and $W = B_0(0,\delta_0) \cap p^{-1}(V')$, open in E_0;
5. $\Gamma(a) = u(a)$ for $a \in V$.

Then $\Gamma(a) \in O \cap W$ (on level 0, $|\Gamma(a)|_0 \le \|\iota_1\|(1+\varepsilon')|a|_1 < \delta_0$), $p\Gamma(a) = a$, $\Gamma(a) - a \in Y$, $|\Gamma(a)|_1 < r$. A point $u_0 \in O \cap W$ satisfies $u_0 = t(u_0)$, hence $|u_0|_1 < r$ by step 3 — *this is how a neighbourhood open on level* 0 *is controlled on level* 1, the sc⁺ property a second time; the contraction estimate between 0 and u_0 and the first constraint on θ then give $p(u_0) \in C$, so $p(u_0) \in V$, and uniqueness gives $\Gamma(p(u_0)) = u_0$. Also $\|Dt(z) - Dt(0)\|_{\mathcal{L}(E_1)} \le \theta_0$ on $C_1 \cap B(0,r)$, and $|(1-p)(u'-u)|_1 \le \frac12|u'-u|_1$

on $O_1 \cap B_1(0,r)$. Consequently $\Gamma(V) = O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.image_V, $\Gamma(0) = 0$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_zero, and p is injective on $O \cap W$ Chapter4/ScPlusGraph · GraphData.Nbhd.injOn_p.

IV.9 The regularity of Γ on the fixed V: the frame formula and the bootstrap

Continuity into every level. $\Gamma(V) \subseteq O \subseteq E_\infty$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_mem_smooth, so Γ has lifts $\Gamma_m : V \to E_m$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl with values in O_m Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_mem_retract. Γ_1 is Lipschitz: with $u = \Gamma(a)$, $u' = \Gamma(a')$, $|u'-u|_1 \le |p(u'-u)|_1 + |(1-p)(u'-u)|_1 \le \|\mathrm{lift}_N\||a'-a| + \frac{1}{2}|u'-u|_1$, so $|u'-u|_1 \le 2\|\mathrm{lift}_N\||a'-a|$ Chapter4/ScPlusGraph · GraphData.Nbhd.norm_Γ_lvl_one_sub_le Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl_one. And $\Gamma_{k+1} = t^1 \circ \Gamma_k$ on level k, because $\Gamma(a) = t(\Gamma(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_lvl_succ; the lift t^1 is continuous from U_k into E_{k+1}, so by induction Γ is continuous into every level Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ_lvl Chapter4/ScPlusGraph · GraphData.Nbhd.continuousOn_Γ.

The frame at a point $u \in O$ near 0. Let $A(u) = p \circ Dt(u)|_N : N \to N$ Chapter4/ScPlusGraph · GraphData.A, $M(u) = A(u)^{-1}$ Chapter4/ScPlusGraph · GraphData.M and $B_u = Dt(u) \circ M(u)$, the *frame* Chapter4/ScPlusGraph · GraphData.frame, with its level-$(k+1)$ version Chapter4/ScPlusGraph · GraphData.frameL. For $n \in N$, $Dt(u)n - n = \iota_1(Dt_1(u) - Dt_1(0))\mathrm{lift}(n)$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_eq. Choose θ_0 with $\|p\|\|\iota_1\|\|\mathrm{lift}_N\|\theta_0 \le \frac{1}{2}$ and $\|1-p\|\theta_0 < 1$ Chapter4/ScPlusGraph · GraphData.FrameSmall Chapter4/ScPlusGraph · GraphData.exists_frameSmall, and let $\|Dt_1(u) - Dt_1(0)\| \le \theta_0$. Then (L1) $|A(u)n - n| \le \frac{1}{2}|n|$, so $A(u)$ is injective Chapter4/ScPlusGraph · GraphData.A_injective, hence an isomorphism of the finite-dimensional N Chapter4/ScPlusGraph · GraphData.Aeq, with $A(u)M(u) = 1$ Chapter4/ScPlusGraph · GraphData.A_M and $p \circ B_u = 1$ Chapter4/ScPlusGraph · GraphData.p_frame. (L2) $T_uO \cap Y = 0$: a vector $v = Dt(u)v$ in Y satisfies $v = (1-p)v = (1-p)(Dt(u) - Dt(0))v$ on level 1, so $|v|_1 \le \|1-p\|\theta_0|v|_1$ Chapter4/ScPlusGraph · GraphData.eq_zero_of_mem_Y_of_scDeriv_eq. Hence for $u \in O_1$: $Dt(u)x - B_u(p\,Dt(u)x)$ lies in $T_uO \cap Y$ and vanishes, $(1 - B_up)Dt(u) = 0$ Chapter4/ScPlusGraph · GraphData.scDeriv_sub_frame; $Dt(u)B_u = B_u$ Chapter4/ScPlusGraph · GraphData.scDeriv_frame; and both hold on every level Chapter4/ScPlusGraph · GraphData.p_toBase_frameL Chapter4/ScPlusGraph · GraphData.lvlDeriv_sub_frameL. On the points of $\Gamma(V)$ the derivative bound holds by the choice of r Chapter4/ScPlusGraph · GraphData.Nbhd.deriv_bound_Γ.

The abstract step. On a set S of a normed space, let s satisfy the mean value estimate with derivative D_0 at u_0 and constant θ, let q, B be continuous linear with $q \circ B = 1$ and $(1-Bq)D_0 = 0$, and $\theta\|1-Bq\| \le \frac{1}{2}$. For fixed points u, u' of s in S, $(1-Bq)(u'-u) = (1-Bq)[s(u') - s(u) - D_0(u'-u)]$, of norm at most $\frac{1}{2}|u'-u|$, so $|u'-u| \le 2\|B\||q(u'-u)|$ Chapter4/ScPlusGraph · norm_sub_le_of_projection. And *every continuous right inverse Γ of q with values in the fixed points of s in S, through u_0, is differentiable at $z_0 = q(u_0)$ with derivative B* Chapter4/ScPlusGraph · hasFDerivWithinAt_of_projection: $\Gamma(z) - \Gamma(z_0) - B(z-z_0) = (1 - Bq)(\Gamma(z) - u_0) = (1-Bq)[(\Gamma(z)-u_0) - D_0(\Gamma(z)-u_0)]$, which is $o(|\Gamma(z)-u_0|)$ by the differentiability of s at u_0, and $|\Gamma(z) - u_0| \le 2\|B\||z - z_0|$ by the Lipschitz estimate.

The frame formula Chapter4/ScPlusGraph · GraphData.Nbhd.hasFDerivWithinAt_Γ_lvl. At $a_0 \in V$ and on level $k+1$, apply the abstract step with $s = t$, $u_0 = \Gamma_{k+1}(a_0)$, $q = p \circ \iota_{k+1}$, B the level-$(k+1)$ frame at u_0, $\theta = 1/(2(\|1-Bq\|+1))$, and $S = C_{k+1} \cap B(u_0, r)$ from the mean value estimate *at* u_0; Γ_{k+1} stays in $B(u_0, r)$ near a_0 by continuity. So

$$D\Gamma(a) = Dt(\Gamma(a))|_N \circ \left(p \circ Dt(\Gamma(a))|_N\right)^{-1} \quad \text{into every level } E_{k+1}.$$

The estimate is used at $\Gamma(a)$ with the projection $1 - B_{\Gamma(a)}p$, not at 0 with $1-p$: this is why the regularity holds at every point of the fixed V.

The C^k-bootstrap. By Proposition 1.18 for the tangent map of the lift t^1, $(x,w) \mapsto Dt_{j+1}(x)w$ is C^k on $U_{j+k+1} \times E_{j+k+1}$, read through the inclusions into level $j+1$ Chapter4/ScPlusGraph · GraphData.contDiffOn_

lvlDeriv_apply. If Γ_{j+k+1} and $w : V \to N$ are C^k, then so is $a \mapsto Dt_j(\Gamma_{j+1}(a))\,\mathrm{lift}(w(a))$ Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_lvlDeriv_Γ; in particular $a \mapsto A(\Gamma(a))$ is C^k when Γ_{k+1} is Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_A_Γ, and so is its inverse, inversion being smooth on the invertible operators Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_M_Γ. Now induct on k, for all levels at once Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl: Γ_{j+1} is C^0 (continuity); if every $\Gamma_{j'+1}$ is C^k, the derivative $a \mapsto \mathrm{frame}_j(\Gamma_{j+1}(a))$ applied to n is $Dt_j(\Gamma_{j+1}(a))\mathrm{lift}(M(\Gamma_1(a))n)$, which is C^k because Γ_{j+k+1} is (the hypothesis at $j' = j+k$) and $M(\Gamma_1(\cdot))n$ is (the hypothesis at $j' = k$, through A); so Γ_{j+1} is C^{k+1}. Derivatives of t are only ever applied to the smooth vectors of the finite-dimensional N, so the loss of levels in Tt is absorbed by Γ being C^k into *every* level and does not accumulate. Hence Γ is C^∞ into every level Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_succ_infty Chapter4/ScPlusGraph · GraphData.Nbhd.contDiffOn_Γ_lvl_infty and sc-smooth on V by the criterion of Proposition 1.19 for a finite-dimensional source Chapter4/ScPlusGraph · GraphData.Nbhd.isScSmooth_Γ.

The sc-diffeomorphism. $(O \cap W, C, E)$ is a retract (Proposition 2.10 (1)) Chapter2/Retracts · proposition_2_10_1 Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_inter_W; $(V, N \cap C, N)$ is a local model, $N \cap C$ being a partial quadrant of N (Proposition 3.22) and V relatively open in it Chapter4/ScPlusGraph · GraphData.Nbhd.V_isRelOpenInQuadrant Chapter4/ScPlusGraph · GraphData.Nbhd.isScRetract_V. p is sc-smooth $E \to N$ as a continuous linear map into a finite-dimensional space (Corollary 1.20) Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional Chapter4/ScPlusGraph · GraphData.p_isScSmooth, so p and Γ are sc-smooth between the retracts Chapter4/ScPlusGraph · GraphData.Nbhd.p_isScSmoothRetractMap Chapter4/ScPlusGraph · GraphData.Nbhd.Γ_isScSmoothRetractMap.

Proposition 4.9 Chapter4/ScPlusGraph · proposition_4_9. *Statement.* Let t be an sc⁺-retraction onto $O \ni 0$, $N = T_0O$ in good position to C, Y a closed good complement of N and $p : E_0 \to N$ the projection along Y. There are an open $W \ni 0$ in E_0, a relatively open $V \ni 0$ of $N \cap C$ and $\Gamma : N \to E_0$, sc-smooth on V, with $p \circ \Gamma = 1$ on V, $\Gamma(a) - a \in Y$, $\Gamma(V) = O \cap W$, $\Gamma \circ p = 1$ on $O \cap W$, $\Gamma(0) = 0$, and p, Γ sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *Proof.* θ_0 from the frame condition, the neighbourhoods of §IV.8 for it, and the regularity above. The finite dimension of N, an instance argument needed to form N as an sc-Banach space, is Lemma 4.7 Chapter4/ScPlusGraph · finiteDimensional_tangentSpace_zero.

IV.10 Theorem 4.2: the characterization

Statement Chapter4/Characterization · MPolyfold.theorem_4_2_general. For every M-polyfold X and every junk value $x_0 \in X_1$ of 1^1_0, the following are equivalent. (i) X has a corner structure. (ii) Every point of X is smooth, $1^1_0 : X \to X^1$ is sc-smooth, and every x lies in a chart ψ of X with $\psi(x) = 0$ whose model is a tame retract and whose tangent space $N = T_0O$ is finite-dimensional and in good position to C. The text's form, for tame X, is the same statement with an unused hypothesis Chapter4/Characterization · MPolyfold.theorem_4_2. Tameness is used neither in the text's proof nor here, and (i) implies it (§IV.1).

Charts composed with an sc-diffeomorphism between retracts. The proof of (ii) □ (i) composes a chart with p, which is an sc-diffeomorphism between retracts, not between relatively open sets, so the construction of Chapter 3 does not apply. For a chart φ onto (O, C, E) and mutually inverse f, g sc-smooth between (O, C, E) and (O', C', E'), $f \circ \varphi$ is a chart onto (O', C', E') Chapter4/Characterization · MChart.compRetract, and a chart of the M-polyfold when φ is Chapter4/Characterization · MChart.compRetract_isChart: for a chart φ' of X the transition maps are $(\varphi' \circ \varphi^{-1}) \circ g$ and $f \circ (\varphi \circ \varphi'^{-1})$, compositions of maps sc-smooth between retracts (Theorem 2.8) Chapter2/Retracts · IsScSmoothRetractMap.comp. This is the compatibility the text's "we obtain a smoothly compatible atlas" skips (Findings, item 55).

(i) □ (ii). Corner charts are modelled on sc⁺-retracts, so every point is smooth and 1^1_0 is sc-smooth by Proposition 4.8 on a cover (§IV.4). The text stops here; (ii) also asks for a centred tame chart with T_0O finite-dimensional and in good position (Findings, item 53). Let x lie in a corner chart χ. Its image is

smooth (§IV.1), so the re-centred chart $\psi = \chi - \chi(x)$ exists; its model is a relatively open subset of $C_{\chi(x)}$ in the same finite-dimensional space `Chapter4/Characterization · image_sub_const_eq_preimage_add' Chapter4/Characterization · MChart.IsCornerChart.exists_centred`. With the identity retraction it is a tame retract; the tangent space of a relatively open subset of a partial quadrant is the whole space, because the derivative of the identity retraction is the identity `Chapter4/Characterization · tangentSpace_relOpen`; and the whole space is in good position to every partial quadrant, with the complement 0, the interior of a partial quadrant being nonempty `Chapter4/Characterization · isInGoodPosition_top Chapter4/Characterization · MChart.IsCornerChart.exists_centred_tame`.

(ii) □ (i). *The data of the corner chart* `Chapter4/Characterization · CornerChartData` for a chart ψ and a finite-dimensional N: a closed good complement Y, the projection p along it, W, V, Γ with p, Γ mutually inverse and sc-smooth between $(O \cap W, C, E)$ and $(V, N \cap C, N)$. *They exist* when the model of ψ is an sc⁺-retract containing 0 and $N = T_0O$ is in good position `Chapter4/Characterization · exists_cornerChartData`: here no complement is prescribed, so one can be chosen — the normal form of Proposition 3.22 gives a continuous projection onto N whose kernel is a good complement, closed, even mapping C into C `Chapter3/GoodComplementProjection · exists_proj_goodComplement` — and Proposition 4.9 is applied to an sc⁺-retraction onto O (Definition 4.6). The issue "$p = Dt(0)$" of §IV.6 does not arise, and nothing is added to Theorem 4.2 (Findings, item 54). *The chart* $p \circ \psi$ on $\psi^{-1}(W)$ `Chapter4/Characterization · CornerChartData.chart` is ψ restricted to $\psi^{-1}(W)$ composed with p by `compRetract`, modelled on $(V, N \cap C, N)$ `Chapter4/Characterization · CornerChartData.model`; it is a corner chart `Chapter4/Characterization · CornerChartData.chart_isCornerChart`, a chart of X `Chapter4/Characterization · CornerChartData.chart_isChart`, contains x when $\psi(x) = 0$ `Chapter4/Characterization · CornerChartData.mem_chart_source` and sends it to 0 `Chapter4/Characterization · CornerChartData.chart_apply_eq_zero`. *At every point* `Chapter4/Characterization · MPolyfold.theorem_4_2_chart`: if all points are smooth and 1^1_0 is sc-smooth, the model of every chart is an sc⁺-retract (§IV.4), so for a chart ψ with $\psi(x) = 0$ and $N = T_0O$ finite-dimensional and in good position the corner chart $p \circ \psi$ exists. With such a chart at every point, X has a corner structure by the local form of Definition 4.1 `Chapter4/Characterization · MPolyfold.hasCornerStructure_of_centred` — whether or not the charts ψ are tame.

IV.11 Smooth finite-dimensional submanifolds; the chart retraction; Remark 4.14

Definition 4.13. $A \subseteq X$ is a *smooth finite-dimensional submanifold* if every $a \in A$ has an open $V \ni a$ and an sc⁺-retraction s on V with $s(V) = A \cap V$ `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold`; the data at a point are a structure `Chapter4/Submanifolds · MPolyfold.SubmanifoldData Chapter4/Submanifolds · MPolyfold.isSmoothSubmanifold_iff`. An sc⁺-retraction is an sc-smooth retraction, so the data are those of Definition 2.20 `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.toSubData`, A is a sub-M-polyfold `Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.isSubMPolyfold` and carries the induced structure of Proposition 2.21 `Chapter2/SubMPolyfold · MPolyfold.subPolyfold`. Its points lie on level 1, as $a = s(a) \in X_1$ `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.mem_levelSet_one`.

The sc⁺ condition read in a chart (Lemma 6.A of the exposition). The text never reads the sc⁺ condition in charts, and all of §§4.2–4.3 rests on it (Findings, item 60). Let φ be a chart of X and $W = V \cap \operatorname{dom}\varphi \cap s^{-1}(V \cap \operatorname{dom}\varphi)$, the open set of the proof of Proposition 2.21 `Chapter2/SubMPolyfold · MPolyfold.SubData.shrink`. (a) The lift of $\varphi \circ s \circ \varphi^{-1}$ into E^1 is sc-smooth from $(\varphi(W), C, E)$ to (O_1, C_1, E^1) `Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScSmoothRetractMap_lift`: it is the sc-smoothness of $1^1_0 \circ s$ read in the charts φ of X and `levelChart` φ of X^1, where 1^1_0 is the lift. (b) For every retraction r onto O: $\varphi(W) = O \cap G$ for an open G; $\rho = \varphi \circ s \circ \varphi^{-1}$ is sc-smooth on the retract $\varphi(W)$ `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_isScSmooth` and idempotent there `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_idem`, with image $\varphi(W \cap A)$ `Chapter2/SubMPolyfold · MPolyfold.SubData.rho_image`; so $t = \rho \circ r$ on $U \cap r^{-1}(G)$ — the restriction of r `Chapter2/Retracts · ScRetraction.restrict` composed with ρ `Chapter2/Metrizable · ScRetraction.compRetraction` — is a retraction onto $\varphi(W \cap A)$, and it is sc⁺: its values $\varphi(s(x))$ lie in E_1 because $s(x) \in X_1$, and its lift is

the map of (a) composed with r Chapter4/Submanifolds · MPolyfold.SubmanifoldData.exists_chartRetraction. This is the text's $t = \varphi \circ s \circ \varphi^{-1} \circ r$ of the proof of Theorem 4.18, with W in place of "$V = \varphi^{-1}(O)$". Hence the model $\varphi(W \cap A)$ of the chart of A induced by φ is an sc⁺-retract Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isScPlusRetract_image, and $\varphi(a)$ is smooth for $a \in A \cap V \cap \operatorname{dom}\varphi$ (Lemma 4.7 (i)) Chapter4/Submanifolds · MPolyfold.SubmanifoldData.apply_mem_smooth.

Tangent spaces of sub-retracts. The tangent space of a retract is a germ: for an open K, $T_z(Q \cap K) = T_z Q$ Chapter4/Submanifolds · tangentSpace_inter_isOpen. For retracts $Q \subseteq O = r(U)$, $T_z Q \subseteq T_z O$: a retraction s onto Q satisfies $s = r \circ s$, so $Ds(z) = Dr(z)Ds(z)$ Chapter4/Submanifolds · tangentSpace_subset_of_subset. And if $F \circ r$ is sc-smooth and F maps the sub-retract Q into a retract Q', then $D(F \circ r)(z)$ maps $T_z Q$ into $T_{F(z)}Q'$ (Proposition 2.6 (3) for $F|_Q$) Chapter4/Submanifolds · scDeriv_mem_tangentSpace_of_subset.

Remark 4.14 in a chart Chapter4/Submanifolds · MPolyfold.SubmanifoldData.remark_4_14. The text: $T_a A = Ts(a)(T_a X)$ is the image of the sc⁺-operator $Ts(a)$, hence finite-dimensional. In a chart, at $a \in A \cap V \cap \operatorname{dom}\varphi$ and the level-1 point z over $\varphi(a)$: z is smooth (Lemma 4.7 (i) for the chart retraction t); the tangent map of ρ at z maps $T_z O = T_z(\varphi(W))$ onto $T_z(\varphi(W \cap A))$ (Proposition 2.21 (4) for ρ) Chapter2/Metrizable · proposition_2_21_4; $T_z(\varphi(W \cap A)) \subseteq T_z O$; and it is the range of $Dt(z)$, finite-dimensional by Lemma 4.7 (ii). The sc⁺-operator is $Dt(z)$, not $Ts(a)$.

Points with $d_X = 0$. The remark adds that near points with $d_X(a) = 0$, A is a classical manifold. Every subspace is in good position to the whole space, any algebraic complement being good Chapter4/Submanifolds · isInGoodPosition_univ, and at a point with $d_C(e) = 0$ the partial quadrant C_e is the whole space Chapter4/Submanifolds · partialQuadrantAtWith_eq_univ_of_degIndex_eq_zero. So at $a \in A$ with $d_X(a) = 0$, in a chart realizing $d_X(a) = 0$ Chapter2/DegeneracyIndex · MPolyfold.exists_isChart_degIndex_eq, A is in good position (Definition 4.16, §IV.13) Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_of_degIndex_eq_zero; and if $d_X = 0$ on all of A, re-centring at each point gives charts modelled in $C_{\varphi(a)} = E$, in which $T_0(\psi(W \cap A))$ is in good position, so the induced structure has a corner structure by the core of Theorem 4.18 (§IV.14) — without tameness of X Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_degIndex_eq_zero. The local statement for the open piece $A \cap \{d_X = 0\}$ of an arbitrary A is not formalized.

The open question after Proposition 4.15 — is a sub-M-polyfold with $A = A_\infty$ and finite, locally constant $\dim T_a A$ a smooth finite-dimensional submanifold; is the "T" $\{(x,0) : |x| < 1\} \cup \{(0,y) : 0 \le y < 1\}$ homeomorphic to a retract $O = O_\infty$ — is a question and is not formalized. Its last sentence, that this is impossible for sc⁺-retracts, is for a retract (O, E, E) without boundary the statement just proved with $A = O$: an sc⁺-retract in E is a manifold; that the "T" is not a topological manifold is not formalized.

IV.12 Proposition 4.15

(1) $A \subseteq X_\infty$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.subset_smoothSet: in a chart at a, $\varphi(a)$ lies in the retract of the sc⁺ chart retraction (Lemma 4.7 (i)). The text does not distinguish this from "$A = A_\infty$ for the induced structure", which holds because the levels of the induced structure are the traces of those of X, computed in the induced charts Chapter2/SubMPolyfold · MPolyfold.subPolyfold_isChart Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.mem_levelSet_subPolyfold_iff Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_1.

(2) *A is locally closed* Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_2. The text's definition is garbled — "a point $b \in V(a)$ belongs to A if $U \cap A \neq \emptyset$ for all open neighbourhoods U of b", with U also the letter of the domain — and is read as $V(a) \cap \overline{A} \subseteq A$, one of the equivalent forms of Mathlib's `IsLocallyClosed` (the intersection of an open and a closed set). Proof: for $b \in V \cap \overline{A}$, b is in the closure of $A \cap V$ (V open), so the filter of neighbourhoods of b within $A \cap V$ is nontrivial; s is continuous at b and equal to the identity on $A \cap V$, so $s(b)$ and b are both limits of the identity along that filter, and $s(b) = b$ (X Hausdorff); hence $b \in s(V) \subseteq A$. A filter replaces the text's sequence $a_k \to b$, so no metrizability is needed.

(3) $\partial A = \{d_A \geq 1\} \subseteq \partial X$ Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_3, by $d_A \leq d_X$ (Lemma 2.26) Chapter2/SubMPolyfold · lemma_2_26. The text's argument for $d_X(a) = 0$ is the proof of that lemma; it needs the chart re-centred at a, since only $C_{\varphi(a)}$, not C, is the whole space.

(4) $1^1_0 : A \to A^1$ *is sc-smooth* Chapter4/Submanifolds · MPolyfold.IsSmoothSubmanifold.proposition_4_15_4. The text says only that "the postulated sc⁺-retraction is the identity on the retract A". The proof: every point of A lies in an induced chart Chapter2/SubMPolyfold · MPolyfold.subChart whose model $\varphi(W \cap A)$ is an sc⁺-retract (§IV.11), and Proposition 4.8 on a cover applies (§IV.4).

IV.13 Good and general position in charts, and their chart independence (Definitions 4.16, 4.17, 4.21)

Definition 4.16 in a chart. For retracts $Q \subseteq O$, a presentation T and a level-1 point z: T_zQ is *in good position to C_z inside T_zO* Chapter4/Submanifolds · IsGoodPositionRetractAt — in the sense of Definition 3.21 inside a subspace Chapter3/GoodPositionGerm · IsInGoodPositionIn: $T_zQ \subseteq T_zO$, $T_zQ \cap C_z$ has interior points in T_zQ, and T_zQ has a complement inside T_zO with a constant ε. The condition depends only on the two tangent spaces Chapter4/Submanifolds · IsGoodPositionRetractAt.congr. For the data d of Definition 4.13, a chart φ and $Q = \varphi(W \cap A)$ it is the chart condition Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGoodPositionChart, and A *is in good position at* a if it holds for some data at a, some chart at a, some presentation and the level-1 point over $\varphi(a)$ Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt. Explicitly, for retractions r onto O and t onto $\varphi(W \cap A)$, the condition says that $Dt(z)E$ is in good position to C_z inside $Dr(z)E$ Chapter4/Submanifolds · MPolyfold.SubmanifoldData.isGoodPositionChart_iff. The text states Definition 4.16 for tame X so that C_aX is a partial quadrant (Proposition 2.36); the chart form does not need it.

Definition 4.21 in a chart. *T_zQ has in T_zO an sc-complement contained in T^R_zO* Chapter4/Submanifolds · IsGeneralPositionRetractAt: an sc-subspace $B \subseteq T_zO \cap E_z$ of E with $(T_zO)_m = (T_zQ)_m \oplus B_m$ on every level, the sc-structure of T_zO being the one induced from E; again it depends only on the tangent spaces Chapter4/Submanifolds · IsGeneralPositionRetractAt.congr. The chart condition Chapter4/Submanifolds · MPolyfold.SubmanifoldData.IsGeneralPositionChart and *general position at a* Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt are defined as for good position. **Definition 4.17**: A is *tame* if its induced structure is Chapter4/Submanifolds · MPolyfold.IsTameSubmanifold.

Transport by a germ of sc-diffeomorphisms. Let F be a germ of sc-diffeomorphisms between retracts $O = r(U)$ and $O' = r'(U')$ with inverse G Chapter2/Boundary · IsDiffeoGerm, mapping subretracts Q, Q' into each other. At $z \in Q_1$, $\mathcal{A} = D(F \circ r)(z)$ restricts to an isomorphism $T_zO \cong T_{F(z)}O'$ with inverse $D(G \circ r')(F(z))$, carrying the partial cones onto each other (Proposition 2.30) Chapter2/Boundary · proposition_2_30_cone and T_zQ onto $T_{F(z)}Q'$ (§IV.11 in both directions). Good position inside a subspace is transported by such an isomorphism (Lemma 3.45) Chapter3/LinearizationCharts · IsInGoodPositionIn.transport, so *good position of Q at z gives good position of Q' at $F(z)$* Chapter4/Submanifolds · IsGoodPositionRetractAt.transport. For general position z must be smooth: then $\mathcal{A}$ is an sc-operator, it carries T^R_zO onto $T^R_{F(z)}O'$ (Proposition 2.30) Chapter2/Boundary · proposition_2_30_reduced, and the image of the sc-complement B is an sc-complement inside $T^R_{F(z)}O'$, level by level Chapter4/Submanifolds · IsGeneralPositionRetractAt.transport.

The chart change Chapter4/Submanifolds · MPolyfold.chart_change_of_transport. Let Φ be a property of $(Q \subseteq O, T, z)$ that depends only on the tangent spaces and is carried by germs of sc-diffeomorphisms at smooth points. If Φ holds for $\varphi(G \cap A) \subseteq O$ at z (smooth), it holds for $\varphi'(G' \cap A) \subseteq O'$ at z', for any presentations: on $H = G \cap G'$ the transition $\varphi' \circ \varphi^{-1}$ is a germ of sc-diffeomorphisms between the overlap retracts Chapter2/Boundary · IsScSmoothRetractMap.isDiffeoGerm carrying $\varphi(H \cap A)$ onto $\varphi'(H \cap A)$ Chapter4/Submanifolds · MChart.image_inter_eq_inter, and the tangent spaces at z are those of the open pieces (the germ property). Applied to the two properties Chapter4/Submanifolds · MPolyfold.isGoodPositionRetractAt_chart_change Chapter4/Submanifolds · MPolyfold.isGeneralPositionRetractAt_chart_change, with G, G' the open sets of

two submanifold charts: *if A is in good position at a, the chart condition holds for every data, chart, presentation and lift* `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isGoodPositionChart`, and good position is equivalent to the condition for all of them `Chapter4/Submanifolds · MPolyfold.isGoodPositionAt_iff_forall`; likewise for general position `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart` `Chapter4/Submanifolds · MPolyfold.isGeneralPositionAt_iff_forall`, the smoothness of the point holding on A. The text takes this independence for granted (Findings, item 66).

IV.14 Lemma 4.19 and Theorem 4.18

Lemma 4.19, the complement kept explicit `Chapter4/Submanifolds · isGoodComplement_sup_range`. Let π be a bounded idempotent with $(1-\pi)E_0$ inside the minimal subspace at 0, $E_0^{\min} = \{\lambda = 0\}$, and $Y \subseteq \pi E_0$ a complement of N inside πE_0 with constant ε. Then $Y \oplus (1-\pi)E_0$ is a good complement of N in E_0 with constant $\varepsilon/(\|\pi\|+1)$: for $y \in Y$, $k \in (1-\pi)E_0$ with $|y+k| \le \frac{\varepsilon}{\|\pi\|+1}|n|$, $y = \pi(y+k)$, so $|y| \le \varepsilon|n|$ and $n \in C \iff n+y \in C$; and adding $k \in E_0^{\min}$ changes no coordinate, so $n+y \in C \iff n+y+k \in C$. The splitting: $N \oplus Y = \pi E_0$ and $\pi E_0 \oplus (1-\pi)E_0 = E_0$. **Lemma 4.19** `Chapter4/Submanifolds · lemma_4_19`: *in the situation of the lemma, if N is in good position to C inside πE_0, then N is in good position to C in E_0, with good complements $Y \oplus (1-\pi)E_0$.* The good-position half is the transport lemma of the proof of Theorem 3.54 `Chapter3/GoodPositionGerm · IsInGoodPositionIn.isInGoodPosition` (§III.23).

The text's proof makes a false step. It concludes $n + y \in C$ from $n + y + z \in C$ "since $z \in Z \subset C$". This needs $-z \in C$: in $C = [0, \infty)$, $z = 1$ and $n + y = -\frac{1}{2}$ give $n + y + z \in C$, $n + y \notin C$. What holds, and what is used, is $Z \subseteq E_0^{\min} \subseteq C \cap (-C)$ — Proposition 2.35 at the corner of the centred chart. The text's norm $|y|_0 + |z|_0$ on $Y \oplus Z$ is not the norm of E; the constant $\varepsilon/(\|\pi\|+1)$ compares them (Findings, item 69).

Good position read in a tame centred chart `Chapter4/Submanifolds · MPolyfold.IsGoodPositionAt.isInGoodPosition_of_tame`. If A is in good position at a and ψ is a chart of X with $\psi(a) = 0$ modelled on a tame retract $O = r(U)$, then $N = T_0(\psi(W \cap A))$ is in good position to C in E: by chart independence the chart condition holds in ψ, i.e. N is in good position to C_0 inside $T_0O =$ range $Dr(0)$; at the corner of the centred chart $C_0 = C$ `Chapter3/GoodPositionGerm · partialQuadrantAtWith_zero`; Proposition 2.35 for the tame r at the smooth point 0 gives $(1 - Dr(0))E_0 \subseteq E_0^{\min}$ `Chapter2/TameRetracts · proposition_2_35`; and $Dr(0)$ is idempotent. Lemma 4.19 with $\pi = Dr(0)$ concludes. Tameness of X enters Theorem 4.18 here and nowhere else.

The corner charts of A `Chapter4/Submanifolds · MPolyfold.hasCornerStructure_of_centred_isInGoodPosition`. If every $a \in A$ lies in a chart ψ of X with $\psi(a) = 0$ in which $T_0(\psi(W \cap A))$ is in good position to C in E, the induced structure has a corner structure. Proof (for nonempty A): every point of A is smooth and $1_0^1 : A \to A^1$ is sc-smooth (Proposition 4.15 (4)); the chart of A induced by ψ is a chart of the induced structure, centred at a, modelled on $\psi(W \cap A)$; its tangent space at 0 is the range of $Dt(0)$ for the sc$^+$ chart retraction, finite-dimensional (Lemma 4.7), and in good position by hypothesis; Theorem 4.2 (ii) □ (i) for the M-polyfold A (§IV.10) concludes.

Theorem 4.18 `Chapter4/Submanifolds · MPolyfold.theorem_4_18`. *Statement.* Let X be tame and A a smooth finite-dimensional submanifold in good position at every point. Then the induced structure on A has a corner structure, and A is tame. *Proof* (the text's route through Theorem 4.2): at $a \in A$ take a tame chart centred at a (§IV.5) — the "sc-smooth tame chart $\varphi : (V, a) \to (O, 0)$" the text takes for granted; good position read in it (above) gives the hypothesis of the previous paragraph; and a corner structure makes A tame (§IV.1). What the text skips, besides the false step of Lemma 4.19: the centred chart; the set W in place of "$V = \varphi^{-1}(O)$" (§IV.11); the sc$^+$ condition in charts; the closed complement and the projection along it in place of "$p = Dt(0)$" (§IV.10); and the compatibility of the charts $p \circ \varphi$ (§IV.10). The text's "$V(a) = \varphi^{-1}(V_0) \cap A$" should be $(p \circ \varphi)^{-1}(V_0) \cap A$; its "Proposition 4.2" is Theorem 4.2 (Findings, item 68). No hypothesis is added.

IV.15 Remark 4.20: good position is an open condition

Statement `Chapter4/Submanifolds · MPolyfold.remark_4_20`. For tame X, if A is in good position at a, there is an open $U \ni a$ with A in good position at every point of $U \cap A$. The text omits "tame" (Definition 4.16 is for tame X, and Lemma 4.19 is used) and argues "since the tangent spaces of A move only slowly, Lemma 3.55 implies…". The formal proof makes this precise.

The tangent spaces along a graph `Chapter4/Submanifolds · tangentSpace_eq_range_graph`. If the retract Q is the image of $\Gamma : (V, N \cap C, N) \to (Q, C, E)$, sc-smooth between retracts with inverse p, then at $v \in V$ and the level-1 point y over $\Gamma(v)$: Γ is differentiable within V with derivative its sc-derivative `Chapter4/Submanifolds · hasFDerivWithinAt_scDeriv_const`, $T_yQ = D\Gamma(v)N$, and $p \circ D\Gamma(v) = 1$ — the tangent maps of the germ of sc-diffeomorphisms Γ, p.

Proof of Remark 4.20. Take a tame chart ψ centred at a with tame retraction r, the sc$^+$ chart retraction t onto $\psi(W \cap A)$, and $N = T_0$, in good position to C in E (§IV.14). The normal form of Proposition 3.22 gives a continuous projection π onto N with closed kernel a good complement `Chapter3/GoodComplementProjection · exists_proj_goodComplement`, and Proposition 4.9 gives W_1, V, Γ over $N \cap C$. Write $\Gamma(v) = v + \sigma(v)$ with $\sigma(v) \in \ker \pi$. Then σ is C^1 on V, $\sigma(0) = 0$, and $D\sigma(0) = 0$: $D\Gamma(0)m$ lies in $T_0 = N$, so $D\Gamma(0)m = \pi D\Gamma(0)m = m$. The tangent space at $\Gamma(v)$ is the graph $\{n + D\sigma(v)n\}$ `Chapter3/GoodPositionOpen · graphSpace`. Lemma 3.55, with the closed complement $\ker \pi$ `Chapter3/GlobalIFTFull · lemma_3_55_relOpen`, gives $\rho > 0$ such that for $|v| < \rho$ this graph is in good position to $C_{\Gamma(v)}$ in E. It lies in $T_{\Gamma(v)}O$ (Remark 4.14 in the chart), and good position in E of a subspace of $T_{\Gamma(v)}O$ gives good position inside $T_{\Gamma(v)}O$, with the complement cut down by the modular law `Chapter3/GlobalIFTNearbyGraph · IsInGoodPosition.isInGoodPositionIn_of_le`. That is Definition 4.16 at $b = \psi^{-1}(\Gamma(v))$, read in ψ. The neighbourhood is $U = W \cap \psi^{-1}(W_1 \cap \{|\pi u| < \rho\})$.

IV.16 Linear algebra at the corner of a centred chart; Lemmas 7.A and 7.B of the exposition

In a chart centred at 0, every coordinate vanishes at 0, so the minimal subspace there is $E_0^{\min} = \{\lambda = 0\}$, the text's "$\{0\} \oplus W$" `Chapter4/LocalStructure · mem_minimalSubspaceWith_zero`, and it lies in every E_x `Chapter4/LocalStructure · minimalSubspaceWith_zero_le`. Adding a vector of $E_0^{\min}$ changes no coordinate `Chapter4/LocalStructure · coord_add_of_mem_zero`, hence neither the degeneracy index `Chapter4/LocalStructure · degIndexWith_add_of_mem_zero` nor membership in C `Chapter4/LocalStructure · add_mem_iff_of_mem_zero`.

Closed complements. If N is finite-dimensional, K closed and $N + K$ the whole space, N has a closed complement $Y \subseteq K$: take a continuous projection q onto the finite-dimensional $N \cap K$ and $Y = K \cap \ker q$ `Chapter4/LocalStructure · exists_isClosed_isCompl_le`. The projection along a closed complement of a finite-dimensional subspace is continuous, with kernel exactly the complement `Chapter4/LocalStructure · exists_proj_of_isCompl`.

Good position from $N + E_0^{\min} = E_0$. Let N be finite-dimensional with $N + E_0^{\min} = E_0$. The coordinates restricted to N `Chapter4/LocalStructure · coordsOn` map N onto $\mathbb{R}^n$ `Chapter4/LocalStructure · coordsOn_surjective`, so $N \cap C$ has interior points, those with all coordinates positive `Chapter4/LocalStructure · interior_nonempty_of_sup_eq_top`. A complement inside $E_0^{\min}$ is good with every ε `Chapter4/LocalStructure · isGoodComplement_of_le_zero`. Hence N is in good position, with a closed good complement $Y \subseteq E_0^{\min}$ `Chapter4/LocalStructure · exists_goodComplement_le_zero`. **The degeneracy index on the slice** `Chapter4/LocalStructure · degIndex_slice`: $N \cap C$ is the partial quadrant of N presented by $v \mapsto (\lambda(v), v - s(\lambda(v)))$, s a linear section of $\lambda|_N$, and $d_{N\cap C}(v) = d_C(v)$ — both count the vanishing coordinates. This replaces the text's splitting $T_0A = N \oplus (T_0A \cap W)$, $E = N \oplus W$ (Findings, item 83).

Lemma 7.A: an sc-complement inside an sc-subspace `Chapter4/LocalStructure · exists_scComplement_le`. Let $N \subseteq E_\infty$ be finite-dimensional, R an sc-subspace and $M = N + R$. Then N has in M an sc-complement

$B \subseteq R$: $K = N \cap R$ is finite-dimensional and smooth, so it has an sc-complement G in E (Proposition 1.6) `Chapter1/ScBanach · ScBanach.proposition_1_6`; put $B = R \cap G$. Then $R = K \oplus B$, $N + R = N + B$, $N \cap B \subseteq K \cap G = 0$, and $M_m = N \oplus B_m$ on every level because $N \subseteq E_\infty$. B is closed with $B_m = (1-P)R_m$, P the sc-projection onto K, so the smooth points of B are dense in every level: for $x \in B_m$ and smooth $u \in R_m$ near x, $u - Pu \in B$ is smooth and near $x - Px = x$.

General position from $T_yQ + E_0^{\min} = E_0$ `Chapter4/LocalStructure · isGeneralPositionRetractAt_of_sup_eq_top`. Let r be tame, $Q \subseteq O$ a retract, $y \in Q_1$ smooth, and T_yQ finite-dimensional, smooth, with $T_yQ + E_0^{\min} = E_0$. As $E_0^{\min} \subseteq E_y$, intersecting with $T_yO \supseteq T_yQ$ gives $T_yO = T_yQ + (T_yO \cap E_0^{\min}) \subseteq T_yQ + T_y^RO$ (modular law). T_y^RO is an sc-subspace by Proposition 2.36 for the tame r `Chapter2/PartialCone · proposition_2_36_partialCone`, and Lemma 7.A gives the sc-complement inside it. This is the step of Theorem 4.22 (3) "implying that T_aA has in T_aO a sc-complement contained in T_a^RO" (Findings, item 85).

Lemma 7.B: general position implies good position `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`. For r tame and a smooth $z \in O_1$: if T_zQ has in T_zO an sc-complement $B \subseteq T_z^RO$, then T_zQ is in good position to C_z inside T_zO. $B \subseteq E_z$ changes none of the coordinates λ_i, $i \in I(z)$, so it is a good complement with every ε. The coordinates $\lambda_{I(z)}$ restricted to T_zO have kernel T_z^RO, whose codimension is $d_C(z) = \#I(z)$ (Proposition 2.36) `Chapter2/TameRetracts · quotDim_reducedTangentSpaceWith_eq`, so they map T_zO onto $\mathbb{R}^{I(z)}$; they kill B, so they map T_zQ onto $\mathbb{R}^{I(z)}$, and the preimage of the open positive orthant gives interior points of $T_zQ \cap C_z$. The text never states this step, which Theorem 4.22 (4) and Theorem 4.23 need for tameness and corner structures.

IV.17 Theorem 4.22 (1)–(3): local structure in general position

$T_0Q + E_0^{\min} = E_0$ from general position at the centre `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top`. If T_0Q has in T_0O an sc-complement $B \subseteq T_0^RO \subseteq E_0^{\min}$ and r is tame, then $E_0 = T_0O + (1 - Dr(0))E_0$ with $(1 - Dr(0))E_0 \subseteq E_0^{\min}$ (Proposition 2.35), so $E_0 = T_0Q + B + (1 - Dr(0))E_0 \subseteq T_0Q + E_0^{\min}$. (The text: "$E = T_0O \oplus Z = T_0A \oplus \widetilde{A} \oplus Z$, so T_0A has in E an sc-complement Y contained in W"; only $T_0A + W = E$ is used afterwards.)

$T_yQ + E_0^{\min} = E_0$ along the graph `Chapter4/LocalStructure · sup_minimalSubspaceWith_eq_top_graph`. If $\Gamma : V \to Q \cap W$ is the graph map of Proposition 4.9 for a projection p onto N whose kernel lies in $Y \subseteq E_0^{\min}$, and $N + E_0^{\min} = E_0$, then at y over $\Gamma(v)$, $T_yQ = D\Gamma(v)N$ (§IV.15) and $D\Gamma(v)n - n \in \ker p \subseteq E_0^{\min}$ because $p \circ D\Gamma(v) = 1$; so $T_yQ + E_0^{\min} = N + E_0^{\min} = E_0$. This is the text's "by means of the particular form of the map $\alpha : h \mapsto h + D\delta(v)h$, also $T_aA + W = E$", which needs $D\delta(v)h \in W$.

Theorem 4.22 in a tame centred chart `Chapter4/LocalStructure · theorem_4_22_chart`. Let r be tame onto $O \ni 0$, t an sc⁺-retraction onto $Q \subseteq O$ with $0 \in Q$, $N = T_0Q$, and suppose Q is in general position at 0. Then: $N + E_0^{\min} = E_0$ (above); N is in good position with a closed good complement $Y \subseteq E_0^{\min}$ (§IV.16) and the projection p along it, with kernel exactly Y; Proposition 4.9 for this Y gives W, V, Γ; (2) $d_{N\cap C}(v) = d_C(\Gamma(v))$ for $v \in V$, because $d_{N\cap C}(v) = d_C(v)$ on the slice and $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate; (3) at every level-1 point y over $Q \cap W$, Q is in general position: y is smooth (Lemma 4.7 (i)), T_yQ is the range of $Dt(y)$, finite-dimensional (Lemma 4.7 (ii)) and smooth — the range of $Dt(y)$ is an sc-subspace at a smooth point `Chapter2/TameRetracts · ScRetraction.isScSplitting_range_scDeriv`, finite-dimensional sc-subspaces are smooth `Chapter1/ScBanach · ScBanach.proposition_1_6_iff` — and $T_yQ + E_0^{\min} = E_0$ along the graph gives general position (§IV.16).

Theorem 4.22 `Chapter4/LocalStructure · MPolyfold.theorem_4_22`. *Statement.* Let X be tame, A a smooth finite-dimensional submanifold, in general position at $x \in A$. There are a chart ψ of X with $\psi(x) = 0$, a tame retraction r onto its model for a presentation with n half-line coordinates, and an open $U \ni x$ inside the domain of ψ, such that: (1) $d_X(x) = n$, and $\psi(x)$ lies in exactly $d_X(x)$ faces of the M-polyfold O; (2) $d_X(a) = d_A(a)$ for $a \in A \cap U$; (3) A is in general position at every point of $A \cap U$; (4) for every

$\sigma \subseteq \{1,\dots,n\}$ the intersection of local faces F_σ is a tame sub-M-polyfold inside U with $d_{F_\sigma} + \#\sigma = d_X$, and $A \cap F_\sigma$ is a tame smooth finite-dimensional submanifold of F_σ, in general position in F_σ at each of its points. *Proof.* x is smooth (Proposition 4.15 (1)), so there is a tame chart ψ centred at x (§IV.5), with a tame retraction r and a presentation T; the chart retraction t of A in ψ is sc$^+$ (§IV.11), and general position at x read in ψ (chart independence, §IV.13) is general position of $t(U)$ at 0. Theorem 4.22 in the chart gives Y, p, W_1, V, Γ; put $U = W \cap \psi^{-1}(W_1)$, W the open set of the submanifold chart.

1. *(1) is a statement about the chart.* At the centre all n coordinates vanish, so $d_C(0) = n$; the tame chart realizes d_X `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameWith`, so $d_X(x) = n$. The face count is Corollary 2.41 for the M-polyfold O `Chapter2/TameRetracts · isFaceStructured_toMPolyfold Chapter2/TameRetracts · LocalModel.degIndex_toMPolyfold_of_isTameWith`. "Precisely d local faces contained in U" is read for the faces $O \cap \{\lambda_i = 0\}$ of the chart model; their identification with global faces of X is not formalized (Findings, item 86).
2. *(2).* The text: "$p : A \to (T_0A \cap C) \cap V$ is a sc-diffeomorphism … by Proposition 2.24, $d_A(a) = d_{T_0A\cap C}(p(a))$". The sc-diffeomorphism is the corner chart $p\circ\psi$ of A: the data of §IV.10 for the chart of A induced by ψ, with the complement $Y \subseteq E_0^{\min}$, give a chart of the induced structure modelled on $(V, N\cap C, N)$ with the identity retraction. That model is tame, so the chart realizes d_A `Chapter2/TameRetracts · MPolyfold.degIndex_eq_chartDegIndex_of_isTameRetract`; ψ realizes d_X; and $d_{N\cap C}(p\psi(a)) = d_C(\Gamma(p\psi(a))) = d_C(\psi(a))$ by (2) of the chart form (Findings, item 84).
3. *(3)* is (3) of the chart form, read back through the chart: at $b \in U \cap A$ the level-1 point over $\psi(b)$ is smooth, and the chart condition in ψ is general position at b.
4. *(4)* is §IV.18.

The text's further claims "$\delta(0) = 0$ and $D\delta(0) = 0$" hold (§IV.15) but are not used.

IV.18 Theorem 4.22 (4): the faces, and $A \cap F_\sigma$

The text: "in view of the arguments in the proof of Proposition 2.43, the sets F_σ are tame M-polyfolds. Moreover, $A \cap F_\sigma$ are smooth submanifolds of F_σ. The previous arguments, but now applied to $A \cap F_\sigma$ conclude the proof." Four things are missing (Findings, item 89): Proposition 2.43 is about one face, and intersections need Proposition 2.44 (second sentence); "smooth submanifold of F_σ" needs an sc$^+$-retraction *in the M-polyfold* F_σ; general position in F_σ needs an argument; and tameness of $A \cap F_\sigma$ needs Theorem 4.18 in F_σ and good position.

Killing coordinates. For $\sigma \subseteq \{1,\dots,n\}$, $a \mapsto a \cdot 1_\sigma$ keeps the coordinates in σ `Chapter4/LocalStructure · maskL`, and for a linear section s of $\lambda|_N$, $\kappa_\sigma(v) = v - s(\lambda(v)1_\sigma)$ `Chapter4/LocalStructure · killOn` has $\lambda_i(\kappa_\sigma v) = 0$ for $i \in \sigma$ and $\lambda_j(\kappa_\sigma v) = \lambda_j(v)$ otherwise `Chapter4/LocalStructure · coordsOn_killOn`, fixes the vectors of the stratum $f_\sigma = \{\lambda_i = 0, i \in \sigma\}$ `Chapter4/LocalStructure · killOn_eq_self`, is idempotent `Chapter4/LocalStructure · killOn_idem` and maps $N \cap C$ into itself `Chapter4/LocalStructure · killOn_mem`. A continuous linear map into a finite-dimensional space is an sc-operator for the constant structure `Chapter4/LocalStructure · opToConst`.

The sc$^+$-retraction onto $A \cap F_\sigma$ in the chart `Chapter4/LocalStructure · exists_scPlus_retraction_corner`. In the situation of the chart form of Theorem 4.22, for every open $B \ni 0$ there are an open $G \ni 0$, $G \subseteq W \cap B$, and an sc$^+$-retraction $t_\sigma = \Gamma \circ \kappa_\sigma \circ p \circ t$ on $U \cap t^{-1}(G)$ onto $Q_\sigma = Q \cap G \cap f_\sigma$: (a) it is sc$^+$, as sc-smooth □ sc$^+$ (§IV.2), p, κ_σ and Γ being sc-smooth; (b) it is a retraction, because $t\Gamma = \Gamma$, $p\Gamma = 1$ and $\kappa_\sigma^2 = \kappa_\sigma$; (c) its image is $Q \cap G \cap f_\sigma$, because $\Gamma(v) - v \in Y \subseteq E_0^{\min}$ changes no coordinate, so $\lambda(\Gamma(v)) = \lambda(v)$; (d) at a point y of Q_σ the tangent vectors of Q lying in f_σ are tangent to Q_σ: for such h, $k = ph$ has $\lambda(k) = \lambda(h)$, so $\kappa_\sigma k = k$ and $Dt_\sigma(y)h = D\Gamma(w)\kappa_\sigma pDt(y)h = D\Gamma(w)k = h$, $w = p(y)$, by the chain rule and $T_yQ = D\Gamma(w)N$.

The corner model `Chapter4/LocalStructure · exists_cornerModel_retraction`. For a retraction S with retract in f_σ, $\pi_\sigma \circ S \circ \iota_\sigma$ on $\iota_\sigma^{-1}(S.U)$, with the corner inclusion and projection of Chapter 2 `Chapter2/`

CornerGerms · cornerIncl Chapter2/CornerGerms · cornerProj, is a retraction of $[0,\infty)^{n-\#\sigma} \oplus W$ onto $\pi_\sigma(S(U))$, sc⁺ if S is.

Theorem 4.22 (4) in the chart Chapter4/LocalStructure · theorem_4_22_four_chart. For every σ and every open $B_0 \ni 0$ there are faces $F_i \ni 0$ of O ($i \in \sigma$) and an open $G \ni 0$, $G \subseteq W \cap B_0$, such that: *F_σ near 0 is a tame retract* — $O \cap G \cap f_\sigma$ is the retract of the face retraction of Proposition 2.44 Chapter2/CornerGerms · corner_retraction restricted to G, whose corner model is the retract of a tame retraction r' with $d_{C'}(\pi_\sigma y)+\#\sigma = d_C(y)$ Chapter2/CornerGerms · corner_tameModel; *$A \cap F_\sigma$ near 0 is an sc⁺-retract* — $Q \cap G \cap f_\sigma$ is the retract of t_σ, and its corner model that of the sc⁺-retraction t', inside the retract of r'; *general position in F_σ* — at a level-1 point y' of $\pi_\sigma(Q \cap G \cap f_\sigma)$, with $y = \iota_\sigma(y')$: $T_yQ + E_0^{\min} = E_0$ (§IV.17) gives for every $b \in \mathbb{R}^{n-\#\sigma}$ a $v \in T_yQ$ with $\lambda(v) = \iota(b)$, zero on σ; $v \in f_\sigma$, so v is tangent to Q_σ, and $\pi_\sigma v$ is tangent to $\pi_\sigma(Q_\sigma)$ at y' (§IV.11), with coordinates b; so the tangent space of the corner model plus its minimal subspace at 0 is everything, and general position follows (§IV.16), in the corner model with the identity presentation.

sc⁺-retractions in a chart are sc⁺-retractions on open subsets Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart — the converse of §IV.11, which the text does not state. For a chart χ and an sc⁺-retraction τ of its model with $\tau(U) \subseteq O$, $s = \chi^{-1} \circ \tau \circ \chi$ is an sc⁺-retraction on the open set $\chi^{-1}(\tau.U)$, with image $\chi^{-1}(\tau(U))$: sc-smoothness of s and of $1_0^1 \circ s$ are read in the charts χ of X and `levelChart` χ of X^1, where they are τ and its lift; the values lie in X_1 since $\tau(U) \subseteq E_\infty$ (Lemma 4.7 (i)).

(4) on the M-polyfold, for one σ Chapter4/LocalStructure · MPolyfold.theorem_4_22_four_of_chart. In the tame chart ψ centred at x with the chart data of §IV.17: the faces F_i and G of the chart form, with G inside the domain of the submanifold chart, give $F_\sigma = \psi^{-1}(G \cap \bigcap_{i\in\sigma} F_i)$ Chapter4/LocalStructure · MChart.cornerPreimage, which contains x and lies in W. It is a sub-M-polyfold, a local property Chapter2/TameFaces · MPolyfold.isSubMPolyfold_of_local; its face chart $\pi_\sigma \circ \psi$, compatible with the induced charts Chapter2/TameFaces · MPolyfold.faceChart_compatible_subChart, is a chart of the induced structure Chapter2/TameFaces · MPolyfold.isChart_subPolyfold_of_compatible modelled on the tame corner model, so F_σ is tame Chapter2/TameFaces · MPolyfold.isTame_of_forall_exists_chart with $d_{F_\sigma} + \#\sigma = d_X$. $A \cap F_\sigma$ is, in the face chart, the retract of the sc⁺-retraction t', which the previous paragraph turns into an sc⁺-retraction on an open subset of F_σ: it is a smooth finite-dimensional submanifold of F_σ. It is in general position at each of its points by the chart form read in the face chart. Its tameness is Theorem 4.18 in the tame M-polyfold F_σ, good position following from general position (Lemma 7.B). F_σ is a germ, chosen inside U — the text's "contained in U" — and G depends on σ. Paracompactness of F_σ and $A \cap F_\sigma$, instance arguments of the induced structures, holds for every subset Chapter2/Metrizable · MPolyfold.paracompactSpace_subtype.

The proof of Theorem 4.23 uses $\dim(A \cap F_\sigma) = \dim A - \#\sigma$, which Theorem 4.22 does not state; it is proved in the model in §IV.21.

IV.19 Remark 4.24

If $T_zO = T_zQ + B$ with $B \subseteq K$, the image of T_zQ spans T_zO/K, so $\dim(T_zO/K) \le \dim T_zQ$ Chapter4/LocalStructure · quotDim_le_finrank_of_eq_sup. **In a chart** Chapter4/LocalStructure · remark_4_24_chart: at a smooth point z of a tame retract, $\dim(T_zO/T_z^RO) = d_C(z)$ (Proposition 2.36), and if T_zQ has in T_zO an sc-complement inside T_z^RO, then $d_C(z) \le \dim T_zQ$. **On the M-polyfold**: for tame X and smooth x, in a tame chart at x the reduced tangent space has codimension $d_X(x)$ Chapter2/TameRetracts · MPolyfold.IsTame.exists_chart Chapter4/LocalStructure · MPolyfold.remark_4_24_codim — the remark's first sentence, which needs X tame and x smooth; if A is in general position at a, then $d_X(a) \le \dim T_aA$ in any tame chart at a Chapter4/LocalStructure · MPolyfold.remark_4_24; and a submanifold in general position with $T_aA = 0$ does not meet ∂X at a Chapter4/LocalStructure · MPolyfold.remark_4_24_not_mem_boundary. The remark's "$x \in A_\lambda$ can only belong to a corner if $d_X(x) \le \dim A_\lambda$" holds for every submanifold in general position, so it applies to the A_λ of Theorem 4.23 once their general position is proved (§IV.24).

IV.20 Sard's theorem, vendored

The proof of Theorem 4.23 applies "Sard's theorem" to maps $A \cap F_\sigma^\circ \to \mathbb{R}^m$ between spaces of arbitrary dimensions. Mathlib has the cases $\dim E \le \dim F$ only. The Milnor project of this repository proves the general theorem by Milnor's induction; its import closure is copied into `Polyfold/Vendor/MilnorSard/` with the namespace renamed and no statement or proof changed (Findings, item 100). The proof, in Milnor's steps:

- *The objects.* Crit $= \{x \in U : Df(x) \text{ not onto}\}$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet`, and C_k, the points where all derivatives of orders $1, \dots, k$ vanish `Vendor/MilnorSard/Basic · MilnorSard.Sard.vanishSet`, with Crit $\subseteq$ (Crit $\setminus C_1$) $\cup \bigcup_{1 \le i < k} (C_i \setminus C_{i+1}) \cup C_k$ `Vendor/MilnorSard/Basic · MilnorSard.Sard.critSet_subset_union`. The statement in dimension n `Vendor/MilnorSard/Basic · MilnorSard.Sard.SardStatement`.
- *Straightening.* A smooth w with $Dw(\bar{x}) \neq 0$ is straightened by $H(x) = (w(x), Px)$, a local diffeomorphism by the inverse function theorem `Vendor/MilnorSard/InverseFunction · MilnorSard.isLocalDiffeomorphAt_of_isInvertible_mfderiv` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.Straightening` `Vendor/MilnorSard/Straighten · MilnorSard.Sard.exists_straightening`.
- *Step 1,* $f(\text{Crit} \setminus C_1)$ null, locally `Vendor/MilnorSard/Step1 · MilnorSard.Sard.step1_local`: in straightened coordinates $g = f \circ H^{-1}$ maps hyperplanes to hyperplanes, the critical points of g on a hyperplane are those of its restriction, the induction hypothesis in dimension $n-1$ applies to the restrictions, and Fubini concludes.
- *Step 2,* $f(C_k \setminus C_{k+1})$ null, locally `Vendor/MilnorSard/Step2 · MilnorSard.Sard.step2_local`: a k-th derivative w vanishes on C_k with $Dw \neq 0$; after straightening, C_k lies in a hyperplane, on which every point of C_k is critical, and the induction hypothesis applies. Local null sets glue along a countable cover `Vendor/MilnorSard/Step2 · MilnorSard.Sard.measure_image_eq_zero_of_locally`.
- *Step 3,* $f(C_k \cap K)$ null for $(k+1)\dim F > \dim E$ and K compact convex `Vendor/MilnorSard/Step3 · MilnorSard.Sard.measure_image_vanishSet_inter_eq_zero`: Taylor's bound $|f(y) - f(x)| \le c|y - x|^{k+1}$ for $x \in C_k$ `Vendor/MilnorSard/Step3 · MilnorSard.Sard.norm_sub_le_of_iteratedFDeriv_eq_zero` makes f Hölder of exponent $k+1$ there, so $f(C_k \cap K)$ has Hausdorff dimension at most $\dim E/(k+1) < \dim F$, hence measure zero `Vendor/MilnorSard/Basic · MilnorSard.Sard.addHaar_eq_zero_of_dimH_lt_finrank` — Milnor's cube count is the standard proof of that estimate.
- *The induction* `Vendor/MilnorSard/Sard · MilnorSard.Sard.sardStatement`, with the zero-dimensional target treated separately (no critical points) `Vendor/MilnorSard/Sard · MilnorSard.Sard.critSet_eq_empty_of_subsingleton`.

Sard's theorem `Vendor/MilnorSard/Sard · MilnorSard.sard`. *For E, F finite-dimensional real normed spaces in the same universe, a Haar measure μ on F, $U \subseteq E$ open and f of class C^∞ on U, $\mu(f(\text{Crit})) = 0$;* and the complement of the critical values is dense `Vendor/MilnorSard/Sard · MilnorSard.dense_compl_image_critSet`.

IV.21 The strata, Sard on the strata, and Theorem 3.78 in C^∞ form

The finite-dimensional model of the proof of Theorem 4.23: a finite-dimensional V (in the application T_0A of the chart of Theorem 4.22) with linear coordinates $\ell : V \to \mathbb{R}^n$ (those of the tame chart, onto), the partial quadrant $C = \{\ell \ge 0\}$ `Chapter4/FamiliesLocal · quadC`, a relatively open $U = O \cap C$ and a map f of class C^∞ on U (in the application the projection to $\mathbb{R}^m$ composed with the graph map Γ).

The strata. The stratum subspace of σ is $\{\ell_i = 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · stratSub` `Chapter4/FamiliesLocal · mem_stratSub`, the tangent space of $A \cap F_\sigma$; $\sigma(v) = \{i : \ell_i(v) = 0\}$ is the set of faces through v `Chapter4/FamiliesLocal · zeroSetOf`, and v lies in the stratum subspace of $\sigma(v)$ `Chapter4/FamiliesLocal · mem_stratSub_zeroSetOf`. **The dimension count** `Chapter4/FamiliesLocal · finrank_stratSub`: if ℓ is onto, the stratum subspace of σ has dimension $\dim V - \#\sigma$ — it is the kernel of $v \mapsto (\ell_i(v))_{i \in \sigma}$, which is onto $\mathbb{R}^\sigma$. This is the count "$A \cap F_\sigma^\circ$ is a smooth manifold … having dimension $\dim(A) - \#\sigma$", which the text attributes to "Theorem 4.23" (it means Theorem 4.22, which does not state it).

Stratified critical points `Chapter4/FamiliesLocal · stratCritSet`: the $v \in U$ at which $Df(v)$ restricted to the stratum subspace of $\sigma(v)$ is not onto — the critical points of the text's $p_\sigma : A \cap F^\circ_\sigma \to \mathbb{R}^m$ for $\sigma = \sigma(v)$. **Sard on the strata** `Chapter4/FamiliesLocal · measure_image_stratCritSet`: *the stratified critical values form a null set for every Haar measure.* The open stratum of σ, $\{k \in O : \ell_i(k) = 0\ (i \in \sigma),\ \ell_j(k) > 0\ (j \notin \sigma)\}$, is an open subset of the stratum subspace and lies in U; f restricted to it is C^∞, with derivative $Df(v)$ restricted to the stratum subspace; so a stratified critical point v with $\sigma = \sigma(v)$ is a critical point of f on the open stratum of σ, and Sard's theorem applies to each of the finitely many strata. The union of the open strata is U, and a point is stratified-regular iff it is a regular point of the p_σ for its own σ; so the stratified critical values are the union of the critical values of all the p_σ.

Theorem 3.78 in C^∞ form. Theorem 3.78 (§III.17) is stated for one j, with j-dependent constants in its proof; the fibre retraction below needs one solution that is C^∞. In the normalized form `Chapter4/FamiliesLocal · ContractionData.exists_solution_smooth`: if the contraction data of the proof hold for every $j \geq 1$, the radii of Lemmas 3.79–3.80 `Chapter3/QuadrantIFT · ContractionData.exists_radii` and the fixed point $\tau(k)$ are those for $j = 1$, independent of j, and only the regularity statement depends on j `Chapter3/QuadrantIFT · Radii.contDiffOn_τ`; so one τ is C^j for every j. **Theorem 3.78, C^∞** `Chapter4/FamiliesLocal · theorem_3_78_smooth`: for U relatively open in a partial quadrant C of a finite-dimensional V, $0 \in U$, f of class C^∞ on U with $f(0) = 0$ and $Df(0)$ onto, $K = \ker Df(0)$ in good position to C with good complement Y: there are a relatively open $V' \ni 0$ of $K \cap C$ and $\tau : K \to Y$, C^∞ on V', with $\tau(0) = 0$, $D\tau(0) = 0$, $k + \tau(k) \in U$ and $f(k + \tau(k)) = 0$, and the uniqueness of $\tau(k)$ among small y — the proof of Theorem 3.78 `Chapter3/QuadrantIFT · theorem_3_78` (the splitting $V \cong K \times Y$, $L = Df(0)$ an isomorphism $Y \to F$, $B(k, y) = y - L^{-1} f(k + y)$) with the C^∞ normalized form in place of the C^j one.

IV.22 The fibre over a stratified regular point

Partial quadrants of the model. $\{\ell_i \geq 0, i \in \sigma\}$ `Chapter4/FamiliesLocal · quadAt` is, for $\sigma = \sigma(v_0)$, the partial quadrant C_{v_0} of Definition 2.33. With a linear section s of ℓ, $v \mapsto (\ell(v), v - s(\ell(v)))$ is a linear isomorphism $V \cong \mathbb{R}^n \times \ker \ell$ `Chapter4/FamiliesLocal · sliceEquiv`, which presents both as partial quadrants `Chapter4/FamiliesLocal · isPartialQuadrant_quadC` `Chapter4/FamiliesLocal · isPartialQuadrant_quadAt`; C is convex `Chapter4/FamiliesLocal · convex_quadC` with nonempty interior `Chapter4/FamiliesLocal · interior_quadC_nonempty`, so derivatives within $O \cap C$ are unique `Chapter4/FamiliesLocal · uniqueDiffWithinAt_quadC`.

The kernel at a stratified regular point is in good position `Chapter4/FamiliesLocal · exists_goodComplement_stratSub` — the argument of Lemma 5.26, which the text needs here and does not give. Let $K + \{\ell_i = 0, i \in \sigma\} = V$ (for $K = \ker Df(v_0)$, $\sigma = \sigma(v_0)$, this is stratified regularity). The stratum subspace is closed, so K has a closed complement Y inside it (§IV.16). Adding a vector of Y changes none of the coordinates ℓ_i, $i \in \sigma$, so Y is a good complement with $\varepsilon = 1$. For the interior points: choose x with $\ell(x) = (1, \dots, 1)$ and write $x = k + y$; then $\ell_i(k) = 1$ for $i \in \sigma$, and the open set $\{k' \in K : \ell_i(k') > 0, i \in \sigma\}$ contains k and lies in $K \cap \{\ell_i \geq 0, i \in \sigma\}$.

The fibre retraction `Chapter4/FamiliesLocal · exists_fibre_retraction`. *Let ℓ be onto, f C^∞ on $O \cap C$ and $v_0 \in O \cap C$ not stratified-critical. There are an open $O' \ni v_0$, $O' \subseteq O$, and ρ, C^∞ on $O' \cap C$, mapping $O' \cap C$ into itself and into $f^{-1}(f(v_0))$, equal to the identity on the fibre there, with $D\rho(v_0)$ of range* $\ker Df(v_0)$ *and equal to the identity on it.* Proof:

1. Stratified regularity gives $Df(v_0)$ onto (its restriction to the stratum already is) and $K + \text{stratum} = V$ for $K = \ker Df(v_0)$; so K is in good position to C_{v_0} with a closed good complement Y in the stratum subspace.
2. Near 0, $v_0 + C_{v_0}$ and C agree: on $G_0 = \{w : \ell_j(v_0 + w) > 0, j \notin \sigma\}$, a point $v_0 + w$ is in C iff $w \in C_{v_0}$, since $\ell_i(v_0) = 0$ for $i \in \sigma$ — the re-centring of §IV.5 in the model. So $U_1 = (O - v_0) \cap G_0 \cap C_{v_0}$ is relatively open in C_{v_0}, contains 0, and $w \mapsto v_0 + w$ maps it into U.
3. $g(w) = f(v_0 + w) - f(v_0)$ is C^∞ on U_1, $g(0) = 0$, $Dg(0) = Df(v_0)$ onto; Theorem 3.78 in C^∞

form gives the fibre near v_0 as the graph $\{v_0+k+\tau(k)\}$ of a C^∞ map $\tau : V' \to Y$, $V' \subseteq K \cap C_{v_0}$, unique for small k and y.

4. With $q(v)$ the projection of $v-v_0$ onto K along Y `Chapter4/LocalStructure · exists_proj_of_isCompl`, put $\rho(v) = v_0 + q(v) + \tau(q(v))$ on $O' \cap C$, O' a small ball around v_0 chosen so that q maps it into the uniqueness region and $\rho(O' \cap C) \subseteq O'$. The projection along Y keeps the coordinates in σ, so q maps C near v_0 into C_{v_0}; ρ lands on the fibre; and on the fibre $\rho(v) = v$ by uniqueness.
5. $D\tau(0) = 0$ gives $D\rho(v_0) = q$, whose range is K and which is the identity on K.

IV.23 Stratified regularity, the sc⁺-retraction onto A_λ, and its general position

The family. For tame X, $\mathbb{R}^m \times X$ with the product charts is tame `Chapter3/ParamPerturbation · prod_rn_isTame` (§III.40 for $\mathbb{R}^m$ as an M-polyfold `Chapter3/ParamPerturbation · rnPolyfold` with its translation charts `Chapter3/ParamPerturbation · rnChart`). The fibre of $A \subseteq \mathbb{R}^m \times X$ over λ is $A_\lambda = \{x : (\lambda, x) \in A\}$ `Chapter4/Families · fibreSet`. $\mathbb{R}^m$ is presented as a partial quadrant without half-line coordinates `Chapter4/Families · rnPres` `Chapter4/Families · rnPres_eq`, and the product chart $\chi = (r \mapsto r-a)\times\varphi$ has the product presentation `Chapter2/FibredProduct · prodPresentation`, whose minimal subspace at z is $E_z = \mathbb{R}^m \times (E_X)_{\varphi(z_2)}$.

Stratified regular points `Chapter4/Families · IsStratRegularPoint`. $z \in A$ is a stratified regular point if, for some data of Definition 4.13 at z, some $a \in \mathbb{R}^m$, some chart φ of X at z_2, some presentation of its partial quadrant and the level-1 point y over $\chi(z)$, every $\mu \in \mathbb{R}^m$ is the first component of a tangent vector v of the chart model of A at y with $v_2 \in E_{\varphi(z_2)}$: the projection maps $T_zA \cap E_z$ — the tangent space of the stratum through z — onto $\mathbb{R}^m$. λ is a *stratified regular value* if every point of A over λ is a stratified regular point `Chapter4/Families · IsStratRegularValue`. This is the text's "regular value of every p_σ". The chart independence is not formalized (§IV.25). *The naive notion does not suffice*: in $\mathbb{R} \times [0,\infty)$, $A = \{(x,x) : 0 \le x < 1\} \cup (-1,1) \times \{5\}$ and $\varepsilon = \frac{1}{2}$ satisfy the hypotheses of Theorem 4.23 (at $(0,0)$, $T_zA = \mathbb{R}(1,1)$ has the complement $\mathbb{R} \times 0 = T_z^R Z$), Dp is onto everywhere, but $A_0 = \{0,5\}$ is zero-dimensional and meets ∂X, so it is not in general position (Remark 4.24); $(0,0)$ is stratified-critical, $T_zA \cap E_z = 0$.

The slice $e \mapsto (c,e)$: the inclusion $h \mapsto (0,h)$ `Chapter4/Families · ScBanach.prodInrOp` followed by the translation by the smooth vector $(c,0)$ `Chapter4/Families · mem_smooth_rn_zero` is sc-smooth `Chapter4/Families · isScSmooth_slice` with derivative $h \mapsto (0,h)$ `Chapter4/Families · scDeriv_slice`. On a finite-dimensional source the sc-derivative is the classical one `Chapter4/Families · hasFDerivWithinAt_scDeriv_const'`.

The sc⁺-retraction onto A_λ `Chapter4/Families · exists_slice_retraction` — Definition 4.13 requires it and the text does not construct it. In the model $\mathbb{R}^m \oplus E$ of a product chart, let t be an sc⁺-retraction (the chart model of A), described near a point as the graph $\Gamma : V \to t(U) \cap W$ with inverse p (Proposition 4.9), and ρ a C^∞ retraction of $O' \cap (\mathbb{R}^m \times C) \subseteq V$ onto the points v with $\Gamma(v)_1 = c$ (the fibre retraction of §IV.22 for $f = (\Gamma)_1$). Then

$$\tau(e) = \big(\Gamma(\rho(p(t(c,e))))\big)_2$$

is an sc⁺-retraction of E (partial quadrant C) onto $\{e : (c,e) \in t(U) \cap W,\ p(c,e) \in O'\}$: it is sc⁺ because t is sc⁺ and the slice, p, ρ (sc-smooth as a C^∞ map on a finite-dimensional set, Corollary 1.20), Γ and the second projection `Chapter2/StrongBundle · ScBanach.prodSndOp` are sc-smooth (§IV.2); and it is a retraction, because for $e' = \tau(e)$, $\Gamma(w) = (c,e')$ with $w = \rho(\cdot)$ on the fibre, so $t(c,e') = \Gamma(w)$, $p\Gamma(w) = w = \rho(w)$ and $\tau(e') = e'$. *Its tangent space*: if $D\rho(v_0) = 1$ on the h with $(D\Gamma(v_0)h)_1 = 0$, then at the level-1 point over $\Gamma(v_0)_2$ the tangent space of the retract of τ contains $(D\Gamma(v_0)h)_2$ for all such h, by the chain rule $D\tau(y)w = (D\Gamma(v_0)D\rho(v_0)p\,Dt(c,y)(0,w))_2$, Dt fixing $T(t(U)) = D\Gamma(v_0)N$ and $p \circ D\Gamma(v_0) = 1$.

General position of the fibre `Chapter4/Families · isGeneralPositionRetractAt_slice`. Let r_X be tame, the chart model of A in general position at $z = (c,y)$ with respect to $\mathrm{id} \times r_X$ and the product presentation, z regular on its stratum (every μ is the first component of some $a \in T_zA \cap E_z$), and τ an sc⁺-retraction

onto $Q \subseteq O$ whose tangent space at the smooth y contains the second components of the tangent vectors of A at z with first component 0. Then Q is in general position at y. For $w \in T_yO$ write $(0,w) = a + b$ with $a \in T_zA$ and $b \in T^R_zZ = \mathbb{R}^m \times T^R_yO$ (general position of A at z); correct a by $a_2 \in T_zA \cap E_z$ with the same first component as a (regularity on the stratum); then $(a - a_2)_2 \in T_yQ$ and $(b + a_2)_2 \in T^R_yO$, a_2 lying in E_z, so $w \in T_yQ + T^R_yO$. Proposition 2.36 makes T^R_yO an sc-subspace and Lemma 7.A gives the sc-complement (§IV.16). This is the text's "$T_{z'}(A \cap F^\circ_\sigma) = T_{z'}((\{\lambda\} \times A_\lambda) \cap F^\circ_\sigma) \oplus \xi_{z'}$ … in view of the assumption (2)", made precise: regularity on the stratum is what puts a_2 in E_z.

IV.24 Theorem 4.23: families and Sard's theorem

The conclusions at a point `Chapter4/Families · FibreLocal`: $z = (\lambda, x)$ is a stratified regular point, A_λ is near x the image of an sc⁺-retraction on an open subset of X, and A_λ is in general and in good position at x.

The local step `Chapter4/Families · exists_local_family`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold in general position at every point, every $z_0 \in A$ has an open neighbourhood U and a Lebesgue-null* CV $\subseteq \mathbb{R}^m$ *such that the conclusions hold at every $z \in A \cap U$ with $z_1 \notin$ CV.* Proof:

1. *The chart.* $z_{0,2}$ is smooth `Chapter3/ParamPerturbation · prod_rn_mem_smoothSet_iff`, so there is a tame chart φ of X centred at it (§IV.5); the product chart $\chi = (r \mapsto r - \lambda_0) \times \varphi$ is a chart of $\mathbb{R}^m \times X$ `Chapter3/ParamPerturbation · isChart_rnChart` `Chapter2/FibredProduct · MPolyfold.IsChart.prod` centred at z_0, with the tame retraction $\mathrm{id} \times r_X$ `Chapter2/FibredProduct · ScRetraction.prod` `Chapter2/FibredProduct · ScRetraction.IsTameWith.prod` and coordinates those of φ ($\mathbb{R}^m$ has none).
2. *Theorem 4.22 in the chart.* The chart retraction t of A in χ is sc⁺ (§IV.11), and general position at z_0 read in χ is general position at 0; Theorem 4.22 in the chart (§IV.17) gives $N = T_0(\chi A)$ with $N + E^{\min}_0 = E_0$, so the coordinates ℓ are onto on N, a closed good complement $Y \subseteq E^{\min}_0$, the projection p, and $\Gamma : V \to \chi(A)$ over $V = O_V \cap \{\ell \geq 0\} \subseteq N$, with $\ell \circ \Gamma = \ell$.
3. *Sard.* $f(v) = \Gamma(v)_1 + \lambda_0$, the projection P read in the chart, is sc-smooth on V as the first projection `Chapter2/StrongBundle · ScBanach.prodFstOp` composed with Γ, hence C^∞ (§IV.1); CV $= f($stratified critical points of $f)$ is null (§IV.21). $U = W_A \cap \chi^{-1}(W)$, with W_A the open set of the submanifold chart (§IV.11) and W that of Proposition 4.9.
4. *A point $z \in A \cap U$ with $z_1 \notin$ CV.* $\chi(z) \in t(U) \cap W$, so $v_0 = p(\chi z) \in V$ with $\Gamma(v_0) = \chi(z)$ and $f(v_0) = z_1$; hence v_0 is not stratified-critical. Since $\ell(D\Gamma(v)h) = \ell(h)$, $D\Gamma(v_0)$ maps the stratum subspace of $\sigma(v_0)$ onto $T_yA \cap E_y$, $y = \Gamma(v_0)$; so z is a stratified regular point, in the chart χ.
5. *The fibre.* The fibre retraction ρ of f at v_0 (§IV.22), the sc⁺-retraction τ onto the chart image of A_{z_1} (§IV.23), the sc⁺-retraction $\varphi^{-1} \circ \tau \circ \varphi$ on an open subset of X (§IV.18), whose image is A_{z_1} near z_2; the tangent space of τ's retract contains the vertical tangent vectors of A, since $D\rho(v_0) = 1$ on $\ker Df(v_0)$; general position of A_{z_1} at z_2 (§IV.23); good position from general position (Lemma 7.B).

The cover `Chapter4/Families · theorem_4_23_of_subset`. *For X tame and $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with compact closure, in general position at every point, and any set $S \subseteq \mathbb{R}^m$ of parameters, there is $\Sigma \subseteq S$ with $S \setminus \Sigma$ null, consisting of stratified regular values, with the conclusions at every point of every A_λ, $\lambda \in \Sigma$.* The text covers $\{z \in A : |p(z)| \leq \varepsilon\}$ "with the finitely many such neighborhoods $U(z_1), \dots, U(z_k)$". That set need not be compact: in $\mathbb{R} \times \mathbb{R}$, $A = (-2\varepsilon, 2\varepsilon) \times \{2\} \cup \{(\varepsilon(1-x), x) : 0 < x < 1\}$ has compact closure and compact non-empty fibres over $|\lambda| \leq \varepsilon$, but $\{|p| \leq \varepsilon\}$ contains the open segment, which accumulates at $(\varepsilon, 0) \notin A$, and neighbourhoods of radius less than half the distance to $(\varepsilon, 0)$ have no finite subfamily covering it. A countable subcover suffices: $\mathbb{R}^m \times X$ is metrizable (Theorem 2.16) `Chapter2/Metrizable · MPolyfold.metrizableSpace`, so the compact $\overline{A}$, and with it A, is separable, hence second countable; countably many of the neighbourhoods of the local step cover A, the countable union of their null sets is null, and $\Sigma = S \setminus \bigcup_i \mathrm{CV}(z_i)$ (Findings, item 103). Some countability is necessary: an uncountable discrete A can have critical values of positive measure; here it comes from the compactness of $\overline{A}$, which the text assumes.

Theorem 4.23 `Chapter4/Families · theorem_4_23`. *Statement.* Let X be tame, $A \subseteq \mathbb{R}^m \times X$ a smooth finite-dimensional submanifold with tame induced structure and compact closure, $\varepsilon > 0$, and (1) for $|\lambda| \leq \varepsilon$ ($|\lambda|^2 = \sum \lambda_i^2$) the set $A \cap \{z_1 = \lambda\}$ is compact and non-empty, (2) A is in general position at every point. Then there is $\Sigma \subseteq \{|\lambda| < \varepsilon\}$ with null complement in $\{|\lambda| < \varepsilon\}$, consisting of stratified regular values, such that for $\lambda \in \Sigma$ the fibre A_λ is compact, a smooth finite-dimensional submanifold of X, in general position at each of its points, with a corner structure (Definition 4.1), and tame. *Proof.* The cover for $S = \{|\lambda| < \varepsilon\}$. A_λ is the image of the compact $A \cap \{z_1 = \lambda\}$ under the projection to X, hence compact. It is a smooth finite-dimensional submanifold and in general position by the conclusions at its points; it is in good position at every point (Lemma 7.B), so Theorem 4.18 gives the corner structure and tameness. Hypothesis (2), misprinted in the text ("there exists a sc-complement $T_z A$ in $T_z Z$ which is contained in $T_z^R Z$"), is Definition 4.21, as the proof uses it. Not used: the tameness of the induced structure of A, the non-emptiness in (1), $\varepsilon > 0$ (Findings, item 101). Every stratified regular value is a regular value — $T_z A \cap E_z \subseteq T_z A$ — so Σ consists of regular values in the text's sense; but the conclusion is not true at every ordinary regular value (§IV.23).

IV.25 What is not formalized

- **Lemma 4.10** (the section $u \mapsto (1-p)(u - t(u))$ of $U \lhd Y$ is sc-Fredholm): no longer used, since Proposition 4.9 is proved directly. It would need $U \lhd Y$ as a strong bundle, Y and $Y \cap F$ as sc-subspaces, a complement B of $Y \cap F$ in Y chosen inside Y_∞ (an arbitrary algebraic complement does not give an sc-splitting), and at each smooth point the re-centred strong bundle chart (§IV.5), the trivial filler, the sc$^+$-section $s(v) = \mathbf{f}(u) + (1-p)[t(u) - t(u+v)]$ and the conjugation of $g - s = (1-p)v$ to a basic germ with contraction part 0. The text's proof has a sign slip in $D\mathbf{f}(u)$, calls $D\mathbf{f}(u)$ the sc$^+$-operator where $Dt(u)$ is meant, argues "a perturbation of the identity" for an operator into Y, writes $L(T_0 O)$ for $L(Y)$, and omits the interior points of $(T_0 O \oplus B) \cap C$ (Findings, item 57).
- **§4.4** (Definitions 4.25, 4.27, Proposition 4.26, the exterior derivative, the relative complex, Theorem 4.28): it needs $\oplus^k TX$ as an M-polyfold and a linear $T_x X$ (gap G.7), the M-polyfolds X^i for $i \geq 2$ (gap G.6), and integration on manifolds with corners of varying dimension. The text's $d : \Omega^k(X^{i+1}) \to \Omega^{k+1}(X^i)$ should presumably run $\Omega^k(X^i) \to \Omega^{k+1}(X^{i+1})$, the bracket of Proposition 4.26 living one level up; $d^2 = 0$ and $dj^* = j^* d$ are asserted; forms on ∂X are not defined; and Theorem 4.28 has no proof (Findings, items 120–124).
- **The open question** after Proposition 4.15 (§IV.11). For **Remark 4.3** the example is formalized (§IV.27); the argument that an example cannot come from a retraction whose level maps are C^1 is not.
- **The chart independence of stratified regularity** (§IV.23), true by the transport argument of §IV.13, since transition maps of product charts are id $\times\, \tau$ up to translation; Theorem 4.23 does not need it.
- The local form of Remark 4.14 near $\{d_X = 0\}$ (§IV.11), the identification of the local faces of a chart with global faces in Theorem 4.22 (1) (§IV.17), and the classical manifold structure behind Definition 4.1 beyond the C^∞ transitions (§IV.1).

IV.26 What the formalization taught us

1. *Centred charts are a construction.* They exist only at smooth points and live in the partial quadrant $C_{\varphi(x)}$, on the set where it agrees with C; four theorems of the chapter and Lemma 5.11 rely on them, and the text never builds them (§IV.5).
2. *Proposition 4.9 is a statement about fixed sets, and its proof is not the implicit function theorem.* The text's route, as we read it, gives a germ over a different complement from the one needed. The direct proof uses only the sc$^+$ property — a mean value estimate on every level and the level-1 continuity of t — and its regularity argument rests on two observations: the estimate is applied at $\Gamma(a)$ with the frame $B_{\Gamma(a)}$, not at 0, and derivatives of t only meet the finite-dimensional smooth tangent

space, so the loss of levels does not accumulate (§§IV.6–IV.9). The projection must be the one along the closed good complement, not $Dt(0)$.

3. *Theorem 4.2 does not need tameness*, and its compatibility claim needs charts composed with sc-diffeomorphisms between retracts (§IV.10).
4. *The sc⁺ condition has to be read in charts*, in both directions: a submanifold's chart retraction is sc⁺ (§IV.11), and an sc⁺-retraction in a chart is one on an open set (§IV.18). Every statement of §§4.2–4.3 passes through these two facts.
5. *One step of Lemma 4.19 is false*, and the repair is the observation that the complement of T_0O lies in $C \cap (-C)$ at the corner of a centred chart (§IV.14).
6. *General position implies good position* (§IV.16), a step the text uses in Theorem 4.22 (4) and Theorem 4.23 without stating it.
7. *Theorem 4.23 needs stratified regular values and a countable cover*, and the fibre's manifold structure needs the implicit function theorem at corner points with the kernel in good position — the argument of Lemma 5.26 — and Theorem 3.78 in C^∞ form (§§IV.21–IV.24). Sard's theorem had to be supplied in all dimensions (§IV.20).

IV.27 Remark 4.3's example: a retraction that loses a level

Remark 4.3 says that "$X^1 \neq X$" is possible as M-polyfolds although the underlying sets agree, and gives no example. By Proposition 4.8 in charts (§IV.4) an example is an sc-retract $O = r(U)$ with $O = O_\infty$ that is not an sc⁺-retract. It cannot come from a retraction whose level maps $U_m \to E_m$ are C^1 — the maps that Proposition 1.19, Corollary 1.20 and their composites produce: then $T_oO = Dr(o)E_0$ is finite-dimensional by a Baire category argument (O is a countable union of the compact sets $\mathrm{cl}_0\{x \in O : |x|_1 \leq n\}$), and the topologies of O in E_0 and E_1 agree near o (Findings 141; an argument, not formalized). So the retraction must lose a level in its first derivative, as the splicings of HWZ do, and its sc-smoothness needs a criterion the project did not have (Findings 140–146).

A criterion with loss. Let V be finite-dimensional and $L(p)_m : Z_m \to W_m$, $p \in V$, be bounded operators compatible with the inclusions, such that $(p,z) \mapsto L(p)_m z$ is continuous on $V \times Z_m$ — strong continuity, not continuity of $p \mapsto L(p)_m$ in the operator norm — and $p \mapsto A_m(p) = L(p)_m \circ \iota$ is C^∞ as a map into the operators $Z_{m+1} \to W_m$: one level of loss buys all the p-derivatives `Chapter4/RemarkFourThreeExample · IsLossFamily` `Chapter4/RemarkFourThreeExample · famLoss` `Chapter4/RemarkFourThreeExample · famDeriv`. *Then $(p,z) \mapsto L(p)z$ is sc-smooth on $V \oplus Z$* `Chapter4/RemarkFourThreeExample · lossFamMap` `Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth`. The map is sc⁰ by the strong continuity `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc0`, and sc¹ with derivative $(\pi,\zeta) \mapsto \partial_p A_0(p)[\pi]z + L(p)\zeta$ at a level-1 point (p,z) `Chapter4/RemarkFourThreeExample · IsLossFamily.differentiable_famLoss` `Chapter4/RemarkFourThreeExample · IsLossFamily.isSc1With`: the increment in p is taken in the operator norm of A_0, which needs $z \in Z_1$, and the derivative family is continuous on the tangent levels by the C^∞ dependence of $\partial_p A_m$ `Chapter4/RemarkFourThreeExample · IsLossFamily.continuous_famDeriv`. The tangent map is again a map of the same kind: after the sc-isomorphism that reorders (p,z,π,ζ) as $((p,\pi),(z,\zeta))$ `Chapter4/RemarkFourThreeExample · famSwap` it is the map of the *tangent family* $\tilde{L}(p,\pi)(z,\zeta) = (L(p)z, \partial_p A(p)[\pi]z + L(p)\zeta)$ on $V \times V$ `Chapter4/RemarkFourThreeExample · famTangent` `Chapter4/RemarkFourThreeExample · tangentMapWith_lossFamMap_eq`, and $\tilde{L}$ is a loss family `Chapter4/RemarkFourThreeExample · IsLossFamily.tangent`. Induction on k gives sc^k for every k `Chapter4/RemarkFourThreeExample · isSck_lossFamMap`. Only one level of loss is allowed: maps whose j-th derivative costs j levels, such as the map $(t,x) \mapsto (\cos(kt)x_k)_k$ of the example after Definition 1.14, are not covered.

The reduction. For a loss family of projections ($L(p)_0^2 = L(p)_0$), $r(p,z) = (p, L(p)z)$ is an sc-smooth retraction of $V \oplus Z$ `Chapter4/RemarkFourThreeExample · IsLossFamily.isScSmooth_famRetr` `Chapter4/RemarkFourThreeExample · IsLossFamily.retraction`; if the ranges of the $L(p)_0$ are smooth points, $O = O_\infty$ `Chapter4/RemarkFourThreeExample · IsLossFamily.retract_subset_smooth`; and if the lift $O \to E_1$ is discontin-

uous, Proposition 4.8 says that O is not an sc$^+$-retract Chapter4/RemarkFourThreeExample · IsLossFamily.not_isScPlusRetract. These are the data of the example Chapter4/RemarkFourThreeExample · RemarkFourThreeData Chapter4/RemarkFourThreeExample · exists_retract_smooth_not_scPlus_of_data.

The family. On the weighted sequence space X of the example after Definition 1.14 ($|x|_m = \sup_k (k+1)^m |x_k|$; Chapter1/ScOneNotNormContinuous · ScOneExample.X) the moving vector must pass between coordinates of comparable weight, or the level-m operators are unbounded, so the transitions must be fast and the gain per step geometric: the indices are $n_k = 2^k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.nk. With nodes $1/(k+1)$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.node and angles $\theta_k(t)$ turning from 0 (for $t \geq 1/(k+1)$) to $\pi/2$ (for $t \leq 1/(k+2)$) through Mathlib's smooth transition, flat at both ends Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition, the coefficients $b_k = \sin\theta_{k-1}\cos\theta_k$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb vanish except $b_k = \cos\theta_k$ and $b_{k+1} = \sin\theta_k$ at the active block $k = \lfloor 1/t \rfloor - 1$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_eq_zero_of_pos Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt Chapter4/RemarkFourThreeFamily · RemarkFourThree.bb_kt_succ, and all vanish for $t \leq 0$. $L(t) = \sum_k$ (block k) Chapter4/RemarkFourThreeFamily · RemarkFourThree.blk Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam is a sum with two non-zero terms, which on level 0 is the projection onto $b_k e_{n_k} + b_{k+1} e_{n_{k+1}}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_of_pos; on level m its entries carry the weight ratios $((a+1)/(b+1))^m \leq 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.elemL. It is a loss family Chapter4/RemarkFourThreeFamily · RemarkFourThree.isLossFamily_fam: compatible with the inclusions Chapter4/RemarkFourThreeFamily · RemarkFourThree.toBase_fam; uniformly bounded, $\|L(t)_m\| \leq 3 + 2^m$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.norm_fam_le; strongly continuous, from the bound and the norm continuity after the dense inclusion by an $\varepsilon/3$ argument Chapter4/RemarkFourThreeFamily · RemarkFourThree.continuous_apply_of_bound; and $t \mapsto L(t)_m \circ \iota$ is C^∞ in the operator norm, by Mathlib's differentiation of series Mathlib/Analysis/Calculus/SmoothSeries · contDiff_tsum: j derivatives of block k are bounded by a polynomial in k of degree $2j$ (Leibniz's rule and the derivatives of the flat transition), while one level of loss gains a factor $(2 + 2^m)2^{-k}$ Chapter4/RemarkFourThreeFamily · RemarkFourThree.contDiff_fam_comp. Every $L(t)_0$ is a projection Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_idem with finitely supported, hence smooth, range Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_mem_smooth. The lift is discontinuous: the points $o_k = (1/(k+1), (1/(k+1))e_{n_k})$ lie in O Chapter4/RemarkFourThreeFamily · RemarkFourThree.fam_node_smul_6v and tend to 0 in E_0, while their level-1 coordinates $(2^k+1)/(k+1)\, e_{n_k}$ have norm at least 1 Chapter4/RemarkFourThreeFamily · RemarkFourThree.not_continuousOn_liftOne. So the data exist Chapter4/RemarkFourThreeFamily · RemarkFourThree.remarkFourThreeData, and there is an sc-retract $O = O_\infty$ that is not an sc$^+$-retract Chapter4/RemarkFourThreeFamily · remark_4_3_retract_example.

The M-polyfold. A retract with its single chart $O \to O$ is an M-polyfold Chapter4/RemarkFourThreeFamily · retractChart Chapter4/RemarkFourThreeFamily · retractPolyfold; if $O \subseteq E_\infty$ and O is not an sc$^+$-retract, every point of it is smooth and $1_0^1 : X \to X^1$ is not sc-smooth, for every junk value (Proposition 4.8 in charts, §IV.4) Chapter4/RemarkFourThreeFamily · retractPolyfold_smooth_not_toLevelOne. *Remark 4.3's example* Chapter4/RemarkFourThreeFamily · remark_4_3_example: there is an M-polyfold with $X_0 = X_\infty$ for which 1_0^1 is not sc-smooth, hence (by the remark's first part, §IV.4) not an sc-diffeomorphism: $X^1 \neq X$ as M-polyfolds. In this example X and X^1 even carry different topologies on the same set, since $o_k \to 0$ in X and not in X^1. "$X^1 \neq X$" is read as "1_0^1 is not sc-smooth"; the M-polyfold is a single retract, which is all the remark needs; its parameter space is $\mathbb{R}$ and its partial quadrant all of $\mathbb{R} \oplus X$, so there is no boundary.

Part V. The Fredholm package for M-polyfolds (HWZ Chapter 5, §§5.1–5.4)

Chapter 5 turns the implicit function theorems of Chapter 3 into the working tools of a Fredholm theory: a way to measure perturbations (auxiliary norms, §5.1), the compactness that keeps the solution sets of small perturbations under control (§5.2), a supply of perturbations built from sc-smooth bump functions (the start of §5.3), and the perturbation theorems themselves — transversality when there is no boundary (Theorem 5.13), general position (Theorem 5.18), good position relative to the boundary (Theorems 5.19, 5.25), and the lemmas of the Morse-type Theorem 5.22. Appendix 5.4.1 proves Theorem 5.5; Appendix 5.4.2 relates sc-smooth bump functions and partitions of unity to the geometry of Banach spaces. This Part retells §5.1 (§§V.1–V.3); §5.2 with Appendix 5.4.1 (§§V.4–V.7), the longest retelling being Theorem 5.5 (§§V.5–V.6); the start of §5.3 — bump functions, the calculus of sections, Lemma 5.11, allowable sections (§§V.8–V.10); Theorem 5.13 (§§V.11–V.15); general position, Lemma 5.26 and Theorem 5.18 (§§V.16–V.20); Theorems 5.25 and 5.19 (§§V.21–V.22); the homotopy lemmas (§V.23); Appendix 5.4.2 (§§V.24–V.26); what is not formalized (§V.27); and what the formalization taught us (§V.28). A fourth addendum adds Corollary 5.8 on the shifted strong bundle (§V.29), the metric space of the allowable sections (§V.30) and weak against pointwise bump functions (§V.31).

What is proved, and what is not. Theorems 5.5, 5.6, 5.7, 5.13, 5.18, 5.19, 5.25, 5.34 and 5.37 are proved, Theorems 5.5–5.7 without tameness; so are Lemmas 5.2, 5.11, 5.23, 5.24, 5.26, Propositions 5.3, 5.10, 5.30, 5.35, 5.38, 5.42, Remark 5.12 and Corollary 5.44, the last with Theorem 5.43 as an explicit hypothesis. Every definition of the chapter is formalized. Corollary 5.8 is proved on the shifted strong bundle $P^1 : Y^1 \to X^1$, which the text uses without defining it, and by iteration for every fixed i, with the filler condition on level 1 that its "it is a priori clear" hides (§V.29). The perturbation theorems take f with *tame germ data* (Definition 3.51), the reading of the tame strong bundle with which §5.3 opens and the hypothesis under which the §3.6 implicit function theorems are proved since their repair (Part III, Findings of Chapter 3, items 190–199); Theorem 5.18 also assumes X tame, which its proof uses through Theorems 4.22 and 4.23. The metric space $(\Gamma^{+,1}_U(P), \rho)$ is built, and the text's "not complete" turns out to need a hypothesis (§V.30). Not formalized: Theorems 5.22, 5.27, Remarks 5.14, 5.16, 5.21, the cited Theorems 5.32, 5.40, 5.43, the Sobolev remark, a dense form of Theorem 5.19, and Corollary 5.8 as one statement for a family indexed by i (§V.27).

Four structural choices. (1) *The linearization is read in charts.* T_xX carries no linear structure in the formalization (gap G.7 of Chapter 2), so $f'(x)$ is the operator $\mathrm{Lin}(z) : T_zO \to K_z$ of a strong bundle chart (Part III), transversality is surjectivity of Lin in a chart, and general position is stated in a tame chart; its chart independence is proved where it matters (§V.18). (2) *"Generic" is "off a null set".* Every perturbation theorem is proved first in a generic form — for λ outside a Lebesgue null set of a small ball, $s_0 + \sum \lambda_i t_i$ does the job — and the text's dense-set form is derived from it. (3) *The parametrized solution set is not built as a manifold.* The Sard step is local: at each zero of $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ in good position, Theorem 3.54 (2) gives a regular tangent germ chart, Sard's theorem is applied on its finite-dimensional model, and finitely many null sets are collected (§V.14). Where Theorem 4.23 needs a submanifold, the *bridge lemma* supplies it (§V.17). (4) *Density is pointwise.* "Within ε" is $N(s(x) - s'(x)) \le \varepsilon$ for all x; this

is $\rho(s,s') \le \varepsilon$ for the metric $\rho(s,s') = \sup_x N(s(x) - s'(x))$, which the perturbation theorems never needed and which was built afterwards (§V.30).

Map.

- `Chapter5/AuxNorm`: the charts $\Phi[1]$, the fibre $(Y_{0,1})_x$ and its chart norms, Definition 5.1, Lemma 5.2, Proposition 5.3.
- `Chapter5/Compactness`: Definition 5.4, Theorems 5.5–5.7, Corollary 5.8 in a chart, Remark 5.12.
- `Chapter5/BumpFunctions`: $\mathbb{R}$ as an M-polyfold, Definitions 5.9, 5.29, 5.33, Propositions 5.10, 5.30, 5.35, Theorem 5.34, the calculus of sections, Lemma 5.11, $\Gamma^{+,1}_U(P)$.
- `Chapter5/Transversality`: Definitions 5.15, 5.17, Lemma 5.26, the parametrized section, the Sard step, Theorem 5.13.
- `Chapter5/GeneralPosition`: the bridge lemma, $T_yS = \ker f'(y)$, chart independence and openness of general position, properties (i)–(ii), Theorems 5.18, 5.19, 5.25.
- `Chapter5/Homotopy`: Definition 5.20, Lemmas 5.23, 5.24.
- `Chapter5/PartitionsOfUnity`: the algebra of sc-smooth functions, Definitions 5.28, 5.31, 5.36, 5.39, 5.41, Theorem 5.37, Propositions 5.38, 5.42, Corollary 5.44, the statements of Theorems 5.40, 5.43.
- Added in the fourth addendum (§§V.29–V.31): `Chapter5/StrongBundleShift` (the shifted strong bundle P^1, raising the index of sc-smooth maps and sections), `Chapter5/Corollary58` (the filler condition on every level, Corollary 5.8), `Chapter5/AllowableSectionsMetric` (the metric ρ, completeness and its failure), `Chapter5/BumpWeakPointwise` (bump points, an sc-Banach space as an M-polyfold).

Throughout, $P : Y \to X$ is a strong bundle over an M-polyfold (both paracompact and Hausdorff), a section is a map $s : X \to Y$ with $P \circ s = \mathrm{id}$, 0_x is the zero of the fibre over x, and $\operatorname{supp} s$ is the closure of $\{s \neq 0\}$ `Chapter5/BumpFunctions · sectionSupport`. A strong bundle chart Φ has base chart $\varphi : V \to O$, $O = r(U)$ a retract, and local bundle $K \to O$; the *principal part* of f is $\hat{f}(u) = \Phi(f(\varphi^{-1}(u)))_2$, and $\mathrm{Lin}(z)$ is its linearization at a level-1 point z (Part III). $|\lambda|$ is the maximum norm on $\mathbb{R}^m$.

V.1 The fibre $Y_{0,1}$ and the charts $\Phi[1]$

§5.1 works on the part $Y_{0,1}$ of biregularity $(0,1)$, the M-polyfold $Y[1]$ of §2.5 (Part II), and assumes two things the project did not have: charts $\Phi[1]$ for *every* strong bundle chart, and a Banach structure on the fibres of $Y_{0,1}$.

$Y_{0,1}$ in any chart. For a strong bundle chart Φ compatible with the atlas — the text's "strong bundle isomorphism $\Phi : Y|V \to K$" — a point y over $\operatorname{dom}\varphi$ lies in $Y_{0,1}$ iff $\Phi(y)_2 \in F_1$: compare with a chart of the atlas through the transition map, which is a strong bundle map and so preserves biregularity `Chapter5/AuxNorm · StrongBundle.mem_oneSet_iff_of_isChart`. On $P^{-1}(V) \cap Y_{0,1}$ the map $w \mapsto \Phi(w)$ lifted to $K[1]$ is then a partial homeomorphism `Chapter5/AuxNorm · StrongBundle.oneChartOf`, compatible with every chart $\Phi'[1]$ of the atlas because the transition is the $[1]$-transition of Definition 2.50 `Chapter5/AuxNorm · StrongBundle.oneChartOf_transition` `Chapter5/AuxNorm · StrongBundle.oneChartOf_compatible`, hence a chart of the M-polyfold $Y[1]$ `Chapter5/AuxNorm · StrongBundle.isChart_oneChartOf`. For a chart of the atlas it is the chart of Part II `Chapter5/AuxNorm · StrongBundle.oneChart_eq_oneChartOf`.

The fibre. The fibres $P^{-1}(x)$ carry linear structures; the points of $P^{-1}(x)$ in $Y_{0,1}$ form a linear subspace $(Y_{0,1})_x$, because the charts are linear on the fibres and $F_1 \subseteq F_0$ is a subspace `Chapter5/AuxNorm · StrongBundle.oneFiber`. Its points are points of $Y[1]$ and conversely `Chapter5/AuxNorm · StrongBundle.toOne` `Chapter5/AuxNorm · StrongBundle.ofOne`; its zero is the point 0_x of $Y[1]$ `Chapter5/AuxNorm · StrongBundle.zeroOne` `Chapter5/AuxNorm · StrongBundle.fiberZero_mem_oneSet`.

The chart norm $|h|_1$. In a chart around x, the fibre component of $\Phi[1]$ is a linear map $(Y_{0,1})_x \to F_1$ `Chapter5/AuxNorm · StrongBundle.liftOneFiber`, injective because Φ is injective on the fibre `Chapter5/AuxNorm · StrongBundle.liftOneFiber_injective`. Its image is the slice $\{h \in F_1 \mid (\varphi(x), h) \in K[1]\}$ `Chapter5/`

AuxNorm · StrongBundle.mem_range_liftOneFiber, the fixed point set of the continuous map $h \mapsto R[1](\varphi(x), h)$, hence closed in F_1 Chapter5/AuxNorm · StrongBundle.isClosed_slice. So a sequence of the fibre that is Cauchy for $|\Phi[1](\cdot)_2|_1$ converges: its images converge in F_1, the limit stays in the slice, and it is the image of a point of the fibre Chapter5/AuxNorm · StrongBundle.complete_liftOneFiber. This is the text's "the fibers Y_x have the structure of Banach spaces". The norm depends on the chart, but only up to equivalence, and locally uniformly: by Lemma 5.2's upper bound (§V.2) applied to $g = |\Phi_1[1](\cdot)_2|_1$, the norms of two charts are comparable with one constant on the fibres over a neighbourhood of x Chapter5/AuxNorm · StrongBundle.exists_norm_liftOne_le_mul, and so are finitely many charts and one more Chapter5/AuxNorm · StrongBundle.exists_uniform_comparison.

V.2 Auxiliary norms and Lemma 5.2

Definition 5.1. A function $N : Y[1] \to \mathbb{R}$ is a *complete norm on the fibre over* x if $N(v) = 0$ only for $v = 0$, $N(tv) = |t|N(v)$, $N(v + w) \le N(v) + N(w)$, and every N-Cauchy sequence of $(Y_{0,1})_x$ converges Chapter5/AuxNorm · StrongBundle.IsCompleteFiberNorm. An *auxiliary norm* is a continuous such N, complete on every fibre, with condition (2): if $P(w_k) \to x$ and $N(w_k) \to 0$, then $w_k \to 0_x$ in $Y[1]$ Chapter5/AuxNorm · StrongBundle.IsAuxNorm. It is non-negative Chapter5/AuxNorm · StrongBundle.IsAuxNorm.nonneg and homogeneous on the fibres Chapter5/AuxNorm · StrongBundle.IsAuxNorm.smul.

Lemma 5.2, the upper bound. Let $g : Y[1] \to \mathbb{R}$ be continuous at 0_x and positively homogeneous on the fibres near x, and Φ a chart around x. By continuity at 0_x — read through the chart $\Phi[1]$, whose neighbourhoods of $(\varphi(x), 0)$ are the sets $|p - \varphi(x)|_0 + |h|_1 < \varepsilon$ — $g < 1$ where $|p - \varphi(x)|_0 < \varepsilon/2$ and $|h|_1 \le \varepsilon/2$; homogeneity scales every vector of a nearby fibre to $|h|_1 = \varepsilon/2$, which gives $g(w) \le (2/\varepsilon)|h|_1$ on the fibres over a neighbourhood V of x Chapter5/AuxNorm · StrongBundle.exists_le_mul_norm_liftOne. This is the text's first half of "(i) □ (ii)".

Lemma 5.2, the lower bound. For an auxiliary norm, $|h|_1 \le A\,N(w)$ near x Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le. Suppose not: then for every k there is w_k over $\varphi^{-1}(B(\varphi(x), 1/(k+1)))$ with $|h_k|_1 > (k+1)N(w_k)$; rescale to $|h_k|_1 = 1$. Then $N(w_k) \to 0$ and $P(w_k) \to x$ — the points of the shrinking balls converge to x because φ^{-1} is continuous on O, so no metrizability of X is needed — hence $w_k \to 0_x$ by condition (2), and $\Phi[1](w_k) \to (\varphi(x), 0)$ by the continuity of $\Phi[1]$, contradicting $|h_k|_1 = 1$. Together: **in every chart around** x there are a neighbourhood V and $0 < a < A$ with $aN(w) \le |h|_1 \le AN(w)$ over V Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_bounds.

Lemma 5.2. For N continuous and a complete norm on every fibre, N is an auxiliary norm iff every x has a chart and such two-sided bounds near x Chapter5/AuxNorm · lemma_5_2. "(i) □ (ii)" is the previous paragraph. "(ii) □ (i)", "obvious" in the text: if $P(w_k) \to x$ and $N(w_k) \to 0$, the bound gives $|h_k|_1 \to 0$, so $\Phi[1](w_k) \to (\varphi(x), 0)$, and $w_k \to 0_x$ because $\Phi[1]^{-1}$ is continuous on $K[1]$ (§V.1). In the statement V is a neighbourhood of x on which the bounds hold; the text's V is the domain of Φ, but its own proof gives them "for $P(w)$ close to x".

V.3 Proposition 5.3

Existence. For the charts Φ_i of the atlas and a partition of unity (ρ_i) subordinate to their domains — it exists since X is paracompact and Hausdorff, hence normal Mathlib/Topology/PartitionOfUnity · PartitionOfUnity.exists_isSubordinate — the *glued norm* is $N(w) = \sum_i \rho_i(P(w))\,|\Phi_i[1](w)_2|_1$ Chapter5/AuxNorm · StrongBundle.gluedNorm Chapter5/AuxNorm · StrongBundle.gluedNorm_eq_sum. It is continuous on $Y[1]$: each term vanishes off the closed set $P^{-1}(\operatorname{supp}\rho_i)$, which lies in the open domain of $\Phi_i[1]$, where the term is continuous, and the sum is locally finite Chapter5/AuxNorm · StrongBundle.continuous_gluedNorm. Near a point x only finitely many ρ_i are non-zero, all with x in the domain of Φ_i and summing to 1 Chapter5/AuxNorm · StrongBundle.sum_partition_eq_one; their chart norms are uniformly comparable with the norm of any one chart Φ' around x (§V.1), so N is two-sided

comparable with $|\Phi'[1](\cdot)_2|_1$ near x Chapter5/AuxNorm · StrongBundle.gluedNorm_bounds. On each fibre N is a convex combination of norms, hence a norm, and comparable with a complete norm, hence complete Chapter5/AuxNorm · StrongBundle.complete_of_bounds Chapter5/AuxNorm · StrongBundle.isCompleteFiberNorm_gluedNorm. Lemma 5.2 "(ii) □ (i)" concludes Chapter5/AuxNorm · proposition_5_3_exists. The text gives this as one sentence.

Comparison. For auxiliary norms N_0, N_1, Lemma 5.2 "(i) □ (ii)" for both in the *same* chart around y — which is why "(i) □ (ii)" was proved in every chart — gives $r_y > 0$ with $r_y N_0 \le N_1$ and $r_y N_1 \le N_0$ over a neighbourhood of y. At each point the admissible constants r form a convex set (r admissible and $0 < r' \le r$ give r' admissible), so a partition of unity glues the local constants into a continuous $f > 0$ with values in these sets Mathlib/Analysis/Convex/PartitionOfUnity · exists_continuous_forall_mem_convex_of_local_const, and $f(x)N_0 \le N_1 \le f(x)^{-1}N_0$ on the fibre over x Chapter5/AuxNorm · proposition_5_3_compare. Tameness plays no role in §5.1.

V.4 Definition 5.4 and proper sections

"$N(y) \le r$" for a point y of Y is read with the text's convention $N(y) = \infty$ off $Y_{0,1}$: it means $y \in Y_{0,1}$ and $N(y) \le r$ Chapter5/Compactness · StrongBundle.AuxLe, a property of the point only Chapter5/Compactness · StrongBundle.AuxLe.congr. A section f is *weakly proper* if $f^{-1}(0)$ is compact and for every auxiliary norm N there is an open $U \supseteq f^{-1}(0)$ such that $\{f = s\}$ is compact for every sc⁺-section s with support in U and $N(s) \le 1$ Chapter5/Compactness · StrongBundle.IsWeaklyProper; *proper* if instead $\mathrm{cl}_X\{x \in U \mid N(f(x)) \le 1\}$ is compact Chapter5/Compactness · StrongBundle.IsProper.

Proper implies weakly proper for a continuous f Chapter5/Compactness · StrongBundle.IsProper.isWeaklyProper. Take the U of properness. A solution of $f(x) = s(x)$ outside U has $s(x) = 0$, so $f(x) = 0$ and $x \in f^{-1}(0) \subseteq U$ after all; inside U, $f(x) = s(x)$ gives $N(f(x)) = N(s(x)) \le 1$. So $\{f = s\}$ lies in the compact closure, and it is closed because f and s are continuous and Y is Hausdorff. Weakly proper sections have compact solution sets by definition Chapter5/Compactness · StrongBundle.IsWeaklyProper.isCompact_solutionSet. The text's "obviously" hides the continuity of f.

V.5 Theorem 5.5: the local form for basic germs

Theorem 5.5 says that for sc-Fredholm sections a compact solution set already forces properness. The appendix proves it in the coordinates of a basic germ, and that local statement is the heart of the matter.

The setting. A basic germ (Definition 3.7, Part III) is $h : \tilde{C} \oplus W \supseteq U_0 \to \mathbb{R}^N \oplus W$ whose W-component is $w - B(a, w)$, where B is an sc⁰-contraction germ: on small balls, $w \mapsto B(a, w)$ contracts with constant $\le 1/2$ on level 0, uniformly in a, and $B(0, 0) = 0$. Part III built the *parametrized fixed point* $\delta(a, z)$, the unique solution w in a closed ball $\bar{B}(r)$ of $w = B(a, w) - z$ for (a, z) in a set Σ of small parameters, continuous in (a, z) on level 0 Chapter3/LocalCompactness · IsSc0ContractionGerm.delta_continuousOn Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero.

The local form Chapter5/Compactness · IsSc0ContractionGerm.exists_isCompact_bdd. Assume $\tilde{C}$ closed. There is an open $O \ni 0$, *independent of c*, such that for every c the points $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) \in W_1$ and $|w - B(a, w)|_1 \le c$ lie in a compact set. The proof:

1. *A small ball.* $B(a, w) - w$ is continuous at 0 within U_0 and vanishes there, so for $|(a, w)|_0 < \rho_1$ it has norm $< r/8$. Let $\rho = \min(\rho_1, \rho'/2, r/2)$, with ρ' the radius of the parameter set, and $O = B(0, \rho)$.
2. *The compact set of parameters.* The inclusion $W_1 \to W_0$ is compact, so the image of the closed level-1 ball of radius c lies in a compact $K_c \subseteq W_0$. The set $L = (\tilde{C} \cap \bar{B}(\rho)) \times (K_c \cap \bar{B}(r/8))$ is compact (here $\tilde{C}$ closed is used) and lies in Σ.
3. *The candidate.* The compact set is the image of L under the continuous map $(a, z) \mapsto (a, \delta(a, z))$.

4. *Every bounded point is hit.* Let $x = (a, w) \in U_0 \cap O$ with $w - B(a, w) = \iota(z_1)$, $|z_1|_1 \le c$. Put $p = (a, B(a, w) - w)$. Then $a \in \tilde{C}$ with $|a| < \rho$; $B(a, w) - w = \iota(-z_1)$ lies in K_c and has norm $< r/8$ by step 1; so $p \in L$. And w lies in the ball $\bar{B}(r)$ and solves $w = B(a, w) - (B(a, w) - w)$, so $w = \delta(p)$ by uniqueness `Chapter5/Compactness · IsSc0ContractionGerm.eq_delta_zero`, i.e. x is the image of p.

Only the level-0 contraction is used. For a basic germ f itself the W-component of $f(x)$ is $w - B(a, w)$, so the same O works with the level-1 bound $|f(x)|_1 \le c$ `Chapter5/Compactness · IsBasicGerm.exists_isCompact_bdd`. This is the appendix's (82)–(90), where the text writes "$a \in \mathbb{R}^n$" for $a \in \tilde{C}$ (B is defined on $\tilde{C} \oplus W$ only), "$\mathbb{R}^n \oplus Y_1$" for $\mathbb{R}^N \oplus W_1$, and in (89) mixes the $\mathbb{R}^N$- and W-components; the t of its "$h + t$" and its estimates (83), (85) disappear in the next step.

V.6 Theorem 5.5: the local step and the covering

Uniform fibre bounds. A strong bundle map Ψ (Definition 2.47) is linear on the fibres and $\Psi[1]$ is continuous; at $(u_0, 0)$, where $\Psi[1](u_0, 0)_2 = 0$ `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_zero`, continuity gives $|\Psi[1](u, k)_2|_1 \le 1$ for $|u - u_0|_0 < \delta$, $|k|_1 \le \eta$, and fibre linearity `Chapter5/Compactness · IsStrongBundleMap.shiftFiber_snd_smul` turns this into $|\Psi[1](u, k)_2|_1 \le \eta^{-1}|k|_1$ for all $k \in F_1$ `Chapter5/Compactness · IsStrongBundleMap.exists_fiber_bound_one`. The text uses this bound silently when it replaces $N(f(x)) \le 1$ by a bound in basic germ coordinates.

The local step `Chapter5/Compactness · exists_nbhd_isCompact_closure_bdd`. Let (f, x_0) be an sc-Fredholm germ (Definition 3.11) — x_0 need not be a zero. Its data, in the text's germ reading (§III.6): a strong bundle chart Φ centred at x_0, a filling germ g of the principal part $\hat{f}$ on a germ inside the domain U of the retraction, an sc$^+$-germ t_0 on a germ G_s, and a strong bundle isomorphism Ψ with base map ψ conjugating $g - t_0$ to a basic germ h: $h(\psi(u)) = \Psi(u, g(u) - t_0(u))_2$ for $u \in U$ near 0. The claim is an open $V \ni x_0$, independent of C, such that for every C the closure in X of $Z_C = \{x \in V \mid |\Phi(f(x))_2|_1 \le C\}$ is compact.

1. *The neighbourhood.* Near 0 the filler agrees with the principal part on O and the conjugation identity holds (a set N_1); t_0 is continuous into F_1 on level 0 of its germ, $(G_s)_0 = O_s \cap C$ with O_s open — not on U — so the neighbourhood is also cut with O_s, and $|t_0(u)|_1 \le |t_0(0)|_1 + 1$ for $u \in U \cap O_s$, $|u|_0 < \delta_2$ (with t_0 sc-smooth on U the cut was not needed); the fibre bound of Ψ holds for $|u|_0 < \delta_3$ with constant M. On the side of the basic germ, the local form (§V.5) gives an open O'; ψ maps a neighbourhood of 0 into a closed ball $\bar{B}(r_0) \cap C'$ inside the domain U' of ψ^{-1} (relative openness of U'). V is the set of points of the chart domain whose image lies in all of these.
2. *The bound transported.* For $x \in Z_C$, $u = \varphi(x) \in O$, $g(u) = \hat{f}(u)$ lies in F_1 with $|g(u)|_1 \le C$, so $|g(u) - t_0(u)|_1 \le C + |t_0(0)|_1 + 1$, and $|h(\psi(u))|_1 = |\Psi[1](u, g(u) - t_0(u))_2|_1 \le M(C + |t_0(0)|_1 + 1) =: c'$. *Subtracting the sc$^+$-part before the conjugation* is what makes the text's t and its estimates unnecessary, and the neighbourhood independent of C.
3. *Compactness on the side of the basic germ.* So $\psi(\varphi(Z_C))$ lies in the compact set $K_{c'}$ of the local form, intersected with the closed ball $\bar{B}(r_0) \cap C'$; its closure A is compact and inside U'.
4. *Back to U.* ψ^{-1} is continuous on U', so $T = \psi^{-1}(A)$ is compact, contains $\varphi(Z_C)$ and lies in U.
5. *Back to O.* The closure of $\varphi(Z_C)$ lies in $T \subseteq U$ and in the closure of O; since O is the fixed point set of the continuous retraction r on U, it is closed in U, so the closure lies in O.
6. *Back to X.* φ^{-1} is continuous on O, so φ^{-1} of that closure is compact and contains Z_C, hence its closure.

The text's "it suffices to find in local coordinates an open neighborhood" leaves steps 4–6 implicit: its set $\{(a, w) \in U \mid |(h + t)(a, w)|_1 \le c\}$ lives in the chart domain U of the filling, which contains points not in the retract.

Theorem 5.5 `Chapter5/Compactness · theorem_5_5`. Let f be sc-Fredholm with compact solution set, and N

an auxiliary norm. At each zero x the germ (f, x) is sc-Fredholm, so the local step gives a chart Φ and an open $V(x)$; Lemma 5.2's lower bound in that chart `Chapter5/AuxNorm · StrongBundle.IsAuxNorm.exists_norm_liftOne_le` gives $V_1(x)$ and A with $|\Phi(w)_2|_1 \le A\,N(w)$ over $V_1(x)$, so on $V(x) \cap V_1(x)$ the condition $N(f(y)) \le 1$ implies $|\Phi(f(y))_2|_1 \le A$, and $\mathrm{cl}\{y \in V(x) \cap V_1(x) \mid N(f(y)) \le 1\}$ is a closed subset of a compact set. Finitely many of these neighbourhoods cover the compact $f^{-1}(0)$; their union is U, and the closure of $\{y \in U \mid N(f(y)) \le 1\}$ lies in the finite union of the compact closures. That is the text's last sentence, "since the solution set of f is compact, the proof is complete". Tameness is not used, as in the appendix's statement of the theorem.

The equivalence. For an sc-Fredholm section, which is continuous, proper □ weakly proper (§V.4) □ compact solution set □ proper (Theorem 5.5) `Chapter5/Compactness · theorem_5_5_iff`.

V.7 Theorems 5.6, 5.7 and Corollary 5.8

Theorem 5.6 is Theorem 3.12 at a zero: a zero of an sc-Fredholm section is a smooth point (regularizing), where (f, x) is an sc-Fredholm germ, and Theorem 3.12 gives nested open $U(i) \ni x$ with $\mathrm{cl}\{y \in U(i) \mid f(y) = 0\}$ the image of a compact subset of X_i `Chapter3/SectionCompactnessG · theorem_3_12` `Chapter5/Compactness · theorem_5_6`.

The space X_∞. The text says "X_∞ is a metric space" but never gives it a topology. It is the set of smooth points with the initial topology of the inclusions $X_\infty \to X_m$ `Chapter2/Metrizable · MPolyfold.smoothLevel_topology_eq`, metrizable by Theorem 2.16 `Chapter2/Metrizable · MPolyfold.metrizableSpace_smoothLevel`, and a sequence converges in it iff it converges on every level `Chapter5/Compactness · MPolyfold.tendsto_smoothLevel_iff`; the inclusions into X and into each X_i are continuous `Chapter5/Compactness · MPolyfold.continuous_smoothVal` `Chapter5/Compactness · MPolyfold.continuous_smoothIncl`.

Theorem 5.7 `Chapter5/Compactness · theorem_5_7`. $S = f^{-1}(0)$ lies in X_∞ because f is regularizing. Take a sequence (x_k) in S; a subsequence converges in X to some x. The limit is a zero, since a limit of zeros of a continuous section is a zero `Chapter3/SectionCompactness · StrongBundle.IsScSmoothSection.isZeroAt_of_tendsto`, so it is smooth and (f, x) is an sc-Fredholm germ. Now *Corollary 3.13 at the point x* gives the convergence $x_k \to x$ on every level `Chapter3/Chapter5BridgeG · corollary_3_13_section`, hence in X_∞. The text cites Theorem 5.6, which gives compactness in X_i, not the convergence of a given sequence. Tameness is not used.

Corollary 5.8, the compact solution set. The zeros in X_i are the image of the compact subset of X_∞ of Theorem 5.7 under the continuous inclusion $X_\infty \to X_i$ `Chapter5/Compactness · isCompact_solutionSet_level`.

Corollary 5.8 in a chart. "It is a priori clear that f^i is sc-Fredholm" is not clear, and $Y^i \to X^i$ is not defined in the text. What is formalized is the shift of the germ data. A basic germ f on $\tilde{C} \oplus W$ shifts to a basic germ f^1 on $\tilde{C} \oplus W^1$ with contraction germ B^1: the shifted neighbourhood germ `Chapter5/Compactness · ScNbhdGerm.shiftParam`, the contraction estimates on the shifted levels `Chapter5/Compactness · IsSc0ContractionGerm.shift`, and sc-smoothness of f^1 by Proposition 1.17 for germs, which needs $\tilde{C}$ a partial quadrant `Chapter5/Compactness · IsBasicGerm.shift`. If k is conjugated to h by Ψ, then k^1 is conjugated to h^1 by the index raise Ψ^1 `Chapter3/BasicClassComposition · IsStrongBundleIso.shift` `Chapter5/Compactness · IsConjugateToBasicGerm.shift`. Hence, for an sc-Fredholm section with compact solution set, the zeros in X_1 are compact, and at every smooth point, in the chart of the germ, $(g - s)^1$ is conjugated to a basic germ and s^1 is an sc⁺-germ on the shifted germ G^1_s (Proposition 1.17 for germs, `Chapter3/GermLevels · IsSckGerm.shift`), the text's setting `Chapter5/Compactness · corollary_5_8_chart`. In the germ reading of Definition 3.11 (§III.48) g is a filling germ and $g - s$ only an sc⁰-germ on the intersection of the two germs; the level shift of the conjugation needs no more, since the conjugation identity is used near 0 only `Chapter5/Compactness · IsConjugateToBasicGerm.shift_germ`. The statement changed with the switch: in the whole-chart reading s^1 was an sc⁺-section of all of $U^1 \lhd F^1$. The filler of g^1 was the gap: its condition (3) on level 1 does not follow from level 0. It is closed, together with the shifted strong bundle and the whole corollary, in §V.29.

V.8 $\mathbb{R}$ as an M-polyfold, and bump functions

$\mathbb{R}$. "An sc-smooth map $f : X \to \mathbb{R}$" needs $\mathbb{R}$ as an M-polyfold: the constant sc-structure `Chapter5/BumpFunctions · realSc` and the single chart id `Chapter5/BumpFunctions · realChart` `Chapter5/BumpFunctions · realPolyfold` `Chapter5/BumpFunctions · isChart_realChart`. Sc-smoothness of a map between M-polyfolds can be checked in one pair of charts around each point, continuity included, since the local representatives determine the map near the point `Chapter5/BumpFunctions · MPolyfold.isScSmoothMap_of_local` `Chapter5/BumpFunctions · isScSmoothMap_real_of_local`; an sc-smooth function read in any chart is sc-smooth on the retract `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.real_chart`; constants are sc-smooth `Chapter5/BumpFunctions · isScSmoothMap_real_const`; a C^∞ function $\sigma : \mathbb{R} \to \mathbb{R}$ is an sc-smooth self-map of $\mathbb{R}$ (Corollary 1.20 in the constant structure) `Chapter5/BumpFunctions · isScSmoothMap_real_of_contDiff`, so $\sigma \circ f$ is sc-smooth `Chapter5/BumpFunctions · MPolyfold.IsScSmoothMap.comp_contDiff`.

Definitions 5.9 and 5.29. *X admits sc-smooth bump functions* if every x and open $U \ni x$ have an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions`. The text has two definitions: Definition 5.9 asks "$f \neq 0$", Definition 5.29 "not identically zero". The proof of Proposition 5.30 starts from "a bump function g with $g(0) = 1$", i.e. $g(x) \neq 0$ at the given point; on an M-polyfold there are no translations to move a bump to x (even on an sc-Banach space, translation by a non-smooth vector does not preserve the levels). So Definition 5.9 is read with $f(x) \neq 0$; Definition 5.29 as written `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctionsWeak` is implied `Chapter5/BumpFunctions · MPolyfold.AdmitsBumpFunctions.weak`. How far apart the two readings are is §V.31.

Propositions 5.10 and 5.30. Given x and U, take a bump g with $g(x) \neq 0$ and let $f = \tau(3g/g(x) - 1)$, τ the smooth transition from 0 on $(-\infty, 0]$ to 1 on $[1, \infty)$ `Mathlib/Analysis/SpecialFunctions/SmoothTransition · Real.smoothTransition`. Then f is sc-smooth, $0 \le f \le 1$, $f(x) = \tau(2) = 1$, $f \equiv 1$ on the open set $\{g/g(x) > 2/3\} \ni x$, and $f \neq 0$ only where $g/g(x) > 1/3$, so $\operatorname{supp} f \subseteq \operatorname{supp} g \subseteq U$ `Chapter5/BumpFunctions · proposition_5_30` `Chapter5/BumpFunctions · proposition_5_10`. The text's σ ("0 for $s \le 1$ and 1 for $s \ge 1$") is not a function.

V.9 The calculus of sections, and Lemma 5.11

$\Gamma^+(P)$ and its algebra. $\Gamma^+(P)$ is the set of sc⁺-sections (Definition 2.52) `Chapter5/BumpFunctions · StrongBundle.scPlusSections`; their values lie in $Y_{0,1}$, because in a chart their principal parts take values in F_1 `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.mem_oneSet`. The text uses, and does not state, the rules that make $\Gamma^+(P)$ a module over the sc-smooth functions. The zero section is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_fiberZero`. *The product rule:* for f sc-smooth and s sc⁺, the fibrewise product $f \cdot s$ `Chapter5/BumpFunctions · sectionSmul` is sc⁺: in a chart its principal part on level 1 is $(f \circ \varphi^{-1}) \cdot \hat{s}$, the product of an sc-smooth function and an sc-smooth map into F^1, sc-smooth by the product rule of the sc-calculus `Chapter3/ParamPerturbation · IsScSmooth.smul` `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.smul_left`. Sums are sc⁺, since principal parts add in a chart `Chapter5/BumpFunctions · StrongBundle.IsScPlusSection.add`, and by induction finite sums `Chapter5/BumpFunctions · sectionSum` `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_sectionSum`.

Cut-offs. Lemma 5.11 builds a section in one chart and "extends it by 0". For a chart Φ of the atlas, $\tau : O \to F_1$ with $z \mapsto (z, \tau(z))$ sc-smooth into $K[1]$, and β sc-smooth with $\operatorname{supp} \beta \subseteq \operatorname{dom} \varphi$, the section $y \mapsto \Phi^{-1}(\varphi(y), \beta(y)\tau(\varphi(y)))$ on $\operatorname{dom} \varphi$, 0_y outside `Chapter5/BumpFunctions · StrongBundle.cutoffSection`, is sc⁺ `Chapter5/BumpFunctions · StrongBundle.isScPlusSection_cutoffSection`: on $\operatorname{dom} \varphi$ its representative in $\Phi[1]$ is $z \mapsto (z, \beta(\varphi^{-1}(z))\tau(z))$, sc-smooth by the product rule; on the open set $X \setminus \operatorname{supp} \beta$ it is the zero section; the two open sets cover X.

Lemma 5.11 `Chapter5/BumpFunctions · lemma_5_11`. Let X admit bump functions, N be an auxiliary norm, x smooth, e a smooth point of Y over x, $\varepsilon > 0$ and $U \ni x$ open. Take any chart Φ of the atlas around x — neither a centred chart nor tameness is needed — and $\hat{e} = \Phi(e)_2$, a smooth point of F because e is smooth. The constant $\hat{e}$ is sc-smooth into F^1 `Chapter5/BumpFunctions · isScSmooth_const_of_mem_smooth`, and

$R[1]$ is sc-smooth, so $t(z) = R(z, \hat{e})$ gives an sc⁺-section near x with $t(x) = e$. Since N is continuous on $Y[1]$ and t is continuous into $Y[1]$, the set $\{y \mid N(t(y)) < N(e) + \varepsilon\}$ is an open neighbourhood of x. Proposition 5.30 gives $\beta : X \to [0,1]$ with $\beta(x) = 1$ and support in that set, in U and in the chart domain, and the cut-off $s = \beta \cdot t$ is sc⁺ with $s(x) = e$ and $\operatorname{supp} s \subseteq U$. Finally, at every y, $N(s(y)) = \beta(y)N(t(y)) \le N(t(y)) < N(e) + \varepsilon$, because N is a norm on the fibre and $0 \le \beta \le 1$; off the support, $N(s(y)) = N(0_y) = 0$. The text states the bound only at x, where it is trivial; its proof gives, and §5.3 needs, the bound everywhere.

V.10 Allowable sections and Remark 5.12

$\Gamma^{+,1}_U(P)$. For an auxiliary norm N and an open U, the allowable sections are the sc⁺-sections with support in U and $N(s(x)) \le 1$ at every point `Chapter5/BumpFunctions · StrongBundle.allowableSections` — a condition at every point because sc⁺-sections take values in $Y_{0,1}$. The text's metric $\rho(s, s') = \sup_x N(s(x) - s'(x))$ is used through its pointwise form: $N(s(x) - s'(x))$, the difference taken in the fibre $(Y_{0,1})_x$ `Chapter5/Transversality · StrongBundle.sectionDistAt`. The metric space itself is §V.30.

Remark 5.12 `Chapter5/Compactness · remark_5_12`. Let $S_f \subseteq U$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact (the U of properness), and let s vanish outside U with $N(s(x)) \le 1$ everywhere — properties that pass to uniform limits of allowable sections, which is all the remark uses of the completion. At a zero x of $f + s$: if $x \notin U$, then $s(x) = 0$ and $f(x) = 0$, so $x \in S_f \subseteq U$, a contradiction; so $x \in U$, and $f(x) = -s(x)$ gives $N(f(x)) = N(s(x)) \le 1$ `Chapter5/Compactness · StrongBundle.IsAuxNorm.auxLe_of_sectionAdd`. So $S_{f+s} \subseteq \{x \in U \mid N(f(x)) \le 1\}$, whose closure is compact. The remark claims the solution set compact, which needs it closed; for an allowable s this holds because $f + s$ is sc-Fredholm (Theorem 3.15) `Chapter3/FredholmSectionsG · theorem_3_15` and solution sets of continuous sections are closed `Chapter5/Transversality · isClosed_solutionSet`. This is how part (1) of Theorems 5.13, 5.18, 5.19, 5.25 is proved: U from Theorem 5.5, then Remark 5.12 and closedness.

V.11 Transversality and the parametrized section

Definition 5.15. g is transversal if at every zero its linearization is onto in a chart `Chapter5/Transversality · StrongBundle.IsTransversal`, hence in every chart `Chapter3/GlobalIFTIndex · StrongBundle.IsLinSurjAt.linAt_surjective` `Chapter3/LinearizationCharts · StrongBundle.linAt_chart_independence`. The kernel of $\mathrm{Lin}(z)$ in a chart is a subspace of E_0 inside $T_z O$ `Chapter5/Transversality · StrongBundleChart.linKer`.

The parametrized section. For a section g and sections $t_1, \dots, t_m$, Theorem 3.18 (Part III) makes $F(\lambda, y) = g(y) + \sum \lambda_i t_i(y)$ a section of the pull-back of P to $\mathbb{R}^m \times X$, sc-Fredholm when g is and the t_i are sc⁺, and with tame germ data when g has them `Chapter3/ParamPerturbationG · theorem_3_18_tame`. Its *slice* at λ is the section $g + \sum \lambda_i t_i$ of P `Chapter5/Transversality · paramSlice` `Chapter5/Transversality · paramSlice_eq_sectionAdd`, and (λ, y) is a zero of F iff y is a zero of the slice `Chapter5/Transversality · isZeroAt_paramSection_iff`. The slices are sc-Fredholm, since $\sum \lambda_i t_i$ is sc⁺ (§V.9, Theorem 3.15) `Chapter5/Transversality · isScFredholmSection_paramSlice`; slices of sc⁺-sections are sc⁺ `Chapter5/Transversality · isScPlusSection_paramSlice` with support in the union of the supports `Chapter5/Transversality · sectionSupport_paramSlice_subset` and vanish off it `Chapter5/Transversality · isZeroAt_paramSlice_of_not_mem_support`. *Allowability:* if $N(s_0) \le 1 - \varepsilon$ and $N(t_i) \le 1$, the triangle inequality on the fibre gives $N(s_0 + \sum \lambda_i t_i) \le 1 - \varepsilon + m|\lambda|$ `Chapter5/Transversality · StrongBundle.IsAuxNorm.toOne_sum_le`, which is ≤ 1 when $m|\lambda| \le \varepsilon$ `Chapter5/Transversality · auxLe_paramSlice` — the text's "$N(\sum \lambda_j t_j(y)) < \varepsilon/2$", with $|\lambda|$ the maximum norm.

The linear algebra of "regular value ⇔ transversal". Let $M(a, u) = Lu + \sum a_i e_i : \mathbb{R}^m \oplus T \to K$ be onto. If the projection $\ker M \to \mathbb{R}^m$ is onto, L is onto: given $k = Lu + \sum a_i e_i$, choose u' with $(a, u') \in \ker M$, so $k = L(u - u')$ `Chapter5/Transversality · surjective_of_forall_exists_ker_fst`. Conversely, if L is onto, every a lifts to $\ker M$ `Chapter5/Transversality · forall_exists_ker_fst_of_surjective`. In the prod-

uct chart $(r \mapsto r - \lambda) \times \Phi$ of a chart Φ centred at y, the linearization of F at (λ, y) is $\mathrm{Lin}_F(0)(a,u) = \mathrm{Lin}_{g_\lambda}(0)u + \sum a_i \hat{t}_i(0)$ `Chapter3/ParamPerturbationG · linAt_paramSection`, the fibre of the product chart over its centre being the fibre K_0 of Φ `Chapter5/Transversality · mem_K_ofProd_zero_iff`. So if Lin_F is onto at (λ, y) and its kernel projects onto $\mathbb{R}^m$, the slice g_λ is transversal at y `Chapter5/Transversality · isLinSurjAt_paramSlice_of_ker_fst`. In the other direction, if in a chart centred at a zero y of g the range of $\mathrm{Lin}_g(0)$ and the $\hat{t}_i(0)$ span K_0, then F is transversal at $(0, y)$ `Chapter3/ParamPerturbationG · linAt_paramSection_zero` `Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_spanning`; and adding sections keeps this, because a combination over a subfamily is one over the family `Chapter5/Transversality · exists_sum_comp_eq` `Chapter5/Transversality · isLinSurjAt_paramSection_zero_of_comp`. Only "regular value ⇒ transversal" is needed in Theorem 5.13.

V.12 Spanning sections

"For every $x \in S_{f+s_0}$ we find finitely many allowable sc⁺-sections such that the range of $(f + s_0)'(x)$ together with the sections span Y_x": the text does not say how. Two ingredients.

Linear algebra `Chapter5/Transversality · exists_fin_mem_of_dense_of_closedComplemented`. Let $R \subseteq K$ be closed, complemented and of finite codimension, and $D \subseteq K$ a dense subspace. With π a continuous projection onto R, $Q = \ker \pi$ is finite-dimensional, and $(1 - \pi)(D)$ is a dense subspace of Q, hence closed, hence all of Q; a basis of Q lifts to finitely many $d_i \in D$, and $K = R + \mathrm{span}\{d_i\}$.

The sections `Chapter5/Transversality · exists_spanning_sections`. Let g be sc-Fredholm, $g(y) = 0, U \ni y$ open. In the chart Φ of the germ, centred at y, $\mathrm{Lin}(0) : T_0 O \to K_0$ is $Dg(0)|_{T_0 O}$ for a derivative family Dg of the filling germ on its germ (the bridge for germs at the centre, `Chapter3/LinearizationChartsG · IsFillingGerm.fillerRestrict_eq_linAt`), hence Fredholm (Proposition 3.52 for germs, which exposes the germ and Dg) `Chapter3/FredholmGermLinearizationG · proposition_3_52`, so its range R is closed, complemented and of finite codimension in $K_0 = \rho(0)F$. The smooth vectors $\rho(0)F_\infty$ are dense in K_0 — $\rho(0)$ is continuous and preserves the levels at the smooth point 0, and F_∞ is dense — so finitely many smooth k_i span a complement. Each k_i is $\Phi(e_i)_2$ for a smooth point e_i of the fibre over y `Chapter5/Transversality · mem_smoothSet_toMPolyfold0_of_chart`; Lemma 5.11 gives s_i with $s_i(y) = e_i$, support in U and $N(s_i) < N(e_i) + 1$, and $s_i/(N(e_i) + 1)$ is allowable, with support unchanged `Chapter5/Transversality · sectionSupport_sectionSmul_const_subset`, N scaled `Chapter5/Transversality · StrongBundle.IsAuxNorm.oneMk_sectionSmul_const` and principal part scaled `Chapter5/Transversality · principal_sectionSmul_const`. The density of the smooth vectors is the point the text omits: Lemma 5.11 only reaches smooth points of the fibre.

V.13 Theorem 5.13: openness, and the omitted compactness step

No boundary on $\mathbb{R}^m \times X$. "The interior case, Theorem 3.19" is applied to F on $\mathbb{R}^m \times X$, so $\partial(\mathbb{R}^m \times X) = \emptyset$ is needed: $d_{\mathbb{R}^m} \equiv 0$ `Chapter5/Transversality · rnPolyfold_degIndex` and $d_{\mathbb{R}^m \times X} \leq d_{\mathbb{R}^m} + d_X$ `Chapter2/FibredProduct · MPolyfold.degIndex_prod_le` `Chapter5/Transversality · isWithoutBoundary_prod_rn`. At a point with $d = 0$ good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`.

Transversality spreads. If g has tame germ data and F is transversal at $(0, x)$, x a zero of g, then $(F, (0, x))$ is in good position, and Theorem 3.54 (1) gives an open $V \ni (0, x)$ all of whose zeros are in good position, hence transversal; $U(x) = \{y \mid (0, y) \in V\}$ is the text's neighbourhood `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'` `Chapter5/Transversality · exists_isOpen_isLinSurjAt_zero_near`. Likewise, if F is transversal at every $(0, y)$, $y \in S_g$, the union of these sets is an open $V \supseteq \{0\} \times S_g$ all of whose zeros of F are in good position in a tame chart `Chapter5/Transversality · exists_isOpen_goodPosition_of_isLinSurjAt`.

The omitted step. The text passes from the neighbourhood U of $\{0\} \times S_{f+s_0}$ to "$U \subset \{|\lambda| < \delta_0\} \oplus X$" without argument; what is needed is that *every* zero of F with $|\lambda|$ small lies in V. The topological core: if

$Z \subseteq \mathbb{R}^m \times X$ is compact and the open V contains $Z \cap \{\lambda = 0\}$, then $Z \setminus V$ is compact, its projection to $\mathbb{R}^m$ is compact and misses 0, and any δ below its distance to 0 works `Chapter5/Transversality · exists_pos_forall_mem_of_isCompact`. The compact Z is the set of zeros of F with $|\lambda| \le \delta_1$: for such λ the perturbation $s_0 + \sum \lambda_i t_i$ is allowable (§V.11), so by Remark 5.12 its zeros lie in the compact $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ of Theorem 5.5 — Z is a closed subset of $\bar{B}(\delta_1) \times$ (that compact set), closed because F is continuous.

V.14 Theorem 5.13: the Sard step

"$\tilde{S} := F^{-1}(0)$ is a smooth manifold"; "taking a regular value λ_0 for the projection $\tilde{S} \to \mathbb{R}^m$, it is easily verified that $f + s_0 + s(\lambda_0, \cdot)$ is transversal". The formalization does not build $\tilde{S}$; it proves the needed statement near each zero `Chapter5/Transversality · exists_null_isLinSurjAt_paramSlice_near`.

Claim. Let g have tame germ data, t_i be sc$^+$, and z_0 a zero of F in good position in a tame chart. There are an open $W \ni z_0$ and a null set $CV \subseteq \mathbb{R}^m$ such that for every zero $(\lambda, y) \in W$ with $\lambda \notin CV$ the slice g_λ is transversal at y.

1. *The chart of $\tilde{S}$.* Theorem 3.54 (2) gives at z_0 a regular tangent germ chart D of the solution set of F (Definition 3.25 (2)): a chart $D.\varphi$ of $\mathbb{R}^m \times X$ with domain W, a finite-dimensional N, a relatively open $V \subseteq N \cap C$ and an sc-smooth graph map Γ with $D.\varphi^{-1} \circ \Gamma$ parametrizing $\tilde{S} \cap W$ `Chapter3/GlobalIFTTheoremsG · theorem_3_54_two`.
2. *The projection.* $q = p \circ D.\varphi^{-1} \circ \Gamma : V \to \mathbb{R}^m$, p the projection to the parameter, is sc-smooth on a finite-dimensional space with the constant structure, hence C^∞ `Chapter1/ExpositionLemmas · isScSmooth_iff_contDiffOn_const`.
3. *Sard on the strata.* V is relatively open in a partial quadrant of N; Sard's theorem on each stratum of the quadrant (Part IV) makes the set CV of stratified critical values of q null `Chapter4/FamiliesLocal · measure_image_stratCritSet`. Since $\partial X = \emptyset$ the points over the zeros are interior and classical Sard would do, but proving that needs the degeneracy index of the induced structure; the stratified form makes the question unnecessary.
4. *The point.* A zero $(\lambda, y) \in W$ is $D.\varphi^{-1}(\Gamma(v))$ for some $v \in V$ with $q(v) = \lambda$. Let Φ be a chart centred at y and $\psi = (r \mapsto r - \lambda) \times \varphi$ the product chart. On a relatively open $V' \subseteq V$ around v, $\mathrm{gmap} = \psi \circ D.\varphi^{-1} \circ \Gamma$ is sc-smooth (the transition between the two charts of $\mathbb{R}^m \times X$ restricted to the overlap), and $q = \mathrm{pr}_1 \circ \mathrm{gmap} + \lambda$ there.
5. *Kernel vectors.* If $\lambda \notin CV$, $Dq(v)$ maps the stratum through v onto $\mathbb{R}^m$. The values of $D\,\mathrm{gmap}(v)$ lie in the tangent space $T_0 O$ of the product chart, and in $\ker \mathrm{Lin}_F(0)$: the principal part of F vanishes along $r \circ \mathrm{gmap}$, and the chain rule gives $\mathrm{Lin}_F(0) \circ D\,\mathrm{gmap}(v) = 0$. So for every $a \in \mathbb{R}^m$ there is a kernel vector with first component a.
6. *Conclusion.* F is transversal at (λ, y) (good position near z_0), and §V.11 gives the transversality of the slice at y.

Only range $D\,\mathrm{gmap}(v) \subseteq \ker \mathrm{Lin}_F$ is used, not "$T\tilde{S} = \ker F'$".

V.15 Theorem 5.13: the assembly

The generic form `Chapter5/Transversality · theorem_5_13_generic`. Let $\partial X = \emptyset$, X admit bump functions, f have tame germ data and compact solution set, N be an auxiliary norm.

1. *U and part (1).* Theorem 5.5 gives $U \supseteq S_f$ with $\mathrm{cl}\{x \in U \mid N(f(x)) \le 1\}$ compact `Chapter5/Compactness · theorem_5_5`; for allowable s, Remark 5.12 and closedness give S_{f+s} compact (§V.10).
2. *The section $g = f + s_0$.* For s_0 sc$^+$ with $\mathrm{supp}\, s_0 \subseteq U$ and $N(s_0) \le 1 - \varepsilon$, g has tame germ data `Chapter3/TameGermsAddG · theorem_3_15_tame`, a compact solution set (part (1)), and $S_g \subseteq U$ (Remark 5.12).

3. *Local families.* At every zero x of g, spanning sections $s_1^x, \dots, s_{k_x}^x$ supported in U (§V.12) make F_x transversal at $(0,x)$, and at $(0,y)$ for the zeros y in an open $U(x)$ (§V.13).
4. *One family.* Finitely many $U(x_j)$ cover the compact S_g; the families are united through an equivalence $\Sigma_j \mathrm{Fin}(k_{x_j}) \simeq \mathrm{Fin}(m)$ into $t_1, \dots, t_m$, allowable and supported in U; adding sections keeps transversality (§V.11), so F, whose slice at $\lambda = 0$ is g `Chapter5/Transversality · paramSlice_zero`, is transversal at every $(0,y)$, $y \in S_g$.
5. *Good position near the zero slice.* An open $V \supseteq \{0\} \times S_g$ of zeros of F in good position (§V.13).
6. *Allowability and compactness.* For $|\lambda| \le \delta_1 = \varepsilon/(m+1)$ the slices $s_0 + \sum \lambda_i t_i$ are allowable, and the zeros of F over $|\lambda| \le \delta_1$ form a compact Z (§V.13).
7. *The omitted step.* $Z \cap \{|\lambda| < \delta_0\} \subseteq V$ for some $\delta_0 > 0$.
8. *Sard.* With $\delta = \min(\delta_0, \delta_1)$, the zeros Z_0 of F over $|\lambda| \le \delta/2$ are compact and lie in V; the Sard step (§V.14) at each point of Z_0 gives an open W_z and a null CV_z; finitely many W_z cover Z_0, and $CV = \bigcup CV_z$ is null.
9. *The conclusions.* For $|\lambda| < \delta/2$, $\lambda \notin CV$: $s_\lambda = s_0 + \sum \lambda_i t_i$ is allowable; every zero y of $f + s_\lambda$ gives a zero $(\lambda, y) \in Z_0$, inside some W_z, so $f + s_\lambda$ is transversal at y; S_{f+s_λ} is compact by part (1); and if it is non-empty, Theorem 3.56 for a transversal section over X without boundary, applied to $f + s_\lambda$ with tame germ data, makes it a sub-M-polyfold whose induced structure is tame, a manifold without boundary of locally constant dimension, with regular tangent germ data of dimension the Fredholm index at every zero `Chapter3/GlobalIFTIndexG · theorem_3_56_of_surjective` `Chapter5/Transversality · StrongBundle.SolutionSetIsManifold`.

The dense form `Chapter5/Transversality · theorem_5_13`. Given an allowable s' and $\varepsilon > 0$, let $\varepsilon' = \min(\varepsilon, 1)$ and $s_0 = (1 - \varepsilon'/2)s'$, so $N(s_0) \le 1 - \varepsilon'/2$. The generic form gives t, δ, CV; since CV is null and the ball $\{|\lambda| < \min(\delta, \varepsilon'/(2m+2))\}$ has positive measure, some λ in it avoids CV. Then $s = s_0 + \sum \lambda_i t_i$ has $N(s(x) - s'(x)) \le (\varepsilon'/2) N(s'(x)) + \sum |\lambda_i| N(t_i(x)) \le \varepsilon'$. The text's "$N(s_0(y) + s(\lambda_0, y)) \le 1 - \varepsilon/2$" is a slip for ≤ 1, its "$N(f(x)) \le 1$" in the statement is $N(s(x)) \le 1$, and its "the germ (f,x) is in good position" is $(f+s, x)$.

V.16 Definition 5.17 and Lemma 5.26

Definition 5.17, in a tame chart `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm`. (f,y), $f(y) = 0$, is in general position if in some tame strong bundle chart around y (Definition 3.50) `Chapter3/TameGerms · StrongBundle.IsTameChart`, at the level-1 point z over $\varphi(y)$, $\mathrm{Lin}(z)$ is onto K_z, and for every presentation T of the partial quadrant the kernel has an sc-complement B in T_zO inside the reduced tangent space $T_z^R O = T_zO \cap E_z$ `Chapter2/Tame · reducedTangentSpaceWith`, on every level: $(T_zO)_m = (\ker \mathrm{Lin}(z))_m \oplus B_m$. The chart is tame because $T_x^R X$ is the reduced tangent space of a tame M-polyfold and the dimension count below needs it. General position includes transversality `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isLinSurjAt`.

The linear algebra of Lemma 5.26 `Chapter5/Transversality · isInGoodPositionIn_of_scComplement_le_reduced`. Let r be a tame retraction, z a smooth point of the retract, and $N, B \subseteq T_zO$ with $T_zO = N \oplus B$ and $B \subseteq T_z^R O$. Then N is in good position to the cone C_z inside T_zO (Definition 3.21): B is a good complement, because adding a vector of E_z does not change membership in C_z (the coordinates λ_i, $i \in I(z)$, vanish on E_z); and $N \cap C_z$ has non-empty interior in N, because the coordinates λ_i, $i \in I(z)$, map T_zO onto $\mathbb{R}^{I(z)}$ — $\dim T_zO/T_z^R O = d_C(z)$ in a tame retract (Proposition 2.36) — and vanish on B, so they map N onto $\mathbb{R}^{I(z)}$, and a vector of N with all these coordinates equal to 1 is interior. The text omits the interior condition; this is the argument of Part IV for tangent spaces `Chapter4/LocalStructure · IsGeneralPositionRetractAt.isGoodPositionRetractAt`, for a subspace not given as one.

Lemma 5.26 `Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGerm`. For f sc-Fredholm and (f,y) in general position: in the chart of Definition 5.17 $\mathrm{Lin}(z)$ is onto and its kernel is in good position by the linear algebra; Definition 3.53 then follows `Chapter3/`

LinearizationChartsG · StrongBundle.isGoodPositionGerm_of_linAt_surj. For f with tame germ data, good position holds in the tame chart of the germ data Chapter3/TameGermsAddG · StrongBundle.IsGoodPositionGerm.toTame Chapter5/Transversality · StrongBundle.IsGeneralPositionGerm.isGoodPositionGermTame — the form the §3.6 theorems take. X itself need not be tame. (The text places the lemma after Theorem 5.25 and calls it "clear" after Theorem 5.18; it is used in Theorems 5.18 and 5.25 here.)

V.17 The bridge lemma, and $T_yS = \ker f'(y)$

Theorem 5.18 applies Theorem 4.23 to the parametrized solution set $\tilde{S}$, which needs $\tilde{S}$ to be a smooth submanifold in the sense of Definition 4.13 — locally the image of an sc⁺-retraction. The implicit function theorem gives tangent germ data (Definition 3.25), and the text never connects the two.

sc⁺ on a finite-dimensional domain Chapter5/GeneralPosition · isScPlus_of_isScSmooth_const. On a finite-dimensional space with the constant structure all levels of the domain coincide, so an sc-smooth f is C^∞ into every F_m, and its lift into F^1 is sc-smooth: f is sc⁺.

The bridge lemma. Let D be regular tangent germ data for M at x (Definition 3.25 (2)), with graph map $\Gamma : V \to E$ over the finite-dimensional N, projection π onto N, and r a retraction onto the chart model. Then Γ is sc⁺ on V Chapter5/GeneralPosition · MPolyfold.TangentGermChart.Γ_isScPlus, so $\tau = \Gamma \circ \pi \circ r$, on the set of points whose image under $\pi \circ r$ lies in V, is an sc-smooth retraction (as in the proof of Theorem 3.27) Chapter3/TangentGerm · theorem_3_27 which is sc⁺ (sc⁺ □ sc-smooth), with $\tau \circ \Gamma = \Gamma$ on V and retract the chart image of M Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_scPlusRetraction. Transported to X it is an sc⁺-retraction on an open set with image $M \cap V$ Chapter4/LocalStructure · MPolyfold.isScPlusRetractionOn_of_chart Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_isScPlusRetractionOn. Hence a set with regular tangent germ data at each point is a smooth finite-dimensional submanifold Chapter5/GeneralPosition · MPolyfold.isSmoothSubmanifold_of_isRegular. For solution sets: if all zeros of f in an open W are in good position in a tame chart, Theorem 3.54 (2) gives regular data at each, on charts inside W, and $S_f \cap W$ is a smooth submanifold Chapter5/GeneralPosition · StrongBundle.isSmoothSubmanifold_solutionSet_inter.

$T_yS \subseteq \ker f'(y)$ Chapter5/GeneralPosition · tangentSpace_subset_linKer. In a chart, let $Q \subseteq O$ be a retract on which the principal part $\hat{f}$ vanishes, S a retraction onto Q, z a level-1 point of Q. Then $T_zQ = \operatorname{range} DS(z) \subseteq T_zO$, and $\hat{f} \circ r \circ S = 0$ gives $D(\hat{f} \circ r)(z) \circ DS(z) = 0$ by the chain rule, i.e. $T_zQ \subseteq \ker \mathrm{Lin}(z)$ Chapter5/GeneralPosition · StrongBundleChart.mem_linKer_iff.

Equality, by counting Chapter5/GeneralPosition · exists_tangentSpace_eq_linKer. Let f have tame germ data and (f, x_0) be a zero in good position in a tame chart. The index form of Theorem 3.54 gives an open $V \ni x_0$ whose zeros are in good position, and regular data D of a fixed dimension d at each zero, with $\dim \ker \mathrm{Lin}(z) = d$ in *every* chart Chapter3/GlobalIFTIndexG · theorem_3_54_index Chapter3/GlobalIFTIndex · StrongBundle.IsSurjLinOfIndex. On the other side, the tangent space of the bridge retraction τ at the point over $\Gamma(v)$ contains the d vectors $D\Gamma(v)b_i$, b_i a basis of N: $D\tau \circ D\Gamma(v) = D(\tau \circ \Gamma)(v) = D\Gamma(v)$, and $\pi \circ D\Gamma(v) = 1$ makes them independent Chapter5/GeneralPosition · MPolyfold.TangentGermChart.exists_linearIndependent_tangentSpace. "Having d independent tangent vectors" Chapter5/GeneralPosition · HasIndepTangent is carried by germs of sc-diffeomorphisms, since $D(F \circ r)(z)$ maps T_zQ into $T_{F(z)}Q'$ and is injective on T_zO Chapter5/GeneralPosition · HasIndepTangent.transport, so it passes from the chart of D to any chart Φ through the transition maps Chapter4/Submanifolds · MPolyfold.chart_change_of_transport. A d-dimensional subspace of the d-dimensional $\ker \mathrm{Lin}(z)$ — tangent spaces of retracts are germs, so the retract $\varphi(G \cap S_f \cap V)$ of any small open G has the same tangent space — is all of it.

V.18 General position: chart independence and openness

Definition 5.17 is Definition 4.21 `Chapter5/GeneralPosition · exists_isGeneralPositionGerm_iff`. Near a zero x_0 in good position (with f of tame germ data) there is an open V such that every zero in V is in good position, $S = S_f \cap V$ is a smooth submanifold (§V.17), and at every zero $y \in V$: (f, y) is in general position iff S is in general position at y (Definition 4.21, Part IV). Since $T_z S = \ker \mathrm{Lin}(z)$ in every chart, an sc-complement of $\ker \mathrm{Lin}(z)$ inside $T^R_z O$ *is* an sc-complement of $T_z S$ inside $T^R_z O$. Definition 4.21 is chart independent (Part IV) `Chapter4/Submanifolds · MPolyfold.IsGeneralPositionAt.isGeneralPositionChart`, so Definition 5.17 is too, at these zeros.

From submanifold general position to good position `Chapter5/GeneralPosition · isGoodPositionGermTame_of_isGeneralPositionAt`. If $f'(y)$ is onto and a smooth submanifold $A \subseteq S_f$ is in general position at y, then in the tame chart of the germ data $T_z A \subseteq \ker \mathrm{Lin}(z)$ (§V.17) `Chapter5/GeneralPosition · StrongBundleChart.linKer_le` and $T_z A + T^R_z O = T_z O$; so $\ker \mathrm{Lin}(z) + T^R_z O = T_z O$, the kernel has an algebraic complement inside $T^R_z O$ `Chapter5/GeneralPosition · exists_compl_le_of_sup_eq`, and §V.16 gives good position.

Openness `Chapter5/GeneralPosition · exists_isOpen_isGeneralPositionGerm`. Let X be tame, f with tame germ data, (f, x_0) in general position. By Lemma 5.26 it is in good position; the previous paragraph turns its general position into general position of the submanifold $S_f \cap V$ at x_0; Theorem 4.22 (3) for tame X (Part IV) makes that open along the submanifold `Chapter4/LocalStructure · MPolyfold.theorem_4_22`; Definition 4.21 depends only on the germ of the submanifold, since tangent spaces of chart models agree on open sets `Chapter5/GeneralPosition · MPolyfold.IsGeneralPositionAt.congr_nhds`; and the equivalence turns it back into general position of (f, y) at the nearby zeros. The text uses this without proof ("they will also hold at $(0, y)$ for $y \in S_f$ close to x"; "the properties (i)–(ii) hold for all $(\lambda, y) \in \tilde{S}$").

V.19 Properties (i)–(ii)

In a chart centred at y, with a presentation T, the properties of the proof of Theorem 5.18 are `Chapter5/GeneralPosition · StrongBundleChart.IsParamGP`: (i) $\mathrm{Lin}_g(0)(T_0 O) + \mathrm{span}\{\hat{t}_i(0)\} = K_0$ — $F'(0, y)$ onto; (ii) every $w \in T_0 O$ is $u + e$ with $e \in E_0$ and $\mathrm{Lin}_g(0)u + \sum a_i \hat{t}_i(0) = 0$ for some a — $\ker F'(0, y)$ transversal to $\mathbb{R}^m \oplus T^R$. Adding sections keeps both, by linear algebra `Chapter5/GeneralPosition · StrongBundleChart.IsParamGP.comp`.

(i)–(ii) give general position `Chapter5/GeneralPosition · isGeneralPositionGerm_paramSection_of_isParamGP`. In the tame product chart $(r \mapsto r) \times \Phi$ of a tame chart centred at y `Chapter5/GeneralPosition · pc0`, whose tangent space at the centre is $\mathbb{R}^m \times T_0 O$ `Chapter5/GeneralPosition · ofProd_mem_rangeS` and whose reduced tangent space is $\mathbb{R}^m \times T^R_0 O$, the minimal subspace of $\mathbb{R}^m$ being everything `Chapter5/GeneralPosition · minimalSubspaceWith_rnPres`: (i) is the surjectivity of $F'(0, y)$; (ii) says $\ker F'(0, y) + T^R = T(\mathbb{R}^m \times O)$ on level 0, so the kernel has an algebraic complement inside T^R, and §V.16 gives good position of $(F, (0, y))$. *Then* $\ker F'(0, y) = T_{(0,y)} \tilde{S}$ (§V.17), which is finite-dimensional and smooth; a finite-dimensional smooth subspace has an sc-complement inside any sc-subspace it complements algebraically `Chapter4/LocalStructure · exists_scComplement_le`, which gives Definition 5.17 for F. The text's "at this point the combined system of sections already satisfies (i) and (ii)" compresses these two steps.

General position gives (i)–(ii) in every tame chart centred at y `Chapter5/GeneralPosition · isParamGP_of_isGeneralPositionGerm`: (i) is surjectivity read in the product chart; for (ii), $\tilde{S}$ is in general position at $(0, y)$ (§V.18), and in the product chart its tangent space is $\ker F'(0, y)$, which therefore has a complement inside $\mathbb{R}^m \times T^R$.

Achieving (i)–(ii) `Chapter5/GeneralPosition · exists_sections_isParamGP`. Spanning sections give (i) (§V.12) `Chapter5/GeneralPosition · spanning_of_isLinSurjAt`. For (ii), "take a linear subspace L complementing $T^R_x X$ in $T_x X$ and add sections which at x span $f'(x)L$": the sections must come from Lemma 5.11, which needs *smooth* points of the fibre, so L must consist of smooth vectors. The coordinates $\lambda_{I(0)}$ map $T_0 O$

onto $\mathbb{R}^{I(0)}$ with kernel $T_0^R O$ (Proposition 2.36), and already map the dense smooth vectors of $T_0 O$ onto it (a dense subspace of a finite-dimensional image is all of it); so there are smooth $l_i \in T_0 O$ with $\lambda(l_i) = e_i$. $\mathrm{Lin}_g(0) l_i$ is smooth (Lin is an sc-operator at the smooth point), Lemma 5.11 gives allowable sections through $\Phi^{-1}(0, \mathrm{Lin}_g(0) l_i)$ supported in U, and with these added every $w \in T_0 O$ is $u + e$: subtract $\sum \lambda_i(w) l_i$, whose image under $\mathrm{Lin}_g(0)$ is cancelled by the new sections.

V.20 Theorem 5.18

From a family `Chapter5/GeneralPosition · theorem_5_18_of_family`. Let X be tame, f with tame germ data and compact solution set, N auxiliary, U from Theorem 5.5. Fix s_0 with $\operatorname{supp} s_0 \subseteq U$, $N(s_0) \le 1 - \varepsilon$, and allowable $t_1, \dots, t_m$ with F in general position at every $(0, x)$, $x \in S_g$, $g = f + s_0$.

1. *General position near the zero slice.* $\mathbb{R}^m \times X$ is tame `Chapter3/ParamPerturbation · prod_rn_isTame` and F has tame germ data `Chapter3/ParamPerturbationG · theorem_3_18_tame`, so openness (§V.18) gives, around each $(0, x)$, an open set on which the zeros of F are in good and general position; their union $W \supseteq \{0\} \times S_g$.
2. *Compactness and the omitted step* as in §V.13: the zeros Z of F over $|\lambda| \le \delta_1$ are compact, and $Z \cap \{|\lambda| < \delta_0\} \subseteq W$.
3. *The submanifold.* $A = \tilde{S} \cap W \cap \{|\lambda| < \delta\}$ is a smooth finite-dimensional submanifold of $\mathbb{R}^m \times X$ (the bridge lemma, §V.17), in general position at each of its points (Definition 4.21, by §V.18 and locality), with compact closure (inside Z).
4. *Theorem 4.23.* In its countable-cover form (Part IV) `Chapter4/Families · theorem_4_23_of_subset`, it gives a set Σ of parameters with null complement over which the fibre A_λ is a smooth submanifold of X in general position.
5. *Transversality.* The Sard step of §V.14 on the compact zeros over $|\lambda| \le \delta/2$ gives a further null set off which the slices are transversal at their zeros.
6. *The conclusions for λ.* For $|\lambda| < \delta/2$ off both null sets, with $h = f + s_\lambda$: the fibre of A over λ is S_h; h is transversal; S_h is a submanifold in general position, so (h, y) is in good position (§V.18), and then in general position by the equivalence of §V.18 applied at y together with the locality of Definition 4.21; Theorem 4.18 gives the corner structure and tameness of S_h `Chapter4/Submanifolds · MPolyfold.theorem_4_18`, and Theorem 4.22 (2) gives $d_{S_h} = d_X$ on S_h `Chapter4/LocalStructure · MPolyfold.theorem_4_22`. These are the conclusions collected in `Chapter5/GeneralPosition · StrongBundle.SolutionSetInGeneralPosition`.

The generic form `Chapter5/GeneralPosition · theorem_5_18`. At every zero x of g choose a tame chart centred at x with a presentation; sections with (i)–(ii) there (§V.19) make F_x general at $(0, x)$, hence (openness, §V.18) at $(0, y)$ for the zeros y in an open $U(x)$. Finitely many $U(x_j)$ cover S_g; the united family t keeps (i)–(ii) at every $(0, y)$ — read (i)–(ii) off general position in the chart of x_j, add sections, go back — so F is in general position at every $(0, y)$, and the previous theorem applies. "Replacing $f + s_0$ by f and N by cN" is not needed; the proof keeps s_0.

The dense form `Chapter5/GeneralPosition · theorem_5_18_dense` is derived exactly as for Theorem 5.13 (§V.15).

The text's $\tilde{S}$ omits the equation $F = 0$, and its "$p^{-1}(\{|\lambda| \le \varepsilon\})$ is compact" is the omitted step of §V.13.

V.21 Theorem 5.25

`Chapter5/GeneralPosition · theorem_5_25`. Let X be tame with bump functions, f proper with tame germ data. Properness gives a compact solution set; Theorem 5.18 with $s_0 = 0$ gives U, t, δ and a null CV `Chapter5/GeneralPosition · StrongBundle.IsAuxNorm.fiberZero_eq` `Chapter5/GeneralPosition · sectionAdd_fiberZero`; the ball $\{|\lambda| < \delta\}$ has positive measure, so some λ avoids CV, and $s = \sum \lambda_i t_i$ is allowable with $f + s$ in general position at every zero and all the conclusions of Theorem 5.18; Lemma 5.26 turns

general into good position in the tame chart of the germ data (§V.16). The text's proof says "in view of Theorem 3.54 … in general position", which is the openness of §V.18 (Theorem 3.54 is about good position), and writes $x \in S_f$ for $x \in S_{f+s_0}$. Here U is the neighbourhood Theorem 5.18 produces; the text lets the reader choose "an associated" one.

V.22 Theorem 5.19

`Chapter5/GeneralPosition · theorem_5_19`. Let X admit bump functions, f have tame germ data and compact solution set, and (f, x) be in good position at the zeros $x \in \partial X$. The text's "as before" applies the implicit function theorem to F at the boundary zeros $(0, x)$, where all t_i vanish and $\ker F'(0,x) = \mathbb{R}^m \times \ker f'(x)$; this is in good position to $\mathbb{R}^m \times C_x$ only if $\ker f'(x) + E_x \supseteq T_xX$, a general position condition that good position does not imply ($\mathbb{R}(1,1)$ in $[0,\infty)^2$; checked on paper only). The formal proof avoids those points:

1. *Near the boundary zeros.* Theorem 3.54 (1) at each zero of f on ∂X gives an open set of zeros in good position `Chapter3/GlobalIFTNearbyOntoG · theorem_3_54_one'`; their union V_{bd}, and an open V_1 with $S_f \cap \partial X \subseteq V_1$ and $\operatorname{cl} V_1 \subseteq V_{bd}$.
2. *Away from them.* The zeros of f outside V_1 are not on ∂X, so interior ($d = 0$). Spanning sections at these zeros are taken with supports in $U \setminus (\partial X \cup \operatorname{cl} V_1)$, an open set containing them; finitely many families cover the compact $S_f \setminus V_1$, and F is onto at every $(0, y)$, $y \in S_f$ outside V_1.
3. *Good position of F off the boundary.* At the interior points good position is surjectivity `Chapter3/GlobalIFTIndexG · StrongBundle.isGoodPositionGerm_of_isLinSurjAt`; Theorem 3.54 (1) for F gives an open $W \supseteq \{0\} \times (S_f \setminus V_1)$ of zeros in good position, off $\mathbb{R}^m \times \partial X$.
4. *Compactness and Sard* as in §V.15 for the zeros of F over small λ outside $\mathbb{R}^m \times V_1$.
5. *The conclusions.* For small λ off the null set, $h = f + s_\lambda$. On V_1 all t_i vanish, so $h = f$ there, and the good position of (f, y) at the zeros in V_1 passes to (h, y) because good position depends only on the germ of the section `Chapter3/Chapter5BridgeG · StrongBundle.IsGoodPositionGermTame.congr_nhds`. Off V_1 the zeros of h are interior, h is transversal there by the Sard step, hence in good position. Theorem 3.57 makes S_h, if non-empty, a sub-M-polyfold whose induced structure is tame and a manifold with boundary with corners `Chapter3/GlobalIFTTheoremsG · theorem_3_57`.

$s_\lambda = \sum \lambda_i t_i$ has support disjoint from ∂X, so it vanishes on the open neighbourhood $X \setminus \operatorname{supp} s_\lambda$ of ∂X, and it is "arbitrarily small" as $\lambda \to 0$. The text's tameness of X is not used, and its "(f, x) in good position for every $x \in S_{f+s}$" is $(f + s, x)$. No dense form is stated.

V.23 Definition 5.20, and Lemmas 5.23 and 5.24

Definition 5.20 `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy`. X tame, f with tame germ data and compact solution set, $t : X \to \mathbb{R}$ sc-smooth with $t(X) = [0,1]$, and for every $\tau \in [0,1]$: $X_\tau = t^{-1}(\tau)$ a sub-M-polyfold (Definition 2.19) with tame induced structure, on which the restriction $f|X_\tau$ — the section $a \mapsto (a, f(a))$ of the pull-back of P along the inclusion `Chapter5/Homotopy · restrictSection` `Chapter3/ParamPerturbation · StrongBundle.pullback`, the inclusion being sc-smooth (Proposition 2.21) — has tame germ data. The text never says what $Y|X_t$ is; the pull-back of Proposition 2.54 is the strong bundle structure available. The end problems are the restrictions to $t^{-1}(0)$ and $t^{-1}(1)$; their solution sets are $S_f \cap X_\tau$ `Chapter5/Homotopy · isZeroAt_restrictSection_iff`, closed in the compact S_f, hence compact `Chapter5/Homotopy · StrongBundle.IsGenCompactHomotopy.isCompact_solutionSet_restrict`.

Lemma 5.23 and the hidden hypothesis. The proof of Theorem 5.22 fixes a critical point x of $t|S$ and says "since x is a smooth point, we find a one-dimensional smooth linear sc-subspace $Z \subset T_xX$ such that $T_xX = Z \oplus \ker(dt(x))$". For $dt(x) = 0$, $\ker dt(x) = T_xX$ and the only complement is 0: a non-zero complement exists only if $\ell \neq 0$ `Chapter5/Homotopy · ne_zero_of_isCompl_ker`. Theorem 5.22 does not assume that t is a submersion at the critical points of $t|S$, so this is a hidden hypothesis. When $\ell \neq 0$, some

vector e of any spanning set (e.g. the dense smooth vectors) has $\ell(e) \neq 0$, and $\mathbb{R}e$ complements $\ker \ell$ `Chapter5/Homotopy · exists_isCompl_ker_of_ne_zero`. Lemma 5.23 is then linear algebra: $\ell|_Z$ is an isomorphism onto $\mathbb{R}$, so $b = (\ell|_Z)^{-1} \circ \tau$ is the unique $b : S \to Z$ with $\ell \circ b = \tau$, and adding a with values in $\ker \ell$ does not change $\ell \circ b$ `Chapter5/Homotopy · lemma_5_23`.

The calculus. At a level-1 point z, $D(a \cdot w)(z)v = Da(z)v \cdot w(z) + a(z)Dw(z)v$ for sc-smooth $a : U \to \mathbb{R}$, $w : U \to F$ `Chapter5/Homotopy · scDeriv_smul_apply`, and $D(\ell(\rho(\cdot) - p))(z) = \ell \circ D\rho(z)$ `Chapter5/Homotopy · scDeriv_clm_sub_apply`. The function $g = \beta \cdot \ell(\varphi(\cdot) - \varphi(x))$ on $\operatorname{dom}\varphi$, 0 outside `Chapter5/Homotopy · chartLinCutoff`, is sc-smooth when β is with $\operatorname{supp}\beta \subseteq \operatorname{dom}\varphi$ `Chapter5/Homotopy · isScSmoothMap_chartLinCutoff`.

Lemma 5.24. *The construction* `Chapter5/Homotopy · exists_scPlusSection_linAt`: for a chart Φ around the smooth point x, a smooth point e over x, a continuous linear ℓ on the model space and $U \ni x$, let s_e be the sc⁺-section through e of Lemma 5.11, supported in $U \cap \operatorname{dom}\varphi$, β a bump with $\beta \equiv 1$ near x and support in the chart domain, and $s = g \cdot s_e$ with g as above. Then s is sc⁺ (§V.9), $s(x) = g(x)e = 0$ `Chapter5/Homotopy · principal_sectionSmul`, $\operatorname{supp} s \subseteq U$ `Chapter5/Homotopy · sectionSupport_sectionSmul_subset`, and in the chart, by the product rule, $\mathrm{Lin}_s(z)v = D(g\circ\varphi^{-1}\circ r)(z)v\cdot\hat{s}_e(\varphi(x)) + g(x)\mathrm{Lin}_{s_e}(z)v = \ell(Dr(z)v)\Phi(e)_2 = \ell(v)\Phi(e)_2$ for $v \in T_zO$, since $g \circ \varphi^{-1} = \ell(\cdot - \varphi(x))$ near $\varphi(x)$ `Chapter5/Homotopy · chartLinCutoff_of_mem` and $Dr(z) = 1$ on T_zO. *The lemma* `Chapter5/Homotopy · lemma_5_24`: for a finite-dimensional $K \subseteq T_zO$ (the text's $T_xS = \ker f'(x)$) and a linear $b : K \to T_zO$ with $\mathrm{Lin}_f(z)(bv) = \lambda(v)\Phi(e)_2$, extend $-\lambda$ (linear on the finite-dimensional K) to a continuous linear functional ℓ on the model space (Hahn–Banach) and apply the construction: $\mathrm{Lin}_f(z)(bv) + \mathrm{Lin}_s(z)v = \lambda(v)\Phi(e)_2 - \lambda(v)\Phi(e)_2 = 0$. The text's graph of S over T_xS, its complement L with $j : L \to \mathbb{R}$, and its case "if this operator is the zero operator we can take $s = 0$" are not needed. That $f'(x) \circ b_\tau$ has its image in the line of a *smooth* point is a hypothesis; the text's justification ("the image of any vector in T_xS belongs to Y_∞") is not formalized.

V.24 Bump functions in local models: Theorem 5.34 and Proposition 5.35

Definition 5.33 `Chapter5/BumpFunctions · LocalModel.HasBumpProperty`. For $x \in O$ and W open in E_0 with $\mathrm{cl}(O \cap W) \subseteq O$, an sc-smooth f on O with $f(x) \neq 0$ (as in Definition 5.9) and support in O inside W.

A retract is closed near its points `Chapter5/BumpFunctions · closure_inter_ball_subset`. If $B(z,r) \subseteq W'$ and $U = C \cap W'$ is the domain of a retraction ρ onto O, then $O \cap \mathrm{cl}\,B(z,r/2) \subseteq O$: a limit of points of O in the closed half ball lies in C (C closed) and in $B(z,r)$, hence in U, where ρ is continuous, and it is a fixed point of ρ.

Theorem 5.34 `Chapter5/BumpFunctions · theorem_5_34`. "□": every chart's model has the property — given x, W, a bump of X at $\varphi^{-1}(x)$ supported in $\varphi^{-1}(O \cap W)$, read in the chart. "□", "clearly" in the text, hides a support condition: a local bump f_0 with support in $U(x)$, $\mathrm{cl}_E(U(x)) \subseteq O$, gives $f = f_0 \circ \varphi$ on the chart domain, but $\mathrm{cl}_X\{f \neq 0\}$ may leave the domain. For $X = E$ a Hilbert space, the chart $\varphi(u) = u/(1 - |u|_0^2)$ from the open unit ball onto $O = E$, $U(x) = O$ and $f_0 \equiv 1$, the extension by 0 is the indicator of the ball. The proof chooses $U_1 \ni x$ with $\mathrm{cl}_X U_1 \subseteq U \cap \operatorname{dom}\varphi$ (regularity of X) and r with $B(\varphi(x), r) \cap O \subseteq \varphi(U_1)$, takes the local bump in $W = B(\varphi(x), r/2)$ — admissible by the previous paragraph — and extends $f_0 \circ \varphi$ by 0: its support lies in $\mathrm{cl}_X U_1 \subseteq U$, and it is sc-smooth because it vanishes on the open complement of its support and is sc-smooth in the chart (§V.8).

Proposition 5.35. If $\kappa\|v\|^2 \leq b(v,v)$ for a continuous bilinear b on E_0, then $f(v) = \tau(1 - b(v-x, v-x)/\rho)$ is C^∞ on E_0, equal to 1 at x and supported in a small ball; it is sc-smooth on E by Corollary 1.20 from the base level `Chapter1/Levels · IsScSmooth.of_contDiffOn_finiteDimensional`, and so is its composition with any retraction onto O `Chapter5/BumpFunctions · LocalModel.hasBumpProperty_of_quadratic`. "E_0 is a Hilbert space" is stated as $\|v\|^2 = b(v,v)$ `Chapter5/BumpFunctions · proposition_5_35`: an inner product structure on E_0 would be a second normed structure next to that of the sc-scale. Translating by a non-smooth x is legitimate here

because f is C^∞ on E_0 itself; for a merely sc-smooth function it would not be. With Theorem 5.34: an M-polyfold covered by charts on sc-Hilbert spaces admits sc-smooth bump functions, the remark before Definition 5.9 `Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert`.

V.25 Partitions of unity: Theorem 5.37 and Proposition 5.38

The algebra of sc-smooth functions. Sums, differences and products of sc-smooth functions are sc-smooth, on retracts `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_add` `Chapter5/PartitionsOfUnity · IsScSmoothRetractMap.real_mul` and on X `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_add` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_sub` `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothMap.real_mul`, and so are finite sums and products `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetSum` `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsetProd`. Sc-smoothness is local `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_eventuallyEq`, so a locally finite sum, which near each point is a finite sum, is sc-smooth `Chapter5/PartitionsOfUnity · isScSmoothMap_real_finsum`; and a function supported in a chart domain with sc-smooth representative is sc-smooth on X `Chapter5/PartitionsOfUnity · isScSmoothMap_real_of_chart`.

Definitions 5.28 and 5.36. A partition of unity is Mathlib's (continuous $\rho_i \geq 0$, locally finite supports, $\sum \rho_i = 1$), sc-smooth if every ρ_i is `Chapter5/PartitionsOfUnity · MPolyfold.IsScSmoothPartitionOfUnity`; X admits sc-smooth partitions of unity if every open cover has a subordinate one `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity`. The approximation property: for $V = O \cap W$ with $\mathrm{cl}\, V \subseteq O$, continuous $f : O \to [0,1]$ supported in V and $\varepsilon > 0$, an sc-smooth $g : O \to [0,1]$ supported in V with $|f - g| \leq \varepsilon$ `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty`. Partitions of unity give bump functions: the function attached to U of a partition subordinate to $\{U, X \setminus \{x\}\}$ is 1 at x `Chapter5/PartitionsOfUnity · MPolyfold.AdmitsScPartitionsOfUnity.admitsBumpFunctions` — the easy half of the open problem the text states after Definition 5.29.

"(i) ⇒ (ii)" `Chapter5/PartitionsOfUnity · hasApproxProperty_restrict`. The text calls $f \circ \varphi$ "sc-smooth" (f is only continuous) and asserts $\mathrm{cl}_X(\varphi^{-1}(V)) \subseteq U$; with the chart of the unit ball above, $V = O$ and $f \equiv 1$, the extension by 0 is not continuous. Since (ii) asks only for *some* atlas, the proof restricts each chart to an open s with $\mathrm{cl}_X(s) \subseteq \mathrm{dom}\, \varphi$. Then for closed $A \subseteq \varphi(s)$ the set $\varphi^{-1}(A)$ is closed in X `Chapter5/PartitionsOfUnity · MChart.isClosed_inter_preimage`, so $F = f \circ \varphi$ extended by 0 is continuous. Cover X by $\{F < \varepsilon/2\}$ and the sets $U_x = \{|F - F(x)| < \varepsilon/2\} \cap \varphi^{-1}(V)$, take a subordinate sc-smooth partition of unity (β_0, β_x), and let $\hat{g} = \sum_x \beta_x F(x)$: a locally finite sum of sc-smooth functions with values in $[0,1]$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.finsum_mul_mem_Icc`, supported in $\varphi^{-1}(V)$ `Chapter5/PartitionsOfUnity · PartitionOfUnity.tsupport_finsum_mul_subset`, and within $\varepsilon/2$ of F, since every $F(x)$ with $\beta_x(y) \neq 0$ is `Chapter5/PartitionsOfUnity · PartitionOfUnity.abs_sub_finsum_mul_le`. Whether *every* chart has the property is not settled.

"(ii) ⇒ (i)". Every point of a chart domain has small open neighbourhoods N with $\mathrm{cl}_X N \subseteq \mathrm{dom}\, \varphi$ and $\mathrm{cl}\, \varphi(N) \subseteq O$ — half a ball inside the domain of a retraction, closed by §V.24 `Chapter5/PartitionsOfUnity · MChart.exists_small_nhds` — and the approximation property transports to continuous functions on X supported in such N `Chapter5/PartitionsOfUnity · LocalModel.HasApproxProperty.exists_approx`. *Sc-smooth Urysohn functions* `Chapter5/PartitionsOfUnity · exists_isScSmoothMap_urysohn`: for disjoint closed s, t, cover t by small neighbourhoods W_x inside $X \setminus s$, take continuous $f_x = 1$ near x supported in W_x, a locally finite refinement, approximations $\hat{g}_x$ within $1/2$ — so $\hat{g}_x \geq 1/2$ where $f_x = 1$ — and $F = \tau(4 \sum \hat{g}_x - 1)$, sc-smooth, 0 on s and 1 on t. Then Mathlib's bump coverings produce a covering indexed by the given cover with sc-smooth functions `Mathlib/Topology/PartitionOfUnity · BumpCovering.exists_isSubordinate_of_prop`, and $\rho_i = f_i \prod_{j<i}(1 - f_j)$, locally a finite product, is the partition of unity `Mathlib/Topology/PartitionOfUnity · BumpCovering.toPartitionOfUnity` `Chapter5/PartitionsOfUnity · admitsScPartitionsOfUnity_of_approx`. This replaces the text's refinement "with the same index set", which does not exist in general (the cover $\{X\}$); metrizability is not needed.

Theorem 5.37 `Chapter5/PartitionsOfUnity · theorem_5_37` combines the two directions, "atlas" read as a family

of charts covering X.

Proposition 5.38 Chapter5/PartitionsOfUnity · proposition_5_38 is not "an immediate consequence of Theorem 5.37": that theorem produces some atlas, not the given model, and applied to O would need partitions of unity on an open subset of C. The proof repeats "(i) □ (ii)" in E_0: $A = \{x \in O \mid f(x) \geq \varepsilon/2\}$ is closed in E_0 because $\operatorname{cl} V \subseteq O$; $E_0 \setminus A$ and balls $B(x, r_x) \subseteq W$, $x \in V$, on which f varies by less than $\varepsilon/2$, cover E_0; a C^∞ partition of unity of E_0 gives $g = \sum \beta_x f(x)$, C^∞ on E_0, hence sc-smooth (Corollary 1.20), composed with a retraction onto O.

V.26 Banach space geometry: Definitions 5.31, 5.39, 5.41, Proposition 5.42, Corollary 5.44

A Banach space admits a C^k-bump function if some C^k function, not identically zero, has bounded support Chapter5/PartitionsOfUnity · HasContDiffBump, and C^k partitions of unity if every open cover has a subordinate C^k one Chapter5/PartitionsOfUnity · AdmitsContDiffPartitionsOfUnity. $c_0(\Gamma)$ is the subspace of $\ell^\infty(\Gamma)$ of functions tending to 0 along the cofinite filter Chapter5/PartitionsOfUnity · cZero, equivalently with $\{|f(\gamma)| > \varepsilon\}$ finite for every ε Chapter5/PartitionsOfUnity · mem_cZero_iff, normed by the supremum Chapter5/PartitionsOfUnity · norm_cZero; coordinate-wise C^k embeddings are as in the text Chapter5/PartitionsOfUnity · IsCoordinatewiseContDiffEmbedding, and Toruńczyk's Theorem 5.40 is recorded as a statement, not proved and not used Chapter5/PartitionsOfUnity · TorunczykStatement. WCG means a weakly compact set with dense span Chapter5/PartitionsOfUnity · IsWCG; reflexive, used in Proposition 5.42 and not defined in the text, means $V \to V''$ onto Chapter5/PartitionsOfUnity · IsReflexive.

Proposition 5.42. (i) For reflexive V the weak closure of the closed unit ball is weakly compact — Kakutani's theorem, in Mathlib as Banach–Alaoglu in V'' through the embedding of (V, weak) into $(V'', \text{weak-*})$ Mathlib/Analysis/Normed/Module/DoubleDual · NormedSpace.isCompact_closure_of_isBounded — and its span contains the ball, hence is V Chapter5/PartitionsOfUnity · span_closedBall_eq_top Chapter5/PartitionsOfUnity · proposition_5_42_reflexive. (ii) For separable V, a dense sequence (y_n) scaled by $1/((n+1)(|y_n|+1))$ is a null sequence; with 0 it is norm compact, hence weakly compact, and its span contains every y_n Chapter5/PartitionsOfUnity · proposition_5_42_separable. The text's "$\{0\} \cup \{x_n/n\}$" has its index range missing.

Corollary 5.44 Chapter5/PartitionsOfUnity · corollary_5_44. Let $\|v\|^2 = b(v,v)$ on E_0, and assume Theorem 5.43 for E_0 and $k = \infty$, recorded as a statement Chapter5/PartitionsOfUnity · GodefroyTroyanskiWhitfieldZizlerStatement. A Hilbert space is reflexive: the parallelogram law holds, so the norm comes from an inner product Mathlib/Analysis/InnerProductSpace/OfNorm · InnerProductSpace.ofNorm, and every $\phi \in V''$ is evaluation at the Riesz representative of $x \mapsto \phi(\langle x, \cdot \rangle)$ Chapter5/PartitionsOfUnity · isReflexive_of_hilbert; hence WCG by Proposition 5.42 (i). It has the C^∞-bump $\tau(1 - b(v,v))$ Chapter5/PartitionsOfUnity · hasContDiffBump_of_hilbert. Theorem 5.43 gives C^∞ partitions of unity of E_0, and Proposition 5.38 — not Theorem 5.37, as the text says — the approximation property. With Theorem 5.37, an M-polyfold on such charts admits sc-smooth partitions of unity Chapter5/PartitionsOfUnity · MPolyfold.admitsScPartitionsOfUnity_of_hilbert.

V.27 What is not formalized

- **Corollary 5.8 as one statement for all i.** It is proved for each fixed i (§V.29; gap G.41 closed in the fourth addendum). Not formalized: a family P^i indexed by i, the identification of the iterated base $(X^1)^1$ with X_2 as a topological space (the sets agree; gap G.6), and the tame form of the conclusion, which needs the tameness of the index raises of tame charts. Condition (3) of a filling on level 1 does not follow from level 0 alone — on $E_m = \{x \mid \sum e^{2mk} x_k^2 < \infty\}$ the right shift T makes $1 - T/2$ invertible on E_0 while $(1-T/2)^{-1} e_0 = \sum 2^{-n} e_n \notin E_1$ — so the corollary's proof needs the regularity argument of §V.29, through Lemma 3.17 Chapter3/Filler · lemma_3_17.

- **The non-completeness of $(\Gamma_U^{+,1}(P), \rho)$ in the text's setting** (§V.30), and the identification of the completion of Remark 5.12; **Remarks 5.14, 5.16, 5.21** (gaps G.43, G.44, G.47), and **a dense form of Theorem 5.19** (gap G.45).
- **Weak against pointwise bump functions at a non-smooth point** (§V.31).
- **Finding 88** (gap G.46): the failure of good position of F at the boundary zeros (§V.22) is checked on paper only.
- **Theorem 5.22** (gap G.48): it needs finite-dimensional Morse theory on manifolds with boundary, the cotangent bundle of the solution set, a parametric Sard theorem and a trivialization of the family S_λ, none in Mathlib or the project, and the hidden hypothesis $dt(x) \neq 0$ (§V.23). **Theorem 5.27** (gap G.49): "the proof is left to the reader".
- **Theorems 5.32, 5.40, 5.43 and the Sobolev remark** (gaps G.50–G.53): deep Banach space geometry cited from [9], [12], and Sobolev spaces with their sc-structures (gap G.1). Nothing on the main line uses partitions of unity.

V.28 What the formalization taught us

The tameness hypothesis. The first formal versions of Theorem 5.13 and Lemma 5.26 assumed, as the §3.6 theorems then did, that the base retraction of *every* compatible strong bundle chart — and of every chart of the pull-backs to $\mathbb{R}^k \times X$ — is tame. No strong bundle over a nonempty base satisfies this (a compatible chart $(1, \varphi)$ with base retraction $(a, u) \mapsto (1, r(u))$ on $[0, \infty) \times U$ is never tame), so those statements were vacuous. The repair is the text's own Definitions 3.50, 3.51, 3.53 in a tame chart: the perturbation theorems take f with tame germ data, the parametrized sections inherit them through the tame product chart, and Definition 5.17 lives in a tame chart, where Lemma 5.26's count $\dim T_z O / T_z^R O = d_C(z)$ holds.

What "it is easily verified" costs. The perturbation theorems of §5.3 are sketches. Their formal proofs needed the density of smooth vectors (§V.12), the compactness of the zeros over a small parameter ball (§V.13), a local, stratified Sard step (§V.14), the bridge from tangent germ data to Definition 4.13 (§V.17), $T_y S = \ker f'(y)$ by a dimension count, which gives the chart independence of Definition 5.17 and the openness of general position (§§V.17–V.18), and, in Theorem 5.19, a route around the implicit function theorem at boundary points where its hypothesis fails (§V.22).

V.29 Corollary 5.8 on the shifted strong bundle

Corollary 5.8 says that, for an sc-Fredholm section f with compact solution set, f^i is an sc-Fredholm section of $P^i : Y^i \to X^i$ with compact solution set, and that "it is a priori clear that f^i is sc-Fredholm". The text defines $Y[i]$ for $i = 0, 1$ only, never Y^i, and "a priori clear" hides the filler condition on level 1 (§V.7). Both are supplied here (Findings, Chapter 5, items 140–146).

The reading of Y^1. $Y^1 = Y_{1,1}$ with $(Y^1)_{m,k} = Y_{m+1,k+1}$ — the convention of E^1 and of $(U \lhd F)^1 = U^1 \lhd F^1$ — not $Y[1] = Y_{0,1}$ and not $Y_{1,2}$. It is forced by "f^i is obtained by raising the index": an sc-smooth section maps X_1 into $Y_{1,1}$, and in general not into $Y_{1,2}$.

Fibres of a strong bundle. Through a chart, $P^{-1}(x)$ is linearly homeomorphic to $K_{\varphi(x)} = \{h \mid \Gamma(\varphi(x), h) = h\}$ `Chapter5/StrongBundleShift · StrongBundleChart.fiberK` `Chapter5/StrongBundleShift · StrongBundleChart.fiberCLE`, the fixed points of a continuous projection, a closed subspace of the fibre model `Chapter5/StrongBundleShift · StrongBundleChart.isClosed_fiberK`; so every fibre is a Banach space `Chapter5/StrongBundleShift · StrongBundleChart.isBanachable_fiber`, which Definition 2.51 asks of a strong bundle.

The levels of X^1, and raising the index of maps. $(X^1)_m = X_{m+1}$ and $(X^1)_\infty = X_\infty$ as sets `Chapter5/StrongBundleShift · MPolyfold.mem_shift_levelSet_iff` `Chapter5/StrongBundleShift · MPolyfold.mem_`

shift_smoothSet_iff (their topologies are not compared; gap G.6). An sc-smooth map $f : X \to Z$ between M-polyfolds induces an sc-smooth $f_1 : X^1 \to Z^1$: continuity, and in the level charts the local representatives are the level-1 maps of those of f, sc-smooth by Proposition 1.17 for retracts Chapter5/StrongBundleShift · MPolyfold.levelChart_apply_levelOne Chapter5/StrongBundleShift · MPolyfold.IsScSmoothMap.levelOne.

The shifted strong bundle Chapter5/StrongBundleShift · StrongBundle.shift. Its total space is $Y_{1,1}$, the level 1 of the M-polyfold $Y[0]$ with its level topology, Hausdorff and paracompact by Theorem 2.16 Chapter5/StrongBundleShift · StrongBundle.ShiftSpace; the projection is $P^1 = P|_{Y_{1,1}}$ Chapter5/StrongBundleShift · StrongBundle.shiftProj, continuous and onto Chapter5/StrongBundleShift · StrongBundle.continuous_shiftProj Chapter5/StrongBundleShift · StrongBundle.surjective_shiftProj; the fibre $P^{-1}(x) \cap Y_{1,1}$ is a linear subspace — in a chart it is the preimage of F_1 under the linear fibre map Chapter5/StrongBundleShift · StrongBundle.shiftFiberSubmodule — carries the topology of F_1, not that of $P^{-1}(x)$, and is a Banach space, a closed subspace of F_1 through a chart Chapter5/StrongBundleShift · StrongBundle.banachable_shift. The atlas consists of the *index raises* of the strong bundle charts Chapter5/StrongBundleShift · StrongBundleChart.shift Chapter5/StrongBundleShift · StrongBundle.shiftAtlas: base chart the level chart $\varphi^1 : V_1 \to O_1$, fibre model F^1, retraction R on $U^1 \lhd F^1$ Chapter5/StrongBundleShift · IsStrongBundleRetraction.shift, bundle chart the level chart of Φ in $Y[0]$. Compatible charts have compatible index raises Chapter5/StrongBundleShift · StrongBundleChart.Compatible.shift: on $[0]$ the transitions are level charts of compatible charts, on $[1]$ the transition $(\Phi'^1 \circ (\Phi^1)^{-1})[1]$ is the level-1 map of $(\Phi' \circ \Phi^{-1})[1]$ (Proposition 1.17 for retracts) Chapter5/StrongBundleShift · StrongBundleChart.shift_transition_one, and the F_1-condition of Definition 2.50 holds because the $[1]$-transition is sc^0 Chapter5/StrongBundleShift · StrongBundleChart.shift_compatible_range. Checks: $(Y^1)_{m,k} = Y_{m+1,k+1}$ Chapter5/StrongBundleShift · StrongBundle.mem_shift_doubleLevel_iff, the M-polyfold $Y^1[0]$ is $Y[0]^1$ Chapter5/StrongBundleShift · StrongBundle.isChart_shift_toMPolyfold0_iff, and the index raise of every chart of P is a chart of P^1 Chapter5/StrongBundleShift · StrongBundle.isChart_shift.

The section f^1. $f^1 = f|_{X_1}$ Chapter5/StrongBundleShift · StrongBundle.sectionShift is a section of P^1 Chapter5/StrongBundleShift · StrongBundle.isSection_sectionShift, sc-smooth by the raising of the index of maps Chapter5/StrongBundleShift · StrongBundle.IsScSmoothSection.shift, with the zeros of f in X_1 as its zeros Chapter5/StrongBundleShift · StrongBundle.solutionSet_sectionShift, and regularizing, from $(X^1)_m = X_{m+1}$ and $(Y^1)_{m,m+1} = Y_{m+1,m+2}$ Chapter5/StrongBundleShift · StrongBundle.regularizing_sectionShift. In the index raise of a chart the principal part of f^1 is the level-1 map of that of f Chapter5/Corollary58 · StrongBundleChart.principal_sectionShift, and the raised chart is centred where the chart is Chapter5/Corollary58 · StrongBundleChart.shift_φ_eq_zero.

Condition (3) on every level, at every centre Chapter5/Corollary58 · IsFillingGerm.mem_range_toBase_of_fillDefect. *Let (g, s) be a datum of Definition 3.11 in a chart — g a filling germ of $\hat{f}$ with fill defect $k = [1-\rho(r)]g$ and derivative family Dk, s an sc$^+$-germ with $s(0) = g(0)$, $g - s$ conjugated to a basic germ. If $v \in \ker Dr(0)$ and $Dk(0)v \in F_m$, then $v \in E_m$* — at the centre of the datum, which need not be a zero. At a zero this is the regularity of §III.45 Chapter3/LinearizationChartsScG · IsFillingGerm.mem_range_toBase_of_scDeriv_r_eq_zero. The reduction to a zero needs neither a global sc$^+$-section nor a bump function: $g(0) = \iota s(0)$ is a smooth point of F Chapter5/Corollary58 · IsScSmoothGerm.toBase_zero_mem_smooth; $\hat{s}(z) = \Gamma(z, -g(0))$ is sc-smooth on U Chapter3/GlobalIFTNearbyOntoXi · isScSmooth_const_of_smooth and a section of $K \to O$, so by Lemma 3.17 $g' = g + \hat{s} \circ r$ fills $\hat{f} + \hat{s}$ with the *same* fill defect Chapter3/GermCalculus · IsFillingGerm.add_section, and $g'(0) = g(0) - \Gamma(0, g(0)) = 0$. With the sc$^+$-germ $s' = s + \Gamma^1(r(\cdot), -s(0))$, $g' - s' = g - s$ on U, so Proposition 3.52 in sc form makes $Dg'(0)$ sc-Fredholm Chapter3/FredholmGermLinearizationScG · proposition_3_52_local_sc; and $Dk(0) = (1-\rho(0))Dg'(0)$ Chapter3/GermFillerLinearization · fillDefect_deriv_zero_eq_rho, so the case of a zero applies to g'.

The filling on level 1 Chapter5/Corollary58 · IsFillingGerm.germ_shift. The fill defect of g^1 is the level-1 map of k, an sc^1-germ with derivative family $(Dk)_1$ by Proposition 1.17 for germs Chapter3/GermLevels · IsSc1GermWith.shift; $\iota\, Dr^1(0) = Dr(0)\, \iota$, by the uniqueness of the derivative of $r^1 = r_1$ on the shifted germ Chapter5/Corollary58 · toBase_scDeriv_r_shift, and $\iota\, \rho^1(0) = \rho(0)\, \iota$ Chapter5/Corollary58 · toBase_

ρCLM_shift. So $Dk^1(0) = Dk(0)|_{E_1}$ maps $\ker Dr^1(0)$ to $\ker \rho^1(0)$; it is injective by condition (3) on level 0, onto by the regularity just proved with $m = 1$, and a topological isomorphism by the open mapping theorem. Conditions (1) and (2) of the filling shift directly Chapter5/Corollary58 · StrongBundleChart.shift_filling_eq_on Chapter5/Corollary58 · StrongBundleChart.shift_filling_mem_of_fixed, and g^1 is a filling germ of the principal part of f^1 in Φ^1 Chapter5/Corollary58 · StrongBundleChart.shift_isFillingGerm.

Corollary 5.8. At every smooth point of X^1, the datum (g, s) of f in a chart Φ gives the datum (g^1, s^1) of f^1 in Φ^1: g^1 fills, s^1 is an sc⁺-germ on the shifted germ, $s^1(0) = g^1(0)$, and $g^1 - s^1 = (g - s)_1$ near 0 is conjugated to a basic germ by the index raise of the conjugation (§V.7) Chapter5/Corollary58 · isScFredholmGerm_sectionShift. *For an sc-Fredholm section f, f^1 is an sc-Fredholm section of P^1* Chapter5/Corollary58 · corollary_5_8_isScFredholmSection; *if f has compact solution set, so has f^1*, by the compactness on level 1 (§V.7) Chapter5/Corollary58 · corollary_5_8. Since this holds for every strong bundle, it applies to P^1 and f^1: $f^2 = (f^1)^1$ is an sc-Fredholm section of $P^2 = (P^1)^1$ with compact solution set Chapter5/Corollary58 · corollary_5_8_two, and so on for every fixed i. The unpacked data in the shifted chart are kept for reference Chapter5/Corollary58 · corollary_5_8_data. No hypothesis beyond the text's is added, and tameness is not used. *Not done*: a family P^i indexed by i (a dependent type of strong bundles), the identification of $(X^1)^1$ with X_2 as topological spaces, and the tame form of the conclusion — germ data of f^1 in tame charts of P^1 — which needs the presentation independence of the degeneracy index for C^1 and that $A \cap E_1$ is an sc-complement of T_xO^1 when A is one of T_xO; there is no counterexample either.

V.30 The metric ρ on $\Gamma^{+,1}_U(P)$

The text: $\Gamma^{+,1}_U(P)$ "becomes a metric space" for $\rho(s, s') = \sup_x N(s(x) - s'(x))$, and "is not complete" (p. 174). The first is proved here; the second turns out to need a hypothesis (Findings, Chapter 5, items 150–153).

The metric. $\rho(s, s')$ is the supremum of the pointwise distances of §V.10 Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist. For an auxiliary norm the fibre norm is symmetric and subadditive (Definition 5.1 (1)) Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.neg Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_le Chapter5/AllowableSectionsMetric · StrongBundle.IsAuxNorm.sub_triangle, so for allowable sections $N(s(x) - s'(x)) \le N(s(x)) + N(s'(x)) \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.sectionDistAt_le_two: the supremum is finite and $\rho \le 2$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_two. $\rho(s, s) = 0$, symmetry and the triangle inequality come from the fibre norms Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_self Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_comm Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_triangle, and $\rho(s, s') = 0$ forces $s = s'$ because N vanishes only at 0_x Chapter5/AllowableSectionsMetric · StrongBundle.eq_of_allowableDist_eq_zero. So ρ is a metric Chapter5/AllowableSectionsMetric · StrongBundle.allowableMetric, carried by a type synonym of $\Gamma^{+,1}_U(P)$, since the sections $X \to Y$ carry the product topology Chapter5/AllowableSectionsMetric · StrongBundle.AllowableSpace. Only Definition 5.1 (1) is used. The density statements of Theorems 5.13 and 5.18 (§§V.15, V.20) are $\rho(s, s') \le \varepsilon$ Chapter5/AllowableSectionsMetric · StrongBundle.allowableDist_le_iff.

Not complete, by an example Chapter5/AllowableSectionsMetric · TrivialBundle.not_completeSpace_allowableSections. Take the trivial bundle $\mathbb{R} \times \mathbb{R} \to \mathbb{R}$ of the non-vacuity witnesses of Chapter 3, the fibre norm $N(x, v) = |v|$ — an auxiliary norm, including the condition on the topology of $Y[1]$ Chapter5/AllowableSectionsMetric · TrivialBundle.fibNorm Chapter5/AllowableSectionsMetric · TrivialBundle.isAuxNorm_fibNorm — and $U = \mathbb{R}$. Graphs of C^∞ maps are sc⁺-sections Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_graphSection, so $s_j(x) = (x, \sqrt{x^2 + 1/(j+1)^2}/(2\sqrt{1+x^2}))$ is allowable, with $N \le 1/2$ Chapter5/AllowableSectionsMetric · TrivialBundle.approx Chapter5/AllowableSectionsMetric · TrivialBundle.graphSection_approx_mem, and $\rho(s_i, s_j) \le (\varepsilon_i + \varepsilon_j)/2$ with $\varepsilon_j = 1/(j+1)$, by $|\sqrt{x^2 + a^2} - |x|| \le a$ Chapter5/AllowableSectionsMetric · TrivialBundle.abs_sqrt_sub_abs_le Chapter5/AllowableSectionsMetric · TrivialBundle.abs_approx_sub_limitFn. A ρ-limit s would have, by pointwise convergence, the fibre part $|x|/(2\sqrt{1+x^2})$ Chapter5/AllowableSectionsMetric · TrivialBundle.limitFn, which is not differentiable at 0 Chapter5/

AllowableSectionsMetric · TrivialBundle.not_differentiableAt_limit; but the fibre part of an sc⁺-section of the trivial bundle is C^1, by Proposition 1.18 in the chart $\Phi = \mathrm{id}$ Chapter5/AllowableSectionsMetric · TrivialBundle.differentiable_snd_of_isScPlusSection.

Complete, in two cases. For $U = \emptyset$ every allowable section is the zero section, so the space is a point, for every bundle and auxiliary norm Chapter5/AllowableSectionsMetric · StrongBundle.subsingleton_allowableSections_empty Chapter5/AllowableSectionsMetric · StrongBundle.completeSpace_allowableSections_empty. Over the point $\mathbb{R}^0$ with fibre $\mathbb{R}^k$ every section is the graph of a constant, hence sc⁺ Chapter5/AllowableSectionsMetric · TrivialBundle.isScPlusSection_of_isSection_point, and the space is the closed unit ball of $\mathbb{R}^k$ (or $\{0\}$), complete for every U Chapter5/AllowableSectionsMetric · TrivialBundle.completeSpace_allowableSections_point. So the text's "not complete" is false without a hypothesis and is read as "not complete in general". In the text's setting — U a neighbourhood of the solution set of an sc-Fredholm section on an M-polyfold with bump functions — one expects non-completeness as soon as U contains a non-isolated point over which $Y_{0,1}$ is nonzero, by the same mechanism (bump functions times a non-smooth uniform limit); this is not formalized, and neither is the identification of the metric completion with the sections of Remark 5.12 (§V.10).

V.31 Weak and pointwise bump functions

§V.8 reads Definition 5.9 pointwise, $f(x) \neq 0$, because Proposition 5.30 needs it, and keeps Definition 5.29's "not identically zero" as the weak reading. How far apart are they? (Findings, Chapter 5, items 160–163.)

Bump points. x is a *bump point* if every open $U \ni x$ carries an sc-smooth f with $f(x) \neq 0$ and $\operatorname{supp} f \subseteq U$ Chapter5/BumpWeakPointwise · MPolyfold.HasBumpAt. The pointwise reading says that every point is a bump point Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctions_iff_forall_hasBumpAt, and the weak reading holds as soon as the bump points are dense: a bump at a bump point inside U is not identically zero and is supported in U Chapter5/BumpWeakPointwise · MPolyfold.admitsBumpFunctionsWeak_of_dense. (The converse, weak ⇒ dense bump points, is a nested-neighbourhoods argument with Cantor's theorem in a chart; not formalized.)

On an sc-Banach space. An sc-Banach space E is an M-polyfold with the identity chart and model (E, E, E) Chapter5/BumpWeakPointwise · scChart Chapter5/BumpWeakPointwise · scPolyfold, whose sc-smooth functions are the sc-smooth maps $E \to \mathbb{R}$ Chapter5/BumpWeakPointwise · isScSmoothMap_scPolyfold_iff. *There the weak reading gives a bump point at every smooth point* Chapter5/BumpWeakPointwise · MPolyfold.AdmitsBumpFunctionsWeak.hasBumpAt_of_mem_smooth: given $x \in E_\infty$ and $B(x, \varepsilon) \subseteq U$, a weak bump supported in $B(x, \varepsilon/2)$ is non-zero on a non-empty open set, which contains a smooth point y because E_∞ is dense; $a = y - x$ is smooth, so $v \mapsto f(v + a)$ is sc-smooth Chapter4/Recentre · isScSmooth_add_const, non-zero at x, with support in $B(x, \varepsilon) \subseteq U$. The argument stops at E_∞: translation by a vector $a \notin E_m$ is not even sc⁰, since it sends $0 \in E_m$ out of E_m Chapter5/BumpWeakPointwise · not_isSc0_add_const. On M-polyfolds built on sc-Hilbert spaces both readings hold at every point Chapter5/BumpFunctions · MPolyfold.admitsBumpFunctions_of_hilbert. *Open*: whether the weak reading gives a bump function non-zero at a non-smooth point, on an sc-Banach space or an M-polyfold; a counterexample would need an E_0 without C^1 bump functions.

Part VI. Linearizations, orientations and invariants (HWZ Chapter 6, §§6.1–6.7)

Chapter 6 builds the orientation theory of sc-Fredholm sections. At a smooth point x it collects the linearizations of f into a convex set $\mathrm{Lin}(f,x)$ of Fredholm operators (§6.1); it fixes the conventions for determinant lines and for the isomorphism attached to a four-term exact sequence (§6.2); it defines the determinant $\det(T) = \lambda(\ker T) \otimes \lambda^*(\operatorname{coker} T)$ of a Fredholm operator, compares it with the determinants of PT for good left-projections P and of the stabilizations T_φ, and builds the determinant line bundle $\mathrm{DET}(E,F)$ over the Fredholm operators (§§6.3–6.4); it then restricts that bundle to $\mathrm{Lin}(f,x)$, propagates its orientations along sc-smooth paths of X, first in small neighbourhoods (Theorem 6.35) and then globally, as a functor on the path category (Theorem 6.37), and defines orientability and orientations of f, orients the solution set (Theorem 6.41) and the sections $f+s$ over $X \times [0,1]$ (Remark 6.42) (§6.5); §6.6 states the invariant $\Psi_{(f,o)}$ (Theorem 6.43), and the appendix §6.7 proves Lemma 6.5 and Proposition 6.12. This Part retells §6.1 (§VI.1); §6.2 — determinant lines, Definition 6.3 and Proposition 6.4, the isomorphism Φ_E, Lemma 6.5 and Proposition 6.6 (§§VI.2–VI.3); §6.3 — Definitions 6.7–6.9, Lemma 6.10, the isomorphisms γ_T^P, the stabilization and the direct sums (§§VI.4–VI.6); §6.4 — the local determinant bundles and $\mathrm{DET}(E,F)$ (§§VI.7–VI.8); §6.5 — orientations over $\mathrm{Lin}(f,x)$ (§VI.9), Proposition 6.28 (§VI.10), Lemmas 6.29–6.31 (§§VI.11–VI.13), the push-forward (§§VI.14–VI.15), Lemma 6.33, Lemma 6.34 and Proposition 6.32 (§§VI.16–VI.17), Theorem 6.35 and Definition 6.36 (§§VI.18–VI.19), the path category (§VI.20), the comparison of charts (§VI.21), the transport and Theorem 6.37 (§§VI.22–VI.23), Definitions 6.38–6.40 (§VI.24), Theorem 6.41 (§VI.25) and Remark 6.42 (§VI.26); §6.6 and Theorem 6.43, which are not formalized (§VI.27); the appendix §6.7 (§§VI.28–VI.29); what is not formalized (§VI.30); and what the formalization taught us (§VI.31). Proposition 6.2 was formalized with Chapter 3 and is retold in §III.49. Each item carries the number of its comparison item (V6.1–V6.64) in `Exposition/verification/ch06.md`.

What is proved, and what is not. Theorems 6.35 and 6.37 are proved; so are Propositions 6.4, 6.6, 6.12, 6.15, 6.20, 6.23, 6.24, 6.26, 6.32, Lemmas 6.5, 6.10, 6.11, 6.14, 6.17, 6.18, 6.21, 6.25, 6.31, 6.33, 6.34, Corollary 6.13, Lemma 6.29 in the form the text uses afterwards, the kernel statement of Lemma 6.30, Proposition 6.28 with (1′) in place of (1), the equivalence of Definitions 6.38 and 6.39, Theorem 6.41 in the form of tangent orientations that vary continuously along the paths of zeros in a chart, and Remark 6.42 at a point. Every definition of the chapter is formalized. Three places needed more than a reading. The projection of the proof of Lemma 6.10 does not lie below the two given projections as we read it, and the lemma is proved by another construction, without the hypothesis that T is Fredholm (§VI.4). The isomorphism Φ_E of formula (94) does not vary continuously in the topology of DET where $\dim\operatorname{coker} T$ changes parity, as we read it, so the propagation uses $(-1)^{\dim\operatorname{coker} T}\Phi_E$ (§VI.14); the map $h \mapsto \tilde{h}$ of Remark 6.42 gets the same sign (§VI.26). Proposition 6.28 (1), the sc-smooth contractibility of the neighbourhood, is not established by its proof as we read it; what Theorem 6.35 uses is (1′), the homotopy of two paths with the same end points inside a larger region, and that is proved (§§VI.10, VI.18). Five proofs the text omits are supplied: the orientability of a line bundle over a convex set (§VI.9), the general position of Lemma 6.29 (§VI.11), the "simple computation" of Lemma 6.33 (§VI.16), the covering argument of Theorem 6.37 (§VI.22) and the equivalence of Definitions 6.38 and 6.39 (§VI.24). Not formalized: Theorem 6.43 and the

admissible sections of §6.6 (gaps G.54, G.55), Proposition 6.28 (1) (gap G.61), the orientation of the solution set as an abstract manifold, its chart independence and the perturbation paragraph after Theorem 6.41 (gaps G.67–G.69), Remark 6.42 beyond a point (gap G.66), the general Lemma 6.19, several remarks, and the facts on paths recalled on p. 201 (§VI.30).

Six structural choices. (1) *Chart points.* T_xX carries no linear structure in the formalization (gaps G.6, G.7, G.13 of Chapter 2), so $\mathrm{Lin}(f,x)$, $\mathrm{DET}(f,x)$ and Or_x^f are made at a *chart point* (c,z) of x, with operators $T_zO = Dr(z)E \to K_z = \rho(z)F$ on level 0, and they are identified between charts by explicit transition maps (§VI.21); the text's $f-s$ with $s(x)=f(x)$ is $f+s$ with $(f+s)(x)=0$. (2) *Global sections, from bump functions.* The text assumes sc-smooth partitions of unity from §6.2 on, "to simplify the presentation" (p. 203). The formalization assumes sc-smooth bump functions, which partitions of unity give (§V.25), and uses them only to make global sc⁺-sections (Findings, Chapter 6, item 43). (3) *Families on* $\mathbb{R}^n$. Paths, homotopies and families are sc-smooth maps on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ whose conditions are imposed on $[0,1]$, $[0,1]^2$, $[0,1]^n$, and $X\times[0,1]$ is read as $\mathbb{R}^n\times X$ with the pull-back π^*P of the bundle along the projection (decision D2, as for Proposition 6.2 in §III.49; gap G.65; Findings, Chapter 6, item 102). (4) *The sign* $\alpha(T) = (-1)^{\dim\operatorname{coker} T}$ at both end points of every propagation (§VI.14). (5) *(1'), and orientable neighbourhoods as pairs* $V \subseteq U$, since Proposition 6.28 gives (1') rather than contractibility (§§VI.10, VI.19). (6) *Dyadic subdivisions* in place of the finite coverings of the proof of Theorem 6.37 (§VI.22). An orientation of a line is a ray (Mathlib's `Module.Ray`); an orientation of a topological line bundle is a choice of rays given near every point by a continuous nowhere-zero section.

Map.

- `Chapter6/LinSet`: chart points, T_zO, K_z, sc⁺-operators, Definition 6.1, the index and convexity of $\mathrm{Lin}(f,x)$.
- `Chapter6/DetLine`: $\lambda(E)$, $\lambda^*(E)$, wedges, v^*, $\lambda(\Phi)$, Definition 6.3, Proposition 6.4.
- `Chapter6/ExactSequence`: four-term exact sequences, the choices of (91), Φ_E, Lemma 6.5 with the proof of §6.7.1, (92), Proposition 6.6.
- `Chapter6/FredholmDet`: Definitions 6.7, 6.8.
- `Chapter6/GoodProjections`: Definition 6.9, the order on Π_T, the facts after it, Lemma 6.10 and the example on $\mathbb{R}^2$.
- `Chapter6/ProjectionSequence`: Lemma 6.11, γ_T^P, Proposition 6.12 with the proof of §6.7.2, Corollary 6.13.
- `Chapter6/Stabilization`: Lemma 6.14, formula (94), Proposition 6.15.
- `Chapter6/DirectSumDet`: Definition 6.16.
- `Chapter6/LocalDetBundle`: Lemmas 6.17, 6.18, the frames of Proposition 6.20, Lemma 6.21.
- `Chapter6/DetBundle`: $\mathrm{DET}(E,F)$ as a vector prebundle, Definition 6.22, Propositions 6.23, 6.24, Lemma 6.25, Proposition 6.26.
- `Chapter6/LineBundleOrientation`: orientations of line bundles, the orientation double cover.
- `Chapter6/LinOrientation`: Definition 6.27.
- `Chapter6/OrientableNbhd`: sc-smooth homotopies, Proposition 6.28.
- `Chapter6/Lemma629`: Lemma 6.29.
- `Chapter6/KernelFamily`: the kernels of Lemma 6.30.
- `Chapter6/SubspaceBundle`: continuous families of subspaces, their determinant lines, continuation over $[0,1]$.
- `Chapter6/StabilizationContinuity`: families of kernels and their images, the stabilization sign, the example without the sign.
- `Chapter6/ParamKernel`: the stabilized section in a product chart, Lemma 6.31.
- `Chapter6/Continuation`: continuation over a simply connected base, the propagation $\Phi_{E_1}^{-1}\circ$ continuation $\circ\,\Phi_{E_0}$.
- `Chapter6/Propagation`: stabilized kernels and the push-forward φ_*; Lemmas 6.33 and 6.34 for it.
- `Chapter6/StabKernelsExist`: the stabilized kernels exist.
- `Chapter6/PushData`: push-forwards along paths, their existence, halves and reparametrizations.
- `Chapter6/AddSections`: the linear algebra of Lemma 6.33.

- `Chapter6/FamilySections`: sections over parameter families.
- `Chapter6/PropagationWellDefined`: Proposition 6.32, the homotopy argument, Theorem 6.35.
- `Chapter6/ScPaths`: sc-smooth paths in a chart, concatenation.
- `Chapter6/ContinuousOrientation`: Definition 6.36, continuous orientations.
- `Chapter6/ScHomotopy`: reparametrization homotopies.
- `Chapter6/PathCategory`: homotopy classes and the path category PX.
- `Chapter6/ChartTransfer`: conjugation of determinants, the transfer between chart points.
- `Chapter6/PushTransfer`: the push-forward does not depend on the chart.
- `Chapter6/SectionOrientation`: orientable neighbourhoods as structures, Or_x^f in a chosen chart, Definitions 6.39, 6.40.
- `Chapter6/PathTransport`: transport along paths from local push-forwards.
- `Chapter6/OrientationFunctor`: Theorem 6.37, Definitions 6.38–6.40 and their equivalences.
- `Chapter6/SolutionOrientation`: Theorem 6.41.
- `Chapter6/ProductOrientation`: Remark 6.42.

Throughout, E and F are real Banach spaces, $T : E \to F$ is bounded, $R(T)$ is its range and $\operatorname{coker} T = F/R(T)$; T is Fredholm if $\ker T$ and $\operatorname{coker} T$ are finite-dimensional, with index $\dim\ker T - \dim\operatorname{coker} T$. A *line* is a one-dimensional real space. $P : Y \to X$ is a strong bundle over an M-polyfold, both paracompact and Hausdorff, and from §VI.10 on X is tame with sc-smooth bump functions and f is an sc-Fredholm section with tame germ data (Definition 3.51), the reading of Chapter 5 (Part V). A strong bundle chart c has base chart φ, retraction r on an open set U of a partial quadrant C, retract $O = r(U)$, and fibre projections $\rho(u)$; the *principal part* of a section is its fibre coordinate in the chart. $\mathbb{R}^1$ is the M-polyfold of the parameter t, with t the point of $\mathbb{R}^1$ `Chapter6/OrientableNbhd · pt1`, and (τ, t) the point of $\mathbb{R}^2$ `Chapter6/OrientableNbhd · pt2`; κ is Mathlib's smooth transition function, 0 on $(-\infty, 0]$, 1 on $[1, \infty)$.

VI.1 Definition 6.1: the space $\mathrm{Lin}(f, x)$

§6.1 (pp. 200–201) works at a smooth point x. Since $f(x)$ is a smooth point of the fibre, there is an sc⁺-section s with $s(x) = f(x)$ — in a chart, $s(y) = R(y, f(x))$ — and $(f - s)'(x) : T_xX \to Y_x$ is sc-Fredholm. Definition 6.1 collects these linearizations, perturbed by sc⁺-operators, into $\mathrm{Lin}(f, x)$; the index is the index of any member, and Proposition 6.2 says that it is constant along sc-smooth paths.

The chart point (V6.1). A chart point of x is a strong bundle chart c of the atlas whose base chart contains x, with the level-1 point z of the model over $\varphi(x)$ `Chapter6/LinSet · ChartPoint`. In it T_xX is $T_zO = Dr(z)E$, the range of the derivative of the retraction `Chapter6/LinSet · ChartPoint.T`, and Y_x is $K_z = \rho(z)F$, the range of the fibre projection over z `Chapter6/LinSet · ChartPoint.K`; both are closed, as ranges of continuous projections, hence Banach spaces. An sc⁺-operator $T_xX \to Y_x$ is level 0 of an sc⁺-operator between these sc-subspaces `Chapter6/LinSet · ChartPoint.IsScPlusOp`.

Definition 6.1 (V6.1). $\mathrm{Lin}(f, x)$, at the chart point p, is the set of the operators $(f+s)'(x)+a : T_zO \to K_z$ for an sc⁺-section s with $(f + s)(x) = 0$ and an sc⁺-operator a `Chapter6/LinSet · ChartPoint.linSet` `Chapter6/LinSet · ChartPoint.mem_linSet_iff`. The linearization of $f+s$ at its zero x is the derivative at z of the principal part composed with r, restricted to T_zO — the $\mathrm{Lin}(z)$ of Part III. Replacing the text's s by $-s$ turns $f - s$ with $s(x) = f(x)$ into $f + s$ with $(f + s)(x) = 0$.

Two sections differ by an sc⁺-operator. For two sections s, t with $(f + s)(x) = (f + t)(x) = 0$, the principal parts of $f + s$ and $f + t$ differ by that of the sc⁺-section $s - t$, whose derivative at z, followed by $\rho(z)$, is level 0 of an sc⁺-operator: the derivative of an sc⁺-map is sc⁺ on the shifted levels. So $\mathrm{Lin}(f, x)$ is $(f + s)'(x)$ plus all the sc⁺-operators, for any one s `Chapter6/LinSet · ChartPoint.exists_isScPlusOp_linAt_eq`, and two members differ by an sc⁺-operator `Chapter6/LinSet · ChartPoint.isScPlusOp_sub_of_mem_linSet`. The paragraph before the definition puts this as "$(f - s)'(x)$ is an sc-Fredholm operator if and only if $(f - s)'(x)$ is"; it is read with $(f - t)'(x)$ in the second place (possible misprint; Findings, Chapter 6, item 41).

Not empty. The text makes its section locally, in a chart. The formalization uses global sc⁺-sections: $\mathrm{Lin}(f,x)$ is not empty when P has sc⁺-sections through the smooth points of its fibres, one through $-f(x)$ giving a member `Chapter6/LinSet · ChartPoint.linSet_nonempty`, and bump functions give such sections `Chapter3/IndexLocallyConstantLocal · MPolyfold.AdmitsBumpFunctions.hasScPlusSectionsThrough`. This is the hidden hypothesis of the definition: an sc⁺-section through $f(x)$.

Fredholm, of one index. Every member is Fredholm on level 0, with the index $\mathrm{ind}(f,x)$ of the germ at x (§III.49) `Chapter6/LinSet · ChartPoint.isFredholm_isIndexAt_of_mem_linSet` `Chapter6/LinSet · ChartPoint.exists_index_linSet` `Chapter6/LinSet · ChartPoint.fredholmIndex_of_mem_linSet`. The proof: $f+s$ has germ data at x (an sc-Fredholm germ plus an sc⁺-germ is one), so its linearization at the zero x is Fredholm of the index of the germ, through the conjugation to a basic germ (Proposition 3.52, §III.30); adding an sc⁺-operator, compact on level 0, keeps the Fredholm property and the index; and the index of the germ does not depend on the choices. That the members are sc-Fredholm on every level, as the text says, is not stated: the determinant lines use level 0 only (gap G.64; Findings, Chapter 6, item 40).

Convex `Chapter6/LinSet · ChartPoint.convex_linSet`. $(1-\tau)(L+a)+\tau(L+b)=L+((1-\tau)a+\tau b)$, and the sc⁺-operators form a vector space. The text notes this on p. 219, where Proposition 6.26 needs it.

Proposition 6.2 (V3.98). The constancy of the index along sc-smooth paths was formalized with Chapter 3 and is retold in §III.49: for paths on $\mathbb{R}$, with the text's hypotheses — tame germ data, no bump functions, the locally defined sc⁺-sections of Definition 6.1. It is not repeated here. The facts on paths recalled before it on p. 201 — an M-polyfold is locally path connected, and two smooth points in one path component are joined by an sc⁺-smooth path — are formalized only inside a chart (§VI.18; gap G.56; Findings, Chapter 6, item 42).

VI.2 Determinant lines, Definition 6.3, Proposition 6.4

$\lambda(E)$ and $\lambda^*(E)$ (V6.2). For a finite-dimensional real space E, $\lambda(E)=\Lambda^{\max}E$ is the top exterior power `Chapter6/DetLine · maxPow` and $\lambda^*(E)$ its dual `Chapter6/DetLine · maxPowDual`; both are lines `Chapter6/DetLine · finrank_maxPow`, $\lambda(\{0\})=\mathbb{R}$ being the zeroth exterior power. A family $a_1,\dots,a_n$ with $n=\dim E$ has its wedge $a_1\wedge\cdots\wedge a_n$ `Chapter6/DetLine · wedge`. Two rules carry every computation of the chapter. A change of basis multiplies the wedge by its determinant `Chapter6/DetLine · wedge_eq_det_smul`, also when only one block of a two-block wedge $a\wedge b$ is changed `Chapter6/DetLine · wedge_append_basis_left` `Chapter6/DetLine · wedge_append_basis_right`; and adding to the vectors of one block vectors of the span of the other does not change the wedge `Chapter6/DetLine · wedge_append_add_left` `Chapter6/DetLine · wedge_append_add_right`. For $v\neq 0$ in a line, v^* is the functional with $v^*(v)=1$ `Chapter6/DetLine · dualVec` `Chapter6/DetLine · dualVec_self`, and that value determines it `Chapter6/DetLine · eq_dualVec_of_apply_eq_one`. For $\dim E=\dim F$ and $\Phi:E\to F$, $\lambda(\Phi)(a_1\wedge\cdots\wedge a_n)=\Phi a_1\wedge\cdots\wedge\Phi a_n$ `Chapter6/DetLine · maxPowMap` `Chapter6/DetLine · maxPowMap_wedge`, with $\lambda(\Psi\Phi)=\lambda(\Psi)\lambda(\Phi)$ `Chapter6/DetLine · maxPowMap_comp`, and $\lambda(\Phi)\neq 0$ iff Φ is bijective `Chapter6/DetLine · maxPowMap_ne_zero_iff`; $\lambda^*(\Phi)$ is its dual `Chapter6/DetLine · maxPowDualMap` (pp. 203–204).

Definition 6.3 (V6.3). $\iota:\lambda(E^*)\to\lambda^*(E)$ is $\iota(e_1^*\wedge\cdots\wedge e_n^*)(a_1\wedge\cdots\wedge a_n)=\det(e_i^*(a_j))$ `Chapter6/DetLine · iotaDual` `Chapter6/DetLine · iotaDual_wedge_wedge`. The formula is alternating and multilinear in both families, so it descends to the exterior powers; that is the text's well-definedness. ι is a nonzero map of lines `Chapter6/DetLine · iotaDual_ne_zero`, and for a basis with its dual basis $\iota(e_1^*\wedge\cdots\wedge e_n^*)=(e_1\wedge\cdots\wedge e_n)^*$ `Chapter6/DetLine · iotaDual_wedge_dualBasis`, since the matrix is the identity. The text's matrix $(e_i^*(a_j))$ is the transpose of the one of Mathlib's pairing, with the same determinant (p. 204).

Proposition 6.4 (V6.4) `Chapter6/DetLine · proposition_6_4`. For an isomorphism $\Phi:E\to F$, $\lambda^*(\Phi)\circ\iota=\iota\circ\lambda(\Phi^*)$. Both sides, evaluated at $f_1^*\wedge\cdots\wedge f_n^*$ and $a_1\wedge\cdots\wedge a_n$, give $\det(f_i^*(\Phi a_j))$: the left side by the definition of $\lambda^*(\Phi)$, the right side because $(\Phi^*f_i^*)(a_j)=f_i^*(\Phi a_j)$ (p. 205).

VI.3 Four-term exact sequences: Φ_E, Lemma 6.5, Proposition 6.6

The data of (91) (V6.5). Let $\mathbf{E} : 0 \to A \xrightarrow{\alpha} B \xrightarrow{\beta} C \xrightarrow{\gamma} D \to 0$ be an exact sequence of finite-dimensional spaces `Chapter6/ExactSequence · FourTermExact`. A *choice* `Chapter6/ExactSequence · FourTermExact.Choice` consists of a complement Z of $\alpha(A)$ in B, a complement V of $\beta(B)$ in C, bases a of A, d of D, b' of Z, and the γ-preimages c in V of the basis d. Then $\alpha(a), b'$ is a basis of B, and $c, \beta(b')$ is one of C, because β maps Z isomorphically onto $\beta(B)$ and γ maps V isomorphically onto D. Formula (91) assigns to the choice the value $(\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}) \otimes (c_1 \wedge \cdots \wedge c_l \wedge \beta(b'_1) \wedge \cdots \wedge \beta(b'_{m-n}))^*$ as the image of $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, with the text's two ordering conventions `Chapter6/ExactSequence · FourTermExact.Choice.value`. $\Phi_E : \lambda(A) \otimes \lambda^*(D) \to \lambda(B) \otimes \lambda^*(C)$ is the linear map of lines defined by one fixed choice `Chapter6/ExactSequence · FourTermExact.canonicalChoice` `Chapter6/ExactSequence · FourTermExact.phi`; it is nonzero `Chapter6/ExactSequence · FourTermExact.phi_ne_zero`, hence an isomorphism `Chapter6/ExactSequence · FourTermExact.phiEquiv` (p. 206).

Lemma 6.5 (V6.6) `Chapter6/ExactSequence · FourTermExact.lemma_6_5`. The text: Φ_E does not depend on the choices. The formalization proves the strongest form: Φ_E, defined from one choice, maps the source element $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$ of *any other* choice to the right-hand side of (91) computed with that choice. This is the form every later computation uses — γ_T^P is always evaluated in bases adapted to the problem at hand (Findings, Chapter 6, item 1). The proof is §6.7.1 (§VI.28).

Proposition 6.6 (V6.7). With the contraction $\bar{\iota}(v \otimes b \otimes \varphi) = \varphi(v)\, b$ `Chapter6/ExactSequence · contractDual`, (92) defines $\Psi_E(c \otimes h) = \bar{\iota}(c \otimes \Phi_E(h))$ `Chapter6/ExactSequence · FourTermExact.psi`. The proposition `Chapter6/ExactSequence · FourTermExact.proposition_6_6`: Ψ_E maps the wedge of the basis $c, \beta(b')$ of C, tensored with $(a_1 \wedge \cdots \wedge a_n) \otimes (d_1 \wedge \cdots \wedge d_l)^*$, to $\alpha(a_1) \wedge \cdots \wedge \alpha(a_n) \wedge b'_1 \wedge \cdots \wedge b'_{m-n}$. By Lemma 6.5 in its strong form, Φ_E of the source is the value of (91) for this choice, and the dual wedge evaluated on the wedge is 1. The wedge in C is taken in the order of (91); the statement exchanges the names of its two blocks (possible misprint; Findings, Chapter 6, item 2) (p. 207).

VI.4 Determinants of Fredholm operators, good left-projections, and Lemma 6.10

Definition 6.7 (V6.8). $\det(T) = \lambda(\ker T) \otimes \lambda^*(F/R(T))$ `Chapter6/FredholmDet · linDet` `Chapter6/FredholmDet · fredholmDet`, a line for T Fredholm `Chapter6/FredholmDet · finrank_fredholmDet`. Equal operators have identified determinants `Chapter6/FredholmDet · linDetEquivOfEq`: PQT and PT are the same operator when $P \le Q$, but not the same expression, and Proposition 6.12 compares their determinants. The alternative definition $\lambda(\ker T) \otimes \lambda(\mathrm{coker}(T)^*)$ mentioned after the definition is not formalized (gap G.57) (p. 208).

Definition 6.8 (V6.9). An orientation of T is a ray of $\det(T)$ `Chapter6/FredholmDet · FredholmOrientation`; there is one `Chapter6/FredholmDet · nonempty_fredholmOrientation`, and exactly two `Chapter6/FredholmDet · exists_two_fredholmOrientation` (p. 208).

Definition 6.9 and the facts after it (V6.10). A *good left-projection* for T is a bounded projection $P : F \to F$ with $\dim F/R(P) < \infty$ and $R(PT) = R(P)$ `Chapter6/GoodProjections · IsGoodLeftProj`; Π_T is their set `Chapter6/GoodProjections · goodLeftProjs`. $P \le Q$ iff $P = PQ = QP$ `Chapter6/GoodProjections · ProjLE`, a partial order `Chapter6/GoodProjections · ProjLE.refl` `Chapter6/GoodProjections · ProjLE.antisymm` `Chapter6/GoodProjections · ProjLE.trans`. A good left-projection is Fredholm of index 0 — $\ker P$ is a complement of $R(P)$, of dimension $\dim F/R(P)$ `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm` `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_eq_zero` — so PT is Fredholm with $\mathrm{ind}(PT) = \mathrm{ind}(T)$ when T is, by the index of a composite `Chapter6/GoodProjections · IsGoodLeftProj.isFredholm_comp` `Chapter6/GoodProjections · IsGoodLeftProj.fredholmIndex_comp`. For $P, P' \in \Pi_T$, $R(P) \cap R(P')$ has finite codimension `Chapter6/GoodProjections · IsGoodLeftProj.finiteDimensional_quotient_inf`. And $\Pi_T \neq \emptyset$ for T Fredholm

Chapter6/GoodProjections · exists_isGoodLeftProj — the projection onto $R(T)$ along a finite-dimensional complement is one — which the text uses from Lemma 6.11 on without stating it (Findings, Chapter 6, item 3) (p. 208).

Lemma 6.10 (V6.11) Chapter6/GoodProjections · lemma_6_10. *For $P, P' \in \Pi_T$ there is $P'' \in \Pi_T$ with $P'' \le P$ and $P'' \le P'$*, for any bounded T. The text's P'' projects onto $R(P) \cap R(P')$ along a complement assembled from $\ker P$, $\ker P'$ and a further piece, and $P'' = P''P$ needs $\ker P \subseteq \ker P''$. As we read it, this does not hold in general: on $\mathbb{R}^2$ with $T = 1$, let P project onto the x-axis along the y-axis and P' along the diagonal; $R(P) \cap R(P')$ is the x-axis, and a projection onto it below both would equal both Chapter6/GoodProjections · lemma_6_10_text_reading. The lemma holds with a smaller range. *Step 1.* $K = \ker P + \ker P'$ is finite-dimensional, each kernel having the dimension of a finite-dimensional quotient. *Step 2.* K has a closed complement N (finite-dimensional subspaces are complemented, by Hahn–Banach), and $R'' = R(P) \cap R(P') \cap N$ is closed, of finite codimension, and meets K only in 0. *Step 3.* R'' has a complement $C \supseteq K$, topological because the quotient by R'' is finite-dimensional; P'' is the projection onto R'' along C. *Step 4.* $\ker P \subseteq C = \ker P''$ gives $P''P = P''$, and $R'' \subseteq R(P)$ gives $PP'' = P''$; the same for P'. *Step 5.* $R(P''T) = P''(R(PT)) = P''(R(P)) = R''$, since P'' is the identity on $R'' \subseteq R(P)$. Nothing uses that T is Fredholm. The 2017 version drops the lemma and remarks that a common lower bound need not exist; with the order of Definition 6.9, used by both versions, it always exists, as we read it, so the preprint's route to Lemma 6.25 is sound and is the one formalized (Findings, Chapter 6, item 4) (pp. 208–209).

VI.5 Lemma 6.11, γ_T^P, Proposition 6.12 and Corollary 6.13

Lemma 6.11 (V6.12). For $P \in \Pi_T$, the sequence $\mathbf{E}(T,P) : 0 \to \ker T \to \ker(PT) \to F/R(P) \to \operatorname{coker} T \to 0$, with the inclusion Chapter6/ProjectionSequence · projSeqJ, $x \mapsto Tx + R(P)$ Chapter6/ProjectionSequence · projSeqPhi and $y + R(P) \mapsto (I-P)y + R(T)$ Chapter6/ProjectionSequence · projSeqPi, is exact Chapter6/ProjectionSequence · lemma_6_11, a four-term exact sequence of §VI.3 Chapter6/ProjectionSequence · projSeq. The four checks use only $P^2 = P$ and $R(PT) = R(P)$. At $\ker(PT)$: if $PTx = 0$ and $Tx \in R(P)$, then $Tx \in R(P) \cap \ker P = 0$. At $F/R(P)$: the last map is well defined because $(I-P)R(P) = 0$; it kills $[Tx]$ for $x \in \ker(PT)$, since $(I-P)Tx = Tx$; and if $(I-P)y = Tx$, then $PTx = 0$ and $y - Tx = Py \in R(P)$. Onto: $Py = PTe$ for some e, and then $(I-P)(y - Te) = y - Te$, which has the class of y in $\operatorname{coker} T$ (p. 209).

The isomorphism γ_T^P (V6.13). $\gamma_T^P : \det(T) \to \det(PT)$ is $\Phi_{\mathbf{E}(T,P)}$ followed by the identification of $\lambda^*(F/R(P))$ with $\lambda^*(F/R(PT))$ Chapter6/ProjectionSequence · gammaProj Chapter6/ProjectionSequence · gammaProj_tmul — the text's "recalling $R(P) = R(PT)$" (Findings, Chapter 6, item 5). By Lemma 6.5 it is given on wedges by (91) for any choice Chapter6/ProjectionSequence · gammaProj_wedge_tmul (p. 210).

Proposition 6.12 (V6.14). *For $P \le Q$ in Π_T, QT is Fredholm, $P \in \Pi_{QT}$, and $\gamma_{QT}^P \circ \gamma_T^Q = \gamma_T^P$*, up to the identification of $\det(P(QT))$ with $\det(PT)$ Chapter6/ProjectionSequence · proposition_6_12. $P \in \Pi_{QT}$ because $R(PQT) = R(PT) = R(P)$ Chapter6/ProjectionSequence · IsGoodLeftProj.of_projLE. The identity is a computation in adapted bases Chapter6/ProjectionSequence · comp_comp_eq_of_projLE, the proof of §6.7.2, retold in §VI.29 (p. 211).

Corollary 6.13 (V6.15) Chapter6/ProjectionSequence · corollary_6_13. For $P \le S$ and $P \le R$ in Π_T, $(\gamma_{RT}^P)^{-1} \circ \gamma_{ST}^P = \gamma_T^R \circ (\gamma_T^S)^{-1}$. Proposition 6.12 twice gives $\gamma_{ST}^P \gamma_T^S = \gamma_T^P = \gamma_{RT}^P \gamma_T^R$; rearranging gives the claim. So the left side, the comparison of two local frames through a common lower bound, does not depend on the lower bound (p. 211).

VI.6 The stabilization: Lemma 6.14, formula (94), Proposition 6.15, Definition 6.16

Lemma 6.14 (V6.16). For T Fredholm and $\varphi : \mathbb{R}^n \to F$ with $T_\varphi(e,r) = Te + \varphi r$ onto, the sequence $0 \to \ker T \xrightarrow{j} \ker T_\varphi \xrightarrow{\pi} \mathbb{R}^n \xrightarrow{[\varphi]} F/R(T) \to 0$, with $j(e) = (e,0)$ `Chapter6/Stabilization · stabJ`, $\pi(e,r) = r$ `Chapter6/Stabilization · stabPi` and $[\varphi](r) = \varphi(r) + R(T)$ `Chapter6/Stabilization · stabClass`, is exact `Chapter6/Stabilization · lemma_6_14` `Chapter6/Stabilization · stabSeq`. j is injective; $\ker\pi \cap \ker T_\varphi$ is $\{(e,0) \mid Te = 0\} = j(\ker T)$; for $(e,r) \in \ker T_\varphi$, $\varphi r = -Te \in R(T)$, and conversely $\varphi r = -Te$ puts (e,r) in $\ker T_\varphi$; $[\varphi]$ is onto because T_φ is. A stabilizing φ exists for every Fredholm T `Chapter6/Stabilization · exists_surjective_coprod`: send a basis of $\mathbb{R}^n$ to representatives of a basis of the cokernel. The proof writes $j(e) = (0,e)$, read as $(e,0)$ (possible misprint; Findings, Chapter 6, item 8) (pp. 211–212).

Formula (94) (V6.17). $\psi : \lambda^*(\mathbb{R}^n) \to \lambda^*(\{0\})$ maps $(e_1 \wedge \cdots \wedge e_n)^*$ to 1^* for the standard basis `Chapter6/Stabilization · stdWedge` `Chapter6/Stabilization · stabOne` `Chapter6/Stabilization · stabPsi`. Then $\psi((c_1 \wedge \cdots \wedge c_n)^*) = \det[c_1,\dots,c_n]^{-1}\, 1^*$ for every basis `Chapter6/Stabilization · stabPsi_dualVec`: $c_1 \wedge \cdots \wedge c_n = \det[c]\, e_1 \wedge \cdots \wedge e_n$, and the dual vectors scale inversely (p. 212).

Proposition 6.15 (V6.18). Since T_φ is onto, $\det(T_\varphi) = \lambda(\ker T_\varphi) \otimes \lambda^*(\{0\})$, and $\iota_\varphi = (1 \otimes \psi) \circ \Phi_{\mathbf{E}}$, for the sequence $\mathbf{E}$ of Lemma 6.14, is an isomorphism $\det(T) \to \det(T_\varphi)$ `Chapter6/Stabilization · stabIota` `Chapter6/Stabilization · stabIota_bijective`. The formula `Chapter6/Stabilization · proposition_6_15`: for a basis a of $\ker T$, vectors $\bar c_1,\dots,\bar c_m$ with $[\varphi](\bar c_i)$ a basis of $F/R(T)$, and $\bar b_i = (b_i, r_i)$ completing $(a,0)$ to a basis of $\ker T_\varphi$, $\iota_\varphi(h) = \det(\bar c, r)^{-1}\,((a_1,0) \wedge \cdots \wedge (a_k,0) \wedge \bar b_1 \wedge \cdots \wedge \bar b_l) \otimes 1^*$. These data are a choice of (91) for $\mathbf{E}$ — Z the span of the $\bar b_i$, V the span of the $\bar c_j$ — so Lemma 6.5 gives $\Phi_{\mathbf{E}}(h) = ((a,0) \wedge \bar b) \otimes (\bar c \wedge r)^*$, and (94) turns the second factor into $\det(\bar c, r)^{-1} 1^*$. To form $\det(\bar c, r)$ one needs $m + l = n$: the statement takes it as a hypothesis, and it is proved to hold always `Chapter6/Stabilization · stab_card_eq`; that the r_i and $\bar c_j$ form a basis of $\mathbb{R}^n$, which the text notices, is proved — the r_i span $\pi(\ker T_\varphi) = \ker[\varphi]$, and the $\bar c_j$ a complement of it `Chapter6/Stabilization · isCompl_range_stabPi_span`. The statement calls ψ "φ" (possible misprint). The identification (95) and the maps $V \otimes \lambda(\mathbb{R}^n) \to V$, $V \otimes \lambda^*(\mathbb{R}^n) \to V$ after the proposition are not formalized; the formalized statements use only (94) (gap G.58; Findings, Chapter 6, items 9, 10) (p. 213).

Definition 6.16 (V6.19). For Fredholm T, S, the operator $T \oplus S$ has $\ker(T \oplus S) \cong \ker T \times \ker S$ and $\operatorname{coker}(T \oplus S) \cong \operatorname{coker} T \times \operatorname{coker} S$, through the inclusions of the factors `Chapter6/DirectSumDet · kerInl` `Chapter6/DirectSumDet · kerInr` `Chapter6/DirectSumDet · cokerInl` `Chapter6/DirectSumDet · cokerInr` `Chapter6/DirectSumDet · kerProdEquiv` `Chapter6/DirectSumDet · cokerProdEquiv`; the dimensions add `Chapter6/DirectSumDet · finrank_ker_prodMap` `Chapter6/DirectSumDet · finrank_coker_prodMap`, so $T \oplus S$ is Fredholm `Chapter6/DirectSumDet · isFredholm_prodMap`. The map $\det(T) \otimes \det(S) \to \det(T \oplus S)$, $h \otimes h' \mapsto (a \wedge a') \otimes (d \wedge d')^*$, is defined by the formula in one choice of bases `Chapter6/DirectSumDet · directSumVec` `Chapter6/DirectSumDet · directSumDet`, satisfies it for every choice `Chapter6/DirectSumDet · directSumVec_eq_smul` `Chapter6/DirectSumDet · definition_6_16` — a change of bases multiplies source and target by the same product of determinants, the block wedges by the block determinants — and is an isomorphism `Chapter6/DirectSumDet · directSumDet_bijective`. "Natural" is read as the formula for every choice, as in Lemma 6.5 (Findings, Chapter 6, item 11) (p. 214).

VI.7 Local determinant bundles: Lemmas 6.17–6.19, Proposition 6.20, Lemma 6.21

Lemma 6.17 (V6.20). The Fredholm operators form a subspace of $L(E,F)$ `Chapter6/LocalDetBundle · FredholmOp`. *For a bounded T, $P \in \Pi_T$ and a topological complement Y of* $\ker(PT)$, *there is $\varepsilon > 0$ such that for $\|S - T\| < \varepsilon$, $P \in \Pi_S$ and $PS|_Y : Y \to R(P)$ is a topological isomor-*

phism Chapter6/LocalDetBundle · lemma_6_17. *Step 1.* $PT|_Y$ is injective, since $Y \cap \ker(PT) = 0$, and onto $R(P)$, since $PT(E) = PT(Y)$ and $R(PT) = R(P)$; as a bijection between the Banach spaces Y and $R(P)$ (both closed) it is an isomorphism by the open mapping theorem. *Step 2.* $\|PS|_Y - PT|_Y\| \le \|P\|\,\|S - T\|$ Chapter6/LocalDetBundle · projRestrict, and the invertible operators form an open set. *Step 3.* $R(PS) \supseteq PS(Y) = R(P)$, so $P \in \Pi_S$. That T is Fredholm is not used (Findings, Chapter 6, item 20); a topological complement is an algebraic complement with a continuous projection (p. 214).

Lemma 6.18 (V6.21). A *local determinant chart* is the data (T, P, Y, ε) of Lemma 6.17 with its conclusions Chapter6/LocalDetBundle · LocalDetChart. For $k \in \ker(PT)$, $y(k,S) = -(PS|_Y)^{-1}PSk$ Chapter6/LocalDetBundle · LocalDetChart.ySol is continuous in (k,S), inversion being continuous, and $k + y(k,S) \in \ker(PS)$. The map $(S,k) \mapsto (S, k + y(k,S))$ Chapter6/LocalDetBundle · LocalDetChart.kerMap is onto $\bigcup_{\|S-T\|<\varepsilon}\{S\} \times \ker(PS)$, inverted by the continuous projection onto $\ker(PT)$ along Y Chapter6/LocalDetBundle · LocalDetChart.kerProj: for $w \in \ker(PS)$ write $w = k + y$ with $y \in Y$; then $PSy = -PSk$, so $y = y(k,S)$ Chapter6/LocalDetBundle · LocalDetChart.lemma_6_18. "Is a trivial vector bundle" is stated as this trivialization and its inverse (Findings, Chapter 6, item 21) (p. 215).

Lemma 6.19 and Proposition 6.20 (V6.22). For a basis k of $\ker(PT)$, $k_i(S) = k_i + y(k_i,S)$ is a continuous frame of the kernels Chapter6/LocalDetBundle · LocalDetChart.kerFrame, and with a constant basis e of $F/R(P) = F/R(PS)$, $\omega(S) = (k_1(S) \wedge \cdots \wedge k_n(S)) \otimes (e_1 \wedge \cdots \wedge e_l)^*$ Chapter6/LocalDetBundle · LocalDetChart.frame is a nowhere-zero frame Chapter6/LocalDetBundle · LocalDetChart.frame_ne_zero of $\mathrm{DET}(T,P,\varepsilon) = \bigcup_S \det(PS)$. The frames of two charts with the same P differ by a continuous nowhere-zero factor Chapter6/LocalDetBundle · proposition_6_20: the determinant of the matrix expressing one kernel frame in the other, continuous because both frames are continuous in S, times the constant change of the bases of $F/R(P)$. Lemma 6.19 — exterior powers, duals, sums and tensor products of topological vector bundles — is not formalized in general (gap G.59): it is applied only to trivial bundles, and Proposition 6.20 is proved with the explicit frames; "in a natural way" is read as this relation between frames (Findings, Chapter 6, item 22) (p. 216).

Lemma 6.21 (V6.23). *For $Q \le P$, γ^Q_{PS} maps the P-frame at S to a continuous nowhere-zero multiple of the Q-frame, on the common ball of any two local determinant charts* Chapter6/LocalDetBundle · LocalDetChart.gammaProj_frame_eq Chapter6/LocalDetBundle · lemma_6_21. The text's sketch — extend the kernel sections of PS to sections spanning $\ker(QS)$ — is the proof. For a basis $f_1, \ldots$ of $R(P) \cap \ker Q$, $z_m(S) = (PS|_Y)^{-1} f_m$ is continuous and $PSz_m(S) = f_m$; the $k_i(S)$ and $z_m(S)$ span $\ker(QPS) = \ker(QS)$; and in the choice of (91) for $\mathbf{E}(PS, Q)$ these vectors go with the *constant* classes $[f_m]$. So $\gamma^Q_{PS}(\omega_P(S))$ is $(k(S) \wedge z(S)) \otimes (e \wedge [f])^*$, continuous in S, and its ratio to the Q-frame of the other chart is the determinant of a continuous change of basis. The lemma is stated for two charts with different centres and radii, the form Lemma 6.25 uses (Findings, Chapter 6, item 23) (pp. 216–217).

VI.8 $\mathrm{DET}(E,F)$: Definition 6.22, Propositions 6.23 and 6.24, Lemma 6.25

The prebundle (V6.24). The fibre over a Fredholm S is $\det(S)$ Chapter6/DetBundle · DetFiber. A local determinant chart gives the pretrivialization $(S,h) \mapsto (S, \omega(S)^*(\gamma^P_S h))$ over its ball, with model fibre $\mathbb{R}$ Chapter6/DetBundle · LocalDetChart.pretriv Chapter6/DetBundle · LocalDetChart.coord — the text's $\gamma_{(T,P,\varepsilon)}$ read in the frame of Proposition 6.20 (Findings, Chapter 6, item 24). Every Fredholm operator is the centre of one Chapter6/DetBundle · exists_localDetChart: $\Pi_T \neq \emptyset$ (§VI.4) and the finite-dimensional $\ker(PT)$ has a topological complement. The charts form a Mathlib vector prebundle Chapter6/DetBundle · detPrebundle, which needs exactly that the coordinate changes be continuous, Lemma 6.25 below; its topology and bundle structure are those of $\mathrm{DET}(E,F)$ (p. 217).

Lemma 6.25 (V6.28), proved first. *The transition between two charts is $(S,t) \mapsto (S, g(S)t)$ with g continuous and nowhere zero on the common ball* Chapter6/DetBundle · lemma_6_25_local Chapter6/DetBundle · lemma_6_25. The

proof is the text's: a common lower bound $Q \le P, P'$ (Lemma 6.10, §VI.4); Corollary 6.13 to pass from γ^P to $\gamma^{P'}$ through Q; and Lemma 6.21 for each chart, which turns both frames into multiples of Q-frames (p. 218).

Definition 6.22, Propositions 6.23 and 6.24. $\mathcal{B}$ is the collection of sets lying over the ball of some chart whose image in ball $\times\,\mathbb{R}$ is open (V6.25) `Chapter6/DetBundle · detBasis`. It is a basis of the topology of the prebundle (V6.26) `Chapter6/DetBundle · proposition_6_23`: its members are open, the charts being homeomorphisms onto ball $\times\,\mathbb{R}$, and every open set is the union of its intersections with chart domains, which are members; so the topology $\mathcal{B}$ generates is that one. $\mathrm{DET}(E,F)$ is a topological vector bundle with model fibre $\mathbb{R}$, and every $\gamma_{(T,P,\varepsilon)}$, read in its frame, is a trivialization (V6.27) `Chapter6/DetBundle · proposition_6_24` (p. 218). The remark that $\mathrm{DET}(E,F)$ is not orientable and the classical case at the opening of §6.5 are not formalized (gap G.60; Findings, Chapter 6, item 26).

VI.9 Orientations of line bundles, Proposition 6.26, Definition 6.27

Orientations of line bundles (V6.29). An orientation of a topological line bundle is a choice of rays of the fibres given near every point by a continuous nowhere-zero section `Chapter6/LineBundleOrientation · IsLineOrientation` `Chapter6/LineBundleOrientation · LineOrientation`. Two orientations that agree on one fibre agree over a preconnected base `Chapter6/LineBundleOrientation · LineOrientation.eq_of_apply_eq`: the set where they agree is open and closed. The rays of the fibres form a two-sheeted covering `Chapter6/LineBundleOrientation · OrCover` `Chapter6/LineBundleOrientation · isCoveringMap_orCover`, and an orientation is a continuous section of it; over a simply connected, locally path-connected base the lifting criterion extends every orientation of one fibre `Chapter6/LineBundleOrientation · exists_lineOrientation_apply_eq`, and over a nonempty such base there are exactly two orientations `Chapter6/LineBundleOrientation · exists_two_lineOrientation`. The text gives "since the base space is contractible" as the reason, without the theorem it needs (Findings, Chapter 6, item 25) (p. 219).

Proposition 6.26 (V6.30). Over a nonempty convex set C of Fredholm operators `Chapter6/DetBundle · fredholmIncl`, the restriction $\mathrm{DET}(E,F)|_C$ `Chapter6/DetBundle · DetOn` is a topological line bundle with exactly two orientations `Chapter6/DetBundle · proposition_6_26`: a nonempty convex set of a normed space is contractible, hence simply connected, and locally path-connected. The proposition is stated for every nonempty convex C; $\mathrm{Lin}(f,x)$ is one (§VI.1) (p. 219).

Definition 6.27 (V6.31). At a chart point the members of $\mathrm{Lin}(f,x)$ are Fredholm `Chapter6/LinOrientation · ChartPoint.isFredholm_of_mem_linSet`; $\mathrm{DET}(f,x)$ is the pull-back of $\mathrm{DET}(T_zO, K_z)$ to $\mathrm{Lin}(f,x)$ `Chapter6/LinOrientation · ChartPoint.detLin`, and Or_x^f is the set of its orientations `Chapter6/LinOrientation · ChartPoint.Or`, with exactly two elements `Chapter6/LinOrientation · ChartPoint.exists_two_or`. For each $L \in \mathrm{Lin}(f,x)$, $o \mapsto o(L)$ is a bijection onto the two orientations of $\det(L)$ `Chapter6/LinOrientation · ChartPoint.orEquivRay`: injective because an orientation is determined by one value over the connected base, onto because each ray extends. The category $\mathcal{O}_f$ is read through its objects, the two-element sets, with bijections as morphisms; between two two-element sets there are bijections `Chapter6/OrientationFunctor · nonempty_equiv_of_two`, each determined by one value `Chapter6/OrientationFunctor · equiv_eq_of_apply_eq_of_two`. No negation is set up on the tensor products, so $\{o_x, -o_x\}$ is stated as "exactly two elements" (Findings, Chapter 6, item 44) (pp. 219–220). The comparison of Or_x^f in two charts is §VI.21.

VI.10 Proposition 6.28: the neighbourhoods of the propagation

The text: every smooth point x has an open neighbourhood U which (1) is sc-smoothly contractible, (2) has $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ compact, (3) carries, for every sc-smooth path $\varphi : [0,1] \to U$, an sc$^+$-section $s : U \times [0,1] \to Y$ with $s(\varphi(t), t) = f(\varphi(t))$, and (4) carries sc$^+$-sections $q : U \times [0,1] \to Y$ through

every smooth point of every fibre over U_∞, at every t_0. The proof takes local strong bundle coordinates, $P : K \to O$ with $x = 0$.

The statement formalized (V6.32) `Chapter6/OrientableNbhd · proposition_6_28`. *With sc-smooth bump functions, every smooth x has open $U \subseteq W$ in the domain of one chart of the atlas with* (1′) *any two sc-smooth paths in U with the same end points are sc-smoothly homotopic in W with the end points fixed* `Chapter6/OrientableNbhd · IsScHomotopyRel`; (2) $\{y \in \mathrm{cl}(U) \mid f(y) = 0\}$ *is compact*; (3) *for every sc-smooth family $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n) \subseteq W$ there is an sc⁺-section S of $\pi^*P \to \mathbb{R}^n \times X$ with $(f+S)(q,\Phi(q)) = 0$ for q near $[0,1]^n$, depending on $q \in [0,1]^n$ only through $\Phi(q)$*; (4) *π^*P has sc⁺-sections through all smooth fibre points.* A path "in U" is one with $\gamma([0,1]) \subseteq U$.

The neighbourhoods. Take a chart c of the atlas at x, $u_0 = \varphi(x)$, the retraction r on $U_c \subseteq C$ with $O = r(U_c)$. Choose δ with $B_\delta \cap C \subseteq U_c$ (the domain of the retraction is relatively open); δ_2 with $r(B_{\delta_2} \cap U_c) \subseteq B_\delta$ (continuity of r at u_0 on level 0, $r(u_0) = u_0$); a neighbourhood N of x whose closure meets the zeros of f in a compact set `Chapter6/OrientableNbhd · exists_nhds_isCompact_zeros` — at a zero by the local compactness of Theorem 5.6 (§V.7), elsewhere because the zero set is closed and misses a neighbourhood of x; a closed neighbourhood $T \subseteq N$ of x and δ_3 with $\varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$. With $\delta' = \min(\delta, \delta_2, \delta_3)$, $W = \varphi^{-1}(B(u_0,\delta) \cap O)$ and $U = \varphi^{-1}(B(u_0,\delta') \cap O)$.

(1′). Let γ_0, γ_1 be sc-smooth paths in U with the same end points. Since paths are maps on $\mathbb{R}$ that lie in U only for $t \in [0,1]$, first take a C^∞ map $\sigma : \mathbb{R} \to \mathbb{R}$ equal to the identity near $[0,1]$ with values in the open set $\gamma_0^{-1}(U) \cap \gamma_1^{-1}(U)$ `Chapter6/OrientableNbhd · exists_contDiff_eq_id_near_Icc_mapsTo`. Then $\Phi(\tau,t) = \varphi^{-1}\big(r\big((1-\kappa(\tau))\,\varphi(\gamma_0(\sigma(t))) + \kappa(\tau)\,\varphi(\gamma_1(\sigma(t)))\big)\big)$. The convex combination lies in the convex set $B_{\delta'} \cap C$, hence in U_c; r maps it into $B_\delta \cap O$, so Φ takes values in W; it is sc-smooth as a composite of sc-smooth maps in the chart; at $\tau = 0, 1$ it is γ_0, γ_1 on $[0,1]$, because σ is the identity there and r fixes O; and the end points stay fixed, because γ_0 and γ_1 agree there. The text's contraction $\hat r(t,y) = r(ty)$ contracts $r(V)$ for a convex V; for $U = r(V')$ it needs $r(V') \subseteq V'$, which is not stated, and a convex domain of the retraction is not available in general. As we read it, (1) is not established for U (gap G.61; Findings, Chapter 6, item 45). What Theorem 6.35 uses is (1′).

(2). $\mathrm{cl}(U) \subseteq T \subseteq \mathrm{cl}(N)$, since $U \subseteq \varphi^{-1}(B_{\delta_3} \cap O) \subseteq T$ and T is closed; so the zeros in $\mathrm{cl}(U)$ are a closed subset of a compact set.

(3), sections along families `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_family`. This is the text's $s(y) = R(y, f(x))$ made to depend on the parameter. An sc-smooth map on the finite-dimensional $\mathbb{R}^n$ takes values in the smooth points, and the principal part $q \mapsto \hat f(\Phi(q))$ lifts to a C^∞ map into the level-1 fibre model. In the chart, the section over (q,y) whose fibre coordinate is the projection onto $K_{\varphi(y)}$ of that lift of $-\hat f(\Phi(q))$ — after a reparametrization of the parameter, equal to the identity near $[0,1]^n$, that keeps the family in the chart domain — is an sc⁺-section of the pull-back over the chart domain, and at $y = \Phi(q)$ it is $-f(\Phi(q))$. Cut off by an sc-smooth function on X supported in the chart domain and equal to 1 near a compact set containing $\Phi([0,1]^n)$ `Chapter6/OrientableNbhd · exists_isScSmoothMap_eq_one_nhdsSet`, it becomes a global sc⁺-section, with $f + S = 0$ at the points $(q, \Phi(q))$ where the cut-off is 1 `Chapter6/OrientableNbhd · exists_isScPlusSection_pullback_of_mapsTo_of_bump`. For $q \in [0,1]^n$ it is the projection of $-\hat f(\Phi(q))$, so it depends on q only through $\Phi(q)$. The text's (3) is for paths; it is stated for families because the homotopies of Theorem 6.35 need it (Findings, Chapter 6, item 46).

(4) `Chapter6/OrientableNbhd · hasScPlusSectionsThrough_pullback_rn_of_bump`. Products of bump functions on $\mathbb{R}^n$ and on X are bump functions on $\mathbb{R}^n \times X$, and they give the sections. $\mathbb{R}^n \times X$ `Chapter2/FibredProduct · MPolyfold.paracompactSpace_prod` and the total space of π^*P `Chapter2/StrongBundlePullback · paracompactSpace_pullbackSpace` are paracompact, as the project's M-polyfolds and strong bundles must be. *Hidden hypotheses:* bump functions, for the global sections of (3) and (4), where the text assumes partitions of unity from §6.2 on. The construction uses neither that x is smooth nor that X is tame (pp. 220–221).

VI.11 Lemma 6.29: general position along a family

The text (pp. 221–222): along a path φ with a section $s(\cdot,t)$, the section $F(t,y,\lambda) = f(y) - s(y,t) + \sum \lambda_i s_i(y,t)$ of the bundle over $[0,1]\times\mathbb{R}^k\times X$; Lemma 6.29 asserts that there are $s_1,\dots,s_k$ with property (P): for every t, the slice F_t is in general position to the boundary of $X\times\mathbb{R}^k$ at $(\varphi(t),0)$. The proof is the sentence "using the results from the transversality theory".

The statement formalized (V6.33) `Chapter6/Lemma629 · lemma_6_29`. *Over a tame X with bump functions, for f with tame germ data, an sc-smooth $\Phi:\mathbb{R}^n\to X$ and an sc⁺-section S of π^*P with $(f+S)(q,\Phi(q))=0$ near $[0,1]^n$, there are finitely many sc⁺-sections $s_1,\dots,s_k$ of π^*P such that $F(\lambda,(q,y)) = f(y)+S(q,y)+\sum\lambda_i s_i(q,y)$ is in general position at $(0,(q,\Phi(q)))$ for all q near $[0,1]^n$.*

The proof is that of Theorem 5.18 (§V.20). The points $(0,(q,\Phi(q)))$ are zeros of F `Chapter6/Lemma629 · isZeroAt_paramSection_zero`, and for q in a compact neighbourhood of $[0,1]^n$ they form a compact set. At each of them finitely many sections make the parametrized section satisfy properties (i)–(ii) of §V.19; general position is open (§V.18); finitely many neighbourhoods cover the compact set, and the sections of all of them together do the job `Chapter6/Lemma629 · exists_sections_isGeneralPositionGerm_of_isCompact`. *Hidden hypotheses:* X tame, bump functions, tame germ data, the standing hypotheses of §6.5 and of Theorem 5.18. The conclusion is the general position of F at the points $(t,\varphi(t),0)$, which the text derives from (P) and which Lemmas 6.30 and 6.31 use. (P) itself, the general position of each slice F_t, is not stated (gap G.63); as we read it, it follows, since the tangent vectors of the family of zeros lie in $\ker F'$ and $\mathbb{R}^n$ has no boundary, but this chart computation is not carried out (Findings, Chapter 6, item 47).

VI.12 Lemma 6.30: kernels as continuous families of subspaces

The text (p. 222): near $(t,\varphi(t),0)$ the solution set of F is a manifold with boundary with corners, $\pi(t,y,\lambda)=t$ restricted to it is a submersion, and the kernels L_t form a smooth vector bundle L over $[0,1]$. What the propagation needs is less: the kernels form a continuous family, so that an orientation of one of them can be continued to the others.

The kernels (V6.34). At a zero with surjective linearization and germ data of index d, $\dim\ker = d$ `Chapter6/KernelFamily · StrongBundle.IsSurjLinOfIndex.finrank_linKer`. Near a zero in good position the kernel is the tangent space of the solution set (Theorem 3.54, §III.45), the range of the derivative of an sc-retraction onto the solution set read in the chart. So along a continuous family y_q of zeros in good position, read in one chart at continuously varying level-1 points z_q, near every q_0 the vectors $DS(z_q)b_i$, for a basis b of the kernel at q_0 and that retraction S, are a continuous frame of the kernels `Chapter6/KernelFamily · exists_local_frame_linKer` — continuous because $(z,v)\mapsto DS(z)v$ is, independent near q_0 because independence is open, and spanning because the dimension is constant. The kernels form a continuous family of subspaces of the model space, of dimension the index `Chapter6/KernelFamily · exists_subspaceFamily_linKer`. For an sc-smooth family of zeros $\mathbb{R}^n\to X$ read in a chart `Chapter6/KernelFamily · hasFDerivAt_scDeriv_rn`, differentiating $\hat f(\Phi(q))=0$ shows that the derivative lies in the kernel of the linearization `Chapter6/KernelFamily · fderiv_mem_linKer`. The bundle L is a continuous family of subspaces; the solution set as a manifold with boundary with corners near $(t,\varphi(t),0)$, the map π and the smooth structure of L are not formalized (gap G.62; Findings, Chapter 6, item 48).

Continuous families of subspaces (V6.35). A family of n-dimensional subspaces L_b of a normed space, spanned near every point by a continuous frame `Chapter6/SubspaceBundle · SubspaceFamily`, has the line bundle $b\mapsto\lambda(L_b)$ `Chapter6/SubspaceBundle · SubspaceFamily.det`, with the wedges of the frames as trivializations `Chapter6/SubspaceBundle · SubspaceFamily.Frame.pretriv`; the coordinate change between two frames is the determinant of the matrix expressing one in the other, continuous because that matrix solves a linear system with continuous coefficients `Chapter6/SubspaceBundle · SubspaceFamily.Frame.coordChange_continuousOn`. Families and orientations restrict along continuous maps `Chapter6/SubspaceBundle · SubspaceFamily.comp` `Chapter6/SubspaceBundle · LineOrientation.comp`. Over $[0,1]$ an orientation of $\lambda(L_0)$ extends uniquely to an orientation of $\lambda(L)$ (§VI.9), and its value at 1 is the continuation

Chapter6/SubspaceBundle · SubspaceFamily.transport Chapter6/SubspaceBundle · SubspaceFamily.transport_eq, injective Chapter6/SubspaceBundle · SubspaceFamily.transport_injective and compatible with restriction Chapter6/SubspaceBundle · SubspaceFamily.transport_comp; over a simply connected, locally path-connected base the continuation along a path depends only on its end points. This is the text's "an orientation of the lines … determines by continuation an orientation of all the other lines" (pp. 222–223), and the bundles $\tilde{L}$ of pp. 224–227. "Vector bundle" is read as a family of subspaces of one model space with continuous local frames.

Families of kernels and their images (V6.36). The kernels of a norm-continuous family of surjective Fredholm operators over a connected base form a continuous family Chapter6/StabilizationContinuity · exists_subspaceFamily_ker: their dimension is the index, and near q_0 the kernel of A_q is the graph over $\ker A_{q_0}$ given by Lemma 6.18 with $P = 1$. The images of a continuous family under a jointly continuous family of linear maps injective on its members form one Chapter6/StabilizationContinuity · SubspaceFamily.exists_map, and continuous orientations are carried along Chapter6/StabilizationContinuity · SubspaceFamily.isLineOrientation_map. Cutting a continuous family by the kernel of a fixed ℓ that is onto on every member gives one Chapter6/StabilizationContinuity · SubspaceFamily.exists_inf_ker. "As τ varies, the bundle varies continuously" (pp. 223–226) is read in the operator norm, and for maps into a model space as joint continuity.

VI.13 The stabilized section in a product chart, and Lemma 6.31

The text (pp. 222–223): with $T = (f - s(\cdot,t))'(\varphi(t))$, the kernel L_t of $F_t'(\varphi(t),0)$ fits into the sequence (96) $0 \to \ker T \to L_t \to \mathbb{R}^k \to \operatorname{coker} T \to 0$, and Lemma 6.31 says that it is exact.

The product chart (V6.37). The parametrized section F lives on $\mathbb{R}^k \times (\mathbb{R}^n \times X)$; it is read in the product chart $(\lambda \mapsto \lambda) \times ((q \mapsto q) \times \varphi)$ at $(0,(q,z))$, whose tangent space and fibre are products Chapter6/ParamKernel · PullbackChartData.ofProd_scDeriv_r_apply Chapter6/ParamKernel · PullbackChartData.ofProd_mem_tangent_iff Chapter6/ParamKernel · PullbackChartData.mem_K_ofProd_iff. There the linearization of F is $(\lambda,(\sigma,v)) \mapsto D(\hat{g}_q \circ r)(z)v + \sum \lambda_i \hat{s}_i(q,\cdot)(z) + D_q\sigma$, with $g_q = f + S(q,\cdot)$ and D_q the derivative in the parameter Chapter6/ParamKernel · StrongBundle.scDeriv_ofProd_slice_apply Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_apply Chapter6/ParamKernel · StrongBundleChart.paramDerivAt Chapter6/ParamKernel · StrongBundle.scDeriv_paramSection_ofProd_ofProd. With the projection to σ and the inclusion of the slice $\sigma = 0$ Chapter6/ParamKernel · sliceProj Chapter6/ParamKernel · sliceIncl, the kernel cut by $\sigma = 0$ is the image of $\ker(T \oplus \varphi)$ Chapter6/ParamKernel · linKer_inf_ker_sliceProj_eq_map Chapter6/ParamKernel · exists_sliceOps, where $T = g_q'(x)$ lies in $\mathrm{Lin}(f,x)$ Chapter6/ParamKernel · mem_linSet_sectionAdd_pullbackSlice and $\varphi(\lambda) = \sum \lambda_i s_i(q,x)$. *$T \oplus \varphi$ is onto when F is in general position at $(0,(q,\Phi(q)))$* Chapter6/ParamKernel · surjective_coprod_of_isLinSurjAt: given w, the surjectivity of F' gives (λ,σ,v) with image w; the derivative of the curve of zeros, $(0,\sigma,D\Phi\,\sigma)$, lies in $\ker F'$ (§VI.12); subtracting it gives $(\lambda,0,v - D\Phi\,\sigma)$ with the same image, a preimage under $T \oplus \varphi$. The kernel is then a closed subspace of the product with the right inclusion Chapter6/ParamKernel · exists_sliceKernel. $\ker F_t'(\varphi(t),0)$ is read as the kernel of the linearization of the whole F cut by $\sigma = 0$ (Findings, Chapter 6, item 49) (pp. 221–223).

Lemma 6.31 (V6.38) Chapter6/ParamKernel · lemma_6_31. With these operators, (96) $0 \to \ker T \to \ker(T \oplus \varphi) \to \mathbb{R}^k \to K_z/R(T) \to 0$ is the stabilization sequence of T and φ (Lemma 6.14, §VI.6), hence exact. The text has c onto by (P); here it is onto by the general position of F (§VI.11) and the derivative of the family of zeros, which absorbs the $\mathbb{R}^n$-directions (p. 223).

VI.14 Continuation, the sign, and the push-forward

The text (pp. 223–224): Φ_{E_t} of (94) identifies $\det(T_t)$ with $\lambda(L_t)$; an orientation o_0 of $\det(T_0)$ gives one of $\lambda(L_0)$, continuation carries it to $\lambda(L_1)$, and $\Phi_{E_1}^{-1}$ carries it to $\det(T_1)$: this defines $o_1 = \varphi_* o_0$.

Continuation (V6.39). Over a simply connected, locally path-connected base the continuation of a ray of $\lambda(L_{q_0})$ to $\lambda(L_{q_1})$ is the value at q_1 of the unique orientation extending it `Chapter6/Continuation · SubspaceFamily.cont` `Chapter6/Continuation · SubspaceFamily.cont_eq`. It is the identity from a point to itself `Chapter6/Continuation · SubspaceFamily.cont_self`, composes `Chapter6/Continuation · SubspaceFamily.cont_trans`, is a bijection of rays `Chapter6/Continuation · SubspaceFamily.contEquiv`, is compatible with pull-back `Chapter6/Continuation · SubspaceFamily.cont_comp`, and along $[0,1]$ it is the continuation of §VI.12 `Chapter6/Continuation · SubspaceFamily.cont_eq_transport`. For a stabilization $T \oplus \varphi$ that is onto, $\det(T \oplus \varphi) \cong \lambda(\ker(T \oplus \varphi)) \otimes \lambda^*(\{0\}) \cong \lambda(\ker(T \oplus \varphi))$ `Chapter6/Continuation · detEquivOfSurjective`, and through ι_φ (§VI.6) and an identification of the kernel with a member of L the rays of $\det(T)$ and of $\lambda(L)$ correspond `Chapter6/Continuation · stabRayEquiv`. The propagation is Φ_{E_0}, then continuation, then $\Phi_{E_1}^{-1}$ on rays `Chapter6/Continuation · propagate`, with the identity on a constant family `Chapter6/Continuation · propagate_self`, composition `Chapter6/Continuation · propagate_trans` and pull-back `Chapter6/Continuation · propagate_comp`. It is stated over any simply connected, locally path-connected base, so that one statement serves the paths and the squares; the fixed identification $\lambda^*(\mathbb{R}^k) \cong \lambda^*(\{0\})$ of (94) occurs at both ends.

The sign (V6.40). In a local determinant chart with projection P, the chart coordinate of $\iota_\psi(x) \in \det(S \oplus \psi)$ is $(-1)^{k \dim F/R(P) + \dim \operatorname{coker} S}\, D(S,\psi)$ times that of x `Chapter6/StabilizationContinuity · coord_stabIota`, with D `Chapter6/StabilizationContinuity · stabD` continuous `Chapter6/StabilizationContinuity · continuousOn_stabD` and nowhere zero `Chapter6/StabilizationContinuity · stabD_ne_zero` near the centre. The first summand of the exponent is constant on a chart, the second is not. So where $\dim \operatorname{coker} T$ changes parity along a family, the rays carried by Φ_E jump. With $\alpha(T) = (-1)^{\dim \operatorname{coker} T}$ `Chapter6/StabilizationContinuity · stabSign`: *for a continuous orientation o of* DET*, a norm-continuous family T_q and a continuous stabilization φ_q with $T_q \oplus \varphi_q$ onto, the rays $\alpha(T_q)\,\Phi(o(T_q))$ of $\lambda(\ker(T_q \oplus \varphi_q))$ form an orientation* `Chapter6/StabilizationContinuity · isLineOrientation_stabRay`. Without the sign they need not `Chapter6/StabilizationContinuity · not_isLineOrientation_stabRay_uncorrected`: $E = F = \mathbb{R}$, $k = 1$, $T_q = q/(1+|q|)$ and $\varphi(r) = r_0$, where $\dim \operatorname{coker} T_q$ is 1 at $q = 0$ and 0 elsewhere. As we read it, the orientations carried by Φ_E of (94) along a continuous family are not continuous where $\dim \operatorname{coker} T$ changes parity, and Lemmas 6.33, 6.34 and the homotopy argument before Theorem 6.35 use that they are. The propagation uses $\alpha(T)\Phi_E$, which on each operator carries the text's orientation or its opposite (Findings, Chapter 6, item 50) (p. 212, as used on pp. 223–226).

The push-forward (V6.41). At a chart point p of a point y `Chapter6/Propagation · chartPt`, a section S with $(f+S)(y) = 0$ and sections $s_1, \dots, s_k$ give $T = (f+S)'(y) \in \operatorname{Lin}(f,y)$ `Chapter6/Propagation · ChartPoint.stabT` `Chapter6/Propagation · ChartPoint.stabT_mem_linSet` and $\varphi(\lambda) = \sum \lambda_i \hat{s}_i(z)$ `Chapter6/Propagation · ChartPoint.stabPhi`, with appended sections giving the appended stabilization `Chapter6/Propagation · ChartPoint.stabPhi_finAppend`; $T \oplus \varphi$ is Fredholm `Chapter6/Propagation · isFredholm_stabCoprod`, and the sign is an involution `Chapter6/Propagation · stabSign_inv` `Chapter6/Propagation · toPerm_stabSign_symm`. A family of *stabilized kernels* `Chapter6/Propagation · IsStabKernels` over a parameter space Q consists of a global sc$^+$-section S of the pull-back with $f + S = 0$ at the points $(q, y(q))$, sections $s_1, \dots, s_k$, a continuous lift of $q \mapsto \varphi(y(q))$ to level 1 of the chart, and a continuous family Λ of subspaces of $E_0 \times \mathbb{R}^k$ such that at every chart point of $y(q)$ in the chart $T_q \oplus \varphi_q$ is onto and Λ_q is the image of its kernel. It restricts along continuous maps `Chapter6/Propagation · IsStabKernels.comp`. Its *push-forward* between two parameters a, b `Chapter6/Propagation · IsStabKernels.push`: $o \in \mathrm{Or}^f_{y(a)}$ gives the ray $o(T_a)$, then $\alpha(T_a)\Phi_{E_a}(o(T_a))$, a ray of $\lambda(\Lambda_a)$; continuation carries it to $\lambda(\Lambda_b)$; $\alpha(T_b)\Phi_{E_b}^{-1}$ carries it to $\det(T_b)$; and $\mathrm{Or}^f_{y(b)} \cong \{\text{rays of } \det(T_b)\}$ (§VI.9) gives the image. It depends only on the data `Chapter6/Propagation · IsStabKernels.push_congr`. With $\mathrm{orRay}_q(o)$, the ray of $\lambda(\Lambda_q)$ read from o at q `Chapter6/Propagation · IsStabKernels.orRay`, $\varphi_* = \mathrm{orRay}_b^{-1} \circ \mathrm{cont}_{ab} \circ \mathrm{orRay}_a$ `Chapter6/Propagation · IsStabKernels.push_eq_orRay`. *Along a path with continuously varying data and a fixed point, continuation carries the rays read at one parameter to those read at the next* `Chapter6/Propagation · IsStabKernels.cont_orRay_path`: the operators T_q, φ_q vary continuously; their kernels form a continuous family (§VI.12); the sign makes the rays $\alpha\Phi(o(T_q))$ an orientation; and the inclusion into $E_0 \times \mathbb{R}^k$ carries that orientation

to $\lambda(\Lambda)$. The push-forward is compatible with pull-back `Chapter6/Propagation · IsStabKernels.push_comp` and with concatenation `Chapter6/Propagation · IsStabKernels.push_trans`. For given end-point operators the result differs from the text's by $\alpha(T_a)\alpha(T_b)$; the sections enter pointwise, and the continuity in the parameter is that of Λ (Findings, Chapter 6, item 50) (p. 224).

VI.15 The stabilized kernels exist; push-forwards along paths

Lemma 6.30 for the push-forward (V6.42) `Chapter6/StabKernelsExist · exists_isStabKernels`. *For an sc-smooth $\Phi : \mathbb{R}^n \to X$ with $\Phi([0,1]^n)$ in the domain of a chart, an sc$^+$-section S with $f + S = 0$ along Φ near $[0,1]^n$, and sections $s_1, \dots, s_k$ as in Lemma 6.29, the stabilized kernels of the slices form a continuous family over $[0,1]^n$.* The family $q \mapsto \varphi(\Phi(q))$ has a continuous lift to level 1 `Chapter6/StabKernelsExist · exists_continuousOn_lift_chart`, the values of an sc-smooth map on $\mathbb{R}^n$ being smooth points. The zeros $(0,(q,\Phi(q)))$ of F are in general position, hence in good position; their kernels form a continuous family (§VI.12); cutting it by $\{d\sigma = 0\}$, a fixed functional onto on every member because the derivative of the family of zeros lies in the kernel, keeps it continuous (§VI.12); and §VI.13 identifies the cut with $\ker(T_q \oplus \varphi_q)$. The deviation is that of §VI.12 (gap G.62).

Push-forwards along paths (V6.43). The parameter points of $[0,1]$ `Chapter6/PushData · pt1_mem_Icc`, level-1 lifts at smooth points `Chapter6/PushData · exists_lift_of_smooth` and the chart points they give `Chapter6/PushData · chartPtOf` are the bookkeeping. *$e : \mathrm{Or}^f_{\gamma(0)} \to \mathrm{Or}^f_{\gamma(1)}$ is a push-forward along an sc-smooth path γ in a chart* `Chapter6/PushData · IsPush` if there are global sc$^+$-sections $S, s_1, \dots, s_k$ of $\pi^*P \to \mathbb{R} \times X$ with $(f+S)(t,\gamma(t)) = 0$ for t near $[0,1]$, and a family of stabilized kernels of their slices along γ whose push-forward is e. Equal data give equal push-forwards `Chapter6/PushData · IsStabKernels.push_eq_of_eq`. Pulled back along an affine reparametrization the sections still vanish along the reparametrized path `Chapter6/PushData · isZeroAt_sectionAdd_reparamSection` and give a push-forward along it `Chapter6/PushData · isPush_reparam`; so a push-forward along γ is the composite of push-forwards along its two halves, each reparametrized to $[0,1]$ `Chapter6/PushData · IsPush.split`. *Along every sc-smooth path in a set with property (3) of Proposition 6.28 there is a push-forward* `Chapter6/PushData · exists_isPush`: (3) gives S, Lemma 6.29 gives $s_1, \dots, s_k$, and the stabilized kernels exist. Global sections of the pull-back stand in place of the text's sections over $U \times [0,1]$, and (P) is used only to construct a push-forward (p. 224).

VI.16 Lemma 6.33 and sections over families

Lemma 6.33 (V6.45) `Chapter6/Propagation · IsStabKernels.push_append`. *Appending sc$^+$-sections $\bar{s}_1, \dots, \bar{s}_d$ after $s_1, \dots, s_k$ does not change the push-forward.* The text proves it with a homotopy $\tilde{F}(\tau, \cdot)$ and "one verifies by a simple computation". The computation is this. For $\ell_1, \dots, \ell_d \in \ker(T \oplus (\varphi, \psi))$ whose ψ-coordinates are the standard basis of $\mathbb{R}^d$, the map $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$ `Chapter6/AddSections · wedgeExt` `Chapter6/AddSections · wedgeExt_wedge` does not depend on the ℓ_j — two choices differ by vectors of the image of $\ker(T \oplus \varphi)$, the span of the first block `Chapter6/AddSections · wedgeExt_congr` — and is bijective `Chapter6/AddSections · wedgeExt_bijective`. With the appended stabilization `Chapter6/AddSections · appendStab` and the kernel inclusion $(v,\lambda) \mapsto (v,\lambda,0)$ `Chapter6/AddSections · kerAppend`, *the stabilization isomorphism of (φ,ψ) is that of φ followed by $w \mapsto (w,0) \wedge \ell_1 \wedge \cdots \wedge \ell_d$, exactly, with no sign* `Chapter6/AddSections · detEquiv_stabIota_append`: a choice of Proposition 6.15 for (φ,ψ) is one for φ with the ℓ_j appended to the kernel basis and the standard basis vectors to the r_i, and with the appended sections last the matrix $(\bar{c}, r)$ becomes block triangular with an identity block, so the determinant in (94) is unchanged. Along a family the ℓ_j are chosen continuously `Chapter6/AddSections · SubspaceFamily.exists_continuous_lifts` — the larger kernels project onto $\mathbb{R}^d$ because $T \oplus \varphi$ is already onto — and the wedge extension carries continuous orientations along `Chapter6/AddSections · isLineOrientation_wedgeExt`, through the kernel inclusion `Chapter6/AddSections · kerEquivMap`. So the stabilized rays `Chapter6/AddSections · stabRayEquiv_append` and the propagations `Chapter6/AddSections · propagate_append` agree, and with them the push-forwards. The homotopy of the text's proof is not needed (Findings, Chapter 6, item 51) (pp. 224–225).

Sections over families (V6.46). For g affine, $(q, y) \mapsto S(g(q), y)$ `Chapter6/FamilySections · reparamSection` is again an sc⁺-section of the pull-back `Chapter6/FamilySections · StrongBundle.IsScPlusSection.reparam`, through the sc-smooth product map $g \times 1$ `Chapter6/FamilySections · rnProdMapOp`, with the expected slices `Chapter6/FamilySections · pullbackSlice_reparamSection`. Sc⁺-sections may be checked in one chart around each point `Chapter6/FamilySections · StrongBundle.isScPlusSection_of_local`. Adding sections keeps the general position of F at $(0, y)$ `Chapter6/FamilySections · StrongBundle.isGeneralPositionGerm_paramSection_of_comp`: the linearization only gains columns, and the properties of Definition 5.17 survive. These are the sections $s(y, \tau, t)$ over a square and the text's "adding more sections, the properties will still hold" (pp. 224–226).

VI.17 Lemma 6.34 and Proposition 6.32

Lemma 6.34 (V6.47) `Chapter6/Propagation · IsStabKernels.push_square`. *For a family of stabilized kernels over the square $[0, 1]^2$ with the end points x_0, x_1 fixed and the end-point operators varying continuously, the push-forwards along the edges $\tau = 0$ and $\tau = 1$ agree.* The edges are the restrictions to affine segments of the square `Chapter6/PropagationWellDefined · sqEdge Chapter6/PropagationWellDefined · sqPath`. On the edge $t = 0$ the point is x_0 and the operators vary continuously, so by §VI.14 continuation along it carries $\mathrm{orRay}_{(0,0)}(o)$ to $\mathrm{orRay}_{(1,0)}(o)$; the same on $t = 1$. The square is simply connected, so continuation from $(0, 0)$ to $(1, 1)$ is the same through $(0, 1)$ and through $(1, 0)$. Writing both push-forwards as $\mathrm{orRay}^{-1} \circ \mathrm{cont} \circ \mathrm{orRay}$ gives the claim. The text's convex homotopy s^τ is part of the square of Proposition 6.32 below, and the continuity of the end-point orientations along τ is that of §VI.14, with the sign (p. 226).

Proposition 6.32 (V6.44) `Chapter6/PropagationWellDefined · proposition_6_32`. *Two push-forwards along the same sc-smooth path in the domain of a chart agree.* Let the data be $(S^0; s^0_1, \dots, s^0_{k^0})$ and $(S^1; s^1_1, \dots, s^1_{k^1})$. The text combines Lemma 6.33 (more sections) and Lemma 6.34 (a homotopy of the data); the formalization does both over one square (Findings, Chapter 6, item 52). *Step 1, the square.* $\Phi(\tau, t) = \gamma(t)$, constant in τ, and $S^\tau = (1 - \tau)S^0 + \tau S^1$, a section of the pull-back over $\mathbb{R}^2 \times X$ by §VI.16; $f + S^\tau$ vanishes along Φ since both $f + S^0$ and $f + S^1$ do and the fibres are linear `Chapter6/PropagationWellDefined · isZeroAt_sectionAdd_combo`. *Step 2, the sections.* With $\widehat{s}^i_j = s^i_j$ for $j < k^i$ and 0 otherwise, the sections $(1-\tau)\widehat{s}^0_j + \tau\widehat{s}^1_j$, followed by the sections that Lemma 6.29 gives on the square. The parametrized section is in general position along the square — adding sections keeps it (§VI.16) — so the stabilized kernels exist over $[0, 1]^2$ (§VI.15). *Step 3, the edges $\tau = 0, 1$.* At $\tau = 0$ the sections s^0 are the first k^0 of the family and at $\tau = 1$ the sections s^1 are the first k^1, so Lemma 6.33 compares each datum with the edge of the square without reordering `Chapter6/PropagationWellDefined · IsStabKernels.push_eq_of_prefix`, the edges being data of the same kind `Chapter6/PropagationWellDefined · IsStabKernels.congr_data Chapter6/PropagationWellDefined · IsStabKernels.push_congr_data`. *Step 4, the edges $t = 0, 1$.* There the point is fixed and the operators are affine in τ `Chapter6/PropagationWellDefined · linAt_principal_combo`, hence continuous `Chapter6/PropagationWellDefined · continuous_principal_pullbackSlice`. *Step 5.* Lemma 6.34 equates the push-forwards along the edges $\tau = 0$ and $\tau = 1$, and Step 3 identifies them with the two given ones. *Hidden hypotheses:* the standing ones of §6.5 (X tame, bump functions, tame germ data). "Does not depend on the choices" is read as the agreement of any two data (pp. 224–226).

VI.18 Theorem 6.35, and paths in a chart

Theorem 6.35 (V6.48). U has the *local propagation property* in a chart c `Chapter6/PropagationWellDefined · HasLocalPropagation` if U lies in the domain of c, along every sc-smooth path in U there is a push-forward, and any two push-forwards along paths in U with the same end points agree. `Chapter6/PropagationWellDefined · theorem_6_35`: *over a tame X with bump functions, for f with tame germ data, every smooth point x has an open $U \ni x$ and a chart c with the local propagation property.* The statement names the bundle and the point but not the section (possible omission); it is read for an sc-Fredholm section with tame germ data (Findings, Chapter 6, item 53). *Step 1. $U \subseteq W$* and c from Proposition 6.28; push-forwards exist along paths in W, a fortiori in U, by (3) (§VI.15). *Step 2, homotopy invariance*

Chapter6/PropagationWellDefined · push_eq_of_homotopy. For an sc-smooth $\Phi : \mathbb{R}^2 \to X$ with $\Phi([0,1]^2) \subseteq W$ and the end points fixed, (3) for $n = 2$ gives S, Lemma 6.29 gives sections, and the stabilized kernels exist over the square. The edge $t \mapsto (\tau, t)$ is an affine map, so the restriction to it is a push-forward datum along $\Phi(\tau, \cdot)$ (§§VI.14–VI.15). Along $t = 0, 1$ the point is constant and so is the section $S(q, \cdot)$, because it depends on q only through $\Phi(q)$; the end-point operators are then constant, and Lemma 6.34 applies. With Proposition 6.32, any push-forward along $\Phi(0, \cdot)$ equals any along $\Phi(1, \cdot)$. *Step 3, the reparametrization.* The homotopy of (1′) is γ_0 and γ_1 at its edges only for $t \in [0, 1]$, while a push-forward datum asks for zeros of $f + S$ for t near $[0, 1]$. The proof therefore passes through $\gamma_i \circ \kappa$, which maps all of $\mathbb{R}$ into $\gamma_i([0, 1])$, with three homotopies whose edges are exact for all t: $H(\sigma, t) = \gamma_0((1 - \sigma)t + \sigma\kappa(t))$ from γ_0 to $\gamma_0 \circ \kappa$; the homotopy $(\tau, t) \mapsto \Phi(\tau, \kappa(t))$ from $\gamma_0 \circ \kappa$ to $\gamma_1 \circ \kappa$; and the analogue of H for γ_1. Step 2 applied three times gives the agreement. U is not shown to be sc-smoothly contractible (gap G.61); what the proof uses is (1′) (pp. 226–227).

Paths in a chart (V6.49). Sc-smooth maps that agree near every point of a set are sc-smooth together Chapter6/ScPaths · MPolyfold.isScSmoothMap_of_eventuallyEq; C^∞ maps of the parameter compose Chapter6/ScPaths · pt1_apply Chapter6/ScPaths · contDiff_pt1_comp. *For an open U in the domain of a chart and $x \in U$ there is an open $V \ni x$ in U whose smooth points are pairwise joined by sc-smooth paths in U* Chapter6/ScPaths · exists_nhds_scPath: in the chart, $t \mapsto \varphi^{-1}(r(u_1 + \kappa(t)(u_2 - u_1)))$, for u_1, u_2 in a small ball around $\varphi(x)$, as in (1′). The concatenation Chapter6/ScPaths · concatPath is $\gamma_1(\kappa(3s))$ for $s \le 1/2$ and $\gamma_2(\kappa(3s - 2))$ for $s > 1/2$ Chapter6/ScPaths · concatPath_of_le Chapter6/ScPaths · concatPath_of_gt. It is constant on $[1/3, 2/3]$, with value $\gamma_1(1) = \gamma_2(0)$, so the two pieces glue sc-smoothly Chapter6/ScPaths · isScSmoothMap_concatPath; it has the right end points Chapter6/ScPaths · concatPath_zero Chapter6/ScPaths · concatPath_one and halves Chapter6/ScPaths · concatPath_half₁ Chapter6/ScPaths · concatPath_half₂. Reparametrizing by a C^∞ map keeps sc-smoothness Chapter6/ScPaths · isScSmoothMap_reparam, and push-forwards compose along concatenations Chapter6/ContinuousOrientation · HasLocalPropagation.eq_trans. Paths are joined only locally, inside a chart; the global facts recalled on p. 201 are not formalized (gap G.56) (pp. 227–228).

VI.19 Definition 6.36: orientable neighbourhoods and continuous orientations

Definition 6.36 (V6.50). The text: an orientable neighbourhood of (f, x) is an open, sc-smoothly contractible U with the local propagation property, and a continuous orientation over it is a choice $y \mapsto o_y$ carried into itself by the propagation. Since Proposition 6.28 gives (1′), an orientable neighbourhood is a pair $V \subseteq U$ Chapter6/SectionOrientation · IsOrientableNbhd: U open with the local propagation property in a chart, $V \ni x$ open with any two smooth points joined by an sc-smooth path in U. A *continuous orientation* is a choice $y \mapsto o_y$ on V_∞, read in the chart, carried into itself by the push-forwards along the paths in U Chapter6/ContinuousOrientation · IsContinuousOrientation. *There are exactly two* Chapter6/ContinuousOrientation · exists_two_isContinuousOrientation: given o_x, set o_y to be the push-forward of o_x along any path in U from x to y, well defined by the local propagation property; push-forwards compose along concatenations, so this is a continuous orientation; the two values of o_x give two different ones; and a continuous orientation is determined by its value at x. *Every smooth point has an orientable neighbourhood* Chapter6/ContinuousOrientation · exists_orientableNbhd Chapter6/SectionOrientation · exists_isOrientableNbhd, by Theorem 6.35 and §VI.18. *Hidden hypotheses:* the standing ones of §6.5 (Findings, Chapter 6, item 54) (p. 227).

VI.20 The path category PX

Reparametrization homotopies (V6.51). The values of an sc-smooth path at parameters are smooth points Chapter6/ScHomotopy · mem_smoothSet_of_isScSmoothMap, and C^∞ maps of $\mathbb{R}^2$ compose Chapter6/ScHomotopy · contDiff_pt2_comp. *For a C^∞ θ with $\theta(0) = 0$, $\theta(1) = 1$, $\gamma \circ \theta$ is sc-smoothly homotopic to γ with the end points fixed* Chapter6/ScHomotopy · isScHomotopyRel_reparam, by $H(\tau, t) = \gamma((1 - \tau)t + \tau\theta(t))$, and

inside every set containing $\gamma([0,1])$ when $\theta([0,1]) \subseteq [0,1]$ Chapter6/ScHomotopy · reparam_mem. With the halves of a path Chapter6/ScHomotopy · leftHalf Chapter6/ScHomotopy · rightHalf, the concatenation of the two halves of γ is such a reparametrization Chapter6/ScHomotopy · halvesReparam Chapter6/ScHomotopy · concatPath_halves (p. 228).

The category (V6.52). The tools are the value $\kappa(1/2)$ Chapter6/PathCategory · smoothTransition_half, constant maps at smooth points Chapter6/PathCategory · isScSmoothMap_const_of_mem_smoothSet, maps defined by cases on closed pieces that agree near the seams Chapter6/PathCategory · isScSmoothMap_ite Chapter6/PathCategory · isScSmoothMap_ite₃ and the middle third of a concatenation Chapter6/PathCategory · concatPath_of_le_two_thirds Chapter6/PathCategory · concatPath_of_one_third_le. Sc-smooth homotopy with the end points fixed Chapter6/PathCategory · ScHomotopic is an equivalence relation on the sc-smooth paths from x to y Chapter6/PathCategory · ScHomotopic.refl Chapter6/PathCategory · ScHomotopic.symm Chapter6/PathCategory · ScHomotopic.trans. Transitivity needs care: a homotopy meets its edge conditions only on $[0,1]$ (decision D2), and two homotopies glued in τ need not agree off $[0,1]$, so all paths and homotopies are first reparametrized by κ in t (Findings, Chapter 6, item 71). The paths with smooth end points Chapter6/PathCategory · ScPath Chapter6/PathCategory · ScPath.source_mem_smoothSet Chapter6/PathCategory · ScPath.target_mem_smoothSet have the constant paths Chapter6/PathCategory · ScPath.const, concatenation Chapter6/PathCategory · ScPath.concat and reversal Chapter6/PathCategory · ScPath.reverse. Concatenation is well defined on classes Chapter6/PathCategory · ScPath.concat_congr, has the constant paths as units Chapter6/PathCategory · ScPath.const_concat Chapter6/PathCategory · ScPath.concat_const, is associative Chapter6/PathCategory · ScPath.concat_assoc — the two bracketings differ by an explicit reparametrization Chapter6/PathCategory · assocReparam, joined by a three-piece homotopy Chapter6/PathCategory · isScHomotopyRel_triHomotopy — and the reversed paths are inverses Chapter6/PathCategory · ScPath.concat_reverse Chapter6/PathCategory · ScPath.reverse_concat. The classes Chapter6/PathCategory · ScPathClass with their composition Chapter6/PathCategory · ScPathClass.comp make the smooth points a category PX Chapter6/PathCategory · PathCat Chapter6/PathCategory · PathCat.category, indeed a groupoid Chapter6/PathCategory · PathCat.groupoid, with the morphism of a path Chapter6/PathCategory · PathCat.homMk, composition and identities on representatives Chapter6/PathCategory · PathCat.comp_mk Chapter6/PathCategory · PathCat.id_eq_mk, equality of morphisms as homotopy Chapter6/PathCategory · PathCat.mk_eq_mk_iff, every morphism represented Chapter6/PathCategory · PathCat.exists_eq_mk and inverses by reversal Chapter6/PathCategory · PathCat.inv_mk. The text does not name the identities; they are the classes of the constant paths. The second piece $\psi(\beta(s))$ of the composition on p. 228 leaves $[0,1]$, since $\beta([1/2,1]) = [1,2]$; it is read as $\psi(\beta(s) - 1)$ (possible misprint; Findings, Chapter 6, item 55). Mathlib's $[\varphi] \gg [\psi]$ is the text's $[\psi] * [\varphi]$ (pp. 227–228).

VI.21 Comparing charts: the transfer, and Or_x^f in a chosen chart

The proof of Theorem 6.37 carries an orientation along orientable neighbourhoods of different points of a path, which may lie in different charts. The text takes T_xX, Y_x, $\mathrm{Lin}(f,x)$ and Or_x^f to be the same whichever chart is used (p. 219); here they are made at chart points (§VI.1), so the charts have to be compared (Findings, Chapter 6, item 70).

Conjugation of determinants (V6.53). For isomorphisms A, Γ with $S'A = \Gamma S$, A maps $\ker S$ onto $\ker S'$ Chapter6/ChartTransfer · conjKerEquiv and Γ induces $\operatorname{coker} S \cong \operatorname{coker} S'$ Chapter6/ChartTransfer · conjCokerEquiv; together they give $\det(S) \cong \det(S')$ Chapter6/ChartTransfer · conjDet, on wedges $(a \wedge \dots) \otimes (d \wedge \dots)^* \mapsto (Aa \wedge \dots) \otimes ([\Gamma d] \wedge \dots)^*$ Chapter6/ChartTransfer · conjDet_wedge_tmul, compatible with composition, equal data and the identity Chapter6/ChartTransfer · conjDet_trans Chapter6/ChartTransfer · conjDet_congr Chapter6/ChartTransfer · conjDet_refl. Good left-projections are conjugated, $\Gamma P \Gamma^{-1} \in \Pi_{S'}$ Chapter6/ChartTransfer · IsGoodLeftProj.conj, and the γ^P commute with the conjugation Chapter6/ChartTransfer · gammaProj_conjDet; read in local determinant charts the conjugation has a continuous nowhere-zero coordinate Chapter6/ChartTransfer · coord_conjDet_coordInv, so continuous orientations of DET are carried along Chapter6/ChartTransfer · isLineOrientation_conj Chapter6/ChartTransfer · LineOrientation.conj Chapter6/ChartTransfer · LineOrientation.conj_apply, with the same rules Chapter6/ChartTransfer · LineOrientation.conj_refl Chapter6/ChartTransfer · LineOrientation.conj_trans Chapter6/

ChartTransfer · LineOrientation.conj_congr.

The transfer between two chart points (V6.53). For chart points p, p' of x, A is the derivative at z of the transition of the base charts Chapter6/ChartTransfer · ChartPoint.baseDeriv, with the cocycle rules Chapter6/ChartTransfer · ChartPoint.baseDeriv_self Chapter6/ChartTransfer · ChartPoint.baseDeriv_trans, an sc-operator Chapter6/ChartTransfer · ChartPoint.isScOperator_baseDeriv; Γ is the fibre part of the transition of the strong bundle charts Chapter6/ChartTransfer · ChartPoint.fiberMap Chapter6/ChartTransfer · ChartPoint.fiberMap_self Chapter6/ChartTransfer · ChartPoint.fiberMap_trans Chapter6/ChartTransfer · ChartPoint.isScOperator_fiberMap. They restrict to isomorphisms $T_zO \cong T_{z'}O'$ and $K_z \cong K_{z'}$ Chapter6/ChartTransfer · ChartPoint.transferT Chapter6/ChartTransfer · ChartPoint.transferK, again with the cocycle rules Chapter6/ChartTransfer · ChartPoint.transferT_self_apply Chapter6/ChartTransfer · ChartPoint.transferT_trans_apply. At a zero the linearizations in the two charts are related by them, the chain rule applied to the principal parts Chapter6/ChartTransfer · ChartPoint.scDeriv_principal_baseDeriv, and sc⁺-operators conjugate to sc⁺-operators Chapter6/ChartTransfer · ChartPoint.IsScPlusOp.conj; so $L \in \mathrm{Lin}_p(f,x)$ iff $\Gamma L A^{-1} \in \mathrm{Lin}_{p'}(f,x)$ Chapter6/ChartTransfer · ChartPoint.mem_linSet_transfer_iff. Conjugation of orientations over this bijection of the bases is the transfer $\mathrm{Or}_x^f(p) \cong \mathrm{Or}_x^f(p')$ Chapter6/ChartTransfer · ChartPoint.orTransfer Chapter6/ChartTransfer · ChartPoint.orTransfer_apply, with the cocycle rules Chapter6/ChartTransfer · ChartPoint.orTransfer_self Chapter6/ChartTransfer · ChartPoint.orTransfer_trans.

The push-forward does not depend on the chart (V6.54). The sign is invariant, conjugation preserving $\dim\operatorname{coker}$ Chapter6/PushTransfer · stabSign_conj; the stabilization is natural — $(\Gamma S A^{-1}) \oplus \Gamma\varphi$ is the conjugate of $S \oplus \varphi$, with the conjugated kernel, onto iff $S \oplus \varphi$ is Chapter6/PushTransfer · coprod_conj Chapter6/PushTransfer · ker_coprod_conj Chapter6/PushTransfer · surjective_coprod_conj — and so is its isomorphism on rays Chapter6/PushTransfer · conj_stab_key Chapter6/PushTransfer · stabRayEquiv_conjDet; continuation is natural under jointly continuous families of injective linear maps Chapter6/PushTransfer · SubspaceFamily.cont_map. The principal parts of the sections transform by Γ Chapter6/PushTransfer · ChartPoint.fiberMap_principal, so the operators T and φ of a datum in one chart are the conjugates of those in the other Chapter6/PushTransfer · ChartPoint.conj_of_linAt Chapter6/PushTransfer · ChartPoint.conj_of_principal, and A varies continuously along a path Chapter6/PushTransfer · continuous_baseDeriv. A family of stabilized kernels in one chart is therefore carried by $A \times 1$ to one in the second chart, with the same sections Chapter6/PushTransfer · IsStabKernels.exists_transfer, the rays read from an orientation corresponding under the transfer Chapter6/PushTransfer · IsStabKernels.orRay_transfer; a push-forward in one chart, conjugated by the transfers at the end points, is a push-forward in the other Chapter6/PushTransfer · IsPush.transfer, and the local propagations of two charts agree Chapter6/PushTransfer · HasLocalPropagation.push_compat. The text does not compare charts (pp. 222–228).

Or_x^f in a chosen chart (V6.55). A chart of the atlas at x is chosen for every x Chapter6/SectionOrientation · orChart Chapter6/SectionOrientation · isChart_orChart Chapter6/SectionOrientation · mem_orChart, and Or_x^f is the set of the orientations read there Chapter6/SectionOrientation · Orx, with two elements Chapter6/OrientationFunctor · exists_two_orx. It is identified with the orientations read in any other chart at x by the transfer Chapter6/SectionOrientation · toChartOr Chapter6/SectionOrientation · toChartOr_orChart, compatibly with composition Chapter6/OrientationFunctor · toChartOr_trans, and a map $y \mapsto o_y$ is read in the chart of an orientable neighbourhood through it Chapter6/SectionOrientation · restrictOr. A chosen chart stands for the text's chart-free Or_x^f (pp. 220, 228).

VI.22 Transport along paths from local push-forwards

The proof of Theorem 6.37 (p. 228): for a path φ, "using the compactness of $[0,1]$ and a finite covering" by orientable neighbourhoods, φ_* is the composite of the local propagations; "by a covering argument" it depends only on $[\varphi]$; and "it is obvious that Γ_f is unique". The formalization separates this argument from orientations: it is proved for any family of types over X_∞ (Findings, Chapter 6, item 72).

Local transport data (V6.56) Chapter6/PathTransport · LocalTransport: a family of *good* open sets covering X_∞,

all fibres in bijection, and for every good set U a push-forward along every sc-smooth path in U, such that push-forwards in one good set compose and do not depend on the good set containing the path. Then the push-forward along a loop in a good set is the identity `Chapter6/PathTransport · LocalTransport.push_eq_refl`, in a good set it depends only on the end points `Chapter6/PathTransport · LocalTransport.push_eq_push`, and only on the path on $[0,1]$ `Chapter6/PathTransport · LocalTransport.push_congr`.

The dyadic transport. At level 0 it is the push-forward in a chosen good set containing $\gamma([0,1])$ (an arbitrary bijection if there is none); at level $n+1$ it is the composite of the transports at level n along the two halves `Chapter6/PathTransport · LocalTransport.transportN`. On a path in a good set every level gives the push-forward `Chapter6/PathTransport · LocalTransport.transportN_eq_push`, by the composition rule. The pieces of level N lie in good sets `Chapter6/PathTransport · LocalTransport.IsFine` for some N: the preimages of the good sets cover the compact $[0,1]$, so there is a Lebesgue number δ `Chapter6/PathTransport · LocalTransport.IsLebesgueNumber Chapter6/PathTransport · LocalTransport.exists_isLebesgueNumber`, and $2^N\delta > 1$ suffices `Chapter6/PathTransport · LocalTransport.isFine_of_isLebesgueNumber`. From such an N on the levels agree `Chapter6/PathTransport · LocalTransport.transportN_eq_of_isFine`, since at the next level each piece is split inside its good set. The *transport* is the value at such a level `Chapter6/PathTransport · LocalTransport.transport`.

Its properties. It is the push-forward on a path in a good set `Chapter6/PathTransport · LocalTransport.transport_eq_push` — property (□) — depends only on the path on $[0,1]$ `Chapter6/PathTransport · LocalTransport.transport_congr`, is the identity on paths constant on $[0,1]$ `Chapter6/PathTransport · LocalTransport.transport_const`, and is the composite along the two halves `Chapter6/PathTransport · LocalTransport.transport_halves`. *Homotopy invariance* `Chapter6/PathTransport · LocalTransport.transport_eq_of_isScHomotopyRel` is the covering argument. For a homotopy $\Phi : \mathbb{R}^2 \to X$ the preimages of the good sets cover $[0,1]^2$, with a Lebesgue number δ `Chapter6/PathTransport · LocalTransport.exists_isSqLebesgueNumber`. *The square lemma* `Chapter6/PathTransport · LocalTransport.transportN_square`: for a rectangle of $[0,1]^2$ with sides $< 2^n\delta$, the transports at level n along its vertical, horizontal and diagonal segments `Chapter6/PathTransport · vSeg Chapter6/PathTransport · hSeg Chapter6/PathTransport · dSeg` commute around its boundary — left side then top equals bottom then right side. By induction on n: for $n = 0$ the rectangle lies in one good set, and both composites are the push-forward along the diagonal; for $n + 1$ the rectangle is cut into four quadrants and the four commutative squares are pasted, the interior edges cancelling `Chapter6/PathTransport · equiv_paste_squares`. Applied to $[0,1]^2$ at a level with $2^n\delta > 1$, with the sides $\tau \mapsto \Phi(\tau,0)$, $\tau \mapsto \Phi(\tau,1)$ constant on $[0,1]$ and of trivial transport, it gives the invariance. The transport is invariant under reparametrization `Chapter6/PathTransport · LocalTransport.transport_reparam` (§VI.20) and takes concatenations to composites `Chapter6/PathTransport · LocalTransport.transport_concatPath`, the halves of a concatenation being reparametrizations of the two paths. *Uniqueness* `Chapter6/PathTransport · LocalTransport.transport_unique`: a family of bijections along the sc-smooth paths that is homotopy invariant, takes concatenations to composites and is the push-forward on paths in good sets, is the transport — $G(\gamma) = G(\text{left half}) \circ G(\text{right half})$ through the reparametrization, and induction on the dyadic level. Dyadic subdivisions of $[0,1]$ and of $[0,1]^2$ stand in place of arbitrary finite coverings (p. 228).

VI.23 Theorem 6.37: the functor Γ_f

The text: *for a strong bundle over a tame M-polyfold and an sc-Fredholm section f there is a uniquely determined functor $\Gamma_f : PX \to \mathcal{O}_f$, $x \mapsto \mathrm{Or}_x^f$, with (□): for every smooth x, every sc-smoothly contractible open $U \ni x$ on which f has the local propagation property, and every sc-smooth path $\varphi : [0,1] \to U$, the propagation φ_* and $\Gamma_f([\varphi])$ coincide.*

The local data of f (V6.57). U is a propagation neighbourhood if it is open with the local propagation property in some chart `Chapter6/OrientationFunctor · IsPropagationNbhd`. Along a path in such a U the push-forward, read in the chosen charts of the end points through the transfer (§VI.21), is orPush `Chapter6/OrientationFunctor · orPush`; it equals every push-forward along the path in every chart in which the path lies in a set with the local propagation property `Chapter6/OrientationFunctor · orPush_eq` — by the chart indepen-

dence of §VI.21 and the cocycle rule of the transfer — it does not depend on the neighbourhood `Chapter6/OrientationFunctor · orPush_compat`, and it composes `Chapter6/OrientationFunctor · orPush_trans`. With Theorem 6.35, which makes the propagation neighbourhoods cover X_∞, and §VI.9, which puts every Or_x^f in bijection with every other, these are local transport data `Chapter6/OrientationFunctor · orLocalTransport`, and their transport satisfies (□) `Chapter6/OrientationFunctor · orLocalTransport_transport_eq`.

The functor. $\Gamma_f([\varphi])$ is the transport along φ, well defined on classes by homotopy invariance `Chapter6/OrientationFunctor · orFunctorMap` `Chapter6/OrientationFunctor · orFunctorMap_mk`; it takes identities to identities `Chapter6/OrientationFunctor · orFunctorMap_id` and compositions to composites `Chapter6/OrientationFunctor · orFunctorMap_comp`, by the constancy and concatenation rules of §VI.22; and it is a functor $PX \to$ Type `Chapter6/OrientationFunctor · orFunctor`. Property (□) for a family G on the morphisms of PX, on the members of a family of open sets `Chapter6/OrientationFunctor · IsStarPropertyOn` and on all of them `Chapter6/OrientationFunctor · IsStarProperty`, reads $G([\varphi]) = \mathrm{toChartOr}^{-1}_{\varphi(1)} \circ e \circ \mathrm{toChartOr}_{\varphi(0)}$ for every chart c, every open U with the local propagation property in c, every sc-smooth path φ in U with smooth end points and every push-forward e along φ in c. Γ_f has it `Chapter6/OrientationFunctor · orFunctorMap_isStarProperty`, so the property is satisfiable. *Uniqueness*: a family with the rules for identities and composites and with (□) on the members of a family of such open sets covering X_∞ is Γ_f `Chapter6/OrientationFunctor · orFunctorMap_unique_of_cover` — restrict the local data to the cover `Chapter6/OrientationFunctor · LocalTransport.restrict` and apply the uniqueness of the transport — in particular with (□) everywhere `Chapter6/OrientationFunctor · orFunctorMap_unique`.

Theorem 6.37 `Chapter6/OrientationFunctor · theorem_6_37`. *Over a tame X with bump functions, for f with tame germ data and P with sc⁺-sections through the smooth fibre points, there is exactly one family of bijections $\Gamma_f([\varphi]) : \mathrm{Or}_x^f \to \mathrm{Or}_y^f$ on the morphisms of PX taking identities to identities and compositions to composites, with property (□). Hidden hypotheses:* the standing ones of §6.5, and sc⁺-sections through every smooth fibre point, so that every Or_x^f has two elements (bump functions give them). (□) is asked for every open set with the local propagation property, not only for sc-smoothly contractible ones (gap G.61), and the uniqueness needs it only on a covering family — for instance on the sc-smoothly contractible ones, wherever they cover X_∞. $\mathcal{O}_f$ is read as types and bijections, and "uniquely determined" for the maps on morphisms: on objects there is no choice (p. 228).

VI.24 Definitions 6.38–6.40: orientability and orientations

Definition 6.38 (V6.58). *f is orientable if for every pair of smooth points x, y in the same path component, $\Gamma_f([\varphi])$ does not depend on the class $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · IsOrientableSection`. "In the same path component" is read as "joined by an sc-smooth path" `Chapter6/OrientationFunctor · scPathSetoid`; for points not so joined there is nothing to check (Findings, Chapter 6, item 73). *Equivalently there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for every $[\varphi] : x \to y$* `Chapter6/OrientationFunctor · isOrientableSection_iff_exists`: from left to right, choose a base point in each class of points joined by sc-smooth paths and set $o_y = \Gamma_f([\varphi])o_{\text{base}}$ for any $[\varphi]$, which does not depend on $[\varphi]$; from right to left, $\Gamma_f([\varphi])$ and $\Gamma_f([\psi])$ are bijections between two-element sets with the same value at o_x, hence equal (§VI.9) (p. 229).

Definition 6.39 (V6.59). *f is orientable if there is $y \mapsto o_y$ on X_∞ such that every smooth point has an orientable neighbourhood over which it is one of the two continuous orientations* `Chapter6/SectionOrientation · IsOrientableSection'`. The definition writes "$y \in U_\infty$" before any U is introduced; it is read as $y \in X_\infty$ (possible misprint; Findings, Chapter 6, item 74). *This holds iff there is $y \mapsto o_y$ with $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi]$* `Chapter6/OrientationFunctor · isOrientableSection'_iff_exists`. From right to left: on an orientable neighbourhood $V \subseteq U$ the push-forwards along paths in U are values of Γ_f, by (□), so the restriction is a continuous orientation. From left to right: elements carried into each other by the transport along the paths in the members of an open covering are carried into each other along every path `Chapter6/OrientationFunctor · LocalTransport.transport_apply_eq_of_local` — the transport is computed on dyadic pieces inside orientable neighbourhoods. *So Definitions 6.38 and 6.39*

are equivalent Chapter6/OrientationFunctor · isOrientableSection_iff. The text calls them equivalent without proof (Findings, Chapter 6, item 73) (p. 229).

Definition 6.40 (V6.60). *An orientation of f is a map $y \mapsto o_y$ on X_∞ whose restriction to every orientable neighbourhood of every smooth point is a continuous orientation* Chapter6/SectionOrientation · IsSectionOrientation. Equivalently $\Gamma_f([\varphi])o_x = o_y$ for all $[\varphi] : x \to y$ Chapter6/OrientationFunctor · isSectionOrientation_iff, by the argument of Definition 6.39; so f has an orientation iff it is orientable Chapter6/OrientationFunctor · exists_isSectionOrientation_iff, two orientations that agree at a point agree on the smooth points joined to it by sc-smooth paths Chapter6/OrientationFunctor · IsSectionOrientation.apply_eq_iff, and there are exactly two on each such class Chapter6/OrientationFunctor · IsSectionOrientation.exists_two. An orientable neighbourhood is the pair of §VI.19 (Findings, Chapter 6, item 75) (p. 229).

VI.25 Theorem 6.41: the natural orientation of the solution set

The text: *over a tame X, let f be a proper sc-Fredholm section in general position to the boundary at every zero, and o^f an orientation of f; then $S = f^{-1}(0)$ is a smooth compact manifold with boundary with corners with a natural orientation.* The proof: S is such a manifold "from the transversality discussion"; $\ker f'(x) = T_xS$ and, $f'(x)$ being onto, T_xS is oriented by o^f_x; and "one verifies that the differential geometric local prolongation of the orientation on S is the same as the Fredholm one".

The first assertion (V6.61) Chapter6/SolutionOrientation · theorem_6_41_solutionSet. *For X tame and a proper f with tame germ data in general position (Definition 5.17) at every zero, S is compact, a smooth submanifold in general position with the induced structure of a tame manifold with boundary with corners* — the conclusions of Theorem 5.18 (§V.20), with good position from general position by Lemma 4.19 (§IV.14) Chapter6/SolutionOrientation · MPolyfold.IsGeneralPositionAt.isGoodPositionAt. In a chart, $T_xS = \ker f'(x)$ Chapter6/SolutionOrientation · tangentSpace_solutionSet_eq_linKer (§V.17). "Proper" is the text's hypothesis; the proof uses only the compactness of S, which for sc-Fredholm sections is equivalent to it (Theorem 5.5, §V.6), so nothing of the hypothesis is lost (Findings, Chapter 6, item 90).

The tangent orientation. At a zero, $f'(x)$ — the linearization of $f + s$ with $s = 0$ Chapter6/SolutionOrientation · ChartPoint.linZero — lies in $\mathrm{Lin}(f, x)$ Chapter6/SolutionOrientation · ChartPoint.linZero_mem_linSet and is onto Chapter6/SolutionOrientation · ChartPoint.linZero_surjective, with kernel T_xS in the chart Chapter6/SolutionOrientation · ChartPoint.map_ker_linZero. So an element o_x of Or^f_x gives a ray of $\det f'(x) \cong \lambda(\ker f'(x)) = \lambda(T_xS)$, the *tangent orientation* Chapter6/SolutionOrientation · ChartPoint.tangentOr, read in a chart at a zero Chapter6/SolutionOrientation · tangentOrAt. The sign of a surjective operator is 1 Chapter6/SolutionOrientation · stabSign_of_surjective, and the stabilization by $\mathbb{R}^0$ of a surjective operator is $\det(T) \cong \lambda(\ker T)$ on rays Chapter6/SolutionOrientation · stabRayEquiv_zero, so the tangent orientation is the stabilized ray of §VI.14 for the empty family of sections Chapter6/SolutionOrientation · ChartPoint.stabRayEquiv_eq_tangentOr.

"The local prolongation is the Fredholm one." Along an sc-smooth path γ of zeros in the domain of a chart, the datum $S = 0$, $k = 0$ is a family of stabilized kernels Chapter6/SolutionOrientation · isStabKernels_zero, its kernels being those of $f'(\gamma_q)$ Chapter6/SolutionOrientation · SubspaceFamily.congrL; they form a continuous family of subspaces, by Lemma 6.30 (§VI.12) Chapter6/SolutionOrientation · exists_tangentFamily Chapter6/SolutionOrientation · tangentFamily Chapter6/SolutionOrientation · tangentFamily_L, and the rays read from an orientation are the tangent orientations Chapter6/SolutionOrientation · IsStabKernels.orRay_eq_tangentOr. So along a path of zeros the push-forward is continuation between tangent orientations Chapter6/SolutionOrientation · cont_tangentOr, and *the tangent orientations given by an orientation of* DET *along the path form an orientation of the line bundle $\lambda(\Lambda)$* Chapter6/SolutionOrientation · isLineOrientation_tangentOr. Intermediate parameters are reached by the paths $s \mapsto \gamma(t\kappa(s))$, which stay in $\gamma([0,t])$ for all $s \in \mathbb{R}$. For an orientation o of f and an sc-smooth path of zeros in an orientable neighbourhood $V \subseteq U$, the tangent orientations of o along it form an orientation of $\lambda(\Lambda)$ Chapter6/SolutionOrientation · isLineOrientation_tangentOr_of_isSectionOrientation, and together this is the theorem Chapter6/SolutionOrientation · theorem_6_41

(Findings, Chapter 6, item 91).

Readings and what is left. *Hidden hypotheses:* tame germ data, the standing hypothesis of Chapter 5; the orientation o exists iff f is orientable (§VI.24). "A natural orientation" is read as the tangent orientations with their continuity along the sc-smooth paths of zeros in a chart, and the prolongation sentence as the identity of continuation and push-forward along such paths. Not formalized: an orientation of S as an abstract manifold, of its tangent bundle in the charts of the induced structure (gap G.67); the compatibility of the tangent orientations read in two charts at one zero (gap G.68); and the paragraph after the theorem — for f not generic, a small sc$^+$-perturbation s supported near the zero set, the orientation of S_{f+s} from o^f, and $\mathrm{Lin}(f+s,x)=\mathrm{Lin}(f,x)$ for the orientation of $\det((f+s)'(x))$ (gap G.69) (Findings, Chapter 6, items 92, 93) (pp. 229–230).

VI.26 Remark 6.42: the orientation of $\hat{f}$ at a point

The text: for an oriented (f,o) and an sc$^+$-section s of $Y\to X\times[0,1]$, $\hat{f}(x,t)=f(x)+s(x,t)$ has a natural orientation from o and the standard orientation of $[0,1]$: for $L=(f-t)'(x)$ oriented by $h=(a_1\wedge\cdots\wedge a_n)\otimes(b_1\wedge\cdots\wedge b_l)^*$, orient $\tilde{L}(a,b)=La$ by $\tilde{h}=(a_1\wedge\cdots\wedge a_n\wedge e)\otimes(b_1\wedge\cdots\wedge b_l)^*$, e the standard orientation of $[0,1]$; $\tilde{L}$ lies in the linearization space of $\hat{f}$ at (x,t) (p. 230).

The linear algebra (V6.62). For Fredholm L, $\tilde{L}=L\oplus 0: E\times\mathbb{R}\to F$ `Chapter6/ProductOrientation · extOp` is Fredholm `Chapter6/ProductOrientation · isFredholm_extOp` with $\ker\tilde{L}=\ker L\times\mathbb{R}$ — one dimension more `Chapter6/ProductOrientation · finrank_ker_extOp`, the new direction $e=(0,1)$ `Chapter6/ProductOrientation · extE` — and the same range `Chapter6/ProductOrientation · range_extOp` and cokernel `Chapter6/ProductOrientation · extCoker`. The map $h\mapsto\tilde{h}$, $(a_1\wedge\cdots\wedge a_n)\otimes d^*\mapsto((a_1,0)\wedge\cdots\wedge(a_n,0)\wedge e)\otimes d^*$, is an isomorphism $\det(L)\cong\det(\tilde{L})$ `Chapter6/ProductOrientation · extDet` `Chapter6/ProductOrientation · extDet_wedge_tmul`, the wedge extension of §VI.16 with one vector. In local determinant charts its coordinate is $(-1)^{\dim F/R(P)+\dim\operatorname{coker}L}$ times a continuous nowhere-zero function `Chapter6/ProductOrientation · coord_extDet`. Over the convex $\mathrm{Lin}(f,x)$, where $\dim\operatorname{coker}L$ can change parity, the orientation defined by $\tilde{h}$ at one L could therefore differ from the one defined at another, as we read it. With the sign of §VI.14, $(-1)^{\dim\operatorname{coker}L}\,h\mapsto\tilde{h}$ `Chapter6/ProductOrientation · extDetN` has a coordinate continuous up to a sign constant on a chart `Chapter6/ProductOrientation · coord_extDetN`, agrees with the text's map when L is onto `Chapter6/ProductOrientation · extDetN_eq_of_surjective`, and pulls continuous orientations back along $L\mapsto\tilde{L}$ `Chapter6/ProductOrientation · isLineOrientation_extPull` `Chapter6/ProductOrientation · LineOrientation.extPull` (Findings, Chapter 6, item 95). Transported along isomorphisms A, Γ, $\hat{L}=\Gamma^{-1}(L\oplus 0)A$ `Chapter6/ProductOrientation · extConjOp` gets the isomorphism $h\mapsto\tilde{h}$ followed by the conjugation of §VI.21 `Chapter6/ProductOrientation · extConjDet`; and for a preconnected C and a simply connected, locally path-connected $\hat{C}$ containing the $\hat{L}$, an orientation o of DET over a nonempty C gives exactly one orientation $\hat{o}$ over $\hat{C}$ with $\hat{o}(\hat{L})$ the image of $o(L)$ for every $L\in C$ `Chapter6/ProductOrientation · existsUnique_extConjOr`.

In charts. $X\times[0,1]$ is read as $\mathbb{R}\times X$ (decision D2), with the parameter first; s is an sc$^+$-section S of π^*P, and $\hat{f}=\tilde{f}+S$ with $\tilde{f}(t,y)=f(y)$. At the product chart point (t,z) `Chapter6/ProductOrientation · ChartPoint.prodPt` the tangent space is $\mathbb{R}\times T_zO$ `Chapter6/ProductOrientation · ChartPoint.mem_prodPt_T_iff` and the fibre K_z `Chapter6/ProductOrientation · ChartPoint.mem_prodPt_K_iff`. The text's $\tilde{L}(a,b)=La$ puts the $\mathbb{R}$-factor second, so $\hat{L}=\Gamma^{-1}(L\oplus 0)A$ with A the swap of the factors and $\Gamma=1$ `Chapter6/ProductOrientation · ChartPoint.prodT` `Chapter6/ProductOrientation · ChartPoint.prodK` `Chapter6/ProductOrientation · ChartPoint.prodLin`. Sc$^+$-operators lift to the product `Chapter6/ProductOrientation · ChartPoint.IsScPlusOp.prodPt`, sections of P lift to t-constant sections of π^*P `Chapter6/ProductOrientation · constLift`, and so $\hat{L}\in\mathrm{Lin}(\hat{f},(t,x))$ for every $L\in\mathrm{Lin}(f,x)$ `Chapter6/ProductOrientation · ChartPoint.prodLin_mem_linSet` — the derivative in t of a t-constant lift vanishes, and S only contributes an sc$^+$-operator. *An orientation o of* $\mathrm{DET}(f,x)$ *gives exactly one orientation $\hat{o}$ of*

$\mathrm{DET}(\hat{f},(t,x))$ *with* $\hat{o}(\hat{L})$ *the image of* $o(L)$ *for every* L `Chapter6/ProductOrientation · ChartPoint.existsUnique_prodOr` `Chapter6/ProductOrientation · ChartPoint.prodOr` `Chapter6/ProductOrientation · ChartPoint.prodOr_apply`, and $o \mapsto \hat{o}$ is injective `Chapter6/ProductOrientation · ChartPoint.prodOr_injective` (Findings, Chapter 6, item 96). *Hidden hypotheses:* x smooth and $\mathrm{Lin}(f,x)$ nonempty (bump functions give it, §VI.1); only the orientation of $\mathrm{DET}(f,x)$ at the one point is used, not an orientation of f. That $(t,y) \mapsto \hat{o}$ is an orientation of $\hat{f}$ in the sense of Definition 6.40 — continuity over the orientable neighbourhoods of $\mathbb{R}\times X$, independence of the chart of the pull-back bundle — built from an orientation of f, is not formalized (gap G.66; Findings, Chapter 6, item 97).

VI.27 §6.6 and Theorem 6.43: not formalized

§6.6 (p. 230) recalls, for a proper oriented (f,o) over a tame X with sc-smooth bump functions and an auxiliary norm N, the neighbourhood U of S of Theorem 5.18 and the sc$^+$-perturbations s supported in U with $N(s)\le 1$ for which $f+s$ is in general position at its zeros; it calls them *admissible for the pair* (N,U). Theorem 5.18 is formalized (§V.20) and the orientation of S_{f+s} is the subject of §VI.25, but the notion itself, used only in Theorem 6.43, is not formalized (gap G.55; Findings, Chapter 6, item 101).

Theorem 6.43 (p. 231): a well-defined linear map $\Psi_{(f,o)} : H^*_{dR}(X,\partial X)\to\mathbb{R}$, $\Psi_{(f,o)}([\omega,\tau]) = \int_{S_{f+s}}\omega - \int_{\partial S_{f+s}}\tau$ for every admissible s, with its proof (pp. 231–233) and the case $\partial X=\emptyset$ (p. 233). It is not formalized (gap G.54): it needs the sc-differential forms and their cohomology of §4.4 (gaps G.37–G.40), and integration over oriented compact manifolds with boundary with corners with Stokes' theorem, which Mathlib does not have (decision D1). What the theorem rests on is formalized: the perturbations of Theorem 5.18 (Part V) and the tangent orientations of Theorem 6.41 (§VI.25) (Findings, Chapter 6, item 100).

VI.28 Appendix §6.7.1: the proof of Lemma 6.5

The text's proof (pp. 234–235) has three steps: a change of the basis of the complement of $\alpha(A)$ multiplies both wedges of (91) by its determinant; a change of the bases of A and D multiplies the source by $\det(\tau)/\det(\varepsilon)$; and a change of complements adds to the basis of the complement of $\alpha(A)$ vectors of $\alpha(A)$, and to the γ-preimages vectors of $\beta(B)$, which leaves the wedges unchanged.

The formalized proof (V6.63) does the three steps at once. For two choices χ, χ' there is a scalar μ with $\mathrm{source}(\chi') = \mu\,\mathrm{source}(\chi)$ and $\mathrm{value}(\chi') = \mu\,\mathrm{value}(\chi)$ `Chapter6/ExactSequence · FourTermExact.Choice.exists_smul_source_value`. *Step 1.* Project the basis b'' of the new complement Z' onto Z along $\alpha(A)$; the projections form a basis of Z, related to b' by some σ, and since $\beta\circ\alpha=0$ they have the same β-images as b''. *Step 2.* In B, $\alpha(a')\wedge b'' = \alpha(a')\wedge p(b'')$, adding vectors of the span of the first block; with $a'=\tau a$ this is $\det\tau\,\det\sigma\;\alpha(a)\wedge b'$. *Step 3.* In C, the γ-preimages c' of $d'=\varepsilon d$ in V' differ from the γ-preimages in V by vectors of $\ker\gamma=\beta(B)$, the span of the second block, so the wedge of the basis of C `Chapter6/ExactSequence · FourTermExact.Choice.wedgeC` is $\det\varepsilon\,\det\sigma$ times that of χ. *Step 4.* So the value scales by $\det\tau\,\det\sigma/(\det\varepsilon\,\det\sigma)=\det\tau/\det\varepsilon$, as the source does, and Lemma 6.5 follows from the fixed choice that defines Φ_E. The step "independent of the choices of the basis if and only if $\det(\tau)=\det(\varepsilon)$" is read as this homogeneity of (91) in the bases of A and D, τ being the σ just introduced; the corrections p_i lie in $\beta(B)$, written $\beta(C)$, and are combinations of the β-images, whose names are exchanged with those of the γ-preimages in the last step (possible misprints; Findings, Chapter 6, item 1).

VI.29 Appendix §6.7.2: the proof of Proposition 6.12

(V6.64.) Let $P\le Q$ in Π_T. *Step 1, adapted bases.* Take a basis a of $\ker T$ and classes $[d_i]$ forming a basis of $\mathrm{coker}\,T$ with $d_i\in\ker Q$, possible because $R(QT)=R(Q)$; complete a by b to a basis of $\ker(QT)$, and

(a,b) by $\tilde b$ to a basis of $\ker(PQT)=\ker(PT)$. *Step 2, the three values.* By (91) with these choices (Lemma 6.5, §VI.3), $\gamma_T^Q(h_0)=(a\wedge b)\otimes([d]_Q\wedge[Tb]_Q)^*$; $\gamma_{QT}^P\gamma_T^Q(h_0)=(a\wedge b\wedge\tilde b)\otimes([d]_P\wedge[Tb]_P\wedge[QT\tilde b]_P)^*$, the classes $[d]_P$, $[Tb]_P$ being preimages of $[d]_Q$, $[Tb]_Q$ because $Pd_i=PQd_i=0$ and $PTb\in R(P)\subseteq R(Q)$; and $\gamma_T^P(h_0)=(a\wedge b\wedge\tilde b)\otimes([d]_P\wedge[Tb]_P\wedge[T\tilde b]_P)^*$ `Chapter6/ProjectionSequence · gammaProj_wedge_tmul`. *Step 3, the spanning.* $[T\tilde b]_P-[QT\tilde b]_P=[(I-Q)T\tilde b]_P$, and $(I-Q)T\tilde b\in\ker Q$. If $Qg=0$, then $g=\sum c_id_i+Tx$ with $x\in\ker(QT)$ — write $[g]=\sum c_i[d_i]$ in $\operatorname{coker}T$ and use $Qd_i=0$ — so the classes of the d_i and of the Tb_j span what the text calls $\ker\widetilde{Q}$ `Chapter6/ProjectionSequence · mem_span_append_of_ker`. *Step 4.* The last block of the third value differs from that of the second by vectors of the span of the first two blocks, which does not change the wedge (§VI.2), and with the identification of $\det(P(QT))$ with $\det(PT)$ the two values agree `Chapter6/ProjectionSequence · comp_comp_eq_of_projLE`. The spanning is proved directly in F; the name α_q in the last part is read as α^P (possible misprint; Findings, Chapter 6, items 6, 7) (pp. 235–237).

VI.30 What is not formalized

- **Theorem 6.43** (gap G.54): the invariant $\Psi_{(f,o)}$, its proof and the case $\partial X=\emptyset$ need the forms and cohomology of §4.4 (gaps G.37–G.40), integration and Stokes' theorem (decision D1) (§VI.27). **The admissible sections of §6.6** (gap G.55), used only there.
- **The facts on paths of p. 201** (gap G.56): local path connectedness and sc$^+$-smooth paths within a path component; only the local form inside a chart is formalized (§VI.18), and "in the same path component" is read as "joined by an sc-smooth path" (§VI.24).
- **§§6.2–6.4 remarks**: the alternative determinant $\lambda(\ker T)\otimes\lambda(\operatorname{coker}(T)^*)$ (gap G.57, §VI.4); the identification (95) and the maps after Proposition 6.15 (gap G.58, §VI.6); Lemma 6.19 in general, applied only to trivial bundles (gap G.59, §VI.7); the non-orientability of $\mathrm{DET}(E,F)$ and the classical case at the opening of §6.5 (gap G.60, §VI.8).
- **Proposition 6.28 (1)** (gap G.61): the sc-smooth contractibility of U, and with it the word "contractible" in Theorem 6.35, Definition 6.36 and (□) of Theorem 6.37; (1′) is proved instead and is what the proofs use (§§VI.10, VI.18, VI.19, VI.23).
- **Lemma 6.30 as a statement about manifolds** (gap G.62): the solution set near $(t,\varphi(t),0)$, the submersion π and the smooth bundle L; the kernels are a continuous family of subspaces (§§VI.12, VI.15). **Property (P) for the slices** (gap G.63); the general position of F that the text derives from it is proved (§VI.11).
- **Lin on every level** (gap G.64): the members of $\mathrm{Lin}(f,x)$ are Fredholm on level 0, not shown sc-Fredholm on every level (§VI.1).
- **Paths on** $[0,1]$ (gap G.65): paths, homotopies and families sc-smooth on $[0,1]$, $[0,1]^2$, $[0,1]^n$ only, and $X\times[0,1]$ as an M-polyfold with boundary; they are read on $\mathbb{R}$, $\mathbb{R}^2$, $\mathbb{R}^n$ and $\mathbb{R}^n\times X$ (decision D2), and a path that does not extend sc-smoothly to $\mathbb{R}$ is not covered.
- **Remark 6.42 beyond a point** (gap G.66, §VI.26).
- **Theorem 6.41, the global parts** (§VI.25): the orientation of S as an abstract manifold (gap G.67), the compatibility of the tangent orientations read in two charts (gap G.68), and the perturbation paragraph after the theorem (gap G.69).
- **The chart-free objects** T_xX, Y_x, $\mathrm{Lin}(f,x)$, Or_x^f (gaps G.6, G.7, G.13 of Chapter 2): they are made in chart points and identified between charts (§VI.21).

VI.31 What the formalization taught us

The sign of (94). The main change of substance is a sign. The coordinate of Φ_E in a local determinant chart carries the factor $(-1)^{\dim\operatorname{coker}S}$, which jumps where the parity of the cokernel dimension changes; the orientations it carries along a continuous family are then not continuous, as we read it, and an example on $\mathbb{R}$ shows it (§VI.14). The continuity statements of pp. 223–226 are not written in coordinates

in the text, and it is in coordinates that the sign appears. The normalized $\alpha(T)\Phi_E$ at both ends of every propagation changes nothing on surjective operators — the tangent orientations of Theorem 6.41 are the text's — and the same sign is used in Remark 6.42 (§VI.26).

Lemma 6.10 and the 2017 remark. The common lower bound of two good left-projections exists, but not with the range $R(P) \cap R(P')$: a projection onto a smaller, finite-codimensional piece of it, along a complement containing both kernels, is below both (§VI.4). This keeps the preprint's construction of $\mathrm{DET}(E, F)$ — one lower bound, Corollary 6.13, Lemma 6.21 — intact.

Contractibility was never needed. Proposition 6.28 (1) is not established for the neighbourhood its proof builds, as we read it, but every later use of contractibility is a use of (1′): two paths in U with the same end points are homotopic in a larger region where sections along families exist. Orientable neighbourhoods become pairs $V \subseteq U$, and property ( ) of Theorem 6.37 holds on every set with the local propagation property (§§VI.10, VI.19, VI.23).

Paths on $\mathbb{R}$ cost reparametrizations. With paths and homotopies on $\mathbb{R}$ whose conditions hold on $[0, 1]$ (decision D2), the homotopy of (1′) has the right edges only on $[0, 1]$ while push-forward data need zeros near $[0, 1]$; transitivity of homotopy, associativity of concatenation and Theorem 6.35 all pass through the smooth transition function κ (§§VI.18, VI.20).

Charts must be compared. The text's chart-free Or_x^f becomes a set read in a chosen chart, and the invariance of the local propagation under a change of chart — the naturality of the stabilization isomorphism, of the sign and of continuation — is a file of its own (§VI.21).

"By a covering argument" and "one verifies". The covering argument of Theorem 6.37 is a dyadic transport with a square lemma, proved for any family of types (§VI.22); the "simple computation" of Lemma 6.33 is an exact identity of stabilization isomorphisms, with no sign, once the appended sections come last (§VI.16); and the equivalence of Definitions 6.38 and 6.39, stated without proof, needs the transport computed on dyadic pieces inside orientable neighbourhoods (§VI.24).

Concordance

Every declaration cited in this book, by file, with its line in the source. 3636 declarations in 235 files. The web edition also shows each exact formal statement.

Polyfold/Chapter1/ChainRule.lean. `IsSc1With.norm_sub_le_of_segment` (68); `IsSc1With.exists_delta` (95); `IsSc1With.hasFDerivWithinAt_comp` (125); `IsSc1With.tangentMapWith_comp` (245); `IsSc1With.comp` (256); `theorem_1_21'` (269); `theorem_1_21` (281); `IsSc1.tangentMap_congr` (289); `IsSck.congr` (298); `IsSck.comp` (306); `IsScSmooth.comp` (317); `IsSck.comp_of_isRelOpenInQuadrant` (323).

Polyfold/Chapter1/Degeneracy.lean. `quadrantDeg` (60); `degIndexWith` (65); `coordMap` (77); `coordMap_surjective` (82); `finrank_quotient_ker_coordMap` (91); `mem_ker_coordMap` (99); `eq_zero_of_eventually_nonneg_mul` (110); `eventually_add_smul_mem_quadrant` (122); `map_ker_coordMap_le` (141); `quadrantDeg_eq_of_equiv` (158); `quadrantDeg_zero` (177); `lemma_1_24` (184); `lemma_1_24_dim` (204); `degIndex` (226); `degIndex_eq` (229); `degIndex_quadrant` (237).

Polyfold/Chapter1/ExpositionLemmas.lean. `ScScale.finiteDimensional_of_surjective_incl` (73); `ScBanach.smooth_eq_top_of_finiteDimensional` (94); `ScBanach.toBase_bijective_of_finiteDimensional` (100); `ScBanach.isoConstOfFiniteDimensional` (109); `ScBanach.isoConstOfFiniteDimensional_apply` (115); `ScBanach.IsScSplitting.isTopCompl` (126); `ScBanach.IsScSplitting.scProj` (134); `ScBanach.IsScSplitting.scProj_apply` (150); `ScBanach.IsScSplitting.scProj_idem` (154); `ScBanach.IsScSplitting.range_scProj` (159); `ScBanach.IsScSplitting.ker_scProj` (167); `ScOperator.idem_lvl` (175); `ScOperator.levelSub_range_eq` (180); `ScBanach.isScSplitting_of_idem` (196); `ScBanach.isScSplitting_iff_exists_scProj` (234); `ScBanach.inclBound` (254); `ScBanach.one_le_inclBound` (256); `ScBanach.norm_incl_le` (262); `ScBanach.norm_toBase_le` (265); `ScBanach.norm_inclK_le` (275); `ScBanach.exists_smooth_mem_coset` (293); `ScBanach.mk_mem_quotient_smooth_iff` (363); `continuousOn_apply_iff_uniformOnCompact` (390); `IsSc1With.deriv_uniformOnCompact` (440); `ScBanach.lvlSet_const` (456); `isSc1_iff_contDiffOn_const` (467); `isSc1_iff_fderivWithin_const` (481); `isScSmooth_iff_contDiffOn_const` (492); `isSc1_iff_contDiffOn_of_isRelOpenInQuadrant` (506); `isScSmooth_iff_contDiffOn_of_isRelOpenInQuadrant` (512).

Polyfold/Chapter1/FredholmIndex.lean. `linIndex` (93); `HasFiniteIndex` (97); `finite_of_finite_quotient` (103); `finite_of_finite_ker_range` (111); `finiteDimensional_ker_of_comp` (137); `finiteDimensional_coker_of_comp` (143); `HasFiniteIndex.comp_and_linIndex` (153); `HasFiniteIndex.comp` (215); `linIndex_comp` (218); `HasFiniteIndex.equiv` (231); `linIndex_equiv` (238); `HasFiniteIndex.of_finiteRange_sub_id_and_linIndex` (250); `linIndex_eq_zero_of_finiteRange_sub_id` (306); `HasFiniteIndex.of_bijective_add_and_linIndex` (313); `HasFiniteIndex.of_comp_of_comp` (333); `fredholmIndex` (351); `IsFredholm.hasFiniteIndex` (355); `finiteDimensional_range_sub_id_of_equiv` (362); `bijective_one_add_of_norm_lt` (369); `hasFiniteIndex_comp_add_of_norm_lt` (384); `fredholmIndex_add_of_norm_lt` (401); `fredholmIndex_add_isCompactOperator` (439); `ScBanach.finrank_levelSub_of_le_smooth` (468); `ScFredholmSplitting.C_le_smooth` (480); `ScFredholmSplitting.fredholmIndex_lvl` (488); `IsScFredholm.fredholmIndex_lvl` (498); `scIndex_eq_fredholmIndex` (505); `proposition_1_10_index` (509);

proposition_1_10_index_lvl (517).

Polyfold/Chapter1/FredholmPerturbation.lean. isStrictMap_of_isClosed_range (62); isFredholm_of_banach (73); isCompactOperator_of_finiteDimensional_range (84); finiteDimensional_ker_one_add (103); exists_bound_below_on_compl (130); isClosed_range_one_add (181); exists_unit_far (215); pick (243); pick_notMem (246); chain (255); chain_mono (260); chain_eq_sup (265); chain_ne_top (276); chain_isClosed (285); coFG_range_one_add (292); isFredholm_one_add (333); isCompactOperator_sub_one_of_equiv (344); isFredholm_add_isCompactOperator (354).

Polyfold/Chapter1/Levels.lean. exists_delta_of_continuousOn (117); continuousOn_clm_comp_of_isCompactOperator (144); contDiffOn_clm_comp_of_isCompactOperator (170); ScBanach.shift_lvlSet (310); ScBanach.shift_tangentSet (320); ScBanach.shift_inclK_mem_lvlSet (331); ScBanach.incl_shift_inclK (337); ScBanach.tangent_inclK (343); IsSc0.levelMap_shift (354); IsSc0.shift (364); IsScDomain.shift (377); levelDeriv (389); toBase_levelDeriv (394); levelDeriv_zero (400); IsSc1With.exists_levelDeriv (409); IsSc1With.toBase_levelDeriv (432); IsSc1With.levelMap_tangentMapWith (439); IsSc1With.continuousOn_levelDeriv (455); IsSc1With.continuousOn_levelDeriv_clm (468); IsSc1With.hasFDerivWithinAt_level (483); IsSc1With.contDiffOn_level (592); IsSc1With.levelDeriv_eq (604); IsSc1Levels (644); IsSc1Levels.continuousOn_clm (656); IsSc1Levels.contDiffOn (662); IsSc1Levels.toBase_apply (674); IsSc1Levels.isSc1With (704); IsSc1With.isSc1Levels (734); proposition_1_16 (744); IsSc1With.shift (752); IsSck.shift (769); IsScSmooth.shift (789); IsSck.contDiffOn_levels' (802); IsSck.contDiffOn_levels (856); IsSck.contDiffOn_levels_of_le (862); IsSck.contDiffOn_base (868); IsScSmooth.contDiffOn_levels (874); proposition_1_19 (884); corollary_1_20 (919); corollary_1_20_base (932); IsScSmooth.of_contDiffOn_levels (943); IsScSmooth.of_contDiffOn_finiteDimensional (954).

Polyfold/Chapter1/LevelsWeak.lean. ScBanach.castCLE (96); ScBanach.inclLE (111); ScBanach.toBase_inclLE (114); ScBanach.preimage_inclLE_lvlSet (127); ScBanach.shiftAdd (174); ScBanach.shiftAddZeroIso (185); ScBanach.shiftAddShiftIso (197); ScBanach.toBase_shiftAddShiftIso_symm (209); ScOperator.zeroOp (230); ScOperator.addOp (235); IsSck.zero (244); IsSck.add (248); IsSck.sum (254); cmm_eval_eq (272); hasFDerivAt_cmm_eval (279); IsDerivTower (322); levelDerivTower (336); baseDerivTower (342); contDiffOn_levels_min (348); levelDerivTower_toBase (368); baseDerivTower_toBase (405); isDerivTower_of_contDiffOn_levels (419); IsDerivTower.eq_of_toBase (450); ElemData (467); ElemData.Good (483); ElemData.dom (489); ElemData.map (492); ElemData.mem_lvlSet_dom (496); ElemData.ρ₀ (502); ElemData.ρ (505); ElemData.ρ₀_incl (515); ElemData.map_toBase (532); ElemData.levelMap_eq (552); ElemData.levelMap_incl_eq (565); ElemData.isSc0 (584); ElemData.deriv (629); ElemData.deriv_apply (641); ElemData.deriv_toBase (651); ElemData.hasFDerivWithinAt_level (679); ElemData.continuousOn_deriv (719); ElemData.isSc1Levels (740); ElemData.isSc1With (746); IsScDomain.of_eq (753); IsSck.of_eq_set (763); ScOperator.tangentX (774); ScOperator.tangentH (779); ElemData.shiftData (809); ElemData.derivData (813); ElemData.slotData (818); ElemData.Good.shiftData (823); ElemData.Good.derivData (828); ElemData.Good.slotData (836); ElemData.tangentSet_dom (845); ElemData.tangentMap_eqOn (854); ElemData.isSck (900); ElemData.base (935); ElemData.base_dom (938); ElemData.base_good (942); proposition_1_19_text (949); IsScSmooth.of_contDiffOn_levels_text (973); corollary_1_20_text (984); IsScSmooth.of_contDiffOn_levels_finiteDimensional_text (1005).

Polyfold/Chapter1/Properties.lean. IsSc1With.exists_delta_deriv (69); IsSc1With.continuousOn_deriv_clm (93); IsSc1With.contDiffOn_base (120); IsSc1With.scOperator_deriv (130); proposition_1_19_zero (165); corollary_1_20_zero (233).

Polyfold/Chapter1/ScBanach.lean. ScScale (84); ScScale.toBase (104); ScScale.toBase_injective (112); ScScale.inclK (117); ScScale.toBase_inclK (126); ScScale.smooth (135); ScScale.mem_smooth_iff (137); ScScale.smoothLvl (141); ScScale.exists_inclK_of_mem_smoothLvl (148); ScScale.smoothLvl_submodule_closed_sub (162); ScBanach (171); ScBanach.denseRange_inclK

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